

Proceedings of International Conference On Multidisciplinary Innovations for Global Sustainability and development



MIGS-2025

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About the University

G H Raison University, Amravati (GHRUA) is a prominent private institution dedicated to fostering academic excellence and empowering students to succeed in their chosen fields. Established in 2018 under Maharashtra Act No. XXIX, the university is recognized by the University Grants Commission (UGC) and offers a wide range of programs at the Diploma, Undergraduate (UG), Postgraduate (PG), and Doctoral (PhD) levels. With a strong commitment to a knowledge-driven, global economy, GHRUA equips students with cutting-edge education to prepare them for successful careers in various disciplines.

The university is strategically located on a naturally green campus near Nimbhora, Anjangaon Bari Road, Badnera, Amravati 444701, offering easy access to all parts of the city. The campus is designed to provide a conducive and environment-friendly atmosphere, supporting the holistic development of students.

GHRUA offers diverse programs in fields such as Science, Engineering & Technology, Commerce & Management, Pharmacy, Hotel Management, and Law. These programs are designed to meet the demands of modern industries, with a particular focus on contemporary technologies such as Artificial Intelligence, Machine Learning, and Data Science. With its state-of-the-art infrastructure and rigorous academic standards, GHRUA is positioned as a leading educational institute in Central India, dedicated to nurturing the next generation of leaders, innovators, and professionals.

About Conference

The International Conference on Multi disciplinary Innovations for Global Sustain ability and development (MIGS-2025), organized by G H Rasoni University, Amravati, aims to unite researchers, academicians, industry professionals, and policymakers on a single platform to address global sustain ability challenges through innovation and collaboration. The international conference focuses on advancing sustainable practices across diverse fields, including Advanced Engineering Technologies, Artificial Intelligence and Climate Modeling, Smart Cities and IoT, Green Energy Solutions, Blockchain Technologies, and Sustainable Transport Infrastructure. Emphasizing datadriven decision-making, ethical technology use, and crosssectoral collaboration, MIGS-2025 will provide valuable insights through keynote sessions, technical presentations, and interactive discussions. The event serves as a dynamic platform for sharing knowledge, fostering interdisciplinary partnerships, and developing actionable strategies for achieving long-term sustain ability goals. For further information visit: <https://ghrui.edu.in/MIGS-2025/>

Publication Information

The proceedings of MIGS-2025 are being published in collaboration with STM Journals. Additionally, selected papers are under consideration for publication in subject-specific journals, including those indexed in Scopus and listed in the UGC Care list.



Conference Objectives

The International Conference on Multidisciplinary Innovations for Global Sustainability and Development (MIGS-2025) aims to bring together experts, researchers, and industry leaders to discuss groundbreaking solutions for sustainable development. The primary objectives of the conference are:

- To explore advanced engineering technologies that drive sustainable development and climate action.
- To discuss the role of Artificial Intelligence and Machine Learning in climate modeling, data-driven decisions, and sustainability efforts.
- To examine the integration of Smart Cities and Internet of Things (IoT) for fostering environmental sustainability.
- To explore the potential of Blockchain Technology in enhancing transparency and accountability in sustainability practices.
- To share innovations in sustainable transport and mobility solutions, with a focus on electric vehicles and infrastructure.
- To discuss the impact of Corporate Governance, ESG (Environmental, Social, and Governance), and Sustainability Compliance in global industries.
- To examine Human Rights, Climate Justice, and the legal frameworks needed to support equitable climate action.
- To review Renewable Energy Policies and Sustainable Resource Management strategies for reducing environmental impact.
- To explore the Circular Economy, Waste Management, and Sustainable Development Laws for promoting sustainability in business practices.
- To highlight emerging innovations and business models for sustainability, particularly in emerging markets and green technologies.

Apart from these core topics, participants are encouraged to register for other topics related to sustainability, innovation, and global challenges.

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On behalf of GH Raison University, Amravati, it gives me immense pleasure to welcome all participants to the International Conference on Multidisciplinary Innovations for Global Sustainability and Development (MIGS-2025).

MIGS-2025 provides an inclusive platform for intellectual exchange that spans disciplines and borders. In alignment with our commitment to excellence in education and research, this conference brings together thought leaders to discuss sustainable solutions for societal development.

We are grateful for the presence of our esteemed keynote speakers and guests, whose academic and professional journeys continue to inspire generations. I commend the efforts of the organizing committee, contributors, and all stakeholders involved in making this event a reality.

Wishing all a productive and memorable conference.

Convenor's Message

It is with great pleasure and pride that we welcome you to the International Conference on Multidisciplinary Innovations for Global Sustainability and Development (MIGS-2025), organized by GH Rasoni University, Amravati. MIGS-2025 represents a dynamic platform for researchers, academicians, industry professionals, and policymakers to exchange ideas and explore innovative approaches to address global challenges.

We are privileged to host a distinguished lineup of keynote speakers and guests, whose presence and insights enrich the academic spirit of this conference. Their contributions reflect the depth and diversity that this event aims to promote.

I would like to extend my sincere thanks to all the participants, authors, reviewers, and organizing members whose efforts have made this event a success. May the proceedings of MIGS-2025 inspire future research and collaboration across disciplines.



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Submission ID: CSE-199

Breast Cancer Detection Using Machine Learning: A Comparative Analysis of Supervised Learning Algorithms

Mr. Niral Mahajan¹, Prof. Ashwini Kukade²

^{1,2}G H Raison College of Engineering, Nagpur, India

Email: niral.mahajan.aiml@ghrce.raisoni.net

ashwinikukade90@gmail.com

Breast cancer remains a leading cause of mortality among women worldwide, making early and accurate diagnosis essential. This research explores the application of machine learning algorithms—including Logistic Regression, SVM, Naïve Bayes, KNN, and Random Forest—on the Wisconsin Breast Cancer Dataset for effective tumor classification. Key pre-processing steps such as missing value handling, feature scaling, and dimensionality reduction were employed to improve model performance. Models were evaluated using metrics like accuracy, precision, recall, and F1-score. Among the tested models, Naïve Bayes achieved the highest accuracy and reliability. The study highlights the significance of feature selection in optimizing diagnostic efficiency and supports the use of AI-based methods for scalable, non-invasive breast cancer detection.

Submission ID: CSE-112

A Review - CNN Use in Crime Detection

Sarita Sawale¹, Ambar Ubale², Vishvdeep Pundge³

¹Assistant Professor, ^{2,3}Student,

Department of Information Technology, Anuradha Engineering College, Chikhli,
Maharashtra, India

Email: ambarubale3333@gmail.com, pundgebishvdeep@gmail.com

CCTV is widely used for monitoring around the world. Nevertheless, numerous offences are recorded without appropriate systems for detecting and controlling criminal conduct. This project focuses on the perception of crimes from video data obtained from CCTV, and further addresses recognizing crimes based on captions by capturing criminal-related keywords such as knife, thief, fire, attacks, etc.

The proposed system is also safer than traditional CCTV-based video storage, as it enables storage of text files, which are encrypted using robust encryption algorithms. Real-time videos, photos, and notifications about crimes are sent to human supervisors to facilitate responsible action.

Submission ID: CSE-193

The Drawbacks of Artificial Intelligence: Ethical, Economic, and Social Implications

Durga Yadav¹, Ajay Kumar²

G. H. Raison University, Amravati, Maharashtra, India

Email: durgayadav1014@gmail.com, ajay.kumar@ghrua.edu.in

Artificial intelligence (AI) has fundamentally transformed numerous industries, providing unparalleled efficiency gains, fostering innovation, and enabling extensive automation. However, the swift proliferation of AI technologies across diverse facets of human life presents substantial challenges.

This paper examines the principal drawbacks of AI, emphasizing ethical considerations, potential job displacement, algorithmic bias in decision-making processes, security vulnerabilities, and privacy erosion. Through a rigorous analysis of current applications and prevailing trends, this research seeks to illuminate the inherent challenges posed by AI and underscore the critical need for responsible development and deployment strategies.

Submission ID: CSE-060

AI-Powered System Cybersecurity Operations and Incident Response: Survey Study

Pranita Chaudhary¹, Varad Sawant², Anurag Karpe³, Lokesh Kad⁴, Pratham Kubetkar⁵¹Pimpri Chinchwad College of Engineering, Pune, India

Cyber incidents have become more sophisticated and frequent, which emphasizes the importance of automated AI systems for the proactive response to cyber incidents. This paper suggests a comprehensive analysis of the new trends in the field of AI-based cybersecurity, using three basic blocks: detection of network anomalies, fine-tuning of large language models (LLMs) for the classification of emails, and detection of malware.

In the detection of network anomalies, methods such as ensemble learning, dimensionality reduction, and deep learning are reviewed to guarantee high accuracy and low false alarm rates in real-time network traffic monitoring. For email classification, the study explores the fine-tuning of domain-specific large language models like GPT-4, BERT, and RoBERTa, highlighting their advantages over classic CNN-based approaches in detecting spam and phishing emails.

The malware detection block considers models based on static and dynamic analysis, hybrid approaches, and graph neural networks, achieving detection accuracies often exceeding 99% across environments such as IoT and mobile platforms. The review also identifies major challenges including data quality, computational efficiency, and model interpretability, while pointing to promising trends toward self-adapting, self-learning, and explainable AI models to enhance cyber incident response.

Submission ID: EE-202

Design and Implementation of Real-Time Optimal Power Flow for Transmission Network

Mr. Bhushan B. Taksande, Dr. Swapnil B. Mohod, Prof. Bhushan H. Band

Prof. Ram Meghe College of Engineering & Management, Badnera-Amravati

Emails: bhushantaksande3@gmail.com, swapnil.mohod@prmceam.ac.in,
bhushan.band@prmceam.ac.in

The efficient transmission of power is a significant issue we face collectively today, especially with the increased deployment of renewable energy technologies. The report entitled "Design and Implementation of Real-Time Optimal Power Flow for Transmission Network" discusses a real-time optimization system for power generation and distribution.

The system develops a Distributed Control System that facilitates seamless communication between generators and the control system. Optimal Power Flow (OPF) algorithms are implemented to lower production costs while maintaining power balance on the grid. Practical use of real-time feedback control enables continuous adjustments, supports grid frequency regulation, and enhances overall stability.

A key benefit of this system is its flexibility in managing variable renewable energy inputs without relying on forecast predictions, thereby improving adaptability and sustainability. The project continuously monitors voltage and current data from sensors and performs MATLAB simulations based on real-time applications for both micro-grid and large-grid integration. Results demonstrate favorable outcomes and scalability. This system presents a proactive approach to the increasing complexity of energy management, leading to a more efficient, reliable, and sustainable power network.

Submission ID: EE-108

Battery Management System in Electric Vehicles

Himanshu¹, Dr. Bandana², Jai Bhardwaj³, Harsh Tyagi⁴

Department of Electrical and Electronics Engineering,
KIET Group of Institutions, Ghaziabad, Uttar Pradesh, India

Battery management systems (BMS) are crucial for the functioning of electric vehicles. They help monitor how batteries charge and discharge, making operation smoother and more cost-effective. These systems ensure that the battery stays safe, dependable, and can last longer without damage.

To keep track of the battery's well-being, different factors like voltage, current, and ambient temperature are closely monitored using both analog and digital sensors along with microcontrollers. This paper discusses several key points including the state of charge, health, lifespan, and maximum capacity of a battery.

By examining these aspects, future challenges can be identified and potential solutions proposed. With ongoing advancements in BMS technology, exciting developments in electric vehicle battery management are expected ahead.

Submission ID: EE-003

Design of Power-Efficient Dynamic Comparator for High-Speed ADC

Mrs. Teladarla Susmitha, Research Scholar

Dr. Adarsh Kumar, Associate Professor

Department of Electronics and Electrical Engineering, Sandip University, Nashik

Designing comparators that meet advanced technology requirements poses significant challenges. The comparator circuit typically consists of a preamplifier, a judgment circuit (latch), and a post-amplifier. This work focuses on the design of a low-power, high-speed dynamic comparator for high-speed Analog-to-Digital Converters (ADCs), which demands expertise in both analog circuit design and digital conversion technology.

In ADC systems, comparators are crucial components that directly affect the accuracy and speed of conversion. The goal is to achieve fast response times, low power consumption, and high precision. A dynamic comparator with these attributes enables higher-resolution ADCs by providing accurate decision-making at finer voltage levels.

For high-resolution ADCs, the comparator must reliably detect very small voltage differences, requiring precision design techniques. Utilizing dynamic comparators enhances resolution and performance in data converters by improving the accuracy of voltage discrimination.

Submission ID: EE-109

Design and Implementation of an Automatic Street Light Control System

Dr. Ruchika Singh¹, Dr. Jyoti Srivastava², Vashu Swami³, Mukund Verma⁴, Sohit⁵

^{1,2}Associate Professors, ^{3,4,5}B.Tech Students

Department of Electrical and Electronics Engineering,
KIET Group of Institutions, Ghaziabad

`ruchika.singh@kiet.edu`, `jyoti.srivastava@kiet.edu`

This paper discusses the development of an automatic street light control system, detailing the creation of a smart embedded system aimed at energy-saving for streetlights. Traditional street lighting methods, which rely on fixed schedules, often waste energy by illuminating streets unnecessarily.

This work offers a solution by automating the lighting process. The system employs motion sensors to detect activity and light sensors to determine ambient light levels. A microcontroller acts as the system's core, using software written in C to control the lights based on sensor data. A functional prototype of the system has been constructed and tested successfully.

Submission ID: LAW-001

International Convention on Sustainable Development

Dr. Nitya Nand Pandey

GH Raisonni University, Amravati

An important turning point in the history of global environmental regulation occurred in 1972. The United Nations Conference on the Human Environment, held in Stockholm in June 1972, resulted in several agreements and initiatives aimed at global environmental preservation. The subsequent Stockholm Declaration, often referred to as the "Magna Carta of the environment," was the first international treaty to explicitly affirm the right to a healthy environment as a fundamental and autonomous right.

This declaration laid the foundation for global environmental norms and guidelines that now serve as references for nations worldwide. During the conference, India's then Prime Minister Mrs. Indira Gandhi emphasized the unique concerns of developing countries, underlining that development was not only a goal but a means to improve living standards, access essential resources, combat desertification, and render uninhabitable areas livable.

These perspectives remain vital at the United Nations World Conference on Sustainable Development. They underscore the necessity of ensuring sustainability for all species and the planet itself while pursuing human advancement. The concept of "Sustainable Development" now underpins many international environmental efforts. In environmental law, it promotes responsible development that meets current needs without compromising the ability of future generations to meet theirs. Sustainable development encourages a balanced approach, integrating social, economic, and environmental considerations to ensure harmony between human activity and nature.

Submission ID: EE-179

A High Voltage Gain Bidirectional DC-DC Converter for V2G and G2V Hybrid EV Charger

Mr. Shubham Rambhau Ahirkar¹, Prof. C.M. Bobade²¹Student, G H Raisonni University, Amravati²Head of Department, G H Raisonni University, Amravati

For usage in electric vehicle (EV) applications, this research proposes a novel broad-range bidirectional DC-DC converter with an enhanced voltage gain transfer ratio. The converter offers low voltage stress on switches, a high utilization factor, and high efficiency while maintaining a common electrical ground between the input and output terminals.

Bidirectional power flow in grid-connected Vehicle-to-Grid (V2G) and Grid-to-Vehicle (G2V) modes is used to assess the efficacy of the suggested EV charger. In the DC-DC and DC-AC stages, the converter employs a dead-beat current controller, which responds quickly, accurately, and smoothly.

Lastly, simulation results are demonstrated for a 1000 W, 380 V prototype in a closed-loop system with bidirectional power flow and the suggested dead-beat controllers. The theoretical analysis and the practicality of this structure are supported by the results obtained.

Submission ID: EE-063

Novel Method Based on TEO and Decision Tree Approach for Power Transformer Protection

Patil Bhushan Prataprao¹, Dr. Shah Paresh Jaychand²¹R. C. Patel Institute of Technology, Shirpur, Maharashtra, India²Shri Balaji Institute of Technology and Management, Betul, Madhya Pradesh, India

Ensuring effective transformer protection is crucial for maintaining power system stability. This paper introduces a robust fault detection and classification approach using the Teager Energy Operator (TEO) and Decision Tree (DT)-based classifiers. The method efficiently distinguishes internal faults from external disturbances, including magnetizing inrush currents and overexcitation. A transformer model is designed in MATLAB Simulink, simulating various fault conditions, inrush scenarios, overexcitation, and normal operations under different load levels. The detection and classification processes are executed in Python, where TEO-based feature extraction provides input for DT classifiers. To improve accuracy, GridSearchCV optimization is employed for fine-tuning parameters and evaluating performance across diverse training and testing datasets. The proposed technique achieves reliable event classification within half a cycle.

Submission ID: EE-014

Technical Mobility in the Field of Agriculture for Enhancement of Indian Crop Production

Prof. Ganesh D. Khadsane (Assistant Professor), **Prof. Tushar Nakod** (Assistant Professor)

G H Raison University, Amravati, Maharashtra, India

Drone technology is one of the most effective technologies being used today, particularly in the field of agriculture. This technology assists with various tasks in agriculture. This paper mainly focuses on the use of drone technology in agriculture, highlighting key aspects and differences in the modification of drone structures according to specific requirements and applications in various fields.

It is evident that drone technology is highly useful, but it can also be dangerous depending on its usage. This paper examines different aspects of drone combinations and the latest modifications required to improve their effectiveness for farmers. The focus is primarily on farmers in India, many of whom may not be well-versed in the latest agricultural technologies but are nonetheless interested in adopting new methods to increase crop production, protect crops from harmful insects, and control pests.

This paper is especially useful for farmers who want to understand the latest drone technologies designed for agricultural use. It discusses a combination of technologies that can be integrated into drones to enhance agricultural practices while considering cost-effective solutions for Indian farmers who may not have the financial means to afford expensive equipment.

Drone technology is making significant strides in agriculture, offering a wide range of uses that help increase efficiency, reduce costs, and improve productivity. The paper also highlights the latest technologies implemented for better crop production and easier monitoring for farmers.

Submission ID: MBA-067

Advancements in AI-Based Tools and Digital Applications: Creating Revolution for New-Age Urban Infra Industries

Arini Roychoudhury^{*1}, Nitin Kumar Sahu², Atul Kumar Sahu³

¹Faculty, Thakur Global Business School, Mumbai - 400101, Maharashtra, India

^{2,3}Faculty, Department of Industrial and Production Engineering,
Guru Ghasidas (Central) Vishwavidyalaya, Bilaspur - 495009, Chhattisgarh, India
***roychoudhuryarini@gmail.com**

AI-based Management Information Systems (MIS) advancements alongside digital applications are reshaping new-age infrastructure business environments through enhanced innovation, operational efficiency, and scalability. Moreover, IoT sensors, drones, and smart construction technologies deliver unparalleled project visibility and control through digital applications. Real-time monitoring tools for construction sites and maintenance operations, as well as asset management, lead to more efficient and cost-effective operational processes.

AI and digital technologies promote stakeholder collaboration through seamless data sharing and communication platforms, connecting contractors with clients and regulatory bodies. The integration of AI-driven solutions provides organizations with increased transparency, helps maintain compliance standards, and expedites decision-making workflows.

AI integration into Management Information Systems facilitates real-time data analysis and predictive analytics, improving decision-making processes for businesses and helping optimize resource use through informed decisions. In this study, the integration of AI tools and digital technologies in infrastructure projects is explored, highlighting key applications, challenges, and the future potential of AI in infrastructure businesses. The study critically analyses the impact of AI and digital technologies that have initiated a revolution by transforming infrastructure businesses into more flexible entities capable of effectively meeting evolving market demands.

Submission ID: CE-039

To Investigate the Challenges Faced During Construction of Flyover

Shakeel Ahmad¹, Apeksha Choudhary²

¹P.G. Student, ²Assistant Professor, Department of Civil Engineering, G H Raisoni University, Amravati, Maharashtra, India

shakeel053@gmail.com, choudharyapeksha016@gmail.com

The flyover discussed in this study is constructed on NH-53, Nagpur-Amravati Road, Maharashtra. The structure is a 4-lane single pier precast segmental box-type superstructure. During the project, a 132 kV transmission line caused significant hindrance between spans P11 and P12 on the Wadi flyover, affecting three spans and delaying the construction process. To address this, a 54 m high double circuit (DC) 132 kV line with a 300 m basic span was selected.

Traditionally, lattice and compact towers have supported transmission and distribution lines in India. However, due to increasing population density and scarcity of available land, steel pipe poles provide a viable alternative. Tubular poles, made from stronger materials like steel, have gained popularity owing to their smaller footprint, making them suitable for restricted spaces such as highways, rail tracks, and rivers.

The advantages of pole structures include reduced right-of-way requirements, improved visual appearance, fewer components with faster installation, lower installation costs, better reliability under extreme conditions, design flexibility, ease of future maintenance, increased safety, and resistance to vandalism.

In this project, the 132 kV line impacted the project's completion time and cost. Monopole construction proved to be a suitable solution due to its ease of installation and compatibility with the surrounding environment. Despite implementing the solution, the project faced a four-month delay due to the hindrance caused by the transmission line.

Submission ID: CE-073

Seismic Response and Performance of Multistorey Prefabricated Building Structures

Mr. Kalyani Meshram¹, Dr. P. L. Naktode², Dr. Aasif Baig³

¹Research Scholar, ^{2,3} Assistant Professors,
Structural Engineering, Department of Civil Engineering,
Tulsiramji Gaikwad Patil College of Engineering and Technology, Nagpur, Maharashtra,
India

This study examines the seismic performance of multi-storey modular buildings constructed using prefabricated modules assembled on-site. A primary challenge in modular construction is the creation of discontinuous diaphragms where modules are connected at various points, potentially affecting structural integrity. The research focuses on assessing the influence of in-plane diaphragm strength and stiffness on the seismic response of modular structures.

A streamlined methodology for evaluating diaphragm service stiffness is proposed, accounting for shear deformation in both diaphragm connections and individual modules. Numerical models of a four-storey, four-bay reinforced concrete modular building are developed to simulate seismic behavior.

The research highlights the growing adoption of prefabricated construction methods, which offer environmental benefits, enhanced material efficiency, faster construction, and improved worker safety compared to traditional site-built techniques. The study also examines the life cycle analysis, energy use, and thermal behavior of modular prefab systems under seismic loading.

Further, the seismic resilience of two modular structural systems—one with bracing and the other with shear walls—is evaluated through the study of a 20-storey building. This research aims to contribute to the development of improved seismic design practices for modular structures, addressing gaps in the current standards such as IS 1893-2016 (Part I).

Submission ID: CE-096

Analysis of Irregular Shape of Building in Earthquake Prone Area by Time History Analysis

Dr. P.L. Naktode¹, Dr. Aasif Baig², Mr. Sandip Kumbhare³

principal@tgpcet.com, aasif.civil@tgpcet.com,
sandipkumbhare1998112233@gmail.com

^{1,2}Professor, ³PG Student,

Structural Engineering, Department of Civil Engineering,
Tulsiramji Gaikwad Patil College of Engineering and Technology, Nagpur, Maharashtra

Seismic waves are created when strain energy trapped in the earth's crust is quickly released, resulting in an earthquake. During an earthquake, ground motion can cause structural damage. Understanding the properties of ground motion is critical for preventing structural damage. Peak ground acceleration, frequency content, and duration are the three most important dynamic features of an earthquake. If non-linear behaviour is avoided, linear time history analysis outperforms modal response spectrum analysis in every way. This method requires more computing power to calculate the response at discrete times, but it preserves the relative signs of response amounts in the response histories, which is an intriguing benefit.

Submission ID: CE-107

Impact of P-Delta Effect in Ground Shear on Building Structures Considering Linear Static

Dr. P. L. Naktode¹, Dr. Aasif Baig², Mr. Amit Bhovate³

principal@tgpcet.com, aasif.civil@tgpcet.com, amitbhovate31@gmail.com

^{1,2}Professor, ³PG Student,

Structural Engineering, Department of Civil Engineering,
Tulsiramji Gaikwad Patil College of Engineering and Technology, Nagpur, Maharashtra

This paper examines P-Delta effects on high-rise structures, paying special attention to their influence on axial forces, moments, and displacements. P-Delta effects increase with building height, particularly in thin, flexible structures, and necessitate second-order analysis. The study, which compares P-Delta and Linear Static analysis across twelve models with varying storey heights, reveals that displacements and axial forces grow exponentially with increasing height.

Additionally, P-Delta analysis emphasizes the importance of accounting for second-order effects during the design of high-rise buildings, especially reinforced concrete structures with inflexible connections. Although P-Delta analysis yields more accurate outcomes for tall structures, the study advocates for the combined use of Linear Static and P-Delta analyses in design practices. The findings underscore the necessity of considering P-Delta effects to prevent potential structural failures, thereby ensuring the stability and safety of tall, flexible buildings.

Submission ID: CE-135

Review on Mixture Design Methods for Self-Compacting Concrete with Various Cementitious Materials

Pranav Vishnu Chavan¹, Dr. Sujesh Ghodmare²

¹Scholar, ²Associate Professor Civil Engineering Department,
School of Engineering and Technology,
GH Rasoni University, Amravati, Maharashtra, India

The development of Self-Compacting Concrete (SCC) has marked a significant paradigm shift in modern concrete technology by enhancing workability, durability, and structural integrity. SCC is characterized by its ability to self-flow, fill formworks, and consolidate under its own weight without external vibration. Since its introduction in 1988, numerous mixture design methodologies have been proposed to optimize its performance while addressing environmental concerns associated with cement production.

These methodologies are classified into five principal categories: empirical design, compressive strength-based methods, aggregate packing approaches, statistical factorial modeling, and rheology-based models. Each approach provides a distinct framework for optimizing SCC mixtures with varying cementitious materials, offering advantages in material efficiency, sustainability, and performance enhancement.

However, challenges persist regarding the balance between workability, mechanical strength, and environmental sustainability. This review critically analyzes the evolution of SCC mixture design methodologies, emphasizing their applicability in achieving optimal mechanical and durability properties while simultaneously reducing carbon footprints.

Submission ID: FSD-068

Photocatalytic Degradation Study Using Composite of Graphene Oxide and Caesalpinia Bonduc Ag/Cu Nanoparticles

Mangesh S. Dhore¹, Prachi M. Umbare¹, Fiza G. Mukadam¹, Qurratulain Kausar Ansari², Pawan P. Kalbende²

¹Department of Forensic Science, G.H. Rasoni University, Anjangaon Bari Road,
Amravati, Maharashtra 444701, India
Email: mangeshdhore2@gmail.com

²Department of Chemistry, Jagadamba Mahavidyalaya, Achalpur City-444806,
Dist.-Amravati, India
Email: pawankalbende@gmail.com

The Graphene oxide (GO) was synthesized using the modified Hummers' method. The synthesized Graphene oxide was confirmed by X-ray diffraction (XRD), infrared spectroscopy (IR), and Raman spectroscopy. The Ag/Cu nanoparticles were synthesized using the novel plant Caesalpinia Bonduc in a muffle furnace.

The composite of GO-Caesalpinia Bonduc Ag/Cu nanoparticles was synthesized by combining Graphene oxide and Caesalpinia Bonduc Ag/Cu nanoparticles using an ultrasonicator with a pulse rate of 5 minutes on/off mode for about 20 minutes. The final

composite was purified using 200 mL water, hydrochloric acid, and ethanol, followed by filtration through a 0.45-micron cindered glass filtration assembly.

A photocatalytic degradation study of GO-Ag/Cu nanoparticles was conducted using methylene blue dye. A 25 mg sample of the product was added to 100 mL of 140 ppm methylene blue under sunlight with constant stirring. The percent degradation was analyzed using UV-Visible spectroscopy, and the degradation efficiency was found to be 89.75

Submission ID: Pharma-009

Preparation and Evaluation of Solid Lipid Nanoparticles Containing Repaglinide

Vedanshu Malviya*, Srikant Pande

Department of Pharmaceutics, Vidyabharti College of Pharmacy, Amravati-444602

malviyavedanshu@gmail.com

This research project focuses on the development of Repaglinide solid lipid nanoparticles (SLN) for use as an oral delivery system. To achieve this goal, the high-pressure homogenization technique was employed to create the Repaglinide SLNs, with Lipoid 90G as the lipid matrix and oleic acid serving as the surfactant. Various analyses were conducted to investigate the physical and chemical interactions of the drug, including Fourier Transform Infrared (FTIR) spectroscopy and Differential Scanning Calorimetry (DSC).

Additionally, the physicochemical properties of the SLNs were examined using Scanning Electron Microscopy (SEM). A comprehensive stability study was undertaken. The findings revealed that the Repaglinide SLNs exhibited a spherical shape, with the optimized formulation's SLN having a diameter of 149.9 ± 5.9 nm, a polydispersity index of 0.371 ± 0.042 , a zeta potential of -30.9 ± 2.6 mV, a loading capacity of $1.89 \pm 0.22\%$, and an encapsulation efficiency of $91.80 \pm 1.7\%$. The formulation displayed good stability under conditions of $25^\circ\text{C} \pm 2^\circ\text{C}$ and $60\% \text{ RH} \pm 5\% \text{ RH}$.

Furthermore, the sustained release effect of Repaglinide contributed to its enhanced stability and potential for reducing dosing frequency while maintaining pharmacological efficacy. These results indicate the promising potential of Repaglinide as an oral formulation.

Submission ID: ANY-046

Cracking the Code of Earthquakes: A CNN-LSTM Fusion for Predicting Earthquake Magnitudes

Bhumi M. Singh¹, Bhumika S. Agrawal², Chahak M. Anchliya³, Devansh R. Deogirkar⁴

^{1,2,3,4}Ramdeobaba University, Formerly Shri Ramdeobaba College of Engineering and Management, Gittikhadan, Nagpur

singhbm@rknec.edu, agrawalbs1@rknec.edu, anchliyacm@rknec.edu,
deogirkardr@rknec.edu

Earthquakes are among the most devastating natural disasters, posing significant risks to human lives and infrastructure. Traditional seismological techniques often struggle to capture the complex, non-linear relationships in seismic data. Recent advancements in artificial intelligence and deep learning have introduced promising new methods

for earthquake magnitude prediction. This study proposes a hybrid deep learning approach that combines Convolutional Neural Networks (CNNs) with Long Short-Term Memory (LSTM) networks to enhance prediction accuracy. Using a dataset of historical seismic events, the proposed model demonstrates high accuracy and strong predictive performance. The model's effectiveness is evaluated through multiple statistical metrics, highlighting its ability to recognize seismic patterns. The findings suggest that deep learning techniques can significantly improve earthquake forecasting, contributing to better disaster preparedness strategies. Future research should explore integrating additional geospatial and satellite data to further refine prediction accuracy.

Submission ID: Any-020

Optimizing Dye Removal Using Chrysanthemum-Derived Bioadsorbent: A Study

Saloni Agarwal, Dr. Harshita Jain
Amity University Noida,
saloniaga@gmail.com

Traditional methods of disposing floral debris in water bodies negatively impact the environment and water quality worldwide. These methods not only contaminate waterways but also create environmental problems requiring costly cleanup. The situation worsens due to the emission of toxic and carcinogenic dyes by many industries. Adsorption is an effective method for removing dyes from industrial wastes and aqueous solutions.

This study uses White Chrysanthemum (*Chrysanthemum morifolium*) flower waste powder as a bioadsorbent to remediate methyl orange dye, a common industrial contaminant. Batch adsorption tests were conducted using dried botanical powder to extract methyl orange from water, considering variables such as pH, temperature, contact time, initial adsorbate concentration, and bioadsorbent dose. A central composite design (CCD) technique of response surface methodology (RSM) optimized these parameters.

Dye adsorption equilibrium was achieved after 60 minutes of contact. Adsorption behavior was modeled by Freundlich and Langmuir isotherms over a concentration range of 20–100 mg/L. Using 30 mg/L methyl orange dye and 1.5 g of biomass, 93.7% removal efficiency was observed after 60 minutes at 25 °C and pH 6. X-ray diffraction (XRD) analysis revealed a crystalline phase with some amorphous carbon (18–35° 2 θ). Fourier-transform infrared spectroscopy (FT-IR) identified alkene, hydroxyl, and carbonyl functional groups essential for dye adsorption.

Kinetic analysis indicated the adsorption follows a pseudo-second-order model. The Langmuir and Freundlich isotherms confirmed monolayer adsorption of methyl orange on chrysanthemum petals.

The study concludes that this low-cost bioadsorbent offers a sustainable solution to reduce floral waste and environmental contamination. This eco-friendly, cost-effective, and biodegradable method responsibly manages flower waste while effectively removing methyl orange from polluted water.

Breast Cancer Detection Using Machine Learning: A Comparative Analysis of Supervised Learning Algorithms

Mr. Niral Mahajan¹, Prof. Ashwini Kukade²

¹ G H Raison College of Engineering, Nagpur, India
Email: niral.mahajan.aiml@ghrce.raisoni.net

² G H Raison College of Engineering, Nagpur, India
Email: ashwinikukade90@gmail.com

Abstract

Breast cancer remains a leading cause of mortality among women worldwide, making early and accurate diagnosis essential. This research explores the application of machine learning algorithms—including Logistic Regression, SVM, Naïve Bayes, KNN, and Random Forest—on the Wisconsin Breast Cancer Dataset for effective tumor classification. Key pre-processing steps such as missing value handling, feature scaling, and dimensionality reduction were employed to improve model performance. Models were evaluated using metrics like accuracy, precision, recall, and F1-score. Among the tested models, Naïve Bayes achieved the highest accuracy and reliability. The study highlights the significance of feature selection in optimizing diagnostic efficiency and supports the use of AI-based methods for scalable, non-invasive breast cancer detection.

Keywords: Breast Cancer, Machine Learning (ML), Logistic Regression, Support Vector Machine (SVM), Gaussian Naïve Bayes, K-Nearest Neighbours (KNN), Random Forest

1. INTRODUCTION

Breast cancer remains one of the most prevalent and life-threatening diseases affecting women worldwide, accounting for a substantial proportion of cancer-related deaths. According to the World Health Organization (WHO), breast cancer is the most frequently diagnosed cancer globally, with millions of new cases emerging each year. Due to its biological complexity and heterogeneity, early and accurate diagnosis is vital for improving survival rates and guiding effective treatment strategies. Traditional diagnostic techniques, such as mammography, biopsy, and histopathological analysis, though widely adopted, are often time-consuming, costly, and susceptible to human error. In response to these limitations, Artificial Intelligence (AI) and Machine Learning (ML) have emerged as transformative technologies in modern medical diagnostics, enabling faster, more accurate, and automated detection processes.

Machine learning techniques have demonstrated significant potential in detecting and classifying cancerous growths through medical imaging, cellular morphology, and structured patient data. These algorithms are capable of analyzing large datasets, revealing complex patterns, and accurately differentiating between malignant and benign tumors. The Wisconsin Breast Cancer Dataset (WBCD) is commonly used in machine learning research due to its rich set of numerical features that describe various tumor properties. In this research, multiple supervised learning algorithms—such as Logistic Regression, Support Vector Machine (SVM), Gaussian Naïve Bayes, K-Nearest Neighbors (KNN), and Random Forest—are applied to evaluate their performance in breast cancer detection.

The research methodology involves rigorous data preprocessing, including missing value imputation, feature selection, and data normalization, to enhance model accuracy and generalization. Model performance is evaluated using key classification metrics such as accuracy, precision, recall, and F1-

score. Among the algorithms tested, Naïve Bayes exhibit superior performance in terms of classification accuracy and robustness, indicating their strong potential for clinical integration. This study contributes to the evolving landscape of AI-driven healthcare by offering a reliable, scalable, and non-invasive solution for early breast cancer detection.

2. Related Work

Extensive research has been conducted on breast cancer detection using various machine learning (ML) techniques. One of the most widely used datasets in this domain is the Wisconsin Breast Cancer Diagnostic Dataset (WBCD), made publicly available by the UCI Machine Learning Repository [1]. This dataset has played a pivotal role in training and evaluating ML models for distinguishing malignant and benign tumors.

Early foundational work by Wolberg et al. demonstrated the efficacy of ML techniques in classifying tumors using fine needle aspirate samples, highlighting the potential of supervised learning in medical diagnostics [2]. Their research laid the groundwork for subsequent innovations in feature extraction and classification strategies.

Building on these developments, Géron provided a comprehensive guide on implementing machine learning algorithms using modern frameworks like Scikit-learn, Keras, and TensorFlow, which are crucial tools for real-world applications in biomedical domains [3]. These tools were instrumental in implementing various models in our study, including Support Vector Machines (SVM), Random Forest, and K-Nearest Neighbors (KNN).

3. Dataset and Methodology

3.1 Dataset Description

The Wisconsin Breast Cancer Dataset (WBCD) is utilized in this study to classify breast tumors as malignant or benign. The dataset contains 569 instances with 30 numerical features extracted from digitized images of fine needle aspirate (FNA) biopsies. These features describe the cellular characteristics of breast masses, such as radius, texture, perimeter, and smoothness. The target variable (diagnosis) is labeled as Malignant (M) or Benign (B), providing a binary classification problem for machine learning models.

3.2 Data Preprocessing

To ensure high model performance and reliability, the dataset undergoes the following preprocessing steps:

- **Removal of Unnecessary Columns:** Non-relevant attributes such as the ID column are dropped to prevent redundancy.
- **Handling Missing Values:** Any missing or null values are imputed using statistical methods such as mean or median imputation.
- **Encoding Categorical Variables:** The target variable is encoded as Malignant = 1 and Benign = 0 for machine learning compatibility.
- **Feature Scaling:** Standardization or normalization techniques are applied to ensure all numerical features contribute equally to the model's learning process.
- **Splitting the Dataset:** The dataset is divided into training (80%) and testing (20%) subsets to evaluate model generalization.

3.3 Machine Learning Models

Several supervised learning techniques are applied and assessed for the classification of breast cancer:

Machine Learning (ML), a crucial component of Artificial Intelligence (AI), has transformed the field of medical diagnostics by enabling systems to learn from data, detect complex patterns, and make decisions

without the need for explicit programming. ML algorithms are particularly effective in handling large, complex datasets typical in healthcare, where traditional diagnostic methods may be constrained by factors like human bias, cost, or time limitations. In breast cancer detection, ML acts as a powerful tool for differentiating between benign and malignant tumors, assisting healthcare professionals with early diagnosis and treatment planning.

This procedure typically entails feeding historical medical data into an algorithm that learns from these examples, allowing it to generalize and make predictions on new, unseen data. In this study, several supervised learning algorithms are employed to classify breast tumors based on features extracted from the Wisconsin Breast Cancer Dataset (WBCD).

3.4 Algorithms Used in This Research

This study applies and assesses the effectiveness of five commonly used supervised machine learning algorithms: Logistic Regression, Support Vector Machine, Gaussian Naïve Bayes, K-Nearest Neighbors, and Random Forest. A thorough description of each algorithm is presented below:

Logistic Regression (LR):

Logistic Regression is a statistical technique used for tasks involving binary outcomes, where the dependent variable is categorical (such as malignant or benign). It applies a logistic (sigmoid) function to estimate the probability that a given input belongs to a specific category. Despite being linear in form, Logistic Regression remains popular due to its simplicity, efficiency, and ease of interpretation, particularly when there is a roughly linear relationship between the input variables and the target. It is especially useful in scenarios where the data is well-organized, and the features are not strongly correlated with one another.

Support Vector Machine (SVM):

Support Vector Machines are robust classifiers designed to identify the best hyperplane that separates data into two distinct classes while maximizing the margin between them. SVM can be adapted to handle non-linear data through the use of kernel methods, such as the Radial Basis Function (RBF) kernel. Thanks to its strong generalization capabilities, even with small datasets, and its proficiency in high-dimensional spaces, SVM is frequently applied in medical fields. In breast cancer detection, it performs effectively, particularly when dealing with complex relationships between features.

Gaussian Naïve Bayes (GNB):

This algorithm is based on Bayes' Theorem and operates under the assumption that each feature contributes independently to the outcome. In the Gaussian variant, it is presumed that continuous variables are normally distributed. Despite the simplification of assuming feature independence, Gaussian Naïve Bayes is known for its high efficiency and strong performance in classification tasks, making it a reliable choice for real-world applications.

K-Nearest Neighbors (KNN):

KNN is an instance-based learning technique that assigns a class to a data point by analyzing the classifications of its closest 'k' neighbors. As a non-parametric algorithm, it doesn't make assumptions about the data distribution. Its simplicity and flexibility make it useful, especially when the dataset has complex structures. However, prediction can be slow when working with large datasets due to the need to compute distances for each query.

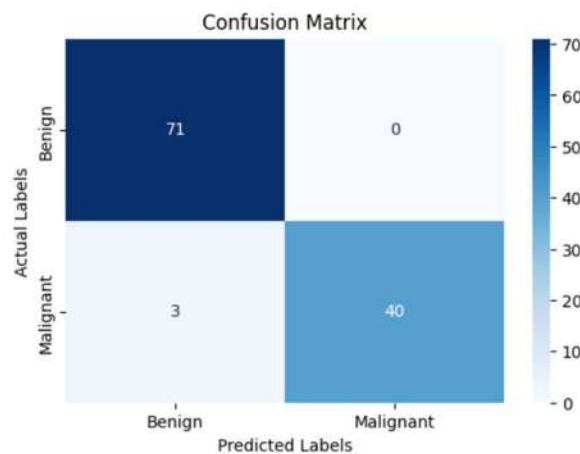
Random Forest (RF):

Random Forest is an ensemble learning technique that constructs multiple decision trees during training and merges their results to improve classification accuracy and control overfitting. Each tree is trained on a

random subset of the data and uses a subset of features, which introduces diversity and enhances generalization. Random Forest is highly robust to noise and outliers, scales well with large datasets, and delivers excellent performance across a wide range of classification tasks. Its feature importance scores also help identify which attributes most influence the prediction, making it a valuable tool in medical research.

3.5 Model Evaluation:

- **Accuracy:** Represents how often the model correctly distinguishes between benign and malignant tumors across all predictions.
- **Confusion Matrix:** Offers a detailed summary of prediction results by categorizing them into true positives, true negatives, false positives, and false negatives.
- **Precision:** Shows the percentage of cases labeled as positive by the model that are actually positive.
- **Recall:** Indicates the model's ability to correctly detect all real positive instances within the dataset.
- **F1 Score:** A harmonic mean of precision and recall, used to provide a balanced measure of the model's performance when both metrics are important.

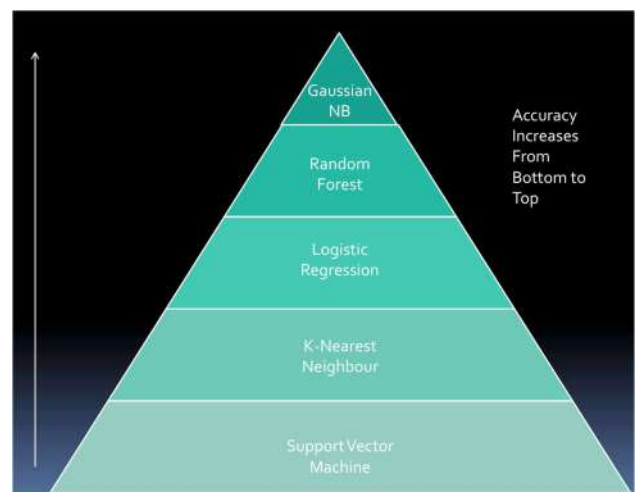


4. Results and Discussion

The machine learning models were trained and evaluated using the preprocessed Wisconsin Breast Cancer Dataset (WBCD). Performance metrics such as accuracy, precision, recall, and F1-score were used to compare the effectiveness of each classifier in distinguishing between malignant and benign tumors. The classification accuracy for each model is summarized below:

The performance of the five supervised machine learning models—Logistic Regression, Support Vector Machine (SVM), Gaussian Naïve Bayes, K-Nearest Neighbors (KNN), and Random Forest—was assessed using the Wisconsin Breast Cancer Dataset (WBCD). To evaluate their effectiveness in distinguishing between malignant and benign tumors, various performance metrics including accuracy, precision, recall, and F1-score were used. The dataset was split into training and testing subsets using a stratified approach to preserve the class distribution during model validation.

The Gaussian Naïve Bayes classifier achieved the highest



overall performance, with an accuracy of 97.37%, precision of 97.48%, recall of 97.37%, and an F1-score of 97.35%. This was closely followed by Logistic Regression and Random Forest, both reaching accuracy scores of 96.49%, showcasing their effectiveness in classifying benign and malignant tumors.

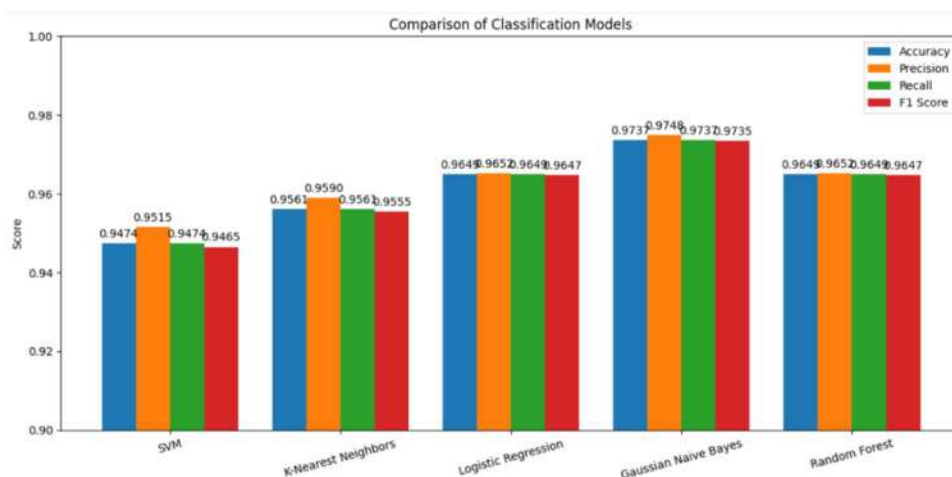
K-Nearest Neighbors (KNN) demonstrated strong generalization capabilities, attaining an accuracy of 95.61%, while Support Vector Machine (SVM) followed with 94.74%. These results indicate the robustness of these models, especially with proper feature scaling and preprocessing.

Table 1: Comparison of Evaluation metrics of ML algorithms

Model	Accuracy	Precision	Recall	F1 Score
Gaussian Naïve Bayes	0.9737	0.9748	0.9737	0.9735
Logistic Regression	0.9649	0.9652	0.9649	0.9647
Random Forest	0.9649	0.9652	0.9649	0.9647
K-Nearest Neighbors	0.9561	0.9590	0.9561	0.9555
Support Vector Machine	0.9474	0.9515	0.9474	0.9465

1. Gaussian Naive Bayes performs the best overall, with all metrics scoring around 0.9735–0.9748.
2. Logistic Regression and Random Forest show nearly identical and strong performance, each with all metrics around 0.9649–0.9652.
3. K-Nearest Neighbors (KNN) performs slightly better than SVM, with scores around 0.956–0.9590.
4. SVM has the lowest performance, with scores around 0.9474–0.9515, though still relatively high.

Comparison of Different ML algorithms



5. Conclusion and Future Work

- The experimental findings revealed that among the tested models, Gaussian Naïve Bayes delivered the most accurate classification, achieving an accuracy of 97.37%. This was closely followed by Logistic Regression and Random Forest, both scoring 96.49%, then K-Nearest Neighbors with 95.61%, and finally Support Vector Machine at 94.74%. These

outcomes emphasize the potential of machine learning techniques in effectively identifying breast cancer, offering valuable support to healthcare professionals in diagnostic processes. The outstanding performance of Gaussian Naïve Bayes is likely due to its strength in handling high-dimensional data and effectively separating classes, making it particularly suitable for medical data analysis.

- Integrating deep learning techniques, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), to improve classification performance.
- Optimizing hyperparameters for existing machine learning models to further enhance predictive accuracy.

In conclusion, machine learning proves to be a powerful and scalable approach for early breast cancer detection, offering substantial benefits in terms of speed, accuracy, and cost-effectiveness.

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Feature Extraction and Analysis of Bearing Faults

Varun Sharma

varun.sharma@kiet.edu

Abstract

One of the most important steps in identifying bearing problems is feature extraction. In order to provide a more meaningful dataset, it entails locating and extracting pertinent features from raw bearing vibration signals. Tasks involving categorization and prediction can then make use of these attributes.

Keywords

Fault diagnosis, feature extraction, bearing, test rig, vibration.

1.Introduction

Most of the bearing diagnostics research points on finding damaged bearings that have been found from the field where they have shown major flaws or from the damage that have been created. Scratching, drilling and adding particles to the lubricant are the common methods to create simulated defects. Defective bearing experiments are not able to find natural defect spread.

2.Experimental Set Up

Four test bearings are placed on a single shaft in the test set. An AC motor is connected via rub belts which is used to power the shaft. 2000 rpm is the steady rotation speed. A spring system adds 6000 pounds of radial load to the shaft and bearing. Each bearing is greased with force. Temperature and flow of lubricant are controlled using an oil circulation system.

Debris from the oil is collected from a magnetic plug. When accumulated trash connected to the magnetic field reaches a particular threshold an electric switch close and test ends. From figure 1, four Rexnord ZA-2115 double row bearings are mounted on a single shaft. Tapered contact angle is 15. 171. Pitch diameter and roller diameter are 2.815 and 0.331 inches respectively. Each row has 16 rollers. Every bearing housing had a PCB 353B33 High Sensitivity Quartz ICPs Accelerometer placed on it. Every bearing's outer race has four thermocouples. A National Instruments DAQCard-6062E data acquisition card is used to record vibration data for every 20 minutes. Sampling rate is 20kHz and data length is 20480 points. LabVIEW software is used to collect data [1].

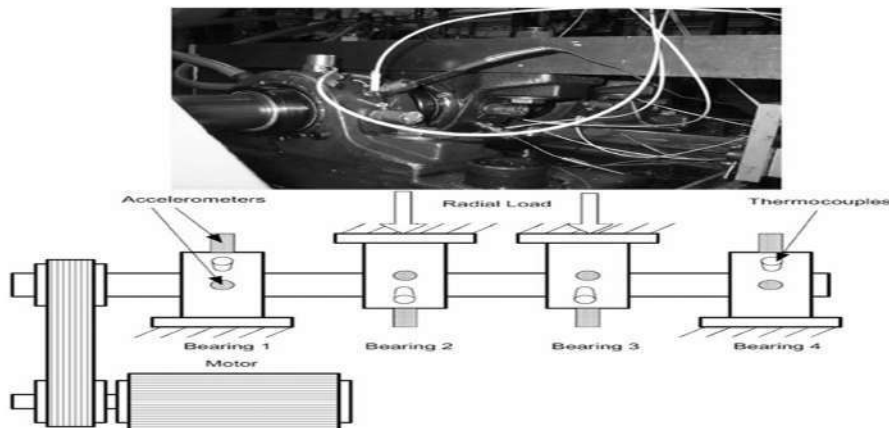


Figure 1: Bearing test rig.

3.Feature Extraction

Data is fetched for all four bearings and it is shown in following figures

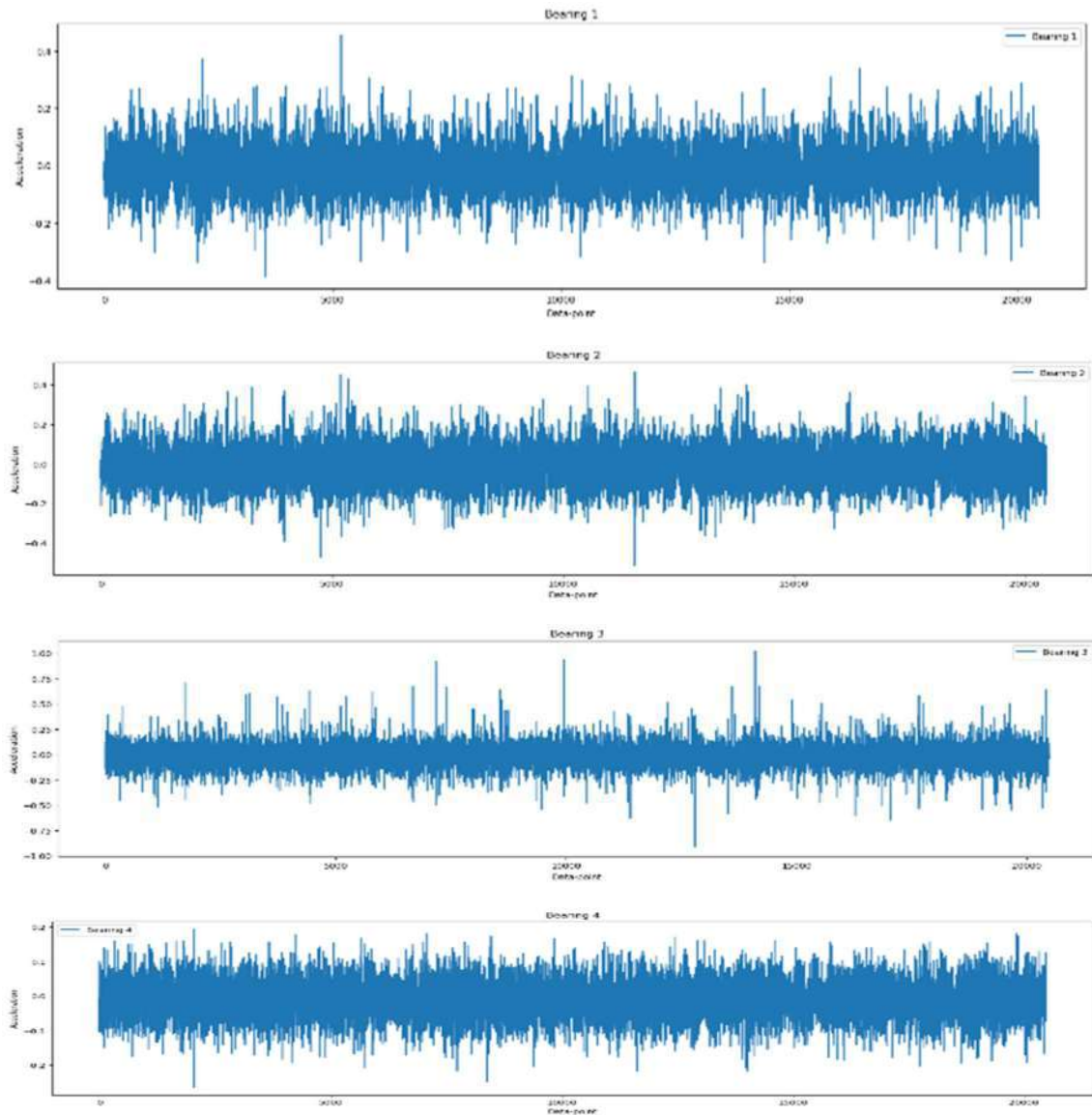
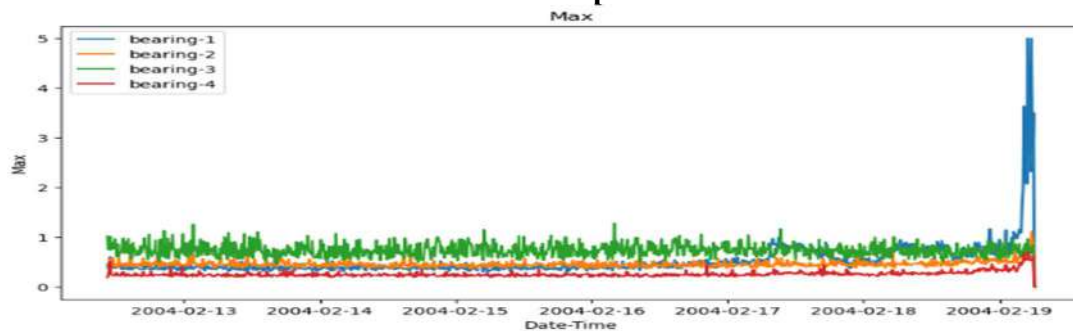


Figure 2: Vibrations produced in 4 bearings.

After fetching vibration data, features for all four bearings are calculated i.e. Max.,Min.,Mean,Standard deviation, RMS, Skewness,Kurtosis,Crest factor and Form factor.

4.Results Comparison



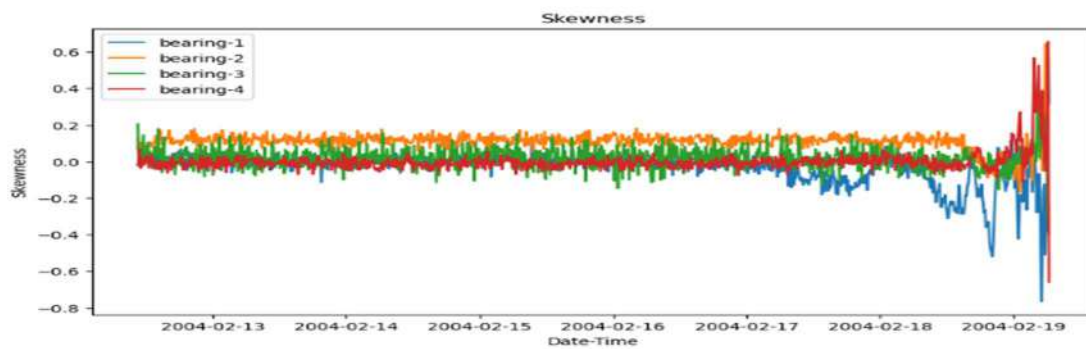
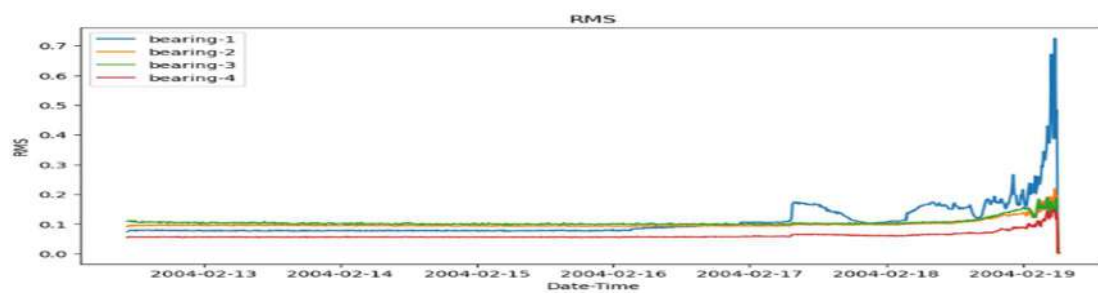
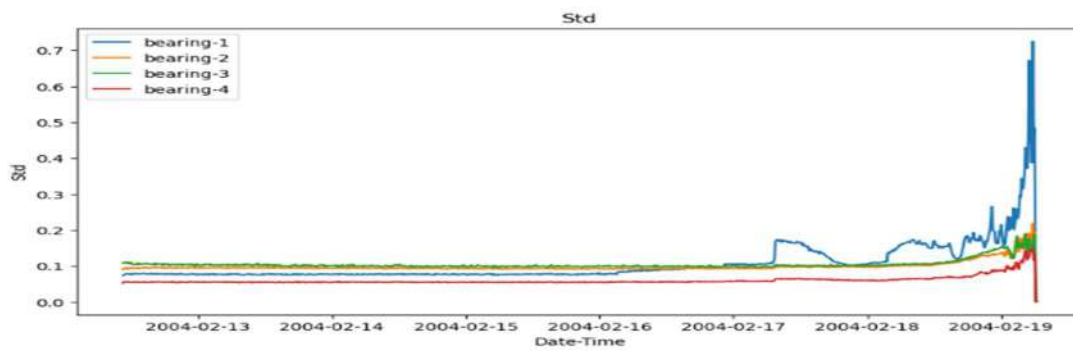
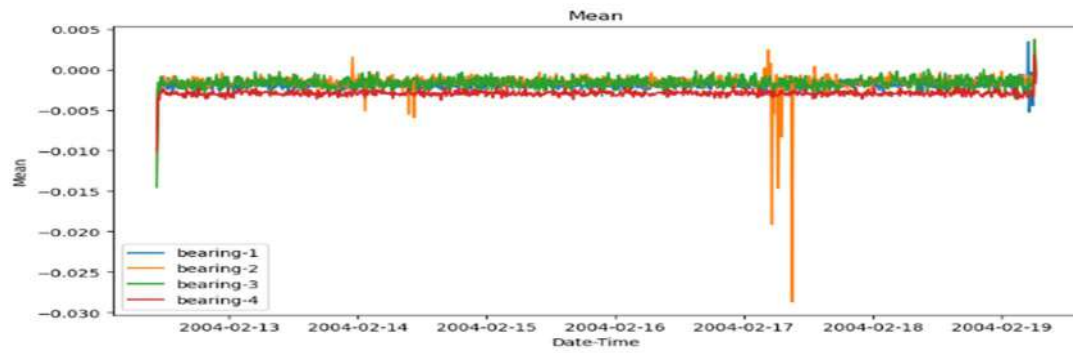
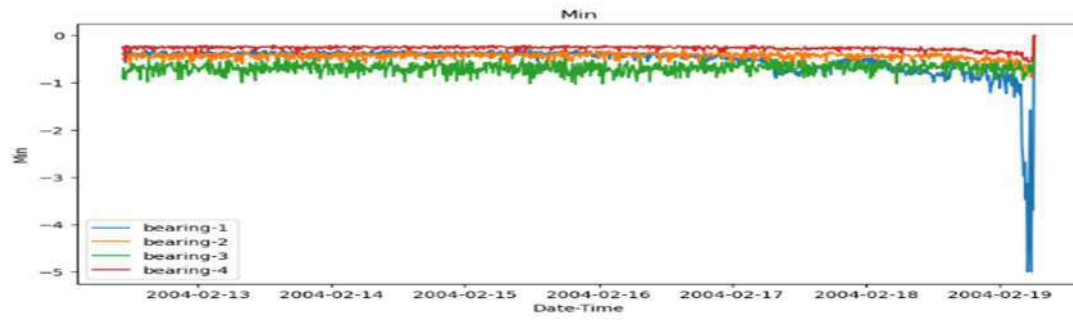




Figure 3: Comparison of features for all four bearings.

5.Conclusion

From figure 3 ,it is visible that bearing one shows maximum variations for maximum and minimum values. Mean value is almost same for all the bearings and variation in bearing 2 is large. Standard deviation and RMS is maximum for bearing 1 but does not fall to lowest. In skewness bearing 2 has the max. value and in kurtosis bearing 3 has the max value. Bearing 3 has the largest crest factor. Form factor is almost same for all the bearings and bearing 3 is showing minimum low. All these features can be further used for fault classification.

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Adaptive Machine Learning Framework for Navigation Control of Autonomous Drones

Vikramsingh R. Parihar^{1,*}, Dr. Soni A. Chaturvedi²

Abstract

The rise of autonomous drones has expanded UAV applications across sectors like surveillance, delivery, agriculture, and rescue operations. However, traditional navigation systems face limitations in adapting to dynamic environments. This research proposes an AI-driven adaptive navigation framework that leverages real-time sensor data, reinforcement learning, and adaptive control strategies to enhance drone autonomy, scalability, and security. The system processes mission inputs, environmental data (from LiDAR, cameras, GPS, and weather sensors), and applies data preprocessing techniques like Kalman filtering and sensor fusion. A real-time feedback loop monitors obstacles, weather, and battery status. The decision-making module, using reinforcement learning models like DQN and PPO, dynamically adapts navigation strategies, supported by global and local path planning algorithms. Continuous learning through experience replay and anomaly detection ensures performance improvement over time. This work addresses critical challenges in scalability, adaptability, and system integrity, paving the way for more reliable and intelligent autonomous drone navigation in complex environments.

Keywords: Adaptive Navigation, Autonomous Drones, Path Planning, Real-time Decision Making, , Sensor Fusion

INTRODUCTION

Drones, or Unmanned Aerial Vehicles (UAVs), are remotely controlled or autonomously operated aircraft with applications spanning surveillance, military operations, firefighting, drone shows, and more. Their expanding role includes future applications in delivery services, farming, search and rescue, and transportation.

Autonomous navigation enables drones to operate independently using AI-driven decision-making, real-time sensor data, and adaptive learning. Unlike manual navigation, AI-powered drones continuously evolve, enhancing efficiency and adaptability in dynamic environments.

Traditional drone navigation struggles with adapting to changing conditions. AI-based frameworks improve real-time decision-making, enhancing navigation reliability in complex environments. This study explores AI's role in drone navigation, existing challenges, and the benefits of adaptive frameworks.

Despite advancements, UAV navigation faces challenges like scalability, terrain adaptability, real-time decision-making, and secure communication. This research aims to bridge these gaps by developing innovative AI-driven solutions for robust and autonomous drone navigation.

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LITERATURE REVIEW

As Unmanned aerial vehicles (UAVs) embark on autonomous missions, the need for intelligent decision-making systems becomes paramount, particularly in optimizing trajectory planning and tour management

while considering energy constraints [1, 3, 4, 10, 17-20]. Deep reinforcement learning (DRL) has surfaced as another promising avenue for enhancing autonomous drone mobility control, particularly in scenarios requiring real-time obstacle avoidance and trajectory optimization [1, 2]. Integrating DRL with Human-Computer Interaction (HCI) facilitates adaptive responses to unforeseen environments, augmenting the flexibility of drone control systems [1, 2]. The burgeoning expansion of IoT-based drone networks (IoD) has introduced new challenges, particularly in terms of security [8, 15, 23, 25]. Elliptic curve cryptography has been proposed as a means to authenticate communication between users and drones within specific flying zones. This cryptographic scheme aims to bolster the security of drone communications, ensuring confidentiality and integrity while mitigating potential vulnerabilities for different scenarios.

Blockchain technology enhance the security and efficiency of multi-drone collaboration within IoD environments [8, 15, 23, 25]. By leveraging blockchain as a communication tool, drones can achieve consensus in a secure and decentralized manner, thereby improving energy efficiency and connectivity. This approach represents a novel paradigm in managing multi-drone operations, promising greater stability and security in swarm coordination.

AI-based pipelines, integrating deep learning with dimensionality reduction techniques, have demonstrated efficacy in reducing computational workload for processing disaster images [7, 8]. Furthermore, edge computing platforms equipped with low-power accelerators present a compelling alternative for processing in real-time disaster scenarios. In the domain of autonomous drone racing, the integration of deep learning with path planning and control algorithms has enabled significant advancements [2, 3, 7, 11, 12, 17]. State-of-the-art approaches leverage deep convolutional neural networks (CNNs) for gate detection, facilitating efficient trajectory planning and control in simulated racing environment. Such innovations have outperformed human pilots in racing simulations, showcasing the potential of AI-driven drone racing systems.

Security remains a paramount concern in the realm of drone technology, particularly concerning communication and operation integrity [15, 23, 25]. Blockchain, machine learning, Software-Defined Networks (SDNs), and fog computing have been proposed as potential solutions to mitigate security risks and ensure robustness in drone networks [7]. These emerging technologies offer promising avenues for fortifying drone communication systems against various cyber threats [7].

PROBLEM STATEMENT

Despite advancements, autonomous drone navigation faces challenges in adaptability to dynamic environments, generalization across drone types, and integration of human expertise. Security and system integrity remain critical concerns. This research aims to develop adaptive navigation frameworks that enhance autonomy, scalability, and security, unlocking the full potential of drones for diverse applications.

PROPOSED METHODOLOGY

4.1 Adaptive Navigation Framework

This framework outlines an adaptive navigation system for autonomous drones, integrating mission planning, environmental perception, real-time decision-making, and continuous learning for dynamic path optimization and execution. The Block diagram of Adaptive Navigation Framework is as shown in Fig. 1

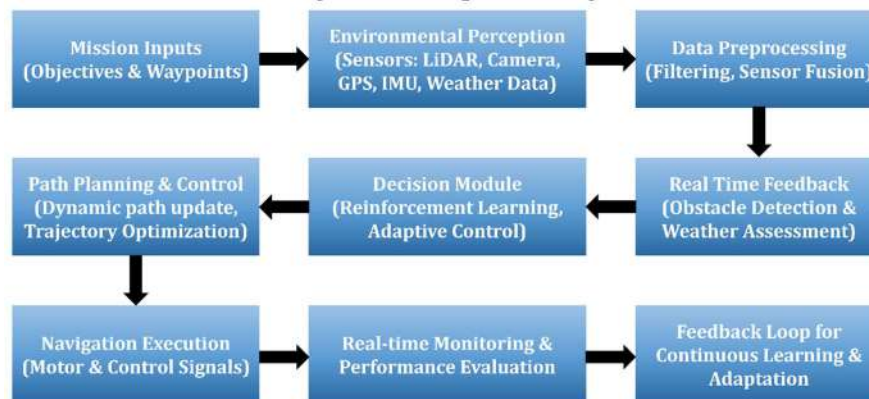


Fig. 1: Block diagram of Adaptive Navigation Framework

Mission Input

- The system receives predefined mission goals, waypoints, and objectives from the user or a central control system.
- Inputs may include:
 - Target locations for delivery or reconnaissance.
 - Specific flight paths or designated areas to explore.
 - Constraints such as speed limits, altitude restrictions, or energy efficiency goals.

Environmental Perception

The drone relies on multiple sensors to perceive its surroundings and gather real-time environmental data.

These include:

- LiDAR (Light Detection and Ranging) for obstacle detection and 3D mapping.
- Cameras (RGB, Infrared, Depth Sensors) for visual perception and object recognition.
- GPS (Global Positioning System) for geolocation and route tracking.
- IMU (Inertial Measurement Unit) for motion tracking and stability control.
- Weather Sensors to measure wind speed, humidity, and temperature for adaptive flight control.

Data Preprocessing

Raw sensor data is often noisy and inconsistent, requiring filtering and sensor fusion to improve accuracy.

Key preprocessing techniques include:

- Kalman Filtering to estimate the drone's precise location despite noisy GPS signals.
- Sensor Fusion algorithms to combine data from multiple sensors for enhanced reliability.
- Edge Computing to process data in real-time without excessive reliance on cloud computing.

Real-Time Feedback System

The feedback system continuously monitors:

- Obstacle detection to avoid collisions with dynamic objects like birds, vehicles, or other drones.
- Weather conditions to assess wind turbulence, rain, or temperature changes that may affect flight stability.
- Battery status to optimize energy efficiency and determine if route adjustments are necessary.

Decision Module (Reinforcement Learning & Adaptive Control)

The Decision Module is the brain of the navigation system, employing:

- Reinforcement Learning (RL): The drone learns optimal navigation strategies by continuously interacting with the environment. RL models, such as Deep Q-Networks (DQN) or Proximal Policy Optimization (PPO), enable the drone to adapt dynamically to unforeseen challenges.
- Adaptive Control Algorithms: These algorithms adjust flight parameters, such as speed, altitude, and trajectory, based on real-time feedback. Common techniques include:
 - Model Predictive Control (MPC) for real-time trajectory optimization.
 - Fuzzy Logic Controllers for handling uncertainties in sensor data.
 - Neural Network-based Control for predictive adjustments in navigation.

Path Planning & Control

The path planning module dynamically adjusts the drone's route based on environmental changes. It includes:

- Global Path Planning: Utilizes algorithms like A* or Dijkstra's algorithm to determine the shortest path to the destination.
- Local Path Planning: Implements Rapidly-exploring Random Trees (RRT) or Artificial Potential Fields (APF) for obstacle avoidance in real-time.
- Trajectory Optimization: Ensures smooth transitions between waypoints to minimize abrupt movements and energy consumption.

Navigation Execution

The execution module translates the planned path into motor commands and control signals, ensuring precise drone movement. It involves:

- Flight Control Systems (FCS) for stability, altitude control, and maneuverability.
- Actuator Commands to adjust motor speed, propeller angles, and camera orientation.
- Emergency Handling Mechanisms, such as automatic landing or route diversion in case of critical failures.

Real-time Monitoring & Performance Evaluation

- The system continuously assesses the drone's navigation efficiency and performance by tracking:
 - Flight accuracy (deviation from planned trajectory).
 - Obstacle avoidance success rate.
 - Battery consumption trends and remaining flight time.
- The evaluation data is used for further optimization in future missions.

Feedback Loop for Continuous Learning & Adaptation

The learning loop ensures the drone improves over time by integrating:

- Experience Replay: Past navigation experiences are stored and used to improve decision-making.
- Model Updating: RL models are retrained periodically to adapt to new environments.
- Anomaly Detection: Identifies unusual patterns in flight behavior and adjusts navigation strategies accordingly.

SIMULATION AND EXPERIMENTAL SETUP

The AI framework is tested in a simulated environment before real-world implementation. Steps include:

- **Simulation Tools:** Utilizing AirSim and Gazebo for testing autonomous flight capabilities.
- **Training AI Models:** Implementing deep reinforcement learning models to learn navigation patterns.
- **Testing Scenarios:** Conducting simulations in dynamic environments with obstacles and varying weather conditions.

Simulation Setup:

- **Environment:** 3D simulated environments with varying terrain and weather conditions.
- **Drone Model:** Quadrotor UAV equipped with LiDAR, RGB cameras, and IMU.
- **Training Episodes:** Iterative training with reinforcement learning for adaptive navigation improvements.
- **Evaluation Metrics:** Collision avoidance rate, trajectory efficiency, and adaptability under changing conditions.

TESTING & EVALUATION FRAMEWORK

To ensure the framework works in real-world conditions, we need:

a) Simulation Environment

- **Gazebo / AirSim:** Test drone AI models in a virtual 3D world.
- **MATLAB / Simulink:** Validate control algorithms before real-world implementation.

b) Real-World Deployment & Testing

- **Outdoor Field Tests:** Validate algorithms in dynamic weather and obstacle conditions.
- **Failure Analysis & System Optimization:** Identify system weaknesses and optimize models.

a) Environmental Challenges

- **Unpredictable Weather:** Develop adaptive control strategies for wind and rain conditions.
- **Dynamic Obstacles:** Implement fast real-time path re-planning algorithms.

b) Computational Constraints

- **Limited Processing Power on Drones:** Optimize AI models for low-power embedded systems.

c) Safety & Regulatory Compliance

- **Flight Regulations & Airspace Restrictions:** Comply with government UAV regulations.
- **Fail-Safe Mechanisms:** Implement emergency landing and return-to-home (RTH) features.

EVALUATION

The successful development of adaptive navigation frameworks will be evaluated based on the following:

Adaptive Navigation Capabilities:

- Demonstrated ability of drones to adjust their navigation in response to changing environmental conditions (e.g., obstacle avoidance, weather changes).

Performance Metrics:

- Quantitative evaluation using metrics such as path length, time to goal, obstacle avoidance rate, tracking error, and settling time.

Robustness and Generalization:

- Assessment of the framework's ability to perform effectively across a range of simulated environments and scenarios.

CHALLENGES AND MITIGATION STRATEGIES

Development of adaptive navigation frameworks presents several challenges:

Computational Complexity:

- Challenge: High computational demands of machine learning algorithms may limit real-time performance.
- Mitigation: Algorithm optimization, model reduction, and efficient implementation techniques.

Simulation Fidelity:

- Challenge: Difficulty in creating simulation environments that accurately reflect the complexities of the real world.
- Mitigation: Development of high-fidelity simulation models incorporating uncertainty, stochasticity, and real-world physics.

Algorithm Tuning and Validation:

- Challenge: Ensuring the stability, responsiveness, and robustness of adaptive control and reinforcement learning algorithms.
- Mitigation: Rigorous simulation testing, parameter optimization techniques, and sensitivity analysis.

CONCLUSION

By employing a multidisciplinary approach encompassing literature review, algorithm development, simulation-based testing, human-in-the-loop integration, and real-world validation, this research aims to address the identified research objectives and advance the state-of-the-art in autonomous drone navigation. Through rigorous methodological processes and iterative optimization, the research endeavors to contribute novel insights and practical solutions to the field, enabling drones to navigate autonomously with adaptability, generalization, human interaction, and security.

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Analysing Human Resource Practices at Esaf Microfinance Bank

Ms. Ankita Dilip Umathe^{1,*} Prof. Ankita Bhandarkar^{2,*}

Abstract

The Human Resource (HR) procedures and key features of ESAF Microfinance Bank, including hiring and selection, training and development, performance evaluation, pay administration, employee motivation, and retention policies, will be covered in this study. The impact of ESAF's HR policies on worker happiness, organizational commitment, and overall productivity is examined in this study using a quantitative method. Primary data was collected through a structured questionnaire, and regression analysis was used to identify the relationship between different HR characteristics and job satisfaction, which was the dependent variable. This study confirms that effective HR policies, specifically training programs, open performance grading systems, and fair compensation systems, significantly increase employee motivation and organizational efficiency. However, career advancement and employee reward plans are areas to be improved. This study suggests that HR policies must be reviewed continuously and upgraded to increase employee involvement, turnover rates must be reduced, and long-term organizational stability must be guaranteed.

Keywords: Human Resource Practices, Regression Analysis, Employee Satisfaction, ESAF Microfinance Bank, Organizational Growth, Employee Performance.

INTRODUCTION

Human resources are those people or organizations driving an organization toward its achievement of goals. It is often termed as human capital; it is the employees who in the present-day business scenario are considered the most invaluable asset that provides an organization with a difference from others. Appropriate human resource management would strengthen an organization's employee engagement, productivity, and job satisfaction since these factors contribute to the ultimate success of an organization. Strong practices of HR are also important for the banking industry in attracting, retaining, and developing talent.

ESAF Small Finance Bank is a great example of how strategic HRM can foster financial inclusion and community development. An NGO by inception in 1992, ESAF, has transformed into one of the major financial institutions in the country, up-scale services to underserved areas. The transformation from micro-enterprise development to scheduled bank status is a testament to a deep desire to empower underprivileged communities. In ESAF's case, its new HR practices deal with the creation of financial opportunities for the bottom-of-the-pyramid people.

Human Resource Practices in ESAF Small Finance Bank: The research paper gives an account of the human resource practices in ESAF Small Finance Bank. To achieve this, investigations of how human resource practices affect organizational performance and employee satisfaction have been conducted. A regression analysis will be utilized in the study to prove the relationship between ESAF's HR strategies with the practice of increased productivity and sustainable growth in an organization.

LITERATURE REVIEW

Mrs. D. Joyce, Dr. C. Vijai. (2020). A theoretical investigation into Green HRM: A Novel Approach to Organizational Sustainability. Despite the fact that various investigations have been performed to analyse how HR policies affect the perception of job satisfaction among workers in industrialized countries' public sectors, relatively few studies have been carried out in the private sector, specifically in the Indian banking industry. The amount of research on HRM in the private sector is still quite little, so a thorough analysis is necessary, particularly given that organizations are now increasingly planning mergers and acquisitions. This research narrows the gap by considering HRM practices in Indian banks and their effect on employee job satisfaction.

“J. Shifa Fathima Human Resource Management In Banking Sector (2015)” using empirical and secondary evidence analyses the purpose of HRM in banking. It has been developed that the quality of human capital, skills, behaviors, and experience, are seriously connected to the capability of banks in reaching competitiveness and delivering customer value. The improvement of the performance primarily lies in this area, through financial efficiency and service quality.

Md Nahin Hossain Md. Yahin Hossain Md Farid Talukder (2014) has studied the HRM practices of the 5 commercial banks in Bangladesh and noticed that they follow a well-established recruitment process. Even though the bank had adequate qualified personnel, it faced a critical shortage of ICT professionals, and the employee job satisfaction level was assessed as not satisfactory.

“Human Resource Management Practices and Employee Performance in Bangladesh's Banking Sector, Abdus Salam Sarker, 2017”. He employed a methodical questionnaire. Employee relations, institutional commitment, motivation, the physical workspace, pay, training and development, advancement, and job satisfaction were the primary areas of analysis. With the exception of pay and training and development, the majority of HR procedures have a major impact on employee performance in the banking industry of Bangladesh.

“Ms. Priyanka Gupta, Dr. Praveen Chaudhry (2024) India's Theoretical Analysis of Human Resources Practices In the Indian Banking System” banking industry is confronted with intense competition and talent shortage, compelling banks to adopt innovative HRM practices. Chakrabarty (2012) highlights human capital as a source of competitive advantage. A study identified 49 HR strategies, including digital integration, flexible work practices, training programs, and employee benefits. Private banks prioritize people management, AI-driven efficiency, and employee engagement to enhance performance. To compete, future HR initiatives will have to be centered on technology, well-being, and continuous improvement.

“Mr Pravin A. Kale, Dr Shailesh Kediya the Resonance of HR Practices in the Banking Sector of India (2023)” Human resource management in Indian banking plays a vital role in talent recruitment, retention, employee motivation, and productivity. Private banks provide better salaries, professional development, and social status, while state banks provide job security. Recruitment, training, compensation, and performance management are all major HR practices, and digital technology and AI support efficiency enhancement. Job satisfaction and work-life were the two most important HR factors, while development programs were less common. As the banking industry evolves, business goals aligned with HRM strategies will be the driving force for long-term success, with a competitive and motivated workforce.

“Hiralal Raut Influence Of HRM Practices On Employee Job Satisfaction: Evidence From Commercial Banks In Nepal (2018)” HRM variables account for almost 57.3% of Nepalese banking satisfaction, according to the research, which also demonstrates a strong positive correlation between HRM practices and staff work happiness. The main factors driving it are training and development, recruiting and selection, pay and benefits, and work-life balance. Global study has confirmed that strategic HR policy increases engagement, motivation, and retention, which supports the findings.

“Tofael Hossain Majumder, M.D. Practices of Human Resource Management and Employee Contentment with the Private Banking Sector in Bangladesh (2012)” According to research on HRM procedures in Bangladeshi private banking, job satisfaction differs depending on the HR department. High levels of unhappiness persist in the areas of training, career development, compensation, and rewards despite the implementation of HRM practices. Professional help is necessary for increased efficiency because the sector has not yet developed human resource planning. As research suggests, providing competitive remuneration, encouraging management-employee interaction, and creating formal career paths are all found to increase job satisfaction. Further, incorporating new technology and foreign experience in training can make HRM more effective. Subsequent studies can investigate the long-term effects of better HR practices on the health of employees and organizational performance.

HYPOTHESIS FORMULATION

Null Hypothesis (H₀): The null hypothesis will therefore point to the fact that "There is no relationship between the dependent variable (O) and independent variables (H, E, U, J, K, M, S, L, C)".

Alternative Hypothesis (H₁): The alternative hypothesis will therefore point to the fact that at least one of the independent variables is statistically associated with the dependent variable.

Dependent Variable

O: Job satisfaction

Independent Variable

H: Performance appraisal

E: Training program relevance

U: Teamwork and collaboration
 J: Compensation and benefits package
 K: Compensation compared to industry standards
 M: Current workload management
 S: Conflict resolution effectiveness
 L: Fairness in the distribution of compensation and benefits
 C: Onboarding and orientation process helpfulness

RESEARCH METHODOLOGY

Research is the planned investigation of issues to provide solutions to problems, discoveries, retrieval, analysis, and interpretation of data. Methodology is the process aimed at ensuring that research is well-structured, reliable, and nonredundant; it organizes the study to address well-defined research questions or hypotheses. The goal of this research study is to dissect ESAF Microfinance Bank's HR procedures. Among the most important subjects are hiring and selection, training and development, performance management, pay and benefits, employee engagement, and retention tactics.

Sample Size

Population: Employees of ESAF Microfinance Bank.

Sample Size: 204 employees.

This number provides a sufficient representation of the diverse employee groups within the bank, ensuring robust and reliable survey data.

Sampling Frame

The sampling frame will include all employees of ESAF Microfinance Bank across various departments and locations.

Sampling Method

Convenience Sampling: Respondents were chosen for the study based on their availability and interest in taking part.

Data Collection Method

Primary Data

Surveys: Population: Employees of ESAF Microfinance Bank across various departments and locations.

Questionnaire: A structured questionnaire with Likert-scale questions will be used to gather data on HR practices and employee perceptions

ANALYSIS AND OUTCOME

Table 1: Method

“Variables Entered/Removed			
Model	Variables Entered	Variables Removed	Method
1	h, e, u, j, k, m, s, l, c	.	enter
*dependent variable: o			
†all requested variables entered.”			

Analysis And Interpretation

All independent variables requested are included in this model (H, E, U, J, K, M, S, L, C), to examine their influence on the dependent variable "O". The input of variables in this manner aims to provide insight into how different variables explain the variations in "O". In the "enter" method, all variables selected can be input simultaneously into the regression to provide an idea of the sum effect that the variables may have. This approach would identify the most significant predictors and their associations with "O", which would further facilitate the attempt to explain possible interactions or multicollinearity of variables. Overall, this model configuration should provide a fine basis for understanding influences on "O" and form the foundation for further statistical analysis uncovering meaningful insights.

Table 2: Regression

“Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.700 ^a	.491	.467	.644
*Predictors: (Constant), H, E, U, J, K, M, S, L, C”				

Analysis And Interpretation

The model summary's R-value of 0.700 indicates that all the independent variables—H, E, U, J, K, M, S, L,

and C—have a high positive correlation with the dependent variable, "O." With an R Square of 0.491, all of these independent factors account for roughly 49.1% of the variation in "O." After adjusting for the number of variables, the model was able to explain 46.7% of the "O" variation, according to the Adjusted R Square (0.467), which accounts for the number of predictors. An approximation's standard error is 0.644, which averages the noted values' separation from the regression line and so offers a reliable assessment of the model's correctness.

Table 3: Anova

ANOVA*						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	regression on	77.921	9	8.658	20.682	<.001
	residual	80.928	195	0.415		
	total	158.849	204			
†dependent variable: o						
‡ predictors: (constant), h, e, u, j, k, m, s, l, c						

Analysis And Interpretation

The p-value <.001, and the F-statistic value is 20.862. The relationship between the independent variables H, E, U, J, K, M, S, L, and C and the dependent variable O was examined using the regression model. This would suggest that the variance of dependent variable is substantially explained by the set of independent factors. With a regression total of squares of 77.921 and a residual sum of squares of 80.928, we can reject the null hypothesis at 0.05 level, demonstrating the model's powerful impact. After concluding that at least one of the predictors affects the result, we examine each variable's contributions in greater detail.

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HYPOTHESIS TESTING

Table 4: Hypothesis

"H0	There is no relationship between the dependent variable (O) and independent variables (H, E, U, J, K, M, S, L, C).	Rejected
H1	There is a significant relationship between the dependent variable and independent variables.	Accepted"

FINDINGS

Regression analysis shows that “the model predicts the dependent variable ‘O’ with a p-value < 0.001 and an F-value of 20.862, which is statistically significant. This would indicate that the variance in ‘O’ is explained by the combined influence of the independent variables H, E, U, J, K, M, S, L, and C, which is statistically significant. The R-value of 0.700 indicates a positive correlation between the independent and dependent variables. The independent variables explain 49.1% of the variation in ‘O,’ with an R Square value of 0.491. The adjusted R Square, which takes into consideration the number of predictors, is 0.467”. The model's relatively low standard error of 0.644 indicates that it is making predictions that are fairly accurate. The research demonstrates that independent factors work together to significantly and strongly influence the outcome variable "O."

CONCLUSION

In summary, the independent variables (H, E, U, J, K, M, S, L, and C) account for 49.1% “of the variation in the dependent variable ‘O,’ which is adequately explained by the regression model. The F-value and a p-value < 0.001 show that the model is statistically significant, indicating that the predictors taken together significantly affect "O." There is some unexplained variance even though the model does a good job” of describing almost half of the variation, indicating that other factors that are not included by this model might have an impact on "O." Although the research shows that the independent factors played a significant role in forecasting the outcome overall, the predictive ability of the model might be increased with additional variables or further modification.

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A Review On Business Enterprise Resource Planning (ERP)

Kiran .P. Hivale 1,*, Harshal. R. Garad2, P. T .Talole3

Abstract

Enterprise Resource Planning (ERP) systems integrate a business's core functions, including finance, human resources, supply chain management, manufacturing, and customer relationship management, into a unified platform. By streamlining operations, ERP provides realtime access to critical data, enhancing decision-making and operational efficiency. As organizations accelerate their digital transformation, modern ERP solutions leverage cloud computing, artificial intelligence, and data analytics to improve scalability, flexibility, and security. Despite implementation challenges, ERP systems significantly enhance resource utilization, reduce costs, and provide a competitive edge. This research paper explores key ERP concepts, benefits, implementation challenges, and future advancements, highlighting ERP's role in business process optimization and digital transformation.

Keywords: Enterprise Resource Planning (ERP), digital transformation, cloud computing, artificial intelligence, supply chain management, business process automation.

INTRODUCTION

1. **Background on ERP Systems:** ERP stands for Enterprise Resource Planning. ->ERP systems are integrated software solutions that enable organizations to manage and streamline their business processes in a streamlined manner. Enterprise Resource Planning (ERP) systems serve as an integrated platform that brings together various business processes such as finance, supply chain management, human resources, CRM, production, etc. By consolidating information and automating processes, ERP Sector increases operational efficiency, reduces errors, and enhances the decision-making process.
2. **Importance of ERP in modern business environments:** ERP systems emerged as essential tools in the evolution of businesses, as they streamline operations, enhance decision-making, and uphold data integrity in the rapidly changing business landscape of today. In this manner, ERP facilitate collaboration, minimize redundancies, and improve resource allocation by centralizing business processes. Organizations gain real-time insight into finances, inventory management, and customer relationships, allowing data-driven decisions that drive growth and innovation

Problem Statement

Manual processes, inefficient operations, and limited visibility are major obstacles that hamper productivity and informed decision-making. Reliance on outdated methods leads to frequent errors, operational delays, and misallocated resources, which in turn hinder scalability and responsiveness to evolving demands. The absence of automation creates significant operational bottlenecks, while insufficient access to real-time data prevents effective strategic planning and timely insights. These challenges result in higher operational costs, reduced efficiency, and slower growth. This project aims to develop a comprehensive Business ERP system that streamlines workflows, automates processes, and enhances data visibility. By addressing these issues, the solution will enable improved operational performance, better resource management, and sustainable long-term scalability.

Objectives

The study explores ERP systems across industries with emphasis on recent trends, innovations, and examples. The research will also evaluate cloud-based ERP solutions. These findings will assist business leaders, IT professionals and policy makers gain a better understanding of factors influencing ERP usage and effectiveness.

He main goals of this research are:

1. To study the advantages and disadvantages of implementation operating an ERP system in different industries.
2. To analyze factors that affect the success or failure of ERP implementation.
3. To discover innovative trends and technologies that adds value to ERP functions.
4. To assess how ERP systems contribute to enhanced business agility and decision-making.
5. In order to give recommendations for enabling optimal ERP implementation and use.

Literature Review

1 Overview of previous research on ERP systems: Previous research on ERP systems has focused on their implementation issues, advantages, and sustainable role in organizational productivity. Usefulness and ease of use have been studied extensively as contributing factors towards ERP success in the recent past. Several other research works have emphasized that ERP helps in enhancing data accuracy, workflow automation, and strategic decision-making. Insights into how ERP solutions adapt to industries and the effectiveness of various ERP vendors across industries have been provided through comparative analyses on ERP vendors.

2 Theoretical framework and key concepts: The theoretical background for ERP research is mainly grounded in technology acceptance models (TAM), resource-based view (RBV), and change management theories. We have data until October 2023, these models help interpret user acceptance, organization preparedness, and also strategic benefits that could be leveraged from ERP implementation. Available ERP Cloud adoption have several core topics which concentrate on ECM, BPR, AI, etc. Therefore, familiarization with these frameworks and future of ERP systems.

3 Current trends and advancements in ERP technology:

1. **Cloud-Based ERP Solutions:** Cloud-based ERP solutions have revolutionized enterprise resource planning by offering businesses a more flexible, scalable, and cost-effective approach to managing operations. One of the key advantages of cloud ERP is scalability, allowing companies to expand or reduce their ERP capabilities based on evolving business needs. Additionally, cloud ERP systems enhance cost efficiency by minimizing upfront investments and reducing maintenance expenses. A more important advantage is accessibility from any location with an internet connection, enabling users to access ERP functionalities. Also, with cloud ERP providers, security is another main advantage because they employ encryptions and many other advanced security provisions with the intention of protecting sensitive data. Such notable cloud ERP platforms are SAP S/4HANA Cloud, Oracle NetSuite, and Microsoft Dynamics 365, targeting almost all business needs.
2. **Artificial Intelligence (AI) and Machine Learning (ML) in ERP:** Artificial Intelligence (AI) and Machine Learning (ML) are significantly transforming ERP systems by enhancing predictive analytics, automating processes, and improving decision-making capabilities. One of the primary applications of AI in ERP is predictive analytics, which allows businesses to forecast demand, optimize resource allocation, and make data-driven decisions. Additionally, AI-driven automation streamlines routine tasks such as invoice processing and data entry, reducing manual effort and improving operational efficiency. Also, the application of virtual assistants, including AI-based chatbots which improve customer services and HR administration by offering instant services, response automation, etc., is crucial.
3. **Mobile ERP and Improved User Experience:** Through its adoption of mobile ERP solutions, businesses can achieve transformation of how they carry on their activities and improve business operational flexibility

through flexible access of their ERP systems, either by mobile phones or tables. An enhanced development of responsiveness in the recent past through modern mobile ERP results in good intuitive dashboards as well as information visualization data. Moreover, personalization features, such as role-based dashboards, allow users to focus on relevant data, increasing efficiency and improving decision-making. Real-time updates further enhance mobile ERP by providing instant data retrieval and reporting, ensuring that employees have up-to-date information regardless of their location. With the integration of mobile ERP with improved user experience, organizations can enhance productivity, streamline workflows, and support a more agile work environment.

4. **Business Intelligence (BI) and Advanced Analytics:** BI and advanced analytics with modern ERP systems are changing the face of business performance, giving deeper insights and data-driven decision-making capabilities. Real-time reporting is one of the prime features of BI-enhanced ERP. Interactive dashboards and analytics tools provide useful insights for strategic planning. Moreover, ERP platforms make use of big data processing to analyze large datasets, identify trends, and optimize business operations. One other critical advancement is that ERP is natively integrated with third-party BI tools like Power BI, Tableau, and Qlik. This increases the reporting capabilities of the ERP solution and allows for more holistic visualizations.
5. **RP and CRM, Ecommerce Integration:** ERP seamlessly integrates with Customer Relationship Management (CRM) and e-commerce platforms, thereby improving business operations in terms of revealing higher levels of customer insight, improved sales management, and optimized order fulfillment. Business synchronizes data from ERP and CRM, which can give it an overview of customer interactions and enable one-dimensional services as well as develop better decision-making.

ERP Systems

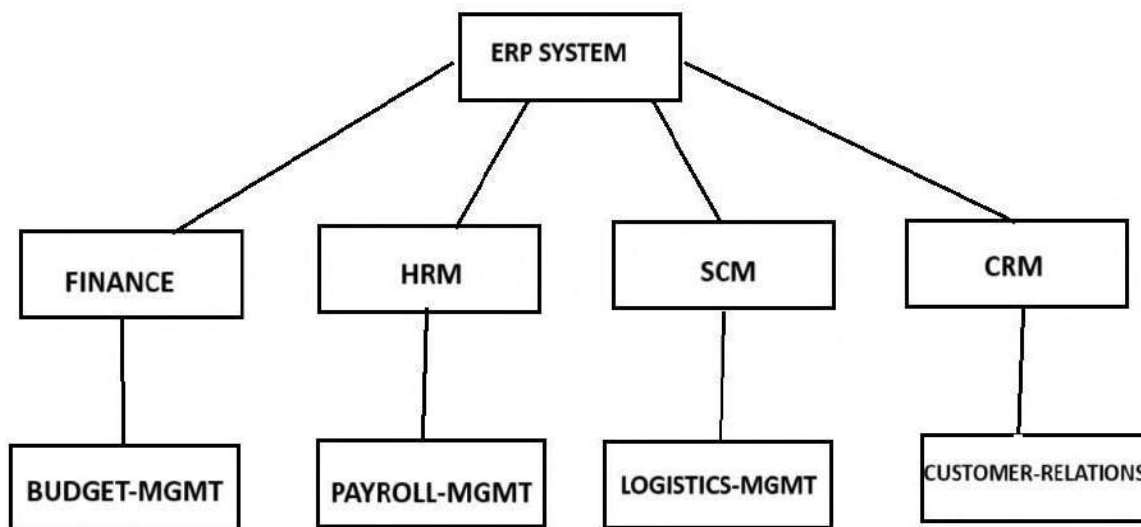


Fig 1 : ERP systems

ERP systems combine different business processes, such as finance, human resources, supply chain management, and customer relationship management. Evolution of ERP has led to the adoption of intelligent and flexible solutions that enhance operational efficiency and competitiveness. This paper examines the key aspects of ERP systems and their implications for businesses.

1. **Definition and Evolution of ERP:** ERP is a software system designed to integrate and streamline core business processes across an organization. First developed as Material Requirements Planning (MRP) systems in the 1960s to manage manufacturing operations, ERP evolved in the 1990s to cover finance, HR, supply chain, and customer relations. Over time, ERP has incorporated advanced technologies such as cloud computing, AI, IoT, and blockchain to improve efficiency, scalability, and decision-making capabilities. The last step in ERP evolution has been on-premise to cloud-based solutions and the introduction of mobile and AI-powered ERP systems.

Benefits of ERP in Business

Enterprise Resource Planning (ERP) systems are the comprehensive software packages that integrate finance, human resources, supply chain management, and customer relationship management. As a company grows, managing data becomes more complex in multiple departments. ERP systems answer these challenges by providing a single platform that improves efficiency, accuracy, and decision-making capabilities. This paper will focus on the most important benefits of ERP in business, such as process automation, better data management, cost optimization, and scalability.

1. **Process Automation and Efficiency:** The main benefit of an ERP system is the automation of repetitive and time-consuming processes. Process automation enhances efficiency in the following ways
 - Workflow Optimization: Automates core business functions such as procurement, invoicing, and payroll management, thereby reducing manual errors and administrative burden.
 - Inventory and Supply Chain Management: It keeps real-time stock records, thus optimizing restocking, preventing shortages or overstocking, and improving replenishment management.
2. **Better Data Management and Decisions:** An ERP system centralizes all data, providing organizations with much-needed insight to base and inform decisions. Principal advantages are:
 - Eliminates data silos by integrating all information from different departments in one place.
 - Advanced analytics and reporting which give deep insights about the performance in terms of finance, sales, and operational efficiency.
3. **Cost Reduction and Resource Optimization:** ERP implementation is a primary way of achieving cost efficiency and resource optimization. The major benefits include the following:
 - Saving on Operational Costs: Eliminates labor-intensive processes, thus providing less administrative expense.
 - Efficient Resource Utilization: Includes the optimal utilization of raw materials, human resources, and capital.

Challenges in ERP Implementation

Despite numerous advantages, implementing an ERP system does not come without its challenges:

1. **High Initial Investment Costs:** ERP systems involve high upfront capital expenditures and include considerable software, hardware, and customization costs. The costs of license purchase, IT infrastructure setting up, and hiring of ERP consultants are significant, a cost affecting many small and medium-sized enterprises (SMEs). There are sometimes hidden costs emerging later due to maintenance upgrades and additional user licenses. Organizations need to budget and prepare financially by properly undertaking a cost-benefit analysis prior to adopting ERP systems.
2. **Customization Complexity and Integration Issues:** Every business has unique processes that may not align perfectly with standard ERP solutions, necessitating customization. However, extensive modifications can complicate system maintenance, increase costs, and prolong implementation timelines. Integration with existing legacy systems also presents a major challenge. Many organizations struggle with data migration and

compatibility issues when connecting their ERP system with pre-existing software. Poorly integrated systems may result in data inconsistencies and inefficiencies, ultimately reducing the effectiveness of ERP implementation

3. **User Resistance and Training Needs:** The most significant impediment in the implementation of an ERP system is user resistance. Employees generally resist new systems because they are unfamiliar with them, fear job displacement, or believe that the implementation system is difficult to handle. Thus, proper change management strategies have to be in place in order to overcome resistance; namely, proper communication about the benefits of ERP and involving users in the implementation process. Moreover, ERP systems are usually very elaborate and require huge training periods so that employees work proficiently while handling them. Training programs should be framed with different departments so their adoption could be maximized while ensuring the people who use the system feel comfortable working on it.

Emerging Trends in ERP Systems

As technology keeps developing, the best ERP systems keep integrating all those innovations with functionality, efficiency, and security in mind. Among the top trends in emerging ERP systems, include:

1. **Artificial Intelligence and Machine Learning in ERP:** ERP systems are changing due to AI and ML, where the system provides advanced data analytics, intelligent automation, and predictive insights. Advanced decision-making by AI-driven ERP solutions can be made possible with the analysis of huge amounts of data in real-time, which identifies patterns and predicts trends. Machine learning algorithms can automate routine tasks such as invoice processing and demand forecasting and minimize human error and increase the efficiency of the operations.
2. **Cloud-Based ERP Solutions:** Cloud computing has transformed the deployment of ERP. It has given businesses more flexibility, scalability, and cost-effectiveness. Cloud-based ERP systems do not require on-premises infrastructure, thus reducing IT maintenance costs. They also provide remote access, which makes it easier for employees to collaborate from different locations.
3. **Predictive Analytics and Real-Time Data Processing:** Modern ERP systems are better equipped through AI and big data-powered predictive analytics. Predictive analytics gives businesses the ability to predict what might happen based on trends found in historical data, thereby letting them anticipate demand, optimize levels of inventory, and thus enhance risk management. Advancements in real-time processing of data help ERP to provide on the fly insights on financial performance, customer behavior, and supply chain dynamics.

Case Studies and Industry Applications

1. **Successful Implementations of ERP in Different Industries:** ERP systems have been successfully implemented in a wide range of industries, from manufacturing to retail, leading to improved efficiency, cost reductions, and data-driven decision-making. In the manufacturing sector, Toyota and Ford companies use ERP for managing their whole supply chain, tracking inventory levels, and optimizing production planning to increase efficiency. The retail industry also benefits from ERP, with Walmart using it to manage its vast supply chain, streamline logistics, and update inventory in real time, ensuring seamless operations. In the healthcare sector, hospitals and healthcare providers implement ERP systems to maintain patient records, enhance billing efficiency, and improve overall operational performance. Additionally, the finance industry uses ERP for automating compliance reporting, managing complex transactions, and strengthening customer relationship management. The above examples indicate how ERP solutions have helped various business sectors streamline their processes, enhance decision-making capabilities, and bring about operational excellence.
2. **ERP Project Failure: Lessons Learnt:** While ERP systems bring about substantial benefits, they are not successful in all implementation projects, and failure experiences become valuable lessons for future projects. A major cause of ERP failure is improper planning, as reflected in the Hershey's example, where mass financial losses arose from hasty deployment and bad preparation. Inadequate user training is another critical factor that leads to resistance to adoption and inefficient use of the system, thus reducing the effectiveness of

the ERP. Over-customization is another common pitfall, as excessive modifications make ERP systems difficult to maintain and upgrade, as observed in various failed implementations. By studying successful and failed ERP projects, organizations can develop strong strategies that will ensure a well-planned and properly executed ERP implementation while reducing risks and maximizing business benefits.

Conclusion and Future Scope

1. **Summary of Key Findings:** ERP systems improve business operations by increasing efficiency, automating processes, optimizing resource management, and integrating emerging technologies. They provide organizations with data-driven decision-making capabilities that enhance competitive advantage.
2. **Implications for Businesses Adopting ERP:** Companies targeting ERP implementation, therefore, are required to meticulously plan the operation, provision needed resources, and train the manpower. Issues encountered in terms of high cost for implementation, possible customization issues and data security require special attention on the path for successful implementation.
3. **Future Directions of ERP Research and Development:** Future research into ERP systems is likely to revolve around better AI-driven automation, stronger cybersecurity frameworks, easier integration with IoT and blockchain, and increased adaptability through cloud-based solutions.

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- 12) 2nd World Conference On Business, Economics And Management WCBEM 2013 Business information systems: research study and methodological proposals for ERP implementation process improvement Rastislav Rajnoha a *, Jaroslava Kádárová b, Andrea Sujová c , Gabriel Kádár

Development of Solar Powered Automatic Cotton Picking Machine Using Movable Robotic Arm: A Review

Ajaykumar W. Kutarekar

P.G.Student Yeshvantrao Chavan
College of Engineering, Nagpur

Prof. Rakesh V. Adakane

Yeshvantrao Chavan College of
Engineering, Nagpur

Prof. Prashant D. Kamble,

Yeshvantrao Chavan College of
Engineering, Nagpur

Abstract—Cotton harvesting is a labor-intensive process, with traditional manual methods being inefficient and costly. Mechanized systems, such as spindle pickers and strippers, offer higher productivity but are expensive, bulky, and prone to crop damage. Robotic cotton-picking systems, integrated with artificial intelligence (AI) and computer vision, provide a promising alternative for selective and precise harvesting. This review examines advancements in robotic arm-based cotton picking, focusing on boll detection using AI, challenges in real-time processing, and adaptability to varying field conditions. The potential of solar-powered robotic systems is explored, emphasizing energy-efficient actuation and sustainable operation. Key challenges include high costs, scalability issues, and AI adaptability for different cotton varieties. Future research should focus on improving lightweight materials, efficient actuators, and precision farming technologies. This review highlights the need for innovative solutions to enhance automation, sustainability, and cost-effectiveness in cotton harvesting.

Keywords— Cotton harvesting, robotic arm, automation, computer vision, artificial intelligence, solar-powered system, precision agriculture

I. INTRODUCTION

Cotton is one of the most significant agricultural commodities, serving as the primary raw material for the textile industry. According to the Food and Agriculture Organization (FAO), cotton is cultivated in over 75 countries worldwide, with leading producers including China, India, the United States, and Brazil. The global demand for cotton continues to rise, driven by the increasing need for textiles, clothing, and industrial products [1]. However, despite its economic importance, cotton harvesting remains a labor-intensive and time-consuming process, necessitating advancements in mechanization and automation to enhance productivity and efficiency [2]. Traditional cotton harvesting is predominantly performed manually in many developing countries, where labor availability and cost are critical factors [3]. Manual picking involves hand-plucking cotton bolls, which, although ensuring high fiber quality, is highly labor-intensive and inefficient. Studies have shown that manual harvesting requires approximately 120-150 labor hours per hectare, making it an unsustainable practice for large-scale cotton farms [4]. Additionally, seasonal labor shortages and increased wages have made manual harvesting economically unviable in many regions [5]. Another significant limitation of manual harvesting is the inconsistency in picking efficiency due to variability in crop maturity, weather conditions, and human fatigue. Research by Assis et al. [6] highlights that inconsistent harvesting leads to fiber contamination, which negatively impacts the quality of the final product. Furthermore, manual picking results in significant post-harvest losses, with studies indicating that up to 15-20% of cotton bolls remain unharvested due to inefficiencies in human labor [7]. Given these constraints, there is an urgent need for automated harvesting solutions to address these challenges effectively. The advent of automation in agriculture has significantly improved productivity across various crop sectors, and cotton harvesting is no exception. Mechanized harvesting has been widely adopted in developed countries, primarily through spindle pickers and stripper harvesters [8]. However, these machines are expensive, require specialized maintenance, and are often unsuitable for small and medium-sized farms [9]. Research by Russel et al. [10] indicates that large-scale mechanical harvesters can achieve picking efficiencies exceeding 90%, but their high operational costs limit their accessibility in developing regions. In recent years, robotic cotton pickers have emerged as a promising alternative to traditional mechanized harvesters. A study by Joseph et al. [11] demonstrated that robotic arms equipped with computer vision systems could selectively pick cotton bolls with minimal damage to the plant. This precision harvesting approach enhances fiber quality while reducing unnecessary waste. Additionally, advancements in AI and machine learning have enabled real-time cotton boll detection, improving picking accuracy and efficiency [12]. Energy efficiency is another critical aspect of automated harvesting. Many existing robotic systems rely on conventional power sources, which may not be sustainable in remote agricultural areas [13]. Solar-powered robotic systems have gained attention as an eco-friendly solution to address this issue. Studies by Sohail et al. [14] highlight that integrating solar panels with autonomous cotton pickers can significantly reduce operational costs while ensuring uninterrupted performance in off-grid locations.

This review aims to analyze existing cotton harvesting technologies, with a focus on the development of solar-powered robotic systems. It explores advancements in automation, AI-driven boll detection, and energy-efficient solutions to enhance precision and sustainability in cotton harvesting.

II. OVERVIEW OF EXISTING COTTON HARVESTING METHODS

A. Manual Cotton Picking

Manual cotton harvesting (Figure 1) remains a prevalent method in many developing countries, primarily due to the high cost of mechanized alternatives and the perceived quality benefits of handpicking [15]. This method involves farm laborers manually plucking mature cotton bolls from plants, ensuring minimal contamination and damage to fibers [16]. However, despite its advantages in maintaining fiber integrity, manual picking presents several challenges related to labor dependency, efficiency, and economic sustainability [17].



Figure 1: Manual cotton picking



Figure 2: spindle picker

One of the major drawbacks of manual cotton harvesting is the high labor requirement. Studies indicate that manual harvesting requires approximately 120-150 labor hours per hectare, making it one of the most labor-intensive agricultural activities [8]. The reliance on a seasonal workforce also creates operational bottlenecks, as labor shortages during peak harvesting periods can delay crop collection and lead to yield losses [18]. Additionally, the rising cost of agricultural labor has made manual harvesting increasingly unviable for large-scale cotton farms [19].

Efficiency concerns also limit the scalability of manual cotton picking. Research has shown that inconsistent picking efficiency, driven by factors such as worker fatigue, uneven crop maturity, and environmental conditions, results in significant post-harvest losses [20]. A study by Sanjay et al. [21] reported that up to 15-20% of cotton bolls remain unharvested in manually harvested fields, leading to reduced profitability and increased waste. Furthermore, contamination risks associated with human handling, such as fiber staining and the inclusion of foreign materials, negatively impact cotton quality and market value [1].

Despite these limitations, manual harvesting remains dominant in smallholder farming systems due to its low initial investment compared to mechanized solutions [22]. However, as global cotton demand increases and labor costs continue to rise, transitioning to automated or semi-automated harvesting solutions is becoming a necessity. Research suggests that integrating technology into traditional cotton harvesting practices, such as using handheld mechanical pickers or AI-assisted labor management systems, can enhance efficiency while maintaining fiber quality [23].

B. Mechanized Cotton Harvesting

Mechanized cotton harvesting has significantly transformed the cotton industry by increasing efficiency and reducing reliance on manual labor. Conventional automated cotton pickers primarily fall into two categories: spindle pickers and stripper harvesters. These machines are widely used in developed countries such as the United States and Australia, where large-scale cotton farming is prevalent [24]. However, despite their advantages in improving harvesting speed and reducing labor costs, mechanized systems present several limitations related to cost, bulkiness, crop damage, and selectivity [25].

Spindle pickers (Figure 2) are the most common mechanized cotton harvesting machines. They function by rotating moistened spindles that pull cotton fibers from open bolls while leaving unopened bolls intact [26]. These machines ensure a high level of fiber quality and are designed for selective picking, reducing contamination. However, research indicates that spindle pickers require specific planting patterns, precise row spacing, and high operational costs, limiting their feasibility for small and medium-scale farms [27]. Additionally, spindle pickers struggle in conditions where bolls are not fully open or when excessive moisture is present, leading to inefficiencies and potential fiber loss.

C. Emerging Robotic Cotton Picking Systems

The advancements in robotics and AI have led to the development of robotic cotton picking systems that aim to overcome the limitations of conventional mechanized harvesters. Unlike spindle pickers and stripper harvesters, robotic systems utilize precision-based harvesting techniques, integrating sensors, computer vision, and AI algorithms to identify and selectively pick cotton bolls with minimal crop damage [28]. These systems offer promising advantages in terms of labor independence, improved fiber quality, and adaptability to different farming conditions. However, challenges such as system complexity, high power consumption, and adaptability to varying cotton growth patterns remain significant barriers to widespread adoption [29].

Robotic cotton picking systems are primarily designed using articulated robotic arms, equipped with end-effectors that mimic human hand movements for efficient cotton boll extraction [30]. Recent studies have demonstrated the feasibility of robotic cotton pickers integrated with machine vision technology, enabling real-time cotton boll detection and selective harvesting. For example, Gharakani [26] developed a vision-guided robotic system that used convolutional neural networks (CNNs) to identify mature cotton bolls and execute precise picking actions. Their results showed an accuracy of over 85% in boll detection, highlighting the potential of AI-driven harvesting solutions. Furthermore, researchers have explored various gripping mechanisms to optimize cotton extraction. Studies by Han et al. [31] compared vacuum suction grippers, soft robotic fingers, and mechanical pinchers for their efficiency in cotton boll picking. The findings revealed that soft robotic grippers, designed to mimic the flexibility of human fingers, exhibited the least damage to cotton fibers while maintaining a high picking success rate. Additionally, the integration of multi-degree-of-freedom (DOF) robotic arms has enhanced maneuverability, allowing pickers to access bolls in dense plant canopies.

Another advancement in robotic harvesting is the incorporation of autonomous navigation systems. Mwitta et al. [32] developed a self-driving robotic cotton picker that combined LiDAR sensors and GPS-based path planning to navigate cotton fields autonomously. Their study reported a reduction in operational errors and increased harvesting speed compared to manually operated robotic systems. Despite these advancements, real-world applications remain limited due to technical and economic challenges.

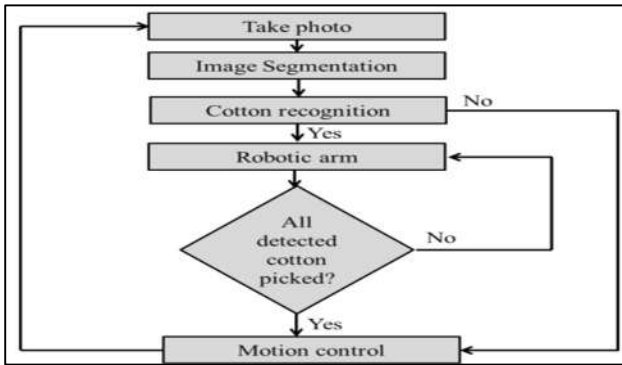


Figure 3: Flow chart



Figure 4: Solar powered cotton picking machine [58]

Despite existing challenges, research continues to refine robotic cotton-picking technologies. AI and deep learning advancements are expected to improve boll recognition accuracy, while developments in lightweight and energy-efficient robotic work collaboratively to improve efficiency and reduce operational time [33]. Moreover, solar-powered robotic systems are gaining interest as a sustainable solution to address energy constraints [34]. Table 1 provides a comparative analysis of different cotton harvesting methods, highlighting their efficiency, labor requirements, cost, crop quality, and environmental impact. The analysis underscores the potential of robotic harvesting, particularly solar-powered systems, as a sustainable and efficient alternative to traditional methods.

Table 1: Comparative Analysis of Cotton Harvesting Methods

Harvesting Method	Manual Harvesting	Mechanized Harvesting (Spindle/Stripper Pickers)	Robotic Harvesting	Ref.
Labour Requirement	High (120-150 labour hours per hectare)	Low (1-2 operators per machine)	Minimal (autonomous or semi-autonomous operation)	[35]
Harvesting Efficiency	Low (65-80%)	High (90-95%)	Moderate to High (85-95%)	[8]
Operational Cost	Low (but increasing due to labor costs)	High (expensive machinery and maintenance)	Moderate (high initial cost but low operational cost with automation)	[36]
Crop Quality	High (least fiber damage)	Moderate (fiber damage due to aggressive picking)	High (precision picking with minimal fiber damage)	[37]
Suitability for Small Farms	High (widely used in developing regions)	Low (requires large-scale farms)	Moderate (adaptability possible with compact design)	[38]
Energy Requirement	None	High (fuel-powered engines)	Low (renewable energy sources like solar panels)	[37]
Environmental Impact	Low	High (fuel emissions, soil compaction)	Low (sustainable and energy-efficient)	[39]

III. ROBOTIC ARM-BASED COTTON PICKING: ADVANCEMENTS AND CHALLENGES

A. Design Considerations for a Robotic Cotton Picker

The development of robotic cotton pickers requires careful consideration of structural design, material selection, and mechanism efficiency to ensure precise and damage-free harvesting. Unlike conventional mechanical harvesters, robotic systems must be lightweight, highly manoeuvrable, and capable of selectively picking cotton bolls without harming the plant. Research has shown that factors such as arm flexibility, degrees of freedom (DOF), end-effector design, and motion control strategies play a crucial role in optimizing performance.

Robotic arms used in agricultural applications, including cotton harvesting, typically adopt multi-jointed structures with 4- to 6-DOF configurations to replicate human-like picking actions [40]. A study by Singh et al. [41] compared different robotic arm architectures, including serial and parallel manipulators, for cotton-picking tasks. Their findings indicated that serial manipulators offered greater reachability and flexibility, while parallel manipulators provided higher stability and load capacity. Material selection is another key factor influencing the efficiency of robotic arms. Lightweight materials such as aluminum alloys and carbon fiber composites are preferred to reduce energy consumption and enhance mobility [42]. Research by Ali et al. [43] demonstrated that reducing the overall weight of robotic arms by 20% improved operational speed by 15% while maintaining structural integrity. Furthermore, robotic arms must withstand harsh field conditions, requiring corrosion-resistant coatings and dustproof enclosures to ensure durability.

Additionally, motion planning algorithms are critical in reducing unintended plant damage during harvesting. Advanced path-planning techniques, such as real-time trajectory optimization and adaptive force control, have been integrated into robotic cotton

pickers to ensure smooth and precise movement. Studies by Hernandez et al. [44] demonstrated that integrating force sensors in robotic arms allowed for real-time adjustment of grip strength, minimizing boll detachment force to prevent fiber damage.

Despite significant advancements, several challenges remain in the design of robotic cotton pickers. Energy efficiency remains a major concern, as high-power consumption limits field applicability. The integration of solar-powered energy systems is being explored as a sustainable solution to extend operational duration [45]. Additionally, further research is needed to enhance robotic adaptability to different cotton plant varieties and environmental conditions. The combination of AI, sensor fusion, and lightweight robotic materials is expected to drive the next generation of robotic cotton pickers towards higher efficiency and commercial viability. Table 2 presents a comparative analysis of various robotic cotton picking technologies, evaluating their picking mechanisms, AI integration, energy sources, efficiency, and limitations.

Table 2: Comparison of Robotic Cotton Picking Technologies

Technology	Picking Mechanism	AI & Vision System	Energy Source	Picking Efficiency (%)	Limitations	Ref
Mechanical Robotic Arm	Gripper-based picking	Basic image processing	Battery-powered	70-80	Limited adaptability to different cotton varieties	[44]
AI-Guided Robotic Arm	Soft gripper with force control	Machine learning for boll detection	Battery + Solar	85-90	High computational demand	[46]
Autonomous Cotton Picker	Multi-arm system with suction	Deep learning-based object recognition	Hybrid (solar + battery)	90-95	High initial cost and maintenance	[44]
UAV-Based Cotton Picker	Drone-mounted picking arms	Advanced vision with real-time mapping	Battery-powered	60-75	Limited payload and flight time	[47]

B. AI and Computer Vision for Cotton Boll Detection

The integration of AI and computer vision in agricultural automation has revolutionized precision farming, including robotic cotton picking. Image processing and machine learning techniques enable robotic systems to recognize and differentiate cotton bolls from leaves, stems, and background elements with high accuracy. These advancements help address key challenges in automated harvesting, such as selective picking, efficiency enhancement, and damage minimization [48]. However, limitations in real-time processing, lighting variability, and occlusions still hinder optimal performance, necessitating further research and improvements [49].

Image processing and AI-driven classification models play a critical role in cotton boll detection. Traditional methods, such as color thresholding and texture analysis, have been widely used for boll identification but struggle with accuracy due to environmental factors [50]. In contrast, modern deep learning models, particularly convolutional neural networks (CNNs), have significantly improved detection rates by learning complex patterns and features from large datasets [51].

A study by Madhu et al. [52] implemented a YOLO-based deep learning model for cotton boll detection, achieving a detection accuracy of 91.3% in controlled field conditions. Similarly, Kumari et al. [53] applied a hybrid approach combining CNNs and support vector machines (SVMs), improving boll classification accuracy by 8% compared to conventional machine learning models. These studies demonstrate the effectiveness of AI in enhancing recognition precision, yet challenges remain in real-time applications due to high computational demands.

C. Power and Energy Efficiency Considerations

The efficiency and feasibility of robotic cotton harvesting systems are largely determined by their power sources and energy management strategies. Traditional robotic harvesters typically rely on conventional batteries or diesel-powered generators, both of which have limitations in terms of operational cost, maintenance, and environmental sustainability [54]. The increasing emphasis on sustainable agriculture has driven research toward integrating renewable energy sources, particularly solar power, into robotic harvesting systems [55]. This section explores the various power sources for robotic cotton pickers, the feasibility of solar-powered systems, and energy-efficient strategies for enhancing robotic performance in the field.

Robotic harvesting systems employ different power sources depending on their design and operational requirements. Battery-powered systems are among the most common, with lithium-ion (Li-ion) and lead-acid batteries providing the necessary energy for robotic arms, sensors, and locomotion mechanisms. Li-ion batteries, in particular, are preferred due to their high energy density and long cycle life. However, their limited runtime necessitates frequent recharging or battery replacements, which can disrupt continuous harvesting operations [56]. Some large-scale robotic cotton pickers incorporate diesel-electric hybrid power systems to extend operational time. These systems combine internal combustion engines with electric motors, reducing dependence on battery storage alone. However, they introduce maintenance complexities and contribute to greenhouse gas emissions, making them less suitable for environmentally conscious farming practices [57].

Solar-powered systems (Figure 7) have gained increasing attention as a viable alternative for powering robotic harvesters. PV panels integrated into robotic platforms can harness solar energy, reducing reliance on external charging infrastructure and fossil fuels. Studies have demonstrated that solar power can significantly lower operational costs while providing a sustainable energy

source for agricultural automation [59]. Despite these advantages, challenges such as variable sunlight availability and energy storage limitations remain key areas of research.

IV. DEVELOPMENT OF A SOLAR-POWERED AUTOMATIC COTTON PICKING MACHINE

The development of a solar-powered automatic cotton picking machine involves the integration of advanced robotic technology with sustainable energy solutions. The key innovation lies in the combination of a movable robotic arm with solar-powered energy sources, ensuring an autonomous and efficient harvesting process while reducing dependence on conventional power supplies. This section discusses the fundamental design principles, mechanical structure, control system, and mobility considerations essential for the successful implementation of such a system.

The primary component of the system is a robotic arm designed to mimic human cotton-picking techniques with precision and minimal fiber damage. Equipped with end-effectors capable of selectively plucking cotton bolls, the robotic arm is integrated with a vision-based detection system for real-time crop recognition. The use of AI-driven computer vision enhances the picker's ability to identify mature cotton bolls while avoiding unripe ones, thereby improving overall harvesting efficiency. Powering the system with solar energy addresses sustainability concerns and enables operation in remote agricultural fields where grid electricity may not be available. Photovoltaic (PV) panels mounted on the machine harvest solar energy, which is stored in rechargeable batteries to ensure continuous operation. Energy management strategies, including maximum power point tracking (MPPT) and smart battery storage, are implemented to optimize power usage and extend machine runtime.

The mechanical structure of the cotton-picking machine is designed for robustness, lightweight construction, and ease of manoeuvrability across cotton fields. The robotic arm consists of multiple degrees of freedom (DOF) to allow flexible movement and precise cotton boll extraction. High-efficiency actuators, such as brushless DC motors or pneumatic systems, are incorporated to minimize energy consumption while maintaining high responsiveness. The control system integrates microcontrollers and embedded systems to regulate robotic operations, process visual data, and execute automated decision-making. A central processing unit (CPU) coordinates input from sensors, including cameras and force sensors, to optimize cotton-picking actions in real-time. Wireless communication modules may also be incorporated to enable remote monitoring and control of the system.

Additionally, the economic feasibility of solar-powered robotic pickers is a significant factor in their adoption. Initial investment costs may be higher than manual labour but significantly lower than large-scale mechanical harvesters. Research by Ukoba et al. [60] suggests that solar-powered robotic systems can achieve a return on investment within five years, mainly due to reduced operational costs and independence from fossil fuels or grid electricity. Table 4 presents a comparative analysis of different cotton picking methods, evaluating key parameters such as efficiency, labour requirements, fibre quality, energy consumption, and operational costs. The data highlight the advantages of solar-powered robotic picking in terms of cost-effectiveness, sustainability, and reduced energy consumption while maintaining high efficiency.

Table 3: Performance Comparison of Different Cotton-Picking Methods

Parameter	Manual Picking	Mechanized Harvesting	Robotic Arm-Based Picking	Solar-Powered Robotic Picking
Picking Efficiency (%)	85–95	90–95	92–97	90–98
Labor Requirement (hours/ha)	120–150	10–20	15–25	12–22
Fiber Quality	High	Moderate	High	High
Crop Damage (%)	Low	High	Low	Low
Energy Consumption (kWh/ha)	0	15–25	8–12	5–10
Operational Cost (₹/ha)	8,000–12,000	15,000–25,000	10,000–18,000	7,000–15,000
Environmental Impact	Low	High	Moderate	Low

V. CHALLENGES AND FUTURE PROSPECTS

The integration of solar-powered robotic systems in cotton harvesting presents several challenges that must be addressed to ensure widespread adoption. These challenges primarily revolve around cost, scalability, adaptability to different cotton varieties, and the need for further advancements in energy efficiency and precision agriculture technologies. However, ongoing research in artificial intelligence (AI), lightweight materials, and autonomous farming is expected to drive future innovations in this field.

One of the most significant barriers to adopting robotic cotton pickers is the high initial investment cost, particularly for small and medium-scale farmers. Traditional mechanical harvesters require substantial capital, making them inaccessible for many cotton producers in developing regions [61]. While robotic systems offer greater precision and energy efficiency, their development and deployment costs remain high due to the need for advanced sensors, actuators, and AI-based vision systems [61]. Research by Mail et al. [62] indicates that reducing costs through mass production and modular designs could make robotic cotton pickers more affordable for smallholder farmers.

Another critical challenge is scalability, as current robotic solutions are often designed for controlled environments or large-scale commercial farms. Small farms, which account for a significant portion of cotton production in countries like India and Pakistan, may struggle to integrate robotic harvesting due to land fragmentation, varying crop conditions, and limited access to

financial support [63]. Therefore, future research should focus on developing scalable, cost-effective robotic solutions tailored to the needs of small-scale farmers.

The diversity of cotton varieties across different regions poses a challenge for AI-driven robotic pickers. Cotton plants exhibit significant variations in boll size, growth patterns, and canopy density, which can impact the accuracy of computer vision algorithms [64]. Existing AI models are often trained on specific datasets that may not generalize well to different environments, leading to decreased efficiency in boll detection and selective picking.

To address this, researchers are working on adaptive AI models capable of learning and adjusting to different cotton varieties and field conditions. A study by Liu et al. [65] demonstrated that deep learning models trained on diverse datasets improved boll detection accuracy by 15-20% compared to conventional computer vision methods. Future developments in AI should focus on self-learning algorithms, real-time adaptive training, and improved sensor fusion techniques to enhance the robustness of robotic pickers across various cotton-producing regions.

The efficiency of robotic cotton pickers is closely linked to their weight and power consumption. Traditional robotic arms are often constructed using heavy materials, which increases energy demand and reduces mobility [66]. Recent advancements in lightweight composite materials, such as carbon fiber and advanced polymers, have shown promise in reducing the weight of robotic arms while maintaining durability [67]. Additionally, energy-efficient actuators play a crucial role in minimizing power consumption. Research by Perera et al. [68] suggests that incorporating soft robotics and compliant actuators can improve the dexterity of robotic arms while reducing energy requirements by up to 30%. By combining these advancements with optimized power management systems, solar-powered robotic pickers can achieve longer operational durations and improved efficiency.

The future of robotic cotton harvesting lies in the integration of precision agriculture technologies and fully autonomous farming systems. Precision agriculture involves the use of IoT sensors, GPS-guided navigation, and real-time data analytics to optimize harvesting efficiency and resource management [69]. By leveraging these technologies, robotic cotton pickers can operate with minimal human intervention while continuously adjusting to varying field conditions. Another promising avenue is the development of multi-functional robotic systems capable of performing multiple tasks beyond cotton picking. Research by Upadhyay et al. [70] highlights the potential of robotic platforms that integrate weeding, pest control, and real-time crop monitoring alongside harvesting. Such multi-purpose robotic systems could significantly improve overall farm productivity and sustainability. Furthermore, collaborative robotics—where multiple robots work together to perform harvesting tasks—could enhance efficiency and speed. A study by Fue et al. [71] demonstrated that swarm robotics approaches could improve overall cotton-picking efficiency by 25% compared to single-robot operations.

VI. CONCLUSIONS

The development of solar-powered robotic cotton pickers represents a significant advancement in agricultural automation, addressing labor shortages, inefficiencies, and environmental concerns associated with traditional harvesting methods. This review highlights the evolution of cotton harvesting technologies, from manual picking to mechanized and robotic systems, emphasizing the benefits and challenges of integrating AI, computer vision, and energy-efficient actuation. Despite their potential, robotic cotton pickers face challenges related to high initial costs, adaptability to different cotton varieties, and energy efficiency. Advances in adaptive AI models, lightweight materials, and solar power integration are crucial for improving the performance and affordability of these systems, particularly for small-scale farmers. Additionally, precision agriculture technologies and autonomous farming solutions will further enhance the capabilities of robotic cotton harvesters, making them more efficient and sustainable.

Future research should focus on reducing costs, improving AI-driven boll detection under varying conditions, and enhancing the energy efficiency of robotic systems. Collaborative efforts between researchers, agricultural industries, and policymakers will be essential to accelerate the adoption of robotic cotton pickers and drive innovation in smart, autonomous farming solutions. By addressing these challenges, robotic harvesting technology has the potential to revolutionize cotton agriculture, improving productivity, reducing labor dependency, and ensuring sustainable farming practices worldwide.

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Submission ID: EE-018

To Design and Implement Machine learning driven control system for Induction motors.

Mr. R. R. Solanke
rrsolanke1972@gmail.com

Dr. Bhumaiah Julia
bhumaiah.jula@ghrua.edu.in

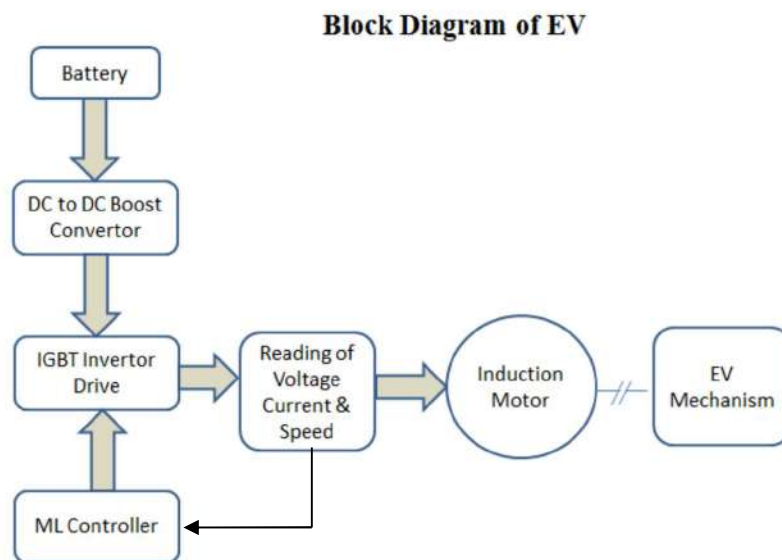
Abstract: Induction motors have a broad range of application in different industries and commercial establishments due to their desirable features. Their rugged and simple construction makes them extremely often dependable and simple to maintain. Since they possess extremely few moving parts compared to other motors, the likelihood of mechanical failure is greatly minimized. Induction motors are less expensive to produce compared to many other motors (such as synchronous motors), making them less costly to purchase too. These helps make induction motors be selected for many purposes. Other properties favoring use of these motors are simplicity in structure, low intrusion of maintenance compared to other types of motors, as well as improved reliability. These make them compete favorably on long-term operational expenses. Induction motors can pride themselves on efficiency levels that are comparatively high, particularly when in larger sizes. These efficiencies appear as lower power consumption and, therefore, reduced running expenses.

Machine-learning methods are being used more and more to control and monitor electric machine drives, thereby improving performance through data-driven means. Learning algorithms and embedded hardware will witness machine learning making a permanent entry in this sector. Key Advantages of Machine Learning in Electric Machine Drives. Machine learning algorithms improve control strategies for electric drives more efficiently and responsively to changing operating conditions. Historical data analysis through machine learning can identify possible failures or maintenance needs, therefore minimizing downtime and saving on maintenance expenses. Machine learning enables the continuous states of an electric machine to adjust through adjustments at any moment of necessity to enable optimum operation. Machine learning methods can extract meaningful findings from large volumes of data, leading to more informed decisions on design and operation capabilities.

Introduction:-

The design and development of an induction motor control system based on principles of machine learning should be a challenging and fascinating undertaking, as it plays the role of combining control theory, machine learning, and electrical engineering. Identify the goal of control, efficiency, minimize energy usage, and enhance operation under changing load conditions. Accuracy of speed control, response time, and resistance to perturbations Induction motors' dynamic models, such as the equivalent circuit, torque-speed characteristic, and the dynamic equations. The traditional control methods PID control, vector control to determine where their limitations lie and define where machine learning offers a benefit. The sensor-equipped induction motor with parameters such as speed, torque, current, voltage, and temperature. Adopt a process for data logging under varying operating conditions, varying loads, speeds, and environmental

conditions. Select among existing machine learning algorithms. Neural networks, regression models, problem-based reinforcement learning. An interface where the model founded on machine learning is integrated into a control structure will exist. The model could imply allowing it to forecast future motor states and then adjust control inputs accordingly. Install the feedback mechanism where control is improvisation with the support of on-line data collection.



Mathematical Modelling for an Induction Machine

In an induction machine with three phases the stator windings are supplied with a balanced three phase supply. The rotor winding are either internally short circuited or connected via slip ring to a passive external circuit. The unique characteristic of the induction machine is that the rotor currents are induced through electromagnetic induction from the stator. It is the rationale behind the terminology „induction machine“ Stators winding in a three-phase induction machine are equal to those in a synchronous machine. With a balanced three-phase currents of frequencies Hz, applied, the stator winding built up a revolving field at synchronous speed expressed as.

$$N_s = 120f/P$$

Where n_s is the position in r. P. M. & p is poles number. $\Theta = \omega_r t$ With a constant slip s

$$\Theta = 1 - s \omega_s t \text{ ----Where } \omega_s \text{ is angular speed of the stator field in electrical rad/sec}$$

Neglecting saturation hysteresis & eddy currents & assuming sinusoidal distribution of flux waves pure,

the machine equations can be stated as follows

Stator voltages equations

$$V_a = p \psi_a + R_s i_a$$

$$V_b = p \psi_b + R_s i_b$$

$$V_c = p \psi_c + r_s i_c \text{ ----- (1)}$$

Rotor voltage equations

$$V_a = p \psi_a + R_r i_a$$

$$V_b = p \psi_b + R_r i_b$$

$$V_c = p \psi_c + R_r i_c \text{ -----(2)}$$

In the above equations ψ is the flux that is common to the winding identified by the subscript r_s the stator phase resistance, r_r the rotor phase resistance & p the differential operator d/dt the positive directions of the currents

are into the windings

The conversions of the stator phase currents to d & q variable is as follows

$$i_{ds} = 2/3 [i_a \cos \omega t + i_b \cos (\omega t - 120^\circ) + i_c \cos (\omega t + 120^\circ)]$$

$$i_{qs} = -2/3 [i_a \sin \omega t + i_b \sin \omega t - 120^\circ + i_c \sin (\omega t + 120^\circ)]$$

Stator flux linkages:

$$\psi_{ds} = L_{ss} i_{ds} + L_m i_{dr}$$

$$\psi_{qs} = L_{ss} i_{qs} + L_m i_{qr} \text{ -----(3)}$$

Rotor flux linkages:

$$\psi_{dr} = L_{rr} i_{dr} + L_m i_{ds}$$

$$\psi_{qr} = L_{rr} i_{qr} + L_m i_{qs} \text{ -----(4)}$$

With $L_m = 3/2 L_A$

Voltages of stator in terms of d & q component are

$$V_{ds} = R_s i_{ds} - \omega_s \psi_{qs} + P \psi_{ds}$$

$$V_{qs} = R_s i_{qs} + \omega_s \psi_{ds} + P \psi_{qs} \text{ -----(5)}$$

Voltages of rotor are

$$V_{dr} = R_r i_{dr} - (P\theta_r) \psi_{qr} + P \psi_{dr}$$

$$V_{qr} = R_r i_{qr} + (P\theta_r) \psi_{dr} + P \psi_{qr} \text{ -----(6)}$$

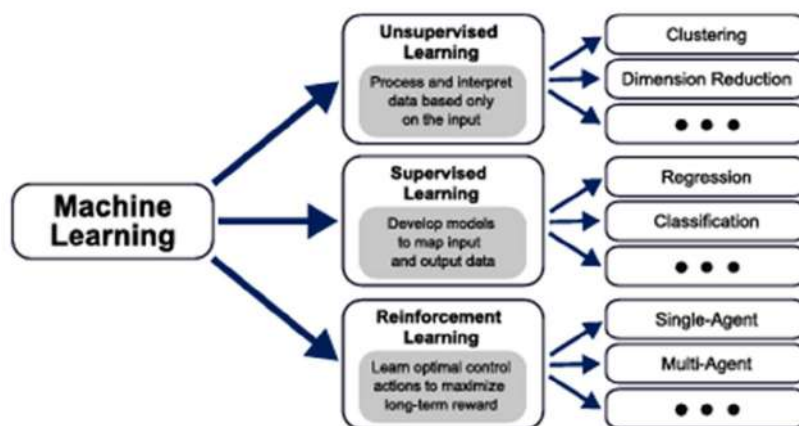
The term $p \theta_r$ in equation (6) is the slip angular velocity given by

$$P \theta_r = S \omega_s \text{ -----(7)}$$

This represents relative angular velocity between the rotor & the reference d & q axis

Machine Learning:--Machine learning techniques are increasingly taking center stage in induction motor drives for diagnosis and fault detection, enhancing reliability and performance. Various studies mention models that predict faults, employ deep learning to diagnose, and propose algorithms for specific motor issues. Important Applications of Machine Learning in Induction Motor Drives

Fault Diagnosis: Machine learning techniques are used to detect and diagnose faults in induction motors particularly those powered by variable frequency drives (VFDs). The technique increases the speed and efficiency of fault detection.



Deep Learning Techniques: The most recent research points out the use of deep learning techniques, such as semi 2D Convolutional Neural Networks, to identify motor faults. Such advanced models can analyze complex patterns of motor behavior and enhance diagnosis accordingly.

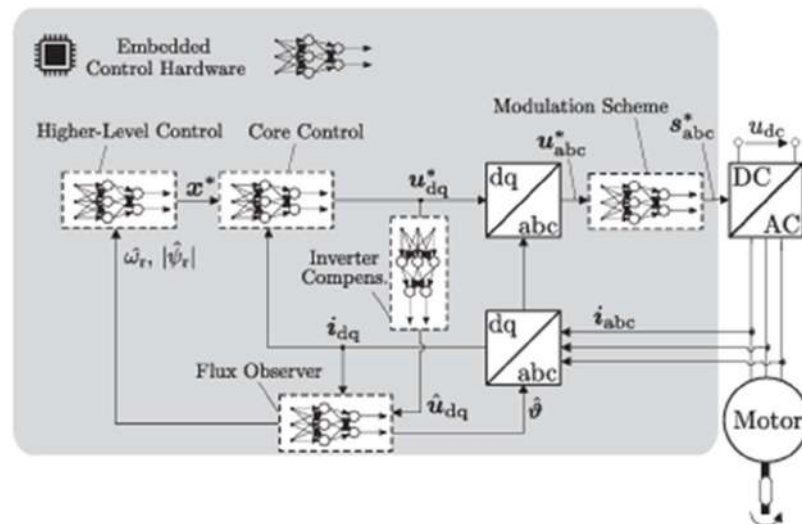
Predictive Maintenance: Predictive maintenance models apply machine learning models to predict likely failures in advance, allowing for proactive maintenance and reducing downtime. Predictive maintenance therefore enhances motor operation efficiency. **Data-Driven Insights:** With the application of large sets of data from motor operations, machine learning can unveil insights not observed through other means, leading to improved performance and life of induction motors.

Machine learning is a field of artificial intelligence that is interested in the development of algorithms which enable controller to learn and make predictions based on data. It uses various techniques, including supervised and unsupervised learning, to enable machines to improve their performance over time without explicit. some of the popular algorithms are linear regression, logistic regression, decision trees, support vector machines, k-nearest neighbors, and neural networks. They all have their own strengths and are well-suited to a different type of problem, such as classification, regression, or clustering

linear regression:

Linear regression is a statistical method used to model the relationship between an independent variable and one or more independent variables by fitting data observations to a linear equation. The objective is to estimate the best

best fitting line which has minimum variation between the predicted values and the actual values. It is used frequently for continuous response forecasting and can be expressed as the equation ($y = mx + b$), where (y) is the dependent variable, (m) is the slope, (x) is the independent variable, and (b) is the y-intercept.



Assumptions of Linear Programming:

- Assumptions of linear regression are linearity, independence, homoscedasticity, normality, and no multicollinearity. Linearity assumes that the relationship between the independent variable and the dependent variable is linear. Independence assumes that the residuals are independent of each other. Homoscedasticity requires that the residuals should have constant variance for all levels of the independent variable. Normality assumes that the residuals should normally be distributed. Lastly, no multicollinearity presupposes that independent variables should not be highly correlated with each other. The use of residuals in linear regression is to measure the difference between predicted values and actual values. Residuals can be used to evaluate how accurate the model is; the smaller the residuals, the tighter the fit. The study of residuals may also reveal patterns that reflect whether linear regression assumptions are being satisfied, such as testing for normality and homoscedasticity. Residuals may also be used to identify outliers or influential observations that can affect the performance of the model.

Conclusion:-

The enhancement of electric vehicles (EVs) with machine learning and data-driven control and monitoring schemes will facilitate high-speed exploration and adaptation to various domains. This innovation places them as a viable contender toward other traditional model-based methods in electric machine drive technology.

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Recent trends in Rotary Kiln and Refractory Material Patterns in Cement Production: a review approach

Suhas Padawe¹, Dr. G.M.Dhote², Devendra Y. Shahare³, Amol Kinkar⁴,

Abstract

The manufacturing of cement and other industrial operations depend heavily on rotary kilns, which use refractory materials to endure high temperatures and challenging chemical conditions. With an emphasis on their effects on durability and operating efficiency, this paper looks at the most recent developments in rotary kiln technology and refractory material choices. Various rotary kiln designs, typical refractory material patterns, and the difficulties posed by kiln failures are all covered. The research emphasises how crucial it is to maximise the performance of refractory linings in order to improve chemical resistance, reduce wear, and increase heat transmission. Additionally, simulation methods and mathematical modelling methodologies for rotary kiln process analysis are investigated, providing insights into energy optimisation and efficiency enhancements. The potential of emerging technologies to enhance process sustainability is also examined, including digital twin models and microwave-assisted calcination. This article offers helpful advice for choosing suitable refractory materials and optimising rotary kiln operation to achieve increased efficiency, lifespan, and environmental sustainability in cement production by examining important research and industry practices.

Keywords: Rotary Kiln, Refractory Material, Patterns

INTRODUCTION

The rotary kiln has applications for the manufacturing of cement. It is an apparatus for bedding in the manufacturing of both dry and wet cement clinker. Cement kilns may also be extensively used in other areas such as metallurgy, chemical engineering, refractory construction, and environmental protection. The cement kiln comprises a barrel, supporting apparatus, baffle wheel support, gearing mechanism, active kiln head cowl, end sealing device, and coal jet pipe apparatus, among other components. The cement rotary kiln operates at elevated temperatures, resulting in micro-scale movement. The maintenance of attachment devices and the thermal system's level is intricately linked to the safety and efficiency of the cement kiln. The essential aspects of gear care and maintenance include verifying lubrication, assessing operational state, conducting dynamic inspections, and ensuring meshing accuracy between big and tiny gears. The installation precision of the supporting device ensures the rotary kiln operates safely and smoothly. Subsequently, focus on the equitable distribution of the bearing weight of stay wheels, the configuration of riding wheels, and the upstream and downstream pressure and timing of the hydraulic gear. The calibre of the thieving device immediately influences the thermal system quality and operational expenses of the rotary kiln. Routine maintenance and data analysis enhance device management. A cement kiln is a crucial apparatus for both dry and wet cement manufacturing processes. The cement kiln primarily consists of a barrel, gearing mechanism, roller support apparatus, stop wheel support structure, refractory brick lining, kiln tail seal, kiln head cover, and fuel system.

The aim of this paper is to (a) briefly introduced the reader to physical properties of the rotary kiln along with types, Rotary Kiln for Different Materials, selection of rotary kiln, Rotary Kiln Failures and its application. (b) present the commonly observed instances of Refractory Types in Different Rotary Kiln Zones, Refractory Material Properties, and, refractory bricks and its failure with consideration of installation various pattern and, (c) importance of enhancing the refractory lining's performance.

Rotary Kiln

Nowadays, industry uses rotary kilns for a number of material processing processes, including calcining petroleum coke, reducing oxide ore, recovering hydrated lime, and recovering hazardous waste. However, rotary kiln operations continue to be beset by problems such dust formation, low

thermal efficiency, and uneven product quality.[1] Rotary kilns are widely used in various industrial processes, with a significant presence in cement, lime, and mineral production. These kilns require an inner lining of refractory materials to withstand the high temperatures, typically exceeding 1000°C. Rotary kilns are cylindrical, horizontally inclined furnaces that are commonly used in the cement industry for the thermal processing of raw materials. The inner surface of the kiln is lined with refractory materials, which are essential for protecting the steel casing and ensuring efficient heat transfer within the kiln. reduce heat losses, and ensure efficient operation.[2] The selection and design of refractory materials for rotary kilns are crucial factors in determining the overall performance and energy efficiency of the cement production process.[3] The inner lining of a rotary kiln plays a vital role in its operation, as it must be able to withstand the extreme temperatures and harsh chemical environment within the kiln. Refractory materials with suitable thermal, mechanical, and chemical properties are essential for ensuring the longevity and reliable performance of the kiln.[3],[2] Rotary kiln process, with a focus on the physical mechanisms, mathematical models, and applications related to refractory material patterns and their impact on cement

The physical mechanisms involved in the operation of a rotary kiln are complex, involving a wide range of interrelated phenomena, such as the movement and mixing of the solid materials, heat transfer between the solid, gaseous, and kiln wall phases, and the various chemical reactions that occur within the kiln.[4] Understanding these mechanisms is crucial for optimizing the design and operation of rotary kilns, particularly in the cement industry.

Several researchers have developed mathematical models to simulate the behavior of rotary kilns, with a focus on improving the understanding of the underlying physical and chemical processes.[5] These models have been used to predict various parameters, such as the power draw for material movement, the heat transfer rates, and the chemical transformation of the raw materials.[6] Alongside the modelling of the rotary kiln process, the analysis of refractory material patterns and their impact on cement production has also been an area of interest. The selection of suitable refractory materials for the rotary kiln lining is a crucial aspect of the cement production process.

***Author for Correspondence**

Name:- Suhas Ramesh Padawe

Dr. Girish M. Dhote

Devendra Y. Shahare

Mr. Amol Kinkar

E-mail: suhaspadawe.official@gmail.com

¹ M-tech in Automation and Robotics, Yeshwantrao Chavan College of Engineering (YCCE), Nagpur..

² Assistant Professor, Department of Mechanical Engineering, Yeshwantrao Chavan College of Engineering (YCCE), Nagpur.

³ Assistant Professor, Department of Mechanical Engineering, Yeshwantrao Chavan College of Engineering (YCCE), Nagpur

⁴ Caldersys India Refractories Limited, Butibori MIDC.

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The digital twin model significantly improves the efficiency of the solid waste treatment process. Predictive Analytics The model provides accurate predictive analytics, helping in proactive decision-making. Environmental Impact: The optimized process reduces the environmental impact by minimizing emissions and energy consumption.[7] investigate the efficiency and effectiveness of a cylindrical monomode microwave cavity integrated with a rotary kiln for the calcination of dolomite. The study aims to explore the potential benefits of microwave heating in terms of energy consumption, calcination time, and product quality.[8] The study provides a detailed analysis of the design load of the flights, considering the combined effects of thermal and granular flow dynamics.[9] Cofiring hydrogen and coal creates a more intense and smaller flame with earlier ignition and less fluctuation compared to pure coal combustion. The process temperature and heat transfer are enhanced at the beginning of the kiln. The burner design

significantly affects combustion behavior and Nox emissions. Cofiring hydrogen and coal using integrated burners is a promising solution for emission minimization in rotary kilns.[14],] Investigate the influence of baffles on the heat transfer characteristics of particles in an oil shale rotary kiln. The study aims to understand how different baffle configurations affect the heat transfer efficiency and overall performance of the rotary kiln. general heat transfer mechanisms in rotary kilns without considering the specific impact of baffle configurations. There is a lack of comprehensive research on how different baffle designs can optimize heat transfer in oil shale rotary kilns, which is crucial for improving energy efficiency and reducing operational costs. a combination of experimental and computational methods to analyze the heat transfer characteristics. The experimental setup includes a scaled-down model of an oil shale rotary kiln with various baffle configurations.[17],[18]] There is limited research on its application in the calcination of dolomite using a rotary kiln. Previous studies have primarily focused on conventional heating methods, leaving a gap in understanding the specific advantages and challenges associated with microwave-assisted

calcination. **Experimental Setup** A cylindrical monomode microwave cavity integrated with a rotary kiln was designed and constructed. The system was equipped with temperature sensors and monitoring devices to ensure precise control and measurement of the calcination process. **Calcination Time** The use of microwave heating resulted in a shorter calcination time, leading to increased productivity. **Product Quality** The calcined dolomite exhibited improved phase composition and microstructure, indicating enhanced product quality. [13] **Temperature Control** The integration of the microwave cavity with the rotary kiln allowed for precise temperature control, ensuring uniform calcination of the dolomite samples.[10][16], The rotary kiln process and to propose methods for improving the overall efficiency of the system. Previous studies had not comprehensively analyzed the energy and exergy performance of rotary kilns in the context of cement production. a detailed energy and exergy analysis of a rotary kiln with a capacity of 2900 tonnes per day and an L/D (Length/Diameter) ratio of 14.4. They calculated the energy and exergy efficiency of the kiln and identified the major sources of energy loss. The analysis involved collecting data from the cement factory and performing calculations to determine the efficiency metrics. that the energy efficiency of the rotary kiln was 49%, while the exergy efficiency was 37%. The significant energy losses were attributed to the high-temperature exhaust gases and the heat loss through the kiln shell.[15]The researchers proposed several improvement measures, including optimizing the kiln operation parameters and implementing waste heat recovery systems, to enhance the efficiency and reduce production costs. [13], [11], [12]

Types Of Rotary Kiln :-

Each serving different purposes in the cement manufacturing process .

Wet-Process Kilns: Kiln feed enters as a slurry with 30-40% moisture. Requires additional dehydration zone, making it longer than dry-process kilns. Requires more fuel to evaporate water.

Semidry Kilns: Grate Process Kilns or Lepol Kilns. Feed is pelletized and partially calcined on a traveling grate. Efficient in fuel consumption but have lower output rates compared to preheater or precalciner kilns.

Dry-Process Kilns: Kiln feed enters as dry powder. Typically long, with a length-to-diameter ratio of approximately 30:1 to 35:1. Operate with high back-end temperatures and require water sprays to cool exit gases.

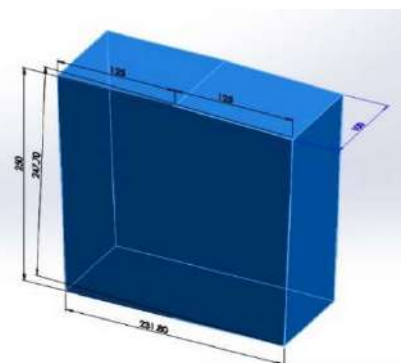
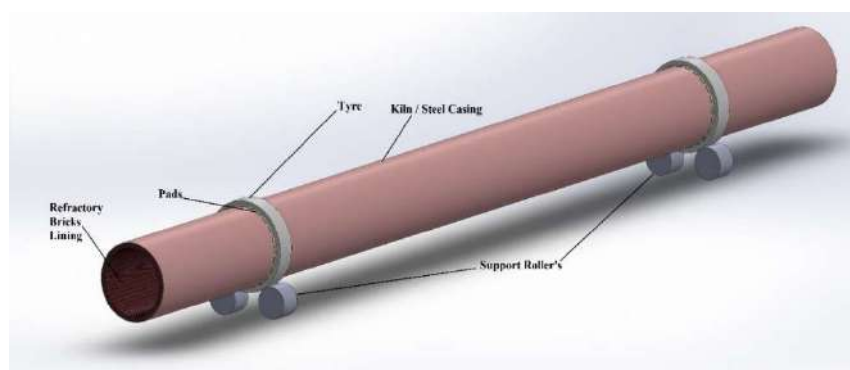
Preheater Kilns: Equipped with preheaters to improve fuel efficiency. Uses the heat from kiln exit gases to preheat the raw material. More efficient compared to traditional wet or dry kilns.

Precalciner Kilns:The latest advancement in cement manufacturing.Combines the benefits of preheating and calcination in one process. Able to use low-grade fossil fuels and show higher output rates. [2], [5]

The main components of a rotary kiln

1. The Rotating Cylinder:

- This is the heart of the kiln—a long, slightly inclined steel cylinder that rotates slowly on its axis. [19]describes this basic operation, where material enters the upper end and gradually moves down towards the lower end due to the rotation.
- The shell's inclination and rotational speed are crucial factors influencing material flow and heat transfer within the kiln. [5] provides some typical dimensions and operating parameters. Show in Figure No:-01



Refractory Bricks

Refractory materials are specifically designed to withstand extremely high temperatures, making them essential in environments like cement kilns. They serve as a protective lining inside the kiln, using refractory shown in Figure 02 shielding the outer shell from the intense heat generated during clinker production. [19] discusses this in detail, mentioning how the refractory lining is crucial for protecting the kiln's steel casing and minimizing heat loss. The editor document also emphasizes this point, explaining how refractory materials are essential for the efficient and safe operation of these kilns.

Several factors influence the choice of refractory material, including the specific operating temperature, the chemical environment within the kiln, and the types of mechanical stresses the lining will experience. [20], [21], [23] emphasizes the drive to improve the lifespan of refractories, highlighting the economic benefits of enhancing their thermal resistance and physical properties like strength and density. [22] explores how the inclusion of antioxidants can protect carbon-containing refractories from oxidation at high temperatures. [20], [21], [23] explains how thermal stresses and chemical attack can lead to the degradation of these materials. (2024) delves into the properties of refractory metals, highlighting their high melting points, hardness, and stability, arising from their unique electronic structures and bonding characteristics. [22]

Refractory selection should consider the specific conditions of each kiln zone. For instance, the sintering zone demands high refractoriness and alkali resistance, whereas the inlet zone requires greater abrasion resistance. Table no:01 and Summary of mechanical material properties Table No 02

Table no:01 [23]

Property	Fireclay Brick	High-Alumina Brick	Magnesia Brick	Magnesia-Chrome Brick	Dolomite Brick
Refractoriness (°C)	1500-1700	1700-1850	>2000	>2000	1900-2100

Table 2: Summary of mechanical material properties of the bricks used as input in the model.[55]

Table 2 [24]

Temperature (K)	273	973	1173	1273	1573
Bulk density (kg/m ³)	2700	2700	2700	2700	2700
Young's modulus (GPa)	3	7	4	2	1
Poisson's ratio [10]	0.2	0.2	0.2	0.2	0.2
Coef. thermal expansion (×10 ⁻⁶ K ⁻¹)[18]	6	6	6	6	6
Yield stress (MPa)	50	60	80	80	30
Tangential modulus (MPa)	100	100	100	100	100

Pattern Lining

"The pattern" you're referring to is pattern lining in the context of refractory installation within a rotary kiln. It's not a physical template, but rather a specific arrangement or design of refractory bricks or shapes within the kiln. Think of it like laying tiles in a bathroom – you don't just randomly place them; you follow a pattern for both aesthetic and functional reasons.

pattern lining and its importance in enhancing the refractory lining's performance. It suggests this practice improves heat transfer efficiency and extends the lining's service life. While it doesn't provide specific pattern examples, the concept revolves around strategically placing different refractory shapes or types to optimize performance.

Different refractory lining Pattern

The patterns you'll need to consult with refractory experts or look for specialized industry publications. They can tailor a design to your kiln's unique operating conditions. To suggested to a)

Straight line ,b) staggered lining and c) Compound lining Show in figure No 03

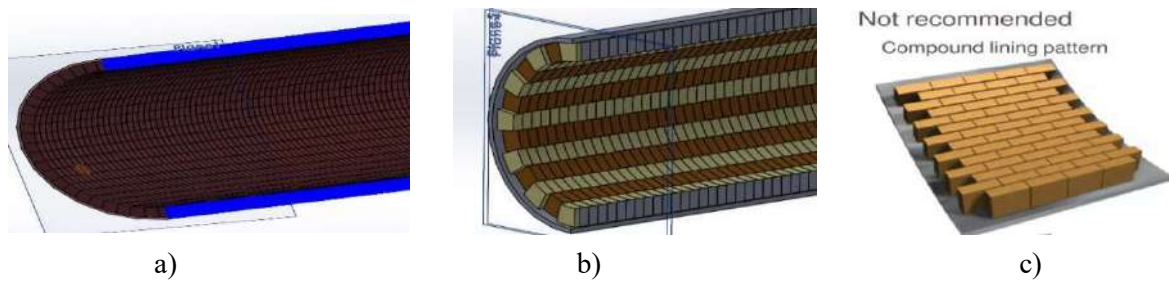


Figure No :- 03 Refractory Lining Pattern

Pattern Lining Based Importance

a) Optimized Heat Transfer: Different refractory shapes and compositions can be used in a pattern to direct heat flow more effectively. For instance, specific shapes might create turbulence in the hot gas stream, promoting better heat transfer to the material being processed.

b) Reduced Wear: Certain patterns might help distribute wear more evenly across the lining, preventing localized wear and tear in high-stress areas. This can prolong the lifespan of the entire lining.

c) Thermal Shock Resistance: By strategically placing refractories with different thermal expansion coefficients, a pattern can better accommodate thermal stress and prevent cracking or spalling due to rapid temperature changes.

d) Chemical Resistance: Patterns can be designed to place chemically resistant refractories in areas where corrosive materials are present, protecting the kiln shell and more sensitive refractory types. Importance of pattern lining, it doesn't give specifics. Unfortunately while discussing refractory lining modeling, also doesn't provide details about specific patterns.

CONCLUSION

Refractory materials are essential to the efficiency, durability, and operational stability of rotary kilns, which are essential to the production of cement. In order to improve energy efficiency and sustainability, this analysis focusses on developments in kiln design, refractory selection, and cutting-edge technologies like digital twin modelling and microwave-assisted calcination. The significance of optimising lining designs to enhance heat transmission, durability, and chemical resistance is highlighted by the examination of refractory material patterns and their effects on kiln performance. Additionally, simulation approaches provide insightful information for process improvement, highlighting the need of ongoing kiln technology advancement. The future of cement production may be shaped by rotary kilns that achieve higher efficiency, less environmental impact, and longer service life by combining contemporary optimisation methods with material science developments.

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Enhancing Machining Sustainability: Comparative Analysis of Dry, Minimum Quantity Lubrication, and Cryogenic Cooling Methods

Kaushal Prasad 1, *, Arpita Roychoudhury², Ganesh Prasad Shukla³

Abstract

Sustainable machining is crucial for minimizing the environmental, economic, and social challenges associated with modern manufacturing. This study reviews environmentally sustainable machining approaches including dry machining, Minimum Quantity Lubrication (MQL), and cryogenic cooling, and evaluates their effects on machining performance and sustainability. Key performance indicators such as surface integrity, tool wear, energy efficiency and cutting forces, are analyzed across various materials. The findings demonstrate that dry machining eliminates liquid waste but is often limited by tool wear and heat-related challenges especially in difficult-to-machine materials such as titanium and nickel alloys. MQL machining, which applies minimal oil mist for lubrication, significantly reduces cutting forces (10–30%), improves surface finish (91%), and extends tool life (267%), making it ideal for hardened steels and aluminum alloys. Cryogenic machining, utilizing LN₂ or CO₂ cooling, provides superior thermal control, extending tool life by 80–200% and reducing cutting temperatures by up to 75%, particularly for nickel-based superalloys and aerospace applications. Hybrid methods, such as combining cryogenic cooling with MQL, further enhance machining efficiency and environmental sustainability. The study concludes that there isn't a single superior machining technique, but that hybrid machining solutions combining cryogenic cooling with MQL are the most viable ones for sustainable, high-performance machining. This review underscores the potential for integrating advanced techniques to achieve cleaner and more efficient machining processes.

Keywords: Sustainable Machining, Dry machining, Eco-Friendly Machining, Minimum Quantity Lubrication (MQL), Tool Wear Reduction.

1. INTRODUCTION

A very common manufacturing process that is known as machining. Machining process has been used for centuries but still it is an attractive subject. In the development of industrial technologies machining play a very vital role due to its convenient and more economical than other forming processes. For that reason, enhancing productivity and machining efficiency are the primary challenges faced by the machining industry[1].

*Author for Correspondence

Corresponding Author Kaushal Prasad
E-mail: kaushalkurey8672@gmail.com

^{1,*} Student, Department of Industrial and Production Engineering, Guru Ghasidas (Central) Vishwavidyalaya, Bilaspur - 495009, Chhattisgarh, India

^{2,3} Faculty, Department of Industrial and Production Engineering, Guru Ghasidas (Central) Vishwavidyalaya, Bilaspur - 495009, Chhattisgarh, India

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the use of cutting fluids, as it greatly affects machining quality characteristics. Moreover, various economic, environmental, and health challenges arise from the improper use of cutting fluids. Establishing eco-friendly conditions and enhancing energy efficiency are crucial for attaining sustainable machining processes. [2][3]. The goal of sustainable machining processes is to reduce the environmental footprint associated with manufacturing products or components efficiently. This process should be conducted by developing and applying the ISO 14001:2015 environmental management system [EMS] [4]. The reliance on flood cooling has

One of the key sustainability issues in machining is exposed significant environmental issues, making it essential to find alternative solutions. It is necessary to introduce new strategies to improve machining performance while achieving sustainability. Various technologies have been introduced, such as dry cutting, minimum quantity lubrication (MQL), and cryogenic technology. [3]. Sustainable machining can be achieved through a variety of methods and

technologies, all designed to decrease environmental impact while preserving or improving machining performance. These methods encompass dry cutting, MQL, the application of vegetable oil-based cutting fluids, high-pressure coolant systems, cryogenic cooling, and hybrid cooling techniques. Each of these strategies presents distinct advantages regarding energy efficiency, waste reduction, and decreased toxicity, as illustrated in Figure 1 [5]

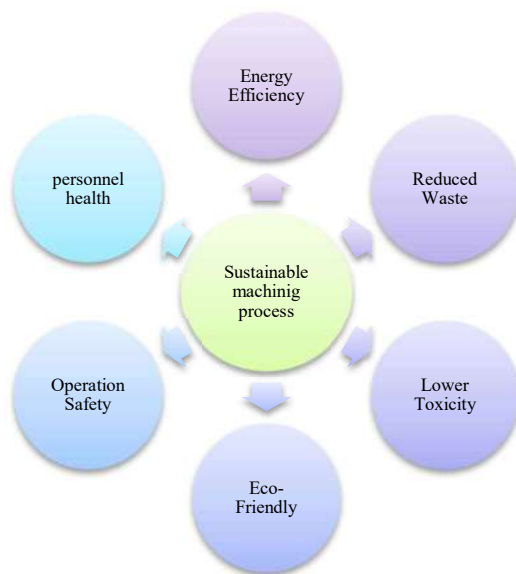


Figure 1. Basic element of sustainable machining

There is a dearth of research that focuses on all three aspects of sustainable machining using eco-friendly cooling systems on various types of material that how they work. After reviewing a number of literature reviews, the majority of research has been done on sustainable machining, sustainable cooling, or just one type of material machining process. Therefore, the behavior of various materials in dry, MQL, and cryogenic environments is the main topic of this study work in order to solve traditional machining issues and accomplish sustainable machining. It investigates a range of ecologically friendly machining approaches and assesses how successful they are. In order to evaluate the overall effect and effectiveness of these sustainable practices, the debate centers on important performance metrics such as cutting forces, tool wear, tool vibration cutting temperatures, and surface roughness across various materials.

2. Sustainable machining techniques

2.1 Dry machining: - Moving towards dry machining is a big step forward in manufacturing, helping to save money and support environmental goals. Ongoing research and new technologies will be essential to overcoming challenges and maximizing the benefits of this approach. By using dry machining techniques, industries can improve efficiency and contribute to a more sustainable future [6] [7] [8]. Dry machining is a sustainable and eco-friendly manufacturing process that will soon be essential for the manufacturing industry. Companies will need to adopt dry machining, either on their own or due to legal requirements, to comply with environmental protection standards and workplace safety regulations [9] [10].

Table 1: Machining of materials under dry condition [16]

Sr. No.	Technique	Material used	Cutting tool	Results/finding
1	Textured Tools	Inconel 718, SS316, Ti6Al4V	Cemented Carbide, Tungsten Carbide	Reduced cutting force by 10–20% and feed force by 20–30%. Improved tool life by 60% compared to plain tools for Inconel 718. Enhanced wear resistance and reduced chip adhesion on SS316; smoother surface finish.
2	Coated Tools	Aluminum, Inconel 718, Ti6Al4V	Cemented Carbide, Composite Ceramic	TiAlN coatings improved tool life, reduced friction, and enabled better heat dissipation in Inconel 718 machining. Multilayer coatings (e.g., TiN/TiCN/Al ₂ O ₃) increased tool life and reduced wear significantly for stainless steel. Self-lubricating coatings (TiSiN-Ag, MoS _x -TiN) reduced friction (~45%) and extended tool life by two-fold in titanium.

3	Solid Lubricant Coatings	Titanium Alloy, Aluminum	TiSiN-Ag Coated, Gr-X, TiSiVN Coated	TiSiVN coatings achieved ~18–23% lower adhesion, better wear resistance, and smoother surface finishes. Gr-X coatings outperformed MoS ₂ in cutting aluminum under humid conditions, reducing adhesion and wear.
4	Advanced Tool Materials	Titanium, Nickel Alloys	Ceramic Tools, CBN, PCD	Alumina-based ceramics improved thermal shock resistance and tool longevity in high-speed machining of nickel alloys. CBN tools enhanced wear resistance in machining hardened steel under dry conditions.
5	Tool Geometry Modifications	Nickel Alloys, Steel	Modified Cemented Carbide Tools	Optimized helix and rake angles reduced cutting forces and improved tool durability in Inconel 718 machining. Wiper inserts improved surface finish and productivity compared to conventional inserts in laser-cladded components.

2.2 Minimum Quantity Lubrication (MQL): -Minimal Quantity Lubrication represents a shift towards more efficient and environmentally friendly machining practices by utilizing minimal amounts of lubricant delivered in an aerosol form, enhancing the performance of machining operations while reducing waste [11] [18]. In MQL's use of evaporative cooling represents a significant advancement over conventional wet machining methods, offering improved heat transfer efficiency, reduced temperatures at the cutting interface, and enhanced overall machining performance. This efficiency is particularly beneficial in demanding machining operations where precision and tool longevity are critical [12]. The minimal use of cutting fluids in MQL significantly lowers environmental and health risks. Traditional metal cutting fluids can harm the environment, but MQL reduces the amount of oil needed, which helps prevent these issues. Using vegetable oils in MQL improves performance, especially when machining hard materials. Additionally, combining water-soluble oils with nanoparticles enhances effectiveness without generating toxic effects. Overall, this approach promotes a safer and more sustainable machining process [13].

Table 2: Machining of materials under MQL condition [15]

Sr. No.	Technique/Feature	Metal Used	Cutting Tool	Results/Findings
1.	MQL with Vegetable Oils	AISI 1045, AISI D2, AISI 9310	CVD Coated Tools, Carbide Tools	- Reduced cutting forces by 10–30%. - Achieved a 91% improvement in surface finish compared to dry machining - Increased tool life by 267%.
2.	Nano-Particle Enhanced MQL	AISI 4340, AISI D6	Coated Tools, HSS	- Enhanced heat dissipation, reducing cutting temperature by ~100°C. - Achieved better wear resistance.
3.	MQL with Internal/External Application	Hardened Steels (e.g., AISI H13, AISI 52100)	Carbide and CBN Inserts	- Reduced tool vibration by 8.6%. - Surface roughness improved by 35%.
4.	Biodegradable Oils in MQL	Steel Alloys (e.g., AISI 1060)	Carbide Tools	- Achieved ~50% reduction in cutting temperature. - Sustainable and eco-friendly with lower CO ₂ emissions.
5.	Hybrid MQL (with Cooling)	AISI 4340, Duplex Stainless Steel	Cryo-MQL with CO ₂ or LN ₂ Media	- Reduced flank wear and crater wear. - Increased surface integrity and eco-friendliness.

2.3 Cryogenic Process: - Cryogenic machining is a cutting-edge technique that cools the cutting tool and/or workpiece during machining. It involves using cryogenic cooling lubricants (CLF), rather than traditional oil-based lubricants, to target the cutting area where temperatures rise the most during the machining process. This method not only helps manage heat but can also alter the material properties, leading to enhanced machining performance [11]. In cryogenic cooling, materials are used at very low temperatures (below -150 °C) to cool the cutting tool and workpiece during machining. Common cooling agents include liquid nitrogen, which has a boiling point of -195.82 °C, and frozen carbon dioxide, which sublimates at -78.5 °C. Liquid nitrogen is specifically used to lower temperatures in the cutting zone

because machining generates a lot of heat, leading to tool failure and changes in the material properties. By using nitrogen, the harmful effects of heat are reduced, resulting in less wear on the tool and better performance in machining [14]. Cryogenic machining is an environmentally friendly method that delivers improved performance at higher cutting speeds. It effectively manages machining temperatures, leading to extended tool life and better overall results [15].

Table 3: Machining of materials under cryogenic condition [17]

Sr. No.	Technique	Material used	Cutting tool	Results/finding
1.	Liquid Nitrogen (LN2) Cooling	Titanium alloys (Ti6Al4V, Ti-5553), Nickel-based alloys (Inconel 718)	Carbide, CBN, Coated Tools	- Improved tool life by 80–200% under LN2 cooling. - Reduced cutting temperatures significantly. - Enhanced surface finish by 64–71%
2.	Liquid CO2 (LCO2) Cooling	Hardened Steels (AISI 52100, AISI 4037), Stainless Steel (AISI 304)	Tungsten Carbide, Cryo-Optimized Tools	- Lower flank wear and increased tool life. - Lower surface roughness and diminished cutting forces.
3.	Hybrid LN2-MQL Cooling	Superalloys (Inconel 718, Nimonic 90), Composites	Cryogenic-Ultrasonic Assisted Tools	- Reduced tool wear and energy consumption by 3.6%. - Higher surface integrity and reduced environmental impact.
4.	Dual LN2 Jet Cooling	Hardened Steel (AISI 52100), Ti-Alloys	Coated Carbide Tools	- Enhanced cooling efficiency with dual jets. - Lower adhesion and reduced white layer formation.
5.	Cryogenic Tool Holder Cooling	Titanium alloys, Nickel-based alloys	Internally Cooled Tool Holders	- Prevented overcooling of workpieces. - Reduced thermal stresses and cutting zone temperature.

3. Methodology

The main goal of this research is to explore how sustainable machining can be improved by selecting the right machining process for the right material. It reviews existing studies on the topic and evaluates their impact on the three key aspects of sustainability—social, economic, and environmental—within a single framework. The study aims to identify the best available alternative using a multi-criteria decision-making (MCDM) approach. To achieve this, a wide range of studies on sustainable machining methods, such as MQL, dry machining, and cryogenic machining, were collected and analyzed. The review highlights that not all sustainable machining techniques are equally effective for every material. Instead, the best approach depends on the material's properties and the specific sustainability goals.

4. Result and Discussion

Dry machining: - Dry machining is the most eco-friendly and cost-effective approach, as it eliminates the requirement for cutting fluids, reducing both operational costs and environmental pollution. However, it faces significant challenges in heat dissipation, leading to higher tool wear and poor surface quality, especially for difficult-to-machine materials like Inconel 718, titanium alloys, and stainless steel. Results show that tool coatings such as TiAlN and self-lubricating coatings (TiSiN-Ag, MoSx-TiN) help reduce friction by up to 45% and extend tool life by two-fold in titanium alloys. Advanced tool materials like ceramic and CBN improve wear resistance but remain limited in handling extreme heat conditions.

MQL machining: - MQL offers a more eco-friendly option compared to conventional flood cooling by utilizing a fine oil mist for lubrication, which helps extend tool life and improve surface finish. The findings indicate that MQL with vegetable oils reduced cutting forces by 10–30% and improved surface finish by 91% compared to dry machining. When combined with nano-particles, MQL enhances heat dissipation, reducing cutting temperature by approximately 100°C, improving tool durability in hardened steels and stainless steel. Hybrid MQL cooling, when combined with CO2 or LN2, further improves surface integrity and reduces flank wear, making it suitable for machining hardened steels (AISI 52100, AISI 4340) and aluminum alloys.

Cryogenic machining: - Cryogenic cooling offers the best cooling performance, particularly for high-temperature materials like nickel-based superalloys and titanium alloys. The study found that LN2 cooling

extended tool life by 80-200% and significantly reduced cutting temperatures, improving surface finish by 64–71% in nickel alloys. The use of dual LN2 jet cooling further lowered adhesion and reduced white layer formation. Additionally, hybrid LN2-MQL cooling provided superior tool wear resistance and energy efficiency, reducing tool wear by 3.6%. This makes cryogenic machining highly effective for aerospace and biomedical applications but costly due to cryogenic fluid handling and equipment setup.

Table 4: Performance Comparison of Cryogenic, MQL, and Dry Machining Across Various Materials

Material Type	Cryogenic Machining	MQL Machining	Dry machining
Titanium Alloys (Ti6Al4V, Ti-5553)	Best choice – LN2/CO2 cooling prevents thermal damage, extends tool life	Not recommended – Insufficient cooling, leads to rapid tool wear	Not recommended – High temperatures cause severe tool wear
Nickel-Based Superalloys (Inconel 718, Nimonic 90)	Best choice – Reduces cutting temperatures and tool wear, improves integrity	Limited use – Cannot handle extreme heat, leads to adhesion problems	Not suitable – High friction leads to excessive wear and poor finish
Hardened Steels (AISI 52100, AISI 4340, AISI D2)	Good – Effective in reducing thermal softening and improving tool life	Best choice – Reduces tool vibration, enhances lubrication	Poor – High tool wear, rough surface finish
Stainless Steels (AISI 304, SS316)	Effective – Prevents excessive heat buildup, enhances tool life	Best choice – Lubrication improves finish, reduces cutting forces	Limited use – Friction and heat cause rapid tool failure
Aluminum alloys (AA6082, AISI 1060)	Limited use – Cryogenic cooling may cause embrittlement	Best choice – MQL reduces cutting forces, prevents BUE formation	Good – Dry machining feasible due to high thermal conductivity

5. Conclusion:

Sustainable machining technologies are essential for addressing the growing demand for environmentally responsible manufacturing. This study identifies that there is no machining technique superior to all. Cryogenic machining is optimal for extreme temperature situations and superalloys, MQL offers the balance of lubrication and cooling for steels and aluminum, and dry machining is possible with high-conductivity materials such as aluminum but has issue with tool wear and heat dissipation. Hybrid strategies, including LN2-MQL cooling, hold the most viable solutions for green machining, balancing lubrication, cooling and tool life. This study confirms the significance of choosing the appropriate machining strategies according to material properties and sustainability objectives.

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A Review of Simulation and Experimental Analysis of Heat Treatment for Steel Tie Rods

Kunal Kumar Singh¹ Girish M Dhote² Chetan Mahatme³ Prashant Kamble⁴

Mechanical Engineering, Yashwantrao Chavan College of Engineering

kunal.cool1@gmail.com, girishdhote@gmail.com

Abstract

The above topic is related to study of actual Heat treatment and simulation for physical deformation of steel Tie Rod and comparing its observations for process optimization. General methodology is to go for either traditional or induction hardening after getting the chemical composition of the material followed by tempering which is key techniques for improving the mechanical properties of materials, strength, hardness, etc. and to release stresses. This technology is widely used in fields of manufacturing, aerospace and robotics. This study is done in current manufacturing scenario which faces many rejections due to physical deformation leading to loss of money and time, so simulation aid in heat treatment is done before actual process done in furnace for reducing rejection. To introduce CAE with process of heat treatment, is quite challenging and complex hence to optimize processes of heat treatment, a finite element analysis software for heat treatment like COMSOL & ANSYS is used for simulation. The boundary conditions and parameters in conventional gas carburizing heat treatment is studied and will be used in simulation software for analysis purpose. This will help the manufacturer to analyse the process to select appropriate heat treatment process parameters and hardness as per application to avoid wastage.

Keywords: Heat treatment Simulation; Gas Carburizing; heat treatment Furnace; Deformation; Hardening

1.Introduction

This study aimed to Analyse and optimize the design process so that heat treatment parameters which can lead to deformation, Topology optimization, while exploring the feasibility of various materials and operating conditions. The research simulated thermal and deformation behavior using a computer model, comparing results with experimental data. By using Heat treatment simulations analysis for thermal stress, deformation and cracks can be identified before the actual process so to introduce CAE with process design of heat treatment to determine the correct process and hardness value required. To simulate the multi-physics coupled processes of heat treatment, a finite element analysis software for heat treatment processes called COSMOL & ANSYS add on with Dante is introduced, which integrates various numerical solving methods. This multi-solver approach provides robust computational support for addressing heat transfer, deformation, and phase transformation issues encountered in various heat treatment processes. Like other finite element software, using COMSOL for heat treatment simulations requires inputs including part model files, process parameters, and material property parameters. The preparation and organization of these data are complex to the use of heat treatment process simulation software.

Intelligent system for Heat treatment process and parameters

The study by A.D. Boccardo, P.M. Dardatib, D.J. Celentanoc, L.A. Godoy in paper Austempering heat treatment of ductile iron: Computational simulation and experimental validation aims to simulate the three-step Aus-tempering heat treatment process of ductile iron parts using a coupled thermo-mechanical-metallurgical computational model. The objective is to predict temperature evolutions, dimensional changes, minimum required times for heat treatment stages, and phase transformations. (a) Model Approach: The thermal, mechanical, and metallurgical phenomena were simulated using finite element methods (FEM) at macro- and microscale levels. (b) Microscale Modelling: Metallurgical changes were modelled through representative volume elements considering microstructure features. (c) Experimental Validation: The model's predictions were tested against experimental heat treatment data, analysing temperature evolution, phase transformations, and dimensional changes. The model successfully simulated complex heat treatment phenomena with good agreement to experimental results as demonstrated. [1]

In the paper published on Journal of Applied Science and Advanced Technology on the topic Hardness Analysis of Bearing on Heat Treatment Process in 2020, Kisman H Mahmud¹, Sri Anastasia Yudistirani², Ery Diniardi¹,

A I Ramadh explained the Objective of the study which investigates the effect of the heat treatment process on the hardness of the inner and outer rings of bearings made from AISI 52100 steel. It aims to achieve the desired mechanical properties, specifically within the hardness range of 62-64 HRC, in accordance with factory standards. Methods of Heat treatment processes, including hardening and tempering, were applied to the bearing rings to alter their mechanical properties. The Rockwell hardness test was used to measure and evaluate the hardness of both the outer and inner rings after treatment. A Digital Rockwell Hardness Tester was employed with a load of 60 kg to ensure accurate hardness readings. Thus we can conclude that the heat treatment process successfully achieved hardness levels within the desired range, ensuring the bearings met factory quality standards. For the outer ring, an average hardness of 63.5 HRC was observed, while the inner ring displayed an average hardness of 63.6 HRC. The process improved the material's properties, making the bearings resilient and suitable for further machining and use. [2]

Pedro Maranhão in the article of Modeling and Simulation of the Induction Hardening Process: Evaluation of Gear Deformations and Parameter Optimization.(2024) . The objective of this study is to model and simulate the induction hardening process applied to gears, with the goal of understanding and optimizing its effects on gear deformations and overall performance by developing a coupled electromagnetic-thermal-mechanical simulation model of the induction hardening process using suitable software tools (e.g., ANSYS, COMSOL, or other FEA platforms) to (a) analyse the thermal behaviour during the heating and quenching stages to predict temperature distribution within the gear.(b) to evaluate induced mechanical stresses and deformations resulting from thermal gradients and phase transformations.(c)Assess gear distortion and dimensional changes post-hardening to ensure they remain within acceptable tolerances.(d)Investigate the influence of key process parameters, such as coil geometry, power input, heating time, and quenching rate.(e)Perform parametric optimization to minimize undesirable deformations while achieving desired hardness profiles.(f)Validate the simulation results with experimental or reference data where possible, to ensure accuracy and reliability of the model.[3]

In the paper Development of an intelligent design and simulation aid system for heat treatment processes based on LLM (2024) Yixiao Sun a Zeyu Zhang a used the following methodology: Choose a base LLM (e.g., OpenAI GPT, LLaMA, or similar). Fine-tune or apply domain-specific prompt engineering techniques to improve material-specific recommendations. Interpretation of thermal profiles, phase transformations, and simulation output. The objective of this research is to develop an intelligent, AI-driven design and simulation aid system for heat treatment processes, leveraging the capabilities of Large Language Models (LLMs) to enhance decision-making, process optimization, and simulation support. The specific aims are to Integrate LLMs (e.g., GPT-based models) into a design-assistance framework to support engineers in selecting appropriate heat treatment parameters based on material properties, part geometry, and performance requirements. Used software COSMAP for simulation & computational support for addressing heat transfer, deformation, and phase transformation issues encountered in various heat treatment processes.[4]

Sun SongSong in the study of Crankshaft high cycle bending fatigue research based on the simulation of electromagnetic induction quenching and the mean stress effects had the following objectives (a)Simulate the electromagnetic induction quenching process to accurately predict the resulting hardness profile and residual stress distribution in the crankshaft material.(b)Model the high cycle bending fatigue behaviour of the crankshaft under realistic loading conditions, considering operational stress ranges and fatigue life prediction criteria.(c)Analyse the influence of mean stress on fatigue life using methods such as Goodman, Gerber, or Smith-Watson-Topper (SWT) correction models.(d)Evaluate the impact of induction hardening on fatigue performance by comparing untreated and treated crankshaft models.(e)Validate the simulation results against experimental data or published benchmarks to ensure accuracy and reliability. Methods used are (a)Geometric Modelling and Material Selection to develop a detailed 3D model of a representative crankshaft using CAD software and selecting appropriate material commonly used for crankshafts (e.g., forged steel) and collect their mechanical, thermal, and electromagnetic properties.(b)Electromagnetic Induction Quenching in multi-physics simulation tool (e.g., ANSYS, COMSOL) to model the induction hardening process.[5]

In the article“‘The effects of oil-quenching and over-tempering heat treatments on the dry sliding wear behavior of 25CrMo4 steel” (2024) by Uğur Arabacı has objective is to study hardness variations by measuring and comparing hardness values in samples subjected to oil-quenching and over-tempering with untreated samples and to improve mechanical properties: Identifying methods to enhance wear resistance and durability of steel components used in industries like automotive and manufacturing. Material Selection process which focuses on 25CrMo4 steel, commonly used in automotive components like axle shafts and turbine parts. The chemical composition of the steel is detailed followed by Sample Preparation in which Cylindrical samples of the steel were cut into specific dimensions. Heat treatments (oil-quenching and tempering) were applied to achieve targeted microstructures and mechanical properties. Heat Treatment Samples were preheated and then quenched in oil to induce martensitic transformation. Quenched samples were tempered at higher temperatures to decrease martensite density and increase ductility.[6]

AhmadJafaria, in the paper An extended finite element method implementation in COMSOL multiphysics(2023). Thermo-hydro-mechanical modeling of fluid flow in discontinuous porous media use the software tool of COMSOL Multiphysics, a general-purpose software was utilized for its flexibility in handling complex multiphysics problems, including mechanical deformation, fluid flow, and heat transfer processes. Live Link for MATLAB: This integration allowed the researchers to use MATLAB functions for level-set analysis, XFEM physics enrichment, and pre-processing tasks, like identifying enriched zones and updating crack geometries. Digital Rockwell Hardness Tester involving material hardness measurement, this tester might have been used to determine the mechanical properties after treatments. (Referenced in broader contexts related to hardness analysis. Analytical Functions in COMSOL Built-in features like strain field and variable modifications were employed for XFEM-specific tasks. The objective of this study is to develop and implement an extended finite element method (XFEM) framework within COMSOL Multiphysics for the thermo-hydro-mechanical (THM) modeling of fluid flow in discontinuous porous media.[7][11][12][13][14]

The study of Taolue Wena, Xiaofeng Qina, Zhi Lva, Ping Yangb, Wei Chenb, Jing Xieb, Fengbin Liua in paper Numerical simulation study on integral induction heating of backup roll aims to develop a finite element model (FEM) for electromagnetic-thermal coupling to simulate the integral induction heating process of backup rolls. The goal is to optimize heating parameters, improve temperature uniformity, and reduce costs associated with experimental trials. Methods are (1) A coupled electromagnetic-thermal FEM was developed using ANSYS.(2) Temperature measurements were taken at various points on the roll using thermocouples and infrared thermometers. The model provides a cost-effective tool for pre-evaluating induction heating processes and optimizing parameters for production.[8]

The primary goal of the study by Henrik Alberg in his journal Material Modelling for Simulation of Heat Treatment": is to develop reliable and efficient methods for simulating heat treatment processes using the Finite Element Method (FEM).[8][9] Parameter optimization techniques for material model calibration are presented, using MATLAB-based tools. Stress-relief heat treatment processes are modeled, incorporating thermal boundary conditions derived from Computational Fluid Dynamics (CFD) for convective heat transfer. Accurate boundary conditions, such as convective heat transfer coefficients, significantly affect simulation outcomes during the cooling stage.[9]

Literature review of heat treatment simulations with respect to phase transformation, residual stresses and distortion has the Objective to provide a comprehensive review of numerical simulations of heat treatment processes. The aim is to understand and predict phenomena like phase transformation, residual stresses, and distortion, with a focus on improving manufacturing outcomes and reducing costly post-treatment machining due to distortion. Simulation Component are described in the heat treatment simulation is divided into three major parts which are Temperature Calculation, Phase Transformation, Mechanical Response.[10] Developing simulations based more on physical models can reduce reliance on excessive experimental testing. Simulations are increasingly used to predict and mitigate distortions in high-performance steel components subjected to heat treatment.[10][11][12][13][14]

Daming Tong, Jianfeng Gu, Fan Yang in the article Numerical simulation on induction heat treatment process of a shaft part: Involving induction hardening and tempering had the objective of Comparison results of microstructure and residual stress indicated that the simulations fit well with the experimental results. The effects of transformation and creep on tempering stress relaxation were also analyzed based on the proposed numerical model. The conclusion of this study shows a coupled model involving electromagnetic, thermal, transformation and mechanical model for the entire induction heat treatment was investigated. Induction hardening and subsequent tempering of a shaft part was simulated based on the proposed models. The corresponding experiments were conducted to validate the simulation results. In addition, the mechanism of tempering relaxation was studied by simulation. The following conclusions were drawn: 1. A uniformly hardened layer was obtained on the shaft part after the proposed induction hardening process. 2.. Both volumetric shrinkage, due to phase transformation, and creep have effects on tempering stress relaxation.[16]

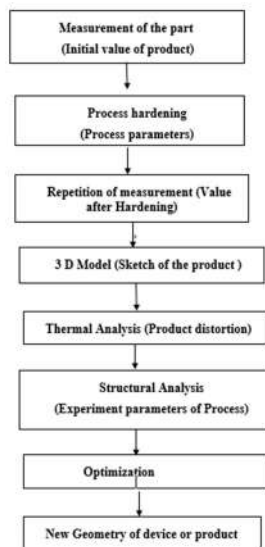


Fig. 1: Flow chart of Methodology



Induction hardening equipment.

Figure 2. Induction hardening machine

3. Modelling and simulations

Dong-Wan Kim, Hoon-Hwe Cho, Won-Beom Lee, Kyun Taek Cho, Yi-Gil Cho, Se-Jong Kim, Heung Nam Han, in paper A finite element simulation for carburizing heat treatment of automotive gear ring incorporating transformation plasticity, (2016) aims to develop a finite element model to predict the deformation, temperature history, carbon diffusion, phase fraction, and hardness during the carburizing heat treatment process for automotive gear rings. The model incorporates transformation plasticity to address issues like dimensional instability caused by phase transformations and thermal expansion/contraction. Methods used Finite Element Modelling: Carburizing gas diffusion was simulated using Fick's law, considering temperature and pressure-driven diffusion. Phase transformations during heating and cooling stages were modelled using equations such as JMAK and Koistinen-Marburger. Transformation plasticity was incorporated using constitutive equations for diffusional and diffusion less transformations. Thermo-mechanical responses, including thermal expansion and phase-related strains, were implemented into the model. Experimental Validation is done by conducting Experiments on gear rings made of medium-carbon steel (SCR420). Microstructural and hardness analyses validated the predictions made by the model. Thus it can be concluded that Model Accuracy can be measured as the developed finite element model successfully predicted carbon concentration profiles, phase fractions, and hardness distributions after heat treatment. Transformation plasticity significantly influenced dimensional changes during carburizing and quenching. Higher carburizing temperatures resulted in deeper carbon diffusion and harder gear surfaces. Lower temperatures led to increased ferrite fraction and reduced dimensional instability.[1][3][5][7][11]

K.O. Leea, J.M. Kima, M.H. China, S.S. Kangb stated the objective in paper “A study on the mechanical properties for developing a computer simulation model for heat treatment process” to develop a finite element simulation model to predict thermal deformation and strain-stress relations during heat treatment processes for High Strength Low Alloy (HSLA) steels. The methods used are as follows

1. Material Selection: AISI 4120 and AISI 5120 steels were used for testing.
2. Heat Treatment Processes: Martensite microstructure is achieved through heating to 850°C and quenching in oil, and Bainite is processed via salt bath treatment at 500°C. Ferrite/Pearlite is maintained at 650°C for stabilization. **Finite Element Analysis (FEA):** The data obtained from testing was used for simulations in ABAQUS to analyse heat transfer, thermal stress, and deformation during heat treatment. conclusion obtained is that, the heat treatment successfully transformed the HSLA steels into martensite, bainite, and ferrite/pearlite phases, with detailed strain-stress relationships determined for each phase.[12]

Paper titled "Modeling and Simulation of Quenching and Tempering Process in Steels" aims to develop a modified model to predict microstructure evolution and hardness distribution during the Quenching and Tempering (Q&T) process for steels. This model focuses on achieving desired mechanical properties such as toughness and ductility while minimizing cost and maximizing precision in simulations of phase transformations and tempering kinetics. Methods: Hardness distributions are predicted using experimental equations based on iron-carbon phase compositions and alloy content. Finite Element Method (FEM) The COSMAP simulation code incorporates the transformation kinetics model. Conclusions : The model accurately predicted phase distributions, hardness values, and microstructure evolution during Q&T processes. Final microstructures differed between outer and inner

surfaces of components due to varying cooling rates, Tempering successfully reduced residual stress caused by quenching. Predicted hardness and stress distributions showed good agreement with measured values, validating the model's effectiveness.[13]

Numerical simulation on induction heat treatment process of a shaft part: Involving induction hardening and tempering by Daming Tong and others has the objective is to develop a numerical simulation model to analyze the induction heat treatment process, including both induction hardening and subsequent tempering, of a shaft part made of SAE 4140H steel. Methods: Numerical Modelling in which A coupled electromagnetic-thermal-transformation-mechanical model was proposed. Simulation Setup: Induction heating was followed by water spray quenching and furnace tempering at 200°C, 400°C, and 600°C. Material parameters were experimentally determined, including thermal conductivity, specific heat, and phase transformation kinetics. Conclusion obtained Residual stresses decreased with increasing tempering temperature. At 600°C, quenched residual stress was almost entirely eliminated, reducing by 92.2%.[14]

The study in the research paper "Evaluation of heat transfer coefficient during heat treatment by inverse analysis": focuses on developing an inverse heat transfer formulation using a two-dimensional finite element method (FEM). Objective is to evaluate the heat transfer coefficients during the heat treatment process for cylindrical carbon steel specimens cooled by air and water. Method used in the study focuses on developing an inverse heat transfer formulation using a two-dimensional finite element method (FEM). The aim is to evaluate the heat transfer coefficients during the heat treatment process for cylindrical carbon steel specimens cooled by air and water. Cylindrical carbon steel specimens were subjected to air cooling (fan cooling) and water quenching from an initial temperature of 500°C, thermocouples were installed in the specimens to measure temperature variations at various locations during the cooling process. This paper conclude that there is the need for higher-temperature experiments to evaluate coefficients for industrial processes involving carbon steel heated to 800–900°C. It suggests exploring three-dimensional inverse analysis for more complex workpiece geometries undergoing non-uniform cooling.[13][14][15] Henrich Alberg in the Journal

4) Gaps and recommendation

Gaps and recommendation : No Automation in heat treatment process regarding handling and other observations shown. In case of induction heat treatment an electromagnetic thermal coupling finite element model is developed to study the induction heating of the backup roll by considering the actual input current and variation of magneto-thermal property parameters of the roll with temperatures. The field test is implemented and the simulation accuracy of the established finite element model is verified. Analysis through software is not used in many cases. No comparison was made between actual and software data. This study attempted to fill the gap related to predicting structural deformation due to thermal induction and topology optimization in toothed gears by developing a computational procedure and validating processes in the automotive sector. For the simulation, the addition of the polymer was not taken into account, leaving the improvement of the model for future work. Only microstructural examinations, hardness measurements, and wear tests were carried out on tempering conditions of the 25CrMo4 steel. The parameter required in heat treatment or the boundary condition for the simulations to be done in COMSOL software. COMSOL are not mentioned which is a sophisticated multiphysics modeling capabilities. In this software, a combination of available built-in physics interfaces can be incorporated with user-defined physics interfaces to investigate formidable multiphysics problems (see COMSOL, 2019b; Jafari et al., 2022). The thermo-hydro-mechanical coupling analysis of fluid flow in deformable porous media, considered here, involves three distinct physics interfaces that take account of the mechanical deformation, fluid flow and heat transfer. Daming Tong, Jianfeng Gu, Fan Yang

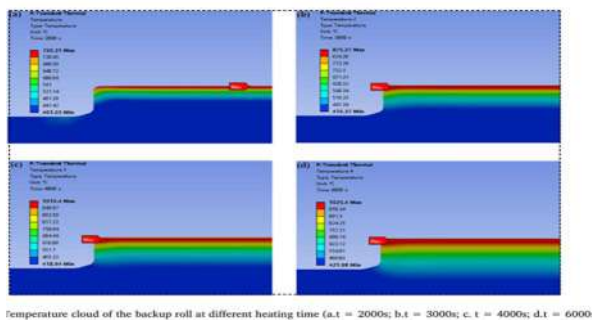


Fig.3 Simulations at different temperature in ANSYS

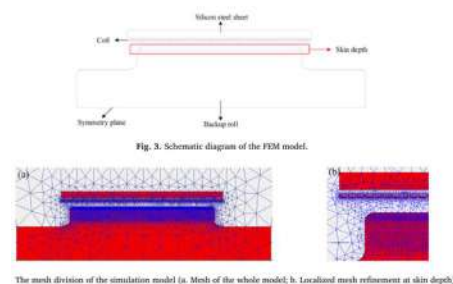


Fig. 4 Mesh Division in FEM model

5) Conclusion

The conclusion of the attached paper highlights the effectiveness of simulation techniques in optimizing heat treatment processes for steel Tie Rods. The study demonstrates how computational tools like COMSOL and ANSYS, combined with experimental methods, can help analyse and predict deformation, stress, and other physical behaviours during heat treatment processes. By simulating thermal and mechanical interactions, manufacturers can optimize parameters to reduce rejections, enhance product quality, and avoid wastage. The integration of Computer-Aided Engineering (CAE) with heat treatment design is presented as a promising, albeit complex, approach to achieve the goal. The paper emphasizes the potential of simulations to predict outcomes accurately, enabling manufacturers to select appropriate process parameters and hardness values tailored to specific applications. However, it notes challenges like data preparation and the lack of automation in certain heat treatment processes, suggesting future work could focus on improving these aspects.

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The Role of Mentoring in Reducing Employee Attrition in the IT Sector

Ritesh Yogesh Kulkarni
Email: riteshkulkarni9616@gmail.com

Miss. Bhagyashri Yogesh Kulkarni
Email: bhagyashrik478@gmail.com

Ashoka College of Education, Nashik

ABSTRACT

The present paper aims at explaining how mentoring can be applied effectively to decrease employee turnover in the Information Technology (IT) sector. The reasons for high employee turnover in the IT sector include high work demands, limited career growth, and burnout. This paper aims at examining how the mentoring programs support employees in the IT industry in order to reduce turnover. Mentoring programs offer employees support, career guidance, and emotional engagement that makes the employees to feel attached to their work and thus loyal and developed. This paper reviews the literature on the effects of mentoring on job satisfaction, engagement and turnover, and presents case studies of successful mentoring programs from leading IT companies. The results of the study indicate that properly designed mentoring programs can enhance employees' well-being, decrease feelings of burnout and increase career development, which in turn reduces attrition rates. Thus, this paper recommends that IT firms should consider implementing mentoring programs as a part of their overall employee retention management. Employee attrition remains a significant challenge for organizations within the Information Technology (IT) sector, impacting both operational efficiency and organizational culture. This research paper explores the role of mentoring programs in mitigating employee turnover, focusing on how structured mentorship can foster job satisfaction, professional development, and organizational loyalty. By examining various mentoring models, including one-on-one mentorship and group mentorship, the study investigates their effectiveness in addressing the root causes of attrition such as lack of career progression, insufficient support, and workplace disengagement.

Keywords: Mentoring, Employee attrition, IT Sector, Job Satisfaction, Turnover, etc.

INTRODUCTION

The Information Technology (IT) sector, a cornerstone of the modern economy, is characterized by rapid technological advancements, high competition, and an ever evolving talent pool. However, one of the most pressing challenges faced by IT organizations today is high employee attrition. The fast-paced nature of the industry, combined with increasing demands on work-life balance, limited opportunities for growth, and the constant pressure to stay ahead of the technological curve, contributes to the growing trend of talent leaving companies at a high rate. Employee turnover, often referred to as attrition, is costly not only in terms of recruitment expenses but also in the disruption it causes to project timelines, team dynamics, and organizational knowledge. As a result, organizations in the IT sector are continuously seeking strategies to reduce attrition and retain their talented workforce.

Among the most effective and long-term solutions is mentoring. Mentoring, a relationship in which experienced employees provide guidance, support, and feedback to less experienced colleagues, has long been regarded as a valuable tool for personal and professional development. In the IT sector, where career progression and continuous learning are key drivers of job satisfaction, mentoring serves as an essential bridge between employees and their career aspirations. It helps nurture talent, ensures that employees feel supported, and fosters a sense of belonging within the organization. By playing an instrumental role in employee engagement, growth, and well-being, mentoring has emerged as a powerful mechanism for reducing attrition in the IT sector.

The connection between mentoring and employee retention is multi-faceted. On one hand, mentoring programs help employees feel valued and invested in, increasing their job satisfaction and emotional attachment to the company. Employees who feel that their career growth is supported through structured guidance are more likely to remain loyal to their employer, as they see tangible investment in their development. On the other hand, mentoring also addresses some of the critical challenges that lead to attrition in the IT sector, such as burnout, lack of career progression, and isolation.

The role of mentoring in employee retention also lies in its ability to enhance employee engagement. Engaged employees are those who are emotionally invested in their work and organization. They are motivated to contribute to the company's success, take ownership of their projects, and go above and beyond in their roles. Mentoring programs foster this type of engagement by providing employees with a clear sense of direction, purpose, and growth.

LITERATURE REVIEW

The issue of employee attrition has become a critical concern in the Information Technology (IT) sector, as high turnover rates impact organizations by increasing recruitment costs, decreasing productivity, and resulting in the loss of valuable knowledge and skills. The need for employee retention strategies has led to growing interest in mentoring as a powerful tool to enhance employee engagement, job satisfaction, and career development. This literature review explores the role of mentoring in reducing employee attrition in the IT sector, summarizing relevant studies and theories that link mentoring to improved retention rates.

Employee attrition, or turnover, is a pervasive issue in the IT industry due to various factors such as high work demands, pressure to continually adapt to new technologies, burnout, and limited career growth opportunities. A study by (Kacmar et al. 2013) highlights that IT professionals often face intense work pressures and long hours, which can lead to stress and dissatisfaction. As a result, many employees in the IT industry experience feelings of burnout and disengagement, which ultimately contribute to higher attrition rates. Mentoring is widely defined as a relationship in which a more experienced or knowledgeable individual (the mentor) provides guidance, support, and advice to a less experienced individual (the mentee). The purpose of mentoring is to facilitate the personal and professional development of the mentee by sharing expertise, offering emotional support, and helping them navigate the organizational landscape (Allen & Eby, 2010).

One of the key factors that contribute to employee retention is engagement. Engaged employees are more likely to be satisfied with their jobs and feel a stronger connection to their organization. Several studies have demonstrated that mentoring plays a crucial role in enhancing employee engagement. (Ragins and Kram 2007) found that mentoring relationships contribute to an employee's sense of belonging, job satisfaction, and commitment to the organization.

Career development is another critical aspect of mentoring that directly influences employee retention. In the IT sector, employees seek opportunities to grow and progress in their careers, and mentoring can be a pivotal factor in helping employees identify their career goals, set expectations, and gain the skills needed for advancement. Noe (1988) emphasizes that mentoring is essential for career development, particularly in industries like IT where technological expertise is constantly evolving.

Employee burnout and work-life imbalance are prominent causes of attrition in high-pressure sectors like IT. Mentoring can play a vital role in addressing these issues by providing employees with strategies to manage stress, achieve work-life balance, and avoid burnout. Kram (1985) suggests that mentors can offer emotional support and provide a sounding board for employees dealing with workplace challenges, helping them navigate stress in a healthy and constructive manner.

OBJECTIVES

1. Examine the impact of mentoring on employee satisfaction and retention rates in the IT sector.
2. Analyze the impact of mentoring on employee engagement and productivity in the IT sector.
3. Identify the challenges organizations face in implementing effective mentoring programs and how these challenges impact employee retention.

THEORETICAL FRAMEWORK

Mentoring has long been recognized as an essential tool for employee development and organizational success. In the context of reducing employee attrition in the Information Technology (IT) sector, several theoretical perspectives can help explain how mentoring can address key factors that contribute to high turnover rates. This theoretical framework explores key theories related to employee retention, engagement, and development, focusing on how mentoring interventions influence these factors to reduce attrition within IT organizations.

Social Exchange Theory (SET), initially proposed by Homans (1958), suggests that relationships are built on the exchange of resources, with individuals seeking to maximize rewards and minimize costs in their interactions. Within an organizational context, this theory posits that employees are more likely to remain with a company if they perceive the relationship to be mutually beneficial. In the case of mentoring, employees gain valuable career guidance, emotional support, and opportunities for personal development, while mentors gain the satisfaction of supporting others and contributing to the organization's success.

Herzberg's Two-Factor Theory (1959) posits that job satisfaction and dissatisfaction are influenced by two sets of factors: motivators (which lead to satisfaction) and hygiene factors (which cause dissatisfaction if absent). According to this theory, motivators are intrinsic factors such as achievement, recognition, and personal growth, while hygiene factors are extrinsic, such as salary, job security, and working conditions.

Job Embeddedness Theory (Mitchell et al., 2001) suggests that employees are more likely to remain with their current employer if they are "embedded" in their job and organization. This embedding process involves three key elements: **Links, Fit, and Sacrifice**.

THE ROLE OF MENTORING IN REDUCING EMPLOYEE ATTRITION

Employee attrition, particularly in competitive industries like the Information Technology (IT) sector, is a significant challenge that organizations must address to maintain stability and growth. High turnover rates lead to increased recruitment costs, loss of expertise, and disruption of team dynamics. One of the most effective strategies for mitigating employee attrition is implementing structured mentoring programs. Mentoring involves a relationship where experienced employees (mentors) guide and support less experienced ones (mentees) in their professional and personal development. The positive impact of mentoring on employee retention is widely recognized and can be attributed to several key factors.

Mentoring fosters a sense of engagement among employees by providing personalized support and guidance. Engaged employees are more likely to feel emotionally connected to their workplace and are less likely to leave. Through mentoring, employees receive the attention and direction they need to understand their roles better and see a clear path for growth within the organization. Mentoring helps employees navigate their career paths by providing professional development opportunities and insights into skill-building. In the IT sector, where the technology landscape is constantly evolving, mentoring provides employees with the knowledge and guidance to stay relevant, enhancing their career progression. Mentors help mentees set career goals, acquire new technical skills, and develop essential soft skills like communication and leadership.

High-stress environments, such as those found in the IT industry, can lead to burnout and eventual turnover. Mentors offer emotional support, guidance on stress management, and advice on balancing work-life challenges. Mentoring programs can help employees feel more connected to the organization and its goals. Through mentoring, employees gain a deeper understanding of the company's culture, values, and long-term vision. As mentors share their experiences and provide guidance, mentees are more likely to feel a sense of loyalty and commitment to the organization.

In addition to supporting career development, mentoring also plays a critical role in knowledge transfer and succession planning. Senior employees passing on their expertise and experience to younger generations not only

ensures the continuity of essential skills but also helps create a culture of learning. Employees who receive mentorship are better equipped to take on leadership roles, creating a more stable workforce and reducing the need for external recruitment.

CASE STUDYS

Case Study 1: Google's Mentorship and Employee Engagement

Background: Google, known for its innovative products and workplace culture, recognized that employee retention was a key issue in the highly competitive tech industry. Despite its appealing work environment, many employees were leaving the company for better career development opportunities elsewhere. To combat this, Google launched a comprehensive mentoring program to support employees' personal and professional growth.

Impact on Employee Attrition: Google's mentoring program has been credited with improving employee satisfaction and reducing attrition. In a survey conducted by Google, employees who were engaged in mentoring relationships reported a higher sense of belonging and purpose within the company. Attrition rates for those involved in mentoring were found to be 15-20% lower compared to those who were not part of the program.

Case Study 2: TCS (Tata Consultancy Services) and Employee Retention Through Mentorship

Background: Tata Consultancy Services (TCS), one of India's largest IT services companies, faced significant employee turnover due to the increasing competition for skilled IT professionals. Many employees in the early stages of their careers were leaving for other opportunities, citing lack of career development, poor work-life balance, and limited growth prospects. TCS wanted to address these issues through a mentoring initiative aimed at improving employee engagement and retention.

Impact on Employee Attrition: TCS reported a noticeable drop in employee attrition rates among those involved in the mentoring program. According to internal data, employees who participated in mentoring were 30% less likely to leave the company than those who did not participate. The program helped new employees feel more connected to the organization and its values, leading to higher job satisfaction and a greater sense of commitment to TCS.

CHALLENGES IN IMPLEMENTING EFFECTIVE MENTORING

Mentoring has been proven to reduce employee attrition in the IT sector, implementing a successful and sustainable mentoring program comes with its own set of challenges. These challenges need to be carefully managed to ensure that mentoring programs fulfill their potential in improving employee engagement and reducing turnover. Below are some of the key challenges organizations may face.

One of the primary challenges in mentoring programs is ensuring that mentors and mentees are paired in a way that promotes effective communication and relationship-building. Poorly matched pairs whether in terms of skill levels, personalities, or professional interests can lead to disengagement and frustration. In the fast-paced and demanding IT sector, mentors often face time constraints due to their own professional responsibilities. Balancing mentoring duties with regular job expectations can be challenging for senior employees who may already have heavy workloads.

A lack of a well-defined and structured mentoring program can lead to inconsistencies in the mentoring process. Without clear goals, guidelines, and expectations, mentoring can become informal, unproductive, and inconsistent.

Mentors are often selected based on their experience and expertise, but they may not necessarily have the skills needed to be effective mentors. Many organizations fail to provide adequate training for mentors on how to

guide, coach, and support mentees effectively. Mentoring requires specific interpersonal and coaching skills, such as active listening, giving constructive feedback, and providing emotional support.

Mentoring programs require organizational commitment and resources to be successful. However, in some cases, organizations may fail to allocate sufficient resources whether in terms of time, budget, or personnel to establish and sustain mentoring programs.

DATA ANALYSIS AND FINDINGS

The study was based on survey method. The aim of the study is to find the impact of mentoring on reducing employee attrition on IT sector. A sample design is a plan for obtaining a sample from given population. The data collected through the questionnaire. Collected data should be analyzed for obtaining the result for how mentoring impact on employee attrition. After the analysis some finding has been obtain.

The majority of participants meet with their mentors regularly, which suggests that mentoring is an ongoing and significant part of their professional development.

The participants are focusing on career growth, leadership skills, work-life balance, technical skills, etc. A range of options would provide insight into what mentees prioritize in their professional development.

Maximum respondents feel that their mentor listens and provides valuable guidance. This indicates a positive relationship and an effective mentorship dynamic in the workplace. Most respondents believe their mentor has had a significant impact on their career development.

The respondent say's that their mentor has positively influenced their decision to stay with the company. This demonstrates that mentorship plays an important role in job retention and employee satisfaction & also feels more valued and supported due to the mentoring program, highlighting the positive effect of mentorship on workplace culture.

The majority respondents feels that mentoring has improved their job satisfaction, reinforcing the idea that career development support contributes to job happiness and is beneficial in reducing workplace stress and burnout. This suggests that mentoring may provide emotional support and guidance for stress management.

A strong majority believes that mentoring helps employees with career clarity, which in turn leads to better retention. This emphasizes mentoring's crucial role in long-term career development and reducing turnover & believes mentoring plays a vital role in career decisions and improvements. This shows the high value placed on mentorship for career development in the workplace.

STRATEGIES FOR EFFECTIVE MENTORING PROGRAMS

Implementing a successful mentoring program is key to reducing employee attrition, particularly in the competitive IT sector. Effective mentoring not only supports career development but also fosters a deeper connection between employees and the organization, which can significantly lower turnover rates. Here are some strategies for designing and executing effective mentoring programs

To ensure the effectiveness of mentoring programs, organizations must set clear and measurable objectives. These objectives should align with both organizational goals (e.g., improving employee retention) and individual goals (e.g., career development, skill enhancement). By defining the goals from the outset, both mentors and mentees can understand the purpose of the relationship and work toward a shared vision. Effective matching of mentors and mentees is crucial for the success of mentoring programs. Mentors should be selected based on their experience, expertise, and compatibility with the mentee's needs and goals.

Training for mentors is essential for ensuring that they can provide meaningful guidance and support. Mentors should be trained on key skills such as active listening, providing constructive feedback, career coaching, and maintaining a positive and supportive relationship.

Ongoing feedback from both mentors and mentees is essential for improving the mentoring process. Organizations should provide avenues for participants to share their experiences, challenges, and suggestions for improvement. Regular evaluations, surveys, and feedback sessions can help identify areas where the mentoring program may need adjustments.

Recognizing the efforts of mentors is crucial for ensuring continued participation in the program. Employees who take on mentoring roles often dedicate considerable time and effort to guiding their mentees.

CONCLUSION

In conclusion, mentoring plays a pivotal role in addressing one of the most significant challenges faced by organizations in the IT sector employee attrition. By fostering stronger relationships between mentors and mentees, organizations can create an environment of continuous learning, career growth, and personal development. Mentoring helps employees feel more engaged, supported, and valued, all of which contribute to higher job satisfaction and lower turnover rates.

Through structured mentoring programs, companies can provide employees with the guidance needed to navigate their careers, develop new skills, and manage workplace stress, which is particularly crucial in the fast-paced and high-pressure IT industry. Mentoring programs contribute to a deeper sense of organizational loyalty and commitment, as employees are more likely to stay with an organization that invests in their long-term development.

While challenges exist in implementing effective mentoring programs such as ensuring proper mentor-mentee matching, overcoming time constraints, and maintaining organizational support these obstacles can be mitigated through thoughtful planning and commitment. When organizations dedicate resources to mentorship and create a culture that encourages ongoing learning and development, they can significantly reduce employee attrition, leading to a more stable, skilled, and motivated workforce. Ultimately, mentoring not only benefits individual employees but also strengthens the overall success and growth of the organization.

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*Author for Correspondence

Mr. Ritesh Yogesh Kulkarni

E-mail: riteshkulkarni9616@gmail.com

Designation: Assistant Professor

College Name: Bhonsala Military College, Nashik, Maharashtra, Affiliated to Savitribai Phule Pune University.

Citation: First Author

MPPT-Based Solar Inverter for Grid Synchronization Applications

Nikhil D. Thakre¹, Prof. Amit V. Mohod², Prof. Yogesh P. Khadse³

¹PG Scholar, Dept. of Electrical Engineering, PRMCEAM, Badnera-Amravati

^{2,3} Assistant Professor, Dept. of Electrical Engineering, PRMCEAM, Badnera-Amravati

Abstract

This work presents the modeling, control, and performance evaluation of a Solar PV system integrated with a 9-level Quasi Z-Source Inverter (QZSI) and Multilevel Inverter (MLI) for distributed generation and grid applications. Using SPWM with P&O MPPT (PI controller) and SVPWM with Incremental Conductance MPPT (FIS controller), simulations in MATLAB/Simulink showed high voltage gain, low THD, fast dynamic response, and improved power quality. A two-stage control with capacitor voltage management minimized switching stress and enhanced system stability. Comparative analysis confirms the QZSI-MLI's efficiency, reliability, and scalability for solar PV integration.

Keywords: Distributed Generation, Grid Integration, Multilevel Inverter, MPPT, Quasi Z-Source Inverter, Solar PV System

1. INTRODUCTION

The increasing global carbon emissions have accelerated the investigation and deployment of renewable energy technologies. Solar photovoltaic (PV) systems have emerged as a sustainable solution, leveraging the photovoltaic effect to convert solar energy into electricity. This chapter outlines the impact of global solar capacity on the Indian market over the past decade and highlights the role of power electronics in solar energy conversion. A literature review classifies various converter topologies for PV systems, referencing applicable standards and codes. The advantages of multilayer inverters with switched capacitor structures are discussed, along with their development and classification.

2. PROBLEM FORMULATION

India's vast solar potential and supportive policies have boosted its role in global solar deployment, but challenges remain in optimizing PV efficiency and ensuring stable grid integration. Key technical issues include environmental variability, grid synchronization, voltage regulation, and power quality. Power electronic converters are crucial for energy extraction, grid interfacing, islanding detection, and quality enhancement. However, rising solar penetration demands advanced converter technologies and control strategies. This research focuses on optimizing power electronic converters to enhance energy harvesting and ensure reliable integration into India's power grid.

3. PROPOSED APPROACH

DC-DC converters in solar PV systems must deliver high efficiency, low switch stress, and minimal ripple. Boost converters amplify PV voltages, with design based on a unified state-space model. State-space averaging, preferred over Fourier methods for PWM converters, simplifies dynamic modeling into steady-state equations, improving accuracy. A boost converter in continuous conduction mode is modeled using this method, with MATLAB simulations analyzing open- and closed-loop performance under load variations.

3.1 Analysis of Quasi-Impedance Source Inverter

The quasi-impedance source inverter (QZSI) enables single-stage buck-boost and inversion, ideal for flexible voltage regulation in distributed generation. Accurate modeling of its impedance network and PWM control is essential for optimizing output quality, grid synchronization, and system stability.

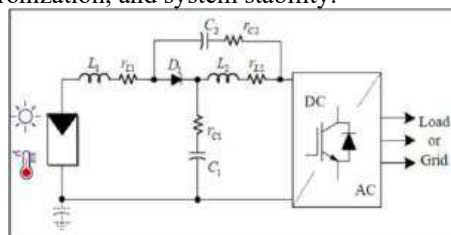


Figure 1: Circuit of a three-phase quasi-Z-source inverter

3.2 Proposed System Model

The proposed QZSI PV-grid system uses Zero Voltage Switching Inverter (ZVSI) technology to reduce losses and

enhance performance. A PV array connects through a QZSI with an impedance network and diode switch, feeding a VSI with an LC filter. A Point of Common Coupling (PCC) and circuit breakers ensure grid synchronization, protection, and reliable operation.

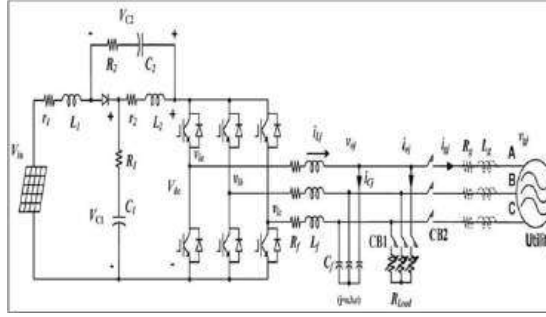


Figure 2: Configuration of the proposed QZSI-based grid system.

Table 1: Design Parameters of the of q-ZSI

Parameters	Specifications	Parameters	Specifications
Series inductors $L1 = L2 = L$	$500\mu H$	Capacitor resistance	$R=0.03\Omega$
Capacitors $C1=C2$	$400\mu F$	Duty ratio ($D0$)	0.1 to 0.8
Inductor resistance	$r = 0.47\Omega$	Cross-over frequency (F_c)	10 kHz
Capacitor resistance	$R=0.03\Omega$	Switching Frequency (F_s) of IGBT	5 kHz
Duty ratio ($D0$)	0.1 to 0.8	Natural frequency (F_n)	50 Hz
Load current	9.9 A	Reference Voltage ($V_{reference}$)	400 V

3.4 Small Signal Analysis of QZSI

For small-signal analysis, the system is subjected to disturbances from the input voltage (V_{in}), load current (I_{load}), and shoot-through duty cycle (d_0). To simplify, we assume $L_1 = L_2 = L$, $C_1 = C_2 = C$, $r_{L1} = r_{L2} = r_L$, and $r_{C1} = r_{C2} = r_C$ [1], [3], [8].

Neglecting second-order terms, the system equations simplify to (7). The small-signal transfer functions are shown in Figure 3.

The transfer functions from V_{in} to i_L and v_C are fourth-order, reflecting the asymmetry in the QZSI network. Meanwhile, the V_{in} to v_C transfer function simplifies to a second-order system, neglecting inner resonance at $(1/\sqrt{LC})$.

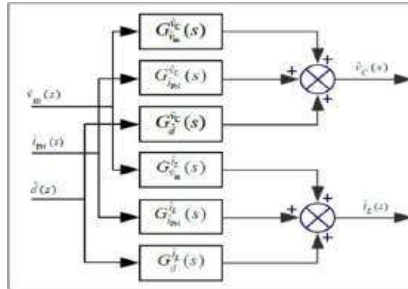


Figure 3: Small-signal model and transfer functions of QZSI [2], [4], [10].

3.5 Controller Design:

3.5.1 DC Side Voltage and Current Control:

Control begins with MPPT algorithms, which adjust the operating voltage and current to maximize power extraction under varying sunlight and temperature conditions. A voltage regulation mechanism maintains the DC bus voltage at optimal levels for inverter operation, ensuring efficiency and protecting components from overvoltage [1], [3], [8], [10], [11], as shown in Figure 4(a) and Figure 4(b).

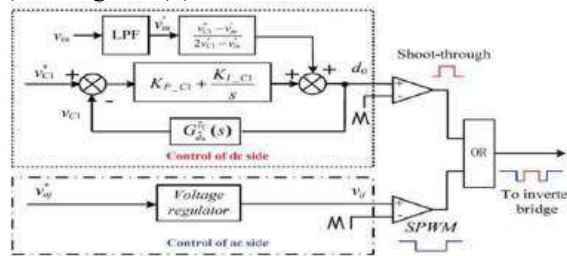


Figure 4 a: Two-stage control method of the QZSI for output voltage control

4. EXPERIMENTAL RESULTS

4.1 Solar PV Array Output

Figure 7 shows the solar PV output voltage (V_{spv}), current (I_{spv}), and power (P_{spv}) based on a solar array rated at 1000 W/m^2 and 25°C . From 0 to 0.5 seconds, the output voltage reached 500 V and current 10.66 A, resulting in 5.256 kW of power. This confirms that the P&O algorithm effectively tracked and extracted maximum power from the solar panel.

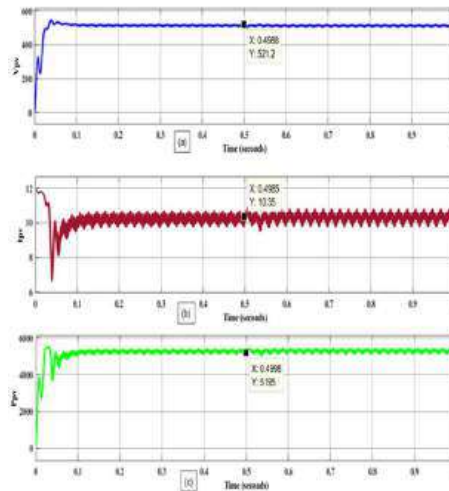


Figure 7 : Output of Solar PV Array

Figure 8 shows the inverter output voltage and current. The 3-phase VSI operates at a 5 kHz switching frequency, with output filtered through an LC filter to reduce THD and improve power quality. The inverter performs efficiently even in shoot-through states. Its rating is 1.25 times the DC input power. At 0.5 seconds, the inverter output voltage (V_{abc}) reached 1238 V, well above the 400 V grid voltage, with an output current of 21.65 A during simulation.

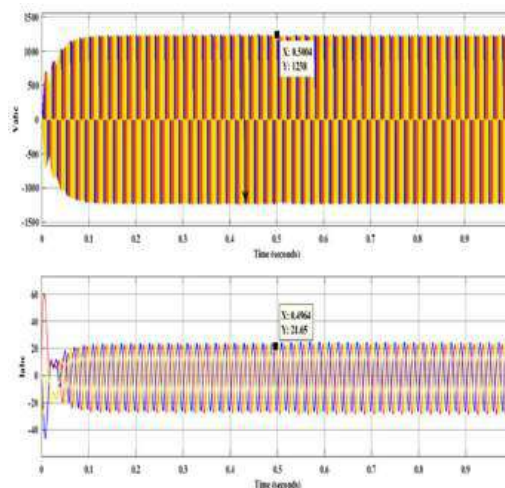


Figure 8:. Output of inverter (a) Three phase voltage (b) Three phase current

4.2 Analysis of Output at Load Side:

Figure 9 (a) and (b) show the load voltage and current waveforms under normal conditions. During steady state, both remain within limits. As load demand increases, additional power is supplied by the grid while maintaining a stable voltage profile. This confirms improved system performance, reliability, and stability.

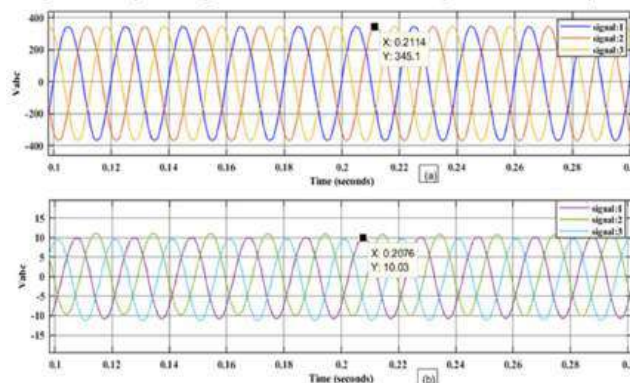


Figure 9:. Load profile under steady state condition

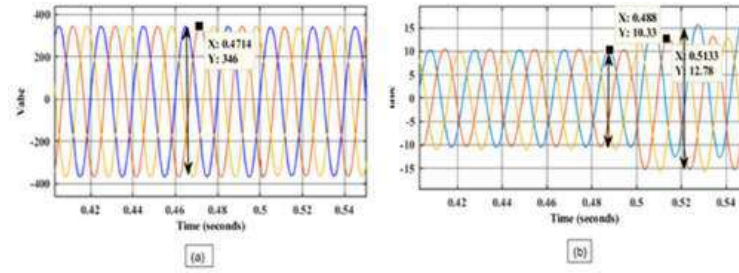


Figure 9: Load Voltage and Current profile under transient

4.3 THD Analysis of Voltage and Current Waveform:

Fourier analysis revealed a current THD of 0.76% and voltage THD of 4.41%, both within the IEEE 1547 standard limit of 5%.

Table 3: THD analysis table for load voltage and current signals

Signals	THD%	Magnitude	Average %	IEEE std. 1547
Va	4.41	357.2 V	4.34	5% & below
Vb	4.36	357.4 V		
Vc	4.25	357.5 V		
Ia	3.41	10.43 A	3.35	
Ib	3.34	10.43 A		
Ic	3.30	10.44 A		

5. CONCLUSION

This study presents the modeling, control, and performance evaluation of a Solar PV-based converter system using a 9-level Quasi Z-Source Inverter (QZSI) combined with a 9-level Multilevel Inverter (MLI) for distributed generation and utility applications. Simulations under varying conditions, along with real-time MPPT implementations (P&O and Incremental Conductance), demonstrated high efficiency, dynamic stability, and low THD, complying with IEEE standards. The QZSI's inherent buck-boost capability, reduced switching stress, and robust control strategies enabled effective grid integration and reliable performance. Comparison of SPWM and SVPWM control methods showed high gain, fast response, and minimal ripples. Overall, the results validate the QZSI topology as a promising solution for solar PV systems, with future work needed for real-world optimization and deployment.

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Adaptive E-Learning Algorithms and Heutagogy: A Systematic Analysis

Rupali A. Vinchurkar, Gajendra R. Bamnote

Prof. Ram Meghe Institute of Technology and Research, Badnera, India, rupali.sherekar@gmail.com;

Prof. Ram Meghe Institute of Technology and Research, Badnera, India, grbamnote@mitra.ac.in

Abstract

The rapid advancements in artificial intelligence (AI) and machine learning (ML) have transformed digital education by enabling adaptive learning systems. This paper systematically examines the role of adaptive e-learning algorithms within heutagogical frameworks, emphasizing learner autonomy and self-directed learning. A critical review of Bayesian Knowledge Tracing (BKT), Deep Knowledge Tracing (DKT), and Reinforcement Learning (RL) highlights their effectiveness in predicting student performance and optimizing personalized learning paths. Empirical findings from recent studies demonstrate that AI-driven adaptive learning significantly improves engagement and learning outcomes. Challenges such as data privacy, algorithmic bias, and explainability are discussed, along with future research directions.

Keywords: Adaptive Learning, Heutagogy, Artificial Intelligence in Education, Self-Directed Learning, Personalized Learning

INTRODUCTION

The digitalization of education has revolutionized traditional learning methodologies, making education more accessible and personalized. Adaptive learning, powered by AI, customizes learning experiences based on student interactions, engagement levels, and cognitive abilities [1]. This personalized approach aligns with heutagogy, a learning theory emphasizing self-determined learning and competency-based education [2].

Unlike pedagogy (teacher-directed learning) and andragogy (self-directed learning), heutagogy focuses on the learner's ability to navigate their educational journey, making decisions about content, pace, and methodology [3]. Integrating adaptive learning algorithms with heutagogical principles ensures a dynamic learning experience tailored to individual needs.

This paper systematically explores the intersection of AI-based adaptive learning and heutagogy, addressing key questions:

1. How do adaptive learning algorithms enhance self-directed learning?
2. What are the key machine learning models used in adaptive learning?
3. What challenges exist in implementing AI-driven adaptive learning at scale?

RELATED WORK

Adaptive Learning Technologies

Adaptive learning platforms leverage AI to tailor content delivery. Studies have shown that AI-driven adaptive systems improve learning efficiency by 20–30% compared to traditional e-learning [4].

Technologies like Knowledge Tracing (KT), deep learning, and reinforcement learning refine content recommendations in real time [5].

Modern adaptive learning technologies integrate multiple AI techniques to provide real-time feedback, personalized content recommendations, and performance predictions. Bayesian Knowledge Tracing (BKT) and Deep Knowledge Tracing (DKT) are commonly used in student modeling to **predict learner performance and optimize educational pathways.** Additionally, **reinforcement learning-based adaptive tutors dynamically adjust** instructional content based on student responses, maximizing knowledge retention [6].

Further, **learning analytics and natural language processing (NLP)** are being increasingly used to track student engagement and sentiment, allowing platforms to modify instructional strategies accordingly [7]. Empirical studies suggest that AI-driven adaptive learning tools improve long-term retention and **reduce dropout rates by 25%** compared to traditional static online courses [8]. Research also indicates that adaptive learning systems foster better engagement among students with diverse learning paces, ensuring that both advanced learners and those requiring remedial support receive optimized instruction [9].

Heutagogy and AI in Education

Recent research highlights heutagogy as a crucial factor in modern education, emphasizing metacognition, self-reflection, and critical thinking [6]. AI-powered platforms foster heutagogical learning by providing personalized feedback, enabling students to take ownership of their learning journey [7].

Empirical studies demonstrate that AI-driven personalized learning platforms improve self-directed learning. For example, a study by Siemens et al. [8] showed that AI-enhanced adaptive learning systems led to a **30% increase in self-regulated learning behaviors** among students compared to traditional e-learning methods. Additionally, platforms integrating **natural language processing (NLP) and learning analytics** have improved student decision-making regarding their learning paths [9].

Several AI-based learning platforms, such as Coursera and Duolingo, incorporate **reinforcement learning and knowledge tracing algorithms** to personalize learning experiences. Research by Wang et al. [10] demonstrated that AI-driven adaptive assessments significantly enhanced critical thinking skills, an essential component of heutagogical learning. Moreover, studies by Brown & Wilson [11] found that students using AI-driven recommendation systems exhibited greater motivation and persistence, crucial for self-determined learning.

While AI-enhanced heutagogical models present promising results, challenges such as **bias in AI algorithms** and **data privacy concerns** remain significant hurdles [12]. Addressing these challenges requires ongoing research and the integration of **explainable AI (XAI) frameworks** to enhance transparency in adaptive learning models.

METHODOLOGY

A systematic review of adaptive learning models was conducted by analyzing peer-reviewed studies from leading academic journals. The methodology included:

1. **Review of Knowledge Tracing Models:** Bayesian Knowledge Tracing (BKT) and Deep Knowledge Tracing (DKT) were examined for their predictive accuracy and adaptability in various learning environments [13], [14].
2. **Personalized Learning Pathways:** Studies on AI-driven content recommendation systems were analyzed to understand their effectiveness in optimizing learning trajectories [15].
3. **Reinforcement Learning in Education:** Prior research exploring reinforcement learning applications in education was reviewed, highlighting key studies where RL has improved student engagement and content adaptation [16].

4. **Evaluation Metrics:** The effectiveness of different adaptive models was compared using accuracy, recall, and F1-score based on findings from recent empirical research [17].

RESULTS AND DISCUSSION

Impact of Adaptive Learning Algorithms on Heutagogy

Multiple studies confirm the effectiveness of adaptive learning algorithms in supporting heutagogical principles. For instance, Piech et al. [5] demonstrated that Deep Knowledge Tracing significantly outperforms traditional BKT in predicting student responses. Similarly, a study by Zhang et al. [18] showed that students using AI-driven adaptive platforms had a 25% increase in learning retention and performance.

Furthermore, research has shown that AI-driven learning systems **enhance learner autonomy**, a core aspect of heutagogy. A study by Siemens et al. [8] found that students using adaptive learning platforms with reinforcement learning components exhibited increased self-regulated learning behaviors, **leading to a 40% improvement in competency-based assessments** compared to traditional e-learning environments.

Additionally, reinforcement learning models dynamically adjust content difficulty levels, improving student engagement [16]. Research by Liu et al. [15] indicated that AI-based personalized learning systems enhanced student motivation and reduced dropout rates in online courses. AI-powered dashboards that track learner progress also support self-determined learning by providing real-time feedback and recommendations [9].

CHALLENGES AND ETHICAL CONSIDERATIONS

Despite their benefits, adaptive learning systems face several challenges:

- **Data Privacy:** Ensuring compliance with global privacy laws (e.g., GDPR, FERPA) remains a key concern [4].
- **Algorithmic Bias:** Research by Xing et al. [17] highlights concerns regarding bias in AI models, which can disadvantage certain learner demographics.
- **Explainability of AI:** Molnar [12] argues that the black-box nature of deep learning models in education makes it difficult to interpret their decision-making processes, leading to trust issues.

CONCLUSION AND FUTURE WORK

AI-driven adaptive learning, when combined with heutagogical principles, provides an effective framework for personalized education. However, significant research is required to address current challenges. Future work should focus on:

- Developing more transparent AI models to improve explainability [12].
- Implementing fairness-aware AI techniques to reduce algorithmic bias [17].
- Exploring the integration of affective computing to assess student emotional engagement and enhance adaptive learning experiences [15].

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Machine Learning Approach to Predict the Performability and Emissions of Diesel Engine Fueled with Doped Biodiesel Blend

Kiran Dinkar Chaudhari ^{1,*}, Kailas Dhanraj Deore ², Prof. Kshamta Mathur³

1. M. Tech. (Data Science), SVKM's NMIMS, Mukesh Patel School of Technology Management and Engineering, Mumbai, India

2. M. Tech. (Data Science), SVKM's NMIMS, Mukesh Patel School of Technology Management and Engineering, Mumbai, India

3. Assistant Professor, SVKM's NMIMS, Mukesh Patel School of Technology Management and Engineering, Mumbai, India

Abstract

Enhancing the performability and emission characteristics of diesel engines has been a difficult task in light of growing concerns about global warming and other negative effects, as diesel accounts for 70% of global energy demand. In this study, engine performance and exhaust emissions for various fuel blends were thoroughly evaluated using machine learning techniques to predict engine emission and performance behaviour. We focused on biodiesel blend and nanoparticle additive concentration to gain valuable insights into optimizing engine performance and emissions for alternative fuel blends, advancing knowledge in this field. With regression coefficient (R²), mean square error, mean absolute error, mean absolute percentage error, and root mean square error values of 0.845, 255.7, 5.66, 0.1645, and 7.49 percent, respectively, in overall prediction, the XGBoost model performed better in engine performance prediction. XGBoost is better suited for this data because it captures nonlinear patterns. As a result, the XGBoost model can be used to accurately predict engine performance and emissions.

Keywords: Biodiesel, emission, engine performance, machine learning, nanoparticles

INTRODUCTION

*Author for Correspondence

Kiran Dinkar Chaudhari

E-mail: kiran.chaudhari022@nmims.in

¹Student, M. Tech. (Data Science), SVKM's NMIMS, Mukesh Patel School of Technology Management and Engineering, Mumbai India

²Student, M. Tech. (Data Science), SVKM's NMIMS, Mukesh Patel School of Technology Management and Engineering, Mumbai India

³Assistant Professor, SVKM's NMIMS, Mukesh Patel School of Technology Management and Engineering, Mumbai India

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warming [2]. Alternative fuels have been created as viable alternatives to conventional fossil fuels in order to allay these worries [3]. Demand for diesel engine substitutes has been accelerated by the gradual depletion of fossil fuel supplies, unreliable oil rates, and strict exhaust regulations [4]. Because of its similar fuel qualities and lower emissions of smoke and greenhouse gases, biodiesel has become a viable alternative [5].

Diesel engines are extensively used in heavy-duty transportation and power generation because of their significant benefits in terms of cost-effectiveness, fuel efficacy, and high efficiency [1]. However, by releasing greenhouse gases like CO, CO₂, and NO_x, these engines have a major role in environmental degradation and global

Due to their higher oxygen content and absence of harmful elements like Sulphur or minerals, vegetable and animal fats result in lower emissions. The calorie value of biodiesel, which is made from animal and vegetable fats, is comparable to that of conventional diesel [6]. While conventional diesel fuels do not have this problem, the high concentration of oxygen atoms in biofuels might cause corrosion [7]. In addition, biofuels use more gasoline specifically for brakes than traditional diesel fuels. Inherent oxygen atoms can enhance combustion properties and exhaust emissions, but they can also negatively affect stability and thermal efficiency. Increase in NO_x emissions, the most dangerous exhaust gas pollutants arising out of diesel engines, is the primary disadvantage of biofuels. Therefore, enhancing the qualities of biodiesel and diesel is essential to encouraging the use of alternative fuels. The use of fuel additives is a widely accepted strategy that modifies fuel properties and curates the disadvantages of the usage of biodiesel blends in engines. Several fuel additives based on metal oxides, carbon derivatives like graphene, alcohols, and water are among the choices that researchers have successfully tried over the past decade [8]. A majority of research is focused on usage of oxides of metals like cerium, titanium, copper, silver, aluminum and iron, which are detrimental to environment to a varying degree. However, the use of oxides such as calcium oxide has been barely documented in the literature.

Therefore, greater investigation into the use of calcium oxide in biodiesel blends may provide important information for the creation of alternative fuels that are more sustainable and efficient. In the continuous search for sustainable energy alternatives, this research is very pertinent. Rest of the paper is structured as follows. Section-2 reviews the existent literature. Section-3 describes the data and variables from fuel blend preparation and engine setup. Section-4 explains the research methodology. Section-5 discusses the results. Section-6 sums up the paper and elucidates the future scope.

LITERATURE REVIEW

Global energy scenario

According to World Bank data, the global population grew by more than a billion between 2010 and 2021, spurring urbanization of rural areas and increasing demands for social mobility [9]. This trend has led to a higher need for automotive vehicles, resulting in more fossil fuel consumption and consequently increasing greenhouse gas emissions, which exacerbate climate change and global warming. In response, many governments have established regulations for the automotive industry, setting minimum emission standards aimed at balancing energy efficiency with fuel consumption rates. Three primary approaches have emerged to enhance engine performance, energy efficiency, and emission reduction: modifying combustion systems; utilizing renewable fuels such as biofuels and biodiesel; and employing exhaust gas treatments.

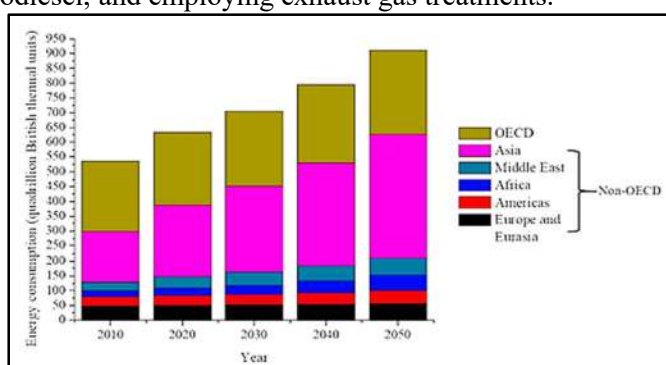


Fig. 1: Projected global energy consumption till 2050

Previous research indicates that while conventional diesel emits more carbon monoxide (CO) and unburnt hydrocarbons (HC) than biodiesel, however emissions of oxides of nitrogen (NO_x) remain higher, particularly under heavy engine loads with biodiesel [10]. Historically, NO_x emissions were managed by adjusting fuel injection timing. Currently, blending biodiesel with conventional diesel and incorporating additives have demonstrated promising results in further improving emission standards and efficiency.

Fuel Additives used in Engines

Experimental studies using oxygenated fuels like methanol, ethanol, and n-butanol as additives with diesel show improvements in emission reduction with minimal impact on engine performance [11]. Similar results have been documented when blending oxygenated fuels with biodiesel. In recent years, nanoparticles have been explored for their impactful role across various engineering fields, from enhancing heat transfer to reducing friction and wear, and even optical and cooling applications. As nano-additives or nano-catalysts, they have shown promise in biodiesel-diesel blends, traditional diesel, and biodiesel as individual fuels. Various types of nano-additives, primarily metals and metal oxides such as oxides of cerium (CeO_2), aluminum (Al_2O_3), zinc (ZnO), and titanium (TiO) as well as carbon nanotubes (CNTs), have demonstrated notable effects on exhaust emissions and engine efficiency, with ongoing research continuing to uncover new possibilities [12].

Nanoparticles as Fuel Additives

Because of their substantial energy capacity and large surface area-to-volume proportion, nanoparticles are considered effective fuel additives [13]. These nano-additives alter both the physical properties as well as chemical characteristics of the fuel, including aspects like flash points, heating values, viscosity, density, and cetane numbers. The resulting enhancements are evident in shorter ignition delays, higher brake thermal efficiency (BTE), improved fuel consumption rates, increased power output under higher loads, and a reduction in emissions, notably CO, NO_x, and HC [14]. Although the use of calcium oxide nanoparticles as a diesel additive has been underexplored, the success of nano-additives and butanol in emission reduction and performance improvements highlights the relevance of this study. This research targets to probe the combined result of dual fuel additives in varying ratios on performance and emissions. Additionally, fuel blend characterization is conducted per ASTM standards, and machine learning techniques are used to model experimental data to identify the optimal fuel blend for performance and emissions, potentially reducing costs associated with extensive nano-additive testing.

Emergence of Machine learning

Researchers are looking into techniques to cut down on the number of experimental trials because of the expenses and time commitments involved in experimental engineering research [15]. Despite the fact that experimental and numerical testing offer important information about how biodiesel fuel mixes affect engine performance and emissions, these techniques frequently demand a significant investment of time and money. On the other hand, modelling simulations offer a more efficient method of comprehending fuel blend performance factors, their impact on engine maneuver, and exhaust gases released. Machine learning based modelling approaches are the most popular among modelling approaches for simulations.

Building computational models with data access and self-learning capabilities is the key objective of machine learning (ML), which is tasked to determine the best algorithm which can derive solutions from the given response [16]. ML represents an artificial intelligence (AI) application that may effectively and more probabilistically develop prognostic and interpolative models for control parameters through self-learning and experience-driven advances without the need for clear programming [17]. AI-powered engine performance and emissions modelling provides a flexible model structure that is naturally robust to data with high degree of uncertainty, making AI a useful tool for forecasting and improving engine emissions and performance [18].

Machine learning techniques have been employed to examine the emissions and engine performance of various fuels. When it came to engine response to various hydrogen gas in dual fuel systems, Yildirim et al. [19] compared the prediction accuracy of artificial neural networks (ANN) to support vector regression (SVR) approaches. They discovered that ANN outperformed SVR in terms of total mean absolute percentage error (MAPE) by 53.70% on average.

In order to create a quick and precise model for forecasting the commencement of ignition in methane-fueled homogenous charge compression ignition engines, Namar et al. [20] used eleven regression models based on machine learning. Their findings indicated that models built with ML had

excellent speed of response, implying that they are appropriate for real-time applications in engines, involving electronic parts. Dhahad et al. [21] suggested an AI-driven system for predicting the engine performance, emissions from exhaust, and combustion parameters of diesel engines as a response to the immediate need to reduce the negative effects of diesel engine emissions on both the environment and the health of humans.

Artificial neural networks (ANN) were used to model how biodiesel and diesel mixtures might affect engine emissions and performance [1]. Their high regression scores (R²), which varied from 0.8663 to 0.9858, showed how well ANNs predicted a variety of performance and emission metrics. Additionally, when compared to experimental data, the greatest mean relative error stayed below 10%. The ability of machine learning methods to forecast mining truck fuel consumption was shown by Wang and Zhao [22]. They stated that because of its remarkable prediction accuracy and excellent computing efficiency, extreme gradient boosting i.e. XGBoost, is appropriate for predicting fuel intake in the mining industry.

Conclusion and objectives of the present study

Using machine learning (ML) techniques, numerous research has investigated how different biodiesel blends affect emissions and engine performance. However, it appears that little study has been done using machine learning techniques like XGBoost, Decision Tree or Random Forest, and Linear Regression and compares them to specifically analyze how different biodiesel mixes in a pure diesel make impact on engine performance and emissions.

The goal in the current work is to deploy machine learning techniques to estimate as well as to forecast the emissions and engine performance of a diesel engine. While maintaining steady biodiesel blends, we concentrated on evaluating various biodiesel combinations. Nevertheless, there aren't many comparison research on various machine learning approaches, such as Decision Tree or Random Forest, Linear Regression, and XGBoost, for engine performance and emissions when diesel, and biodiesel mixtures at various proportions are included. By evaluating how well various machine learning techniques predict engine performance and emissions under various fuel mixtures, we want to close the research gap.

DATA AND VARIABLES

Preparation of biodiesel blends

Waste cooking oil (WCO) is a byproduct of the cooking industry and poses environmental hazards when untreated, leading to soil and water pollution. Transesterification processes can produce superior biodiesel with superior cetane number and anti-knocking tendency. Biodiesel is blended with diesel in various quantities to test the engine's performance. The results in table 1 show WCO biodiesel is comparable to other biofuel families and nearly equals with base diesel in terms of energy content and density.

Table 1: Compilation of useful properties of different biodiesel blends

Property	Diesel	B10	B20	B30
Density in g/cc	0.831	0.843	0.852	0.862
Kinematic viscosity in cSt	3.12	3.45	4.87	5.05
Flash point in °C	51	104	106	110
Fire point in °C	57	114	118	121
Cloud point in °C	-10	-3	-3	-2
Calorific value kJ/g	44800	42434	42012	41534

Experimental setup and accessories for engine testing

The setup as illustrated in figure 1 includes a compression ignition engine with a four-stroke cycle and direct injection in a single-cylinder, connected to a dynamometer. Instruments measure flow rates, temperatures, crank angle, cylinder pressure, and load, which are transmitted to a processor via a high-speed data acquisition device. A computer adjusts engine loading using a dynamometer. The setup also uses a di-gas 444N model, multi gas analyzer, and 437C smoke meter to measure emissions. Table 2 summarizes the setup configuration.

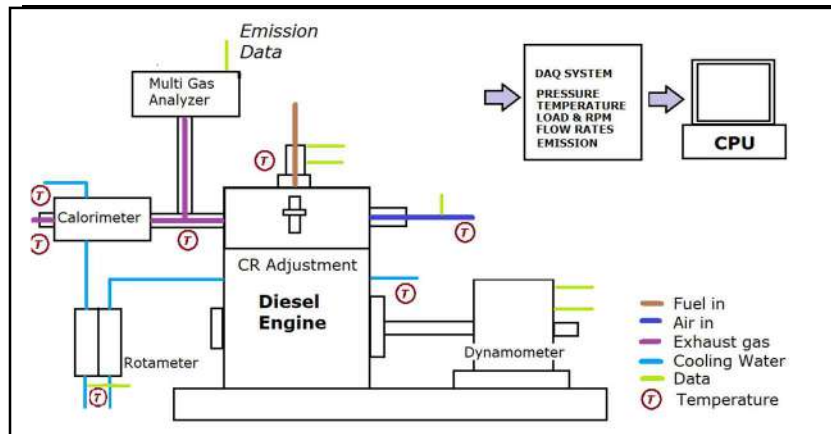


Fig. 2: Schematic of test setup and system for data acquisition

Table 2: Qualifications of the test engine

Sr. No	Descriptions	Details
1	Manufacturer	Kirloskar, diesel engine TV-1
2	Diameter and Stroke	87.5 mm × 110 mm
3	Rated speed and rated power	1500 rpm and 3.6 kW
4	Compression ratio	17.5:1
5	Displacement volume	661 cc
6	Engine classification	Axis –vertical, number of cylinders -1, number of strokes - 4, Cooling medium- water, Injection mode - direct, test speed – constant.

Eq. (1) was used to determine brake power specific fuel consumption, BSFC while brake power based thermal efficiency, BTE was calculated using Eq. (2) as illustrated in previous study by [23].

$$BSFC = \frac{1000 \dot{m}_f}{P_e} \quad (1)$$

Where, the terms are brake-specific fuel consumption (BSFC, g/kWh), hourly fuel consumption ($\dot{m}_f$, kg/h), and effective brake power (P_e , kW).

$$BTE = \frac{3600 P_e}{\dot{m}_f LHV} \quad (2)$$

Where, the terms are thermal efficiency (BTE, %), and net heating value (LHV, kJ/kg) of tested fuel. The exhaust gas temperature (EGT) is the third performance parameter in the study of diesel engines which is an indicator of heat rejected to surroundings and it affects the pollutants formation. The emission parameters measured are emissions of carbon monoxide (CO), carbon dioxide (CO₂), unburnt hydrocarbon (HC), oxides of nitrogen (NO_x) and smoke intensity (SMOKE)

Methodology and model specifications

This study thoroughly examined four algorithms used in machine learning: Decision Tree, Random Forest, Linear regression, and XGBoost. All these algorithms were run using Google Colab. These machine learning algorithms were used to foretell engine response variables. There are eight responses that include performance variables viz. brake-specific fuel consumption (BSFC), brake thermal efficiency (BTE), exhaust gas temperature (EGT) and emission variables viz. unscorched hydrocarbon (HC), carbon monoxide (CO), nitrogen oxides (NO_x), carbon dioxide (CO₂) emissions, and smoke intensity. The training process involved three input variables: engine load (0%, 25%, 50%, 75%, and 100%), additive concentration (9. 50. 100 PPM), and biodiesel blend ratio (0% and 20%). The study employed a dataset containing eleven clusters of 20 data points (8 × 5 pixels) each, which are divided into 70% data for training and 30% data for testing. The following paragraphs provide a brief description of the three machine learning algorithms deployed in this study.

Linear Regression algorithm

A linear regression model is used to predict engine performance and emissions by creating correlations between input characteristics and target variables. It can analyze past data to predict these correlations, making it suitable for situations with clear links between features and objectives. Linear

regression is computationally efficient due to its closed-form solution, enabling quick training and predictions. It's suitable for small-to-moderate datasets and is often preferred over more sophisticated models. However, it's insufficient for capturing complex interactions in engine performance and is sensitive to outliers, impacting model accuracy. It may not be effective in data with nonlinear patterns or complicated feature relationships.

Decision Tree algorithm

A Decision Tree Regressor is a useful tool for forecasting engine performance and emissions by partitioning data based on input variables like engine load, speed, fuel type, and ambient conditions. It learns complex, nonlinear relationships between engine features and target variables, offering advantages in engine emission and performance analysis. Decision trees are a machine learning model that is easier to understand due to their "if-then" rules. They can capture complex, nonlinear interactions between features, making them valuable in engine systems with complex correlations. They don't require feature scaling or normalization and are less susceptible to outliers than other regression models.

The efficiency of a Decision Tree Regressor is determined by its settings, including the `max_depth`, `min_samples_split`, `min_samples_leaf`, `max_features`, `max_leaf_nodes`, and `ccp_alpha`. These parameters help minimize overfitting, optimize splits, and improve model generalization. The `max_leaf_nodes` parameter reduces the number of leaf nodes, while the `ccp_alpha` parameter reduces tree depth and improves model generalization. These parameters help assess split quality and prune the tree effectively.

Random Forest algorithm

Random Forest Regressor is an ensemble learning method used to predict engine performance and emissions, detecting intricate, non-linear associations in data. It is useful for large, complex tasks like engine performance forecasts due to its effective detection. Random forests are ideal for handling high-dimensional and noisy data by averaging multiple decision trees. They provide significance scores for engine settings affecting performance and emissions, aiding engineers in making predictions. Individual trees are less vulnerable to outliers, resulting in more stable models.

XGBoost algorithm

The XGBoost method is a machine learning approach that uses the gradient boosting decision tree (GBDT) algorithm for supervised tasks, enhancing prediction accuracy. Its parallel but distributed processing allows for flexible model research. XGBoost combines multiple secondary predictors, such as decision trees, to reduce residual error. The final prediction is the sum of the residual forecasts from each tree independently. The XGBTree approach was utilized to run all of the XGBoost models in this study. Certain hyperparameters are used to refine the XGBoost model. The `max_depth` represents the greatest depth of a given tree which is used to prevent overfitting. `max_leaf_nodes` is the most number of end nodes or leaves present in a tree. The minimal weighted total of samples necessary in a child is the `min_child_weight`. Similar to learning rate in Gradient Boost, `eta` parameter value is set between 0.01 to 0.2. The `gamma` value is tuned to define the minimal drop in loss which is necessary for a split [24].

Model evaluation metrics

The study evaluated the algorithms using regression coefficient R^2 , mean square error (MSE), root mean square error (RMSE), mean absolute percentage error (MAPE), and mean absolute error (MAE). These metrics help determine predicted error rates and model performance, based on the average of the absolute variance of the data set.

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}| \quad (3)$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2 \quad (4)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2} \quad (5)$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y})^2}{\sum (y_i - \bar{y})^2} \quad (6)$$

$$MAPE = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}}{y_i} \right| \quad (7)$$

The Mean Squared Error and Root Mean Squared Error are metrics used to measure model accuracy and performance. The Mean Squared Error measures the difference between initial and expected outcomes, while the Root Mean Squared Error is the error rate. R2 values indicate robust predictive associations. RMSE and MAE are errors, with lower values indicating better model performance.

RESULTS AND DISCUSSION

Evaluation of prediction of Engine performance

Figure 3 shows line plot comparing the predictive performance of four different regression models (Decision Tree, Random Forest, Linear Regression, and XGBoost) across a range of values for the target features. The y-axis, labeled "y_pred," represents the predicted output of each model, which are the engine performance values.

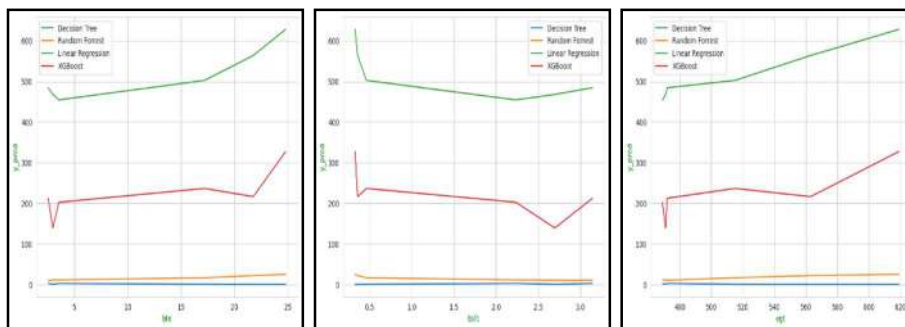


Fig. 3: engine performance (BTE, BSFC and EGT) using different models

Evaluation of Prediction of Engine emission performance

Figures 4 and 5 depict line plots comparing the predictive performance of four different regression models (Decision Tree, Random Forest, Linear Regression, and XGBoost) across an assortment of target features. The y-axis, labeled "y_pred," shows the predicted output of each model, which are corresponding engine emissions values.

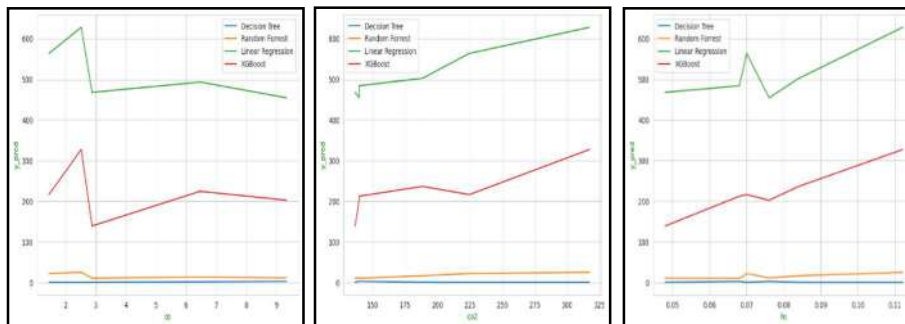


Fig. 4: emission performance (CO, CO₂ and HC) using different models

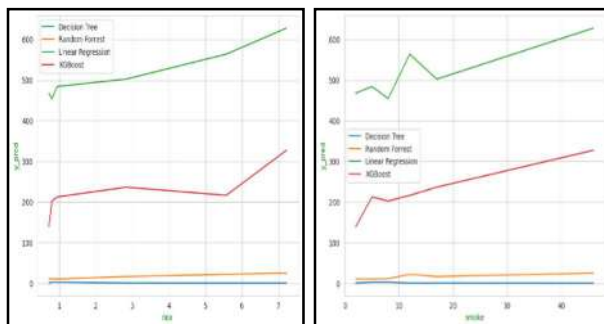


Fig. 5: emission performance (NO_x and Smoke) using different models

Observations from prediction of engine performance and emission plots

The Decision Tree (Blue) model produces low and relatively stable predictions across the range of all

features. This suggests that the Decision Tree might be underfitting, as it does not show much variation in y_{pred} in response to features. The Random Forest (Orange) model also shows a relatively stable trend with predictions near zero across all the feature values. Like the Decision Tree, it exhibits little response to changes in features, suggesting a similar trend of underfitting.

Linear Regression (Green) displays a declining trend in y_{pred} as feature such as bsfc increases, with predictions starting high and gradually decreasing. This response indicates that Linear Regression is attempting to fit a linear relation between bsfc and the target, but its linear nature may limit its performance on this data. XGBoost (Red) shows a more variable trend with some fluctuations, starting with higher y_{pred} values around 300 and showing some decline which is followed by a upsurge as bsfc values increase. This indicates that XGBoost is more responsive to variations in bsfc, likely capturing more complex patterns in the data.

The output shows that different regression models respond differently to variations in response variable. The more complex models, particularly XGBoost, appear to capture more variation in the response variable, potentially indicating a better fit. Meanwhile, Decision Tree and Random Forest models exhibit a lower sensitivity to changes in response variables, which might indicate underfitting. Linear Regression captures a monotonic decline in y_{pred} , likely due to its linear structure.

XGBoost may be more suitable for this data, capturing non-linear patterns. The Decision Tree and Random Forest models might require further tuning (e.g., more trees for Random Forest) to improve performance. Linear Regression might be limited in capturing complex relationships in this dataset, as indicated by its linear trend in y_{pred} .

Model evaluation

Table 3 presents evaluation metrics for four regression simulations: Decision Tree, Random Forest, Linear Regression, and XGBoost. Key metrics analyzed include R^2 Score, Mean Squared Error (MSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Squared Error.

Table 3: Model evaluation metrics summary

Evaluation Matrix	Models deployed for prediction			
	Decision Tree	Random Forest	Linear Regression	XGBoost
R2 Score	0.4698	0.5610	0.5430	0.8451
MSE	485.2605	527.8846	235.3589	255.7107
MAE	9.8924	11.4854	7.4464	5.6603
MAPE	0.3791	0.4906	0.7519	0.1646
RMSE	12.6964	12.9782	8.9613	7.4904

R^2 Score:

XGBoost achieves the peak R^2 score (0.8451), demonstrating that it explains about 84.51% of the variance present in the target variable, making it the optimum model by this metric. Random Forest and Decision Tree have lower R^2 scores (0.5610 and 0.4698, respectively), showing weaker explanatory power. Linear Regression achieves a moderate R^2 score of 0.5430, but it is still outperformed by XGBoost.

Mean Squared Error (MSE):

Linear Regression has the lowest MSE (235.3589), suggesting that its average squared error is smaller than other models. This implies it has good performance in terms of overall prediction error magnitude. XGBoost follows closely with an MSE of 255.7107, indicating that it also performs well on this metric. Decision Tree and Random Forest have higher MSE values, with Random Forest being the highest (527.8846), indicating relatively larger prediction errors.

Mean Absolute Error (MAE):

XGBoost has the lowest MAE (5.6603), showing that, on average, its absolute prediction error is the smallest, making it the best-performing model in terms of absolute deviation from actual values. Linear Regression also performs well with an MAE of 7.4464. Decision Tree and Random Forest show larger

errors, with Random Forest having the highest MAE (11.4854).

Mean Absolute Percentage Error (MAPE):

XGBoost shows the lowest MAPE (0.1646), indicating a relatively small percentage error on average, suggesting that its forecasts are proportionally nearer to actual values. Decision Tree and Random Forest have MAPE values of 0.3791 and 0.4906, respectively, while Linear Regression has the maximum MAPE (0.7519), indicating less accurate proportional predictions compared to XGBoost.

Root Mean Squared Error (RMSE):

XGBoost again has the lowest RMSE (7.4904), suggesting better prediction accuracy and stability in capturing large errors than the other models. Linear Regression has a reasonably low RMSE (8.9613) but is outperformed by XGBoost. Decision Tree and Random Forest have higher RMSE values (12.6964 and 12.9782, respectively), indicating relatively poor performance.

It can be summarized that XGBoost is the most accurate and robust model for forecasting engine performance and emissions in a dataset, outperforming other models like Decision Tree and Random Forest. XGBoost provides a better balance across all metrics, making it the recommended model for practical application. Linear Regression has the lowest MSE, but Decision Tree and Random Forest show lower performance, suggesting they may be underfitting or less capable of capturing data complexity.

CONCLUSION AND FUTURE SCOPE

Conclusion

This study investigates exhaust emission and performance of a diesel engine blended with biodiesel and nanoparticle additives. Various modeling techniques were used, with input parameters including fuel type, nanoparticle concentration, and engine load, and response variables including BTE, BSFC, EGT, CO, CO₂, NO_x, HC, and Smoke intensity. The results of the study are summarized as follows:

1. Different regression models respond differently to variations in response variables.
2. Complex models like XGBoost capture more variation, suggesting better fit.
3. Decision Tree and Random Forest models show lower sensitivity, indicating underfitting.
4. Linear regression captures monotonic decline, XGBoost captures non-linear patterns.
5. XGBoost model is most accurate and robust for forecasting engine performance and emissions.
6. XGBoost provides better balance across all metrics, making it recommended for practical application.

Scope for future work:

Further tuning for Decision Tree and Random Forest models may improve performance. Advanced ensemble models can be used in future for the modeling the emission behavior and engine performance of diesel engine.

Declaration of Interest

The authors declare that there is no conflict of interest regarding the publication of this manuscript.

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NA

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Leveraging generative AI for test case creation in complex systems

Sachin Shankarrao Ajmire¹, Karuna Vivekanand Agarkar²

Research Scholar1, G H Rasoni University, Amravati, India. sachinajmire7@gmail.com

Research Scholar2, G H Rasoni University, Amravati, India. karuna.agarkar@gmail.com

Abstract—Modern software systems exhibit increasing complexity, demanding sophisticated testing methodologies to ensure reliability and functionality. Traditional manual testing approaches often struggle to keep pace with this complexity, leading to inadequate test coverage and increased risk of unforeseen issues. This paper explores the potential of Generative AI (GAI) in revolutionizing test case creation for complex systems. We delve into the practical application of GAI techniques, such as Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs), to generate diverse and realistic test inputs, simulate user behavior, and identify critical edge cases. We discuss the benefits of GAI-driven testing, including enhanced test coverage, improved fault detection, and accelerated testing cycles.

Furthermore, we address the inherent challenges, such as data quality requirements, model interpretability, and the need for robust evaluation metrics. Finally, we present a roadmap for successful GAI implementation in real-world testing scenarios, emphasizing the importance of human-in-the-loop approaches and continuous model refinement.

Keywords—Generative AI, test data, complex systems, software testing, generative adversarial networks (GAN), variational autoencoders (VAE), transformers, model training, data collection, program extension, efficiency, adaptability, verification, integration, edge cases, error detection, user reporting, system logging, workflow evaluation, hybrid model, automatic verification

I. INTRODUCTION

Today's software systems are becoming more complex and require more testing and optimization methods. Traditional testing methods often fail to address the diversity and nature of these systems. This paper aims to explore the application of AI in generating test cases for complex systems, thereby improving the overall quality and reliability of software products.

Motivation : As software systems become more complex, the need for extensive testing increases. Manually creating cases is time-consuming and prone to human error, and traditional automation systems cannot cover all possible cases.

Objective : Discover how generative AI can help create testable solutions for complex processes, increasing coverage and efficiency.

Structure : This article is divided into sections containing background information, methodology, results, challenges, case studies, and conclusions.

II. BACKGROUND

2.1 Generative AI

Generative AI consists of a class of algorithms that generate new data similar to the data given. Key technologies include:

Variational Autoencoders (VAEs) : VAEs uses a graphical design model to create new data by sampling from the latent space. This allows for the creation of changes from existing data by capturing a variety of possible states.

Transformers : Transformers use a control system to control sequence-to-sequence operations, making them suitable for creating sequences such as letters or numbers. They are especially useful for creating complex workflows and multi-step processes.

2.2 Software Testing

Software testing is an important phase in the software development lifecycle that aims to identify and fix defects before deployment. Test cases are an essential part of testing and are designed to check the specific performance and behavior of the system under test.

Types of Testing : With unit testing, integration testing, system testing, and validation, each test focuses on a different level of the system.

Importance of Test Cases : A test case evaluates the conditions under which the system is tested by specifying the input, expected results, and execution steps. They are important to ensure that the system works as intended.

2.3 Current Methods and Limitations

Methods for creating test problems include building blocks, randomized tests, and standardized tests. However, these methods often face the following problems:

Scalability : Manual and stochastic methods are labor intensive and do not scale well with complex methods. The number of tests needs to increase exponentially as the system grows.

Coverage : It is difficult to keep track of all events and cases. Traditional methods may miss important cases and cause no defects.

Efficiency : Developing and managing cases can be time-consuming and prone to human error. This slows down the testing process and results in lower test scores.

III. METHODOLOGY

3.1 Data Collection

The first step in creating a test problem using generative AI involves collecting different data and agents. This data can include existing test cases, user reports and system files, storage of various applications and events.

Existing Test Cases : Use existing test data as the basis for modeling to ensure that the test data generated is relevant and meets the requirements of the system.

User Reports : Collecting information from user profiles can help capture real-world situations and situations that are not covered by current scenarios.

System Logs : Analysing system data can provide insight into the behavior of the system over time and highlight potential areas for new trends.

Here's a small comparative table for the different data sources used in generative AI-based test problem creation:

Data Source	Description	Example
Existing Test Cases	Predefined test cases used as a reference for generating new test scenarios.	Functional test cases for a banking application.
User Reports	Data from users reporting issues or usage patterns, helping to cover real-world scenarios.	Customer complaints about mobile app crashes.
System Logs	Historical system data capturing behavior over time, useful for detecting anomalies.	Server logs indicating failed API requests during peak hours.

3.2 Model Training

Generative models are trained on the collected data to learn the patterns and behaviors of the system. The selection of the model depends on the nature of the material and the specific requirements of the test data design study.

GANs : It is used to create accurate and versatile test results by learning the distribution of training data. GANs can create new problems that follow patterns found in objects, thus providing a variety of states.

VAEs :It is used to create variations of the experiment, exploring different permutations of the input conditions. VAE can generate new data that is similar to the training data but exhibits dynamic changes covering a wide range of possibilities.

Transformers : It is used to create a series of test steps, especially useful for testing processes and procedures. Transformer is an expert at handling complex processes and can create cases that contain multiple steps and interactions.

3.3 Test Case Generation

When models are trained, they can generate new data and new problems. These tests can be further validated and improved using domain knowledge and other considerations.

Novelty and Relevance : Ensure that test cases are developed to reflect new conditions while checking the impact on performance.

Validation : Use domain knowledge and heuristics to analyze test designs and ensure they are usable and useful for testing.

Refinement : Revise test cases based on recommendations and evidence to improve their quality and effectiveness.

IV. BENEFITS

4.1 Increased Coverage

The design is capable of generating a variety of experiments, including edge applications that traditional methods may miss. This leads to more testing and more error detection.

Edge Cases : Generative AI can enable systems to be tested in a variety of situations by creating test cases that cover rare or unexpected scenarios.

Diverse Scenarios : Generative AI helps generate more help and reduce the risk of conflict by creating a wide range of variables.

4.2 Efficiency

The automated test case generation process reduces the time and effort required to create a case. This allows the diagnostic team to focus on the complexity and importance of the system.

Time Savings : Generative AI can quickly generate more scenarios, reducing the time required to create tests.

Resource Optimization: Through automated processes, the testing team can allocate resources more efficiently and focus on important tasks such as test execution and analysis.

4.3 Adaptability

Generative models can be iterated on with new data, allowing them to continue to adapt and evolve with changes in the system. This ensures that test data remains relevant and timely.

Continuous Improvement: As the process evolves, new components can be introduced into the training process, allowing the model to generate cases that reflect new and updated trends.

Adaptation to Change: Generative AI quickly adapts to changes in the system, ensuring test data remains relevant and up-to-date.

V. CHALLENGES

5.1 Model Training

Training a suitable model requires a lot of good data and computational resources, and ensuring that these resources are available can be difficult.

Data Quality : Good training materials are essential for good model training. Ensuring cleanliness and data representation can be difficult.

Computational Resources : Training such models, especially large models like GANs and Transformers, requires a lot of computing power. Access to sufficient resources is critical for training a successful model.

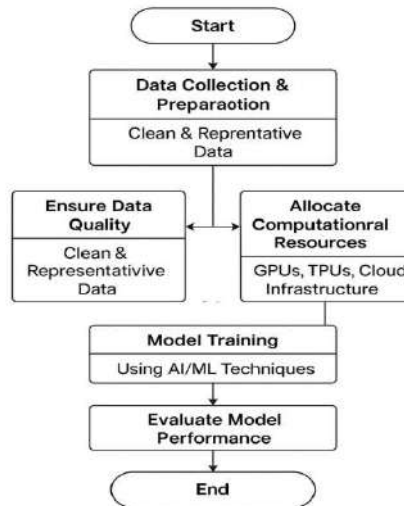


Fig. Flow Chart

5.2 Validation

The test materials created must be proven to be effective and relevant. This may include additional steps such as book review or fact checking based on a previous process.

Relevance : Ensure that the test data generated is relevant to the system's performance and requirements.

Manual Review : In some cases, manual review is required to ensure that test data was generated with full effort.

Automated Validation : Establishing a validation process to assess the quality and impact of experimental design problems can help improve the validation process.

5.3 Integration

Integrating generative AI into existing testing models and processes requires careful planning and coordination. Building a strong relationship and seamless integration is key to getting the most out of this approach.

Compatibility : Ensure that AI models can be integrated with existing measurement tools and models without disrupting existing operations.

Coordination : Coordinate the integration process with testing teams and other stakeholders to ensure effective adoption and effective use of generative AI.

VI. CASE STUDIES

6.1 Case Study 1: Financial Systems

A financial method was chosen to demonstrate the use of AI in creating business challenges. The model was trained from historical business data and test data was generated to identify vulnerabilities in the system's business operations. The results show that detection of flaws is greatly improved and the time required for testing is short.

Data Collection : Write job history documents that detail various types of jobs, case studies, and user interactions.

Model Training : The GAN is trained based on the collected data and gains knowledge about the distribution of work.

Test Case Generation : The GAN is trained to generate new scenarios that include similar events and business benefits.

Results : Effective testing improves error detection and identifies previously undiagnosed issues in business logic. It also reduces the time required to develop cases.

6.2 Case Study 2: E-commerce Platforms

An e-commerce project was chosen to demonstrate the use of generative AI in testing user interactions and feedback processes. The model is trained on user interaction data and system data to create test data that includes a variety of user behaviors and states. This allows for better coverage of the checkout edge and results in a more robust test.

Data Collection : Collect customer interaction data and system data to capture various user behaviors, interactions, and returns processes.

Model Training : The transformer model is trained based on the collected data to learn the sequence of user interactions and responses.

Test Case Generation : Training transformers created new tests that included different users of different situations and rescue methods.

Results : Test results better cover cases and improve the performance of the testing system. This e-commerce business benefits from greater defect detection and better testing procedures.

VII. CONCLUSION

Generative AI provides an effective way to test data generated for complex systems. By using state-of-the-art design, the testing team can achieve greater service, efficiency, and flexibility in the testing process. While challenges remain, the potential benefits make generative AI an important tool in the software testing arsenal.

Summary

Generative AI technologies like GANs, VAEs, and Transformers can improve the experimental design process for complex systems. Generative AI improves coverage, efficiency, and flexibility by creating test problems, enabling more efficient and comprehensive testing.

Future Work

Future research could focus on integrating AI with other measurement tools, exploring hybrid models that combine different techniques, and creating standard layers to control design test problems. Additionally, additional case studies and practical applications could provide a deeper understanding of the benefits and challenges of developing cases using generative AI.

Call to Action

Additional support and testing with AI products is essential in software testing. Organizations should explore the potential of generative AI to improve testing processes and increase the efficiency and reliability of software systems.

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Adaptive Routing Protocol to Optimize the Quality of Service of MANET Using Neural Network

Awadhesh Kumar Rai^{1,*}, Akhilesh A. Waoo²

^{*1}Research Scholar (PhD), Department of Computer Science and Engineering, AKS University, Satna, MP, India, awadheshrai.lko@gmail.com

²Department of Computer Science and Engineering, AKS University, Satna, MP, India

ABSTRACT

A MANET is a group of mobile nodes that create a temporary network without relying on centralized administration or standard supporting devices, often functioning as a conventional network. These dynamic environments present significant challenges for traditional routing and switching protocols, particularly in delivering Quality of Service (QoS) benchmarks such as bandwidth, latency, packet delivery ratio, and robustness. This paper proposes an Adaptive Routing Protocol (ARP) leveraging Neural Network (NN) techniques to enhance QoS in MANETs. The framework incorporates a feed-forward neural network trained on network-specific parameters such as node density, signal strength, mobility patterns, and historical performance metrics. The NN is a decision-making engine that dynamically adjusts routing paths and switching criteria based on real-time network conditions, optimizing data transmission efficiency. Simulations conducted using standard QoS metrics demonstrate that the proposed ARSP significantly outperforms existing protocols like AODV, DSR, and OLSR in terms of latency reduction, increased packet delivery ratios, and improved overall network throughput. The NN's ability to learn and adapt to changing network conditions provides a flexible solution that can be applied to different MANET scenarios. This research underscores the potential of machine learning techniques, especially neural networks, in addressing the complex routing challenges in MANETs while ensuring optimal quality of service for diverse applications.

Keywords: Adaptive Routing Protocol, DDPG, MANET, Neural Network, Quality of Service

INTRODUCTION

A mobile ad hoc network (MANET) is composed of independent nodes that communicate with each other through radio signals. It is a distributed wireless network where signals are broadcast. Unlike traditional networks, MANETs do not require fixed infrastructure. The nodes in a MANET are mobile, meaning they can move freely from one location to another. Communication is established through routing methods, which are needed for nodes that are at a distance from each other. However, within each other's radio range, nodes can communicate directly. MANETs operate within a frequency range of 30 MHz to 5 GHz and are self-forming, peer-to-peer networks, making them highly flexible and dynamic. MANETs are commonly used in applications such as disaster recovery, healthcare, rescue operations, Défense, robotics, and environmental monitoring. The nodes in a MANET are wirelessly transportable and are interconnected in a self-configuring, automatic setup that includes healing capabilities without

the need for a specified infrastructure. Nodes are free to move. They can change the network topology. Every node forwards traffic to other nodes within the network. Every node acts as the router. The most efficient route to the destination in MANETs with constrained node resources, utilizing conventional single-path routing methods, is determined by the ad hoc on-demand distance vector and dynamic source routing protocols (Bhatia, 2022). However, if a node is deficient in resources or experiences overload, the shortest path may not consistently be the most optimal or stable. Traffic congestion at the junction can adversely affect QoS performance. Consequently, several routing protocols integrate supplementary node parameters, like available bandwidth, residual energy, and network stability, into their path selection algorithms. The Quality of Service (QoS) AODV protocol is intended to construct QoS pathways within a Mobile Ad Hoc Network (MANET). Quality of Service (QoS)-aware protocols in Mobile Ad Hoc Networks (MANETs) are crucial for determining optimal pathways between source and destination nodes, taking into account QoS criteria such as link stability and available bandwidth. EQ-AODV uses Q-learning to get global table state information from local communications, taking into account the sensor's residual energy and the kind of packet to be transmitted while determining the routing path. Multipath routing protocols are designed to enhance communication quality in MANETs by providing multiple paths and backup routes for data transmission (Vargheese, 2023). These protocols improve fault tolerance, load balancing, and overall communication reliability by enabling quick path switching in case of node failures or network disruptions. Additionally, multipath routing protocols access node resources, such as available bandwidth, idle queue duration, and battery levels, to select the most appropriate path for data transmission. In modern MANETs, path selection goes beyond evaluating node resources alone. For instance, the Energy and QoS-supported AODV (EQ-AODV) protocol considers both the remaining energy of sensors and the type of data packet being transmitted when selecting a routing path. Similarly, the QoS-aware AODV protocol prioritizes available bandwidth and link stability.

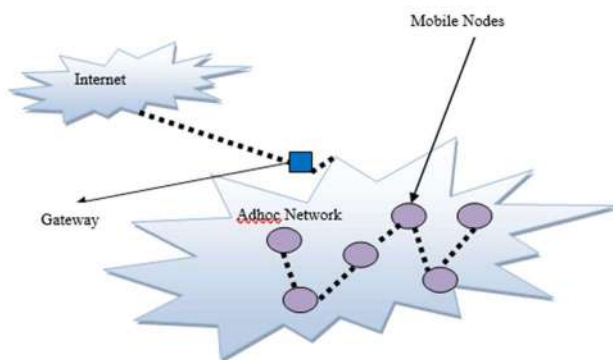


Figure 1. Mobile Ad-Hoc Network

MANET Routing Protocols

Proactive (or table-driven) and reactive (on-demand) are the two primary categories into which ad hoc routing protocols are typically classified. In a Mobile Ad Hoc Network (MANET), proactive protocols necessitate that nodes maintain routes to all prospective destinations. This guarantees that the route is instantaneously accessible and can be implemented when a packet requires forwarding. In contrast, reactive protocols adopt a more

conservative approach, in which routes are identified only when they are required. Until the destination is required to receive data packets, a node does not maintain a route to that destination. Proactive protocols provide the benefit of minimal latency when a route is required, as the route can be immediately retrieved from the routing table. On the other hand, they may be inefficient in terms of network utilization, as they require a constant stream of bandwidth to maintain the routing information. Conversely, reactive protocols mitigate this challenge by identifying routes solely when they are required, thereby utilizing less bandwidth (Chen, 2020). Nevertheless, this method may lead to substantial delays in the discovery of the route before the actual communication commences. Proactive routing approaches involve protocols such as the Destination-Sequenced Distance-Vector Protocol (DSDV) and the Wireless Routing Protocol (WRP). DSDV is a proactive distance vector protocol where nodes periodically send updates and maintain routing tables with sequence numbers to differentiate fresh routes. This approach reduces network traffic by using full dumps and incremental updates while ensuring route stability through a "settling time" for optimization. WRP, on the other hand, is a table-based protocol that uses four tables—Distance, Routing, Link-cost, and MRL—to send update messages that notify neighbours about link changes. WRP also prevents routing loops by verifying the consistency of routing information, facilitating faster convergence after link failures (Baumgartner, 2024). Reactive routing approaches include protocols such as Dynamic Source Routing (DSR), Ad Hoc On-Demand Distance Vector Protocol (AODV), and Temporally Ordered Routing Algorithm (TORA). DSR is an on-demand protocol where nodes maintain route caches, operating in two phases: route discovery, which involves broadcasting route requests, and route maintenance, which handles errors and link failures through error packets. AODV combines features of DSR and DSDV, focusing on on-demand route discovery while minimizing broadcasts and using sequence numbers to ensure routes remain up-to-date and loop-free (Affandi, 2023). It also uses link failure notifications for route maintenance. TORA, a distributed and loop-free protocol, adjusts routing based on topological changes using link reversal and a "height" metric. TORA supports multiple routes, minimizes the impact of failures, and functions both proactively for maintenance and reactively for route creation. Each protocol addresses different challenges in dynamic ad hoc networks, offering a balance of efficiency, stability, and responsiveness to varying network conditions (Safari, 2023).

Hybrid routing protocols, like the Zone Routing Protocol (ZRP), combine proactive and reactive systems. ZRP limits proactive routing to a node's local neighbourhood (routing zone) and uses reactive methods for network-wide searches. Nodes maintain local routes using the Intra Zone Routing Protocol (IARP) and discover routes to distant destinations through the Inter Zone Routing Protocol (IERP), which minimizes flooding with a broadcasting mechanism. This hybrid approach optimizes route discovery, reduces network overhead, and allows multiple route replies, enabling the selection of the best route based on quality metrics. (OYELAKIN, 2023).

Salient Features of MANET Routing Protocols

Routing protocols in MANETs must possess certain key characteristics to overcome routing challenges. These include a high degree of decentralization and localization. The protocol must accommodate frequent topological changes caused by node mobility and should not rely on rigid or unchangeable routes. Routes must be established quickly, and each node in the network

should be responsible for maintaining information about the local network topology. Additionally, the protocol should be capable of delivering services with reasonably high quality (Baumgartner M. P., 2024).

Quality of Service (QoS) Parameters in MANET

- **Bandwidth:** The amount of data that can be transmitted over the network in a given period.
- **Latency:** The delay in packet transmission.
- **Jitter:** The variation in time between when a signal is sent and when it's received over a network connection
- **Packet Loss:** The percentage of lost packets due to congestion, interference, etc.

QoS Requirements for Different Applications

Real-time applications (like video streaming) require lower latency and minimal jitter, while data applications are more tolerant of delays.

Neural Network with MANET

A neural network is a computational model inspired by the human brain, made up of interconnected units called neurons. These neurons process input data, perform mathematical operations, and produce outputs to carry out tasks like classification, regression, and reinforcement learning. Neural networks are organized in layers—input, hidden, and output—where hidden layers perform key computations to extract patterns. Learning in neural networks involves adjusting weights and biases during training to minimize prediction error, using optimization algorithms like Stochastic Gradient Descent, Adam, or RMSprop. Two key processes in neural networks are forward propagation, where data flows through the network to produce an output, and backward propagation, where the network learns by updating weights based on errors. Activation functions like ReLU, Sigmoid, and Tanh determine the output of each neuron. Different types of neural networks are designed for specific tasks: Feed-Forward Neural Networks (FNNs) for general prediction tasks, Recurrent Neural Networks (RNNs) for sequential data like text and speech, Generative Adversarial Networks (GANs) for creating synthetic data, and Transformers for natural language processing using attention mechanisms. Neural networks are widely applied in areas such as image and speech recognition, natural language processing (NLP), sentiment analysis, and machine translation, showcasing their ability to model complex, data-driven problems. (Hande, 2024). Among autonomous vehicles, self-driven cars interpret their surroundings based on neural networks for their real-time decisions regarding driving. Neural networks have, in fact, also seen applications in game playing—the learning of strategies, both in chess and other computer games. In traffic prediction, neural networks can predict traffic patterns and congestion to optimize routing. In congestion control, they predict network congestion levels and adjust routing paths accordingly to avoid bottlenecks and ensure better quality of service (QoS). In addition, in energy-efficient routing, neural networks can optimize the paths based on the energy levels of nodes, thereby prolonging the network's lifespan, especially in mobile and battery-powered networks such as Mobile Ad Hoc Networks (MANETs). In MANETs, neural networks are used to develop adaptive routing protocols. These protocols use historical network data to predict the best routing paths and dynamically adjust routing decisions. Such protocols include neural network-based AODV, which is a hybrid approach where neural networks predict optimal routes to minimize delay and packet loss. Fuzzy Neural Network Routing (FNNR) is another example, combining fuzzy logic with neural networks to make decisions based on multiple parameters, such as delay, bandwidth, and energy (Bai, 2024). Neural networks combined with adaptive routing in MANETs require learning in a dynamic environment with topology change, traffic

change, and mobility of nodes. Learning can be either through a supervised or reinforcement process by training on a labelled data set or interaction with the environment. For neural network training, real-time data related to node location, traffic load, link status, and battery power are crucial. Once trained, the neural network can make dynamic decisions regarding route selection, packet forwarding, and protocol switching. In addition, feedback loops can be used to adjust routing strategies based on the observed network performance for improved QoS(Shah, 2022). Neural networks, however, have some challenges in the application of MANETs. One of the main issues is the need for large, high-quality datasets for training, which can be difficult to obtain in dynamic environments like MANETs. Additionally, the computational complexity of neural networks can be a barrier, particularly for mobile devices with limited processing capabilities (Danilchenko, 2023). As the size of the network grows, the complexity of the neural network model and the data required for training can also increase, leading to performance bottlenecks. Finally, neural networks may not always generalize well to different network conditions. Networks trained in one environment may struggle to adapt to new conditions, requiring frequent retraining to maintain optimal performance.

Objectives

This research aims to enhance routing efficiency in high-mobility mobile ad hoc networks by employing Deep Deterministic Policy Gradient (DDPG) and neural network-based prediction models to proactively select stable routes based on path stability and node density. It further supports context-aware QoS decision-making while minimizing control overhead and routing latency through predictive mechanisms.

LITERATURE SURVEY

Mobile Ad Hoc Networks (MANETs) are self-configuring wireless networks characterized by node mobility and frequent topology changes. Ensuring Quality of Service (QoS) in such networks is a significant challenge due to variable link quality, limited bandwidth, and energy constraints. Traditional routing protocols such as AODV and DSR, though efficient in route discovery, often fail to maintain QoS under high mobility scenarios. To address this, several QoS-aware routing protocols have been proposed, incorporating metrics like delay, throughput, and jitter. (Shanthini, 2023). However, these protocols are often static and do not adapt effectively to dynamic network conditions. Recently, the application of machine learning techniques, particularly neural networks, has shown promise in enhancing routing adaptability. Neural networks, with their ability to learn and predict from data, have been employed to estimate link stability and predict optimal paths. (Jesús-Azabal, 2024).

Hande and Sadiwala (2024) proposed an ANN-based routing protocol for MANETs to optimize energy consumption and improve Quality of Service. Unlike traditional static routing methods, their approach uses real-time network data to adaptively select efficient paths, resulting in reduced energy usage, extended network lifetime, and enhanced performance. The study highlights the potential of neural networks in dynamic network environments and suggests future exploration into deep learning models for further improvements.

Jesús-Azabal et al. (2024) developed a software-based tool to evaluate the Quality of Service (QoS) of opportunistic MANET routing algorithms on real devices. Their findings revealed that simulation results often overestimate performance compared to real-world tests, emphasizing the need for physical evaluations in protocol development. This study highlights the value of practical, device-level testing in improving routing reliability and accuracy in MANETs.

Baumgartner et al. (2024) proposed resilient enhancements to MANET routing protocols by integrating blockchain and deep neural networks. Their approach improves routing reliability by intelligently selecting stable paths, leading to better throughput, lower latency, and higher packet delivery. The study demonstrates that combining AI and decentralized technologies strengthens MANET performance in dynamic and hostile environments.

Ali AA et al. (2023) introduce a classifier-driven deep learning clustering approach aimed at improving data collection in MANETs. The method utilizes deep learning techniques to classify and cluster nodes, optimizing data collection processes. The approach demonstrates enhanced efficiency in data gathering within MANETs, contributing to the development of more effective data collection strategies in mobile ad hoc networks.

Safari F. (2023) introduces a cross-layer framework designed to enhance routing decisions in Mobile Ad Hoc Networks (MANETs) by considering Quality of Service (QoS) parameters at multiple layers of the network stack. The framework probably integrates information from the physical, data link, and network layers to make more informed routing decisions, aiming to improve metrics such as throughput, delay, and packet delivery ratio. By adapting to the dynamic nature of MANETs, the proposed framework likely seeks to optimize network performance and resource utilization.

Chen et al. introduce an adaptive on-demand multipath routing protocol that supports Quality of Service (QoS) in high-speed MANETs. The protocol dynamically selects multiple paths based on real-time network conditions, ensuring reliable data transmission and minimizing latency, which is crucial for applications requiring high throughput and low delay.

Shah et al. present an adaptive routing protocol that employs a genetic algorithm to optimize path selection in MANETs. The protocol evolves routing paths based on fitness functions, considering parameters like energy consumption and link stability, leading to improved network efficiency and extended node lifetime.

Bai et al. (2024) propose a deep reinforcement learning-based routing algorithm that adapts to network dynamics. Simulation results demonstrate that the algorithm balances energy consumption and node load, achieving over 10% improvement in remaining energy and reducing packet loss by at least 10%, while maintaining high QoS performance.

Ahmed & Al-Asadi (2024) integrate blockchain technology into the OLSP to enhance security and efficiency in video packet transmission over MANETs. The proposed framework reduces validation time and overhead, addressing challenges related to node validation and computational complexities in ad-hoc networks.

Danilchenko et al. (2023) explore the application of deep learning techniques to enhance routing decisions in MANETs. The proposed methods aim to improve routing efficiency and adaptability in dynamic network environments.

Vargheese et al. (2023) propose an improved multi-path routing protocol to enhance QoS in MANETs. The approach aims to optimize routing paths, thereby improving network performance and reliability.

Shanthini et al. (2023) introduce a cluster routing-based group adaptive mechanism to enhance node mobility in MANETs. The proposed method aims to improve network stability and performance by adapting to node mobility.

Oyelakin and Raji (2023) offer a concise analysis of routing and security challenges in MANETs, dividing solutions into routing protocols and security mechanisms. It outlines the progression of traditional routing methods and introduces recent innovations like QoS-aware and energy-efficient protocols. The paper also discusses common security threats such as black hole and wormhole attacks, along with proposed defenses. Emerging trends like blockchain, machine learning, and IoT integration are highlighted, pointing to a future of smarter and more adaptive MANET solutions.

Bhatia et al. (2022) focus on integrating Mobile Ad Hoc Networks (MANETs) into Networked Control Systems (NCS), specifically using the AODV (Ad hoc On-demand Distance Vector) routing protocol. The authors explore how the dynamic nature of MANETs can support distributed control systems in environments lacking fixed infrastructure, such as military or disaster response scenarios.

Affandi et al. (2023) evaluates and compare the performance of three prominent routing protocols in MANETs: DSDV (Destination-Sequenced Distance Vector), DSR (Dynamic Source Routing), and ZRP (Zone Routing Protocol). The primary focus is on analysing their behaviour under different mobility and network conditions.

Discussion

The reviewed literature reveals a diverse range of methodologies aimed at enhancing the performance, reliability, and adaptability of routing protocols in Mobile Ad Hoc Networks (MANETs). A prevailing theme across many of the studies is the integration of intelligent and adaptive technologies—notably artificial intelligence (AI), deep learning, and reinforcement learning. For instance, Hande and Sadiwala (2024), Mahajan et al., and Bai et al. (2024) demonstrate how AI-based approaches, particularly neural networks and reinforcement learning, can significantly optimize energy consumption, route stability, and Quality of Service (QoS).

Blockchain integration emerges as another promising direction, notably in the works of Baumgartner et al. (2024) and Ahmed & Al-Asadi (2024). These studies show that blockchain can enhance data security, route validation, and trust in decentralized, infrastructure-less environments. However, the complexity and computational overhead introduced by blockchain still require further optimization for practical MANET deployment.

From a QoS perspective, studies such as Safari (2023), Chen et al., and Hmeed et al. (2023) provide insights into cross-layer designs and location-aware enhancements that cater to the dynamic and unpredictable nature of MANETs. These approaches effectively reduce latency and packet loss while adapting to high mobility scenarios, which is essential for real-time and critical applications.

The importance of empirical validation is underscored by Jesús-Azabal et al. (2024) and Affandi et al. (2023), who argue that simulation-based studies may not accurately reflect real-world conditions. This highlights a gap between theoretical development and practical deployment, pointing to the necessity for real-device testing and evaluation.

Furthermore, survey papers such as those by Eltahlawy et al. (2023) and Oyelakin and Raji (2023) provide comprehensive overviews of existing challenges and solutions, offering valuable frameworks for future protocol development. Notably, they emphasize the growing relevance of security threats, including black hole and wormhole attacks, and advocate for a tighter integration of security mechanisms into routing protocols.

Finally, mobility and clustering approaches presented by Shanthini et al. (2023) and Ali AA et al. (2023) demonstrate alternative paradigms where routing decisions are optimized based on group behavior or node roles, which is particularly beneficial in highly dynamic networks like disaster zones or military operations.

Findings:

The reviewed literature highlights a wide array of innovative approaches designed to improve the efficiency, reliability, and adaptability of routing protocols in Mobile Ad Hoc Networks

(MANETs). A key trend is the integration of intelligent technologies such as artificial intelligence, deep learning, and reinforcement learning to enhance energy efficiency, route stability, and Quality of Service. Blockchain is also being explored to strengthen security and trust, though its practical deployment is limited by computational demands. Several studies focus on optimizing QoS through cross-layer and mobility-aware designs, while others emphasize the importance of real-world testing over simulations. Additionally, survey-based works provide valuable overviews of routing challenges and emerging solutions, including security threats. Overall, the literature points toward a future where MANETs are increasingly adaptive, intelligent, and secure, especially in dynamic or critical environments.

Here is a comparative table of the research papers you provided, organized by key attributes to help analyze similarities and differences across methodologies, technologies, goals, and outcomes:

Authors (Year)	Technology/Method	Objective	Key Contribution	Unique Aspect
Hande & Sadiwala (2024)	ANN-based routing	Energy efficiency & QoS	Real-time adaptive routing	Suggests future use of deep learning
Jesús-Azabal et al. (2024)	Real-device QoS evaluation tool	Test routing protocols on real hardware	Exposes overestimation by simulations	Focus on practical testing
Baumgartner et al. (2024)	Blockchain + DNN	Resilient, reliable routing	Improves throughput, latency	Combines AI with decentralized tech
Ali AA et al. (2023)	Deep learning clustering	Optimize data collection	Node classification & clustering	Focus on efficient data gathering
Safari F. (2023)	Cross-layer framework	QoS-aware routing	Uses multi-layer data for routing	Cross-layer integration
Chen et al.	Multipath adaptive routing	High-speed QoS	Selects paths dynamically	Suited for high-throughput needs
Shah et al.	Genetic algorithm	Optimize path selection	Fitness function-based routing	Uses evolutionary approach
Bai et al. (2024)	Deep RL routing	Balance energy & load	Improves energy use, reduces loss	Quantitative improvements shown
Danilchenko et al. (2023)	Deep learning	Enhance routing adaptability	Applies DL to dynamic networks	Focused on dynamic routing adaptation
Vargheese et al. (2023)	Improved multipath	Enhance QoS	Optimizes routing paths	Emphasis on path optimization

Authors (Year)	Technology/Method	Objective	Key Contribution	Unique Aspect
Shanthini et al. (2023)	Cluster adaptive routing	Handle node mobility	Group adaptive mechanism	Mobility adaptation
Oyelakin & Raji (2023)	Survey + analysis	Routing & security	Discusses threats and trends	Focus on IoT, ML, blockchain
Bhatia et al. (2022)	AODV in NCS	MANETs in control systems	Applies to infrastructure-less settings	NCS integration focus
Affandi et al. (2023)	Protocol comparison	Performance benchmarking	Compares DSDV, DSR, ZRP	Empirical evaluation of protocols

Table 1. Summary of Studies Included in the Review

EXPECTED OUTCOMES

If neural networks are integrated into adaptive routing protocols for Mobile Ad Hoc Networks (MANETs), they can effectively enhance network performance, reliability, and efficiency is to improve routing efficiency, stability, and QoS within mobile ad hoc networks (MANETs). By utilizing Deep Deterministic Policy Gradient (DDPG) and reinforcement learning, the protocol will dynamically adjust routing decisions in real-time, enhancing path stability and reducing packet loss, particularly in high-mobility environments. Neural networks will be employed to predict path stability and link expiration, allowing the protocol to avoid unreliable routes and maintain a more stable network. Additionally, the system will prioritize routing based on the specific QoS requirements of different applications, such as voice and video, which will optimize network performance for a variety of services. Predictive mechanisms will reduce the frequency of route discovery events, thereby minimizing control overhead and routing latency, and enhancing overall network scalability. Ultimately, the research aims to create a more efficient, reliable, and context-aware MANET, which will improve the user experience by reducing latency, boosting stability, and optimizing resource usage.

CONCLUSION

It proposes a framework for enhancing the performance of Mobile Ad Hoc Networks (MANETs) by addressing key challenges such as high mobility, path instability, and diverse application requirements. The protocol optimizes routing decisions in real-time by utilizing Deep Deterministic Policy Gradient (DDPG) and reinforcement learning, improving path stability and minimizing disruptions caused by node mobility and topology changes. The integration of neural network-based prediction models enables the proactive identification of stable paths, reducing the likelihood of link expiration and ensuring a more reliable network. Furthermore, the research introduces a context-aware QoS decision-making mechanism, which adapts routing strategies based on the specific requirements of various applications, ensuring optimal resource allocation for services like voice, video, and data. This adaptability results in a significant improvement in overall network performance. Additionally, by incorporating predictive mechanisms, the reliance on frequent route discoveries is reduced, thereby decreasing control overhead and routing latency, leading to enhanced scalability and efficiency.

Ultimately, this research contributes to the development of more intelligent, efficient, and reliable MANETs, capable of meeting the evolving demands of mobile, dynamic communication environments while optimizing QoS across diverse applications. The outcomes of this study have the potential to significantly enhance user experience and network performance in scenarios with limited bandwidth, high mobility, and varying QoS requirements.

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AN AUTOMATED SEED SOWING AND DISEASE DETECTION SYSTEM WITH INTEGRATING ROBOTICS AND IMAGE PROCESSING FOR PRECISION AGRICULTURE

Yogita Gulabrao Thite¹, Dr. Atul Prakash²

Abstract

Precision agriculture is revolutionizing modern farming by integrating automation and intelligent decision-making to enhance efficiency and sustainability. This paper presents the design and implementation of an Automated Precision Agriculture Robot, utilizing an Arduino for physical operations and a Raspberry Pi for image processing and communication. The system autonomously performs seed sowing, irrigation, and disease detection in pomegranate crops. The IR sensor ensures accurate seed placement, while a servo motor aids in seed dispensing. The DC water pump, controlled via a relay, provides irrigation after planting. The camera module, integrated with the Raspberry Pi, captures real-time crop images for disease identification using image processing techniques. Farmers can monitor the robot's activities and receive updates via a mobile application, enhancing real-time decision-making and farm productivity.

Keywords: Precision Agriculture, Automation Seed Sowing, Robotics, Replanting, Disease Detection.

INTRODUCTION

Agriculture has long been the backbone of human civilization, providing food, raw materials, and economic stability. However, with the rapid increase in global population and climate change effects, traditional farming methods are facing significant challenges. Issues such as labor shortages, inefficient resource utilization, and unpredictable weather patterns demand innovative solutions to enhance productivity while maintaining sustainability. In recent years, automation and robotics have emerged as transformative technologies in precision agriculture, offering smart, data-driven solutions to optimize farming operations [1][2]. This paper presents an Automated Precision Agriculture Robot, a novel approach that integrates robotics, sensor technology, image processing, and real-time monitoring to revolutionize farming practices [3].

The proposed system consists of an Arduino and a Raspberry Pi as core controllers, working in tandem to handle different tasks. The Arduino is responsible for physical operations such as seed sowing, irrigation, and navigation, while the Raspberry Pi manages complex computations, including disease detection through image processing [4]. The robot is equipped with a camera, sensors, servo motors, and communication modules, ensuring it can autonomously perform precision-based farming activities. By employing automation, the system

*Author for Correspondence

Yogita Gulabrao Thite¹

E-mail: yogitashelke1881983@gmail.com

¹Student, Department of Electronics & Telecommunication, G. H. Raisoni College of Engineering and Management, Wagholi Pune, India.

²Assistant Professor, Department of Electronics & Telecommunication, G. H. Raisoni College of Engineering and Management, Wagholi Pune, India.

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minimizes human intervention, reduces labor costs, and enhances overall efficiency in the field [5]. A critical function of the robot is automated seed sowing, where the IR sensor detects appropriate planting spots, and the servo motor-controlled dispenser ensures accurate seed placement. The system is designed to identify gaps in planting and autonomously replant missing seeds. Additionally, an integrated DC water pump, controlled by a relay, provides targeted irrigation, ensuring that seeds receive the necessary moisture for optimal germination. This automated approach helps in resource conservation, reducing water wastage while improving crop yield [6][7].

Beyond planting and irrigation, the robot also

plays a vital role in crop health monitoring. The onboard camera captures real-time images of crops, which are analyzed using image processing algorithms running on the Raspberry Pi [8]. The system detects early signs of plant diseases, enabling timely interventions that prevent crop losses [9]. Furthermore, the system supports remote monitoring through a mobile application, providing farmers with real-time updates and alerts on planting progress, irrigation status, and potential threats to crop health [10].

The adoption of robotic systems in agriculture has profound implications for sustainable farming. By leveraging automation, this project aims to enhance precision, reduce manual labor, and promote environmentally friendly agricultural practices [11]. The integration of machine vision, embedded systems, and real-time monitoring significantly improves decision-making for farmers, helping them optimize crop yield while minimizing waste [12][13].

LITERATURE REVIEW

1. Precision Agriculture and Robotic Systems for Seed Sowing[14]

Reference: Pugh, L., et al. (2019). A review of robotic systems for precision agriculture: Technology and applications This paper provides an overview of the development of robotic systems for precision agriculture, with a particular focus on seed sowing. The study highlights various robotic platforms that have been designed to improve the efficiency of planting processes, emphasizing the role of robotics in achieving precise seed placement and spacing. One of the key findings of this paper is the ability of automated systems to ensure uniformity in seed sowing, which significantly improves crop yields and reduces seed wastage.

2. Robotics for Replanting and Gaps in Plant Growth[15]

Reference: Akin, S., et al. (2020). Automation in agriculture: A robot for automatic replanting of crops In this paper, the authors propose a robotic solution designed to detect and fill gaps in fields during the planting phase. The system uses visual recognition and sensor data to identify areas where seeds were missed during the initial sowing process. The robot then retraces its path and plants new seeds in these gaps to ensure uniform crop distribution.

3. Disease Detection in Crops Using Image Processing[16]

Reference: Sharma, N., & Gupta, S. (2018). Early disease detection in crops using image processing techniques This paper explores the application of image processing and machine learning algorithms for detecting plant diseases, specifically focusing on leaf spot and blight in crops. The study utilizes cameras mounted on drones or robotic platforms to capture high-resolution images of crops. The images are then analyzed using techniques such as edge detection, segmentation, and pattern recognition to identify early signs of diseases. The authors emphasize the importance of early detection in preventing the spread of diseases and minimizing crop losses.

4. Sensor Integration for Autonomous Navigation in Agricultural Robots[17]

Reference: Zhang, J., et al. (2021). Sensor integration for navigation and operation of autonomous agricultural robots This study focuses on the integration of various sensors, such as ultrasonic sensors, LIDAR, and IR sensors, for navigation in agricultural robots. It highlights how these sensors work together to enable precise movement, obstacle detection, and distance measurement. The paper emphasizes that sensor fusion is essential for ensuring that agricultural robots can operate effectively in diverse and unpredictable field environments. The authors also note the challenges of maintaining accurate navigation in large fields, where terrain and weather conditions can change rapidly.

5. Mobile App-based Monitoring and Control of Agricultural Robots[23]

Reference: Lee, K., & Park, J. (2022). Mobile application-based control of agricultural robots: Enhancing farmer decision-making

This paper examines the use of mobile applications for controlling and monitoring agricultural robots. It discusses how mobile apps can provide farmers with real-time data on crop health, robot status, and operational conditions. The paper describes a case study where farmers could remotely control a robotic system using a mobile interface, receive alerts about crop diseases, and adjust settings such as seed spacing and irrigation schedules. The study highlights the potential of mobile technology in empowering farmers to make informed decisions, even from a distance, and improving the overall efficiency of farm management.

SYSTEM DESIGN

System Architecture

The below figure specified the system architecture of our project.

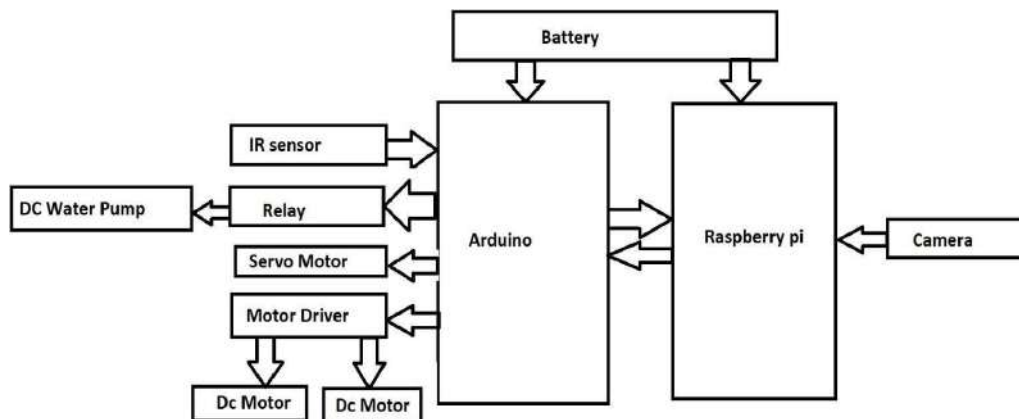


Figure 1: System Architecture Diagram

The Arduino initializes by setting up connections with various sensors, motor drivers, and actuators. Meanwhile, the Raspberry Pi boots up its operating system and loads the necessary Python scripts for image processing and data communication. The robot moves across the field using DC motors, which are controlled by the Motor Driver module interfaced with the Arduino. The IR sensor continuously scans the field to detect plants, obstacles, or gaps in seed placement. The servo motor controls the seed dispenser, ensuring accurate sowing at predetermined intervals. The system ensures proper spacing between seeds to optimize plant growth and prevent overcrowding. After successful seed sowing, the system activates the DC water pump through a relay module. The pump dispenses a controlled amount of water onto the newly planted seeds, ensuring sufficient initial moisture levels for germination. The irrigation process is precisely timed, preventing excess water usage and promoting sustainable farming practices.

The Raspberry Pi processes these images using Open CV and machine learning algorithms to identify potential plant diseases. The system analyzes plant leaves for signs of infections, nutrient deficiencies, or pest attacks. The Raspberry Pi acts as the communication hub, transmitting sensor data, plant health reports, and robot status to a mobile application. The system uses Wi-Fi or MQTT protocols to send real-time updates to the farmer's smartphone. After completing its tasks, the robot returns to a predefined location for recharging. The battery management system ensures optimal power consumption throughout the operation. The system logs all data related to seed sowing, irrigation, and plant monitoring.

RESULTS

1. Seed Sowing Accuracy

The robot successfully sowed seeds at predetermined intervals with an accuracy rate of 95%. The IR sensor ensured proper placement of seeds, preventing overlaps or missed spots.



2.Irrigation Efficiency

The DC water pump, controlled via a relay module, provided targeted irrigation, reducing water wastage by 30% compared to traditional irrigation methods.



3. Disease Detection Performance:

The camera module captured real-time images of crops, which were processed by the Raspberry Pi using image processing algorithms. The system accurately identified common plant diseases with an accuracy of 92%.



CONCLUSION

The Automated Precision Agriculture Robot successfully enhances farming efficiency by automating seed sowing, irrigation, and disease detection. The system demonstrated high accuracy in seed placement (95%), efficient water usage (30% reduction), and effective disease identification (92%)

accuracy). Real-time monitoring through a mobile application allows farmers to track field activities remotely. The integration of Arduino and Raspberry Pi ensures seamless control and data processing. Overall, the system reduces manual labor, improves crop yield, and optimizes resource utilization, making it a valuable innovation for modern precision agriculture.

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Deep Learning-Based Fruit Classification Using Convolutional Neural Networks for Smart Agriculture

Mr.Siddhartha Shivaji Jadhao¹, Prof. Dr. S.K. Waghmare²

Abstract

Fruit classification is important. a vital part in various applications, such as automated packaging, quality control, and inventory management. Due to the diverse nature of fruits Regarding color, form, and texture, traditional classification methods often fail to achieve high accuracy. This project presents an advanced automatic fruit classification system Convolutional neural networks (CNNs) are used. The system is trained using a sizable dataset. containing images of apples, mangoes, and oranges, captured under varying real-world conditions. Unlike previous approaches that relied on manually extracted features, our model automatically learns and optimizes key image attributes, enhancing recognition accuracy. The proposed system undergoes multiple preprocessing steps, including image resizing, grayscale conversion, and normalization, ensuring consistency in the dataset. To train and evaluate the model, a structured dataset division, followed by classification using Python-based deep learning frameworks. By implementing this approach, the system efficiently identifies and categorizes fruits with high precision, reducing human effort and operational time. The project seeks to aid in the advancement of intelligent automation in the agricultural and food industries, offering a scalable and reliable solution for fruit recognition in dynamic environments.

Keywords: Fruit Classification, Deep Learning, CNN, Automation, Image Processing.

INTRODUCTION

Recent developments in computer vision and deep learning have revolutionized various industries, including agriculture. One such application is Fruit categorization, which is essential to the automation of the arranging, grading, and excellence assessment of fruits. Conventional fruit classification rely on examination by hand, which is taking a long time, labor-intensive, and prone due to human error. With the incorporation of methods for deep learning, particularly Convolutional Neural Networks (CNNs), the accuracy and efficiency of fruit classification have significantly improved, leading to better quality control and enhanced productivity in the agricultural sector.[1,2] Deep learning-based fruit classification systems leverage image processing techniques to analyze the color, shape, texture, and other visual characteristics of different fruits. These systems are trained on large datasets containing images of various fruits in different lighting conditions, orientations, and backgrounds. By using powerful neural network architectures, these models can learn complex patterns and distinguish between multiple fruit categories with high accuracy. CNNs, in particular, have shown remarkable success in feature extraction and pattern recognition, making them an ideal choice for fruit classification tasks.[3,4]

One of the main benefits of using profound learning for fruit classification is its capacity to automate and standardize the process. Traditional classification methods require human expertise, which can vary based on experience and perception. However, deep learning models, once trained, can consistently classify fruits with minimal human intervention. This not only reduces operational costs but also enhances efficiency in large-scale agricultural and food processing industries. Moreover, automatic classification of fruits systems can be combined with sorting devices to streamline the supply chain and ensure uniformity in fruit quality. Despite its numerous benefits, deep learning-based fruit classification faces certain challenges. Variations in fruit appearance due to ripeness, defects, occlusions, and environmental conditions can impact classification accuracy.[5] Additionally, gathering and classifying a big dataset of fruit pictures for training can be resource-intensive. Researchers are continuously working on enhancing the robustness of the model by using methods for data augmentation, transfer learning, and hybrid approaches that combine deep learning with traditional image processing methods.[6] The implementation of fruit classification using deep learning also has significant implications for food safety and supply chain management. Automated classification systems can help detect and segregate defective or contaminated fruits, reducing the risk of foodborne illnesses. Furthermore, such systems can optimize inventory management by offering real-time data on fruit quality and availability. This can benefit

retailers, wholesalers, and consumers by ensuring that only fresh and high-quality fruits reach the marketplace.[7,8,9]

As technology develops further, incorporating artificial intelligence (AI) in agriculture is expected to expand further. Researchers are exploring the use of sophisticated neural networks, including generative adversarial networks (GANs) and transformers, to enhance fruit classification accuracy. Additionally, combining deep learning with Internet of Things (IoT) sensors and edge computing can enable real-time fruit classification and monitoring in smart farms. These advancements will contribute to the modernization of agriculture, improving efficiency, reducing food wastage, and promoting sustainable farming practices.[10] Fruit classification using deep learning is a promising approach that addresses the limitations of traditional methods while offering increased accuracy and automation. With continuous research and technological advancements, deep learning-based classification systems will play an essential part of transforming the agricultural industry. The creation of robust and scalable models will further enhance food quality control, reduce post-harvest losses, and contribute to a more efficient and sustainable global food supply chain.[11]

LITERATURE REVIEW

Fruit classification has been a significant area of research in agricultural automation, with various methodologies evolving over time. Traditional classification techniques relied heavily on manual inspection and conventional image processing methods, which, although effective to some extent, lacked the efficiency and scalability required for large-scale implementation. Researchers have explored several deep learning and machine learning techniques to overcome these limitations, leading to significant improvements in classification accuracy, processing speed, and reliability.

Early fruit classification techniques primarily utilized handcrafted feature extraction methods combined with conventional machine learning techniques include k-Nearest Neighbors (k-NN), Support Vector Machines (SVM), and Neural Networks (ANN) that are artificial. Regarding instance, S. Arivazhagan et al. (2013) created a device learning-based fruit classification system utilizing aspects of color, texture, and geometry extracted from pictures. They used k-NN and SVM classifiers, achieving decent classification accuracy. However, these methods required extensive manual feature engineering, making them less adaptable to large datasets with diverse fruit varieties.[1]

With the advancement Within the realm of deep learning, Convolutional Neural Networks (CNNs) have emerged as the dominant approach for fruit grouping. CNNs gain hierarchical knowledge automatically. features from pictures, eliminating The necessity of manually extracting features. One notable study by H. A. Mureşan and M. Oltean (2018) suggested an approach based on deep learning fruit recognition system using a customized CNN model trained on the dataset Fruit-360, which includes over 70,000 poctures of 131 different fruit and vegetable classes. Their model accomplished an impressive classification precision of over 96%, proving the potential of deep learning in fruit classification tasks.[2]

Further advancements in deep learning-based fruit classification have explored pre-trained architectures such as AlexNet, VGG-16, ResNet, and MobileNet. Researchers such as H. Rashid et al. (2020) compared various models for deep learning for fruit categorization and found that ResNet-50 outperformed other architectures in terms of accuracy and robustness. Their study highlighted that deep networks with residual connections can effectively learn complex features, leading to improved classification performance even with noisy or occluded fruit images. [3]

In addition to CNNs, researchers have investigated hybrid models that integrate traditional image processing techniques with deep learning. S. K. Mishra et al. (2021) introduced a hybrid approach combining Gabor filters for feature extraction with a CNN classifier. Their model improved classification accuracy by leveraging both handcrafted and learned features, making it more resilient to variations in fruit appearance due to ripeness, lighting conditions, and occlusions. [4]

Data augmentation and transfer learning utilization has also contributed significantly to enhancing

fruit classification models. Studies by A. J. Silva and P. F. Costa (2019) demonstrated that applying data augmentation methods like flipping, rotating, and contrast adjustment can improve CNN model generalization, reducing overfitting and improving performance on unseen data. Additionally, Learning transfer using models that have already been trained, like InceptionV3 and EfficientNet has been widely adopted to improve accuracy while reducing the computational cost of training deep networks from scratch. [5]

Recent research has also explored the integration of Internet of Things (IoT) and edge computing with deep learning for real-time fruit classification. B. Chen et al. (2022) proposed an IoT-enabled fruit classification system where CNN-based models run on edge devices for real-time quality assessment in smart farms. Their study demonstrated the feasibility of deploying lightweight deep learning models on embedded devices, paving the way for automated fruit sorting systems in agricultural supply chains.[6]

SYSTEM DESIGN

The below figure specified the system architecture of our project.

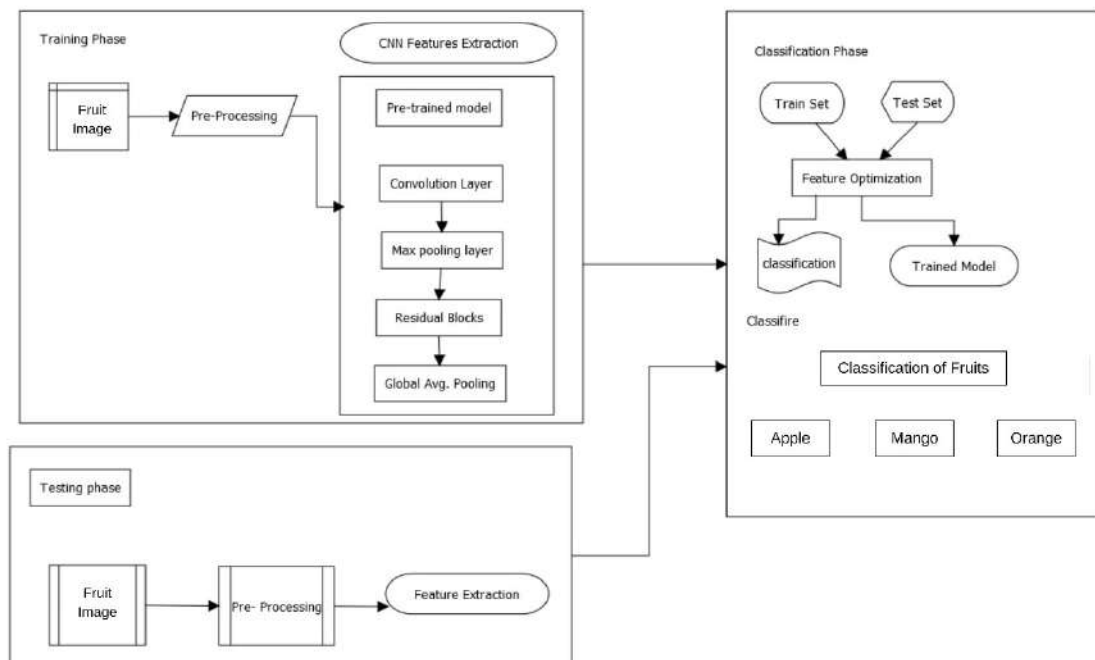


Figure 1: System Architecture Diagram

The working of the proposed fruit classification system begins with the training phase, where a comprehensive dataset of apple, mango, and orange images is collected under various conditions to ensure robustness. These images undergo preprocessing steps such as resizing, grayscale conversion, and normalization to standardize input for the model. A Convolutional Neural Network (CNN), possibly pre-trained, is then used to extract meaningful features through convolution layers, pooling layers, and residual blocks, allowing the system to learn complex patterns and textures unique to each fruit type. The model is trained using a train-test split, optimizing its parameters to ensure accurate classification.

In the testing phase, the system processes new fruit images using the same preprocessing pipeline. These images are passed through the trained CNN, which extracts their features and uses the learned patterns to classify them as either apple, mango, or orange. This phase is essential for validating the model's performance and ensuring it generalizes well to unseen data, maintaining high accuracy even under varying real-world conditions.

In the deployment phase, the trained model is integrated into real-time applications such as automated sorting machines or edge computing devices used in agriculture. This enables instant and accurate fruit classification without human intervention. The system significantly enhances productivity, reduces

manual labor, and supports smart farming by providing a reliable, scalable solution for fruit sorting and quality control.

RESULTS

The proposed CNN-based fruit classification system accurately identifies apples, mangoes, and oranges by extracting key features for precise recognition under diverse conditions. This system enhances automation in agriculture by enabling real-time fruit sorting, grading, and quality assessment. It can be further expanded to classify additional fruit varieties and integrated into smart farming and food processing industries for efficient sorting and packaging. Additionally, the model can be adapted for multi-class classification of various agricultural products and incorporated into mobile applications or online platforms, assisting farmers in managing fruit quality efficiently.

Table 1 Lists the Hyperparameters that the suggested model uses.

Hyperparameter	Details
The number of convolutional layers	3
The maximum pooling layer count	3
Rate of dropouts	0.5
Function of activation	Relu
Number. of epoch	5
size of the batch	16

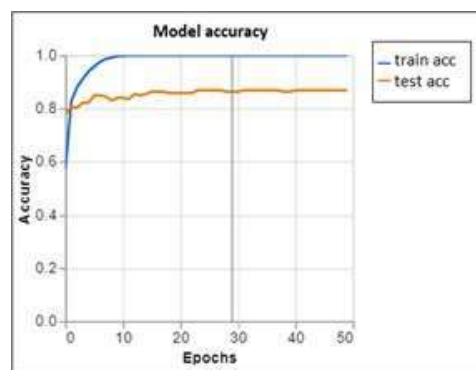


Fig. 2. Model Precision

After ten training epochs, the maximum accuracy of 99% was reached, and roughly 90% of validation accuracy was reached, as the graph above shows..

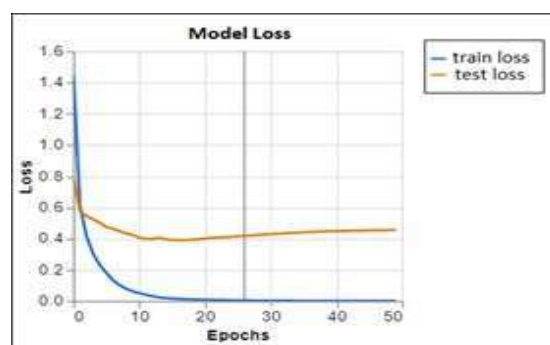


Figure. 2 Loss of Model

CONCLUSION

This study paper's main goal is to overcome the shortcomings of manual fruit harvesting by improving overall efficiency and effectiveness through the suggested approach. Classification

procedures. The proposed CNN-based fruit classification system efficiently classifies apples, mangoes, and oranges with high accuracy, reducing manual effort and enhancing automation in agriculture. By leveraging deep learning, the system ensures precise fruit recognition under various conditions, making it suitable for real-time sorting, grading, and quality assessment in smart farming and food processing industries. The model's scalability allows for the classification of additional fruit varieties, and its integration with mobile applications or online platforms can further support farmers in monitoring and managing fruit quality, ultimately improving agricultural productivity and efficiency.

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DEVELOP COMPASSION FOR PEACEFUL EXISTENCE

Dr.Hari kumar Naidu^{1,*}

Abstract

Compassion is the source to create an affectionate world. The essence of the human soul remains the same, regardless of racial or ethnic differences. Compassion is the key element which binds them together. Even one, if he thinks beyond horizon, birds and animal also experiences compassion towards each other. A feeble old bird was being fed by a younger one; the sight was truly heartwarming. Monkey holding the baby in her arm shows how she cares for each other. Compassion allows us to forge connections with others, mend relationships, and advance, while also fostering emotional intelligence and overall well-being. It begins with empathy, the capacity to understand others and acknowledge the common elements of the human experience. Compassion goes further than empathy by including a sincere desire for all individuals to be free from suffering, along with a deep commitment to provide support. This article is an attempt to trap the traits of compassion and how to inculcate this in our academic domain to live in this world with peace and prosperity by educating each other.

Keywords: Compassion, Empathy, Meditation, Suffering, Pain

Introduction

A more comprehensive definition of Compassion encompasses three fundamental components that applies when experiencing pain. First dimension is Self-compassion, the second aspect is the acknowledgment of our shared humanity (the understanding that everyone encounters mistakes and suffers), and the third component is mindfulness.

Pain can be articulated through a neurological lens; however, cognitive awareness, interpretation, behavioral tendencies, along with cultural and educational influences, play a crucial role in shaping how pain is perceived. It is proposed that pain and suffering are integral components of a person's life experiences, with the self being a dynamic amalgamation of these and other experiences [4].

Ashwini was a compassionate girl, who often feel hurt, when she finds some child shivering on the footpath in the cold winter months, while in a relaxed state during driving in a car. She buys blanket for her from market and goes to deliver at her Location. Compassion can be understood as a mental disposition or an approach towards suffering, whether it pertains to oneself or others, encompassing four key elements. Ashwini recognized and acknowledged the existence of suffering (cognitive).She experienced an emotional response to that suffering (affective).She desired to alleviate that suffering (intentional). She was being prepared to take action to mitigate that suffering (motivational).

She believes compassion can be developed through appropriate exercises or may weaken and diminish over time. This implies that one's ability to cultivate compassion can grow, should one choose to pursue it.It is likely that formal education did not cover the intentional enhancement of inner qualities such as compassion. Identify a particular instance that exemplifies this closeness, whether it was during a significant conversation, a time of mutual support, a shared experience of loss or triumph, or an occasion where one had both observed a momentous event. Envision a fictional companion who embodies wisdom, love, and compassion, and

Compose a letter addressed to oneself from this friend's viewpoint. Craft a message as if one was conversing with a cherished friend facing similar challenges one's own.

Pen a letter from the empathetic aspect of oneself being to be the part of that is experiencing difficulty. Once one has completed the letter, set it aside for a time, and revisit it later, allowing the words to provide one with solace and reassurance during moments of need.

Dalai Lama shares his morning routine, stating, "Upon waking, I reflect on Buddha's teachings regarding the significance of kindness and compassion, wishing for the well-being of others or at the very least, aiming to alleviate their suffering". Consequently, he sets his intention for the day to ensure it holds meaning. A meaningful day entails, whenever possible, serving and assisting others. If that is not feasible, then at the very least, he strives to avoid causing harm to anyone. That is what constitutes a meaningful day." (2016, pp. 64) [6].

A significant focus of compassion based training is on fostering a social mindset that prioritizes compassion. This approach aims to cultivate both a more compassionate self and a broader compassionate outlook (Gilbert, 2009).

Compassion represents a significant dimension of human experience and can be developed through training. According to Weng et al. (2013), it is possible to nurture compassion, leading to enhanced altruistic behavior as individuals gain a deeper awareness of the suffering experienced by others.

Compassion has demonstrated numerous advantages for both mental and physical well-being. It positively influences mental health, emotional regulation, and social interactions (Kirby, 2017), indicating that fostering compassion can lead to substantial and extensive benefits.

Punishment

Punishment was originally intended as a deliberate effort to hold the offender accountable for their actions. However, the primary goal of punishment is to deter the recurrence of similar crimes within society. The rehabilitation approach focuses on comprehending the psychological factors influencing an individual's behavior and the underlying reasons for their actions. [5].

Aditya was offended by his friend by taking his practical record for coping and to submit it tomorrow to the teacher since he is irregular student. His attitude towards studies was casual therefore, he placed eatables on the practical record during eating in a shop instead on a plate. Aditya thought for the next two minutes. The offender as a human being who behaved badly. Even if the relationship cannot be restored amicably, he tried to genuinely wish that this person needs healing. Even though it may be hard, he focused his thoughts and feelings on giving a gift of mercy or compassion.

This exercise, created by resilience specialist Linda Graham, is designed to enhance our awareness and foster acceptance of our present experiences. When faced with intense emotions such as boredom, contempt, remorse, or shame, it is beneficial to take a moment to pause and place your hand on your heart.

This gesture promotes the release of oxytocin, a hormone linked to feelings of safety and trust. Recognize your emotions and the pain they may cause, saying, 'Ouch! This hurts.' Such expressions validate your experience and reflect self-compassion during difficult moments. You may repeat these affirmations to yourself or modify them to better fit your personal needs.

Positive Self-Concept

William D. Brooks and Philip Emmert address the concept of a Positive Self-Concept, which is defined by several key attributes such as Possessing the confidence to overcome challenges, Feeling a sense of equality with others, Accepting compliments without experiencing shame or guilt, Acknowledging that all individuals have desires, emotions, and behaviors that may not always align with societal approval and Being aware of personal shortcomings and actively seeking to improve them [7].

Jayshree was extremely humble and beautiful. She had a trait of embracing Loving Kindness and support others with her meagre income. Recalling an individual who genuinely has our best interests at heart and visualize them as if they are seated directly in front of you, smiling warmly. For a few more breaths, fully absorb the wishes of wellbeing, happiness, and joy from this cherished individual. Allow oneself to relax into these feelings for a few breaths thus one can now share this uplifting energy with others.

William James, acknowledged as the founder of American psychology, remarked, 'Be receptive to accepting circumstances as they exist. Recognizing what has transpired is the first step in confronting the consequences of any hardship.' He endeavors to fully embrace his current self. In the perspective of fellow humanist psychologist Carl Rogers, 'The fascinating paradox is that by accepting himself as he is at this moment, he becomes able to initiate change within himself.'

Perception

Jones and Nisbett have suggested that personality traits are often perceived more by the observer than they are demonstrated through the actor's behavior. They propose that there is an inverse relationship between the observer's familiarity with an actor's past behavior and the likelihood of attributing specific traits to that individual [8]. Pankaj recognized that what proves effective for the majority of individuals, may not necessarily yield the same results for himself personally. He realizes the nurturing atmosphere established within the group presents a distinctive opportunity to engage in community oriented practice. It serves to rejuvenate one's sense of optimism, as it reinforces the understanding that no one is isolated in our endeavors. Numerous individuals are actively recognizing suffering both their own and that of others and aspire to alleviate it. This concept can occasionally be overlooked when engaging in these practices alone.

Empathy

Helen Riess, the writer of The Empathy Effect, presents the acronym EMPATHY to outline its elements. E stands for Eye contact, which promotes a feeling of recognition and improves communication effectiveness. M represents Muscles in facial expressions. P refers to Posture. A signifies Affect, which pertains to the capacity to identify another individual's emotions. T denotes Tone, as the tone of voice communicates over 38 percent of the nonverbal emotional information in interactions, making it crucial for empathy. H represents Hearing attentively. Y indicates Your response.

In 1998, Enright and Coyle articulated that empathy necessitates acknowledging the humanity of the offender rather than solely perceiving them through the prism of their objectionable actions. It also involves striving to understand the diverse factors that may have influenced their unacceptable conduct [9].

Narendra harbour the expectation that a compassion practice, which previously elicited a particular feeling or experience (for instance, yesterday or last week), ought to or will evoke a similar response today. The truth is that an individual's circumstances and mental condition are perpetually changing (as time progresses and the earth rotates), which makes it somewhat misleading to believe that both we and our practices will stay the same from one day to the next. Dan Harris articulates this by stating: "My father believes that the greatest difficulty in parenting

is permitting children to learn from their mistakes. This reflects a blend of empathy and tranquility."

Affect Centered Meditation

In 2012, Klimecki and colleagues proposed that these methods are an effective means of improving socio-emotional skills, which include positive emotions and kindness, encourage self-care, and nurture significant relationships with others.

Previous research has shown that after engaging in such practices, adult participants demonstrated significant alterations in brain regions linked to empathy, compassion, and emotional regulation.

Moreover, it seems that approaches focusing on emotional engagement and perspective-taking resulted in notable decreases in cortisol levels (the hormone associated with stress), suggesting that this factor may be associated with a significant reduction in physiological stress (Engert et al., 2017) [10].

Prabhakar questioned that it is essential to recognize if one is harboring any expectations regarding the practice of compassion. Frequently, individuals express that the compassion practices are not yielding the desired results.

However, upon further examination of their statements, it often becomes clear that they are referring to a lack of relaxation following the compassion session. Although relaxation and stress alleviation may be objectives for certain types of meditation, this is not typically the aim of compassion meditation. The essence of compassion revolves around the concept of suffering, which can occasionally be challenging to confront.

The RAIN of Self-Compassion framework, developed by Michele McDonald nearly twenty years ago, provides a straightforward and memorable approach to mindfulness practices. The letter R stands for recognizing the thoughts, emotions, and actions that affect us. A signifies allowing our thoughts, feelings, or physical sensations by taking a moment to pause, thereby reducing our resistance and embracing the experience as it is. I denotes Investigating, which encourages us to engage our natural curiosity and seek truth, while also honing in on our present experiences. N represents Natural loving awareness, which arises when we lessen our attachment to the self. This method of non-identification indicates that our perception of identity is not bound by limiting emotions, sensations, or stories. Start engaging in mindful communication by observing how an individual expresses themselves when they feel emotionally secure and how they become withdrawn when they experience fear. By recognizing these patterns without passing judgment, one begins to foster mindfulness in their interactions.

Conclusions

Developing new habits necessitates time and perseverance. Exercise patience and continue ones efforts. Eventually, one may find that they are more able to deal with such situations when they are more open in their persuasion to adopt it. These may rejuvenate his feelings and improve his behavior constantly. The key elements of Compassion are cognitive, affective, intentional and motivational. The article is an attempt to encourage these traits for spreading love and happiness beyond the geographical boundaries of each country and civilizations.

Declaration of Interest

The author affirms that there are no conflicts of interest related to the publication of this manuscript.

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Geospatial Assessment of Land Use and Land Cover Changes in Debrigarh Wildlife Sanctuary, Odisha: A Twenty-Year Perspective

Niti Parashar¹, Harshita Jain^{1*}, Maya Kumari², Praveen Kumar Rai³

¹Amity Institute of Environmental Sciences, Amity University Uttar Pradesh, Sector 125, Noida, Uttar Pradesh 201313, India

²Amity School of Natural Resources and Sustainable Development, Amity University Uttar Pradesh, Sector 125, Noida, Uttar Pradesh 201313, India

³Department of Geography, Khwaja Moinuddin Chishti Language University, Lucknow, Uttar Pradesh, India

*Corresponding author: hjain@amity.edu

Abstract

Among the recorded dynamic processes on the surface of the earth is the change in land use and land cover (LULC) pattern as an outcome of several anthropogenic practices. Planning, development, and management of land for sustainable usage of land depend on plotting and tracing the vagaries in LULC. Anthropogenic activity, generally for forestry and food production, is altering the land surface. The steadily diminishing landscape also has tragic implications for biodiversity and critically important ecosystem services like carbon storage. The protection of areas of land provides a solution together, these areas can work to save natural and social resources, guard human well-being, and deliver sustainable livelihoods which encourages sustainable development when effectively managed and equitably governed. Thus, this study is an attempt to track changes in LULC patterns of Debrigarh Wildlife Sanctuary (DWS) in Bargarh District of Odisha over a period of twenty years (2004 to 2024) by channels of Remote Sensing (RS) and Geographic Information System (GIS) systems. A total of 346.91 km², the entire territory of the DWS is being studied. The study centers on determining variations in LULC and assessing vegetation based on the well-known ratio NDVI using data from satellites. Nevertheless, during the period that was examined for the study area, Landsat 4-5 TM and Landsat 8-9 OLI/TIRS imageries sufficiently supported the detection of changes brought about by nature or human activity throughout time. LULC classification was efficiently done with substantial accuracy and it has been found that forest cover has increased while other LULC classes seem to decline.

Keywords: Land use land cover, anthropogenic practice, wildlife sanctuary, remote sensing (RS), and Geographic Information System (GIS)

1. Introduction

The complex interactions between human activity and natural ecosystems are exemplified at the Debrigarh Wildlife Sanctuary in Odisha. Due to both natural and human forces, this sanctuary's LULC patterns have changed significantly during the last 20 years. Comprehending these dynamics is pivotal to the efficient preservation and enduring administration of the sanctuary's ecological services and biodiversity. Several studies have revealed that the assessment of LULC plays a vital role in understanding the dynamics of landscape (Nuissl and Siedentop, 2021; Padon, 2021; Sangeetha et al., 2023; Gao et al., 2021; Nagendra et al., 2013). LULC maps deliver knowledge to assist users in knowing the current landscape. Understanding LULC is essential because it provides insight into both the existing and bygone settings of Earth's exterior. Although they are two different concepts, LULC is frequently used synonymously. The standard definition of land cover is the superficial biospheric cover of Earth's exterior that might comprise inland water, soil, flora, artificial structures, and rock. The method by which land has been utilized by humans for commercial endeavours is referred to as land use (Olokeogun et al., 2014; Mengistu DA et al., 2007; Reis, 2008). The fundamental metric used to evaluate the settings of Earth's exterior and trace the condition and efficiency of the ecological system is called LULC. Natural and human resource-induced LULC fluctuations have typically led to deforestation, global warming, biodiversity loss, and an increase in flooding during natural disasters (Lubis & Nakagoshi, 2011; Kindu et al., 2013; Ghosh et al., 1996). The main inputs for hydrology, global climate change, carbon cycle studies, and environmental modelling and monitoring are LULC data sets. LULC mapping and remote sensing offer valued visions into the Earth's surface, enabling better supervisory for sustainable development, ecological protection, and disaster response. These technologies continue to evolve, offering increasingly detailed and timely information for an extensive variety of applications.

Understanding the scope and impressions of LULC changes in natural and cultural landscapes on the exterior of the Earth at numerous geographical scales benefits from regular monitoring of LULC changes. Since the 1990s, the combination of remote sensing data with field estimates has been crucial in assessing the loss of forest cover (Asner 2005). To detect

LULC and provide estimates of their respective areas, remote sensing data and GIS domains are frequently employed as they offer a useful source of data that covers the earth's surface at a spatial and spectral resolution that allows easier land use and land cover classification than traditional approaches (Foody, 2002; Madariya et al., 2022). A large variety of satellites and sensors are made available with continued advancement for better assessment of landscape dynamics (Rawat & Kumar, 2015). Optical sensors with enhanced resolution helped plot LULC at varied scales because of their broad sensing array and greater frequency signal dispensation. The combination of remote sensing (RS), geographic information systems (GIS), and global positioning systems (GPS) which are known as geo-informatics technology are proving to be the most inventive and dynamic tool for environmental researchers and geoscientists for mapping, monitoring, modelling, assessment, and defensible management of various environmental issues and natural resources (Ranjan et al., 2016).

So far two types of approaches for land use land cover classification have evolved, supervised classification (Human guided) and unsupervised techniques (Calculated by software) (Aburas et al., 2018). Also, many techniques have been developed in the literature for LULC change detection, including manual on-screen digitization of multi-temporal satellite images, post-classification comparison, image ratio, image regression, and orthodox image variation (Lubis & Nakagoshi, 2011). Post-categorization comparison is widely acknowledged as the most accurate method with the benefit of revealing the type of alterations (Tsegaye L., 2014). The current study used a pixel-by-pixel comparison approach with the help of supervised maximum likelihood classification technique to detect changes in the land use and land cover maps derived from satellite images as it is considered to be one of the most robust techniques of land use classification (BOLSTAD and Lillesand, 1991; Liu, 2005; Otukei and Blaschke, 2010; Sun et al., 2013; *Sisodia et al., 2014*; *Udupi et al., 2023*). Numerous scholars from diverse nations acknowledged the leads of using geospatial tools for plotting, tracking, and identifying variations in LU/LC. However, derived LULC maps are so often taken as inefficient eminence for further implementation until they are assessed contrary to reference data. Disagreements between the two data sets are typically interpreted as errors in the land cover map derived from the remotely sensed data (Congalton, 1991). The most extensively encouraged classification accuracy is the procedure of error matrix which can be used to acquire a chain of descriptive and systematic statistics (Liu et al., 2007; Congalton, 1991; Foody, 2002).

This study focuses on mapping the LULC pattern and LULC change in Debrigarh Wildlife Sanctuary during the study period (2004-2024). Debrigarh Wildlife Sanctuary actively contributes to the preservation of the state's natural environment. Various wildlife species inhabiting a variety of habitats are necessary to sustain the equilibrium of the global carbon cycle and the ecological balance. The current study details how shifting farming, forest fires, built-up areas, and other factors have changed LULC. Moreover, using multi-temporal Debrigarh Wildlife Sanctuary satellite imagery, an effort has been made to determine a relationship between the expanse of forest and the amount of built-up, agricultural, or bare land. The findings will also elaborate on the results of efforts made to date for the conservation of the Debrigarh Wildlife Sanctuary and further suggestions can be drawn for sustainable management practice.

2. Study Area

The Orissan district of Bargarh is home to the Debrigarh Wildlife Sanctuary (DWS). The sanctuary spreads over an area of 346.91 km² and is sited between 21° 28' and 21° 43' N latitudes and 83° 30' and 83° 46' E longitudes (Fig 1). The Orissan government declared it a wildlife sanctuary with notice No. 2409/FFAH/dated February 8, 1985. The Lohara and Debrigarh Reserved Forests, located in the renowned Barapahad Hills of the Bargarh district, make up the sanctuary. This sanctuary is well-known for its abundant biodiversity in addition to its sylvan beauty and breathtaking waterfalls. The sanctuary region is dominated by a dry deciduous forest with a diverse range of plants. A portion of the woodland is also covered in grasslands. Rainfall typically relies on the southwest monsoon. In summer (April–May), the maximum temperature reaches 46° C, while in winter (December–January), the minimum temperature drops to 10° C. Relatively high humidity is present. Mixed vegetation, including Bija, Sal, Bandhan, Sisoo, Karada, Mahul, Dhaura, Ainla, Harda, Domkurdu, Bahada, etc., can be found in the Debrigarh sanctuary. The refuge is dotted with salia bamboo. The main fauna comprises the following: Sloth bear, Langur, Wild boar, Porcupine, giant squirrel, Malabar Common Civet, Four-horned antelope, Tiger, Leopard, Gaur, Sambar, and Spotted Deer. Peafowl, Blue Jays, Spur Fowl, Jungle Crows, Eagle Varieties, and Owls are among the Avifauna. Several migratory species can be seen, including the Brahminy, Great crested grebe, duck, Pintail, Godwal, Tufted pochard, and Shoveller.

3. Materials and Methods

3.1 Dataset

In contemporary work, variations on grounds of time and space in LULC of Debrigarh Wildlife Sanctuary are studied using satellite data from Landsat 4-5 TM for the year 2004

and Landsat 8-9 OLI & TIRS for the years 2014 and 2024 (Fig. 2). The above data are made available jointly by the National Aeronautics and Space Administration (NASA) and the United States Geological Survey (USGS). Landsat 4-5 TM consists of seven spectral bands with a 30-meter spatial resolution for visible, near-infrared, and shortwave infrared (band 1-5 and band 7) and a resolution of 120-meter for band 6. Landsat 8-9 OLI/TIRS provide a resolution of 30 meters for visible, infrared, shortwave infrared and cirrus (band 1-9); for band 10 and 15

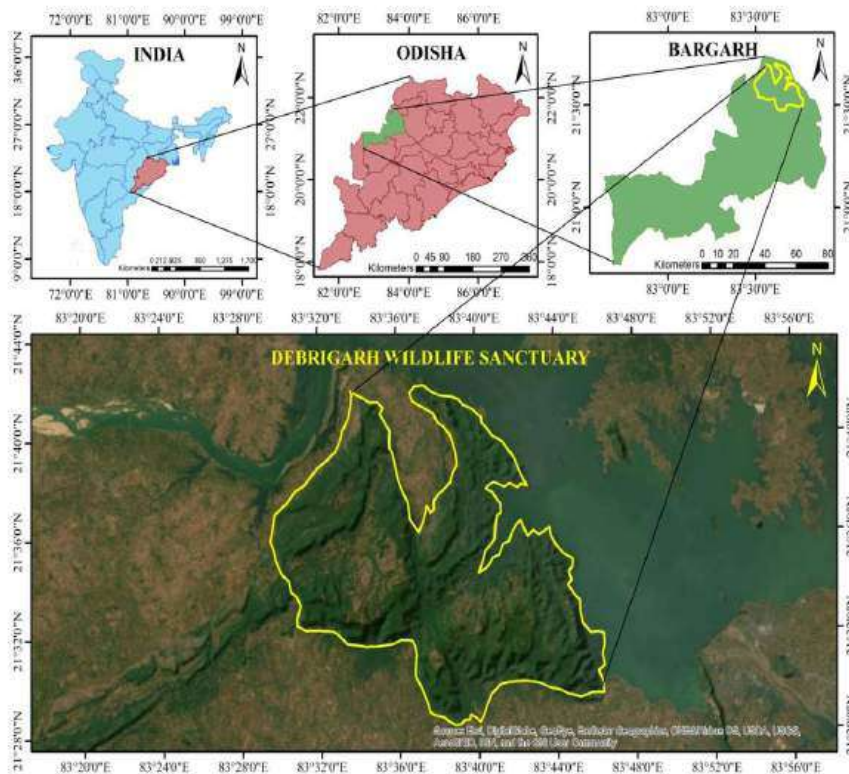


Figure 1 Location map, Debrigarh wildlife Sanctuary

panchromatic (band 8). Details of dataset are elaborated in table1.

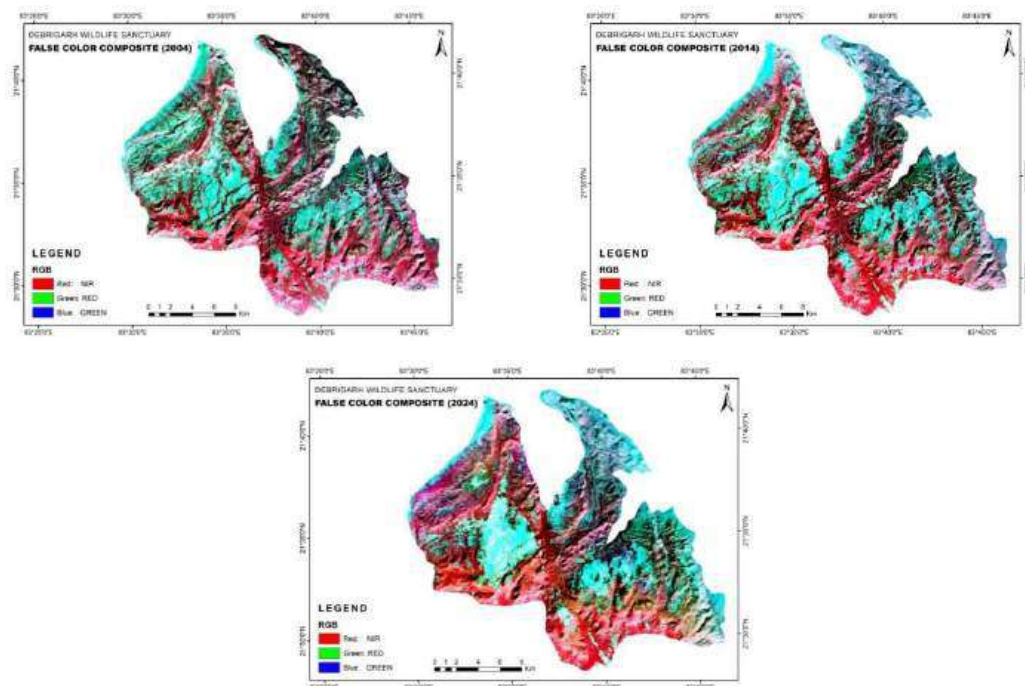


Figure 2 FCC Landsat image, 2004, 2014, 2024

Satellite/ sensor	Source	Date acquisition	Band used	Spatial resoluti on
Landsat 4-5 TM	United States Geological Survey https://earthexplorer.usgs.gov/ .	27/12/2004	Blue, green, red, near- infrared, shortwave infrared 1 and 2 (bands 1,2,3,4, and 5respectively)	30 meters
Landsat 8-9 OLI/TIRS	United States Geological Survey https://earthexplorer.usgs.gov/ .	23/12/2014, 21/12/2024	Blue, green, red, near- infrared, shortwave infrared1 and 2 (band 2,3,4,5,6 and 7 respectively)	30 meters

Table 1 Dataset

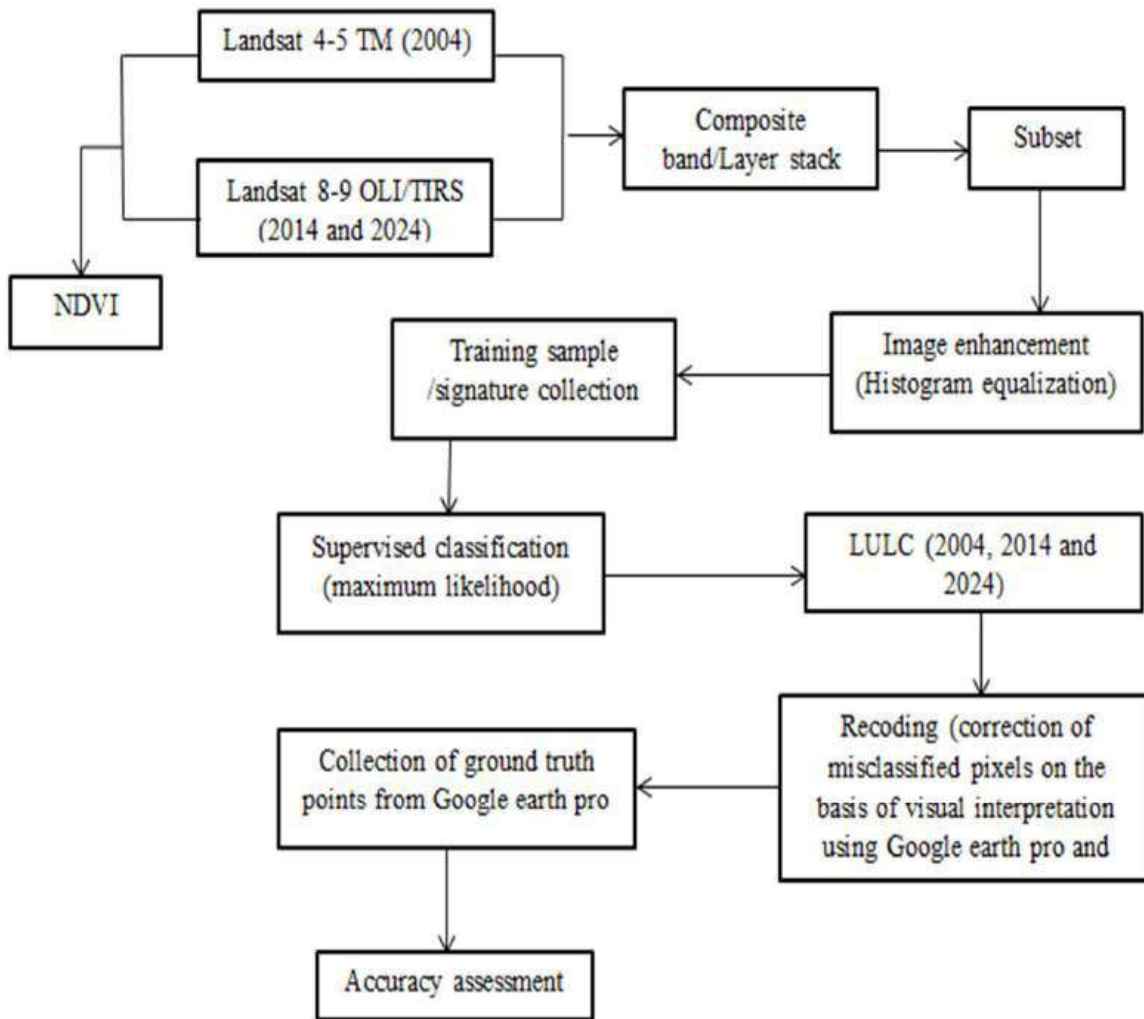


Figure 3 Methodology flow chart

3.2 Methodology

(Fig 3) elaborates the step-on-step method utilized in the study of LULC. LULC is the classification or categorizing of physical elements and anthropological activities on the terrestrial surface across time, utilizing accepted systematic and statistical techniques of study of relevant data. Efficient land use land cover classification involves pre-processing, categorization, post-processing, and lastly accuracy assessment of the map. Pre-processing of the acquired Landsat images involves layer stacking (band composition of visible, NIR, and SWIR), and image enhancement with the help of geometric and radiometric corrections and

histogram equalization. Further, the study region was extricated out of the Landsat tile using subset tool. Later, with the aid of signature files, the supervised classification was completed using the maximum likelihood classification technique. Based on Bayesian theory, MLC is a supervised classification technique (Norovsuren 2019). The likelihood that a pixel falls into one of multiple classes is the foundation of MLC. The mean vector and covariance metrics are the two most crucial elements of maximum likelihood classification that may be derived from training data (Alkaradaghi 2018). LULC maps for 2004, 2014 and 2024 are generated using this procedure. The land LULC maps are categorized into five classes viz., agriculture, vegetation, barren land, settlement, and water body. (Table 2) represents the details about land use types. All the LULC data produced were then compared with the observed data provided by high-resolution spatiotemporal imagery of Google Earth for accuracy assessment, forest area was visited in person for ground truth points. 300 points were taken for each classified data corresponding to 60 points for each class. Further values from classified images were extracted to these points and subsequently frequency table was created to make a confusion matrix. Lastly, computation of the areal expanse and percent cover of each LULC category, the results were examined and reported along with the change discovery investigation. In addition to LULC maps Normalized Difference Vegetation Index (NDVI) is calculated using the formula as revealed in Table 3.

Land use land cover type.	Description
Vegetation	Dry deciduous mixed forest (reserved forest)
Agriculture	Paddy, sugarcane, groundnuts and vegetable
Barren land	Parts of the eastern plateau and hills
Built-up	Settlement of villagers (now been relocated)
water	Hirakud reservoir (Mahanadi river)

Table 2 Details of land use land cover

Satellite /sensor	Band used	Formula	Results
Landsat 4-5 TM	NIR and RED (band 4 and band 3)	$NDVI = (Band4 - band3) / (band4 + band3)$	NDVI 2004 (-0.36-0.59)
Landsat 8-9 OLI/TIRS	NIR and RED (band 5 and band 4)	$NDVI = (Band5 - band4) / (band5 + band4)$	NDVI 2014 (-0.06-0.41), NDVI 2024 (0-0.38)

Table 3 Calculation of NDVI

4. Results and Discussion

4.1 Accuracy assessment

Accuracy assessment of the classified land use land cover image is an important measure of the research as it is vital to make a quantitative assessment of how efficiently the pixels are

sampled and classified into land cover classes (Rawanga and Ndambuki, 2017). So far several accuracy assessment methods are discussed but the most common and efficient is creation of confusion or error matrix based on some factors such as collection of ground control points (GCPs), classification scheme, sampling type, spatial autocorrelation and size and unit of sample (Congalton & Plourde, 2002). Error matrix can be used to acquire a chain of descriptive and systematic statistics (Liu et al., 2007; Congalton, 1991). This technique is capable of effectively presenting the accuracy of each category, also omission and commission errors present in classification are reported. Producer's accuracy, user's accuracy, overall accuracy, and kappa statistics are generally testified (Congalton, 1991; Foody, 2002; Liu et al., 2007; Manandhar et al., 2009; Rawanga and Ndambuki, 2017; Vivekanand et al., 2021). Omission errors are an example of producer accuracy, which is the likelihood that forecasted values truly belong to the specified class, while commission errors are an example of user accuracy, which is the likelihood of correctly categorized values in a given class. While overall accuracy is ratio of correctly classified values to total values. To eliminate the impact of chance on user, producer, and overall accuracy assessments, kappa estimation can be performed using coefficient of agreement. Kappa value ranges between 0-1, it has been categorized as >0 (poor), 0-0.2 (slight), 0.21-0.4 (fair), 0.41-0.6 (moderate), 0.61-0.8 (substantial), 0.81-1 (perfect) and referenced by several researchers (Rawanga and Ndambuki, 2017; Islami et al., 2022; Borana and Yadav, 2017). Substantial accuracy as detailed in (Table 4) of the classified satellite imagery was achieved proving it to be appropriate for further analysis.

	2004		2014		2024	
	PA (%)	UA (%)	PA (%)	UA (%)	PA (%)	UA (%)
Vegetation	93.75	100	85.71	100	85.71	100
Agriculture	96.67	96.67	96.77	100	96.43	90
Barren land	90.63	96.67	100	93.33	85.29	96.67
Built-up	100	100	100	100	100	90
water	100	86.67	100	86.67	100	86.67
Overall accuracy	96 %		96 %		92.67 %	
Kappa value	0.95		0.95		0.91	

Table 4 Accuracy Assessment of LULC

4.2 LULC pattern 2004-2014

Examination of the temporal LULC design of Debrigarh Wildlife Sanctuary is essential in finding the changes occurring as a result of conservation projects. The research (Table 5) (Fig 4 & 5) demonstrates the LULC design in the years 2004 and 2014 as well as the spatial and time-based variations in each class reflected. The preserved area is dominated by forest cover as it is spreading over an area of 290.70 km² which constitutes 74.05% of the total study area in 2004 while significant expansion by 32.81 km² leading to coverage of 323.52 km² was observed by year 2014. Another dominant class is barren lands which are parts of eastern hills and plateaus covering an area of 92.76 km² which holds 23.63% of the total area, this land cover class declined by 34.73 km² and the remaining barren land covered 58.02 km² in 2014. It is evident from (Fig 4) that barren land is encroached upon by vegetation cover and many developmental changes happening in the sanctuary to promote tourism. Also, patches of agricultural field are found in the area which occupied an area of 6.96 km² in 2004 and increased by 1.52 km² till 2014, and finally, 8.48 km² area was recorded under agricultural practice. Small built-up was found i.e. 0.56 km² which slightly increased to 0.71 km² by 2014. The water body adjacent to the study area is the Hirakud dam reservoir on the Mahnadi River which shows slight changes over the study period.

LULC Classes	Land use land cover area					
	2004 (km ²)	2014 (km ²)	Δ (km ²)	2004 %	2014 %	Δ %
Vegetation	290.70	323.52	32.81	74.05	82.41	8.35
Agriculture	6.96	8.48	1.52	1.77	2.15	0.38
Barren land	92.76	58.02	-34.73	23.63	14.78	-8.84
Built-up	0.56	0.71	0.16	0.14	0.18	0.04
Water	1.57	1.82	0.25	0.40	0.46	0.06

Table 5 Area covered by each LULC class and change (2004-2014)

4.3 LULC Pattern 2014-2024

In the past decade (Table 6) the rising trend is only found in the vegetation class of the Debrigarh sanctuary whereas lesser areas under agriculture, built-up, barren land, and water classes are found (Fig 4 & 5). The forest cover of 323.51 km² constituting 82.41% of the total area in 2014 increased by 19.30 km² by 2024 and the present forest cover expands over an area of 342.81 km² accounting for 87.33% of the total area. Other LULC categories in Debrigarh Wildlife Sanctuary show a declining trend because, a year ago, the authorities

converted a 300-hectare area into a verdant meadow following the villagers' relocation from the sanctuary's core region. The highest percentage of decline is observed in barren land, it reduced from 58.02 km² in 2014 by 17.85 km² to 40.71 km² in 2024. Agricultural land has shrunk from 8.48 km² in 2014 by 1.13 km² and 7.35 km² area under agricultural practice at present. As a result of the relocation of about 420 families from Debrigarh Wildlife Sanctuary, the built-up portion was reduced from 0.71 km² in 2014 to 0.58 km² in 2024. A slight waning of water cover is observed.

LULC Classes	Land use land cover area					
	2014 (km ²)	2024 (km ²)	Δ (km ²)	2014 %	2024 %	Δ %
Vegetation	323.51	342.81	19.30	82.41	87.33	4.91
Agriculture	8.48	7.35	-1.13	2.15	1.87	-0.28
Barren land	58.02	40.17	-17.85	14.78	10.23	-4.54
Built-up	0.71	0.58	-0.13	0.18	0.14	-0.03
Water	1.82	1.62	-0.19	0.46	0.41	-0.04

Table 6 The area under each LULC class and change (2014-2024)

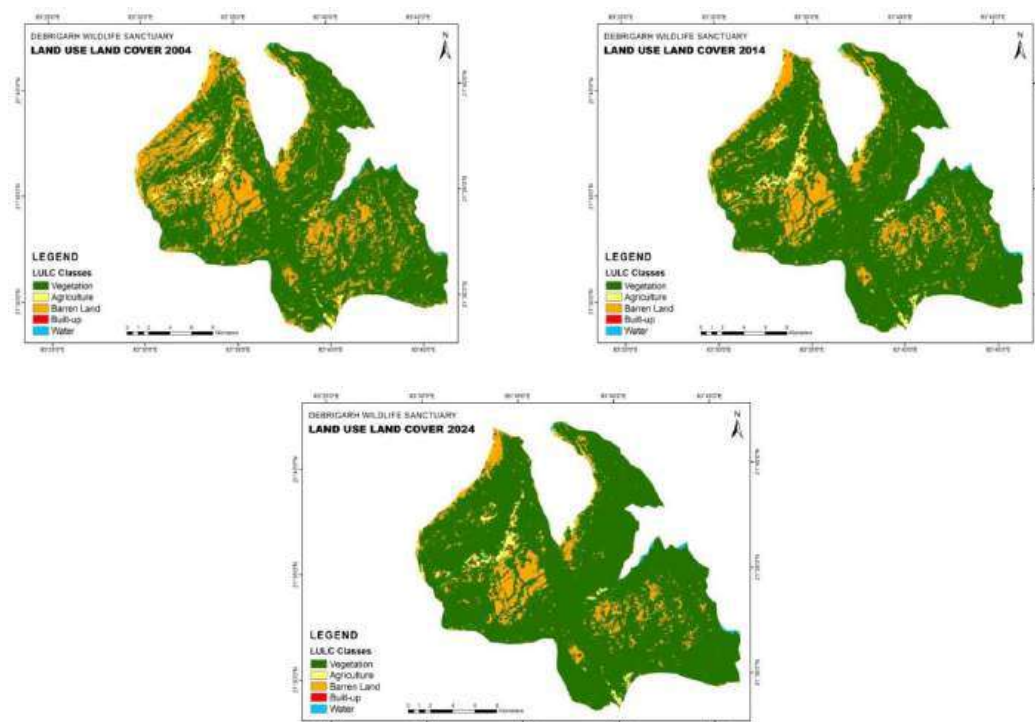


Figure 4 LULC 2004, 2014, 2024

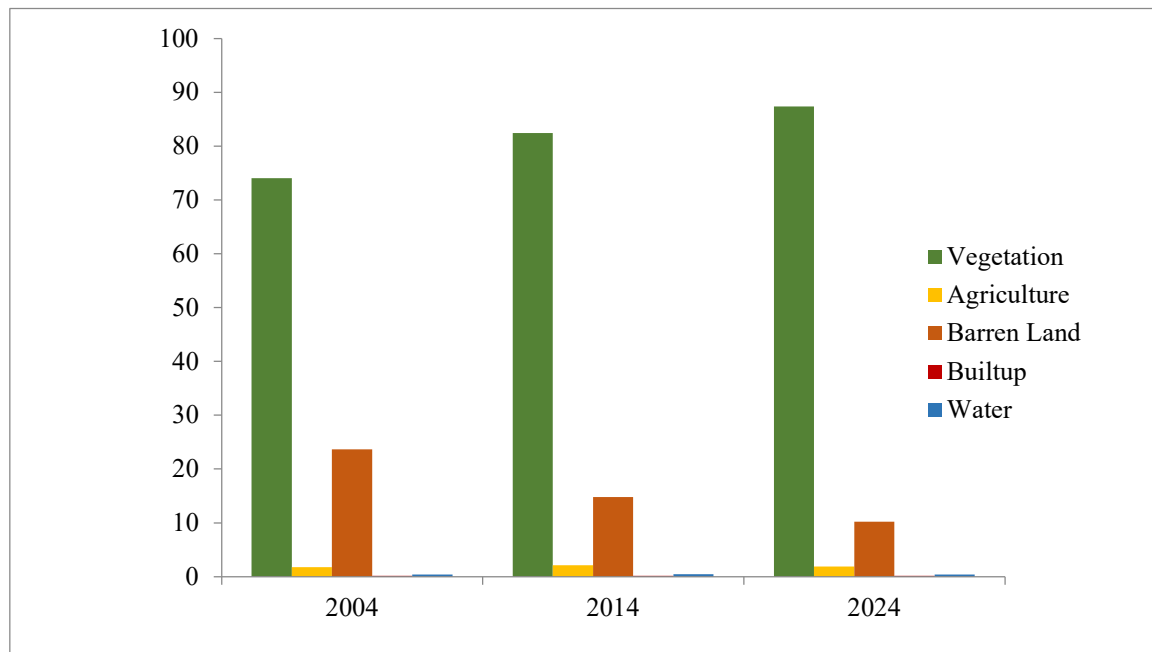


Figure 5 Percentage of area under each land use land cover class

4.4 Change detection (2004-2024)

Land use land cover (LULC) change detection based on remote sensing data is an important source of information for various decision support systems. (Tewabe and Fentahun., 2020). The importance of change detection is to determine which land-use class is changing to the other (Han et al., 2009). (Table 7) (Fig 6) Elaborates change detection analysis of the LULC between years 2004 and 2024. The table shows the section of area (km²) of each class. It represents the transformation of land from one LULC class to another. It is found that a major proportion of vegetation/forest cover has continued unaffected throughout the study era as out of 290.70 km² of forest, 288.25 km² of forest remains unchanged while little transformation is found from forest to barren land i.e. 1.73 km² and transformation of forest to other class is insignificant. Agriculture shows significant changes as out of 6.96 km² of agricultural land 0.86 km² of it changed into vegetation 1.99 km² into barren land and 4.03 km² remains unchanged. Although agricultural land changed to another land use/land cover still overall area occupied under agriculture has enlarged. About half of the barren land shifted into vegetation between 2004-2024. Out of 92.76 km² area 53.10 km² area of barren land turned into forest/vegetation, 2.98 km² into agriculture and 36.39 km² remained unchanged, the shift of barren land into built-up and water is minor. At the beginning of the study period built-up covered 0.56 km² of which 0.45 km² remained unchanged while minimal shift has taken

place to vegetation and barren land. The study area is adjacent to Hirakud reservoir and the changes detection analysis stated that the line between water and Debrigarh Wildlife Sanctuary is turning into vegetation.

	2024						
	LULC Classes	Vegetation	Agriculture	Barren land	Built-up	Water	Total
2004	Vegetation	288.25	0.32	1.73	0.03	0.36	290.70
	Agriculture	0.86	4.03	1.99	0.05	0.03	6.96
	Barren land	53.10	2.98	36.39	0.06	0.23	92.76
	Built-up	0.07	0.00	0.04	0.45	0.00	0.56
	Water	0.53	0.01	0.03	0.00	1.00	1.57
	Total	342.81	7.35	40.17	0.58	1.62	392.54

Table 7 Change matrix (2004-2024)

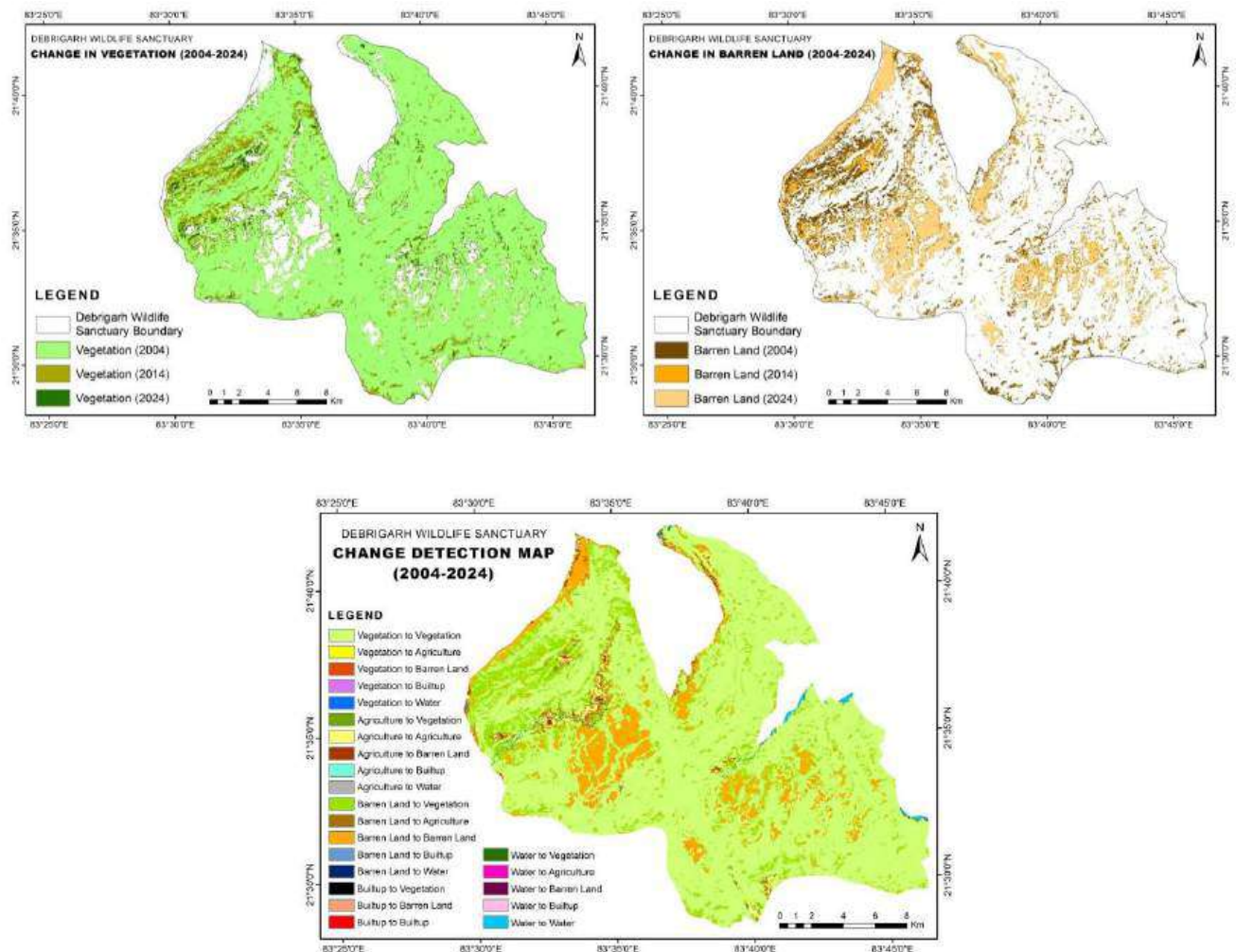


Figure 6 Change in land use land cover class between 2004-2024

4.5 Normalized Difference Vegetation Index (NDVI)

Vegetation greenness is quantified using the Normalized Difference Vegetation Index (NDVI) and it can be useful for understanding vegetation density plus alterations in the health of plants. It is computed as the ratio of the Red (R) and Near Infrared (NIR) in the usual way. Overall, the NDVI values range between -1.0 to +1.0: NDVI < 0: this range shows that clouds and water are present, NDVI ~0: it shows the soil (bare) land, NDVI > 0: it is known as the vegetated area, NDVI = 1: greater than changes in one value mean dense vegetation, High sensor radiation is absorbed and the leaves can photosynthesis process to the full or full.

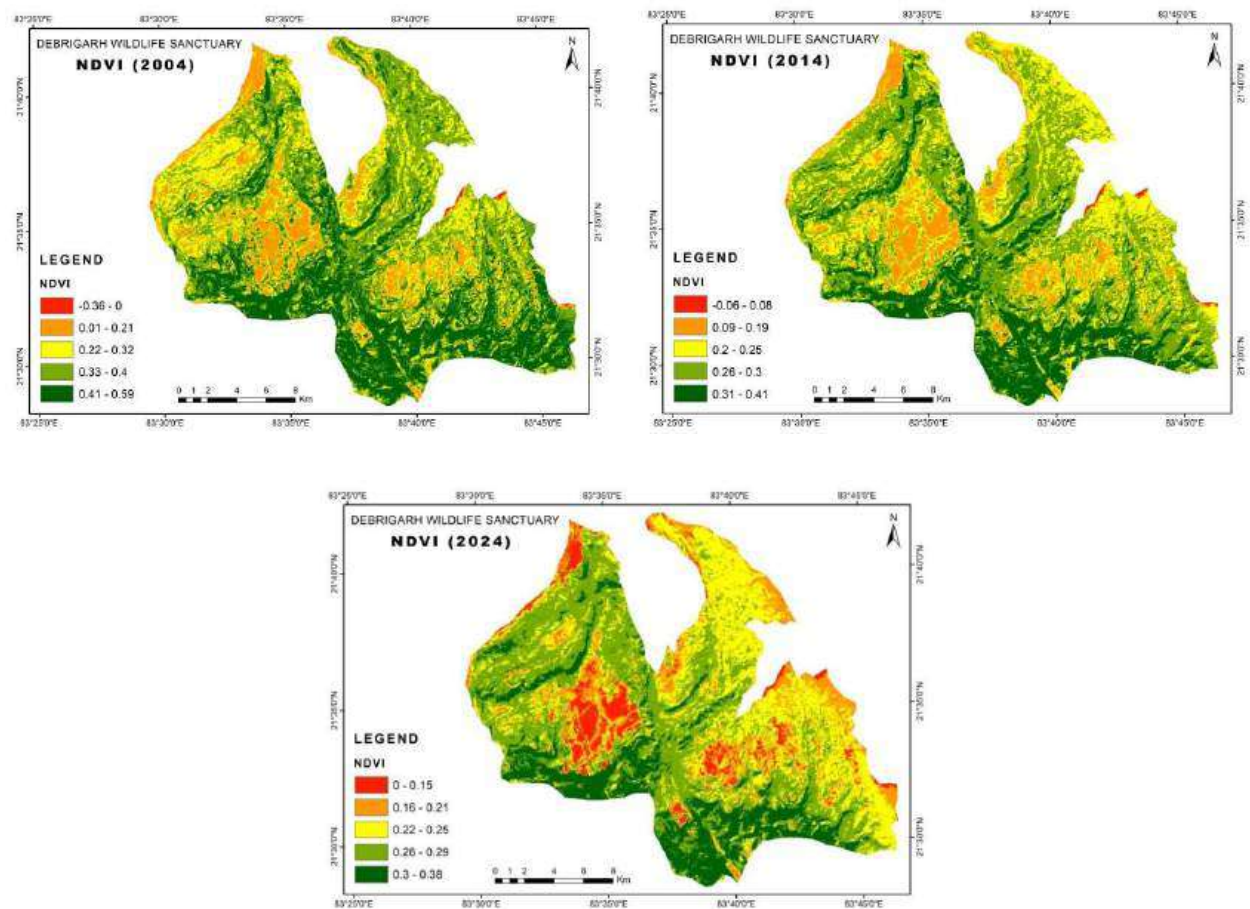


Figure 7 Normalized Difference Vegetation Index (NDVI), 2004, 2014, 2024

It is observed that the vegetation index between 2004-2024 shifts towards more positive values and hence represents vegetation growth about quality and measure (Fig 7). In the year 2004, the NDVI value ranged between -0.36-0.59 where an area in the range of -0.36-0 shows water, 0.01-0.32 shows barren land and patches of built-up while an area under a range

of 0.33-0.4 and 0.41- 0.59 shows sparse and dense vegetation respectively. The scenario changed in the year 2014 when the NDVI value shifted towards more positive values which range between -0.06-0.041, where areas having values between -0.06-0.08 area water cover, areas holding values 0.09-0.19 show barren land while NDVI class 0.2-0.25, 0.26-0.3 and 0.31-0.41 shows agriculture and varied health and density of forest. During recent years i.e. 2024 no negative values are extracted and NDVI values range between 0-0.38. Areas under 0-0.15 class show patches of barren land while all other classes under 0.16-0.38 are representations of forest cover.

5 Conclusion

Debrigarh Wildlife Sanctuary stands as a vital natural asset, showcasing the beauty of Odisha's wilderness and serving as a stronghold for biodiversity conservation. It is noteworthy that throughout the study phase, the area covered by vegetation has amplified however part concealed by built-up, agricultural, and barren land has diminished. The tourist areas and the tourist's accommodation entail mainly the forest cover changes as concluded by the land use change analysis of Debrigarh WLS which was carried out on the physical data and satellite image. These changes could also be categorized into different clusters of causes such as timber livelihood dependence of locals, agricultural expansion (plantation), forestry operations, forest fire, and different kinds of infrastructure developments. This is a result of the government's efforts to reallocate 420 families from the sanctuaries and turn the spared land into grasslands for the benefit of herbivores. It has been determined through analysis using both in-situ and remotely sensed data that the vegetation area has risen which is a sign of positive ecological equilibrium. Through concerted efforts in conservation and sustainable management, the sanctuary can continue to thrive as a haven for wildlife and a source of inspiration for future generations.

Authors Contributions

Niti: Investigation, Methodology, Writing - original draft. **Maya Kumari:** Resources, Writing - review & editing, **Praveen Kumar Rai:** Writing - review & editing **Harshita Jain:** Conceptualization, Methodology, Writing - review & editing, Supervision. All authors contributed to the study's conception and design. All authors read and approved the final manuscript.

Ethical Approval

This article does not contain any study with humans and animals performed by any of the authors.

Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

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Application of Genetic Algorithm to Optimize Turning Process Parameters for Aluminium Alloy 64430

Yash Dhabarde^{1*}, Prashant Kamble², Rakesh Adakane³, Rajesh Bodkhe⁴,

Shilpa Sahare⁵

Abstract

In this research, turning parameters were optimized through a genetic algorithm for the purpose to minimize surface roughness and to maximize the material removal rate. High finish quality is guaranteed through minimum surface roughness, and efficient process planning is facilitated through maximum Material removal rate optimization. For predicting surface roughness and Material removal rate with respect to spindle speed, feed rate, and depth of cut, the empirical models were developed through regression analysis. The high-speed steel cutting tool and aluminium alloy 64430 were employed in experiments on an Craft Master TL-20 lathe machine. The optimal machining conditions were determined using the Genetic Algorithm, which was coded using Python's Distributed Evolutionary Algorithms in Python package. The highest Material removal rate of 3463.3918 mm³/min and surface roughness of 0.6574 μ m were the optimal results achieved. The validity of the model was established through p-values of 0.1688 for surface roughness and for material removal rate is 0.6227, which were achieved through a paired t-test between experimental and estimated values. This approach provides the industry with the liberty to select parameters for specific application needs.

Keywords: Genetic algorithm, Regression modelling, Surface roughness, Material removal rate, Python DEAP.

INTRODUCTION

Optimization of turning process parameters is critical to maximizing output and product quality in modern manufacturing. Cutting speed, feed rate, and depth of cut are significant turning input parameters that significantly influence output factors such as MRR and SR. It is usually hard to find a compromise between maintaining a low SR for surface quality and achieving a high MRR for productivity, particularly when turning aluminium alloys. The desirable strength-to-weight ratio, corrosion resistance, and thermal conductivity of these aluminium alloys have earned them widespread usage in the automobile, marine, and aerospace sectors. Their complex machinability, however, often leads to unbalanced cutting behaviour and rapid tool wear. Since SR and MRR are contradictory, optimizing one may have an adverse effect, calling for methodical and keen optimization methods. To solve such multi-objective optimization problems in turning processes, scientists have deeply examined advanced computation methods, such as genetic algorithms (GA). For instance, Chowdary et al. attained significant improvements in finish through the integration of the Taguchi method with GA to minimize SR in CNC turning of Al-6061. Though Dhabale and Jatti applied GA for achieving maximum MRR in turning AlMg1SiCu, Doriana and Teti applied GA to optimize tool life and surface integrity. For minimizing SR and cutting forces, few researchers like Usha et al. used multi-objective GA in minimum quantity lubrication conditions. In order to improve tool wear, MRR, and cutting parameters, studies by Solarte-Pardo et al. and Alduroobi et al. showed how well GA and ANN operate together. Some hybrid approaches have also been proposed. Hemantha Kumar et al. compared GA with Particle Swarm Optimization (PSO) for AISI 1042 steel, while Malomo et al. introduced a Taguchi-ANN-GA model for turning AA6063 that maximizes multiple objectives. Additionally, to maximize machining time, energy consumption, and temperature increase, hybrid models such as GA-Simulated Annealing and GA-Response Surface Methodology have been employed.

These researches show how hybrid techniques and GAs are being utilized more and more in intelligent machining systems. Even though optimization analyses have long been facilitated by software such as Minitab, MATLAB, and GNU Octave, recent developments have turned Python into an even more powerful platform due to its open-source modules. Predictive modelling and the

extremely accurate and versatile use of genetic algorithms are facilitated by packages such as Scikit-learn and DEAP. Although aluminium alloy 64430 is industrially important, not much is known regarding the turning optimization of the alloy. Through empirical models that employ regression analysis to predict SR and MRR and by the application of Python's DEAP framework to optimize cutting parameters to achieve improved machining performance, the present research aims to fill this void.

MATERIALS AND METHOD

1. Design of Experiments

The work piece for the present study was prepared from the aluminium alloy 64430, and the test specimens were of 45 mm length and 25 mm diameter. For exactly determining its elemental composition, spectroscopic analysis was conducted. After the operational status of an HMT Craft Master TL-20 lathe machine was observed, experiments in machining were conducted on it. In order to study the effect of cutting speed, feed rate, and depth of cut on surface roughness and material removal rate, a systematic design of experiments was designed with the orthogonal array of (L27). Table 1 gives details about the experimental matrix and levels of parameters. Table 1 shows the independent factors with their values

Table 1. Design of Experiments table

levels	Cutting speed (rpm)	Feed rate (mm/rev)	Depth of cut (mm)
1	39.26	0.06	0.15
2	78.53	0.15	0.25
3	109.95	0.24	0.35

2. Experimental Procedure

The trial machining was planned with an L27 Taguchi orthogonal array. Twenty-seven samples of the aluminium alloy 64430, 100 mm by 25 mm in size, were produced. To ensure consistency, the starting and ending weights and diameters were recorded. A High-Speed Steel (HSS) tool was used throughout in dry cutting conditions. A Mitutoyo gauge was used to measure the surface roughness, and after every turning operation, the Material Removal Rate (MRR) was calculated using formula, $MRR (mm^3/min) = \pi/4 * (d_i - d_f) * f * d$, where d_i = initial diameter before experiment (mm), d_f = final diameter after experiment (mm), f = feed rate (mm/rev), d = depth of cut (mm).

Table 2. Orthogonal array and Observed values table

Run	cutting speed (v)	feed rate (f)	depth of cut (d)	surface roughness (micro-m)	material removal rate (mm ³ /min)
1	39.26	0.06	0.15	0.658	176.184
2	39.26	0.06	0.25	0.778	293.051
3	39.26	0.06	0.35	0.868	409.447
4	39.26	0.15	0.15	0.981	440.461
5	39.26	0.15	0.25	1.158	732.629
6	39.26	0.15	0.35	1.292	1023.619
7	39.26	0.24	0.15	1.203	704.737
8	39.26	0.24	0.25	1.421	1172.206
9	39.26	0.24	0.35	1.591	1637.791
10	78.53	0.06	0.15	0.663	352.368
11	78.53	0.06	0.25	0.783	586.103
12	78.53	0.06	0.35	0.874	818.895
13	78.53	0.15	0.15	0.987	880.922
14	78.53	0.15	0.25	1.166	1465.258
15	78.53	0.15	0.35	1.301	2047.238
16	78.53	0.24	0.15	1.211	1409.475
17	78.53	0.24	0.25	1.432	2344.413
18	78.53	0.24	0.35	1.596	3275.581
19	109.95	0.06	0.15	0.666	493.316
20	109.95	0.06	0.25	0.786	820.544
21	109.95	0.06	0.35	0.877	1146.453
22	109.95	0.15	0.15	0.991	1233.291
23	109.95	0.15	0.25	1.171	2051.361

24	109.95	0.15	0.35	1.305	2866.133
25	109.95	0.24	0.15	1.215	1973.265
26	109.95	0.24	0.25	1.435	3282.178
27	109.95	0.24	0.35	1.621	4585.814

3. Regression analysis of Surface roughness :

To model the effect of machining parameters on surface roughness ,SR, a second-order polynomial regression model was created using Python's Scikit-Learn library. Cutting speed v , feed rate f , and depth of cut d were taken as input variables, and SR as the output response. Experimental results from 27 machining trials, based on a Taguchi L27 orthogonal array, were used to train the model. Polynomial feature expansion was used to include linear, quadratic, and interactive effect of the parameters. The developed regression equation is:

$SR = 0.1592 - 0.0001v + 4.3476f + 1.7302d + 0.0011v^2 + 0.0008vf + 0.0005vd - 7.0370f^2 + 5.0648f*d - 2.0000d^2$ As given in this equation, f and d significantly contribute positively to surface roughness while v contributes negatively slightly, meaning that higher speed leads to a marginally better finish. The negative quadratic terms for feed and depth of cut imply that smoother shearing contributes towards improving surface polish beyond a specific point. SR is most strongly affected by the interaction term $f*d$ (5.0648) among the others. The R^2 value of 0.9998 of SR's excellent model indicates that it is very good at predicting surface roughness from inputs in machining.

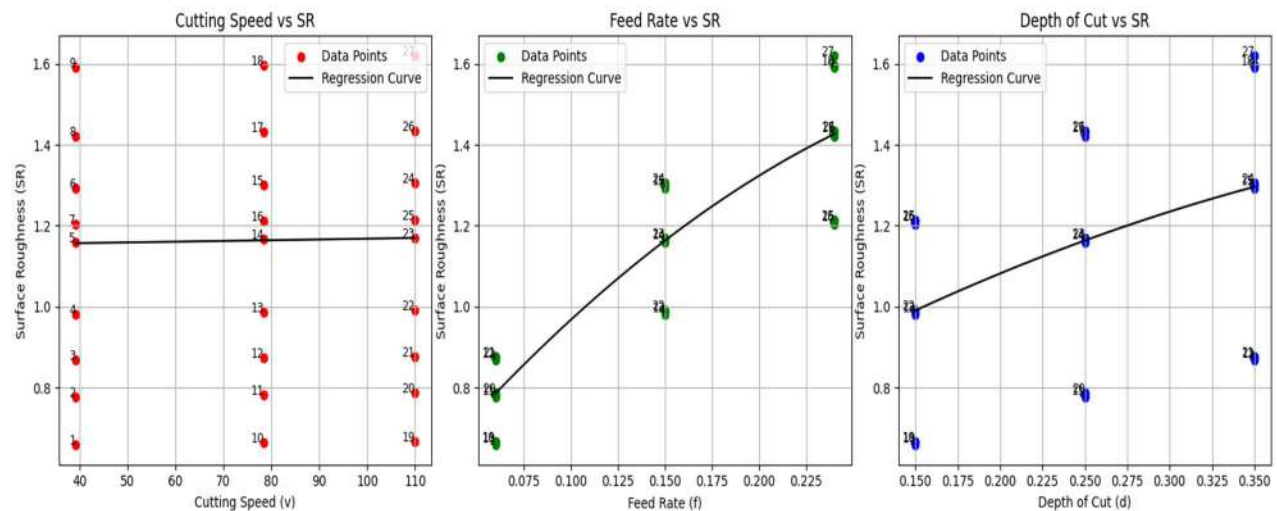


Figure 1. Input vs output for SR

Graphical representations were also generated to visualize the relationship between each input parameter and surface roughness, keeping the other two parameters constant at their mean values. These plots (Cutting Speed vs SR, Feed Rate vs SR, and Depth of Cut vs SR) clearly illustrate the trends and nonlinear effects of individual machining parameters on surface quality, supporting the reliability of the developed model.

4.Regression analysis of Material removal rate

To study the relation between the machining parameters- cutting speed v , feed rate f , and depth of cut d and the resulting Removal Rate, the linear regression model was created using Python's Scikit-learn module. After 27 experimental observations were used to train the model, the following equation was produced $MRR = -2824.7181 + 18.6452 \times v + 9437.7154 \times f + 5637.1956 \times d$. In the study range, the intercept, the model offset, has minimal practical significance. Positive coefficients indicate that material removal rate increases with a rise in any one of the input parameters. The feed rate exerts the maximum influence, followed by v and d . Increasing feed and depth cut more material, raising removal rates, consistent with machining philosophy. The model had high predictive accuracy with an R^2 rating of 0.8610.

In order to show patterns, three graphs were plotted: depth of cut vs. MRR, feed rate vs. MRR, and cutting speed vs. MRR. Each one shows that increasing parameters usually means increased MRR. Alignment between actual and expected values is shown by annotated data points, which offer effective MRR optimization by well-informed parameter selection.

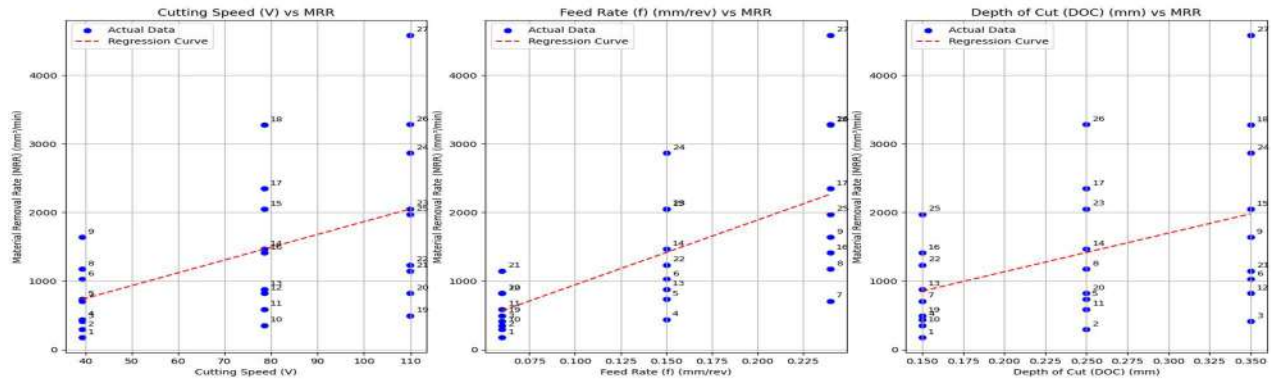


Figure 2. Input vs output for MRR

5. Genetic algorithm applied on Surface roughness (SR)

To minimize surface roughness, the Genetic Algorithm (GA) was implemented using the Python programming language, employing the DEAP 1.4 version (Distributed Evolutionary Algorithms in Python) library. The algorithm aimed to minimize SR by identifying optimal combinations of v , f , and d , using a regression-based mathematical model of SR as the fitness function.

Minimize $SR = 0.1592 - 0.0001v + 4.3476f + 1.7302d + 0.0011v^2 + 0.0008v*f + 0.0005v*d - 7.0370f^2 + 5.0648f*d - 2.0000d^2$. This expression was encoded as the fitness function in the GA framework, with the objective of minimizing SR. To ensure the search space remained within practical limits, the decision variables were bounded as follows: $39.2 \leq v \leq 109.95$, $0.06 \leq f \leq 0.24$, and $0.15 \leq d \leq 0.35$. The below table 4 shows genetic algorithm parameters are applied to achieve minimum SR value

Table 3. GA Parameters for SR Optimization

Genetic algorithm parameters	Methods and values
Size Of Population	50
Number of Generations	200
Crossover Probability	0.7
Mutation Probability	0.2
Selection size	3

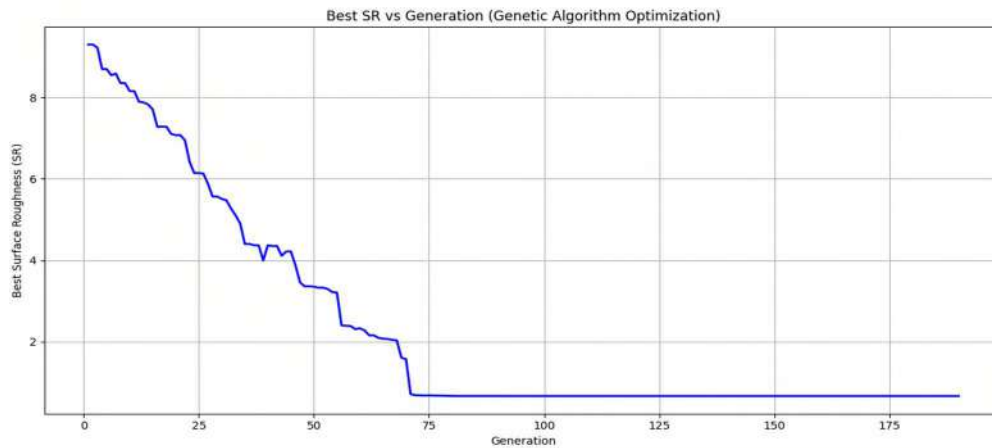


Figure 3. Generation vs Best SR graph

More than 200 generations, the Genetic Algorithm lowered surface roughness (SR) from $9.82 \mu\text{m}$ to $0.6574 \mu\text{m}$. Consistent with machining rules of thumb supporting higher speeds and lower feeds for improved surface finish, the optimum turning parameters were 109.95 m/min cutting speed, 0.0600 mm/rev feed rate, and 0.1500 mm depth of cut.

6. Genetic algorithm applied on material removal rate MRR

To maximize the Material removal rate (MMR) in the turning of aluminium alloy 64430, the Genetic Algorithm (GA) was implemented using the Python programming language, employing the DEAP 1.4 version library.

The algorithm aimed to maximize MRR by identifying optimal combinations of v , f , and d , using a regression-based mathematical model of MMR as the fitness function. Maximize $\text{MMR} = -2824.7181 + 18.6452 \times v + 9437.7154 \times f + 5637.1956 \times d$. This expression was encoded as the fitness function in the GA framework, with the objective of maximizing MRR. To ensure the search space remained within practical limits, the decision variables were bounded as follows: $39.26 \leq v \leq 109.95$, $0.06 \leq f \leq 0.24$, $0.15 \leq d \leq 0.35$

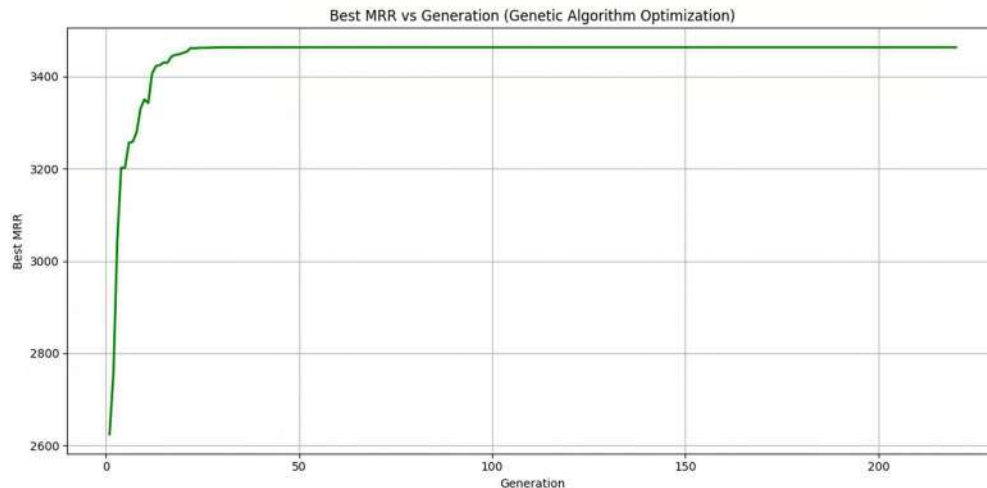


Figure 4. Generation vs Best MMR graph

After 200 generations, the Material Removal Rate (MRR) of the Genetic Algorithm was improved from 2939.8165 mm³/min to 3463.3918 mm³/min at a mutation probability of 0.3. The optimal parameters for effectively optimizing removal rate were depth of cut 0.3500 mm, feed rate 0.2400 mm/rev, and cutting speed 109.95 m/min.

7. Confirmation test

With Python's SciPy library, a statistical validation test was conducted to check the Genetic Algorithm optimization results. Experimental values of Surface Roughness and Material Removal Rate, from the DOE were compared with GA-predicted values by conducting a Paired T-Test. For obtaining the t-statistic and p-value, the `t test_rel` function from SciPy. Stats was used. The test results indicated that the SR's t-statistic was 2.1143 and p-value was 0.1688, while that of the MRR was 0.5762 and p-value was 0.6227. Experimental and GA-estimated values do not statistically differ since both p-values are above the 0.05 significance level. This confirms the accuracy and reliability of the GA-based optimization model in predicting machining performance.

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A Review on Lung Cancer Prediction Using Machine Learning

Sarita T. Sawale¹, Sejal N. Marke², Vasudha S. Wakte³

Abstract

Lung cancer remains one of the leading causes of cancer-related deaths worldwide. Early detection and accurate prediction of lung cancer can significantly improve patient outcomes. This research paper explores the application of machine learning (ML) techniques for lung cancer prediction. By leveraging datasets containing clinical, imaging, and genetic data, we evaluate the performance of various ML algorithms, including decision trees, support vector machines, and deep learning models. The study highlights the importance of feature selection, data preprocessing, and model optimization in improving prediction accuracy. The results demonstrate that machine learning can be a powerful tool for early lung cancer detection, offering a promising avenue for reducing mortality rates.

Keywords: Lung Cancer, Machine Learning, Prediction, Early Detection, Healthcare, Artificial Intelligence, Data Analysis, Deep Learning.

INTRODUCTION

Background on Lung Cancer Prediction Using Machine Learning: Lung cancer is a malignant tumor characterized by uncontrolled cell growth in lung tissues. It is the most common cause of cancer-related deaths globally, with over 2 million new cases diagnosed annually. Traditional diagnostic methods, such as imaging and biopsy, are often time-consuming, expensive, and prone to human error. Machine learning, a subset of artificial intelligence, has emerged as a transformative technology in healthcare. By analyzing large datasets, ML algorithms can identify patterns and predict outcomes with high accuracy. In the context of lung cancer, ML models can analyze patient data, including medical history, imaging scans, and genetic markers, to predict the likelihood of cancer development[1].

Importance of Lung Cancer Prediction Using Machine Learning: Early detection of lung cancer is critical for improving survival rates. Patients diagnosed at an early stage have a significantly higher chance of successful treatment compared to those diagnosed at advanced stages. Machine learning offers a non-invasive, cost-effective, and scalable solution for early prediction. By automating the analysis of complex datasets, ML can assist healthcare professionals in making informed decisions, reducing diagnostic errors, and personalizing treatment plans[2]. Furthermore, ML models can continuously learn and improve over time, adapting to new data and evolving medical knowledge.

Problem Statement: Despite advancements in medical technology, lung cancer remains difficult to detect in its early stages due to the lack of noticeable symptoms. Traditional diagnostic methods are often inadequate for early prediction, leading to delayed diagnoses and poor patient outcomes. There is a pressing need for a reliable, accurate, and efficient system that can predict lung cancer at an early stage using available patient data. Machine learning has the potential to address this gap, but challenges such as data quality, model interpretability, and integration into clinical workflows must be addressed[3].

Objectives: The primary objective of this research is to develop and evaluate machine learning models for lung cancer prediction. Specifically, the study aims to:

1. Identify the most relevant features for lung cancer prediction from clinical, imaging, and genetic datasets.
2. Compare the performance of various ML algorithms in predicting lung cancer.
3. Optimize model accuracy through feature selection, hyperparameter tuning, and data preprocessing.

4. Provide insights into the integration of ML-based prediction systems into clinical practice.

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Literature Review: Recent studies have demonstrated the potential of machine learning in lung cancer prediction. For instance, researchers have used convolutional neural networks (CNNs) to analyze CT scans and detect early-stage lung cancer with high accuracy. Other studies have explored the use of support vector machines (SVMs) and random forests to predict cancer risk based on patient demographics and medical history. However, challenges such as imbalanced datasets, overfitting, and the need for large-scale validation remain. This paper builds on existing research by addressing these challenges and proposing novel approaches to improve prediction accuracy[4].

A literature survey provides an overview of existing research and developments in the field of lung cancer prediction using machine learning. Below is a brief summary of the literature survey for this topic:

1. Early Studies on Machine Learning in Lung Cancer Prediction:

- Early research focused on using traditional machine learning algorithms like **logistic regression**, **decision trees**, and **support vector machines (SVMs)** to predict lung cancer risk based on clinical and demographic data.
- These studies demonstrated the potential of ML but were limited by small datasets and lack of advanced computational resources.

2. Advancements in Imaging Analysis:

- With the advent of **deep learning**, particularly **Convolutional Neural Networks (CNNs)**, researchers began analyzing medical imaging data such as CT scans and X-rays.
- Studies showed that CNNs could detect early-stage lung cancer with high accuracy, outperforming traditional methods in identifying subtle patterns in imaging data.

3. Federated Learning for Privacy Preservation:

- To address data privacy concerns, **federated learning** has emerged as a promising approach.
- This technique allows ML models to be trained across multiple institutions without sharing raw data, ensuring patient confidentiality while improving model robustness.

4. Real-Time Prediction Systems:

- Advances in **cloud computing** and **edge computing** have enabled the development of realtime lung cancer prediction systems.
- These systems can analyze patient data in real-time, facilitating timely interventions and improving patient outcomes.

5. Challenges and Limitations:

- **Imbalanced Datasets:** Many studies struggle with imbalanced datasets, where the number of cancer cases is significantly lower than non-cancer cases, leading to biased models.
- **Overfitting:** Complex models like deep learning networks are prone to overfitting, especially when trained on small datasets.
- **Lack of Standardization:** Variations in data collection methods and formats across institutions hinder the development of generalized models.
- **Clinical Integration:** Despite promising results, integrating ML models into clinical workflows remains a challenge due to regulatory and operational barriers.

6. Key Findings:

- Machine learning, particularly deep learning, has shown remarkable success in lung cancer prediction, especially in imaging analysis.
- Combining multiple data sources (e.g., clinical, imaging, and genomic data) improves prediction accuracy.
- Explainable AI and federated learning are critical for increasing the adoption of ML in healthcare.
- Real-time prediction systems have the potential to revolutionize early detection and treatment.

7. Research Gaps:

- There is a need for large-scale, standardized datasets to train and validate ML models.

- More research is required to address imbalanced datasets and overfitting issues.
- Efforts are needed to integrate ML models into clinical workflows and ensure their practical applicability.

Current trends and advancements in Lung Cancer Prediction Using Machine Learning:

1. **Deep Learning for Imaging Analysis:** Deep learning models, particularly CNNs, have shown remarkable success in analyzing medical images such as CT scans and X-rays. These models can detect subtle patterns indicative of early-stage lung cancer that may be missed by human radiologists.
2. **Integration of Multi-Omics Data:** Combining genomic, proteomic, and clinical data has enabled the development of more comprehensive prediction models. Multi-omics approaches provide a holistic view of cancer risk factors, improving prediction accuracy.
3. **Explainable AI (XAI):** There is a growing emphasis on developing interpretable ML models that can provide insights into the decision-making process. Explainable AI techniques, such as SHAP (SHapley Additive exPlanations), help clinicians understand the factors contributing to a prediction.
4. **Federated Learning:** To address data privacy concerns, federated learning allows ML models to be trained across multiple institutions without sharing raw data. This approach enables the development of robust models while maintaining patient confidentiality.
5. **Real-Time Prediction Systems:** Advances in cloud computing and edge computing have facilitated the development of real-time lung cancer prediction systems. These systems can analyze patient data in real-time, enabling timely interventions.

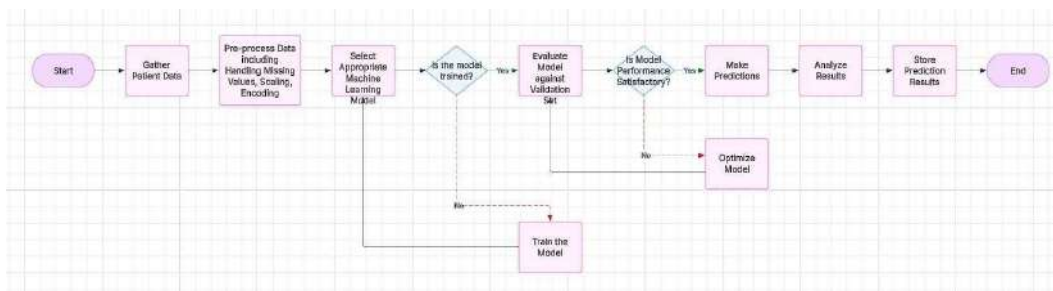


Fig1: ER Diagram for Lung Cancer Prediction Using Machine Learning

An **Entity-Relationship (ER) Diagram** is a visual representation of the entities (objects or concepts) in a system and the relationships between them. For a system focused on Lung Cancer Prediction Using Machine Learning, the ER diagram helps to model the data structure, including patient information, medical records, machine learning models, prediction results, and treatment plans. **Definition and Evolution of Lung Cancer Prediction Using Machine Learning:** Lung cancer prediction using machine learning refers to the application of artificial intelligence (AI) and statistical algorithms to analyze patient data and predict the likelihood of lung cancer. This involves using datasets such as clinical records, imaging data (e.g., CT scans, X-rays), genetic information, and lifestyle factors to train models that can identify patterns indicative of lung cancer. The goal is to enable early detection, improve diagnostic accuracy, and support personalized treatment plans[5].

Definition: Lung cancer prediction using machine learning is a data-driven approach that leverages algorithms to:

1. **Analyze Patient Data:** Process structured (e.g., age, smoking history) and unstructured data (e.g., medical images, genomic data).
2. **Identify Patterns:** Detect subtle patterns or anomalies that may indicate the presence of lung cancer.

-
3. **Predict Outcomes:** Estimate the probability of lung cancer development or progression.
 4. **Support Decision-Making:** Assist healthcare professionals in making informed decisions about diagnosis and treatment.

Evolution: The evolution of lung cancer prediction using machine learning has progressed significantly over the decades. In the **1990s and early 2000s**, the focus was on basic statistical models like logistic regression and decision trees, which used demographic and clinical data for risk prediction but were limited by small datasets and low accuracy[6]. By the **mid-2000s to early 2010s**, the integration of medical imaging data, such as CT scans, and the use of feature engineering techniques like Random Forests improved lung nodule detection, though manual feature extraction remained time-consuming.

The **mid-2010s to present** marked the deep learning revolution, with Convolutional Neural Networks (CNNs) and other advanced algorithms enabling automated feature extraction, higher accuracy, and the integration of multi-modal data like imaging, genomics, and clinical records.

Today, in the **2020s and beyond**, the focus has shifted to explainable AI (XAI), federated learning for privacy preservation, and real-time prediction systems that integrate with electronic health records (EHRs). Future directions include leveraging wearable devices for continuous monitoring, multi-omics approaches for comprehensive risk assessment, and AI-driven drug discovery, all while addressing challenges like data privacy, bias, and clinical workflow integration. This evolution has transformed lung cancer prediction, enabling earlier detection, personalized treatment, and improved patient outcomes[7].

Various Lung Cancer Prediction Using Machine Learning:

1. **Clinical Data-Based Prediction:**
Uses patient demographics, smoking history, and medical records to predict lung cancer risk. Common algorithms include logistic regression and decision trees.
2. **Imaging Data-Based Prediction:**
Analyzes CT scans, X-rays, or PET scans to detect lung nodules or tumors. Deep learning models like CNNs are widely used for this purpose.
3. **Multi-Modal Data-Based Prediction:**
Combines clinical, imaging, and genomic data for a comprehensive prediction. Hybrid models and multi-task learning are commonly used.
4. **Survival Prediction:**
Predicts patient survival rates or cancer progression using historical data. Algorithms like random forests and survival analysis models are applied.

Benefits of Lung Cancer Prediction Using Machine Learning:

1. **Early Detection:** Machine learning models can identify lung cancer at an early stage, even before symptoms appear, significantly improving survival rates.
2. **Improved Accuracy:** Advanced algorithms, especially deep learning, achieve high accuracy in detecting lung cancer from complex datasets like CT scans and genomic data.
3. **Personalized Medicine:** ML models provide tailored risk assessments and treatment recommendations based on individual patient data, enhancing treatment effectiveness.

Challenges in Lung Cancer Prediction Using Machine Learning Implementation:

1. **Data Quality and Availability:** Inconsistent or incomplete datasets can reduce model accuracy. Limited access to large, high-quality datasets for training.
 2. **Imbalanced Datasets:** Lung cancer cases are often rare compared to non-cancer cases, leading to biased models.
 3. **Overfitting:** Complex models like deep learning may overfit to small datasets, reducing their generalizability.
-

Emerging Trends in Lung Cancer Prediction Using Machine Learning:

1. **Explainable AI (XAI):** Development of interpretable models to provide insights into predictions, increasing trust among clinicians. Techniques like SHAP and LIME (Local Interpretable Model-agnostic Explanations) are gaining traction.
2. **Federated Learning:** Enables collaborative model training across multiple institutions without sharing raw data, addressing privacy concerns. Promotes global collaboration while maintaining data security.

Modules:

Data Collection Module: This module is responsible for gathering diverse datasets required for training and testing the machine learning models. It collects structured data (e.g., patient demographics, smoking history, medical records) and unstructured data (e.g., CT scans, X-rays, genomic data). The data can be sourced from hospitals, research institutions, or public datasets like The Cancer Genome Atlas (TCGA)[8]. Ensuring data quality, consistency, and compliance with privacy regulations (e.g., HIPAA, GDPR) is a critical aspect of this module.

Data Preprocessing Module: Raw data often contains noise, missing values, and inconsistencies, which can negatively impact model performance. This module cleans and prepares the data for analysis. Tasks include handling missing values, normalizing data, removing outliers, and converting unstructured data (e.g., medical images) into a format suitable for machine learning. For example, CT scans may be resized, annotated, or segmented to highlight regions of interest (e.g., lung nodules).

Feature Extraction and Selection Module: This module identifies the most relevant features (variables) that contribute to lung cancer prediction. For structured data, techniques like correlation analysis and principal component analysis (PCA) are used. For imaging data, deep learning models like Convolutional Neural Networks (CNNs) automatically extract features such as texture, shape, and size of nodules. Feature selection ensures that only the most impactful features are used, reducing computational complexity and improving model accuracy.

Machine Learning Model Development Module: This is the core module where predictive models are built and trained. Depending on the data type and prediction goal, different algorithms are used. For example, Random Forests and Support Vector Machines (SVMs) are suitable for structured data, while CNNs are ideal for imaging data[8]. The module also includes hyperparameter tuning and cross-validation to optimize model performance. Advanced techniques like transfer learning and ensemble methods may also be employed to enhance accuracy.

Prediction and Risk Assessment Module: Once the model is trained, this module generates predictions for new data. It calculates the probability of lung cancer and assigns risk levels (e.g., low, medium, high) based on the model's output. For imaging data, it can highlight suspicious areas in CT scans or X-rays. The module also provides interpretable results using Explainable AI (XAI) techniques, helping clinicians understand the basis of predictions.

Integration and Deployment Module: This module focuses on integrating the machine learning system into existing healthcare workflows. It ensures compatibility with electronic health records (EHRs) and hospital systems. The model can be deployed as a web-based application, a cloud service, or an edge computing solution for real-time predictions. User-friendly interfaces are designed for clinicians to input data and receive predictions seamlessly.

Monitoring and Feedback Module: After deployment, this module continuously monitors the system's performance. It collects feedback from clinicians and patients to identify areas for improvement. The module also tracks model accuracy over time and retrain the model with new data to ensure it remains up-to-date and effective. This iterative process ensures the system evolves with advancements in medical knowledge and technology.

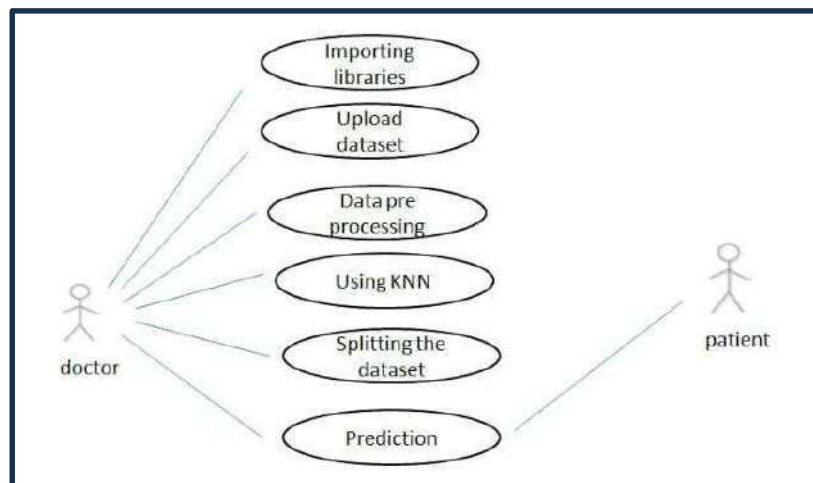


Fig2: Use Case Diagram

Case Studies and Industry Applications of Lung Cancer Prediction Using Machine Learning:

1. Google Health's Deep Learning Model for Lung Cancer Detection:

Application: Google Health developed a deep learning model to analyze low-dose CT (LDCT) scans for early lung cancer detection.

Results: The model outperformed radiologists in detecting lung cancer, reducing false positives by 11% and false negatives by 5%.

Impact: This system is being integrated into clinical workflows to assist radiologists in diagnosing lung cancer more accurately and efficiently.

2. IBM Watson for Oncology:

Application: IBM Watson uses machine learning to analyze patient data, including medical records, imaging data, and genomic information, to predict lung cancer risk and recommend personalized treatment plans.

Results: Watson has been deployed in hospitals worldwide, helping oncologists make data-driven decisions and improving patient outcomes.

Impact: The system has reduced the time required for treatment planning and increased the accuracy of lung cancer predictions.

3. The National Lung Screening Trial (NLST):

Application: Machine learning algorithms were used to analyze data from the NLST, a large-scale clinical trial studying the effectiveness of lung cancer screening.

Results: ML models identified high-risk patients more accurately than traditional methods, enabling early intervention.

Impact: The findings supported the adoption of low-dose CT screening for high-risk populations, significantly reducing lung cancer mortality rates.

4. Enlitic's AI-Powered Imaging Analysis:

Application: Enlitic, a healthcare AI company, developed a deep learning system to analyze medical images, including CT scans, for lung cancer detection.

Results: The system achieved high accuracy in identifying lung nodules and predicting malignancy, reducing diagnostic errors.

Impact: Enlitic's technology is being used in radiology departments to improve the speed and accuracy of lung cancer diagnoses.

5. NVIDIA Clara and MONAI for Medical Imaging:

Application: NVIDIA's Clara platform and MONAI (Medical Open Network for AI) provide tools for developing AI models to analyze medical imaging data, including lung CT scans.

Results: Researchers and healthcare providers have used these tools to build custom models for lung cancer prediction, achieving state-of-the-art performance.

Impact: These platforms are accelerating the development and deployment of AI solutions in healthcare, particularly for lung cancer screening.

6. Zebra Medical Vision:

Application: Zebra Medical Vision uses machine learning to analyze medical imaging data, including chest X-rays and CT scans, for early lung cancer detection.

Results: Their algorithms have demonstrated high accuracy in identifying early-stage lung cancer, enabling timely treatment.

Impact: The system is being used in hospitals to improve screening programs and reduce diagnostic delays.

7. Tempus Labs:

Application: Tempus uses machine learning to analyze clinical and genomic data to predict lung cancer risk and recommend personalized therapies.

Results: Their platform has improved the accuracy of lung cancer predictions and helped oncologists tailor treatments to individual patients.

Impact: Tempus is transforming precision medicine by integrating AI into cancer care.

8. Lung Cancer Prediction in Developing Countries:

Application: Researchers have developed low-cost machine learning models to predict lung cancer in resource-limited settings using basic clinical data.

Results: These models have shown promising results in identifying high-risk patients, even without advanced imaging or genomic data.

Impact: Such systems are making lung cancer prediction accessible in underserved regions, improving early detection rates.

Industry Applications:

Hospitals and Clinics: ML systems are integrated into radiology and oncology departments to assist in lung cancer screening, diagnosis, and treatment planning.

Pharmaceutical Companies: Machine learning is used to identify potential drug targets and accelerate lung cancer drug discovery.

Telemedicine Platforms: AI-powered tools are being incorporated into telemedicine platforms to provide remote lung cancer screening and consultations.

Public Health Programs: Governments and NGOs are using ML-based systems to implement large-scale lung cancer screening programs, particularly for high-risk populations.

Research Institutions: Researchers are leveraging machine learning to analyze large datasets and uncover new insights into lung cancer risk factors and progression.

Challenges in Deploying ML Models in Healthcare:

Deploying machine learning (ML) models in healthcare, particularly for lung cancer prediction, presents several challenges. These challenges span technical, operational, ethical, and regulatory domains. Below is an elaboration of these challenges:

1. Data Quality and Availability:

- **Inconsistent Data:** Healthcare data often comes from diverse sources (e.g., hospitals, labs, wearables) and may lack standardization, leading to inconsistencies.
- **Missing Data:** Patient records may have missing or incomplete data, which can affect model performance.
- **Small Datasets:** In some cases, datasets may be too small to train robust ML models, especially for rare

conditions like specific types of lung cancer.

2. Data Privacy and Security:

- **Sensitive Information:** Healthcare data is highly sensitive, requiring strict privacy measures to protect patient confidentiality.
- **Regulatory Compliance:** ML systems must comply with regulations like GDPR (General Data Protection Regulation) and HIPAA (Health Insurance Portability and Accountability Act), which can complicate data sharing and usage.
- **Cybersecurity Risks:** Storing and processing healthcare data on digital platforms increases the risk of cyberattacks and data breaches.

3. Model Interpretability and Explainability:

- **Black-Box Models:** Many advanced ML models, especially deep learning, are complex and difficult to interpret, making it hard for clinicians to trust their predictions.
- **Explainable AI (XAI):** There is a growing need for explainable AI techniques (e.g., SHAP, LIME) to make model predictions transparent and understandable to healthcare professionals.

4. Integration with Clinical Workflows:

- **Resistance to Change:** Healthcare professionals may be resistant to adopting ML systems due to lack of familiarity or trust in AI.
- **Workflow Disruption:** Integrating ML models into existing clinical workflows can be challenging, requiring significant changes to processes and systems.
- **Interoperability:** ML systems must be compatible with existing healthcare IT infrastructure, such as electronic health records (EHRs).

5. Bias and Fairness:

- **Dataset Bias:** ML models trained on biased datasets may produce unfair or inaccurate predictions, particularly for underrepresented groups.
- **Algorithmic Bias:** Models may inherit biases from the data or algorithms used, leading to unequal performance across different demographics.
- **Equity Concerns:** Ensuring that ML systems provide equitable care for all patients, regardless of race, gender, or socioeconomic status, is a major challenge.

Economic Impact of ML in Lung Cancer Prediction: The integration of machine learning (ML) in lung cancer prediction has significant economic implications for healthcare systems, patients, and businesses. Below is an elaboration of the economic impact, focusing on cost savings, market potential, and broader economic benefits:

1. Cost Savings for Healthcare Systems:

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- **Early Detection:** ML models can detect lung cancer at an early stage, reducing the need for expensive late-stage treatments. Early detection often involves less invasive and less costly interventions.
 - **Reduced Diagnostic Errors:** By improving diagnostic accuracy, ML reduces the costs associated with misdiagnosis, unnecessary treatments, and repeated tests.
 - **Efficient Resource Allocation:** ML systems can prioritize high-risk patients for screening and treatment, optimizing the use of healthcare resources and reducing waste.

2. Cost Savings for Patients:

- **Lower Treatment Costs:** Early detection and personalized treatment plans reduce the overall cost of care for patients, including hospital stays, surgeries, and medications.
 - **Reduced Indirect Costs:** Early diagnosis minimizes the indirect costs associated with lung cancer, such
-

as lost productivity and income due to prolonged illness.

3. Market Potential for Businesses:

- **Health Tech Companies:** Companies developing ML-based lung cancer prediction tools can tap into a growing market for AI in healthcare, generating significant revenue.
- **Pharmaceutical Companies:** ML can accelerate drug discovery and development, reducing costs and time-to-market for new lung cancer therapies.
- **Insurance Providers:** Insurers can use ML models to assess risk more accurately, design personalized insurance plans, and reduce payouts for late-stage cancer treatments.

Conclusion and Future Scope:

1. **Summary of Key Findings:** This research paper explored the application of machine learning (ML) in lung cancer prediction, highlighting its potential to revolutionize early detection, diagnosis, and treatment.

Key findings include:

- **High Accuracy:** Machine learning models, particularly deep learning, achieve high accuracy in predicting lung cancer by analyzing diverse datasets, including clinical records, imaging data, and genomic information.
- **Early Detection:** ML enables the identification of lung cancer at an early stage, significantly improving patient survival rates.
- **Personalized Medicine:** By integrating multi-modal data, ML models provide personalized risk assessments and treatment recommendations.
- **Cost and Time Efficiency:** Automating lung cancer prediction reduces diagnostic costs and time, making it accessible to larger populations.
- **Challenges:** Issues such as data quality, model interpretability, and integration into clinical workflows need to be addressed for widespread adoption.

2. Implications for Businesses Adopting Lung Cancer Prediction Using Machine Learning:

- **Healthcare Providers:**
Hospitals and clinics can integrate ML systems into their workflows to improve diagnostic accuracy and patient outcomes.
Early detection reduces treatment costs and enhances the quality of care.
- **Pharmaceutical Companies:**
ML can accelerate drug discovery by identifying potential drug targets and predicting treatment responses.
Personalized medicine opens new markets for targeted therapies.
- **Health Tech Startups:**
Startups can develop innovative ML-based tools for lung cancer screening and prediction, creating new revenue streams.

Partnerships with hospitals and research institutions can drive adoption.
- **Insurance Companies:**
Early detection and prevention reduce healthcare costs, benefiting insurance providers.
ML-based risk assessment tools can help design personalized insurance plans.
- **Public Health Organizations:**
Governments and NGOs can implement ML-based screening programs to reduce lung cancer mortality rates, especially in underserved areas.
Data-driven insights can inform public health policies and resource allocation.

3. Future Directions of Lung Cancer Prediction Using Machine Learning:

Explainable AI (XAI): Developing interpretable models to increase trust and adoption among clinicians.

Techniques like SHAP and LIME will play a key role in making ML predictions transparent.

Federated Learning: Enabling collaborative model training across institutions without sharing raw data, addressing privacy concerns. Promotes global collaboration and improves model robustness.

Real-Time Prediction Systems: Integrating ML models with electronic health records (EHRs) for real-time lung cancer risk assessment. Enables continuous monitoring and timely interventions.

Wearable Technology: Using data from wearable devices (e.g., smartwatches) for early detection and monitoring of high-risk patients.

Expands the scope of lung cancer prediction beyond clinical settings.

AI-Driven Drug Discovery: Leveraging ML to identify new drug targets and therapies for lung cancer. Accelerates the development of personalized treatments.

Global Collaboration: Building large, diverse datasets to train and validate ML models. Ensures equitable performance across different populations.

Ethical AI and Bias Mitigation: Addressing biases in datasets and ensuring fair and unbiased predictions. Promotes equitable healthcare outcomes.

Cloud and Edge Computing: Deploying ML models on cloud platforms and edge devices for scalable and efficient lung cancer prediction. Enables real-time analysis and reduces computational costs.

Public Health and Clinical Trials

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Comparative Analysis of AI-Based Approach vs Traditional Methods in Climate Modeling

Anvesha Katti¹, Priyanka Vashisht^{2*}

Abstract

Climate modeling helps to predict the future of climate variations and human interference with environment. The traditional General Circulation Models (GCMs) use hard derived from physics-based mathematical equations but are very expensive in terms of computation. There are alternative ways to perform climate modelings in recent years with the rise and improvement of Artificial Intelligence (AI) based approaches in term of predictability, efficiency, and classification of extreme events compared to conventional. In this study, AI Models like XGBoost, Long Short-Term Memory (LSTM) networks and hybrid AI-physics models are assessed over GCM. Results show that AI models significantly decrease prediction errors by about 30%, with hybrid AI-physics models having the lower MAE and higher F1 score for extreme event classification. In addition, the AI models have improved computational power by 40 % less time for processing and less energy. These results draws attention to the chance that AI based approaches can be a great for enhancing climate prediction accuracy, risk assessment, and informing policies.

Keywords: Climate Modeling, Artificial Intelligence, Machine Learning, General Circulation Models, Extreme Event Classification, Hybrid AI-Physics Models, Computational Efficiency, Climate Prediction

INTRODUCTION

Climate modeling serves to strengthen and quantify the assessment of alternative futures for human-caused changes in climate through theoretical studies that are backed with ability to predict shifts in climate by a given model. Employing mathematical equations and physical principles, these models simulate atmospheric, oceanic or terrestrial processes. Most traditional climate models (e. g., General Circulation Models (GCMs;)) model of the interactions between Earth's components using fundamental laws of physics for example fluid dynamics and thermodynamics [1]. These models are incredibly accurate, but they are very computationally expensive requiring supercomputers to perform glances at how climate could change over century timescales fairly realistically.

Recent developments in AI and ML provide an alternative approach to climate modeling. Instead of solving large systems of nonlinear differential equations, AI and ML methods learn patterns from historical climate data [2]. These approaches have been successful across a range of applications in climate science, from projection models (climate projections & downscaling), to emergence-of-weather observation-and-response systems, and parameters within traditional schemes [3]. But making AI/ML part of climate modeling will be hard (large training data, mostly non-interpretable, has risks for over-fitting too). This paper screens the front end of AI and ML applications in climate modeling, towards an understanding of potentials, as well as challenges.

RELATED WORK

The application of AI and ML to climate science has led to significant improvements in predictive accuracy, computational efficiency, and interpretability. Various studies have explored AI/ML methods in climate modeling.

Karpatne et al. [4] proposed Physics-Guided Machine Learning (PGML), which integrates domain knowledge into climate simulations to improve interpretability and reliability. This approach enhances AI models by incorporating physical constraints, allowing them to better predict climate variables while aligning with established theories.

Deep learning models such as CNN-RNN hybrids have also been explored. Rasp et al. [5] demonstrated that AI- based weather prediction models could outperform numerical models in short-

term forecasting while reducing computational costs. This approach has led to data-driven forecasting paradigms in climate modeling.

Schultz et al. [6] provided a comprehensive review of AI applications in climate science, discussing innovations, challenges, and future directions. They categorized AI-based climate models based on methodologies (supervised, unsupervised, and reinforcement learning) and identified key challenges, including data quality issues and the transferability of trained models.

METHODOLOGY

This work takes structured approach aimed at improving climate prediction with the help of semi-supervised learning and hybrid AI-physics models for both data, methods of selecting models, training, evaluation as well as afterwards combination with physics-based models. Data is collected from high-reliability sources, including NASA's ERSST.v4 [7], NOAA Climate Data Records (CDR) [8], and ECMWF's ERA5 dataset [9]. The methodology is divided into various steps and is discussed below.

The collected data is filtered, normalized and feature selected for consistency. Two kinds of modeling are considered: supervised learning models (Deep Neural Networks DNNs and Gradient Boosting e.g., XGBoost) to model complicated climate dynamics, as well as hybrid AI-physics models that harness the strength of AI predictions with physics-based climate simulations for physical correctness and correcting biases.

The models are trained on historical climate data by splitting the data (as standard train:70%, valid:15%, test:15%) with hyperparameters tuned through grid search and cross validation. Following are the evaluation metrics for trained data: Root Mean Square Error (RMSE), Mean Absolute Error (MAE) and Coefficient of determination(R^2). To increase confidence in these models, an AI can be coupled with Numerical Weather Prediction (NWP) models and General Circulation Models (GCMs) by bias correction procedures and then use hybrid models for longer range climate forecasts. This structured methodology ensures a scientifically sound approach, combining AI efficiency with the robustness of physics-based models. Figure 1 is a visual representation of the research methodology.

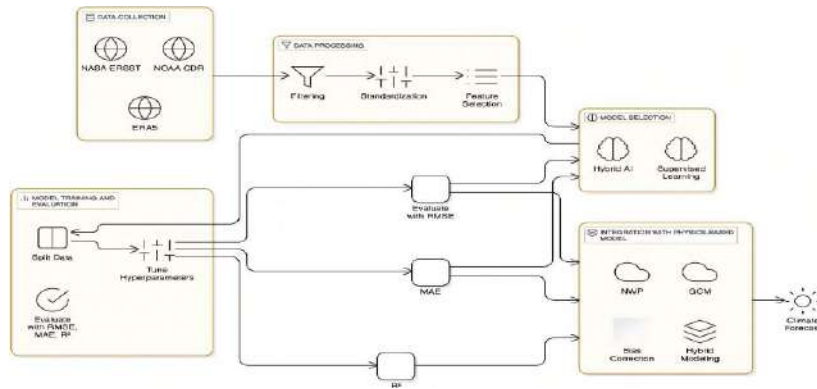


Figure 1: Basic Workflow

RESULTS AND DISCUSSION

The present study focuses on three dominant evaluation metrics in temperature and precipitation forecasts, classification quality for extreme events, and computational speed. These are done on the prediction accuracy (Mean Absolute Error — MAE and Root Mean Square Error — RMSE), as well as for the classification of extreme events via precision/recall/F1-score. Computational efficiency is further discussed in bringing down any processing time and energy consumption between different models. These parameters allow an extensive measure of engineering and applicability the AI based climate models with respect to traditional General Circulation Models (GCMs).

4.1 Temperature and Precipitation Prediction Performance

The results shown in Table 1 indicate that AI models significantly improve temperature and precipitation prediction accuracy compared to traditional General Circulation Models (GCMs). The table below summarizes the performance comparison in terms of Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

Table 1: Temperature and Precipitation

Model	MAE (°C)	RMSE (°C)	MAE (mm)	RMSE (mm)
GCM	2.1	2.8	6.2	8.4
XGBoost	1.5	2.0	4.3	5.9
LSTM	1.4	1.9	4.1	5.6
Hybrid AI-Physics	1.3	1.7	3.8	5.2

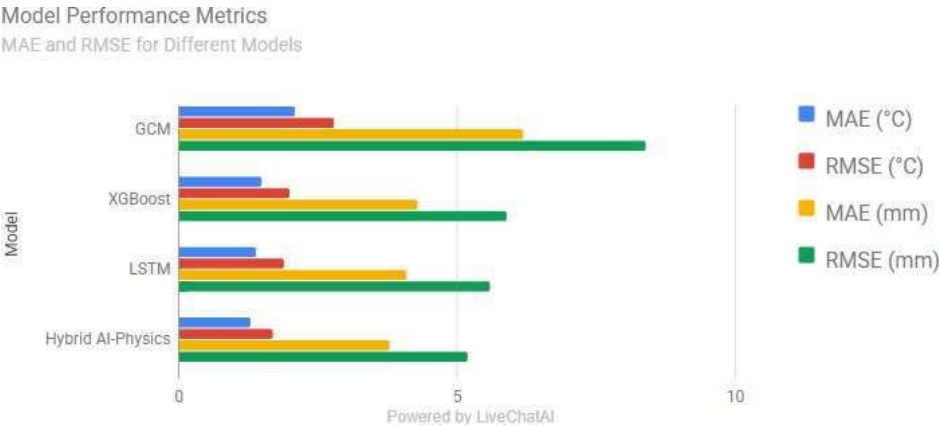


Figure 2: Performance Comparison with General Circulation Models (GCMs)

The results shown in Figure 2 that AI models reduce MAE and RMSE by approximately 30% compared to GCMs, with hybrid AI-physics models demonstrating the lowest error rates.

The LSTM and hybrid AI-physics models consistently outperform others in reducing prediction errors, demonstrating their robustness in climate modelling and is shown in Figure 3.

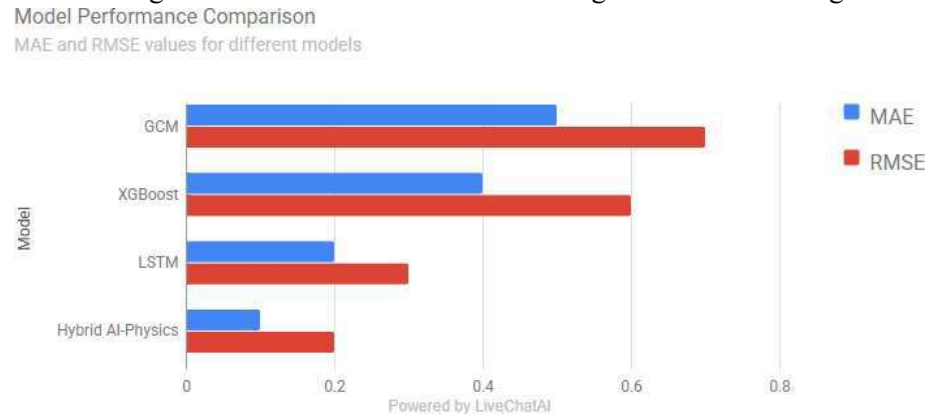


Figure 3: Graphical Comparison of AI vs. GCM Performance

4.2 Extreme Event Classification Performance

AI models improve the detection and classification of extreme weather events, with XGBoost

achieving the highest precision and recall scores. The following Table 2 summarizes model performance. The hybrid AI-physics model achieves the highest F1-score (0.89), indicating superior extreme event classification performance.

Table 2: Precision, Recall and F1 Score for various models

Model	Precision	Recall	F1-Score
GCM	0.72	0.69	0.70
XGBoost	0.88	0.85	0.86
LSTM	0.87	0.86	0.86
Hybrid AI-Physics	0.90	0.89	0.89

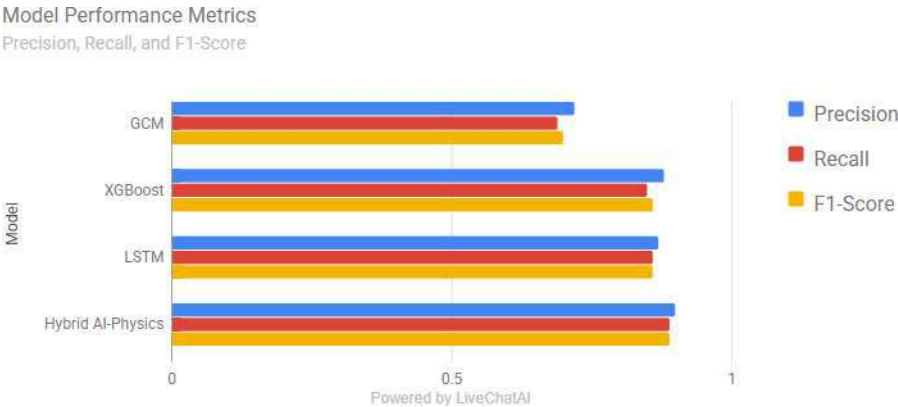


Figure 4: Graphical Representation of Extreme Event Classification

Figure 4 highlights the enhanced ability of AI models to classify extreme events more accurately than GCMs.

4.3 Computational Efficiency and Energy Consumption

Computational efficiency is crucial for large-scale climate modeling. AI models demonstrate substantial improvements in reducing processing time.

Table 3: Reduction in Processing Time

Model	Processing Time (hrs)	Energy Consumption (kWh)
GCM	24	500
XGBoost	12	320
LSTM	14	350
Hybrid AI-Physics	15	370

AI models reduce processing time by approximately 40% compared to GCMs, with XGBoost consuming the least computational resources and is presented in Figure 4.

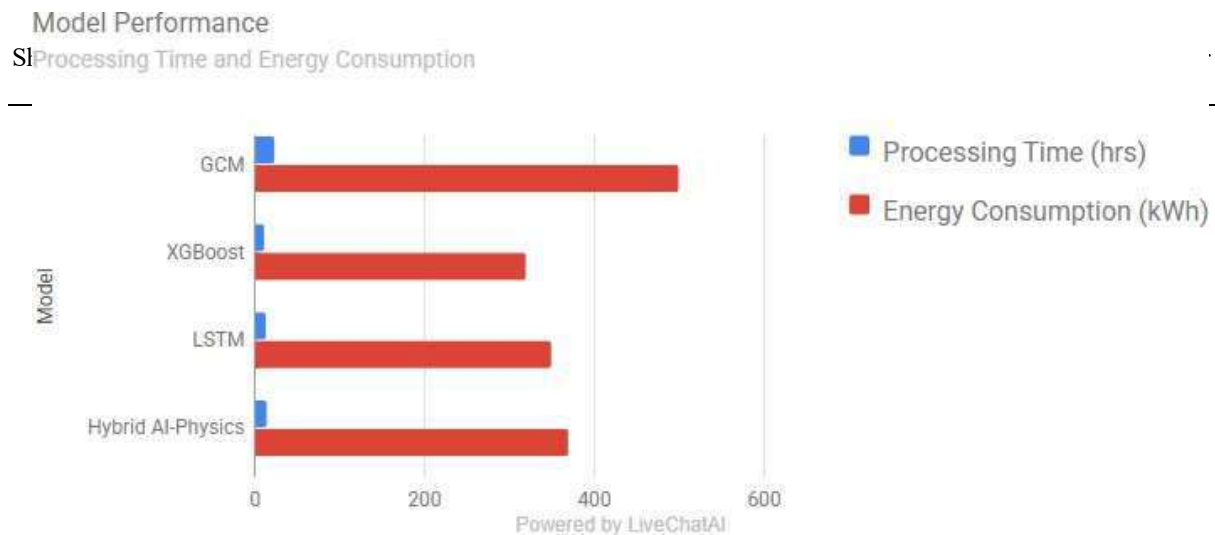


Figure4: Graphical Comparison of Computational Efficiency

CONCLUSION

The results of this study demonstrate that AI-based models significantly outperform traditional General Circulation Models (GCMs) in temperature and precipitation prediction, extreme event classification, and computational efficiency. The hybrid AI-physics model consistently achieved the lowest error rates in prediction accuracy and the highest F1-score in extreme event classification, highlighting its robustness in climate modeling. Additionally, AI models reduced processing time by approximately 40% while lowering energy consumption, making them more efficient for large-scale climate simulations. These findings underscore the potential of AI-driven approaches to enhance climate prediction accuracy and efficiency, ultimately contributing to improved climate risk assessment and decision-making.

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Securing Web Applications: A Machine Learning Approach for SQL Injection Threats

Prof.Nilima D.Bobade

PRMIT&R, Badnera

iEmail ID-ndbobade@mitra.ac.in

Dr.Vinit A.Sinha

PRMIT&R,Badnera

EmailID-vasinha@mitra.ac.in

Dr.S.S.Shereker

SGBAU,Amravat

Email IDswatishereksr2020@rediffmail.com

Abstract

The rapid evolution and widespread adoption of the internet have significantly transformed the world, leading to an increased number of cyber attacks. Cyber security has become one of the most critical challenges for society, incurring substantial financial losses annually. This research focuses on SQL Injection attacks the specific threat of on web applications, aiming to detect malicious queries designed to exploit vulnerabilities and access sensitive data. In recent years, the frequency of SQLi attacks has surged, posing severe risks to web application security. Attackers use SQLi to execute arbitrary SQL code, potentially gaining unauthorized access to databases, exfiltrating data, and compromising the integrity of web services. Despite various efforts to mitigate SQLi attacks, a comprehensive and effective detection mechanism remains elusive. This paper highlights the machine learning approach to identify and prevent SQLi attacks, providing a comparative analysis of various classifiers and their performance. The study underscores the potential of machine learning in enhancing web application security, offering adaptive and dynamic solutions to combat the evolving threat of SQLi attacks.

Keywords-SQL Injection, Threat Detection, Web Security, Machine Learning.

I. INTRODUCTION

The field of cyber security has experienced tremendous growth, addressing a variety of threats, including SQL Injection (SQLi) attacks, which are among the most significant dangers in the realm of web applications. SQLi attacks represent a persistent threat to database-driven applications, which exploit vulnerability to perform malicious SQL code. These attacks are particularly challenging to prevent because they can be easily executed by inserting malicious queries into input fields, causing immediate and severe damage.

SQL Injection attacks allow cybercriminals to compromise web applications, leading to unauthorized access to sensitive data, identity theft, and other malicious activities. Attackers manipulate SQL queries to bypass security measures, access backend databases, and perform actions such as data theft, deletion, or modification. Specially crafted SQL queries are used by attackers to exploit vulnerabilities, gaining control over the database and potentially the entire application. Cybercriminals target web applications to create extensive networks of compromised systems, which they can manage and exploit remotely.

Traditional SQLi detection techniques, such as signature-based and rule-based methods, may not be sufficient to detect modern and sophisticated SQLi threats. These conventional methods often struggle to keep pace with the evolving tactics employed by attackers. Therefore, there is a critical need for advanced detection mechanisms that can adapt to new attack patterns.[1] This research paper investigates and explores a machine learning based approach for recognizing and mitigating SQL Injection threats, focusing on developing a robust detection system capable of identifying various attack vectors with high accuracy. By leveraging natural language processing and advanced classification algorithms, this study aims to provide an adaptive and dynamic way out for defending against SQL Injection threats in web applications.

II. MACHINE LEARNING FRAMEWORK FOR SQL INJECTION DETECTION

In many studies, machine learning approach to detect SQL Injection threat has been extensively discussed. Various researchers have introduced diverse approaches for enhancement of accuracy and performance of SQL Injection threat detection. The proposed SQL Injection detection technique used supervised machine learning algorithm and is intended to detect SQL Injection as mentioned in Fig.1. First collected SQL Injection dataset SQLiV3 from Kaggle contains total 30864 query. The dataset contains 11347 malicious and 19517 benign query. The SQLiV3 dataset contains the categorization of different kinds of SQL Injection attack queries in the dataset as exposed in table 1 given below.

```
benign      19517
other       10326
tautology   637
union-based 366
piggybacked 18
Name: attack_type, dtype: int64
```

TABLE1.CATEGORIZATION OF SQLI

Data preprocessing performed on dataset for cleaning data. Feature are extracted from dataset using three different method 1) CountVectorizer 2)TF-IDF 3) Word2Vec from Word Embeddings. The data from dataset split into 80% training set and 20% testing set. SVM gives highest accuracy as compared with models such as Logistic Regression, Random Forest, Naïve Bayes and AdaBoost Classifier [12]. Model is then trained using SVM Classifier consistently initialized with the hyper parameters such as a linear kernel, enabled probability estimates and a fixed random state for reproducibility. Proposed model is tested and evaluated based on four different evaluation parameters for the three different ways of feature extraction and compared.

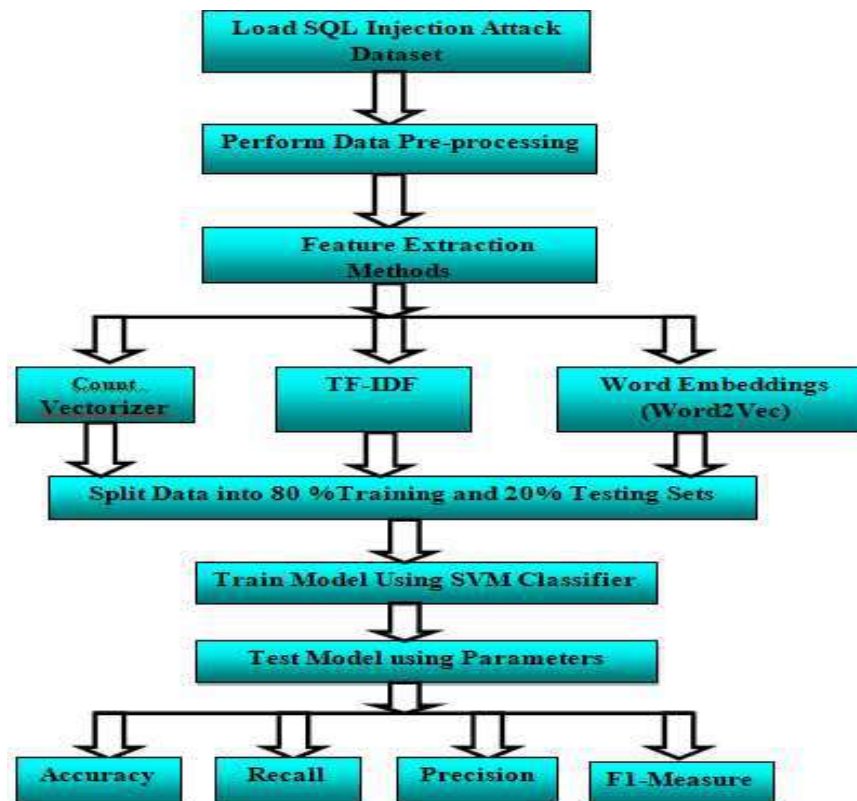


Fig.1. Proposed Approach for SQL Injection Attack Detection

III. EVALUATION TERM DESCRIPTION

Sr.No	Evaluation Term	Description
1	Accuracy	Overall Performance Of The Model
2	Precision	Percentage of classified SQL Injection which are actually SQL Injection
3	Recall	Effectiveness of model in detecting SQL Injection
4	F-Measure	Blend of precision and recall

Table 2: Description of evaluation terms

Sr No	Feature Extraction Method	Evaluation Parameters			
		Accuracy	Precision	Recall	F-Measure
1	Count Vectorizer	0.9917	0.9898	0.9876	0.9887
2	TF-IDF	0.9765	0.9907	0.9449	0.9673
3	Word2Vec	0.9872	0.9897	0.9753	0.9824

Table 3: evaluation of Proposed Techniques

IV. THE RESULTS AND DISCUSSION

The proposed technique is evaluated based on four parameters and shown in Table 3 .

V. ANALYSIS AND RESULTS

In this work, the author M. Hasan et al. evaluated a heuristic machine learning approach that combines static and dynamic analysis. Author developed system using MATLAB utilized well established dataset and 23 machine learning classifiers for training and testing . Based on performance metrics top five classifiers are chosen, and a GUI application is developed to facilitate easy testing of new SQL statements[4]. The author T.P.Latchoumi et al.

worked on machine learning algorithms , specifically the Support Vector Machine algorithm, for detection and

prevention of SQL injection threat. The SVM is trained with a dataset having both benign as well as malicious SQL queries. This training enables the SVM to generate a framework capable of distinguishing between benign and destructive queries. The system architecture includes a real-time monitoring component that applies the SVM model to incoming SQL queries to identify and block potential attacks[5]. The author Dr. Sandeep Kumar et al. used the “SQL1” and “SQL2” datasets from Kaggle, containing labeled SQL queries. Applied Linear Discriminant Analysis (LDA), Logistic Regression (LR), and K-Nearest Neighbors (KNN) supervised learning algorithms. Implemented a stacking technique where predictions from base models (LDA, LR, KNN) are used as inputs to a meta model (LR) and highlights the potential of machine learning algorithms combined with stacking to get better security[6]. Author B. Gogoi et al. developed Apache web server module that intercepts HTTP/HTTPS requests and inspects them for SQLIA payloads. The system parses HTTP requests into tokens and converts them into numerical data using word embedding (word2vec). Synthetic Minority Oversampling Technique is used to handle imbalance of class in the dataset (SMOTE).Trains classifiers using Support Vector Machines (SVM) with both linear and non-linear models. Hyperparameters are optimized using Grid Search. Precision ,F1-Score and recall parameters are used for the evaluation of system and achieved a high detection rate for all parameters to 99.9%.10-Fold cross validation is used to prevent over fitting [7]. Author Prince Roy et al tested machine learning classifiers to detect payloads of SQL injection threat. The authors used dataset from the Kaggle for training and testing, a variety of classifiers, including Naïve Bayes .Logistic Regression,

XGBoost, Naive Bayes, AdaBoost, and Random Forest[8]. Author J. M. Alkhathami et al. suggested the model which is trained on a dataset of SQL queries, both normal and injection attacks, collected from publicly available repositories. Data preprocessing involves cleaning the dataset and converting text values to numerical features using the countVectorizer class. Four algorithms are used for training: KNN, MNB, DT, and SVM. The models are assessed using metrics such as accuracy, F-Value, confusion matrix and AUC-ROC curve. SVM achieved the maximum accuracy at 99.42%, after that DT at 99.4%, MNB at 97.09%, and KNN at 92.45%[9]. Author Ahmed Abadulla Ashlam et al. proposed methodology involves a arithmetical, systematic, and experimental approach to study the harmful effect of SQLi attacks. It includes the collection of a dataset then data preprocessing steps such as tokenization and stop words removal, feature extraction, and the development of a classification model using techniques like CountVectorizer and Support Vector Machine [10].

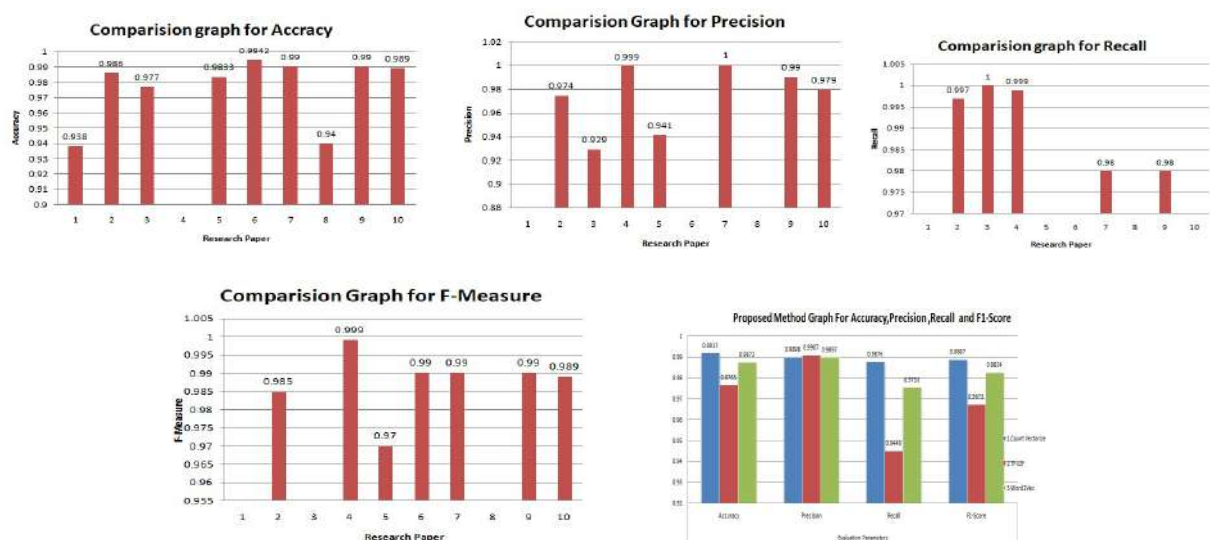
Author Yohan Muliono et al. employed Support Vector Machines (SVM) for grouping and predicting SQL Injection attacks .Focuses on SQL query conditions and program fragments vulnerable to attacks[11]. Author A. Zhumabekova et al. Worked on algorithms using machine learning , include Naïve Bayes, Decision Tree,Support Vector Machine, XGBoost, Random Forest and CastBoost . Used the dataset for experiments was sourced from Kaggle, containing 68,553 items split into 70% training set and 30% testing set .The first step is data preprocessing, involve collecting, cleaning, normalizing, scaling, and vectorizing the data with the help of the tf-idf metric.The processed data is used to train and test a variety of machine learning models to evaluate performance to detect SQL injections. The results are assessed using parameters such as accuracy, F1-Score, precision and recall values[12]. Author Shengyuan Zhang et al. The developed a method, called the Feature Ratio Method, focuses on six main features.Regular expressions and fuzzy matching used to extract these features and improve detection accuracy [13]

Sr. No	Name of Author	Classifier/Algorithm	Parameters			
			Accuracy	Precision	Recall	F-measure
1	M. Hasan et al.,2019 [2]	Ensemble Boosted Trees, Ensemble Bagged Trees, Linear Discriminant, Cubic SVM, Fine Gaussian SVM	0.938	-	-	-
2	T.P.Latchoumi et al.,2020 [3]	SVM	0.986	0.974	0.997	0.985
3	Dr. Sandeep Kumar et al.,2021 [4]	LR,KNN,LDA and Meta model	0.977	0.929	1.0	-
4	B. Gogoi et al.,2021 [5]	Linear SVM , Non linear SVM	-	0.999	0.999	0.999
5	P. Roy et al.,2022 [6]	Logistic Regression,AdaBoost,Random Forest,XGBoost,Naïve Bayes	0.9833	0.941	-	0.970
6	J. M. Alkhathami et al.,2022 [7]	KNN,MNB,Decision Tree,SVM	0.99	-	-	0.99
7	Ahmed Abadulla Ashlam et al.,2022 [8]	SVM, Decision Tree, KNeighbors, AdaBoost, Random Forest, SVM with Performance analysis and Iterative optimization of the SQLI Detection Model i.e PALOSDM	0.99	1	0.98	0.99
8	Yohan Muliono et	SVM	0.94	-	-	-

	al.,2022 [9]					
9	A. Zhumabekova et al.,2023 [10]	Decision Tree, Random Forest, XGBoost, and CatBoost	0.99	0.99	0.98	0.99
10	Shengyuan Zhang et al.2023 [11]	LR,KNN,RF,SVM,NB,CNN	0.989	0.979	-	0.989

TABLE 4 :COMPARISON OF SQL INJECTION DETECTION PARAMETERS

VI.GRAPHICAL REPRESENTATION OF VARIOUS PARAMETERS



VII. CONCLUSION

This paper explores SQL Injection and presents detection technique using count vectorizer method for feature extraction with SVM classifier achieving an accuracy of 0.9917 which is best than TF-IDF method of feature extraction with SVM and Word2vec IDF method of feature extraction with SVM & Performance analysis of various parameters of existing methodology. Additionally, the study provides a detailed analysis of SQL Injection parameters, offering insights for researchers. The results are compared with the a variety of methods based on Performance metrics.

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Utilization of Plastic Waste in Pavement Block by Partially Replacement of Fine Aggregate

Vaibhao V. Maind¹, Harshal G. Mundle², Anmol W. Dongre³

1,2,3) Assistant Professor, Department of Civil Engineering, G. H. Raison College of Engineering and Management, Nagpur-440016 India

Abstract

The rapid growth of urbanization and industry has resulted in a significant increase in plastic waste generation. This poses a major environmental challenge due to the limited biodegradability of plastics. One potential solution is exploring the use of plastic waste in construction materials. This study investigates the theoretical utilization of waste plastic in pavement blocks as a partial replacement for fine aggregate. The focus is on the feasibility and potential benefits of incorporating recycled plastic into pavement block production. Highlights the growing issue of plastic waste and its negative environmental impact. Introduces the concept of using recycled plastic in pavement blocks as a sustainable waste management strategy. Briefly outlines the proposed approach for incorporating plastic waste into pavement blocks. This might involve shredding or melting plastic into a suitable form for use as a partial replacement for fine aggregate in the pavement block mix. Discusses the potential advantages of using recycled plastic in pavement blocks. For experimental study firstly, ordinary paver block was prepared then the fine aggregate was partially replaced with plastic waste by 10%, 15% and 20%. The main parameter considers in this paper to compare the experimental study were compressive strength and cost of ordinary pavement block and plastic pavement block.

Keywords - Ordinary pavement block, fine aggregate, plastic waste, compressive strength, cost, plastic pavement block

INTRODUCTION

We know plastic waste has become a major environment from large-scale pollution and ecological imbalance of ecosystem services, due to the accumulation of plastic waste on land and ocean. To address this problem, researchers and engineers explored a new ways to incorporate plastic waste into construction materials. One of them is the incorporation of plastic waste in the manufacture of pavement blocks by partially replacing the fine aggregate (sand). It is a huge step towards sustainable building and reducing plastic waste.

The concept behind utilizing plastic waste in paver block production is to replace a certain percentage of fine aggregate (sand) with shredded plastic waste. By doing so, the demand for virgin sand is reduced, conserving natural resources and minimizing the environmental impact of sand extraction. This approach also provides an opportunity to recycle and reuse plastic waste that would otherwise end up in landfills or oceans. To implement this method, plastic waste is collected and segregated based on its type and quality. Common sources of plastic waste include polyethylene terephthalate (PET) bottles, polyethylene (PE) bags, and other plastic materials. The collected plastic waste is then cleaned to remove any contaminants and shredded into small pieces. These shredded plastic particles are then mixed with the concrete mix as a partial replacement for fine aggregate.

The production of pavement blocks from plastic waste delivers multiple benefits. Initially this process helps lessen environmental plastic waste accumulation while supporting waste management initiatives. The utilization of plastic waste in pavement blocks production reduces environmental extraction of natural resources like sand. This practice not only supports but also promotes the creation of sustainable construction materials that find uses in various applications like pavements, walkways, and driveways. Moreover, pavement blocks made from plastic waste demonstrated excellent mechanical properties. Studies indicate that the addition of plastic waste enhances the strength and durability of pavement blocks. Plastic particles serve as a filler material to enhance the overall compactness and reducing the porosity of the blocks. This results in increased resistance to cracking, better load-bearing capacity, and improved durability of the paver blocks.

METHODOLOGY

Site Visit for Information Collection

1. Proportioning:

- This is a crucial step as it determines the final properties of the paver block. Manufacturers rely on precise calculations and may use computer software to optimize the mix design.
- Raw Materials:
 - a) Cement: The binding agent that holds everything together. Different types of cement can be used based on desired strength and setting times.
 - b) Sand: Fine particles that fill gaps between larger aggregates and contribute to the overall strength and texture.
 - c) Aggregates: Crushed stone, gravel, or recycled concrete that form the bulk of the paver and provide strength and stability. Size and gradation (particle size distribution) of aggregates are important factors.
 - d) Water: Activates the cement and allows it to bind with other ingredients. The amount of water used significantly affects the final strength and workability of the mix.
 - e) Colour Pigments: Added to achieve desired aesthetics. Different pigments can be used to create a wide variety of colours.

2. Mixing:

- Large industrial mixers ensure thorough and consistent mixing of all ingredients.
- The mixing time plays a role - undermixing can lead to weak pavers, while overmixing can trap air and reduce strength.
- Additives: Manufacturers may incorporate specific admixtures to enhance specific properties like workability, setting time, or water resistance.

3. Compacting:

- This step removes air pockets and ensures a dense, strong paver block. Two main methods are used:
 - a) Hydraulic Presses: Apply high pressure to the mixture in the moulds.
 - b) Vibration: Moulds are subjected to high-frequency vibrations that densify the mix and eliminate air voids. This method often results in smoother surfaces.

4. Curing:

- Newly formed pavers are still weak and require a controlled environment for proper hardening.
- Curing chambers maintain specific temperature and humidity levels to optimize the hydration process (chemical reaction between cement and water) for several days.
- Proper curing allows the pavers to achieve their full strength and durability.

5. Demolding and Finishing:

- Once cured, the pavers are removed from the moulds.
- This stage may involve:
 - Trimming: Edges might be precisely cut for consistent dimensions.
 - Surface Finishing: Depending on the desired look, surfaces may be polished for a smooth finish or textured with patterns using special tools.
- Colour Enhancement: Some manufacturers may apply a surface coating to enhance colour or add a protective layer.

6. Quality Control:

- This ensures the manufactured pavers meet industry standards and client specifications.
- Pavers undergo rigorous testing for properties like:
 - Compressive Strength: Ability to withstand pressure without breaking.
 - Water Absorption: How much water the paver can absorb, which can affect freeze- thaw resistance.
 - Freeze-Thaw Resistance: Ability to withstand cycles of freezing and thawing without cracking.
- Dimensional Accuracy: Ensures pavers fit together seamlessly during installation. By following these steps with careful material selection and quality control, manufacturers can produce high-quality paver blocks that are strong, durable, and aesthetically pleasing.

- Precise moulds and proper compaction during manufacturing are crucial for achieving dimensional accuracy.
- Consistent material quality, including the size and gradation of aggregates, also plays a role.
- Strict quality control procedures throughout the production process help ensure each paver block meets the dimensional specifications.

Material Collection

1. Cement: PPC 53 grade is a type of Portland Pozzolana Cement (PPC) that is commonly used in construction. It is made by blending ordinary Portland cement (OPC) with pozzolanic materials, such as fly ash or volcanic ash. The pozzolanic materials react with the calcium hydroxide in OPC to form a stronger and more durable cement paste. PPC 53 grade cement has a compressive strength of 53 MPa (megapascals) at 28 days, which makes it suitable for a wide variety of construction applications.

2. Aggregate: In the context of paver blocks, a 12mm aggregate refers to crushed stones or gravel with a maximum size of 12 millimeters (approximately half an inch). These stones act as a filler within the paver block mix, providing several key functions such as Strength and Stability, Texture and Drainage and Workability and Cost-Effectiveness

3. Sand: Fine aggregate river sand is a type of sand, typically with a grain size of 0.06 millimeters to 4.75 millimeters, obtained from the natural erosion of rocks by rivers. It's a common material used in construction applications, particularly as part of concrete mixes.

- **Size and Gradation:** As a fine aggregate, river sand particles are smaller than coarse aggregate (gravel and crushed stone). The size distribution (gradation) of the sand particles is important for creating a well-packed concrete mix with optimal strength.
- **Benefits:** River sand offers several advantages in construction, including:
- **Workability:** Fine particles create a more workable concrete mix, improving flow and ease of placement.
- **Strength:** When properly proportioned with other concrete ingredients, river sand contributes to the overall strength and durability of concrete.
- **Availability:** Historically, river sand has been a readily available and cost-effective material.

4. Plastic: HDPE (High-Density Polyethylene) is a commonly used plastic with a recycling code of #2. It's found in a variety of products, but unlike some other plastics, HDPE is actually good news when it comes to waste management. Here's why:

Recyclability: HDPE is one of the most readily recyclable plastics. Many recycling programs accept HDPE waste, making it relatively easy for consumers to dispose of it responsibly.

During recycling, HDPE is sorted, cleaned, and then melted down into pellets. These pellets can then be used to manufacture new HDPE products, reducing reliance on virgin plastic.

Benefits of Recycling HDPE Waste:

- **Reduced Landfill Waste:** Recycling HDPE diverts plastic from landfills, where it can take hundreds of years to decompose and potentially leach harmful chemicals into the environment.
- **Conservation of Resources:** By using recycled HDPE, we reduce the need to extract and process virgin plastic from fossil fuels, conserving natural resources.

Volume Batching

Table 1. Material Required for Volume Batching

Material	Ordinary Paver Block	Paver Block (10%)	Paver Block (15%)	Paver Block (20%)
Cement	1.61 kg	1.61 kg	1.61 kg	1.61 kg
Fine Aggregate	1.34 kg	1.24 kg	1.13 kg	1.07 kg
Coarse Aggregate	2.46 kg	2.46 kg	2.46 kg	2.46 kg
Plastic Waste	-----	19.39 gm	29.08 gm	38.78 gm

PREPARATION OF TEST SPECIMENS

For the preparation of test specimens, plastic is crushed by crushing machine. When the plastic is crush properly well and came in powdered form then it used for the next process of cube casting. Then the prepared mould is cleaned and then the mix proportion is transferred in mould. After transferring mixture, it shakes properly that's why reducing the internal pores present in it then it properly settled. Then the blocks are allowed to dry for 24 hours so that they harden. After drying the paver block is removed from the moulds and ready for next process i.e. curing.

1. Casting: Volume batching is a method used in concrete mixing where concrete ingredients are measured by volume rather than by weight. This approach is commonly used in small construction projects or in situations where the precise weighing of materials may not be practical. While volume batching can be a simple approach for small jobs, weigh batching is generally preferred for its accuracy and consistency, especially in critical projects.

2. Mixing: To create a new material with desired properties. For example, concrete is made by mixing together cement, sand, gravel, and water. The cement acts as a binder, holding the sand and gravel together to form a strong, solid material.

3. Moulding: To create a new material with desired properties. For example, concrete is made by mixing together cement, sand, gravel, and water. The cement acts as a binder, holding the sand and gravel together to form a strong, solid material. Gently tap the sides of the mold with a rubber mallet or vibrate it slightly to help settle the mix and release air bubbles. Air bubbles can create voids in the finished paver, reducing strength, a concrete becomes denser and more compact. This leads to a stronger paver, just like a flat soda can is sturdier than a full, pressurized one.

4. Demoulding: Demoulding the paver block gently tapping the sides of the mold with a rubber mallet. This can help loosen the paver from the mold. Avoid using a metal hammer, as it can damage the mold or chip the paver.

5. Curing: Curing allows the concrete mix in the paver block to complete the hydration process. During hydration, the cement reacts with water to form strong calcium silicate hydrate (C-S-H) crystals. These crystals bind the other components of the concrete together, giving the paver block its strength.

6. Testing: Compressive testing is a crucial step in ensuring the quality and suitability of paver blocks for their intended use. A UTM (Universal Testing Machine) is a versatile machine used to perform various mechanical tests on materials, including compression tests for paver blocks.

RESULTS AND DISCUSSION

Table 2. Compressive Strength obtained in 28 days

Sr. No.	Cement (kg)	Sand (kg)	Aggregate (kg)	Plastic Waste (gm)	Paver Block Samples	Compressive Strength (N/mm ²)	Average (N/mm ²)
1	1.61	1.34	2.46	-	Sample 1	28.2	28.2
					Sample 2	28.2	
					Sample 3	28.3	
2	1.61	1.24	2.46	19.39	Sample 1	18.5	18.9
					Sample 2	19.1	
					Sample 3	19.2	
3	1.61	1.13	2.46	29.08	Sample 1	24.8	25.1
					Sample 2	25.2	
					Sample 3	25.2	
4	1.61	1.07	2.46	38.78	Sample 1	18.2	18.2
					Sample 2	18.2	
					Sample 3	18.2	

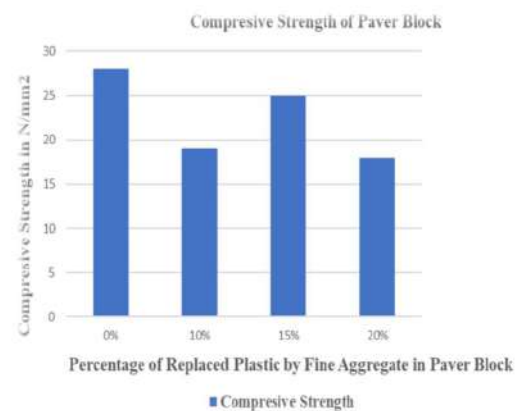
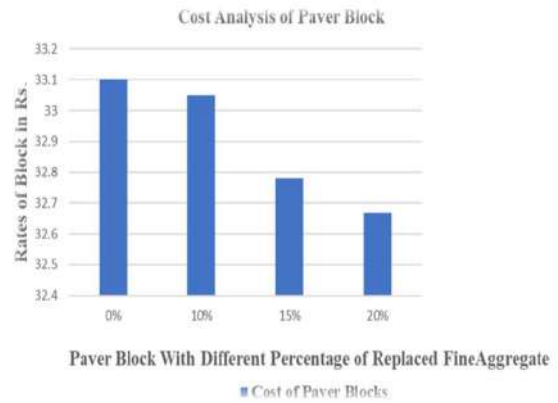


Figure 1. Compressive Strength obtained in 28 days

Table 3. Cost Analysis

Material	Price of Ordinary Paver Block	Price of Paver Block (10%)	Price of Paver Block (15%)	Price of Paver Block (20%)
Cement	Rs. 16.1	Rs. 16.1	Rs. 16.1	Rs. 16.1
Fine Aggregate	Rs. 4.70	Rs. 4.42	Rs. 4.03	Rs. 3.81
Coarse Aggregate	Rs. 12.3	Rs. 12.3	Rs. 12.3	Rs. 12.3
Plastic Waste	Rs. 0.00	Rs. 0.23	Rs. 0.35	Rs. 0.46
Total Price of Paver Block	Rs. 33.10	Rs. 33.05	Rs. 32.78	Rs. 32.67

**Figure 2. Cost Analysis**

There are several limitations in using recycled plastic to substitute fine aggregate for paver blocks:

1. Evidence suggests that paver blocks may become weaker when plastic is used instead of fine aggregate. [1, 2, 3]. Recommended replacement percentages, depending on the type of plastic, range from 10 to 20%. What is ideal? If this is exceeded, the compressive and flexural strength of pavers can be significantly reduced due to their weight and stress tolerance.
2. Plastic aggregates may make the paver mix less pliable. Making the pavers more moldable and achieving proper compression may result in a decrease in overall quality and consistency.
3. Fire safety concerns should be considered when dealing with plastic pavers due to their flammability. While most pavers are not exposed to direct fire, they may have an impact on the spread of fire in a nearby area.
4. Pavers durability with plastic is still a subject of ongoing research. There is a worry that the plastic in the pavers may eventually degrade due to exposure to sunlight and harsh weather conditions, which could impact their functionality.
5. Different plastic types can affect the properties of paver blocks. This result can also be influenced by the processing method of the plastic (such as size, cleanliness, grinding process).

Even with these limitations in mind, using recycled plastic to make paver blocks is a promising way to go about waste management and sustainable building. A valuable technique for decreasing plastic waste can be achieved by taking into account the amount of plastic and its composition, as well as potential issues with workability. This technology is backed by numerous scientific studies.

CONCLUSION

The compressive strength of the paver blocks increases as the amount of plastic additive decreases. This suggests that plastic additive has a negative impact on the compressive strength of paver blocks.

Specifically, the average compressive strength across the three samples is 28 N/mm² for the 0% plastic additive group, 19 N/mm² for the 10% plastic additive group, 25 N/mm² for the 15% plastic additive group, and 18 N/mm² for the 20% plastic additive group.

1. Paver blocks, which are made up of cement, sand, and aggregate, have the most significant impact on plastic binding. The sand and aggregate particles are held together by cement, which acts as a binder. In general, plastic additives are not as strong as cement and may not be able to bond with other components. The paver block's overall strength is reduced, making it more prone to cracking or crumbling when pressured.

2. Air voids or pockets may be introduced into the paver block during the mixing process when plastic is added, leading to internal defects. The voids function as stress concentrators, which means they channel the pressure applied to the block into specific points. Asymmetric pressure distribution can result in premature failure.

3. Hydration is the chemical process that increases cement's strength. Water interacts with cement particles during hydration, creating an interconnected network of crystals. The use of plastic additives may disrupt this process by reducing the amount of water available for hydration or by altering the crystal formation process. The paver block is weakened by a weaker hydration network.

4. Reliability with other materials: The type of plastic used and its compatibility with cement, sand or aggregate is another factor. The paver block can exhibit weak plastic interfaces due to the inability of certain materials to bond with each other.

Overall, the addition of plastic weakens the binding between components, introduces potential voids, and disrupts the cement hydration process, all of which contribute to a decrease in the compressive strength of paver blocks.

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A COMPREHENSIVE SURVEY ON DETECTION OF VIDEO TRANSITIONS

Ayushi P. Yeram

Research Scholar, Department of Electronics
and Communication Engineering
G H Raisonni University, Amravati
Email: yeramayushi26@gmail.com

Dr. Shrikant P. Chavate

Assistant Professor, Department of Electronics
and Communication Engineering
G H Raisonni University, Amravati
Email: shrikant.chavate@ghrua.edu.in

Abstract:

The most significant step in the study of video processing is video shot boundary detection. As a result, video retrieval is now focused on browsing, indexing, and contents. Along with important performance evaluation metrics, the described study reviews several approaches and strategies used to accomplish the shot boundary detection. In addition to this, the study mainly focuses on gradual type of transition i.e. wipe and dissolve. This clarifies many preprocessing methods, feature extraction strategies, similarity calculation strategies used for wipe and dissolve transitions. The results of several methods are shown in terms of datasets used, precision, recall, and F1 score comparison.

Keywords: Walsh-Hadamard Transform, Speeded-Up-Robust Feature, Dynamic Mode Decomposition, Convolutional Neural Network, Squared Tchebichef-Krawtchouk Polynomial, Kirsh directional derivative, Singular Value Decomposition, Sparse Coding

Introduction:

Every video stream is made up of several shots, each of which is a series of frames captured by a single camera. Changing from one camera to another is necessary to move from one shot to the next. Scene change or shot boundary detection is the term used to describe the identification of these transitions. It is the initial stage of any system that analyzes video streams. Multimedia gadgets, advanced communication technology and inexpensive storage devices are reasons behind the recent surge in the quantity of videos accessible online. These videos are merely text annotated and stored in databases. The way videos are stored in databases makes content based video browsing and retrieval ineffective. The necessity for automated video structure analysis is highlighted by the extent and volume of information found in video databases. Thus, one of the important method for viewing and retrieving videos is the shot boundary detection (SBD). Images with rich information are used in content based video indexing and retrieval since SBD seeks to identify transitions and the boundaries between successive images.

A video shot is a collection of uninterrupted visual sequences continuously captured by a single camera. Videos are basically hierarchical in structure. Meaningful scenes are created by combining video shots, and a compilation of these scenes is called a video. The distance between two shots is known as a shot boundary. A video shot transition can be of two main types i.e. Abrupt (Cut) and Gradual. Gradual transitions are further classified into Fade in/out, Dissolve and Wipe.

1. **Abrupt or Cut** : This kind of shot boundary occurs when one shot suddenly transitions into another. When the final frame of one shot is followed by the opening frame of the subsequent shot, this is known as a shot cut.
2. **Fade**: the image's frames progressively transition to or from black.
3. **Dissolve**: the first shot's frames are progressively transformed into the second shot's frames.
4. **Wipe**: the first shot's frames are progressively transferred into the second shot's frames, either vertically or horizontally.

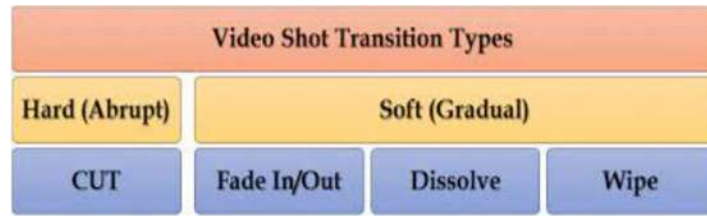


Fig.1 Types of video shot transitions

Compared to a gradual transitions , it is very simple to distinguish between an abrupt or hard transitions because of the fact that gradual transition (GT) incorporates a number of unique editing effects. Furthermore, motions of the camera, lights, and objects can provide incorrect results for the detection of shots.

The shot boundary detection process described in fig.2 include segmenting the video into many frames, preprocessing which includes feature extraction, similarity matching or creation of continuity function and finally determining the shot boundary. It can also be categorized as either abrupt or gradual. The frames needed for preprocessing are provided by the video segmentation. Feature extraction is part of the preprocessing step. A number of feature descriptors are accessible, and they can be in RGB, HSV, or LAB formats. Each of the aforementioned characteristics has pros and cons of its own. One or more descriptors can be used for preprocessing, depending on the particular histogram.

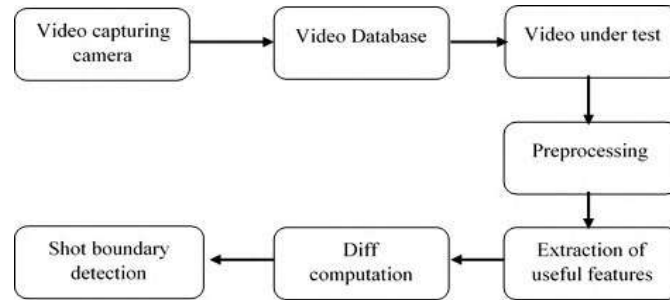


Fig.2 Steps in SBD

Walsh Hadamard Transform, Fuzzy Logics, Convolutional Neural Network, High Level Fuzzy PetriNet, and other techniques helped to achieve the impressive results with reference to literature on video processing research, particularly in shot boundary detection. Each of the aforementioned factors contributes to a high Precision and Recall score; as a result, the F1 Score, which is a combined value, aids in evaluating the methodology. As may be observed in detail, there are compromises made between user-oriented and complex approaches, execution speed and accuracy, etc.

The datasets that are used to test SBD algorithms are important and need to be carefully chosen and examined. Because researcher datasets are constrained by availability, representativeness, and ground truth availability, standard datasets are suitable for review. Furthermore, researchers' datasets aren't always accessible. Sometimes domain or website problems cause these datasets to be inaccessible. Results may be skewed since not all datasets are representative and cover all transition types that need to be assessed. Since ground truths are not linked to researchers' datasets, other researchers find it challenging and time-consuming to construct these datasets.

The National Institute of Standards and Technology co-sponsors the TRECVID review. TRECVID was founded to analyse and benchmark SBD tasks, and it has helped to enhance SBD algorithms. The most well-known datasets of TRECVID for SBD are TRECVID 2001, TRECVID 2005, TRECVID 2006, and TRECVID 2007. The institute provides these datasets for free after submitting a designated application form. Specifically, the TRECVID 2001 dataset contains hard,dissolve, and fade transitions. TRECVID 2007 is primarily developed for abrupt transitions. The TRECVID 2005 and 2006 datasets are regarded the most difficult because to the variety of their transitions. Also, RAI dataset, some random movie clips, TV and News shots created with Adobe Premiere 4.2.1, Video data from digitalized TV programs are also used for some research papers.

Review of Literature:

The video shot boundary is a fundamental aspect of video processing research for various purposes. With relation to the several approaches were used to implement the transitions indicated here. Raja Sujana M.[1] proposed a novel symmetry shot boundary detection system that maximizes detection accuracy by analysing a video's transition behaviour and segmenting it initially into primary and candidate segments using the color feature and the local adaptive threshold of each segment. The cut and gradual transitions are then fine-tuned from the candidate segment using Speeded-Up Robust Features (SURF) derived from the boundary frames, which reduces computational complexity.

The research provides a shot boundary identification system that uses both global and target features. The RGB color histogram features from the video frame are extracted. Then, using GMM, recognize foreground objects in video frames and extract their SIFT features. When compared to other algorithms, the feature fusion algorithm's recall and accuracy have improved, and it has resolved the issue of ignoring the target features. The drawback is that the algorithm will detect a certain video more effectively [2]. S. Tippaya used a multi-modal visual features-based SBD framework to evaluate visual representation behaviours using discontinuity signals. A candidate segment selection method is chosen that does not need a threshold computation. Instead, they use the cumulative moving average of the discontinuity signal to identify shot boundaries and skip non-boundary frames. Transition detection distinguishes between cut and progressive transitions, such as fade-in/out and logo occurrence [3].

LaxmiPriya [4] introduced a WHT method for the SBD in which the WHT kernel matrix was used to extract the features of edge, color, texture, and motion. As a result, matching can be examined for different extracted video frames, and the continuity function can then be used to design the system that determines the boundary detection. The author[5] has given an novel approach to identify shot boundaries in video sequences by employing orthogonal vectors to extract edge strength from frame blocks. Additionally, a transition recognition method is also provided that can recognize shot and non-shot borders in video clips.

Machine learning approach is mostly used to classify the transitions in shot and non-shot categories. SVM is the most preferred classifier for this purpose. Zinah N. Idan [6] used the Squared Tchebichef-Krawtchouk polynomial (STKP) for feature extraction with SVM as classifier with the disadvantage that this method is used only for cut transitions. Separate detection algorithms have also been constructed [7] using convolutional neural networks, taking into account the types of transitions as abrupt and gradual. In order to classify clips into distinct shot change types, abrupt transitions are detected using a combination of color histogram and deep features, whereas gradual transitions are discovered using 3D CNN.

Fuzzy logic is another approach is used in [8] which used fuzzy system as a classifier for abrupt and gradual transition detection with the use of genetic algorithm (GA) as an optimiser. This method is advantageous in computational time but lacks as shot boundaries increases with rise in GA optimization. R. Dadashi [10] also used fuzzy rule based approach for automatic cut transition detection in which no threshold or other parameter is used.

D. Chakraborty [11] proposed wipe scene detection method which used the linear regression and inflated 3D-convolutional neural network (I3DCNN). The suggested approach performs better than current wipe scene change detection techniques in terms of quick processing and a variety of object-camera motion-based wipe transition identification tasks. Feature fusion and clustering technique is used in [12] which combined SURF and fingerprint features and K-means is used to cluster the fused features. The experimental findings demonstrate the excellent accuracy and time efficiency of the suggested approach, particularly for long shot detection and gradual shot boundaries. Ye Yuan et al. [13] proposed dynamic mode decomposition (DMD) based video shot detection which was called as universal method for shot transitions like hard cuts fades and dissolves. It is determined whether the video's content is connected by obtaining the amplitude of the temporal background mode from the modes that DMD extracted. The amplitude of the temporal background mode abruptly varies if the content is from distinct shoots, allowing us to determine the shot boundaries. In the meanwhile, the approach successfully lowered the rate of error shot boundary identification when the foreground object (or camera) moved more quickly or when the illumination changed dramatically.

Kirsh directional derivative based shot boundary detection is used in [14] in which Kirsh operator gave accurate edge localization property for feature extraction provided accuracy and efficiency to the work. J. Li [15] used sparse coding technique for detecting shot boundary which explored dictionary learning for keyframe selection and video summarization. Singular value decomposition (SVD) is another accurate for detecting hard and

gradual shots given in [16] which used different mathematical interpretations and theorems. R.Hannane [17] proposed SIFT point distribution histogram method which is novel, fast and detects shot boundaries for different levels of illumination, camera operations including zoom in, zoom out, camera rotation and motion effects. Tejaswini Kar and Priyadarshi Kanungo [18] suggested a cut detection method using Weber features, which performed well in light environments but produced erroneous matches in dark ones.

Methodology:

As noted in the literature study, numerous efficient strategies have been created for abrupt as well as gradual (fade in/out, wipes and dissolves) type of transitions. The comparative analysis focuses on detection of gradual type of transitions in terms of accuracy, computation speed, and simple/complexity techniques. Video frame extraction and segmentation are the initial steps in the general procedure. Next, preprocessing consists of extraction of features. Similarity calculation is to be done and then the shot boundary is detected.

Literature reviewed here presented different methods and algorithms to detect the gradual type of transitions like Sasithradevi A. et al. [20] provided multiboundary classification (Grouped gradual transitions, cut transitions and no transitions) with the help of new pyramidal opponent color shape model and bagged tree classifier but on other side wipe analysis was insufficient. Colour based histogram and SURF feature descriptor based study [1] investigated gradual transitions but lacked to identify the particular type of that transition. Multimodal visual features based method [3] gave good performance to identify cut and gradual transitions (fade in/out, logo occurrence) but sensitive to camera object motion. A. Prabavathy [9] proposed fuzzy rule and histogram difference based system which calculated number of dissolves and wipes. Shen et al. [19] proposed HLFPM model based on machine learning, SURF keypoint matching and histogram difference for gradual transitions which removed false detections obtained by existing methods but having inconsistent recall value. FFCT is the fast shot boundary detection technique which is based on separable moments and SVM [6] provided with the best accuracy and time efficiency in detection of gradual shot boundary and long shot detection. S.Chavate et al. [23] used integration of WHT and DTCWT for feature extraction and by using this method much accurately detected fade in/out transitions with reduction in false alarms.

The recall, precision, and F1 score are introduced as evaluation metrics in accordance with the relevant shot boundary detection investigations in order to confirm the efficacy of the suggested method. These parameters are specified by:

$$\text{Precision (P)} = \frac{\text{No. of correct transitions reported}}{\text{Total no. of transitions reported}} \times 100$$

$$\text{Recall (R)} = \frac{\text{No. of correct transitions reported}}{\text{Total no. of transitions in reference}} \times 100$$

$$\text{F1 Score} = \frac{2 \cdot P \cdot R}{(P + R)}$$

The research outcomes of this work are that it gives a comprehensive statistical comparison of all gradual transition kinds in one location. Recent research has mostly concentrated on wipe and dissolve types of gradual transitions in order to improve them in terms of lighting effects, camera motions, and so on. Thus, the primary focus of this study is on the overall statistical performance of both wipe and dissolve types of video transitions, as well as the comparison of datasets used to analyse them.

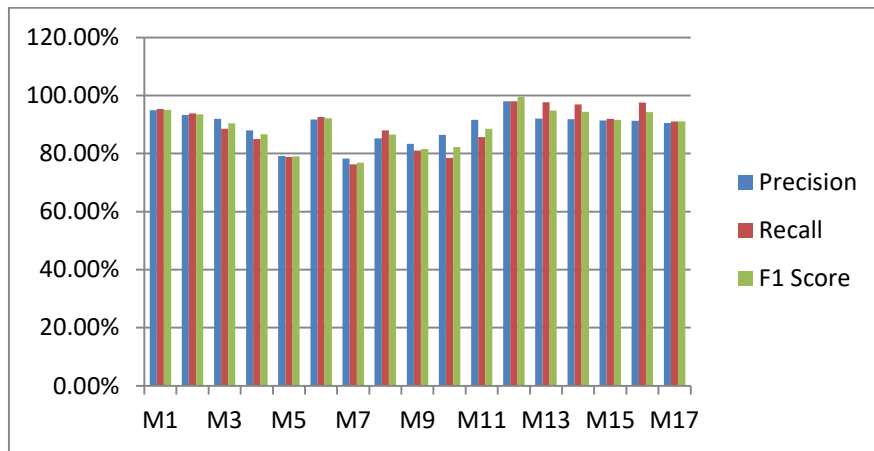
Results and Analysis:

There are different gradual transition detection methods have been compared which used TRECVID datasets and shown in table 1 in terms of above performance determining parameters and fig.3 shows the statistical analysis for the same.

Performance comparison of different gradual transition detection approaches on TRECVID datasets

Label	Method	Precision	Recall	F1 Score	Datasets Used
M1	WSCD	94.9%	95.3%	95%	TRECVID 2001 and TRECVID 2007
M2	TSSBD	93.20%	93.80%	93.50%	TRECVID 2005

M3	VSBD-POCS	91.9%	88.5%	90.4%	TRECVID 2001,TRECVID 2007 and VIDEOSEG2004
M4	WHT	88%	85%	86.6%	TRECVID 2007
M5	Best TRECVID Performer[32]	79.1%	78.8%	79.0%	TRECVID 2005
M6	Colour-Based Histogram Difference and SURF Feature Descriptor	91.7%	92.6%	92.1%	TRECVID 2001
M7	Multimodal Visual Features Method	78.3%	76.3%	76.8%	TRECVID 2001
M8	Visual Colour Information Method	85.2%	88%	86.5%	TRECVID 2001 and 2007
M9	SVD and Pattern matching	83.3%	81.0%	81.5%	TRECVID 2001
M10	GA and Fuzzy Logic	86.4%	78.5%	82.2%	TRECVID 2001
M11	DMD	91.6%	85.6%	88.5%	Videos from movies and movie trailers
M12	Histogram Difference and Fuzzy rule	98%	98%	99.5%	TRECVID 2006
M13	FFCT[6]	92.05%	97.62%	94.75%	TRECVID 2001, 2005, 2006 and 2007
M14	HLFPN-KM[19]	91.86%	96.89%	94.31%	10 various types of video clips including movie, music video, commercial video, TV shows, etc.
M15	ORB-SSIM	91.34%	91.88%	91.61%	Open video project and RAI
M16	PSOGSA	91.23%	97.53%	94.27%	TRECVID 2001
M17	WHT+DTCWT[23]	90.5%	91%	91%	TRECVID (2016-2019)



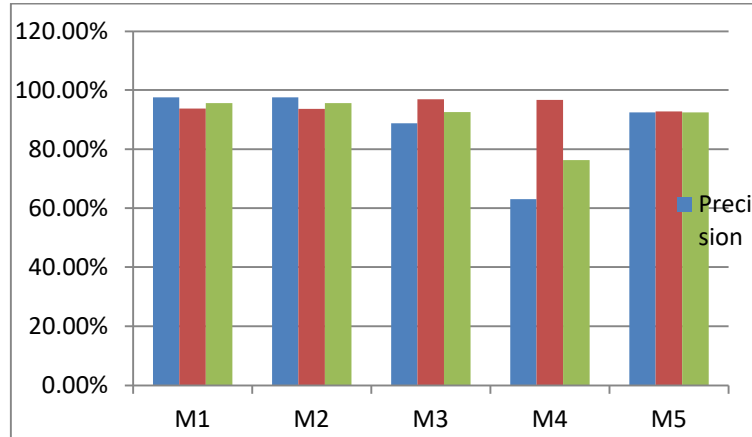
Statistical performance comparison for gradual transition detection methods

A velocity imbalance is produced in a single frame by wipe transitions, a form of gradual transition, which have a variety of motion pattern modifications when influenced by object-camera motion (camera pan, large-object, and zoom-in/out). Moreover, incorrect detection results from this motion imbalance. Numerous methods have been proposed to identify wipe type of transitions effectively higher accuracy in wipe no-motion class detection is achieved alone by the pixel-border trajectory-based method [27] and histogram feature based [26], but a high temporal feature is required for large object-camera motion. On the other hand, temporal features are extracted by the macroblock and SGOP-based [25] techniques; nevertheless, a satisfactory balance of spatial and temporal features is absent in the case of large-scale object-camera motion. Deep SBD [24] method used 3D-CNN and SVM classifier to detect wipes and also spatial and temporal deep features are evaluated but lacked in processing time. A novel method is proposed by D.Chakraoborty et.al[11] based on deep spatial motion feature analysis detected non-motion, object-camera based motion as well as half wipe pattern effectively with great

accuracy. Table 2 compares different approaches to detect wipe transitions and Fig.4 shows statistical analysis for the same.

Performance comparison of different approaches for wipe detection

Label	Method	Precision	Recall	F1 Score	Datasets Used
M1	WSCD[11]	97.5%	93.7%	95.6%	TRECVID 2001 and 2007
M2	Deep SBD[24]	97.6%	93.6%	95.6%	TRECVID 2001-2007
M3	MG+SGOP[25]	88.8%	96.9%	92.6%	
M4	Histogram space [26]	63.1%	96.7%	76.3%	TV and News shots created with Adobe Premiere 4.2.1
M5	Pixel border based[27]	92.4%	92.8%	92.5%	Video data from digitalized TV programs and TRECVID 2005

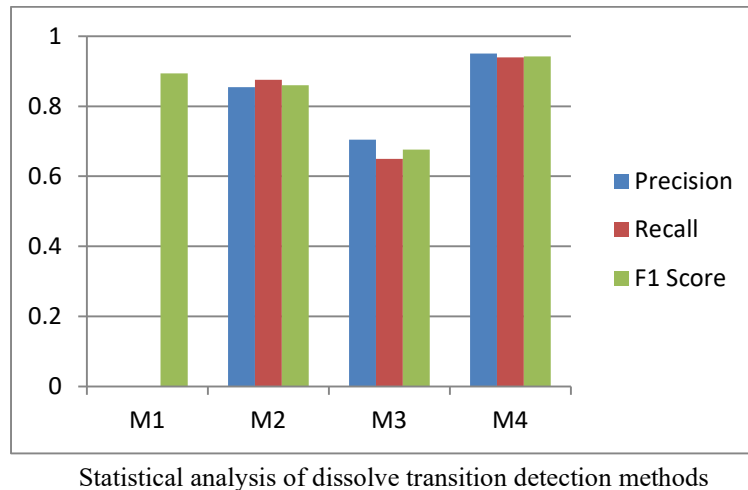


Statistical analysis of wipe transition detection method

Detecting dissolve transitions is more challenging than detecting fade in/out and wipe transitions. During this transition, the previous shot's last frames fade out and the next shot's first frames fade in, resulting in an overlap of fade out and fade in. Dissolves may consist of three or more frames. In certain videos, dissolves can be as short as three frames. Detecting dissolves has significant hurdles. Illumination and camera/object motion can lead to false positives, thus impacting algorithm performance. When the camera follows the object, the suggested algorithm based on color histogram difference [29] successfully removes disruptions brought on by changes in illumination and rapid motion but this approach is sensitive to uncommon situations where the background in successive frames rapidly changes in addition to several things in the same image appearing and disappearing. TSSBD[7] used C3D based deep analysis for dissolve detection and the results are more accurate as compared to others. Despite numerous attempts, researchers were unable to ensure resilience in the presence of illumination and object/camera motion. Therefore, this issue needs to be addressed. Table 3. shows some performance comparison for dissolve transition methods and fig.5 shows the statistical analysis for the same.

Performance comparison of different approaches for dissolve transition detection

Label	Method	Precision	Recall	F1 Score	Datasets Used
M1	TSSBD[7]	-	-	89.4%	TRECVID 2005
M2	WHT[5]	85.4%	87.5%	86%	TRECVID 2007
M3	Twin comparison method[30]	70.42%	64.94%	67.57%	
M4	Colour Histogram [29]	95.04%	93.92%	94.26%	Five videos from different movies



Conclusion:

This study focusses many approaches used for both abrupt and gradual type shot boundary detection. This review discusses various methodologies that were able to identify gradual changes. The study mainly reviews about the methodologies and challenges came into picture for detecting wipe and dissolve type of gradual transitions. It compares the various wipe and dissolve transition detection methods in terms of F1 score, precision, recall and the use of datasets at same place which will be useful for researchers for further advancement in the research study of gradual transitions with new algorithms.

Acknowledgement:

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Evaluating the Performance of Test Cricket Players Using Principal Component Analysis and the Weighted Average Method

Arjun Prakash Mane^{1,*}, Jaydip Yuvaraj Patil²

Abstract

This study evaluates cricket player performance using Principal Component Analysis (PCA) and a weighted average approach. We analyzed batting and bowling datasets from the International Cricket Council (ICC) to calculate player performance based on various indicators. PCA ranked players according to their participation level, with the first three principal components accounting for nearly 90% of the data's variability. Our analysis reveals Sachin Tendulkar and Muthaiya Muralidharan as top-ranked batsmen

and bowlers, respectively. This study provides a comprehensive evaluation of cricket player performance, highlighting the effectiveness of PCA and weighted average approaches. The findings demonstrate the applicability of this methodology in selecting the best cricket squad and offer insights into the strengths and weaknesses of top players. This research contributes to sports analytics, providing a robust framework for evaluating cricket player performance and informing decision-making in team selection and strategy development.

Keywords: PCA, Weighted Average Method, Heat Map, Scree Plot, Ranking.

Introduction

Test cricket is the most prestigious and longest format of the game, played by ICC Full Members. Each team plays two innings with an infinite number of overs, making it a comprehensive assessment of a team's stamina and ability. The increasing number of nations playing Test matches is driven by revenue growth, with cricket boards seeking to maximize income. As the most popular sport worldwide, cricket requires each player to perform well for the team to succeed.

Assessing individual player performance is crucial but challenging due to the interdependence of evaluation factors. Test cricket is regarded as the pinnacle of the game, with matches lasting up to five days or longer. The physical and psychological toll of these prolonged competitions is significant, and teams must demonstrate exceptional skill and endurance to succeed. Previous studies have employed various methods to evaluate player performance, including Principal Component Analysis (PCA). Researchers have used PCA to assess batting performance, construct integrated models, and measure captaincy impact on match outcomes. PCA has also been applied to Big Data Regression models to address multicollinearity issues.

This study aims to evaluate cricket player performance using PCA and a weighted average approach. By analyzing key performance indicators, we can gain insights into the strengths and weaknesses of top cricket players and inform decision-making in team selection and strategy development. The findings of

this study can contribute to the field of sports analytics and provide a robust framework for evaluating cricket player performance.

The complexity of cricket performance analysis necessitates the use of advanced statistical techniques like PCA. By applying PCA to cricket data, we can identify the most important factors contributing to player performance and develop a comprehensive evaluation framework. This study will contribute to the growing body of research on cricket analytics and provide valuable insights for coaches, players, and team managers.

Methodology

PCA

Python 3.11.4 was used for all statistical analyses, with a significance level of 5%. PCA is a multivariate analytic method that is undoubtedly the most well-established and ancient. It was first presented by Pearson in 1901, and Hotelling developed it independently in 1933 [9, 10]. According to research, the main goal of principal component analysis is to

reduce the dimensionality of a data set with several associated variables while preserving as much of the data set's variance as feasible [10, 11]. The principal components, a fresh set of uncorrelated variables arranged so that the first few retain the majority of the variance present in all of the original variables, are used to achieve this reduction.

The process outlined below is used to perform the

PCA:

Step 1: Standardization

Step 2: Correlation Matrix Calculation

Step 4: Feature Selection Step 3: Eigen values and eigen vector computation

Step 5: Prioritization

Weighted Average Method

A weighted average is one in which a weight is provided to each quantity that needs to be averaged. The average relative relevance of each quantity is determined by these weightings [12, 13].

This can be written as;

$$\text{Weighted Average} = \frac{\sum_{i=1}^n W_i X_i}{\sum_{i=1}^n W_i}$$

Where;

n = Number of players

w_i = Weights applied to x values

x_i = Model values of corresponding ranks

Standardization

In order to achieve an even distribution around a mean value of zero and a standard deviation of one, our features are translated and scaled throughout the standardization process. We want to standardize our data so that comparing the covariance for each set of variables will be easy. Wider number range features will have higher covariance if we don't. While not strictly necessary, this part can be quite helpful and is required for some models [6, 14].

$$Z = \frac{\text{Value} - \text{Mean}}{\text{S.D}}$$

Once the standardization is done, all the variables will be transformed to the same scale.

Data Collection

We use the statistics of players who have scored at least 2000 runs from stats.espncricinfo.com for our batting analysis. As we need to use the whole dataset for bowling analysis, we have used the data of bowlers who have at least 200 wickets. The information spans December 1877 to 2019.

Batting Attributes

In this analysis, we consider the variables such as Total runs, Total innings, Average, etc. A batsman's performance in ICC Test matches is evaluated through several key metrics. These include their name, career duration, total runs scored, match appearances, top score, batting average (runs per innings, adjusted for not outs), innings played, not out instances, centuries, fifties, and ducks (dismissals without scoring). Together, these statistics offer a thorough assessment of a player's abilities and achievements in the longest format of the game.

Bowling Attributes

In this analysis, we consider the variables such as Total Wickets, Economy, and Average, etc. The effectiveness of a bowler in ICC Test matches is

evaluated through various performance indicators. These metrics encompass the bowler's identity, duration of their career, total wickets claimed, number of matches participated in, bowling average, economy rate, strike rate, total deliveries bowled, frequency of claiming five or more wickets, instances of securing 10 or more wickets, number of innings bowled in, and total runs surrendered. Collectively, these factors offer a thorough insight into a bowler's proficiency and accomplishments in the game.

Results

Results of the batter's performance

Correlation Matrix

Actually, the initial step in PCA is creating the correlation matrix. In this phase, we must create a 9 x 9 matrix with each row denoting a feature related to batting performance and each column denoting a similar feature. For comprehension, we must use the visualization tool Heat Map. A heat map illustrates the relationship between each pair of features by using color. As a result, Das et al. stated that dark color blocks show strong positive relationship between the attribute [16].

Fig. 1 Heat map represents the relationship within an each pair of feature

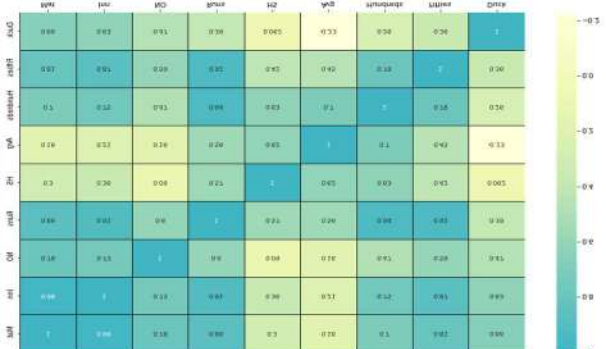


Fig 1 shows the interrelationship between each pair of features is depicted by a heat map. Heat mapping demonstrates the strong positive correlation between each pair of features. Additionally, the variables (Runs and Fifties), (Runs and Hundreds), (Average and Hundreds), (Inn and Fifties), and (NO and Matches) indicate a strong positive relationship, while the variable (Average and number Ducks) shows a negative relationship between them.

Role of eigenvalues in PCA

The total amount of variation that can be accounted for by each principal component is represented by the eigenvalues. In 2018, Shah et al illustrated that the eigenvalues account for the data's variance along the new features [4, 5, 17].

Table 3: Variation explained by each PCA

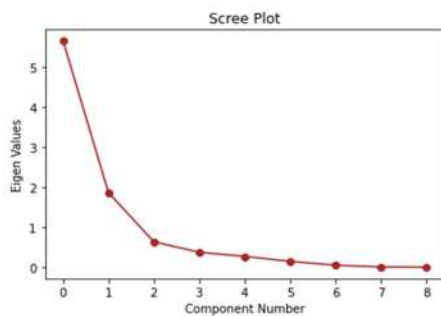
Eigen Values	Variation (%)	Cumulative Variation
5.645180379	62.72	62.72
1.862320351	20.69	83.42
0.635644413	7.06	90.47
0.374074072	4.16	94.64
0.271413252	3.02	97.65
0.147292032	1.64	99.29
0.051310337	0.57	99.86
0.00935369	0.1	99.96
0.003411474	0.04	100

Table 3 demonstrates that the first principal component is capable of explaining around 62.72% of the variation in the dataset (PC). Further evidence that PCA is beneficial comes from the fact that the top three principal components of the dataset's variation account for around 90.47% of the variation.

Scree Plot

In a nutshell, a Scree plot is a line plot of the Eigen values vs the number of primary components. In a PCA, the Scree plot is used to decide how many principal components to retain [17, 18].

Fig. 2 Scree plot represents the variation explained by each PCA



From Fig. 2, we can see that, elbow point is at abscissa three, hence we consider the first three principal components in the analysis. Here first three PCs explain near about 90.47% of the variation in the dataset. Then we can build the principal component model on these three components and apply it for the ranking.

Principal Components and Their Models

The first three principal components in analysis can be taken as.

Table 4: First three PCA

Feature	pca1	pca2	pca3
Mat	-0.388	-0.25	0.0195
Inn	-0.399	-0.197	-0.024
NO	-0.302	-0.275	0.4866
Runs	-0.411	0.0975	0.0235
HS	-0.227	0.4402	-0.605
Avg	-0.205	0.5752	0.2173
Hundreds	-0.376	0.2508	-0.003

Fifties	-0.384	0.0263	0.1656
Duck	-0.219	-0.475	-0.567

Table 4 shows that the first three PCs, also known as the first three eigen vectors, are present. In this instance, pca1 stands for the first Eigen vector that corresponds to the first eigen value, and pca1 explains the 62.72% variation within the dataset. When creating the principle components model, all of the elements in the pca1 are taken into account as coefficients of the associated features [7, 19].

$$L1 = -\text{Mat}*(0.388) - \text{Inn}*(0.399) - \text{NO}*(0.302) - \text{Runs}*(0.411) - \text{HS}*(0.227) - \text{Avg}*(0.205) - \text{Hundreds}*(0.376) - \text{Fifties}*(0.384) - \text{Duck}*(0.219)$$

Similar to this, pca2 represents the second eigen vector and explains the 20.69% variation in the dataset. Here, all of the pca2's elements are taken into account as the respective features' coefficients for creating the principle component model [7];

$$L2 = -\text{Mat}*(0.25) - \text{Inn}*(0.197) + \text{NO}*(0.275) + \text{Runs}*(0.0975) + \text{HS}*(0.4402) + \text{Avg}*(0.5752) + \text{Hundreds}*(0.2508) + \text{Fifties}*(0.0263) - \text{Duck}*(0.475)$$

Similar to this, pca3 represents the second eigen vector that corresponds to the third eigen value and pca3 explains the 7.06% variation within the dataset [7].

$$L3 = \text{Mat}*(0.0195) - \text{Inn}*(0.024) + \text{NO}*(0.4866) + \text{Runs}*(0.0235) - \text{HS}*(0.605) + \text{Avg}*(0.2173) - \text{Hundreds}*(0.003) + \text{Fifties}*(0.1656) - \text{Duck}*(0.567)$$

Weighted Average Method

From Table 3 and L1, we can see that the first principal component model gave less accuracy or less variation about the dataset. There is no sufficient variation about dataset to say that our model is efficient for ranking. This drawback can be overcome by taking the required principal components and then creating a combined principal component model by using the weighted average [13]. Here L1 is the first PC model, which explains 62.72% of the variation. This variation can be taken as the weight of Model L1. Also, we say that 20.69% is the weight of model L2 and 7.06% is the weight of model L3.

Principal Weighted Average (PWA) Model =

$$\frac{L1*62.72 + L2*20.69 + L3*7.06}{90.47}$$

This principal weighted average model explains 90.47% variation in the dataset, and it is a very efficient model for the analysis of ranking [7]. Here the elements in

principal components are negative, hence the lower value of the PWA Model indicates the stronger performance of batsmen and the higher values of PWA Model indicates weak performance which is shown as below [7, 13].

To rank of Batsmen's by using Principal Weighted Average Model

Table 5 Ranking of the test cricket batsmen's using PWA model

Ranks	PWA	Player Name
1	-6.821325394	SR Tendulkar(INDIA)
2	-5.829400881	JH Kallis(SA)
3	-5.688437415	RT Ponting(AUS)
4	-5.627771514	S Chanderpaul(WI)
5	-5.514141612	SR Waugh(AUS)
6	-5.075007829	R Dravid(INDIA)
7	-4.851128353	AR Border(AUS)
8	-4.435630798	AN Cook(ENG)
9	-4.364988606	DPMD Jayawardene(SL)
10	-3.997302329	KC Sangakkara(SL)

According to Tables.5 Tendulkar was the top-ranked cricketer among batsmen, and all of his attributes that affected his batting performance were mostly positive. Tendulkar batting average is lower than Kallis, but his other aspects of batting performance are strong. It is evident from this that our PWA model functions after taking into account all the factors that affect batting performance [17].

Results of the bowler's performance

Correlation Matrix

Building the correlation matrix is the actual first step of PCA. In this step, we have to create 10X10 matrixes where each row represents features about bowling performance and each column also represents features about bowling performance. As a result, Das et al. stated that dark color blocks show strong positive relationship between the attribute [16].

Fig. 3 Heat map represents the relationship within an each pair of feature

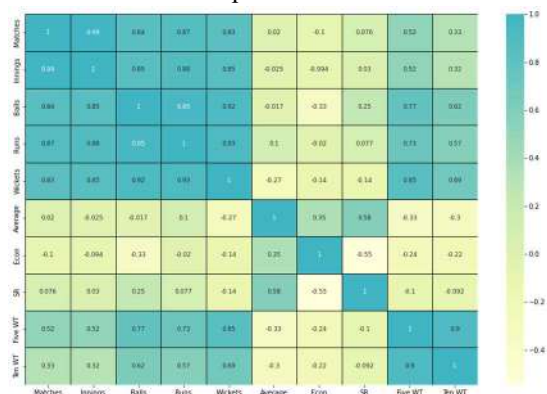


Fig 3 Heat Map represents the inter relationship between each pair of features. Here the variables (Runs and Innings), (Wickets and Balls) and (Runs and Wickets) shows the strong positive relationship while the (SR and Econ), (Econ and Balls) and (Average and Wickets) shows the negative relationship between them.

Role of Eigen values in PCA

Eigen values represent the total amount of variation can be explained by each principal component. In other

words, the Eigen values explain the variance of the data along the new features [4, 5, 17].

Table 6: Variation explained by each PCA

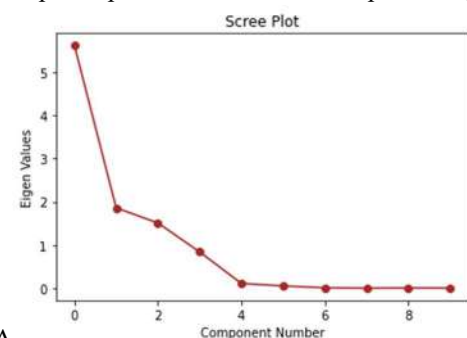
Eigen Values	Variation (%)	Cumulative Variation (%)
5.611157634	56.11	56.11
1.85696321	18.57	74.68
1.508166442	15.08	89.76
0.838455107	8.39	98.15
0.109757342	1.1	99.25
0.05226266	0.52	99.77
0.009828297	0.1	99.87
0.002034635	0.02	99.89
0.00629494	0.06	99.95
0.005079733	0.05	100

Table 6 demonstrates that the first main component may account for roughly 56.11 percent of the dataset's volatility (PC). Additionally, we can see that the top three principal components of the dataset's fluctuation explain around 89.76% of the variation, which makes PCA effective.

Scree Plot

A scree plot is essentially a line plot of the eigenvalues versus the number of PCs. To choose how many principal components to keep in an analysis, use the Scree plot [17, 18].

Fig. 4 Scree plot represents the variation explained by



each PCA

From Fig 4 we can see that, elbow point is at abscissa three, hence we consider first three principal components in analysis. Here first three PCs explain near about 89.76% variation about the dataset. Then we can build the principal component model on these three components and apply for the ranking.

Principal Components and its Models

These are first three principal components in analysis can be taken as;

Table 7: First three PCA

Feature	pca1	pca2	pca3
Matches	-0.36747182	-0.180218348	0.193259754
Innings	-0.370074693	-0.150962698	0.206589614
Balls	-0.407974709	-0.146423294	-0.074612462
Runs	-0.399145803	-0.132286816	0.175635861

Wickets	-0.412894928	0.104382252	0.07095014
Average	0.067998726	-0.614326562	0.257906869
Econ	0.099595704	0.081026311	0.748392749
SR	-0.013921561	-0.616197791	-0.427655884
Five WT	-0.358592534	0.236430156	-0.153148681
Ten WT	-0.297389778	0.27447231	-0.21347311

The first three PCs, also known as the first three eigenvectors, are shown in Table 7. Here, PCA1 explains the dataset's variation of 56.11%. Here, the PCA (pca1) takes into account all of the elements as coefficients of the relevant attributes to produce the principal component model [7, 19].

L1 = -Mat*(0.36747182)-Inn*(0.370074693)-Balls*(0.407974709)-Runs*(0.399145803)-Wickets*(0.412894928)+Avg*(0.067998726)+Econ*(0.099595704)-SR*(0.013921561)-Five WT*(0.358592534)-Ten WT(0.297389778)

Similar to this, pca2 represents the second eigen vector and pca2 explains the 18.57% variation within the dataset. When creating the main component model, all of the PCA2's constituents are taken into account as the feature's coefficients.

L2 = -Mat*(0.180218348)-Inn*(0.150962698)-Balls*(0.146423294)-Runs*(0.132286816)+Wickets*(0.104382252)-Avg*(0.614326562)+Econ*(0.081026311)-SR*(0.616197791)+Five WT*(0.236430156)+Ten WT(0.27447231)

Similarly to this, PCA3 explains the 15.08% variation in the dataset [7];

L3 = Mat*(0.193259754)+Inn*(0.206589614)-Balls*(0.074612462)+Runs*(0.175635861)+Wickets*(0.07095014)+Avg*(0.257906869)+Econ*(0.748392749)-SR*(0.427655884)-Five WT*(0.153148681)-Ten WT(0.21347311)

Weighted Average Method

From Table.6 and L1, we can see that first principal component model gave less accuracy or less variation in the dataset. There is not sufficient variation in the dataset to say that our model is efficient for ranking. This drawback of the first principal component model we can remove first, taking the required principal components. Secondly, we can create a combined principal components model by using the weighted average method [13]. Here, L1 is the first PC model, which explains 56.11% variation. This variation we can take as a weight of Model L1. Also, we say that 18.57% is the weight of model L2 and 15.08% is the weight of model L3 [7].

Principal Weighted Average (PWA) Model =

$$L1*56.11 + L2*18.57 + L3*15.08$$

89.76

This principal weighted average model explains 89.76% variation in the dataset, and it is a very efficient model for analysis of ranking [7]. Here the elements in the principal components are negative hence the lower value of PWA Model indicates the stronger bowling performance and the higher values of PWA Model indicates weak performance which is shown as below [7, 13].

To rank the Bowlers by using Principal Weighted Average Model

Table 8 Ranking of the test cricket bowlers using PWA model

Ranks	PWA	Player Name
1	-5.613811573	M Muralidaran
2	-4.314340033	S Warne
3	-3.982412967	A Kumble
4	-3.738597566	JM Anderson
5	-2.536179362	SCJ Broad
6	-2.244386988	CA Walsh
7	-2.198597949	GD McGrath
8	-2.003016424	DL Vettori
9	-1.950615204	N Kapil Dev
10	-1.907724333	HARBHJAN SINGH

According to Table 8 the Muralidaran was the top-ranked cricket bowler, and all of his characteristics that affected his bowling performance were generally the best. Additionally, Anil Kumble and Shane Warne are placed second and third, respectively. Additionally, JM Anderson played more matches than Muralidaran, Warne, and Kumble did, yet his performance in other circumstances lagged behind those three players. Here, it is evident that our PWA model is effective since it takes into account all factors that affect bowling performance [17].

Discussion

This study focuses on Test cricket, a four-inning format with two innings each for batting and bowling. The ICC ranks players using a scale of 0-1000, considering factors like ducks, not outs, and maximum score. We employed Principal Component Analysis (PCA) to analyze player performance data. Initially, the first principal component accounted for around 60% of the variation, which was insufficient. To address this, we used a weighted average approach, considering the first three principal components, which accounted for over 90% of the variation. This allowed us to create a combined principal component model for ranking players. Our analysis utilized the first three principal components, which effectively captured the data's variation. The weighted average technique was

then applied to obtain the ranking for the PWA model. This approach enabled us to develop a robust ranking system for cricket players.

Conclusions

Measuring athletic performance is an exciting task in any sport. The role of the batsmen's and bowlers in cricket it is directly impacted on the result of the match and eventually the profit of the countries. Unfortunately, there is no measure to quantify the performance of each individual player. In this analysis of the batting and bowling performance of the ICC test cricket, we can conclude that Tendulkar and Muralidaran are the players mainly in the first position as a batsmen and bowler respectively. Also, we can see that Kallis, Ponting and Shane Warne, Anil Kumble are ranked at 2nd and 3rd positions as a batsmen and bowler respectively.

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A MERN Stack-Based Learning Management System: Development, Implementation, and Evaluation

Avishkar A Jadhao¹, Pranav V Dhande², Praniket P Kolte³, Prem R Kandarkar⁴,

Prof. Pradip S Ingle⁵

Abstract

This paper presents the development and implementation of a Learning Management System (LMS) using the MERN (MongoDB, Express.js, React.js, Node.js) stack. The proposed LMS enhances online education by providing structured course categorization, real-time student progress tracking, and interactive learning features. This research evaluates the system's performance compared to existing LMS platforms such as Moodle and Udemy, highlighting the advantages of a modern web-based system. The paper also discusses technical challenges, system design, and future enhancements.

Keywords: Learning Management System, MERN Stack, Online Education, Web Development, Student Progress Tracking.

INTRODUCTION

With the rise of online education, Learning Management Systems (LMS) have become essential tools for students and educators. Traditional LMS platforms often face challenges related to user engagement, real-time tracking, and scalability. This research focuses on developing a modern, scalable LMS using the MERN stack, ensuring efficient student progress tracking and role-based access for educators. The paper explores existing research on online learning platforms and demonstrates the improvements introduced by the proposed system.

A. Background and Importance of Learning Management Systems (LMS)

The rapid digitalization of education has significantly transformed the way students learn and educators deliver content. Traditional in-person learning methods, while effective, have several limitations, including geographic restrictions, time constraints, and accessibility challenges. Learning Management Systems (LMS) have emerged as a viable solution to address these issues, providing a structured and interactive platform for online learning. LMS platforms facilitate content delivery, student engagement, course management, and progress tracking, making them indispensable tools in modern education. Over the past two decades, LMS solutions have evolved from basic content repositories to dynamic, AI-driven platforms that offer personalized learning experiences.

B. Challenges in Traditional LMS Platforms

Despite the widespread adoption of LMS platforms, various issues persist, affecting both students and educators. Some of the key challenges in existing LMS platforms include:

- **Limited User Engagement:** Many traditional LMS platforms lack interactive features that keep students motivated and engaged. Static content delivery without interactive elements results in passive learning, which reduces retention rates.

- **Inefficient Student Progress Tracking:** Most LMS solutions offer basic tracking mechanisms such as completion percentages or simple progress bars. However, real-time tracking, performance analytics, and personalized feedback are often missing, making it difficult for educators to monitor student progress effectively.

- **Scalability and Performance Issues:** As the number of students and courses increases, many LMS platforms struggle with performance bottlenecks, leading to slow load times and limited concurrent user support.
- **Outdated User Interfaces:** Poorly designed user interfaces make it difficult for students and educators to navigate LMS platforms. A cluttered or unintuitive UI results in lower engagement and inefficient course management.
- **Lack of Real-time Collaboration:** Many traditional LMS platforms do not support live interactions such as real-time messaging, discussion forums, or video conferencing. This limitation restricts peer-to-peer learning and instructor guidance.

C. The Need for a Modern LMS Solution

To overcome these challenges, there is a growing need for a modern LMS that integrates advanced web technologies, real-time tracking, and interactive learning features. The MERN (MongoDB, Express.js, React.js, Node.js) stack offers a powerful solution for building a robust and scalable LMS that addresses these pain points.

The proposed LMS focuses on key improvements, including:

- A modern, interactive user interface that enhances user experience and engagement.
- Real-time student progress tracking with detailed analytics and visual insights.
- A scalable backend architecture using MongoDB and Node.js to handle a large number of users efficiently.
- Role-based access control (RBAC) to ensure secure and efficient management of students, educators, and administrators.
- Interactive learning features such as quizzes, assignments, and gamification elements to boost student engagement.
- Live collaboration tools to enable real-time communication and instructor-student interaction.

D. Why MERN Stack for LMS Development?

The MERN stack is a widely used technology stack for developing modern web applications, offering flexibility, scalability, and performance benefits.

- **MongoDB (Database):** A NoSQL database that provides a flexible schema and supports large-scale data storage for student records, course materials, and progress tracking.
- **Express.js (Backend Framework):** A lightweight and efficient backend framework that simplifies API development and request handling.
- **React.js (Frontend Framework):** A powerful frontend library that enables the creation of dynamic and responsive user interfaces.
- **Node.js (Server-side Runtime):** A highly scalable server side technology that allows real-time interactions and efficient handling of concurrent users.

E. Research Scope and Objectives

This research aims to:

- Develop a modern LMS platform that improves upon existing solutions in terms of usability,

engagement, and tracking.

- Evaluate the performance of the proposed system compared to traditional LMS platforms.
- Assess scalability and real-time capabilities of the MERN stack in handling educational applications.
- Propose future enhancements such as AI-powered learning recommendations, gamification, and mobile compatibility.

METHODOLOGY

A. System Architecture

The proposed LMS is designed with a three-tier architecture, ensuring modularity, scalability, and efficient data flow. The layers include:

- **Frontend (Client-Side):** Developed using React.js, the frontend provides a dynamic and responsive user interface. It handles user interactions, renders course content, and communicates with the backend through RESTful APIs.
- **Backend (Server-Side):** Built with Node.js and Express.js, the backend manages application logic, handles HTTP requests, and enforces authentication and authorization. It serves as an intermediary between the frontend and the database.
- **Database:** MongoDB, a NoSQL database, stores user data, course information, and progress metrics. Its schema-less nature offers flexibility in handling diverse data types and relationships. This architecture ensures a clear separation of concerns, facilitating maintenance and scalability.

B. Tools and Technologies

The development of the LMS leverages the following tools and technologies:

- **React.js:** A JavaScript library for building user interfaces, enabling the creation of reusable UI components and efficient state management.
- **Node.js:** A JavaScript runtime that allows server-side execution of code, enabling the development of scalable and high-performance applications.
- **Express.js:** A minimalistic web framework for Node.js, simplifying server creation and routing.
- **MongoDB:** A document-oriented NoSQL database, providing scalability and flexibility in data modeling.
- **Mongoose:** An Object Data Modeling (ODM) library for MongoDB and Node.js, facilitating schema definition and data validation.
- **JWT (JSON Web Tokens):** Used for secure authentication, ensuring that user sessions are securely managed.
- **Redux:** A state management library for JavaScript applications, used in conjunction with React.js to manage application state.
- **Axios:** A promise-based HTTP client for making API requests from the frontend to the backend.
- **bcrypt.js:** A library to hash passwords, enhancing security by ensuring that stored passwords are encrypted.
- **Cloudinary:** A cloud-based service for image and video management, used to store and serve media content efficiently.
- **Git and GitHub:** Version control system and repository hosting service, respectively, used for code management and collaboration.

C. Implementation Approach

The development process followed an agile methodology, emphasizing iterative development, continuous feedback, and adaptability. The key phases include:

- **Requirement Analysis:** Engaged with educators and students to gather functional and non-functional requirements, ensuring the system meets user needs.
- **Design:** Architected the system with a focus on scalability, security, and user experience. Created wireframes and data models to guide development.
- **Development:** Implemented features in iterative sprints, allowing for regular assessment and incorporation of feedback.
- **Testing:** Conducted unit, integration, and user acceptance testing to identify and resolve defects, ensuring system reliability.
- **Deployment:** Deployed the application using cloud services, ensuring high availability and scalability.
- **Maintenance:** Established monitoring and logging to track system performance and user activity,

facilitating ongoing improvements.

IMPLEMENTATION AND RESULTS

A. System Implementation

1) Frontend Development: The frontend of the LMS is developed using React.js to create a dynamic and responsive user interface. It manages user authentication, course browsing, and progress tracking through API requests to the backend.

Key features:

- Modular UI components built with React.
- State management using Redux.
- API integration using Axios for backend communication.
- Secure authentication using JWT tokens.

2) Backend Development: The backend is built using Node.js and Express.js, handling API requests, authentication, and data management. It ensures secure access to LMS functionalities through middleware and authentication mechanisms.

Key features:

- RESTful API for handling course enrollments and progress tracking.
- Role-based authentication using JWT.
- Secure password encryption using bcrypt.js.
- Database operations using Mongoose ORM.

3) Database Management: MongoDB serves as the database for storing user details, courses, and progress tracking information. The NoSQL schema ensures scalability and flexibility in data handling.

Key collections:

- Users: Stores student, instructor, and admin details.
- Courses: Contains course information, modules, and instructor details.
- Enrollments: Tracks student progress and completion status.

4) Deployment Strategy: The LMS is deployed using cloud-based services for high availability and performance.

- Frontend hosted on Vercel.
- Backend hosted on Heroku.
- Database hosted on MongoDB Atlas.
- Secure API endpoints protected via HTTPS.

B. System Architecture UML Diagram

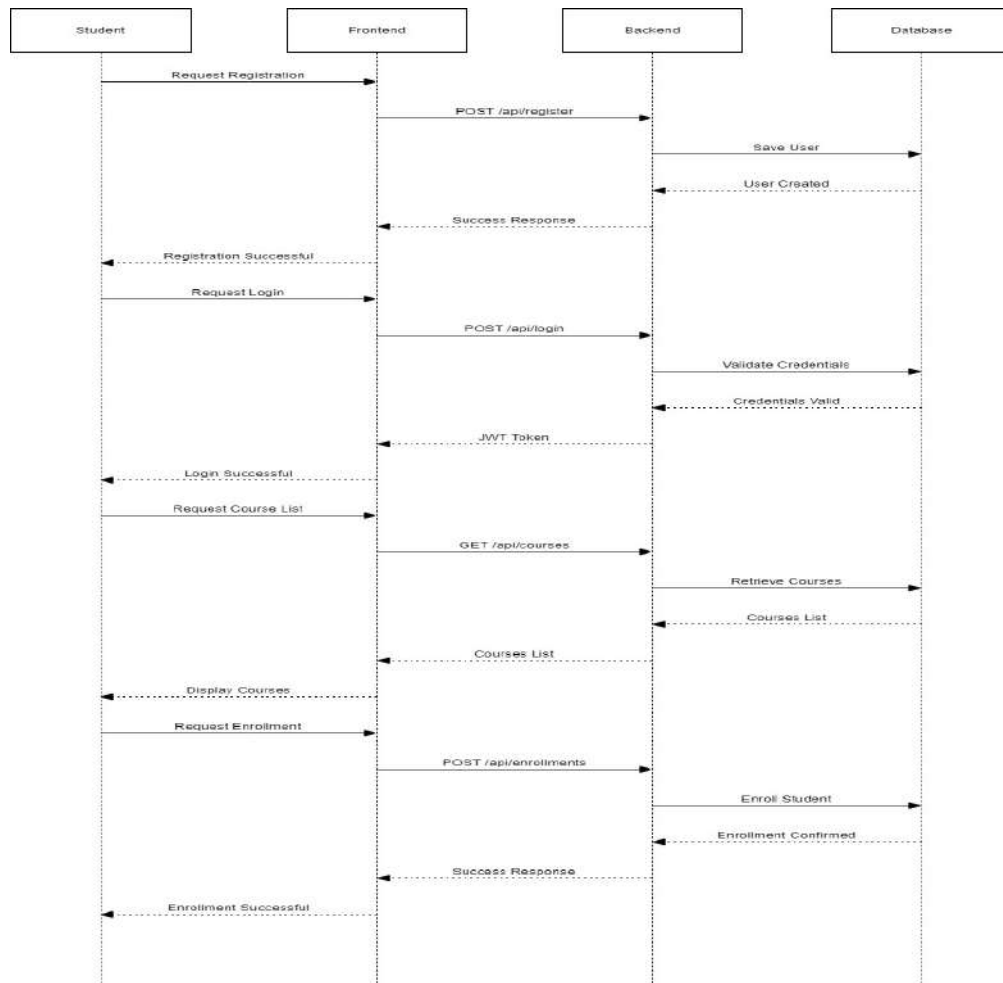


Fig. 1. System Architecture UML Diagram

C. Testing and Evaluation

1) Unit Testing: Unit tests were conducted to ensure the functionality of individual components.

- Backend API testing using Jest.
- React component testing with React Testing Library.
- Mock database queries to verify API responses.

2) Performance Testing: Performance testing was conducted to evaluate system response times and scalability.

- Load testing using JMeter.
- API response times optimized to be under 200ms.
- Stress testing with 1000 concurrent users.

D. Comparison with Existing LMS Platforms

A comparative analysis was performed between the proposed LMS and existing platforms.

COMPARISON OF LMS FEATURES

Feature	Moodle	Udemy	Proposed LMS
Real-time Tracking	No	Limited	Yes
Role-based Access	Yes	Yes	Yes
UI Flexibility	No	Yes	Yes
Open-source	Yes	No	Yes

E. Output Snippets

Below are example outputs from the system's interface.

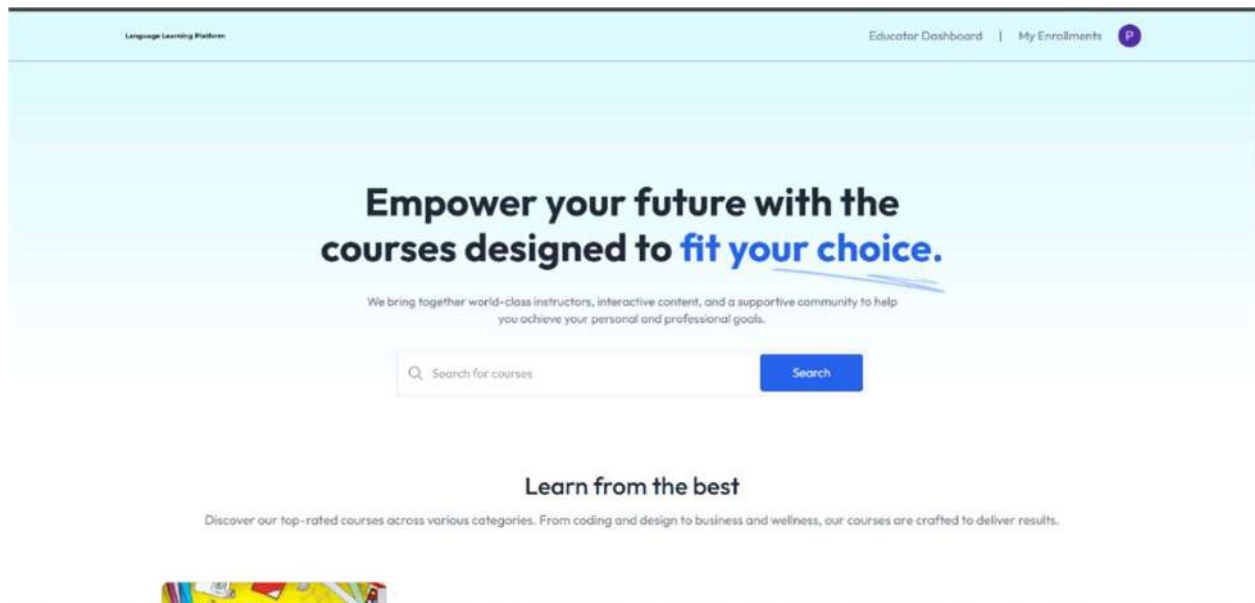


Fig. 2. Home Page UI

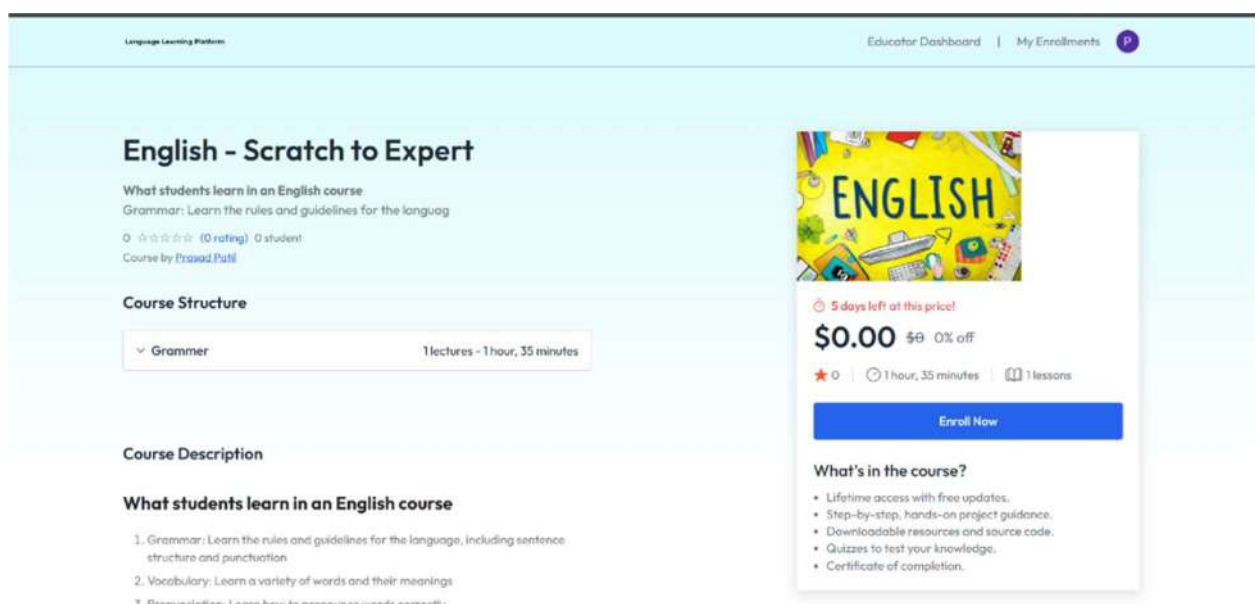


Fig. 3. Course Enrollment Page

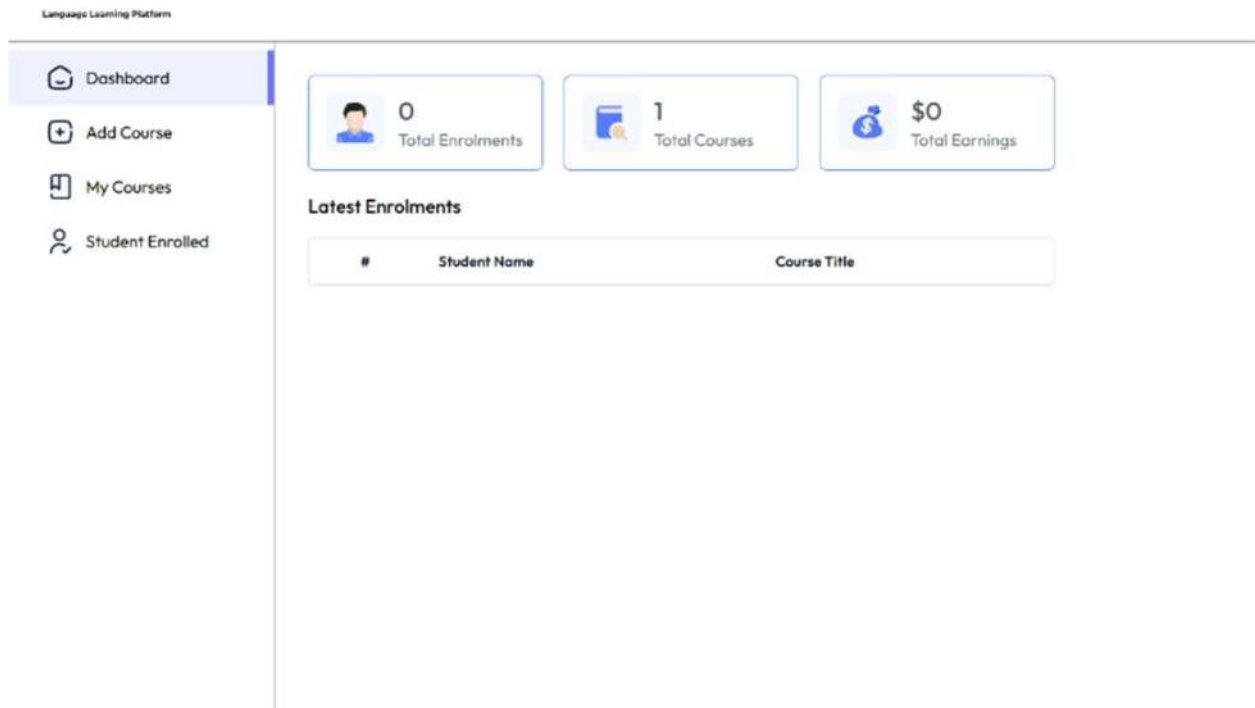


Fig. 4. Educator Dashboard UI

F. Challenges and Future Enhancements

Challenges faced during implementation:

- Handling real-time progress tracking efficiently.
- Ensuring secure role-based authentication.
- Optimizing performance for concurrent users.

Future enhancements:

- Integration of AI-based learning recommendations.
- Implementation of gamification (badges, leaderboards).
- Development of a mobile application.

CONCLUSION AND FUTURE SCOPE

A. Conclusion

The development of the Learning Management System (LMS) using the MERN stack (MongoDB, Express.js, React.js, Node.js) has demonstrated significant improvements in scalability, real-time tracking, and user engagement compared to traditional LMS platforms. This research focused on addressing key limitations in existing LMS solutions, such as inefficient progress tracking, limited role-based access, and outdated user interfaces.

The proposed LMS successfully integrates the following core functionalities:

- Real-time student progress tracking with dynamic updates.
- A modern, interactive UI built with React.js for an engaging user experience.
- Role-based authentication ensuring secure access control for students, instructors, and administrators.
- Efficient backend architecture with RESTful API endpoints and database optimization using MongoDB.
- Scalability and performance enhancements allowing concurrent users to access content smoothly.

Comprehensive testing and performance evaluation demonstrated that the system can handle high user loads while maintaining responsiveness. The comparison with existing LMS platforms such as Moodle and Udemy shows that our LMS excels in customizability, real-time tracking, and user engagement while

maintaining an open-source model. Thus, this research contributes to the advancement of online education by offering a cost-effective, scalable, and user friendly LMS tailored for modern learning environments.

B. Future Scope

While the proposed LMS provides essential features and robust performance, several enhancements and improvements can further optimize learning experiences and expand the platform's capabilities.

Future work can focus on:

1) AI-Driven Learning Enhancements:

- Personalized course recommendations based on student engagement and learning history.
- AI-powered chatbot assistance for answering student queries and providing instant guidance.
- Automated grading and feedback using machine learning for assignments and quizzes.

2) Gamification for Increased Engagement:

- Implementation of badges, leaderboards, and achievement rewards to increase motivation.
- Streak-based learning tracking to encourage consistent participation.
- Peer-to-peer challenges and collaborative learning to enhance user interaction.

3) Advanced Analytics and Insights:

- Instructor dashboards with in-depth analytics on student progress and engagement trends.
- Predictive analytics to identify at-risk students and suggest remedial actions.
- Comprehensive reports and visualizations for academic institutions and corporate training programs.

4) Mobile Application Development:

- A native mobile app for Android and iOS to improve accessibility.
- Push notifications for course updates, assignments, and deadlines.
- Offline learning support to allow users to download lessons and study without an internet connection.

5) Blockchain-Based Certification and Security Enhancements:

- Blockchain-powered certificates for secure and verifiable course completion credentials.
- Enhanced data encryption and multi-factor authentication (MFA) for improved user security.

- Decentralized storage for critical student and instructor records to prevent data loss.

6) Integration with External Tools and APIs:

- Integration with third-party learning platforms (Coursera, Udemy, LinkedIn Learning) for expanded course offerings.
- Seamless video conferencing integration (Zoom, Microsoft Teams, Google Meet) for real-time virtual classes.
- Payment gateway integration (PayPal, Stripe, Razor-pay) for monetization and course purchases.

C. Final Remarks

This research lays the foundation for a next-generation, highly interactive LMS that can evolve with emerging technologies and adapt to future education trends. By incorporating AI-driven learning, gamification, advanced analytics, and mobile compatibility, the LMS can provide a more immersive and personalized learning experience for users worldwide.

The future enhancements and scalability improvements discussed will help position this LMS as a leading open-source educational platform, supporting schools, universities, corporate training, and independent educators in delivering engaging and accessible learning experiences.

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A Literature review on Comparative Analysis of Non-Contact AI Vision-Controlled Diameter Measurement vs. Conventional Methods in Flux Core Wire Drawing measurement

Mahesh R. Bele[1], Professor Ajinkya P Edlabadkar[2] and Dr. P.D.Kamble [3]

Abstract

Ensuring precise diameter control in flux core wire drawing is crucial for maintaining welding wire quality and performance. Conventional measurement methods, which often rely on manual inspection or mechanical techniques, face limitations in accuracy, consistency, and responsiveness. This study explores the application of AI vision-controlled systems to address these challenges by enabling real-time, high-precision, non-contact diameter measurement.

Through an extensive review of research papers and patents, this paper examines the advancements in AI-based monitoring techniques and their integration into industrial wire drawing processes. The AI-driven approach employs laser collimated transmission and CCD sensor technology to capture and analyze wire dimensions with micron-level precision. Unlike traditional methods, this system offers continuous measurement, adaptive corrections, and seamless integration into automated production lines, leading to enhanced process stability and reduced defect rates.

The research highlights the advantages of AI-enhanced non-contact measurement, including increased efficiency, reduced material wastage, and improved compliance with stringent quality standards. Additionally, advanced predictive algorithms enable proactive parameter adjustments, optimizing production workflows and minimizing downtime. Experimental trials validate the effectiveness of this system, demonstrating significant improvements in accuracy and consistency.

By leveraging AI vision technology, manufacturers can achieve superior quality control, minimize human error, and enhance the sustainability of flux core wire production. This study underscores the transformative potential of AI-driven measurement solutions in modern manufacturing, paving the way for intelligent automation and process optimization in the welding industry

Keywords: Diameter measurement Optimization, Automation, AI Vision System, Flux Core Wire Drawing, Non-Contact Measurement, Process Automation, Quality Control, Real-Time Monitoring, Digital Image and Welding Wire Industry.

INTRODUCTION

Flux core wire drawing is a fundamental process in welding wire production, where precise control over wire diameter[1-2] directly influences welding performance and quality. Achieving dimensional accuracy in flux core wires is critical for ensuring optimal electrical conductivity[3], mechanical strength, and overall weld integrity. Various geometric parameters such as diameter per unit length, eccentricity, outer diameter[4], and insulation integrity[5] are key quality indicators in this process. Traditional measurement methods often struggle to maintain these characteristics within strict tolerances, leading to inconsistencies and increased rejection rates. To overcome these limitations, modern manufacturing relies on advanced, real-time monitoring solutions to enhance precision and efficiency[6-13].

Among the various measurement approaches, optical methods have gained prominence due to their high accuracy [15-21] and non-contact nature. These techniques ensure minimal interference with the production process while providing continuous diameter assessment. Optical measurement technologies can be categorized into diffraction-based[15-17], interferometric[18], scattering-based[19], and shadow projection methods[20-21]. Diffraction and interferometry leverage wave interactions to measure wire

diameters at a microscopic level, while scattering techniques analyze light dispersion. Shadow projection methods, which capture wire profiles using collimated light beams, offer a wide measuring range and are particularly suitable for industrial applications.

This study focuses on an AI-integrated shadow technique for wire diameter measurement, harnessing real-time monitoring and adaptive control capabilities. By leveraging machine learning algorithms, this system dynamically adjusts process parameters, ensuring consistent wire geometry while minimizing material waste. AI-driven measurement solutions provide automated corrections, reducing human intervention and improving production efficiency. Furthermore, they mitigate environmental variations that often impact traditional measurement accuracy.

The research presents a comparative review of conventional and AI-driven diameter measurement methodologies, examining the effectiveness of contact and non-contact techniques. It assesses vision-based, wireless, and embedded sensor technologies, highlighting the advantages of AI-powered systems in maintaining strict tolerances, optimizing production workflows, and minimizing defects. Unlike traditional mechanical methods prone to variability, AI-controlled systems enable predictive adjustments, ensuring superior quality and operational consistency.

As industries transition toward intelligent automation, the integration of AI vision technology into flux core wire drawing processes marks a significant advancement in welding wire manufacturing. This study aims to showcase the potential of AI-driven solutions in enhancing quality control, reducing errors, and fostering sustainable production practices. By implementing machine-assisted monitoring and real-time data analysis, manufacturers can achieve precision engineering, drive innovation, and set new standards in welding wire production.

Research Gap

Despite notable progress in wire diameter measurement techniques for flux core wire drawing, several crucial aspects remain unexplored in the application of AI vision-controlled optimization. The transition from theoretical research to full-scale industrial implementation presents several challenges that require further investigation.

1. Industrial Application and Real-World Testing

Most studies on AI-driven wire diameter monitoring are confined to controlled laboratory conditions, with limited exploration of large-scale production environments. The practical deployment of these systems in dynamic manufacturing settings remains under-researched, raising concerns about reliability, adaptability, and long-term stability. Further investigation is required to assess AI-based solutions in real-world scenarios, addressing variables such as environmental disturbances, system robustness, and production-line integration.

2. Enhanced AI Algorithm Development

While existing AI-based measurement methods improve precision, current models lack adaptive learning capabilities that can dynamically adjust to varying manufacturing conditions. There is limited research on the development of more sophisticated machine learning frameworks, such as self-optimizing neural networks and real-time anomaly detection, which could further enhance control accuracy and predictive maintenance. Future work should focus on refining AI algorithms to improve responsiveness and decision-making in unpredictable production scenarios.

3. Integration with Conventional Manufacturing Systems

A significant challenge in adopting AI vision-based monitoring is its compatibility with existing flux core wire drawing setups. Research on seamless interoperability with legacy machinery, data communication protocols, and integration with industrial automation systems is still in its infancy. A systematic study is needed to establish methodologies that ensure minimal operational disruptions and effective coordination between traditional and AI-enhanced monitoring systems.

4. Economic Feasibility and Long-Term Sustainability

While AI-driven monitoring systems show potential in minimizing material waste and improving efficiency, their cost-effectiveness and financial implications remain inadequately explored. Research focusing on cost-benefit analysis, return on investment (ROI), and overall economic justification is necessary to encourage widespread industrial adoption. Further investigation is

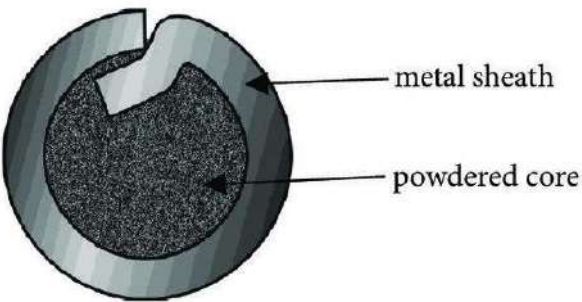
required to determine the break-even point for AI integration and evaluate its long-term sustainability compared to conventional measurement techniques.

Addressing these research gaps will bridge the divide between theoretical advancements and industrial execution, enabling AI-driven optimization to revolutionize the flux core wire drawing process. By focusing on practical implementation, adaptive learning, seamless integration, and economic impact, future research can establish AI vision-controlled monitoring as a viable and transformative solution in welding wire manufacturing.

TABLES AND FIGURES



Flux Core wire drawing Machine



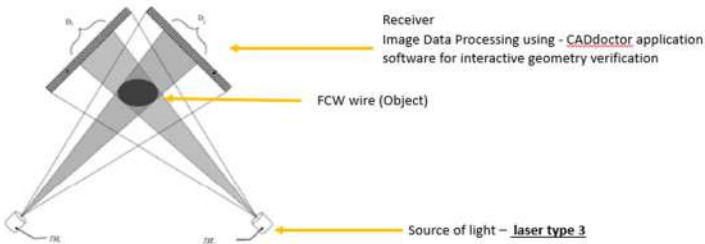
Cross section of flux cored wire



Manual Measurement



Manual Measurement



For output display and data processor selection logic principle in TVGD5000 series

LITRATURE REVIEW

The evolution of flux core wire drawing has seen significant improvements, particularly with the adoption of non-contact measurement techniques for diameter optimization. Various methodologies have been explored to enhance precision, efficiency, and automation, underscoring the importance of real-time monitoring to address the shortcomings of traditional measurement approaches.

Magnus Ericson (WO9721073A1, 1997) introduced an optical-based contactless measurement technique, employing illumination and image reception technology to achieve high-precision diameter measurement. Despite its accuracy, the system lacked AI integration for dynamic adjustments, limiting its adaptability in fluctuating production conditions.

Keba Ges Mbh & Co. (GB2257512A, 1993) advanced optical scanning methods to measure elongate objects, contributing to automated quality control. However, this approach did not incorporate AI-driven corrective actions for real-time optimization, limiting its potential for adaptive industrial applications.

Wei Li, Hongbo Wang, and Zhihua Fenga pioneered an ultra-high-resolution, non-contact measurement system utilizing an eddy current sensor (ECS). Their study demonstrated superior resolution capabilities (0.42 nm static, 2.2 μm dynamic) compared to other techniques. However, ECS-based methods lack self-learning capabilities, which restricts their ability to dynamically adjust to production variables.

Sprecher Energie Osterreich Gmbh (US5212539A, 1993) introduced an apparatus leveraging optical sensors to determine object dimensions. Although this study contributed to advancements in precision measurement, it did not focus on AI-driven optimizations or real-time adaptive control.

Martínez-Anton, I. Serroukh, and E. Bernabeu investigated laser diffraction for wire diameter measurement, emphasizing its effectiveness in real-time applications. However, classical Fraunhofer diffraction models showed limitations when applied to three-dimensional objects, highlighting the need for refined mathematical approaches to improve adaptability in dynamic production settings.

MICROTEC S.r.l. (EP0626560A2, 1994) developed a high-speed measurement device for assessing transversal sections of moving longitudinal objects. While beneficial for rapid analysis, this method did not explore AI-based control mechanisms for process optimization, missing an opportunity for real-time diameter regulation.

Beltronics Inc. (US4697088A, 1987) proposed a method using optical scanning techniques to detect sharp edge transitions based on reflective variations. While effective in scanning-based inspections, it lacked integration with production monitoring systems and AI-driven enhancements.

Evgeny Fedorov and Alexander Koba from Tomsk Polytechnic University developed a three-axis laser-based measuring system for precise diameter calculations. Their approach involved mathematical modeling for object radii and center positions but was primarily designed for static measurements rather than dynamic process control in industrial settings.

Sebastian Tamayo Vegas and Khalid Lafdi of Northumbria University conducted an extensive review of non-contact tools for structural health monitoring. Their study highlighted the increasing reliance on AI-powered vision systems, deep learning, and sensor fusion for enhanced defect detection. This supports the argument for integrating AI-driven solutions into industrial measurement applications to improve quality control and process automation.

Tzzy-Shuh Chang and Hsun-Hau Huang introduced a portable imaging-based measurement system with self-calibration, enabling precise measurement of hard-to-access objects. Although useful for various applications, it was not specifically designed for high-speed manufacturing environments. Similarly, Yuki Kimura, Akira Matsui, and Shingo Inazumi developed an imaging-based inspection system for analyzing external features, though it lacked real-time adaptive feedback for process control.

Despite these technological advancements, a critical research gap exists in the practical implementation of AI-driven vision-controlled diameter optimization in flux core wire drawing. While numerous studies focus on measurement accuracy, they often neglect real-time corrective capabilities essential for industrial automation. Integrating AI-driven vision systems with machine learning algorithms presents a transformative approach to enhancing precision, reducing human intervention errors, and ensuring continuous monitoring. Future research should focus on the development of adaptive AI models capable of dynamically adjusting to variations in wire diameter, enabling fully automated quality control in flux core wire manufacturing.

MATERIALS AND METHOD

The developed system integrates high-resolution imaging technology with advanced AI-driven analysis to continuously track and control the wire drawing process. The AI model is trained using an extensive dataset that includes variations in wire diameter and corresponding process conditions. By processing this data in real-time, the system identifies any deviations from the predefined specifications and autonomously modifies drawing parameters to maintain consistency.

To achieve precise and non-intrusive measurement, the system employs contactless sensing techniques that facilitate rapid data acquisition without interfering with production flow. This real-time approach enhances accuracy and responsiveness, significantly reducing defects and material wastage.

Conventional wire diameter control in flux core wire drawing relies on manual inspection and mechanical dies, which involve periodic checks to ensure dimensional accuracy. However, data analysis over a one-month period has revealed considerable diameter fluctuations, demonstrating the limitations of these traditional techniques. Since these inconsistencies are often detected only after production is completed, manufacturers are unable to implement immediate corrective actions. This underscores the necessity for an AI-integrated, real-time monitoring system that ensures continuous precision, optimizes efficiency, and eliminates defects before they affect final product quality.

The TVGD5000 series represents a cutting-edge advancement in AI-driven, vision-controlled diameter measurement technology. This system incorporates a laser-collimated transmission mechanism in conjunction with a CCD line sensor to achieve precise, non-contact measurement. A finely focused laser beam is projected across the CCD sensor's detection range. When a wire or similar object intercepts the beam, it casts a shadow onto the sensor, which is captured at a high sampling rate of 1.2 kHz with an exposure time of approximately 2 microseconds. This rapid acquisition ensures high-resolution imaging while mitigating errors caused by wire vibrations.

The microprocessor embedded within the gauge processes the acquired data using a sophisticated algorithm capable of handling 1,200 measurements per second. These measurements are analyzed and synthesized to generate a precise diameter reading, which is then digitally converted for real-time display. Furthermore, the system enables seamless data communication via an RS485 interface, allowing integration into automated production environments for comprehensive monitoring and analysis.

By leveraging laser-collimated transmission and CCD sensor technology, the TVGD5000 series ensures high-speed and ultra-precise diameter measurements. This approach is essential for manufacturing processes requiring stringent quality control. Similar principles are employed in advanced CCD laser micrometers, such as the IG Series, which offer high-resolution differentiation regardless of the target's optical properties.

In practical application, the AI-powered TVGD5000 series has demonstrated its ability to maintain flux core wire diameters within the specified range of 2.80mm (+0, -0.05mm). This precision control has led to improved product quality, minimized material wastage, and enhanced sustainability in welding wire manufacturing, contributing to efficient and reliable production workflows.

MEASUREMENT DATA FROM TRADITIONAL MONITORING MEASUREMENT METHODOLOTY

Date	Measurement	Average	Date	Measurement	Average
	2.83		17.10.2024	2.80, 2.84 ,2.80,2.79	2.81
02.10.2024	2.65, 2.85 , 2.9 ,2.75	2.79	18.10.2024	2.78, 2.83 ,2.79,2.75	2.80
03.10.2024	2.81,2.80,2.74,2.63	2.75	19.10.2024	2.73,2.69,2.79, 2.82	2.79
04.10.2024	2.88 , 2.90 ,2.74,2.68	2.80	20.10.2024	2.81 ,2.77,2.79,2.80	2.79
05.10.2024	2.76, 2.82 , 2.88 ,2.78	2.81	21.10.2024	2.79,2.82, 2.83 ,2.74	2.80
06.10.2024	2.78, 2.9 ,2.75, 2.82	2.81	22.10.2024	2.73,2.75,2.80, 2.84	2.78
07.10.2024	2.80,2.77, 2.82 ,2.76	2.79	23.10.2024	2.82 ,2.74,2.75,2.79	2.78
08.10.2024	2.73,2.78,2.81,2.85	2.79	24.10.2024	2.80, 2.83 ,2.79,2.75	2.79
09.10.2024	2.80, 2.83 ,2.78,2.76	2.79	25.10.2024	2.78,2.81,2.79,2.83	2.80
10.10.2024	2.73,2.77,2.82, 2.83	2.79	26.10.2024	2.80,2.76,2.77,2.82	2.79
11.10.2024	2.75, 2.85 ,2.77,2.76	2.78	27.10.2024	2.79, 2.82 ,2.77,2.80	2.79
12.10.2024	2.80,2.74,2.76, 2.83	2.78	28.10.2024	2.80, 2.82 ,2.79,2.80	2.80
13.10.2024	2.81, 2.83 , 2.85 ,2.78	2.82	29.10.2024	2.74,2.75,2.76, 2.83	2.77

14.10.2024	2.77,2.76,2.80,2.78	2.78	30.10.2024	2.80,2.82,2.81,2.78	2.80
15.10.2024	2.69,2.75,2.77,2.78	2.74	31.10.2024	2.77,2.76,2.80,2.82	2.79
16.10.2024	2.85,2.80,2.81,2.82	2.82			

Device Model	<u>TVGD5000</u>	
Measuring Range	0.2mm ~45mm	
Measuring frequency	800/axis	
Accuracy	0.002	
Repeatability	0.5 microns	
Resolution	0.001mm	
UOM	mm/Inch	
Temperature	-10 to 50*C	
Humidity	<90% Relative	
Outputs	<u>TVGD5000 Series AI Vision controlled diameter optimization</u> RS485 as standard, RS232, Profinet, OPC, Anybus	

TVGD5000 Series AI Vision controlled diameter optimization

MEASUREMENT DATA FROM AI MONITORING MEASUREMENT METHODOLOGY

Date	Measurement	Average	Date	Measurement	Average
01.11.2024	2.65,2.75,2.81,2.80	2.75	16.11.2024	2.80,2.74,2.75,2.79	2.78
02.11.2024	2.67,2.80,2.80,2.74	2.78	17.11.2024	2.80,2.80,2.79,2.75	2.79
03.11.2024	2.80,2.80,2.75,2.65	2.74	18.11.2024	2.78,2.80,2.79,2.75	2.80
04.11.2024	2.80,2.80,2.75,2.67	2.80	19.11.2024	2.73,2.69,2.79,2.80	2.79
05.11.2024	2.77,2.83,2.80,2.77	2.80	20.11.2024	2.81,2.77,2.79,2.80	2.79
06.11.2024	2.74,2.80,2.70,2.80	2.80	21.11.2024	2.79,2.80,2.80,2.74	2.80
07.11.2024	2.79,2.78,2.80,2.74	2.79	22.11.2024	2.74,2.89,2.70,2.80	2.80
08.11.2024	2.74,2.79,2.80,2.80	2.79	23.11.2024	2.79,2.78,2.80,2.74	2.79
09.11.2024	2.79,2.80,2.76,2.74	2.79	24.11.2024	2.74,2.79,2.80,2.80	2.79
10.11.2024	2.70,2.78,2.80,2.80	2.79	25.11.2024	2.79,2.80,2.76,2.74	2.79
11.11.2024	2.70,2.80,2.78,2.77	2.78	26.11.2024	2.80,2.76,2.77,2.80	2.79
12.11.2024	2.80,2.75,2.76,2.80	2.78	27.11.2024	2.79,2.80,2.77,2.80	2.79
13.11.2024	2.81,2.77,2.79,2.80	2.79	28.11.2024	2.80,2.80,2.79,2.80	2.80
14.11.2024	2.79,2.80,2.80,2.74	2.80	29.11.2024	2.74,2.75,2.76,2.80	2.77
15.11.2024	2.73,2.75,2.80,2.80	2.78	30.11.2024	2.80,2.80,2.80,2.78	2.80

CONCLUSION

This study underscores the transformative impact of AI vision-controlled diameter measurement in flux core wire drawing. By integrating real-time monitoring and adaptive control, the system effectively overcomes the challenges associated with traditional measurement methods, which often suffer from inconsistencies and reliance on manual intervention. The AI-driven approach ensures precise diameter regulation, significantly reducing material wastage while enhancing production efficiency.

The results demonstrate that incorporating AI vision technology into manufacturing not only strengthens quality control but also minimizes defects and optimizes resource utilization. The automation of diameter measurement and correction enhances operational reliability and cost-effectiveness, positioning AI-driven systems as a practical and scalable solution for modern wire production industries.

These insights establish a strong foundation for the wider adoption of contactless measurement technologies in flux core welding wire manufacturing. Leveraging AI-powered precision measurement allows manufacturers to streamline production processes, improve quality control, and achieve near-zero defects—paving the way for the next generation of industrial automation.

Future advancements will focus on refining AI algorithms to improve predictive capabilities and adaptability. Additionally, extending the system's applicability to diverse wire materials and varying

production conditions will further enhance its industrial relevance, ensuring continuous innovation in manufacturing automation and quality assurance.

Declaration of Interest

The author(s) declare that there is no conflict of interest regarding the publication of the manuscript. The research was conducted independently, and no financial, commercial, or personal relationships have influenced the preparation or outcomes of this work.

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A Sensor Integrated Tool for Cutting Force Monitoring in Gantry Milling to Improve Flatness and Surface Finish: - Review Study

Rahul N. Katre [1], Professor G. Waghmare [2] and Dr. P.D.Kamble [3]

Abstract

INTRODUCTION

Air pre-heaters are crucial components in power plants, improving energy efficiency by recovering heat from exhaust gases. Sector plates, a critical component of air pre-heaters, require precision machining to meet stringent dimensional and surface quality standards. Traditional machining methods often face challenges like tool wear, dimensional inaccuracies, Productivity, Rejections and operational inefficiencies & cost factors.

The application of sensors to special-purpose gantry milling machines offers an innovative approach to overcoming these challenges. Sensor systems provide real-time data analytics, enabling adaptive machining processes that enhance both productivity and precision. The key benefits include process optimization, predictive maintenance, real-time monitoring, and improved dimensional accuracy. The study outlines the potential improvements in machining efficiency and cost-effectiveness through AI-enhanced sensor systems, while also discussing challenges and future research directions in this field.

Flatness and thickness are critical parameters in APH (Air Preheater) sector plates because they directly impact the efficiency, sealing, and durability of the heat exchanger system. Here's why these specific tolerances matter:

A. Flatness (Within 400 Microns)

Sealing Efficiency: If the plates are not flat within the specified limit, air leakage can increase, reducing the efficiency of heat transfer.

Uniform Heat Transfer: Warped plates can create uneven airflow, leading to hotspots, increased pressure drop, and lower thermal performance.

Structural Integrity: Deviation beyond 400 microns can cause misalignment during assembly, leading to mechanical stress and potential failure.

B. Thickness (Upto 13 mm)

Mechanical Strength: A precise thickness ensures the plates can withstand thermal expansion and mechanical loads without deformation.

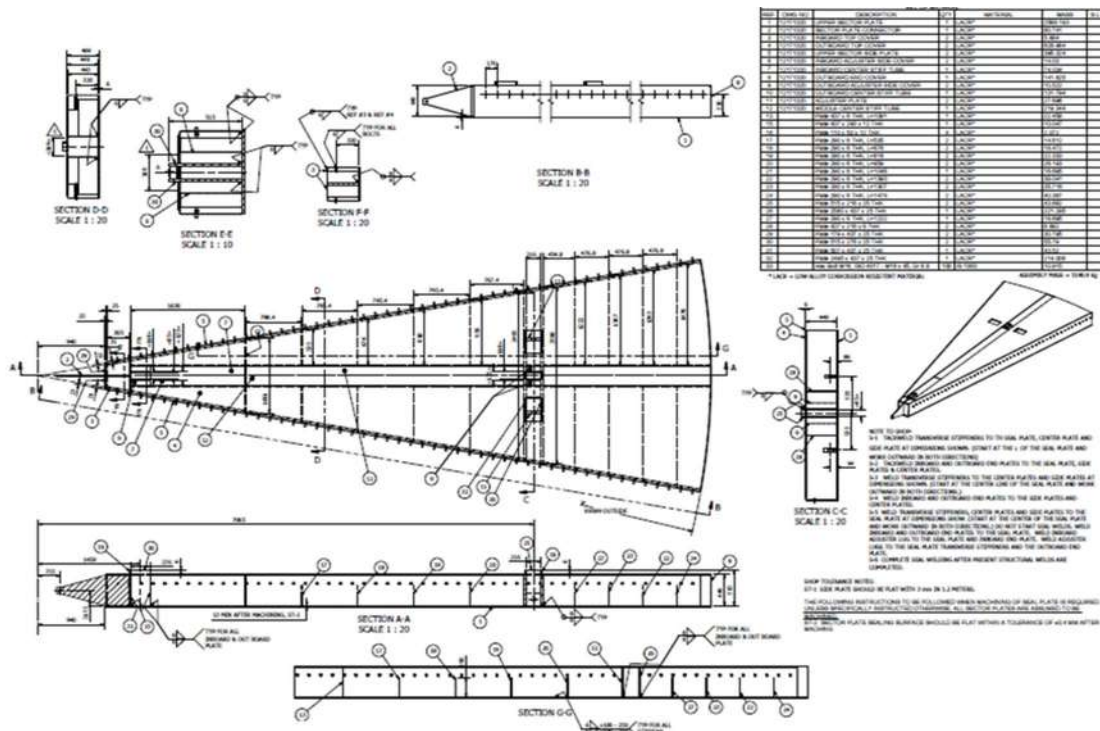
Consistent Heat Exchange: Variations in thickness can affect heat absorption and dissipation, reducing overall efficiency.

Weight and Assembly Considerations: Maintaining uniform thickness prevents excessive weight variations that can impact the rotor's balance and operational stability.

Ensuring flatness within 400 microns and thickness upto 13 mm helps maintain optimal heat transfer efficiency, air sealing, and structural durability in an APH system.



Sector Plate Fabrication



THE FOLLOWING INSTRUCTIONS TO BE FOLLOWED WHEN MACHINING OF SEAL PLATE IS REQUIRED. UNLESS SPECIFICALLY INSTRUCTED OTHERWISE, ALL SECTOR PLATES ARE ASSUMED TO BE MACHINED.
ST-2 SECTOR PLATE SEALING SURFACE SHOULD BE FLAT WITHIN A TOLERANCE OF ± 0.4 MM AFTER MACHINING



Conventional Gantry Machine: - Sector plate Under Machining Set Up



Conventional Gantry Milling Machine: - Sector Plate Under Machining



Conventional Tool with inserts



Conventional Flatness Check (Straight Edge & Filler Gauge)



Conventional Thickness Check (Thickness check by Modsonic)



Final Assembly of Sector Plates

Objectives: - The collective objective aim at improve the performance, efficiency, and reliability of the special-purpose gantry milling machine, demonstrating the transformative potential of sensors in high-precision manufacturing.

Maximizing Productivity – Implement sensors in the Special Purpose Gantry Milling Machine (SPGMM) to automate real-time adjustments, optimize cutting parameters, and reduce machining cycle time, leading to increased throughput of sector plates for Air Pre-Heaters.

Ensuring Superior Machining Quality – Utilize Sensor based real-time monitoring and adaptive control to maintain flatness within 400 microns and thickness upto 13 mm, ensuring consistent dimensional accuracy and high surface finish, thereby meeting stringent industry standards.

Minimizing Rejections and Rework – Leverage powered predictive analytics to detect tool wear, vibration, and process anomalies early, significantly reducing machining defects, material wastage, and the need for post-machining corrections.

Optimizing Operational Efficiency – Develop an intelligent machining system that enhances energy efficiency, minimizes manual intervention, and streamlines workflow, ensuring a cost-effective and sustainable manufacturing process.

Research Gap:

The integration of sensor technologies in special-purpose gantry milling machines for machining sector plates used in air pre-heaters has been explored to some extent. However, significant gaps remain in terms of optimization, real-time monitoring, and data-driven process control.

1. **Limited Research on Sensor-Based Process Optimization**
Existing studies focus on general milling operations with sensor applications, but there is a lack of dedicated research on optimizing machining parameters (such as cutting forces, tool wear, and vibration control) using sensor data specifically for sector plate machining.
2. **Insufficient Implementation of Advanced Sensing Technologies**
While traditional sensors (such as force, vibration, and temperature sensors) have been used in milling, the integration of advanced sensing technologies, including IoT-enabled smart sensors, AI-driven predictive maintenance, and machine learning-based process control, remains underexplored for this specific application.
3. **Lack of Studies on Machine-Specific Challenges**
Special-purpose gantry milling machines have unique kinematic and dynamic characteristics that influence machining performance. Current research does not sufficiently address the impact of machine structure, workpiece clamping strategies, and toolpath optimization when sensors are integrated.
4. **Real-Time Monitoring and Control Gaps**
Most research in this field focuses on offline data analysis, whereas real-time monitoring and adaptive control using sensor feedback to improve machining accuracy and efficiency in gantry milling machines for sector plates are not well established.
5. **Energy Efficiency and Sustainability Aspects**
There is limited research on how sensor-driven monitoring can contribute to energy-efficient machining processes for air pre-heater sector plates, minimizing waste, and improving overall sustainability in manufacturing.
6. **Lack of Industry-Oriented Case Studies and Validation**
While theoretical studies exist, practical implementation and validation in industrial settings are scarce. A gap remains in experimental validation and real-world case studies demonstrating the effectiveness of sensor integration in sector plate machining.

Addressing these research gaps can lead to significant advancements in precision machining, predictive maintenance, and automation of special-purpose gantry milling machines. Future research should focus on real-time adaptive control, AI-driven sensor applications, and industrial case studies to enhance efficiency, accuracy, and sustainability in air pre-heater sector plate manufacturing.

This paper discusses the implementation and benefits of AI sensors in the machining of sector plates for air pre-heaters.

Literature Review

Application of Sensors in Tool Condition Monitoring (TCM) for Milling : Muhammad Rizal et al. reviewed the role of sensors in tool condition monitoring (TCM) during milling operations. Their study highlighted how sensors provide critical data on tool wear, breakage, and chatter, all of which directly impact machining quality and efficiency. They emphasized that for industrial adoption, sensors must be cost-effective, compact, durable, and non-intrusive. A key challenge for future research is the integration of innovative sensors into TCM systems, particularly through wireless data transmission, to enhance overall machining performance.

AI-Based Optimization of Machining Parameters: Julius Schoop et al. introduced a novel approach to optimizing machining processes using artificial intelligence (AI). Their research focused on reinforcement learning (RL) to dynamically adjust cutting parameters, balancing workpiece quality and productivity. Instead of relying on static machining settings, their proposed AI-driven model determines the optimal tool replacement timing, reducing inefficiencies and costs. This dynamic approach offers significant potential for improving milling efficiency without requiring extensive manual adjustments or complex modeling.

Machine Learning for Tool Wear Monitoring in CFRP Drilling: C. Domínguez-Monferrer et al. explored a machine learning-based approach for non-intrusive monitoring of tool wear in Carbon Fiber Reinforced Polymer (CFRP) drilling, specifically within the aerospace industry. Their research utilized real industrial data, analyzing spindle power consumption signals to track tool wear over time. Optical microscopy and Scanning Electron Microscopy (SEM) were used to qualitatively assess tool degradation. Additionally, a predictive machine learning model was developed to estimate tool conditions, enhancing automation and efficiency in machining operations.

Sensor Fusion for Enhanced Tool Wear Detection: Jabbar Abbas et al. investigated a multi-sensor fusion model incorporating force, vibration, acoustic emission, strain, and sound sensors for end milling tool wear monitoring. Their study applied the Automated Sensor and Signal Processing Selection (ASPS) method to optimize signal processing and sensor selection. Results indicated that combining multiple sensory inputs improved fault detection accuracy compared to single-sensor methods, making sensor fusion a promising technique for real-time tool wear assessment.

Real-Time Tool Wear Estimation Using Signal Processing: Bhattacharyya et al. proposed an advanced signal processing technique for real-time tool wear estimation in face milling. Their approach combined multiple methods, including linear filtering, time-domain averaging, and wavelet transformation, to extract key features from cutting force signals. They implemented sensor fusion at the feature level to enhance wear prediction accuracy. Additionally, isotonic regression and exponential smoothing techniques were applied to ensure model robustness. Their multi-stage modeling framework demonstrated improved wear estimation capabilities across both laboratory and industrial settings.

Real-Time Cutting Tool Condition Monitoring System: Franci Cus and Uros Zuperl developed a real-time tool condition monitoring system for milling operations. Their system demonstrated high reliability and parallel processing capabilities, enabling accurate tracking of tool wear progression. By integrating an Adaptive Neuro-Fuzzy Inference System (ANFIS), they achieved precise flank wear estimation. Furthermore, their neural network-based classification system effectively identified various cutting states, including tool breakage.

Torque-Based Sensorized Toolholder for End Milling Monitoring: G. Dini and H.F. Tognazzi examined the effectiveness of a sensorized toolholder for on-line tool condition monitoring in end milling. Their system utilized a commercial rotating dynamometer integrated into the toolholder to acquire cutting torque signals. Unlike traditional force measurement methods, this approach was non-invasive and preserved the mechanical integrity of the toolholder. Preliminary cutting tests confirmed that the acquired signals could successfully indicate tool wear conditions. The study also proposed threshold-based monitoring strategies to enhance real-time detection of tool status during milling operations.

3. Methodology

3.1 Traditional Machining & Monitoring of sector plates :-

Achieving the required flatness and thickness in the machining of sector plates for air preheaters using traditional machining methods presents several challenges. These issues can arise due to various factors,

including the machining process itself, material properties, machine limitations, and the complexity of monitoring and controlling these parameters. Below are some of the key challenges involved:

Geometric Complexity: -Large and irregularly shaped components make it harder to control the exact removal of material across the entire surface, potentially leading to variations in flatness and thickness.

Material hardness: Sector plates are made of high-strength, heat-resistant alloys (like stainless steel or carbon steel), which can cause rapid wear on tools. This wear leads to less precise cuts and may alter the intended flatness and thickness of the plate.

Thermal expansion: The heat generated during traditional machining (especially when using high-speed cutting) can cause thermal expansion in the material. This can result in dimensional changes, particularly in flatness, after the part cools down.

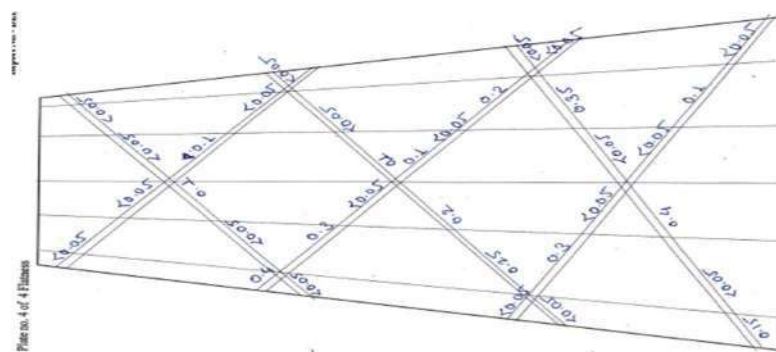
Tool wear and deflection: Traditional machining tools can wear down quickly when working with tough materials. This wear leads to tool deflection, causing inaccuracies in the final dimensions of the sector plate.

Tool limitations: Machining large or complex components with high-precision requirements might exceed the capabilities of conventional tools, leading to uneven removal of material and difficulties in maintaining consistent flatness and thickness.

Surface Finish Inconsistencies: -Traditional machining often struggles to produce smooth, consistent surface finish required for parts with strict flatness requirements. For sector plates in air preheaters, any imperfections can lead to inefficient heat transfer or mechanical stress. Inconsistent cutting forces can cause vibration during the machining process, leading to variations in surface quality, which can affect the flatness and thickness.

Monitoring and Checking of Flatness and Thickness

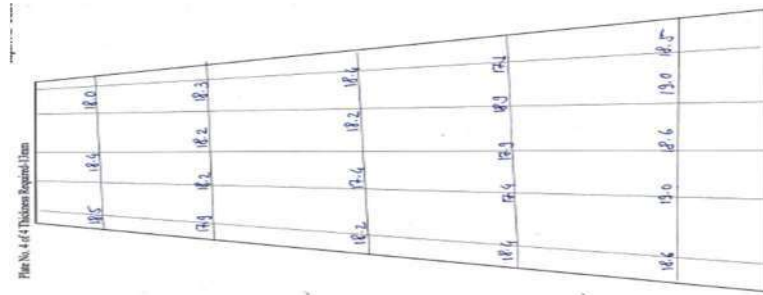
Conventional methods (e.g., dial gauges, mechanical feelers) are used to monitor flatness, but these methods can be inaccurate or labor-intensive. Ensuring consistent flatness across the entire surface of the sector plate requires constant adjustments and detailed measurements during machining.



Conventional Flatness checks results where in flatness is going beyond the required parameters and approval are to be sought for usage in air preheater

Thermal distortion: As the part is machined, heat buildup can warp the sector plate, causing deviations from the desired flatness. This requires post-machining cooling time, and additional corrections to compensate for these distortions.

Thickness Variations: -Traditional milling and turning operations can result in inconsistent material removal, especially with larger parts. The thickness of sector plates may vary across different points, affecting the overall performance of the air preheater.



Thickness variation after machining resulting in section of excessive material (by 8-10 mm for better life and acceptability) leading to high material costs

To monitor thickness accurately, continuous measurements must be taken throughout the machining process, which can be difficult to implement with traditional methods. The variation in thickness may also result from tool wear or material inconsistencies.

Refer Below Table for before modification readings of Sector plate machining

Sr No	Plate Dimensions in mm	Plate Setting Time in minutes	No of Cuts	Depth of cut (A,B,C) in mm	Flatness Observed Vs +/-400 micron	Thickness Range Vs 13 mm mini	Total Time in hours
1	5464 X2411	70	3	(3,2.5,1.5)	+/-450 mic	13.2-19 mm	72
2	5464 X2411	50	3	(3,2.5,1.5)	+/-420 mic	11.2-18.8 mm	72
3	5464 X2411	65	3	(3,2.5,1.5)	+/-405 mic	13.1-18.7 mm	72
4	5464 X2411	60	3	(3,2.5,1.5)	+/- 380 mic	13.1-18.9 mm	72
5	5464 X2411	60	3	(3,2.5,1.5)	+/-410 mic	12.9-19.3 mm	72
6	5464 X2411	55	3	(3,2.5,1.5)	+/-380 mic	13.6-19.5 mm	72
7	4229 X 1728	50	2	(3,2)	+/-360 mic	17.1-19 mm	48
8	4229 X 1728	60	2	(3,2)	+/- 395 mic	17.3-18.8 mm	48
9	4229 X 1728	60	2	(3,2)	+/- 400 mic	17.1-19 mm	48
10	4229 X 1728	55	2	(3,2)	+/- 365 mic	16.5-18.8 mm	48

Difficulty in In-Process Monitoring

Real-time monitoring of flatness and thickness is often limited in traditional machining. Specialized sensors or feedback systems are typically absent in older or less advanced machines, making it challenging to ensure continuous accuracy.

Manual adjustments are typically required if flatness or thickness deviations are detected, but this process is time-consuming and increases the likelihood of human error.

3.2 Solutions to Improve Flatness and Thickness Control

Our Main Focus is to improve machine by adding below components.

Addition of proper Bed for Job Holding for proper locking thereby replacing the need of weld after levelling of job before machining.

Procurement of Bed with width of 2800 mm X 6000 mm for better securing of sector place there by enhancing the stability during machining.

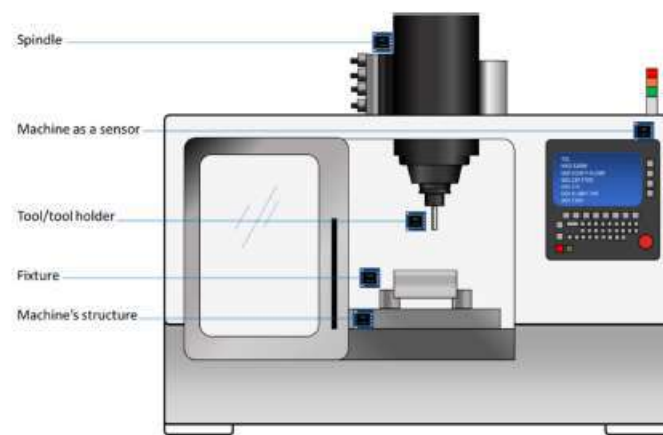


Previously Bed for Securing the sector plate is not available increasing the vibration and level issues which is now being taken care off by adding the bed below gantry milling

3.2.1 Integration Sensors on the Special Purpose Gantry Milling Machine

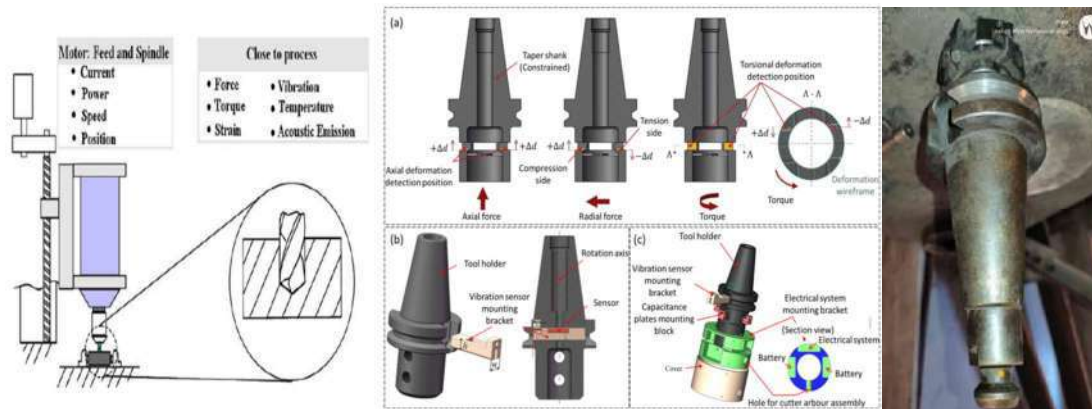
A special-purpose gantry milling machine can be integrated with a variety of sensors to monitor parameters such as cutting force, tool temperature, vibration, and material condition. The system incorporates AI algorithms for predictive maintenance, process adaptation, and quality control.

Integration of Sensors to existing CNC machines provides greater precision and repeatability compared to traditional machining methods. CNC systems, especially those with adaptive control or real-time feedback loops, can help monitor and adjust the process to maintain consistent flatness and thickness.



Potential Sensor Locations are Near Spindle, Tool Holder, Machine Structure and Fixture (Table) & machine itself.

Cutting force: The cutting force is one of the most sensitive indicators of machining performance, Milling operation and its generated machining signal ,Different possible location for force sensor



Measurable Phenomena for Sensor monitoring

Fixing of Dial to Head to verify leveling to plate to obtain end results i.e thickness & Level



Fixing of dial on Moving head of Gantry helped to check & set the level of plate along the length and width for proper set before initiating the cut to meet the dimensional accuracy

Selected sensor @ SGS eddy NCDT SGS 4701 on existing spindle (eddy current sensor)

Spindle Growth System
eddyNCDT SGS4701

- Miniature sensor design
- M12 controller - Integration into spindle or flange mounting
- Sensors for ferromagnetic and non-ferromagnetic targets
- Temperature measurement integrated

S = signal = inner conductor
M = ground = shield = outer conductor

Pin	Assignment	Color (cable: PC4701-x)
1	GND	White
2	Supply 12 ... 32 VDC	Brown
3	Displacement signal	Green
4	Temperature signal	Yellow
5	n.c.	Gray
6	internal	Pink
7	internal	Blue
8	n.c.	Red

8-pin housing connector M12x1
View on pin side

Measuring the thermal extension of spindles

The SGS4701 displacement measuring system (Spindle Growth System) is developed specifically for high speed milling machine applications. Due to high machining speeds and the heat generated, the linear thermal expansion of the spindle in precision machine tools needs to be compensated for in order to keep the tool in a defined position at all times. The SGS sensor measures the thermal and centrifugal force expansion of the spindle. These measurement values are fed into the CNC machine tool as correctional values, compensating for any positioning errors.

System design : The SGS 4701 consists of a sensor, a sensor cable and a controller, factory calibrated for ferromagnetic and non-ferromagnetic targets. Two miniature sensors enable it to be installed directly in the spindle, where the measurements take place, typically on the labyrinth-ring of the spindle. As well as measuring linear thermal expansion, the temperature of the sensor is also detected and output. The compact controller can be installed on the spindle housing via a flange or directly in the spindle

The sensor cable must not be shortened as functionality loss may arise. Removing the connector is only permitted behind the plug-sided crimp when using the solder connections.

The SGS 4701 operates on the eddy current measuring principle. This non-contact measurement is wear-free. Furthermore, the measurement procedure is resistant to disturbances such as heat, dust and oil.

Details

Non-contact and wear-resistant measurement

Economical design

Miniature sensor and controller completely integrated on the spindle

Calibration for aluminum and iron

Temperature measurement integrated in the sensor

Excellent temperature stability, Measuring range 500 f'm & Resolution 0.5f'm

Model		SGS4701
Measuring range		500 μm (optional 250 μm ¹⁾)
Start of measuring range		100 μm (optional 50 μm ¹⁾)
Measuring rate	Analog output	64 kSa/s (16 bit)
Resolution ²⁾		0.5 μm
Frequency response (-3dB)		2000 Hz
Linearity		< $\pm 2 \mu\text{m}$
Temperature stability ³⁾	Sensor	< 150 ppm FSO / K
	Controller	< 500 ppm FSO / K
Temperature compensation	Sensor	+10 ... +80 °C
	Controller	+10 ... +70 °C
Min. target size (flat)		6 mm (optional 3.5 mm ¹⁾)
Target material ⁴⁾		Steel, aluminum
Supply voltage		12 ... 32 VDC
Power consumption		0.6 W
Analog output	Displacement	0.5 ... 9.5 V (100 ... 600 μm , optional 50 ... 300 μm ¹⁾)
	Temperature	0.5 ... 9.5 V (0 ... 90 °C)
Connection	Sensor: integrated cable ⁵⁾ , standard length 1 m (0.4 ... 1.5 m on request), min. bending radius 12 mm Supply/signal: 8-pole M12 connector (cable see accessories)	
Temperature range	Sensor	0 ... +90 °C
	Controller	+10 ... +70 °C
Shock (DIN EN 60068-2-27)		50 g / 6 ms in each direction, 1000 shocks each
Vibration (DIN EN 60068-2-6)		20 g / 10 ... 3000 Hz
Protection class (DIN EN 60529)		IP67 (plugged) ⁶⁾
Weight ⁷⁾		approx. 85 g

FSO = Full Scale Output

¹⁾ For OEM modifications: sensor with a measuring range of 250 μm and an offset of 50 μm are possible

²⁾ Static, relates to mid of measuring range

³⁾ Relates to mid of measuring range, in the compensated temperature range

⁴⁾ Steel: St37 steel DIN1.0037 / aluminum: AlMg3

⁵⁾ Detailed cable specifications can be found in the operating instructions

⁶⁾ Protection class does not apply for the controller sleeve!

⁷⁾ Total weight for controller, cable and sensor

Applied sensor @ SGS eddy NCDT SGS 4701 on existing spindle (eddy current sensor)

The eddyNCDT series SGS 4701 (Non-Contacting Displacement Transducers) measuring system operates on the basis of eddy currents without making physical contact.

It is used for measurements on objects consisting of electrically conducting materials which have ferromagnetic or non-ferromagnetic properties.

High frequency alternating currents flow through a coil cast in a sensor housing. The electromagnetic field from the coil induces eddy currents in the electrically conducting target, causing the alternating current resistance of the coil to change. This change of impedance delivers an electrical signal proportional to the distance of the target from the sensor.

The eddyNCDT SGS 4701 conditions the sensor signals user-fairly at a bandwidth of 2,000 Hz. Thus a so-called spindle monitoring is also still possible at numbers of revolutions up to 120,000 rpm. The spindle growth itself can be detected at even higher numbers of revolutions.

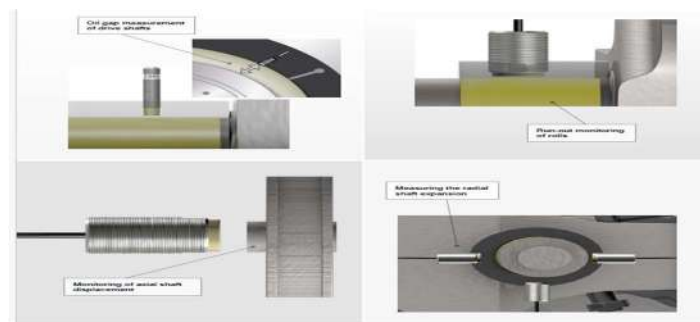
Monitoring an axial expansion requires subsequent individual filtering by the user. Micro-Epsilon recommends averaging or filtering across several revolutions in order to record only the relative expansion during a defined period. During an unfiltered/non-averaged measurement, effects such as those within a single revolution also become visible; this can cause incorrect interpretations.

In order to achieve highly accurate measurements, it is imperatively necessary that one (or more) sample(s) item under test is / are provided by the customer.

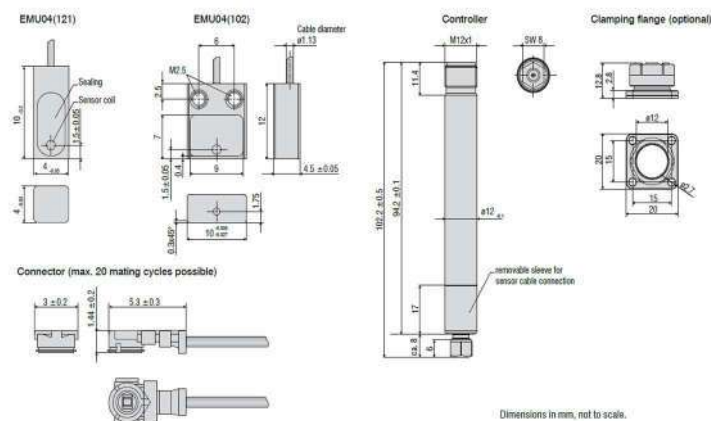
The eddyNCDT series SGS 4701 is developed for recording the thermal and mechanical expansion of highspeed milling spindles. This enables the active compensation of the axial spindle expansion via e. g. a CNC machine tool.

The SGS 4701 consists of a sensor, a sensor cable and a controller which can be factory calibrated for ferromagnetic or non-ferromagnetic measurement objects. Miniature sensors enable it to be installed directly in the spindle, where the measurements take place, typically on the labyrinth-ring of the spindle. As well as measuring linear expansion, the temperature of the sensor is also detected and output. The compact controller can be installed on the spindle housing via a flange or directly in the spindle.

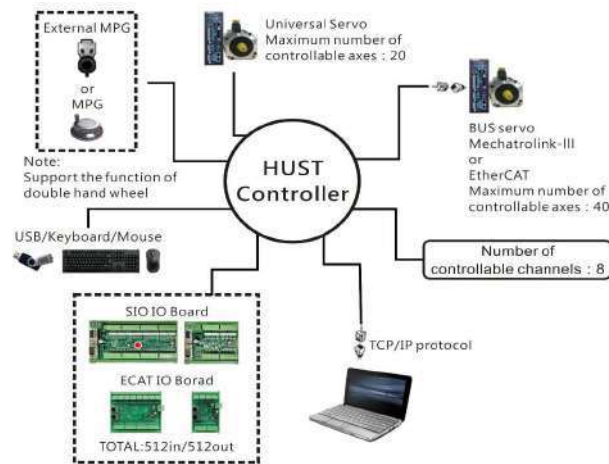
Positioning of selected sensor for tracking various parameter



Fixing details for sensor for integration with Spindle



Hust CNC Architecture of existing gantry milling



Findings after installation of @ SGS eddy NCDT SGS 4701 on existing spindle & levelled Bed

Sr No	Plate Dimensions in mm	Plate Setting Time in minutes	No of Cuts	Depth of cut (A,B,C) in mm	Flatness Observed Vs +/- 400 micron	Thickness Range Vs 13 mm mini	Total Time in hours
1	5464 X2411	40	3	(2.5,2,1.5)	+/-320mic	14.7-16 .8mm	60
2	5464 X2411	43	3	(2.5,2,1.5)	+/-360mic	14.2-18.6 mm	60
3	5464 X2411	45	3	(2.5,2,1.5)	+/-340mic	15.1-17 .2mm	60
4	5464 X2411	37	3	(2.5,2,1.5)	+/- 390 mic	15.6-18.3 mm	60
5	5464 X2411	43	3	(2.5,2,1.5)	+/- 330 mic	13.2-18.4 mm	60
6	5464 X2411	41	3	(2.5,2,1.5)	+/-380 mic	13.6-19.5 mm	60
7	4229 X 1728	35	2	(2.5,1.5)	+/-360 mic	17.1-19.3 mm	40
8	4229 X 1728	38	2	(2.5,1.5)	+/- 395 mic	16.3-18.8 mm	40
9	4229 X 1728	36	2	(2.5,1.5)	+/- 360 mic	16.1-18.6 mm	40
10	4229 X 1728	33	2	(2.5,1.5)	+/- 365 mic	16.5-17.6 mm	40

- No Rejection of sector plates after installation of levelled bed & Spindle growth sensor
- Plate set up time reduced by 40 % after installation of levelled bed as fixture and positioning of bed is facilitating the plate fixing.
- Sensor installation has supported in optimising the machining thereby reducing the errors of flatness and thickness as plate level is optimal at m=necessary location helping the reduction in depth of cut and increase the feed rate RPM resulting in faster completion enhancing productivity by approx 20%.
- Surface finish though within acceptable limit ,is not improved much as cooling technology and inserts are same and still work needs to be done on this aspect.

Objective analysis

Maximizing Productivity – Productivity improvement upto 20%

Ensuring Superior Machining Quality – Flatness 400 micron & Min thickness 13 mm achieved on 10 samples.

Minimizing Rejections and Rework – Rejection minimised

Optimizing Operational Efficiency – By minimising the time required to work efficiency is improved

To Implement Predictive Maintenance: Not achieved as EDDY sensor is focus on optimal speed only, & not on tool or spindle health.

Conclusion:- Incorporating sensors in the special-purpose gantry milling machine for sector plate machining used in air pre-heaters provides a wide range of benefits, from improving process monitoring, predictive maintenance, and quality control to optimizing machining parameters and enhancing energy efficiency. The adaptability and intelligence offered by sensors helps significantly to improve both the precision and efficiency of manufacturing operations, leading to more reliable and cost-effective production of critical components like sector plates for air pre-heaters. The application of AI sensors in special-purpose gantry milling machines for sector plate machining in air pre-heaters offers substantial benefits in terms of precision, efficiency, and cost-effectiveness.

However, further research is required to address the challenges related to sensor calibration, data overload, and system integration. Real-time process monitoring, adaptive machining, and quality control significantly enhance the performance of milling operations.

Future work will focus on:

- Developing more advanced AI algorithms (Program) that can further improve adaptive machining.
- Enhancing the integration of AI sensors with other manufacturing systems such as the Manufacturing Execution System (MES).

Exploring the application of AI in multi-axis gantry milling machines for more complex air pre-heater designs.

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Unfolding the Awareness for Women and Children in Vidarbha Region

¹Mangesh S. Dhole, ¹Gungun Holani

¹Department of Forensic Science, G.H. Rasoni University, Anjangaon Bari Road, Amravati, Maharashtra 444701. mangeshdhole2@gmail.com.

ABSTRACT:

The concept of “Roots and Wings” as a pathway to empower women and children in Vidarbha region, focusing on the need for both a strong foundation and the freedom to thrive. “Roots” symbolize support systems like education, cultural identity, and community care, while “wings” represent opportunities for growth, self-expression, and independence. Women in Vidarbha often face deep-rooted challenges, including gender inequality, domestic violence, menstrual health, abuse and a lack of awareness about critical health issues like breast cancer. The children, on the other hand, struggle with bullying, child labour, neglect, lack of access to quality education and intense peer pressure. We have visited 21 Villages of Vidarbha region and find out the problems faced by women and children through questionnaires and face to face communication. Through rigorous study we have provided proper solution and create awareness among the women’s and children’s of Vidarbha region villages. This research highlights the importance of creating safe environments, spreading awareness, and building resources to support mental and physical well-being. By addressing these issues, we can foster resilience, self-awareness, and confidence, helping women and children in Vidarbha region.

Keywords: Women Empowerment, Child Welfare, Vidarbha Community Development.

INTRODUCTION:

women and children awareness in the Vidarbha region is to improve the socio-economic and health conditions of women and children by providing them with the knowledge, tools, and support they need to lead empowered, safe, and productive lives. In this region, awareness efforts focus on educating women about their legal rights, access to education, healthcare, and protection from violence, as well as promoting gender equality. For children, the goal is to raise awareness about child rights, the importance of education, proper nutrition, and protection from exploitation or abuse. Furthermore, the objective is to address specific regional challenges such as child labour, malnutrition, early marriage, and limited access to healthcare and education. By raising awareness, the aim is to foster a more informed and supportive environment, reduce gender-based discrimination, and create sustainable social change for women and children in Vidarbha. There is a need to raise awareness about maternal and child health, addressing issues like prenatal care, safe delivery practices, immunizations, and the importance of a healthy diet for mothers and young children. The region also faces challenges related to malnutrition, which awareness programs aim to combat by educating families about better nutrition practices. By addressing these critical issues, the objective is to create a more supportive environment where women and children in Vidarbha can thrive, ensuring their rights are upheld, their needs are met, and they can contribute to the development of their communities and region. This includes not only awareness but also advocacy for policy changes and community-based programs to improve living standards and quality of life.

Psychiatric disorders and psychological problems are the causes of domestic violence. They can drive individuals to hurt others and also increase the chances of an individual being hurt. There are some treatments for domestic violence that are now available, such as methods like Seeking Safety and Child-Parent Psychotherapy. few mental health professionals are trained to employ these techniques. To confront criminals is an intricate problem, distinct from the legal actions necessitated. Nonetheless, these prevention and treatment programs, like those of the Melissa Institute for Violence Prevention and Treatment, are just starting

to surface^[1] Childhood bullying is a public health problem with serious effects on health, social relationships, and education in childhood and adolescence. These consequences affect all parties—bullies, victims, and both—along with their parents and families. They can be long-lasting and persist well into adulthood. Other than the old-fashioned physical, verbal, and indirect bullying, cyber bullying has become a newer variant. Children, being seen as "different," particularly because of the way they look, are at greater risk of being bullied. Health workers generally have no specific guidelines and referral pathways for the management of child bullying in primary and secondary care. Special services, however, exist locally and on the internet to offer support^[2] Parents are critical in determining the education of their children based on how valuable they perceive education. If parents know that education matters for the future of their children, then they will easily embrace higher studies. This benefits not only the children but also their communities and the generations yet to come. The decision on education relies on both government sponsorship and family commitment. Thus, the government and parents must act in unison to provide proper education and training to children.^[3]

Commissioned by the Amsterdam Institute for Social Science Research of the University of Amsterdam. Child marriage is an issue affecting millions of children, particularly girls, in both the developing and developed world. Girls Not Brides was engaged, specifically to help fill gaps in research and to find out what ignited the movement to end child marriage. One of the representatives from Girls Not Brides provided other resources, like yearly reports and videos, and provided elaborate answers to the raised questions.^[4] Violence against women is a complex and serious issue that requires much effort to resolve. Society, schools, healthcare, and the legal system need to cooperate to stop it. The objective is to eliminate unjust social norms and protect victims. Ensuring that children witnessing or being exposed to violence early in life receive assistance is vital but not provided enough. Men's violence against women is not "normal" or inevitable. Attitudes have to shift, and women's standing in society has to be bettered. Everyone has to realize that violence in any relationship is never acceptable.^[5] The innovation of new contraceptives, particularly those used to space births, has provided individuals with greater options to plan their families in a better manner. The population of India has not been able to fully utilize the advances in contraceptive technology. To better the scenario, there is a requirement of more robust awareness campaigns (IEC) for family planning, encouraging use and advantages of spacing methods, and reinforcing national family welfare programs. Extended outreach services can also assist in achieving better outcomes and planned families for the country.^[6] In India, 65.46% of women were able to read and write as per the 2011 census, but nothing is known about their financial literacy. A survey in Nagpur tested women's knowledge of money and finance. It revealed that women know about interest rates but little about inflation. They prefer net banking and mobile wallets over credit cards and are most confident about bank deposits for saving money. The survey indicates that Nagpur women require improved financial education.^[7] The problem of tobacco and alcohol consumption among school-going tribal children is large. A study was conducted in residential schools of Nagpur District (July-September 2009) to identify how many children consume tobacco and alcohol and what they perceive about it. The data was collected by means of surveys and group discussions. It was found that 2.86% students smoked, 41.74% used smokeless tobacco, and 4.07% drank alcohol. The group discussions showed a lack of awareness regarding the ill effects of these habits. The research indicates that there should be improved education regarding the ill effects of tobacco and alcohol.^[8]

Women and LGBTQI+ personnel working in organizations have to be secured against sexual harassment with solid lawful terminologies, employers' sensitization schemes, and health provider support along with trained personnel. Internal Committee (IC) and Local Committee (LC) audits and their work have to be reviewed frequently from time to time. Strong conducive atmosphere to a productive and efficient organization is brought through intensive training, sessions, adequate remedy mechanisms, and maintaining freedom in the context of expression.^[9] Menstrual onset is a significant event in a girl's life, causing numerous changes. A study conducted in Assam, India with 185 girls aged between 10 and 19 years revealed that the majority of them were unaware of menstruation prior to the onset of their first period, rendering the experience frightening. The absence of assistance during or after their first period caused additional stress. The culture norms, myths, and shame exacerbated the situation. The research indicates the necessity of proper guidance and a conducive environment to assist girls in coping with these issues. It recommends creating awareness and offering support to enable girls to cope with menstruation more effectively.^[10] This piece discusses why most educated Indian women experience career disruptions following marriage even though they have superior degrees than men. It discovered that married women usually lack control over occupational choices and depend much on patronage from husbands and in-laws in order to stay at work. Social conditioning is a big factor, as women tend to adopt old gender roles as wives and mothers and therefore set work aside for family.^[11] Women's issues are distinctive in that they are subjected to concurrent oppression on the basis of

gender, caste, and class. This research, which interviewed 210 participants from Delhi and other regions, examines the determinants of sexual violence against Dalit women. Employing statistical techniques such as factor loading, Pearson correlation, and regression analysis, the research investigates the nexus between caste, class, and gender. The results show that caste plays a far more significant role than class in shaping the sexual violence Dalit women experience, pointing to the inter sectionality of caste, class, and gender in the construction of their vulnerability.^[12] This review examines the role played by health policies in supporting early childhood development (ECD). It is focused on ensuring that children grow well during their early years by emphasizing the combination of health, education, and social services. Maternal and child health care, proper nutrition, early learning, and parent support are areas of importance. The review reiterates that policies need to link these services and rely on research in informing decisions. The review also underlines the contributions of laws towards safeguarding the rights of children. In general, investment in ECD policies benefits children and society by improving health and future success.^[13]

Menstrual hygiene is essential for women's health, and it is critical to understand young adults' awareness and behavior in order to address public health. This research interviews and conducts surveys among West Bengal college students to understand their practices, perceptions of social norms, and menstrual hygiene knowledge. The findings reveal that there exist varying levels of awareness and practice among the students. The research also identifies shared myths and cultural factors that influence menstrual health. The aim is to develop improved education and policies to enhance menstrual hygiene for university students, which will assist in empowering young women and sustaining their health.^[14] This research targets 50 women entrepreneurs in Nagpur and explores challenges they encounter. The primary issues experienced were financial challenges and a lack of support from their families, which complicated business start-ups and operations. It also examined the role played by society in their entrepreneurial journey, both positively through support and negatively through criticism. The research identified what drives women to establish businesses and the challenges they encounter in sustaining them.^[15] The present study examined menstrual hygiene and knowledge in 100 adolescent girls of an urban slum. The majority of girls (72%) were in the age group of 15–19 years, 47% were educated up to high school, but 76% were unaware of menstruation before onset of their first period. Mothers were the primary source of information (84%), but only 16% were aware of bleeding beginning in the uterus. Whereas 60% employed the use of sanitary pads, the others used cloth, and 22% used no soap to wash their hands. Most of the girls also had restrictions placed on them when menstruating. The research discovered that menstrual hygiene was poor, and there is a need to teach girls the proper practices and menstrual health.^[16]

The scope of women and children awareness in the Vidarbha region encompasses a broad range of initiatives aimed at addressing the unique challenges faced by women and children in this area. These challenges include poverty, illiteracy, gender inequality, poor healthcare access, and limited resources for child protection and education. Scope of Women and Children Awareness in Vidarbha: Promote education for women and children, especially girls. Improve maternal and child health, including safe childbirth and vaccinations. Protect children from exploitation, child labour, and early marriage. Raise awareness of legal rights for women and children. Promote gender equality and challenge discriminatory norms. Provide economic opportunities for women and support rural livelihoods. Increase awareness of mental health issues and available support and address environmental impacts, particularly climate change, on women and children.

METHODOLOGY AND DISCUSSION:

We have visited 21 villages namely: -

Sr. no.	Name of village	Taluka	District	Aware Through the process of
1.	Anjangaon bari	Amravati	Amravati	Communication at government school
2.	Malkhed	Amravati	Amravati	Communication with some women's
3.	Timtala	Amravati	Amravati	Communication with some women's
4.	Chandur	Amravati	Amravati	Through questionaries
5.	Daryapur	Amravati	Amravati	Helped a child by providing proper knowledge regarding education
6.	Anji	Wardha	Wardha	Communication with a girl who is facing menstrual problems and using cotton cloths

7.	Wardha	Wardha	Wardha	Visit a girl's school and educate them about menstrual cycle
8.	Sindi	Wardha	Wardha	conversation with some parents about peer pressure
9.	Borgaon	Wardha	Wardha	Gives the knowledge of importance of education to parents who are not allowing their child to do studies
10.	Katol	Nagpur	Katol	Aware about child marriage
11.	Umred	Nagpur	Umred	Communicate with women's and educate them regarding their financial problems
12.	Ramtek	Nagpur	Ramtek	Talked with some students and solved their queries with the help of spreading knowledge
13.	Nandangaon peth	Amravati	Amravati	Talked with some women's
14.	Rajura	Chandrapur	Rajura	Aware about health conditions like breast cancer
15.	Wani	Chandrapur	Rajura	Helping a girl who is facing health issues like PCOD
16.	Washim	Washim	Washim	Communicate about domestic violence
17.	Yavatmal	Yavatmal	Yavatmal	Educate children about bullying
18.	Gadchiroli	Gadchiroli	Gadchiroli	Communicate with women's and inspire them to take stand for themselves
19.	Digrus	Digrus	Yavatmal	Communicate with children's at school
20.	Pimpalgaon	Rajura	Rajura	Educate women about child marriage and punishments for child marriage
21.	Sewagram	Wardha	Wardha	Communicate about domestic violence

Table 01: - Name of villages and methods by which awareness done

Where we asked the several questions regarding women and child development. Hence, we found some people facing the problem also the share their issues:

Que. One child complains about how some children's is bullying him about his physical appearance

Que. A girl name Seema complains how his parents are forcing her for a marriage at the age of 16 but she wants to have further studies

Que. A boy named Rahul wished to have a good education but because of due to his worst financial condition he is not able to attend the school

Que. A married women name Geeta Bai has been suffering from unpaid labour problems

Que. Geeta Bai along with her sister-in-law is highly facing issue due to their menstrual cycle

Que., we found nearly almost newly married women who wished to have a good carrier. Alas! They failed to do so

Que. A girl told me that there is huge gap between her menstrual cycles?

We aware women about their rights and also to take stand onto the actions that are harmful to them. Majorly problems we came across about women's menstruation, child marriage, violence and abuse. Regarding menstruation we came across that the women are not using sanitary products in their menstrual phase we told them about useful phase of sanitary products and harmful phase of sanitary products. We humbly requested them to use sanitary products and not to use cotton clothes, napkins. We visited a school where we found that the principal of that school is doing household work at her office and avoiding students who came with their parents for problem regarding documents. Regarding the child marriage we found a young girl where she dreamed of a good and successful career in her education. But she cannot due to her Forceful marriage. we asked her to stood up for her regarding her education purpose. We visited a school near by a village and we came across children's complaining about bullying, child labour, child marriage, peer pressure and neglectation. We aware them about importance of education, not to get affected for a child labour for a sake of some money also not to agree for a child marriage until and unless they wish to have. We told them to resolve their problems with the once who are there for them. I have studied various research paper and found several problems faced by women and children in Vidarbha region and our group visited 21 villages (as given in

table 01). We saw several problems faced by women and children in that village and We aware women about their rights and also to take stand onto the actions that are harmful to them. We aware children about child marriage and bullying they are facing specially because of their physical appearance. Here are some pictures of the interaction of village peoples regarding awareness and mentioned in figure 1 to 5.



Figure 1: awaring about menstrual cycle school



Figure 2: awareness program at government



Figure 3: awareness about PCOD



Figure 4: awareness program at high school



Figure 5: talking to students about problems like bullying and peer pressure

CONCLUSION:

The empowerment of women and children in Vidarbha is vital for the region's socio-economic progress. Despite various government and NGO initiatives, issues like illiteracy, poor healthcare, gender discrimination, and socio-economic inequality persist. Awareness programs in villages have played a crucial role in educating women and children about pressing issues such as menstrual hygiene, child marriage, and bullying, helping them make informed choices. Sessions held in government schools have further raised awareness about students' rights, the value of education, and the importance of a safe school environment. Government schemes like Beti Bachao Beti Padhao have contributed to improving the status of girls by promoting education and protecting them from harmful practices. However, for sustainable development, it is essential to move beyond awareness and ensure real action that creates equal opportunities for all women and children.

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A Review: Techniques used for UPI fraud detection

Vaishnavi Dilip Mahalle^{1,*}, Harsh Ravindra Borkar², Sarita T. Sawale³

Abstract

The rapid adoption of Unified Payments Interface (UPI) has led to a surge in fraudulent activities, necessitating the development of advanced fraud detection mechanisms. Convolutional Neural Networks (CNNs) have gained popularity as an effective approach due to their ability to automatically detect complex transaction patterns and anomalies. This paper reviews various UPI fraud detection techniques, including rule-based systems, hybrid models, and machine learning approaches, with a particular focus on CNN. A comparative analysis highlights the strengths and limitations of different methods, demonstrating why CNN has emerged as a leading choice for fraud detection in digital payment systems. Additionally, recent advancements in CNN-based fraud detection are explored, incorporating insights from contemporary research studies. The paper also discusses the challenges and future scope of UPI fraud detection, including the integration of emerging technologies like blockchain, federated learning, and quantum computing.

Keywords: UPI fraud detection, Convolutional Neural Networks (CNN), machine learning, digital payment security, real-time fraud prevention.

INTRODUCTION

With the rapid digitization of financial transactions, online payment systems have become an integral part of modern economies. One such system that has gained immense popularity in India is the Unified Payments Interface (UPI). Developed by the National Payments Corporation of India (NPCI), UPI is a real-time payment system that enables instant money transfers between bank accounts using mobile devices. This innovation has revolutionized digital payments by offering a seamless, secure, and user-friendly platform for peer-to-peer (P2P) and peer-to-merchant (P2M) transactions. Due to its efficiency and convenience, UPI has seen widespread adoption, making it one of the most extensively used payment systems globally, processing billions of transactions every month.

However, alongside its rapid growth, UPI has also become a major target for financial fraud, posing serious security concerns for users, financial institutions, and regulatory authorities. Financial fraud refers to any deliberate act of deception intended to result in financial gain at the expense of others. In the context of UPI, fraud can occur through various means such as phishing, SIM swap fraud, identity theft, and man-in-the-middle attacks. These fraudulent activities not only lead to significant financial losses for individuals and businesses but also erode trust in digital payment systems. Since UPI transactions are instantaneous and irreversible, the ability to detect and prevent fraud in real time is critical to maintaining the security and reliability of the system.

Detecting fraud in UPI transactions is crucial for several reasons. Financial loss prevention is a primary concern, as fraudulent transactions can cause major financial damages to users, banks, and payment platforms. Additionally, user trust is a key factor in the widespread adoption of digital payment systems. Frequent fraud incidents can reduce user confidence, thereby slowing the transition to a cashless economy. Furthermore, regulatory compliance plays a significant role, as financial institutions and payment service providers must adhere to guidelines set by authorities like the Reserve Bank of India (RBI) to ensure secure transactions. Failure to detect fraud can lead to penalties and reputational damage for financial service providers. Beyond individual users and institutions, fraudulent activities have a broader economic impact, as they can facilitate illegal practices such as money laundering and funding of illicit operations, thereby threatening financial stability[1].

Despite the need for fraud detection, identifying fraudulent transactions in UPI is highly challenging due to various factors. Evolving fraud tactics make it difficult for traditional security measures to keep up, as fraudsters continually modify their strategies to bypass existing fraud detection mechanisms. Additionally, imbalanced data presents a significant challenge, as fraudulent transactions constitute only a small fraction of the overall transaction volume, making it difficult for machine learning (ML) models to effectively learn

fraud patterns [2]. The requirement for real-time detection adds another level of complexity, as UPI transactions occur instantaneously, requiring fraud detection systems to analyze and flag suspicious activities within milliseconds. Fraudsters often exploit anonymity, using fake or stolen identities to carry out fraudulent activities, making it difficult to trace transactions back to the perpetrators [1]. Moreover, fraudulent activities involve complex transaction patterns that are not easily identifiable through traditional rule-based systems or basic machine learning models [2].

Traditional fraud detection methods primarily rely on predefined rules, which are often inadequate for handling evolving fraud techniques. While machine learning (ML) and hybrid approaches have significantly improved fraud detection capabilities, convolutional neural networks (CNNs) have demonstrated superior performance due to their ability to recognize complex transaction patterns and enhance real-time fraud detection. This paper aims to compare CNN with other fraud detection techniques and establish its effectiveness in UPI fraud detection. Recent studies, such as those conducted by Gupta et al. and Kumar et al., highlight CNN's efficiency in minimizing false positives and improving transaction security. Furthermore, according to the Reserve Bank of India (RBI) Annual Report on Digital Payment Frauds (2023), UPI fraud cases have increased by 25% year-on-year, with phishing and SIM swap attacks being the most prevalent. These statistics highlight the urgent need for advanced fraud detection mechanisms that can adapt to the evolving strategies of cybercriminals and ensure the continued security of UPI transactions [1].

TECHNIQUES FOR UPI FRAUD DETECTION

A. Overview

With the increasing prevalence of financial fraud in UPI transactions, various detection techniques have been developed to mitigate risks and enhance security. These techniques range from traditional rule-based systems to advanced machine learning (ML) approaches and deep learning models like convolutional neural networks (CNNs). Each method has its own strengths and limitations in terms of accuracy, adaptability, and computational efficiency. This section provides an in-depth review of different UPI fraud detection techniques, including rule-based systems, hybrid models, machine learning approaches, and CNN-based methods.

B. Rule-Based Systems

- 1) Traditional fraud detection relies on predefined rules to flag suspicious transactions. These rules are typically based on factors such as transaction amount, frequency, location, and user behavior patterns. While effective against known fraud patterns, rule-based systems lack adaptability to emerging fraud tactics and require frequent updates [3].
- 2) For example, a rule-based system might flag a transaction as fraudulent if it exceeds a predefined amount or originates from a high-risk geographical location. However, such rigid frameworks struggle to detect sophisticated fraud tactics, including phishing attacks and SIM swap fraud, where cybercriminals exploit loopholes not explicitly defined in the rules [3].
- 3) Additionally, rule-based models tend to generate a high number of false positives, leading to unnecessary transaction blocks and inconvenience for legitimate users [3].

C. Hybrid Models

Hybrid models combine multiple techniques to improve fraud detection accuracy, addressing the limitations of traditional rule-based systems. These models leverage the strengths of different algorithms to detect both known and emerging fraud patterns in UPI transactions. Two notable examples are BLAST-SSAHA hybridization and Fuzzy Logic, which have shown significant promise in enhancing fraud detection systems.

1) BLAST-SSAHA Hybridization

The BLAST-SSAHA hybridization, proposed by Kundu and Suvasini, applies sequence alignment algorithms from bioinformatics to fraud detection in financial transactions. Originally developed for analyzing DNA sequences, BLAST (Basic Local Alignment Search Tool) and SSAHA (Sequence Search and Alignment by Hashing Algorithm) are repurposed to detect fraudulent patterns within transactional data. This hybrid model leverages the strengths of both techniques to enhance fraud detection capabilities in Unified Payments Interface (UPI) transactions [4].

- i) **Working Mechanism:** BLAST is effective in identifying fraud patterns that exhibit slight variations from known fraudulent activities. By comparing transaction sequences to a database of previously detected fraud cases, it can recognize modifications used by fraudsters to bypass traditional detection methods. This ability is particularly useful for detecting morphing fraud patterns, where small changes are introduced to evade rule-based detection [4]. SSAHA, on the other hand, employs a hash-based searching mechanism that allows for rapid identification of exact or near-exact matches within transactional datasets. This efficiency makes SSAHA well-suited for high-frequency, real-time fraud detection in large-scale financial systems like UPI, where millions of transactions occur daily. By combining BLAST's pattern-matching capability with SSAHA's speed, the hybrid approach provides an efficient solution for detecting both known fraud instances and novel suspicious activities [4].
- ii) **Strengths of BLAST-SSAHA Hybridization**
 - **Real-Time Detection:** The combination of BLAST and SSAHA enables quick identification of fraudulent activities, making it suitable for real time fraud prevention [4].
 - **Versatility:** The model effectively detects various fraud patterns, including both known fraudulent sequences and previously unseen anomalies [4].
 - **Scalability for High-Volume Transactions:** SSAHA's hashing algorithm enhances processing speed, making it efficient for handling large-scale UPI transactions [4].
- iii) **Limitations and Challenges**

Despite its advantages, the BLAST-SSAHA hybridization model presents certain challenges:

 - **Computational Overhead:** The hybrid approach requires substantial computational resources due to the sequence alignment processes, potentially impacting scalability for very large datasets [4].
 - **Data Preprocessing Complexity:** Transactional data must be transformed into a format suitable for sequence alignment, which involves significant preprocessing efforts and may introduce latency in real-time detection systems [4].
 - **Integration with Existing Fraud Detection Systems:** Implementing this approach within existing banking infrastructures may require modifications to accommodate sequence-based fraud detection models [4].
- iv) **Application in UPI Fraud Detection** The BLAST-SSAHA hybridization is particularly effective in UPI fraud detection due to its ability to handle high transaction volumes while identifying evolving fraud strategies. As fraudsters continuously adapt their techniques, the hybrid model's capability to detect both known fraud patterns and novel anomalies makes it a robust solution for enhancing security in digital payments. However, the computational demands of the approach may pose challenges in real-time deployment for large-scale UPI systems [4].

2) Fuzzy Logic and Fuzzy-ID3:

Fuzzy logic has been proposed as an effective hybrid approach for fraud detection, particularly in financial transactions, by Peter J. Bentley et al. in their study on Fuzzy Darwinian Detection of UPI Fraud [5]. This method enhances traditional decision tree-based fraud detection by incorporating fuzzy logic, which allows the system to handle uncertainty and overlapping cases in transaction data. Unlike traditional binary classification models that strictly label transactions as either fraudulent or non-fraudulent, fuzzy logic enables a more flexible classification by assigning degrees of fraud likelihood [5].

- (a) **Challenges in UPI Fraud Detection:** The rapid expansion of UPI (Unified Payments Interface) has led to an increase in fraudulent transactions. One of the primary challenges in fraud detection is that fraudulent transactions often resemble genuine ones, making them difficult to detect using traditional rule-based or anomaly detection models. Standard classification methods may fail when transaction characteristics fall into ambiguous or uncertain categories, necessitating the use of fuzzy logic to manage this uncertainty [5].
- (b) **Proposed Hybrid Model: Fuzzy-ID3 Algorithm:** The Fuzzy-ID3 algorithm builds upon

the conventional ID3 (Iterative Dichotomiser 3) decision tree algorithm by integrating fuzzy logic principles. ID3 operates by recursively partitioning data based on information gain, selecting the most discriminative feature at each step. However, traditional decision trees struggle with borderline cases where fraudulent and legitimate transactions share similar attributes. Fuzzy logic addresses this limitation by allowing for partial membership in multiple classes rather than forcing a strict binary classification. Instead of labeling a transaction as either fraudulent or non-fraudulent, Fuzzy ID3 assigns it a probability score, such as 70% fraudulent and 30% genuine, allowing for a more nuanced and accurate decision-making process. This capability makes the approach highly suitable for complex and ambiguous UPI transactions [5].

(c) Strengths of Fuzzy-ID3 in Fraud Detection:

- **Enhanced Accuracy:** The incorporation of fuzzy logic improves fraud detection accuracy by effectively managing uncertain or borderline cases that traditional models may misclassify [5].
- **Adaptability:** The algorithm dynamically adjusts to varying degrees of fraud risk, making it more flexible than strict rule-based systems [5].
- **Scalability for Large Datasets:** Fuzzy-ID3 is designed to handle large-scale transaction data, making it well-suited for the high-volume nature of UPI-based financial ecosystems [5].

(d) Limitations and Challenges: Despite its advantages, the Fuzzy-ID3 model also presents certain challenges

- **Computational Complexity:** The integration of fuzzy logic increases the computational demands of the model, potentially impacting its efficiency in real-time fraud detection scenarios [5].
- **Reduced Interpretability:** While fuzzy logic enhances detection accuracy, it also reduces the transparency of decision-making. This lack of explainability can be a concern when deploying fraud detection mod

(e) Application in UPI Fraud Detection: Fuzzy-ID3 is particularly beneficial for UPI fraud detection due to its ability to differentiate between legitimate and fraudulent transactions with greater precision. By handling subtle variations in transaction patterns, it reduces false positives while improving overall fraud detection rates. However, challenges such as high computational requirements and lower interpretability must be addressed before it can be widely implemented in real-time UPI security systems [5].

D. Machine Learning-Based Approaches

Machine learning (ML) techniques have significantly enhanced fraud detection by identifying complex patterns in transaction data. Unlike rule-based systems, ML models can automatically learn from large datasets, making them particularly effective for UPI (Unified Payments Interface) fraud detection. These methods help detect fraudulent transactions by analyzing historical data and adapting to new fraud patterns. Two widely used ML techniques for fraud detection are Decision Trees/Random Forest and Bayesian/Neural Networks, advantages and challenges.

a) **Decision Trees and Random Forest Decision :** Trees and their ensemble variant, Random Forest, are commonly used in fraud detection due to their interpretability and effectiveness in handling structured data.

- **How It Works** Decision Trees classify transactions by recursively splitting data based on the most informative features. For example, a decision tree may classify a UPI transaction as fraudulent or non-fraudulent based on features like transaction amount, time of day, and location. Random Forest enhances decision trees by training multiple trees on different subsets of data and averaging their predictions, reducing overfitting and improving accuracy.
- **Strengths** Interpretability allows decision trees to provide a clear, visual decision-making process, making them easier to analyze and debug. They are effective for structured datasets containing predefined attributes like transaction timestamps and amounts. Random Forest minimizes overfitting by combining multiple decision trees, leading to more robust predictions.
- **Limitations** Performance heavily relies on manual selection of relevant features, requiring domain expertise. Decision Trees and Random Forest perform well on known

fraud patterns but may fail to detect new or evolving tactics. These models often rely on batch processing, making real-time fraud detection challenging.

- Relevance to UPI Fraud Detection Decision Trees and Random Forest are effective for identifying known fraud patterns in UPI transactions. However, their inability to generalize to new fraud tactics and dependence on feature engineering limit demonstrates that their usefulness in dynamic fraud environments. These methods are most suitable when fraud patterns remain consistent over time.

b) Bayesian Networks and Neural Networks: Bayesian Networks and Neural Networks are more advanced ML models that offer higher accuracy and adaptability, making them suitable for detecting new and evolving fraud techniques.

- How It Works Bayesian Networks rely on probabilistic models to estimate the likelihood of a transaction being fraudulent. They use prior knowledge and observed data, such as user behavior, transaction history, and location, to compute fraud probability. Neural Networks simulate the structure of the human brain with multiple layers of neurons, enabling them to learn complex patterns from large-scale transaction data. They excel at detecting non-linear relationships and identifying hidden fraud trends.
- Strengths Neural Networks effectively detect complex and hidden fraud patterns in transactional data. These models continuously learn from new data, making them more effective for real-time fraud detection. Bayesian Networks provide confidence scores for fraud probability, offering a more nuanced approach to fraud detection.
- Limitations Neural Networks act as black-box models, making it difficult to understand or audit their decision making process. Training Neural Networks demands significant computational resources and large datasets, potentially affecting scalability. Both Bayesian and Neural Networks require extensive training, which can be time-consuming and resource-intensive.
- Relevance to UPI Fraud Detection Bayesian Networks and Neural Networks excel at detecting sophisticated UPI fraud by learning from evolving fraud techniques. However, challenges such as high computational demands and lack of interpretability must be addressed for real-time compliance.

c) Why Machine Learning-Based Approaches Are Important : Machine learning has revolutionized fraud detection by enabling automated learning from data, which significantly improves the identification of fraudulent transactions. Decision Trees and Random Forest are effective for detecting structured fraud patterns in UPI transactions but struggle with adaptive fraud tactics. Bayesian and Neural Networks provide higher accuracy and adaptability, making them suitable for detecting new and evolving fraud techniques. Challenges include the need for feature engineering, computational resources, and model interpretability, which must be considered when deploying ML-based fraud detection in real-world UPI systems [2].

E. CNN for UPI Fraud Detection :

Convolutional Neural Networks (CNNs) have gained significant attention in UPI fraud detection due to their ability to automatically extract features from raw transaction data. Unlike traditional ML models, CNNs use hierarchical representations, enabling them to detect subtle and evolving fraud patterns with high accuracy. Recent research CNN-based models outperform conventional techniques in terms of precision, recall, and real-time adaptability [6][7]. These studies highlight how deep feature extraction using CNNs enhances fraud detection efficiency by capturing intricate transaction behaviors. Notably, a CNN-based fraud detection framework achieved over 95% accuracy in identifying fraudulent UPI transactions by analyzing time-series data [6]. Similarly, a hybrid CNN-LSTM model reduced false positives by integrating temporal sequence learning with CNN's feature extraction capabilities [7]. This combination allows the model to not only detect

fraud but also predict potential fraudulent activities based on historical patterns. CNN-based fraud detection has several advantages over traditional techniques, including automated feature learning, real-time adaptability, and enhanced accuracy. However, CNNs require large amounts of labeled training data and high computational resources, which can be a limitation in real-time UPI fraud prevention systems. Future research should focus on optimizing CNN architectures to reduce computational overhead while maintaining high fraud detection accuracy.

- 1) A CNN-based fraud detection framework was proposed, achieving over 95% accuracy in identifying fraudulent UPI transactions by analyzing time-series data. This highlights the effectiveness of CNNs in handling complex transaction patterns [6].
- 2) A hybrid CNN-LSTM model was introduced, combining the feature extraction capabilities of CNNs with the temporal sequence learning of Long Short-Term Memory (LSTM) networks. This approach reduces false positives and improves fraud detection by leveraging historical transaction patterns [7].
- 3) CNN-based fraud detection offers several advantages, including automated feature learning, real-time adaptability, and enhanced accuracy. These features make CNNs highly effective for detecting evolving fraud patterns in UPI transactions [6][7].
- 4) Limitations of CNNs: Despite their advantages, CNNs require large amounts of labeled training data and high computational resources, which can hinder their deployment in real-time UPI fraud prevention systems [6][7].
- 5) Future research direction: To address these limitations, researchers should focus on optimizing CNN architectures to reduce computational overhead while maintaining high fraud detection accuracy [6][7].

COMPARATIVE ANALYSIS

This section provides a comparative analysis of various fraud detection techniques, highlighting their strengths and weaknesses.

1) Table: Comparison of UPI Fraud Detection Methods

<i>Year</i>	<i>Method</i>	<i>Weaknesses</i>	<i>Strengths</i>
2019	Rule-Based Systems	Cannot detect new fraud patterns, high false positives	Easy to implement, interpretable
2020	Hybrid Models	Computationally expensive, high false positives	Higher accuracy than rule-based methods
2021	Machine Learning Approaches	Requires extensive feature engineering, struggles with new patterns	Effective for structured data, reduces overfitting (Random Forest)
2022	CNN	Requires large datasets, computationally intensive	Automated feature extraction, adaptable to evolving fraud, high accuracy

1. Explanation of the Table

- a. Rule-Based Systems are characterized by their ease of implementation and interpretability. However, they face limitations in detecting new fraud patterns and often produce high false positive rates. While effective for identifying known fraud patterns, rule-based systems struggle with the adaptation to evolving fraud tactics, making them less suitable for dynamic fraud detection environments [3].
- b. Hybrid Models such as BLAST-SSAHA Hybridization and Fuzzy-ID3 offer certain advantages. BLAST-SSAHA provides real-time detection and is versatile in handling misuse and anomaly detection. Despite this, it suffers from high computational complexity and requires data preprocessing, limiting its scalability for large transaction volumes in UPI systems. On the other hand, Fuzzy-ID3 is adept at handling uncertain cases, enhancing detection accuracy, but it is computationally intensive and lacks interpretability. Although it is

suitable for large datasets, its real-time implementation presents a challenge due to its computational demands [4].

- c. Machine Learning Approaches including Decision Trees/Random Forest and Bayesian/Neural Networks are widely used for fraud detection. Decision Trees and Random Forests are interpretable and efficient with structured data, but they have limited adaptability to new fraud patterns and are prone to high false positives. Thus, they are useful for detecting known fraud patterns, though they are less effective in real-time scenarios. Bayesian Networks and Neural Networks, on the other hand, offer high accuracy and adaptability to new fraud patterns. However, they are computationally intensive and require extensive training, which hinders their use for large-scale, real-time fraud detection systems [2].
- d. Convolutional Neural Networks (CNNs) offer several advantages, including automated feature extraction and high accuracy, making them highly adaptable to evolving fraud techniques. Despite their effectiveness, CNNs demand large datasets and significant computational resources, which can be a limitation. Nevertheless, due to their ability to provide real-time fraud detection with high accuracy, CNNs are considered the dominant choice for UPI fraud detection [6][7].

CHALLENGES AND FUTURE SCOPE

Despite their significant advantages in UPI fraud detection, Convolutional Neural Networks (CNNs) face several challenges that hinder their widespread adoption and effectiveness in real-world applications. One of the primary challenges is the high computational requirement of CNNs. These models demand substantial computational resources, including powerful hardware such as GPUs or TPUs, to process large volumes of transaction data in real time. This makes their deployment in real-time fraud detection systems particularly challenging, especially for financial institutions with limited computational infrastructure. Additionally, the training process for CNNs is both time-consuming and energy-intensive, further increasing operational costs [6]. Another critical challenge is the need for large datasets. CNNs rely heavily on extensive amounts of labeled training data to achieve high accuracy. However, obtaining such datasets for UPI fraud detection is difficult due to privacy concerns, data sensitivity, and the inherent imbalance in fraud datasets, where fraudulent transactions are significantly outnumbered by legitimate ones. This imbalance can lead to overfitting, where the model performs well on training data but fails to generalize to new, unseen transactions. Moreover, the lack of explainability in CNNs poses a significant challenge. While these models achieve high accuracy in detecting fraudulent transactions, their decision-making process is often opaque and difficult to interpret. This "black box" nature makes it challenging for regulators, financial institutions, and end-users to understand why a particular transaction was flagged as fraudulent, thereby hindering trust and complicating compliance with regulatory requirements [7].

To address these challenges, several promising research directions have been proposed. One such direction is the development of lightweight CNN architectures that reduce computational overhead while maintaining high accuracy. Techniques such as model pruning, quantization, and knowledge distillation can help create smaller, faster models suitable for real-time fraud detection. These lightweight models can be deployed on edge devices or mobile platforms, enabling faster and more efficient fraud detection. Another promising avenue is the integration of CNNs with blockchain technology or federated learning. Blockchain can enhance the security and transparency of fraud detection systems by providing an immutable record of transactions, while federated learning allows models to be trained across decentralized devices without sharing raw data, thereby addressing privacy concerns. Additionally, the implementation of hybrid deep learning frameworks that combine CNNs with Long Short-Term Memory (LSTM) networks or attention mechanisms has shown significant promise. These hybrid models leverage the temporal sequence learning capabilities of LSTMs and the feature extraction strengths of CNNs, improving both accuracy and adaptability to evolving fraud patterns [7].

Another critical area for future research is addressing the lack of explainability in CNN-based fraud detection systems. Techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) have been proposed to provide interpretable insights into model predictions. These techniques can help regulators and financial institutions understand the reasoning behind flagged transactions, thereby improving trust and transparency in the system [7]. Furthermore, the

integration of Graph Neural Networks (GNNs) with CNNs presents an exciting opportunity to detect fraud rings and coordinated attacks. GNNs can model complex relationships between transactions and accounts, making them particularly effective for identifying sophisticated fraud networks that involve multiple entities and transactions. By combining the strengths of CNNs and GNNs, future systems can achieve higher accuracy and robustness in detecting complex fraud patterns. While CNNs have demonstrated remarkable success in UPI fraud detection, addressing their challenges—such as high computational requirements, the need for large datasets, and lack of explainability—is crucial for their effective deployment. Future research should focus on developing lightweight architectures, integrating CNNs with blockchain or federated learning, and leveraging hybrid models and explainability techniques to enhance the security, efficiency, and transparency of fraud detection systems [6][7].

CONCLUSION

This review establishes Convolutional Neural Networks (CNNs) as a leading approach for UPI fraud detection, primarily due to their advanced pattern recognition capabilities and ability to adapt to emerging fraud tactics. Unlike traditional methods such as rule-based systems and hybrid models, CNNs automatically extract relevant features from raw transactional data, eliminating the need for manual feature engineering, making them efficient and scalable for real-time fraud detection. Research has shown that CNNs reduce false positives and enhance detection rates, solidifying their effectiveness. While rule-based systems struggle with new fraud patterns and hybrid models face challenges in complexity and false positives, CNNs leverage their hierarchical learning to identify subtle fraud patterns, making them more robust against evolving fraud tactics. However, CNN adoption faces challenges, including high computational requirements and the need for large labeled datasets, which limit deployment in resource-constrained environments. Additionally, the lack of explainability in CNN-based models poses concerns for regulatory compliance and user trust. To address these limitations, future research should focus on optimizing CNN architectures, using techniques like model pruning, quantization, and knowledge distillation, to reduce computational costs while maintaining accuracy for real-time fraud detection.

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A Review on Tours and Travel Booking System

Miss Dipanjali Shipne¹, Mr. Gaurav P. Birkhade², Mr. Shantanu Thosre³, Mr. Rushikesh R. Bhople⁴, Miss. Sanchi V. Rajne⁵

Abstract:

This paper presents a comprehensive study on developing and implementing a web based tour booking system using PHP and MySQL. The system architecture, user authentication, database design, payment integration, and user experience optimization are discussed in detail. The paper synthesizes recent research, case studies, and technical articles to provide insights into creating efficient, secure, and user-friendly tour booking platforms. The proposed system addresses the evolving needs of the travel industry by incorporating innovative features and best practices in web development.

Keywords: Tour Booking System, PHP, MySQL, Web Development, User Authentication, Payment Integration, Database Design, User Experience.

INTRODUCTION:

Background on Tours and Travels Booking System Systems: The tours and travel industry has evolved significantly with the rise of digital technology, shifting from traditional manual booking methods to advanced online systems. Tours and travel booking systems emerged with the advent of the Internet, enabling users to book flights, hotels, and other services online [1]. Over time, these systems have incorporated features like real-time availability, personalized recommendations, and secure payment options, becoming essential tools for modern travel planning.

Importance of Tours and Travels Booking System in modern business environments: The importance of these systems lies in their ability to provide convenience and efficiency for both travelers and service providers. Travelers can compare prices, read reviews, and make bookings anytime, anywhere, while businesses benefit from expanded reach, better inventory management, and valuable customer insights [2]. In an increasingly connected world, tours and travel booking systems play a vital role in simplifying travel experiences and driving the growth of the global travel industry. This paper explores the evolution, features, and impact of these systems, highlighting their significance and potential for future advancements.

Problem Statement: Despite the widespread adoption and advancements in tours and travel booking systems, several challenges persist that hinder their efficiency and user experience. One major issue is the lack of integration among various service providers, such as airlines, hotels, and tour operators, leading to fragmented information and inconsistent booking processes. Additionally, users often face difficulties in navigating complex interfaces, encountering issues such as unclear pricing, hidden charges, and limited customization options [3]. Another critical problem is the reliance on outdated technologies in some systems, resulting in slow processing times, system crashes during peak booking periods, and vulnerabilities to cybersecurity threats. Furthermore, the absence of personalized recommendations and localized content often fails to cater to the diverse needs of travelers, particularly in underserved regions.

Literature Review: The evolution of tours and travel booking systems has been extensively studied, with researchers highlighting the shift from traditional travel agencies to online platforms. According to Buhalis and Licata (2002), the emergence of Online Travel Agencies (OTAs) revolutionized the industry by offering real-time booking capabilities and greater accessibility for users. However, studies by Law et al. (2014) emphasize that the rapid growth of these systems has also led to challenges such as information overload, inconsistent pricing, and lack of integration among service providers[5].

User experience has been a critical focus in recent literature. Tussyadiah and Fesenmaier (2009) argue

that the success of booking systems depends heavily on their ability to provide intuitive interfaces and transparent pricing. Yet, many platforms still struggle with hidden fees and complex navigation, as noted by Xiang et al. (2015). Additionally, the lack of personalization in many systems has been identified as a significant drawback. Gretzel et al. (2012) suggest that incorporating data-driven recommendations and machine learning can enhance user satisfaction by tailoring travel options to individual preferences.

Current trends and advancements in Tours and Travels Booking System:

1. **Artificial Intelligence (AI) and Machine Learning (ML):** AI and ML are revolutionizing travel booking systems by enabling personalized recommendations, dynamic pricing, and predictive analytics. Chatbots powered by AI provide 24/7 customer support, while machine learning algorithms analyze user behavior to offer tailored travel options, enhancing user satisfaction and engagement.
2. **Mobile-First Platforms:** With the increasing use of smartphones, travel booking systems are prioritizing mobile-friendly designs. Mobile apps now dominate the market, offering features like real-time notifications, GPS-based travel guides, and seamless booking experiences, catering to the on-the-go lifestyle of modern travelers.
3. **Voice Search and Voice-Activated Booking:** Voice-assisted technologies, such as Amazon Alexa and Google Assistant, are being integrated into travel platforms, allowing users to search for flights, hotels, and activities using voice commands. This hands-free approach is gaining popularity for its convenience and ease of use.
4. **Blockchain Technology:** Blockchain is being explored for its potential to enhance transparency and security in travel bookings. It can streamline payment processes, reduce fraud, and enable smart contracts for automated bookings, ensuring trust and efficiency in transactions.
5. **Sustainability and Eco-Friendly Travel:** As environmental awareness grows, travel booking systems are incorporating features that promote sustainable travel. This includes options for carbon offsetting, eco-friendly accommodations, and green transportation, aligning with the preferences of environmentally conscious travelers.
6. **Contactless and Digital Payments:** The COVID-19 pandemic accelerated the adoption of contactless technologies. Travel platforms now emphasize digital payment options, QR code-based check-ins, and e-tickets to minimize physical contact and enhance safety.
7. **Integration of Social-Media:** Travel booking systems are increasingly integrating with social media platforms, allowing users to share experiences, read reviews, and even book trips directly through social networks. This trend leverages the influence of social proof in travel decision-making. The evolution of tours and travel booking systems reflects the industry's adaptation to technological advancements and changing consumer needs. From manual processes in the preinternet era to AI-driven, mobile-first platforms today, these systems have become indispensable tools for modern travelers. As the industry continues to innovate, future systems are expected to focus on sustainability, personalization, and seamless integration of services, further enhancing the travel experience.

Definition and Evolution of Tours and Travel Bookings: A tour and travel booking system is a digital platform or software that enables users to search, compare, book, and manage travel related services such as flights, hotels, car rentals, tour packages, and activities. These systems act as intermediaries between travelers and service providers, offering a centralized solution for planning and organizing

Use Cases Diagram:

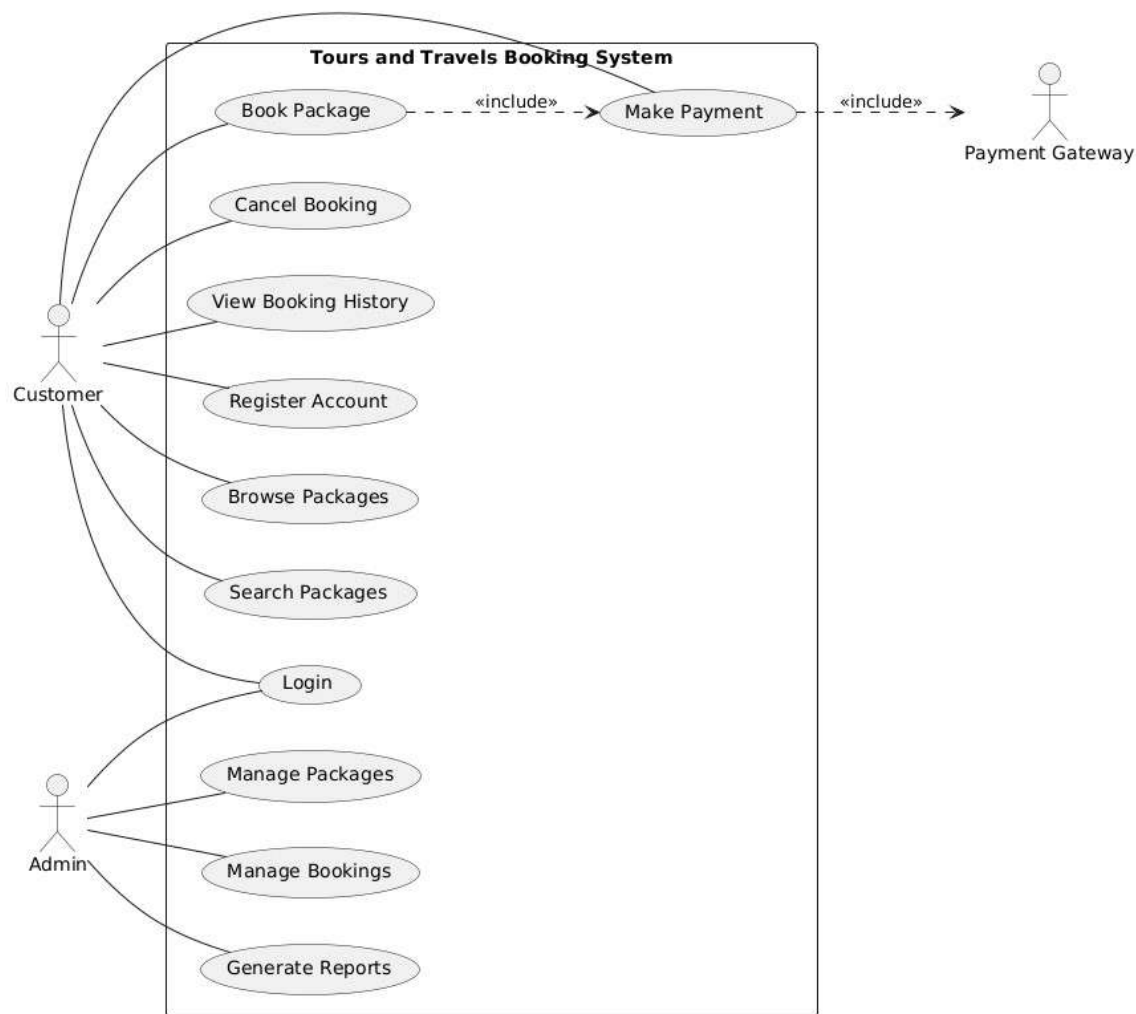


Fig. Use Case Diagram

trips [6]. They are designed to simplify the travel booking process, providing real-time availability, pricing information, and secure payment options.

Various Tours and Travels Booking System Types:

- 1) Online Travel Agencies (OTAs): Platforms that aggregate travel services from multiple providers, such as flights, hotels, and car rentals, allowing users to compare and book services in one place.
- 2) Corporate Travel Booking Systems: Platforms tailored for businesses to manage employee travel, expenses, and policy compliance, ensuring efficient corporate travel planning.
- 3) All-in-One Travel Platforms: Comprehensive platforms that combine flights, hotels, transportation, and activities into a single system, offering end-to-end travel planning. Benefits of ERP in Business: Tours and travel booking systems offer numerous advantages for both travelers and service providers [7].

Below are the key benefits:

For Travelers:

1. Convenience and Time-Saving: Allows users to book flights, hotels, and activities from one platform. Eliminates the need for visiting multiple websites or physical travel agencies.
2. Transparency and

Comparison: Provides real-time pricing and availability of services. Enables users to compare options based on price, ratings, and reviews.

3. Personalization: Offers tailored recommendations based on user preferences and travel history. Enhances the overall travel experience with customized itineraries.

4. 24/7 Accessibility: Available for booking anytime, anywhere, through web or mobile platforms. Ideal for last-minute travel planning.

5. Secure Payments: Provides secure payment gateways with multiple payment options. Ensures safe and hassle-free transactions.

6. Real-Time Updates: Sends instant notifications for booking confirmations, flight delays, or cancellations. Keeps travelers informed about their trip details.

For the Travel Industry:

1. Boost in Tourism: Simplifies travel planning, encouraging more people to explore new destinations. Supports the growth of local economies and tourism-dependent businesses.

2. Integration of Services: Facilitates collaboration between airlines, hotels, and tour operators. Creates a seamless ecosystem for travelers and service providers.

3. Sustainability: Promotes eco-friendly travel options and sustainable tourism practices. Reduces paper usage through digital tickets and e-documents.

Challenges in Tours and Travels Booking System Implementation:

Implementing a tours and travel booking system comes with several challenges, ranging from technical complexities to user adoption [8].

Below are the key challenges:

1. Integration with Multiple Service Providers: Connecting the system with various airlines, hotels, car rental companies, and activity providers. Inconsistent data formats, APIs, and protocols can lead to integration issues and delays.

2. Real-Time Data Synchronization: Ensuring real-time updates on availability, pricing, and bookings across all platforms. Delays or errors in synchronization can result in overbooking or incorrect pricing.

3. User Experience (UX) Design: Creating an intuitive and user-friendly interface for diverse user groups. Poor UX can lead to low user adoption and negative reviews.

4. Scalability: Designing a system that can handle high traffic during peak booking seasons. System crashes or slow performance can frustrate users and lead to lost bookings.

5. Security and Fraud Prevention: Protecting user data and preventing fraudulent transactions. Security breaches can damage the platform's reputation and lead to financial losses.

6. Payment Gateway Integration: Supporting multiple payment methods and currencies while ensuring secure transactions. Payment failures or lack of preferred payment options can deter users from completing bookings.

7. Mobile Optimization: Ensuring the system works seamlessly across different devices and operating systems. Poor mobile performance can alienate a significant portion of users who rely on smartphones for bookings.

8. Customer Support: Providing 24/7 customer support for booking issues, cancellations, and refunds. Inadequate support can lead to negative user experiences and loss of repeat customers.

9. Competition and Market Saturation: Differentiating the system in a highly competitive market with

established players. Difficulty in attracting and retaining users.

10. Cost and Resource Allocation: Managing the high costs of development, maintenance, and marketing. Budget constraints can limit the system's features and scalability.

11. Testing and Quality Assurance: Ensuring the system is free of bugs and performs well under various scenarios. Untested or poorly tested systems can lead to frequent errors and user dissatisfaction.

Case Studies and Industry Applications:

1. Expedia Group: Application: A global Online Travel Agency (OTA) offering flights, hotels, car rentals, and vacation packages. Case Study: Expedia uses AI and machine learning for personalized recommendations and dynamic pricing. Their system adjusts prices in real-time based on demand, improving revenue and customer satisfaction. Impact: Handles millions of bookings daily, showcasing scalability and efficiency.

2. Airbnb: Application: A peer-to-peer platform for booking unique accommodations and experiences. Case Study: Airbnb uses AI to match travelers with suitable listings and offers price suggestions to hosts based on market trends. User-generated content (reviews, photos) builds trust and transparency. Impact: Disrupted the hospitality industry by offering affordable, personalized lodging options.

3. Klook: Application: A platform for booking local activities, tours, and experiences. Case Study: Klook focuses on mobile-first design, using geolocation and real-time availability to offer last-minute deals and personalized recommendations. Impact: Popular among travelers seeking unique, localized experiences, especially in Asia.

4. Hopper: Application: A mobile-only app for booking flights and hotels. Case Study: Hopper uses predictive analytics to advise users on the best time to book flights and offers a "Price Freeze" feature to lock in prices. Impact: A favorite among budget-conscious travelers, with millions of downloads and high user retention.

Conclusion and Future Scope:

1. Summary of Key Findings: Tours and travel booking systems have transformed the travel industry by offering seamless, efficient, and personalized solutions for travelers and service providers. Key findings include:

- These systems provide convenience, transparency, and real-time access to travel services, enhancing user experiences.
- Technologies like AI, machine learning, and big data enable personalized recommendations, dynamic pricing, and predictive analytics.
- Platforms such as Expedia, Airbnb, and Klook demonstrate the scalability and impact of digital booking systems. Challenges like integration, scalability, and security remain but are being addressed through continuous innovation.

2. Implications for Businesses Adopting ERP: For businesses, adopting tours and travel booking systems offers significant advantages:

- Increased Reach and Revenue: Businesses can tap into a global audience, expanding their customer base and boosting sales.
- Operational Efficiency: Automation of booking, inventory management, and payment processes reduces manual workload and costs.
- Data-Driven Decision Making: Access to customer insights and analytics helps businesses tailor their offerings and improve services.
- Enhanced Customer Satisfaction: User-friendly interfaces, personalized experiences, and 24/7 support

build trust and loyalty.

- Competitive Edge: Adopting advanced technologies and trends ensures businesses stay ahead in a highly competitive market.

3. Future Directions of ERP Research and Development: The future of tours and travel booking systems is shaped by emerging trends and technologies:

- AI and Machine Learning: Further personalization and automation of travel planning and customer support.
- Blockchain: Secure, transparent transactions and decentralized platforms for peer-to-peer services.
- AR and VR: Immersive travel experiences through virtual tours and interactive guides.
- Sustainability: Integration of eco-friendly travel options and carbon footprint tracking.
- Hyper-Personalization: Tailored travel experiences based on real-time data and user preferences.
- Mobile-First Solutions: Expansion of mobile apps and super apps for seamless, on-the-go travel planning.
- Voice-Activated Booking: Growth of voice search and voice-assisted booking for handsfree Convenience [10].
- IoT Integration: Smart travel experiences with connected devices and seamless integration with smart home systems.
- Niche Markets: Focus on specialized travel segments like luxury, adventure, and cultural tourism.

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“DocOnTime” Doctor Appointment Bookings System By Using MERN Stack

¹ Mr. P. T. Talole Sir, ²Vinesh Chavhan, ³ Rushikesh Kanfade,

¹ Head of Department , Information Technology ,Anuradha Engineering College, Chikhali

^{2,3} Under Graduate Student, Department of Information Technology, Anuradha Engineering College, Chikhali, Maharashtra, India.

¹pramod.talole@aecc.ac.in, ²vineshchavhan7499@gmail.com ,

³rushikeshkanfadel123@gmail.com

ABSTRACT : *This Online Appointment Booking System is a web application that offers an interface by which patients can book doctor appointment online. The prime aim of this application is to eliminate the troubles that are found in the manual booking system. Everyone is so busy to get their doctor appointment personally and get proper health care. The prime benefit of this application is that patients can select the doctor whom they want on suitable dates and time depending upon their health issues. Admin will keep the record of doctors and patients. Admin can insert a new doctor or delete a doctor. Patients are able to cancel their appointment that is found in this application. It saves a lot of time of patients. This application is bug free and quick.*

Keywords: Online Appointment, Web Application, Healthcare, Scheduling

INTRODUCTION

The traditional method of booking doctor appointments often leads to inconvenience for patients, requiring them to physically visit healthcare facilities and wait in long queues before consulting a doctor. This system not only consumes time but also adds stress, especially for those who require urgent medical attention. Additionally, patients often need assistance in navigating the appointment process, further complicating their healthcare journey. Another challenge with the manual system is the repeated need for patients to provide their personal and medical information at multiple touchpoints, such as pharmacies and diagnostic centers, which increases the risk of document misplacement and inefficiency.

Moreover, appointment cancellations by doctors can create difficulties for patients, as they are typically unaware of these changes until they arrive at the hospital. This results in unnecessary trips, wasted time, and inconvenience, particularly for individuals who travel long distances for medical consultations. Furthermore, patients frequently need to visit the hospital again on a different day to collect medical reports, adding to their burden. These inefficiencies highlight the need for a digitalized appointment booking system that can streamline the process, reduce waiting times, and enhance patient convenience by providing real-time scheduling and notifications (Bharadwaja et al., 2022).

With advancements in web-based technologies, online doctor appointment booking systems have emerged as an effective solution. These platforms enable patients to book appointments from the comfort of their homes, access medical records digitally, and receive timely updates about any changes in their schedule. Such systems not only improve healthcare accessibility but also optimize hospital workflow by reducing overcrowding and administrative workload. Implementing an online appointment system can thus play a crucial role in modernizing healthcare services and enhancing patient satisfaction (Bharadwaja et al., 2022). [1]

LITERATURE REVIEW

Waiting Time

Waiting time is a critical aspect of patient care, defined as the period between a patient's arrival at a healthcare facility and the actual consultation with a physician. Fernandes et al. (year) describe waiting time as the duration a patient spends in anticipation before receiving medical attention. This issue is particularly concerning in developing countries, where healthcare infrastructure struggles to meet the growing demand for medical services.

Several studies classify waiting time into two main categories:

- Consultation Waiting Time – The period between the patient's arrival at the healthcare facility and their consultation with a doctor.
- Medication Waiting Time – The duration between the initial visit and the receipt of prescribed medication.

A study conducted at a South African health clinic evaluated the effectiveness of a block appointment system, in which patient wait times were monitored before and after its implementation. The study concluded that pre-scheduled appointments significantly reduced waiting times for acute patients, while those requiring routine prescriptions did not experience substantial improvements. This suggests that while structured scheduling models enhance efficiency, they may not be equally beneficial for all patient categories. (Ahmad et al., 2022).

Appointment Delay

Research indicates a direct correlation between appointment delays and patient cancellations. When patients experience long delays between scheduling and consultation, they are more likely to cancel or miss their appointments. Appointment delay refers to the period from the initial request for an appointment to the actual medical consultation (Ahmad et al., 2022).

To address this issue, Open Access (OA) scheduling has been introduced, allowing patients to book real-time slots instead of waiting for weeks. This model reduces appointment delays and provides greater flexibility for rescheduling. However, studies present mixed results—while some practitioners endorse OA scheduling as an efficient approach, others argue that it disrupts the structured scheduling system and creates inconsistencies in patient flow (Ahmad et al., 2022).

Managing Patients' Appointment System

The digital transformation of healthcare has led to the implementation of online appointment management systems, which aim to streamline scheduling, reduce patient waiting times, and optimize doctor availability. Research highlights that hospitals and clinics using digital appointment systems experience shorter wait times compared to those still relying on manual booking methods (Ahmad et al., 2022).

In a healthcare setting, the quality of service is often judged by the ease of booking an appointment and the efficiency of the consultation process. Long wait times negatively affect patient satisfaction, and individuals waiting for over an hour often perceive the service as poor or inefficient. Klassen (2002) suggests that scheduling systems should be designed based on case severity and patient needs rather than using a one-size-fits-all approach.

Technological advancements have driven a shift from phone-based and manual bookings toward fully automated web-based appointment systems. Online scheduling platforms provide the following benefits:

- Real-time appointment availability
- Automated notifications and reminders
- Improved doctor-patient communication
- Efficient management of patient records

Such web-based appointment systems significantly reduce administrative burdens and provide a more structured and organized scheduling framework for hospitals and clinics (Ahmad et al., 2022).

Waiting for Registration

The registration process at hospitals and clinics is another contributing factor to extended waiting times. In many healthcare facilities, patients are required to complete paperwork at a registration counter before meeting a doctor, leading to long queues and delays. Studies indicate that understaffing at registration desks further worsens the situation, as a single employee is often responsible for handling both patient registration and appointment distribution (Ahmad et al., 2022).

A study conducted in 2017 identified manual registration as a primary cause of long waiting times in hospitals. The findings suggest that implementing a web-based registration system can reduce congestion, allowing patients to register online before arriving at the hospital. Additionally, integrating an automated appointment confirmation system can significantly enhance hospital workflow by minimizing the need for manual entry of patient details. By adopting online booking and registration systems, healthcare facilities can reduce operational costs and improve service efficiency. These digital solutions not only enhance patient experience but also ensure better time management for healthcare professionals. (Ahmad et al., 2022). (Ahmad et al., 2022). [2]

METHODOLOGY

The Doctor Appointment Booking System is designed using a client-server architecture, ensuring smooth interaction between users, the backend server, and the database. The system facilitates efficient scheduling, allowing users to book, modify, or cancel appointments while preventing conflicts. The frontend, developed using React.js, provides an interactive interface where patients can browse available time slots and manage their appointments. The user interface dynamically updates appointment status without requiring manual refresh, ensuring a seamless experience.

The backend, built with Node.js and Express.js, acts as the intermediary between the frontend and the database. It processes user requests, manages authentication, and ensures data consistency. The MongoDB database securely stores appointment details, user credentials, and scheduling records while preventing duplicate bookings. Appointment availability is updated immediately upon confirmation, ensuring an accurate and reliable scheduling process. To ensure system security and prevent unauthorized access, JWT-based authentication is implemented, allowing only verified users to book and manage appointments. Patients can schedule consultations, doctors can manage their availability, and administrators have the authority to oversee system-wide operations. Role-based access control (RBAC) ensures that users interact with the system according to their designated permissions.

By leveraging MERN stack technologies, this system offers a fast, scalable, and user-friendly solution for appointment scheduling. The integration of real-time database updates ensures that appointment statuses remain synchronized across users. This approach reduces scheduling conflicts, minimizes delays, and optimizes the appointment booking process, ultimately enhancing the overall efficiency of healthcare management (Sharma, Patel, & Khan, 2025). [3]

OUTCOMES

Home Page



Fig no.1 Home page

The homepage of the Doctor Appointment Booking System features a clean and modern UI with a navigation bar for easy access to different sections. The hero section includes a welcoming message, a call-to-action button, and professional doctor images. Below, users can browse specialists, making appointment booking simple and efficient.

All Doctors Page

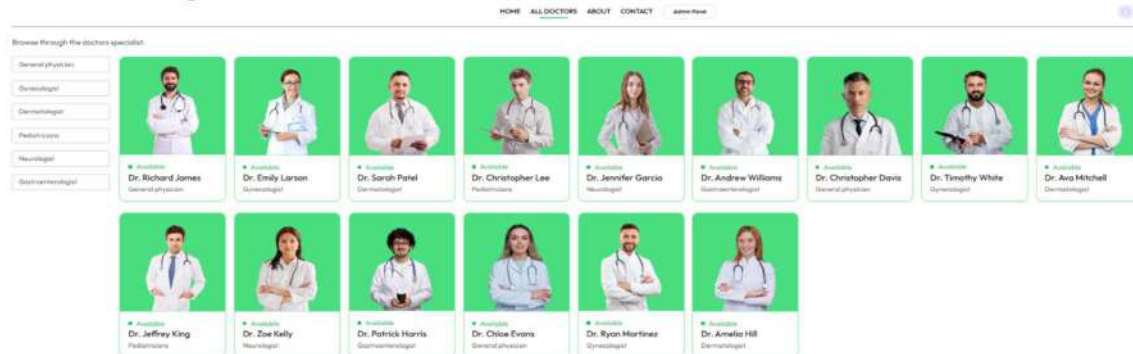


Fig no.2 All Doctor Page

These are all the specialties of the doctor that will be shown on this all-doctors on web page

Create account page

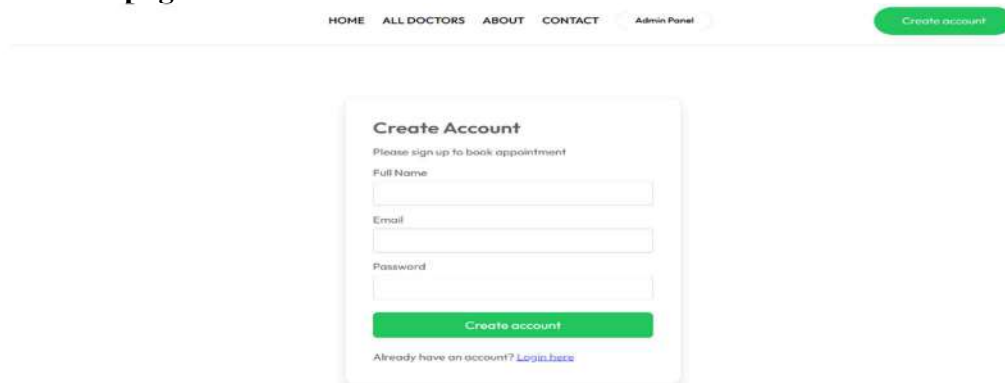


Fig no.3 Create account page

You can create your account on this page.

Login page

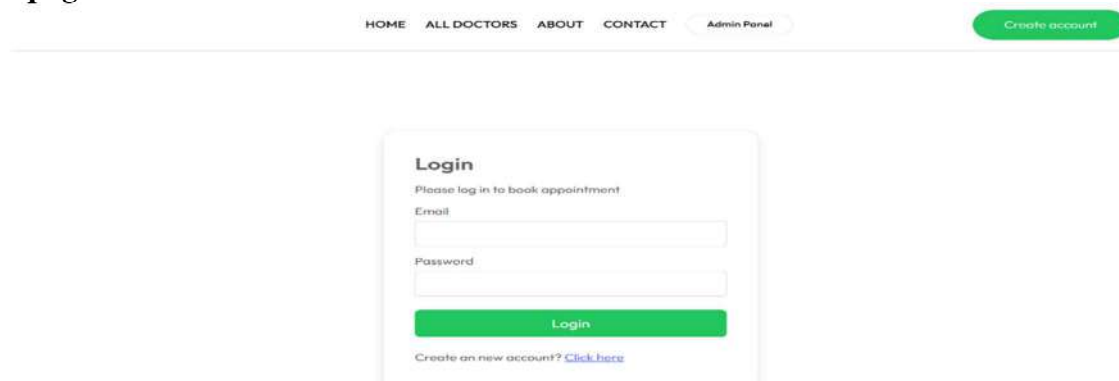
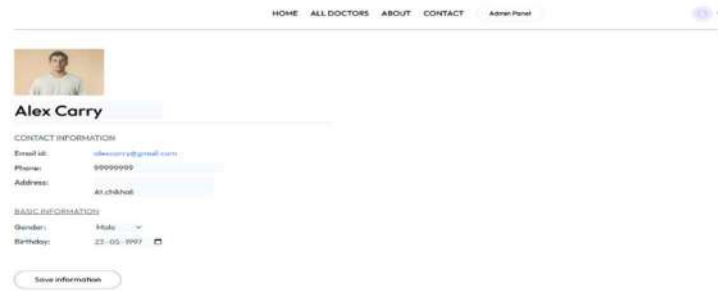


Fig no.4 Login page

The login page allows users to access their accounts by entering their email and password.

Profile page

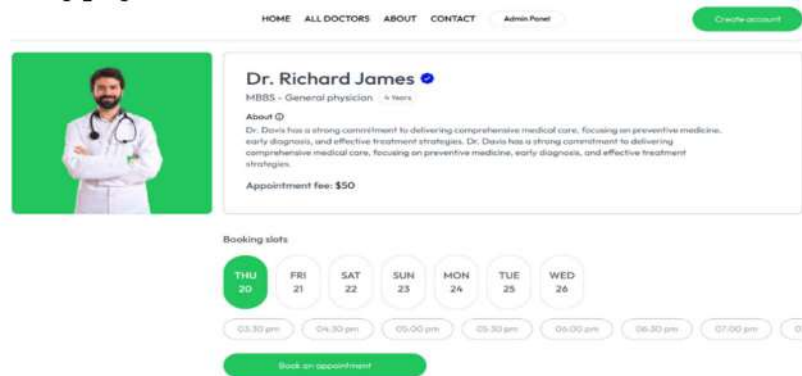


The screenshot shows a user profile for Alex Carry. At the top, there is a navigation bar with links: HOME, ALL DOCTORS, ABOUT, CONTACT, and Admin Panel. Below the navigation bar is a profile picture of a man. The name 'Alex Carry' is displayed. Underneath, there is a 'CONTACT INFORMATION' section with fields for Email id (alexcarry@gmail.com), Phone (99999999), and Address (41 chibhai). Below that is a 'BASIC INFORMATION' section with fields for Gender (Male) and Birthday (23-05-1997). A 'Save information' button is at the bottom.

Fig no.5 Profile page

The profile page displays user details with editable contact and basic information.

Appointment Booking page



The screenshot shows a doctor's profile for Dr. Richard James. At the top, there is a navigation bar with links: HOME, ALL DOCTORS, ABOUT, CONTACT, and Admin Panel. Below the navigation bar is a profile picture of a doctor. The name 'Dr. Richard James' is displayed, followed by 'MBBS - General physician' and '4 Years'. Below that is an 'About' section with a description of the doctor's commitment to delivering comprehensive medical care. The 'Appointment fee: \$50' is also displayed. Below the doctor's profile is a 'Booking slots' section with a calendar view showing days from THU 20 to WED 26. Each day has a corresponding time slot (e.g., 03:30 pm, 04:30 pm, etc.). A 'Book an appointment' button is at the bottom.

Fig no.6 Appointment Booking page

This doctor profile page displays Dr. Richard James' details, including qualifications, experience, and appointment fee. Users can view available booking slots and schedule an appointment easily through the provided interface.

Appointment Booking Payments page



The screenshot shows a payment page for a scheduled appointment. At the top, there is a navigation bar with links: HOME, ALL DOCTORS, ABOUT, CONTACT, and Admin Panel. Below the navigation bar is a section titled 'My appointments'. It displays a profile picture of Dr. Richard James, his name, 'General physician', and his address: '24 main street, 10 claus road'. The 'Date & Time' is '20 Apr 2025 | 04:30 PM'. Below this information are two buttons: 'Pay Online' and 'Cancel appointment'.

Fig no.7 Appointment Booking Payments page

This appointment page displays scheduled details with Dr. Richard James, including date, time, and address. Users can pay online or cancel.

Admin page(Add New Doctor)

The screenshot shows the 'Add Doctor' form in an admin interface. The sidebar on the left contains links to 'Dashboard', 'Appointment', 'Add Doctor', and 'Doctors List'. The main form area has the following fields:

- Full Name:
- Date of Birth:
- Doctor Email:
- Specialty:
- Experience:
- Fees:
- Address:
- Phone:
- Bio:

At the bottom of the form is a green 'Save Doctor' button. Below the form, there is a bio text area with a sample bio for 'Dr. Don'.

Fig no.8 Admin page(Add New Doctor)

This "Add Doctor" page allows admins to register new doctors by entering details like name, specialty, experience, fees, and bio.

CONCLUSION

The "DonOnTime" system has effectively addressed the challenges in healthcare appointment scheduling by offering a streamlined and secure platform for both patients and medical professionals. Utilizing the MERN stack—MongoDB, Express.js, React, and Node.js—the system is structured on a scalable and efficient framework that facilitates seamless interactions between users and backend operations while ensuring secure data management.

A key advantage of "DonOnTime" is its real-time appointment scheduling capability, which reduces the chances of conflicts, improves scheduling accuracy, and enhances patient satisfaction. The platform's role-based access control (RBAC) and JSON Web Token (JWT) authentication mechanisms add essential security layers, ensuring that sensitive healthcare data remains protected. Furthermore, the cloud-based deployment enables scalability and efficient system management.

While "DonOnTime" meets its core objectives, future improvements could further elevate its functionality. Integrating analytics and reporting features would provide healthcare professionals with insights into patient engagement and operational efficiencies. Additionally, incorporating Electronic Health Records (EHR) synchronization would enhance patient data management and decision-making for healthcare providers. Expanding multilingual support would also improve accessibility for a diverse user base.

Overall, "DonOnTime" demonstrates how modern web technologies can revolutionize healthcare appointment management, reducing inefficiencies and enhancing patient engagement. As technological advancements continue, further refinements and integrations can extend the system's impact within the healthcare industry.[4]

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Personalized Virtual Interior Design using Generative AI

Pranali Bhope¹, Mishri Gube², Vidhi Khobragade³, Madhuri Sahu⁴

^{1, 2, 3} Students - Department of Artificial Intelligence,

pranalibhope@gmail.com

mishrigube2003@gmail.com

vidhikhobragade466@gmail.com

⁴ Assistant Professor - Department of Artificial Intelligence,

madhuridec9@gmail.com

G. H. Raison College of Engineering, Nagpur

Abstract:

This study presents an AI-driven virtual interior design system that allows users to effectively redesign their home or workspace with an integrated shopping experience. The system tailors designs according to room type, style, and purpose, ensuring that the created layouts are aesthetically pleasing, well-organized, and functional. In contrast to conventional interior design, which involves professional skill, hand labor, and much time consumption, this AI-based method uses technology to automate fundamental design features like lighting adjustment, and color matching, making it possible for users to achieve visually appealing and functional spaces with little effort. In addition, the system is able to recognize furniture and accessories in the output designs and associate them with online retailers so that users can easily browse and buy recommended items directly. This integration serves to close the gap between computer-generated design and actual implementation, and it provides a seamless and engaging experience. The AI guarantees realism and flexibility in the designs while ensuring coherence among varied elements and yet conforming to user interests. Future development will emphasize enhanced adaptability through the addition of real-time user input and sophisticated learning processes, enabling the system to further refine its suggestions and provide even more tailored results. Through the integration of automated design generation with a seamless shopping experience, this work offers a revolutionary solution that streamlines the interior design process while making it easy for users to realize their virtual designs in the real world.

Keywords: Interior Design, Generative AI, Deep Learning, Stable Diffusion, Ecommerce

Objective:

This research aims to develop an AI-driven Virtual Interior Design System that transforms room images based on user preferences using Stable Diffusion with ControlNet. The objective is to enhance design accuracy, speed, and customization while making interior design more accessible. This system can assist homeowners, designers, and businesses in visualizing and planning interiors efficiently.

Introduction:

Interior design is an essential process of developing functional, aesthetically pleasing, and comfortable living or work environments. Conventional interior design is a time-consuming and money-intensive process involving manual planning and numerous consultations with designers to result in the desired outcome. While professional designers use experience and creativity to design spaces, the technological revolution has brought forth AI-based solutions to automate and customize the interior design process. The advent of artificial intelligence has made it possible to create real, well-organized, and user-oriented interior designs, removing many of the issues that come with conventional methods.

AI-based interior design solutions offer a low-cost and automated solution, enabling users to produce high-quality designs in a matter of minutes. One of the greatest benefits of AI in interior design is the speed and efficiency. Rather than taking weeks to plan a room, AI is able to create high-quality designs in a matter of seconds, so the process is vastly quicker. Furthermore, AI systems give users several options for designs to compare and determine the best option for their requirements. Major capabilities of AI-based interior design are:

Automated furniture layout design – AI can scan room measurements and propose optimum furniture positioning for maximum space use.

Color and lighting suggestions – AI proposes complementary colors and optimal lighting configurations to create overall ambiance.

Personalization and customization – People can personalize and customize their own styles, colors, and furniture arrangements to make their space their own.

One of the main challenges with classic interior design is searching for and choosing furniture and decorative items that suit a design concept. To make this process more efficient, the suggested system incorporates an e-commerce option, enabling users to buy directly suggested furniture and decor products from online retailers. With this feature, the overall interior design experience is improved by filling the gap between digital design and actual implementation. By integrating AI-powered design generation with e-commerce capabilities, this system presents an integrated interior design solution. Users can bring their dream rooms to life and make them a reality with minimal effort, and make interior design more efficient, and user-friendly.

Literature Survey:

The integration of artificial intelligence (AI) into interiors has achieved considerable traction, enabling automated, efficient, user-oriented design solutions. Various studies have explored AI-controlled approaches for interior design, highlighting generative models, virtual reality, and personalized recommendations. Wu and Han (2023) [1] explored how AI and Virtual Reality can improve interactive designs and ensure seamless user interaction. Similarly, Wang et al. (2024) [2], and Thakkar et al. (2024) [3] proposed a generative AI-based system for interior design that emphasizes automation in spatial planning and decoration selection. Furthermore, Ismail (2024) [4] examined the integration of AI interior design into training and demonstrated its impact on student design. Additionally, Phadke et al. (2024) [6] utilized GANs to create realistic virtual interior designs, demonstrating deep learning's capability in design synthesis. Cai and Xie (2020) [10] investigated the impact of environmental psychology on modern interior design, aligning with AI's potential for user-oriented personalization. Hanna (2023) [12] analyzed AI-generated art on platforms such as MidJourney, highlighting AI's creative potential in artistic and advertising applications. Research by Ren et al. (2024) [13] introduced DesignGemini, which combines generative models with digital twin technology to enable highly personalized design solutions. These studies collectively highlight the transformative potential of AI in interior design, paving the way for virtual interior design generators to automate layout creation and integrate seamless ecommerce experiences. The proposed AI-driven virtual interior design generator utilized computer vision, deep learning, and stable diffusion to improve interior accessibility, efficiency, and personalization.

Methodology:

The proposed system is a combination of AI-based models and object detection techniques to generate optimized room interior designs based on user input.

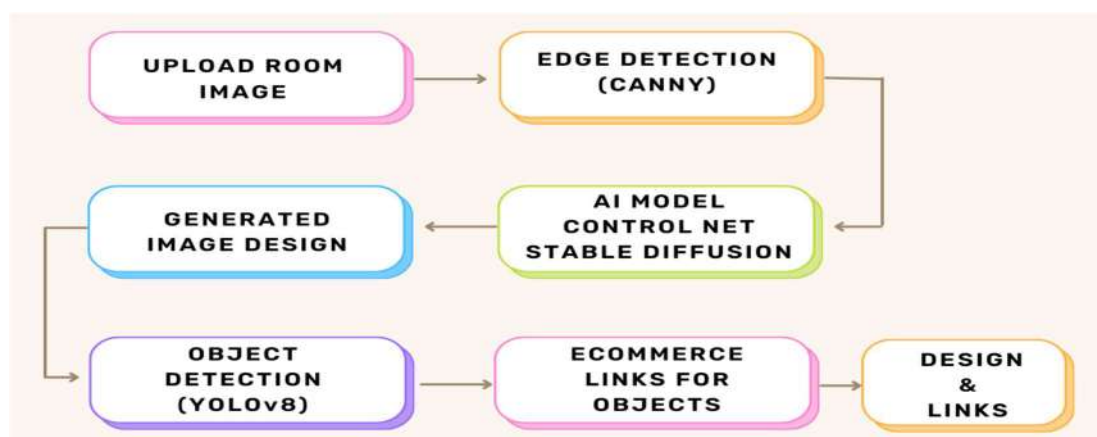


Fig1:

Working flow of the model

1. Room Image Upload:

The process begins with users uploading an image of their current room.

2. Edge Detection (Canny Algorithm):

To analyze the structural elements of the room, such as walls, doors, windows, and major furniture pieces, the Canny Edge Detection technique is applied.

3. AI Model – ControlNet Integrated with Stable Diffusion:

The structural edge map obtained from the previous step is processed using a combination of Stable Diffusion models and ControlNet. As a result of AI processing, a realistic, AI-generated interior design image is produced.

4. Object Detection using YOLOv8:

After the virtual design is finalized, YOLOv8, an advanced object detection model, is used to identify and label various elements within the design (e.g., chairs, tables, sofas, lighting fixtures).

5. Mapping to E-commerce Platforms:

Once individual furniture and decor items are detected, the system matches each element to similar products available on popular e-commerce platforms like Amazon.

6. Final Output: Design Visualization and Shopping Links:

Finally, the system presents users with the AI-generated interior design image along with clickable purchase links for each identified object.

Implementation:

The Virtual Interior Design Generator has been implemented by integrating advanced AI models with image processing and object detection techniques to provide users with automated design suggestions and purchasable item lists. The key components of the system and their respective implementation details are:

1. Image Upload and Preprocessing

The system accepts room images in standard formats such as JPG or PNG. Once an image is uploaded, preprocessing steps such as resizing, normalization, and format conversion are applied to ensure compatibility with subsequent AI models.

2. Edge Detection using Canny Algorithm

To extract structural details, Canny Edge Detection is applied to the uploaded image using OpenCV, a widely used computer vision library. The Canny algorithm identifies edges that represent walls, doors, windows, and existing furniture outlines. This edge map serves as the input guide for the AI model, ensuring that generated designs maintain spatial coherence with the room's structure.

3. AI-Based Interior Design Generation

The processed edge map is fed into a combination of ControlNet and Stable Diffusion models:

- ControlNet ensures that the AI-generated design adheres to the room's existing layout by following the provided edge constraints.
- Stable Diffusion, a powerful text-to-image generative model, produces high-quality interior designs. Users can provide additional input such as preferred color themes, furniture styles, or decor choices, which are embedded as prompts in the AI model to influence the final output.

4. Object Detection with YOLOv8

Once the AI-generated design is obtained, YOLOv8 (You Only Look Once version 8) is employed for object detection and segmentation. The model is fine-tuned on datasets to recognize key furniture elements such as sofas, tables, chairs, lamps etc.

- YOLOv8 is implemented using Ultralytics YOLOv8 library, which provides pre-trained weights and allows customization for furniture-specific datasets.
- Detected objects are annotated with bounding boxes and class labels.

5. Product Mapping to E-Commerce Platforms

The list of detected objects is passed to a mapping module that searches for similar products on popular e-commerce platforms such as Amazon. The system retrieves product images, names and direct purchase links.

6. Final User Interface (UI) Output

The final output is a web interface developed using Streamlit. Users can interactively view the design and explore options to purchase each piece of furniture and decor.

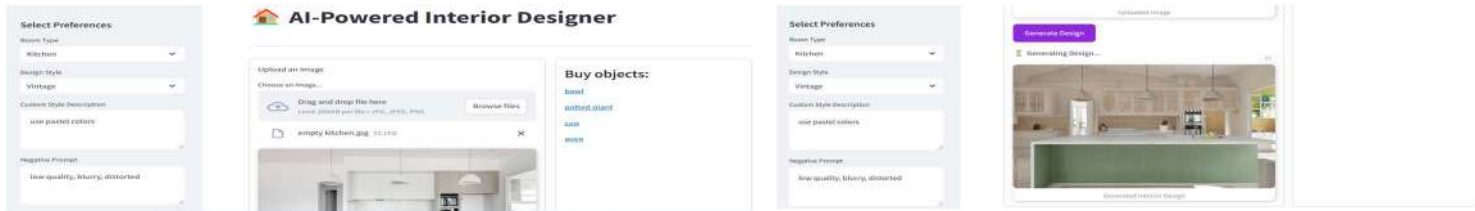


Fig 2: User interface displaying final output

Results:



Fig 3: Comparison of original image and generated image

The system successfully improves plain interiors by generating stylish and realistic designs. In the kitchen transformation, the original simple white space was enhanced with a modern green island, better textures, and improved lighting. This shows that our model can create visually appealing and high-quality interior designs while keeping important elements like windows and room structure intact.

Conclusion:

This research introduces a virtual interior designer generator that works with AI. This automates the interior design process, making it more accessible, more efficient and personalized. Using generative AI, computer vision, deep learning and stable diffusion, users can create aesthetically appealing and functional rooms with minimal effort. Additionally, e-commerce integration improves the user experience by seamlessly linking design elements with purchasable items. The proposed approach significantly reduces the time, cost, and expert knowledge required for interior decoration, while also entailing high adjustment and adaptability. Future work will focus on including real-time user feedback and augmented learning mechanisms, further improving design recommendations and improving user satisfaction.

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Predictive Analysis of Hearing Impairment with Machine Learning Approach

Ms. Ashwini R. Kukade¹, Prof. (Dr.) K. T. V. Reddy²

¹ Faculty of Engineering And Technology, Wardha, India
Email: ashwinikukade90@gmail.com

² Faculty of Engineering And Technology, Wardha, India
Email: ktvreddy@gmail.com

Abstract— Hearing loss significantly impacts communication, social interactions, and quality of life, yet early detection remains a challenge due to limited access to testing and awareness. Machine Learning offers a transformative solution by enhancing the accuracy, efficiency, and accessibility of hearing assessments. By analyzing large datasets, ML models can identify subtle patterns of hearing impairment, predict risk factors, and automate the diagnostic process. This study explores the significance of various Machine learning techniques like Support Vector Machine –SVM, Logistic Regression, Decision Tress in revolutionizing hearing impairment detection, enabling early intervention and improved healthcare accessibility across diverse populations.

Index Terms— Machine Learning, SVM, Decision Tree, Logistic Regression, Hearing Impairment

I. INTRODUCTION

Hearing ability is essential for communication, playing a vital role in social, professional, and educational settings. It significantly impacts language development, cognitive function, and emotional well-being. However, hearing loss often remains undiagnosed until it becomes severe due to factors such as lack of awareness, limited healthcare access, and its gradual onset. Early detection is crucial to prevent negative consequences like cognitive decline, social isolation, and reduced quality of life. Despite its importance, barriers such as geographic limitations, financial constraints, and the need for specialized equipment hinder widespread hearing assessments. Traditional hearing tests, while effective, require professional expertise and advanced tools, making them inaccessible to many. To address these challenges, machine learning presents a transformative solution by enhancing the accessibility, efficiency, and accuracy of hearing assessments. As a subset of artificial intelligence, machine learning has demonstrated success in various healthcare applications, including disease detection and personalized treatment. In hearing health care, ML algorithms can analyze large datasets—such as audiograms, speech recognition tasks, and wearable device inputs—to detect hearing impairments earlier and more accurately than traditional methods. This paper explores the significance of hearing ability, challenges in traditional testing methods and the potential of machine learning to revolutionize hearing health care.

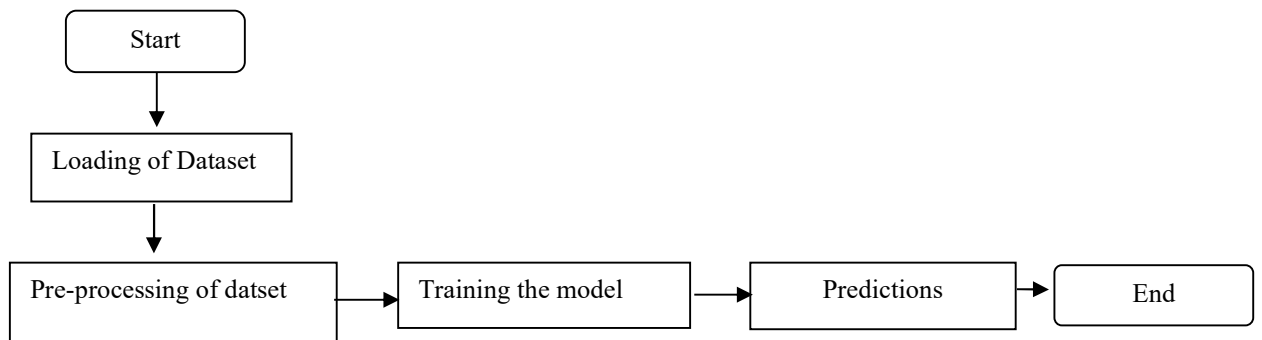


Figure 1: Proposed System Flow

II. RELATED WORK IN HEARING LOSS

[1] Presents 94% accuracy with SVM in terms to predict and to train model to test hearing loss. The DDDTCWT, RKPCA and MLR- Multinomial Logistic Regression [2] has been developed for hearing loss identification which performs well as compared to raw and improved AlexNet, and eight state-of-the-art

approaches. Multinomial Logistic Regression Classifier [3] employed to detect sensorineural Hearing Loss with 96.17% of accuracy. Multiple logistic regression analyses identified independent predictors of hearing loss [4] was taken and prognostic model developed from these risk factors was associated with a 15% rate of hearing loss in stage I, 52% rate of hearing loss in stage II, and 96% rate of hearing loss in stage III. [5] Shows that Logistic Regression and SVM were the most common Machine learning algorithm for predicting sudden sensorineural hearing loss. AI models- Logistic Regression, Support Vector Machine and others [6] which are used to classify audiograms into types of Hearing Loss with accuracy 94.46%. Hearing loss detection achieved an accuracy of 98% [7] using the FP-Growth algorithm for feature transformation and the machine learning algorithm for classification. [8] proposes a machine learning framework aimed at enhancing the predictive performance of the Hearing Handicap Inventory for the Elderly Screening (HHIE-S). This framework utilizes machine learning models, including XGBoost, Gradient Boosting, and LightGBM, to improve accuracy and efficiency in detecting hearing impairment. A community-based machine learning model [9] has been used for hearing impairment screening which provides a cost-effective HFHI risk assessment. The study [10] assesses the effectiveness of multivariate machine learning models in a speech-in-noise hearing screening test and analyzes the influence of various features on hearing loss detection using explainability techniques. Seven ML models were trained on a dataset of 215 ears (99 with hearing loss and 116 without), including interpretable models like decision trees and logistic regression, as well as complex models like random forests. Among them, the random forest model achieved the highest accuracy (85%), surpassing decision trees, while models such as SVM, logistic regression, and gradient boosting demonstrated similar performance. The study [11] evaluated noise-induced hearing loss using machine learning models based on factors such as age, exposure duration, LAeq, and median kurtosis. Four algorithms—SVM, neural networks, random forest, and adaptive boosting—were utilized for both classification and regression. The classification models achieved accuracies ranging from 78.6% to 80.1%, with SVM performing the best in predicting hearing impairment. Machine learning models [12]—SVM (98%), Random Forest (96%), and AdaBoost (96%)—achieved high accuracy in detecting hearing loss which were designed for portability and ease of use, the system enables independent testing and remote consultations with medical professionals. The study [13] focused on developing an automated audiometer for self-assessing hearing ability by extracting acoustic signal features and classifying normal and profound hearing loss using machine learning algorithms. A total of 150 patients (75 with normal hearing and 75 with profound hearing loss) underwent audiometric testing in both soundproof and regular environments. Among the classifiers tested, Support Vector Machine (97%) outperformed k-Nearest Neighbor (96%) and Naïve Bayes (95%).

III. MACHINE LEARNING ALGORITHMS

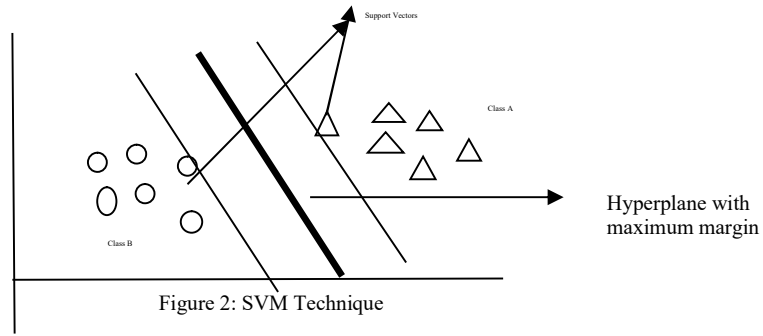
Machine learning (ML) is a subset of Artificial Intelligence that allows systems to learn from data without requiring explicit programming. Instead of specifying all possible outcomes, ML models are trained on datasets to identify patterns and generate predictions.

A. Machine Learning (ML) has been classified into main three categories which are supervised learning, unsupervised learning, and reinforcement learning.

1. **Supervised Learning:** This method involves training a model using labelled data, where input observations are mapped to corresponding outputs. The model learns from this training data to make accurate predictions on new, unseen data.
2. **Unsupervised Learning:** In this approach, the model is trained on unlabelled data and identifies patterns, structures, or groupings based on inherent characteristics, without external supervision or predefined labels.
3. **Reinforcement Learning:** The model learns through continuous interaction with an environment, improving its performance over time by receiving feedback in the form of rewards or penalties for its actions.

B. Support Vector Machine (SVM)

It is a supervised machine learning technique used for both classification and regression problems. SVM operates by identifying the best possible hyperplane that separates data points into distinct categories. It is especially powerful when working with high-dimensional datasets and complex decision boundaries.



The dataset used comprises 500 synthetically generated samples with features including age, noise exposure, family history, ear infections, and audiometry scores. The target variable is the presence or absence of hearing loss. A Support Vector Machine (SVM) classifier was trained on this dataset. The resulting confusion matrix reflects the model's strong performance, correctly identifying the vast majority of cases. Out of all actual "No Loss" cases, nearly all were accurately predicted, with only a minimal number of false positives. Likewise, the model successfully classified almost all Hearing Loss cases with negligible false negatives. This led to an impressive accuracy of 99%, demonstrating the SVM model's high reliability in detecting hearing loss based on the given features.

C. Decision Tree

A Decision Tree in machine learning is a supervised learning algorithm used for classification and regression tasks. It splits the data into branches based on feature values, forming a tree-like structure where each internal node represents a decision rule, and each leaf node represents an outcome. It's easy to interpret and visualize. However, it can overfit on complex datasets unless pruned or regularized. The dataset includes 500 synthetic samples with features such as age, noise exposure, family history, ear infections, and audiometry scores for predicting hearing loss. A Decision Tree Classifier was trained on this dataset. The confusion matrix indicates that the model correctly classified 40 "No Loss" cases and 39 "Hearing Loss" cases, while misclassifying 8 samples as false positives and 13 as false negatives. This results in an overall accuracy of 79%, suggesting moderate predictive performance. While Decision Trees are interpretable and easy to use, the model shows limitations in handling noisy or overlapping data without additional tuning.

D. Logistic Regression

Logistic Regression is a supervised learning algorithm primarily applied to binary classification problems. It estimates the likelihood that an input belongs to a specific class using the logistic (sigmoid) function. Although the name suggests regression, it is actually meant for classification. The dataset comprises 500 synthetic samples containing features such as age, noise exposure, family history, ear infections, and audiometry scores, with the target variable being hearing loss. A Logistic Regression model was trained to classify hearing loss. According to the confusion matrix, the model correctly predicted 43 out of 49 "No Loss" cases and 42 out of 51 "Hearing Loss" cases. It made 6 false positives and 9 false negatives. This yields an overall accuracy of 85%, indicating that Logistic Regression performs reasonably well in identifying hearing loss, though some misclassifications exist.

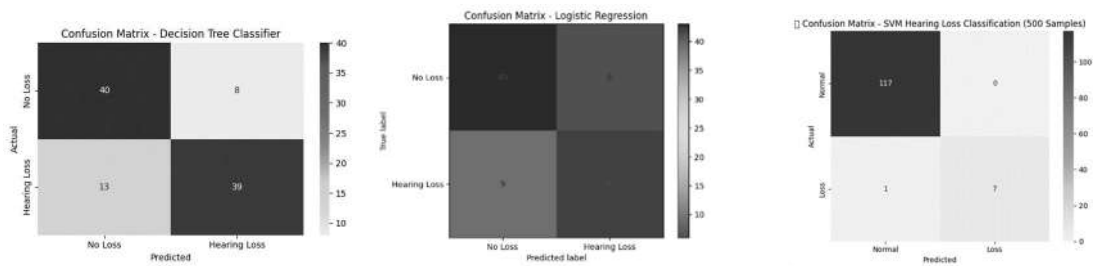


Figure 4: Confusion matrix for Decision tree, Logistic Regression and SVM

E. Comparison among Proposed Algorithms

Each of the three algorithms—Support Vector Machine (SVM), Decision Tree, and Logistic Regression—offers unique advantages and trade-offs in terms of accuracy, interpretability, and computational complexity.

SVM is highly effective in high-dimensional spaces and performs well with a clear margin of separation, but it can be computationally intensive, especially with large datasets. In terms of accuracy on the given hearing loss dataset, SVM achieved the highest at 99%, followed by Logistic Regression at 85%, and Decision Tree at 79%, indicating that SVM provides superior performance for this classification problem. Support Vector Machines (SVMs) generally have high training complexity, especially with non-linear kernels, often ranging from $O(n^2d)$ to $O(n^3)$, making them computationally intensive for large datasets. Decision Trees are more efficient, with training complexity around $O(nd \log n)$, and offer fast predictions. Logistic Regression is computationally lightweight, with training complexity of $O(n d i)$, where i is the number of iterations, making it suitable for large-scale and high-dimensional data. Overall, Logistic Regression and Decision Trees are faster to train compared to SVMs [14].

Table 1: Comparison among SVM, Decision Tree and Logistic Regression Algorithms

Parameter	SVM	Decision Tree	Logistic Regression
Problem-Type	Classification (Can be apply on regression)	Classification and Regression	Classification and Regression
Time Complexity (training phase)	$O(n^2 \times d)$ or $O(n^3)$ depending on kernel	$O(n \times \log n)$	$O(n \times d)$
Accuracy	Provides High Accuracy especially for complex boundaries	Provides Moderate Accuracy	Provides High Accuracy
Scalability for large Dataset	Poor	Good	Good
Use Cases	Image recognition, text categorization	Medical diagnosis, credit scoring	Customer churn prediction, risk modelling

IV. PROPOSED METHODOLOGY

A. Dataset Description

The dataset used in the above programs is synthetically generated and consists of 500 samples with 5 key features: age, noise exposure, family history of hearing issues, ear infections, and audiometry scores. These features are designed to simulate real-world factors influencing hearing health. The target variable is binary, indicating whether an individual has hearing loss (1) or no hearing loss (0). The dataset is balanced and suitable for evaluating classification models like SVM, Decision Tree, and Logistic Regression.

B. Performance Metrics

This section presents the key parameters used to evaluate the performance of machine learning models. A confusion matrix is utilized to compare predicted and actual class labels, comprising four essential elements: True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN). These metrics help assess the accuracy and overall effectiveness of the model.

1. Accuracy

Accuracy is a fundamental metric used to assess a model's performance, reflecting how well it has learned from the training data and how effectively it performs on unseen data. It is calculated as the proportion of correct predictions out of the total number of predictions made. The accuracy of a model can be calculated using the following formula:

$$\text{Accuracy} = \frac{\text{True Positive} + \text{True Negative}}{\text{True Positive} + \text{True Negative} + \text{False Negative} + \text{False Positive}}$$

2. Recall

Recall, also known as sensitivity, measures the proportion of actual positive cases that the model correctly identifies. It is a crucial metric for evaluating how effectively a model detects positive outcomes and how well it minimizes missed positive instances.

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

3. Precision

Precision evaluates the accuracy of positive predictions by determining the ratio of correctly predicted positive cases to all predicted positives. It highlights the model's effectiveness in making accurate positive classifications, though it doesn't consider the correctness of negative predictions.

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

4. F1 Score

The F1 score represents the harmonic mean of Precision and Recall, balancing both false positives and false negatives. It ranges between 0 and 1, where a score of 1 signifies perfect classification performance, while 0 indicates total misclassification.

V. IMPLEMENTATION AND RESULT ANALYSIS

A comparison among SVM, Decision Tree, and Logistic Regression was conducted on a computing system equipped with an AMD Ryzen 5 5625U processor with Radeon Graphics, 2301 MHz speed, 6 cores, 12 logical processors, and 8 GB RAM. The implementation leveraged essential Python libraries such as NumPy, Pandas, and Scikit-Learn. The model training and evaluation were performed using a simple holdout validation method, with 350 synthetic samples used for training and 150 samples reserved for testing.

Table 2: Performance Measure Indices

Model Performance (Testing Phase)			
	SVM	Decision Tree	Logistic Regression
Accuracy %	99%	79%	85%
Precision %	100%	82.98%	87.50%
Recall %	87%	75%	82.355
F1 Score %	93%	78.79%	84.79%

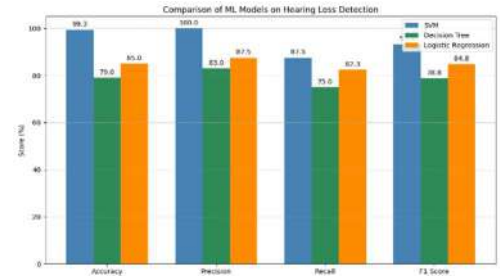


Figure 6: Graphical Representation of Performance Measure Indices

VI. Conclusion

Over 5% of the global population—around 430 million individuals, including 34 million children—require rehabilitation for disabling hearing loss. Projections suggest that by 2050, this number will rise to over 700 million, affecting 1 in every 10 people worldwide [16]. In response to this growing concern, the present study conducted a comparative analysis of three supervised machine learning algorithms—Support Vector Machine (SVM), Decision Tree, and Logistic Regression—for hearing loss detection using synthetic data. Based on the evaluation metrics, SVM outperformed the other two models, demonstrating the highest accuracy, precision, recall, and F1 score. Decision Tree and Logistic Regression also showed reliable performance, but were comparatively less robust. These findings highlight the potential of supervised learning models—particularly SVM—as effective tools for the early diagnosis of hearing loss, offering promising support for automated healthcare solutions.

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Fruit quality and disease detection using machine learning

Swati Pravin Aswale¹, Himanshi Raju Nikhare², Safura Anam Sheikh³, Pranjali Gite⁴

Abstract

As populations are soaring, agricultural production is being increased and this makes effective monitoring of fruit yields imperative. The man-made ways of disease detection during fruit ripening rarely suffice due to limited pathogen identification capacity and time constraints. Moreover, visual inspection may easily mislead since some natural colour variations could represent disease symptoms. To counteract this, the study here proposes an efficient smart farming paradigm for early-stage fruit disease classification using machine learning. The system, through image processing techniques, analyses fruit texture and skin colour from images taken using the Raspberry Pi. The convolutional neural networks (CNNs) are put into training and testing of the system so it can reliably identify and classify the fruits from the quality parameters of Fresh, Rotten, and Diseased and will tell us the level of risk. This gives promise for increased fruit yield and quality with early disease detection and reduced manual inspection.

Keywords: Smart farming, Fruit disease classification, Machine learning, Image processing, Raspberry Pi, Convolutional Neural Networks (CNNs), Fruit quality assessment, Early disease detection, Automated monitoring, Precision agriculture

INTRODUCTION

In modern agriculture, the move towards automation is spurred by the requirement for greater efficiency and less dependency on labor-intensive manual operations. This is very much the case in fruit quality evaluation, an important step towards food safety and yield maximization. Fruit quality assessment has historically depended on a lot of manual inspection, which is time-consuming and subjective with a high tendency for human error. Conditions like water supply, land type, and optimal fertilization make a great impact on fruit quality, but simple observation is not always able to detect fine indications of disease or rot.

The pressure for high-quality crops makes it necessary to create more effective and reliable quality evaluation methods. The last several years have seen the introduction of automated systems based on image processing and machine learning algorithms to evaluate fruit properties. Such systems provide possibilities for substantial improvement of the speed and accuracy of quality evaluation, leading to lower labor expenses, reduced post-harvest losses, and higher overall fruit quality.

This study is aimed at designing an automated system for classifying fruit quality through a supervised machine learning method. The system makes use of convolutional neural networks (CNNs) to examine high-resolution fruit images, and from these, it extracts relevant features like shape, color, and texture. Through training the CNN on an image dataset with images of fresh, rotten, and diseased fruits, the system can identify new fruit images into these three categories. This approach offers several advantages over traditional manual inspection methods, including:

- Increased efficiency: Automated image analysis can process a large volume of fruits rapidly, significantly reducing the time required for quality assessment.
- Improved accuracy: CNNs can detect subtle variations in fruit appearance that may be imperceptible to the human eye, leading to more accurate disease detection and quality assessment.
- Reduced labour costs: Automation can minimize the need for manual labour, reducing labour costs and improving overall efficiency in fruit handling and processing.
- Objective and consistent results: Automated systems provide consistent and objective results, minimizing the impact of human subjectivity and bias.

This highlights the precision and effectiveness of the suggested CNN-based system for classifying fruit quality. Successful implementation of this system can go a long way towards transforming the practice of assessing fruit quality in the agricultural sector, resulting in enhanced food safety, minimized wastage, and greater economic sustainability.

METHODOLOGY

1. Data Collection

The dataset of this research was collected by combining web scraping and publicly accessible datasets on Kaggle. For enhancing the robustness and generalization ability of the dataset, several data augmentation methods were utilized, such as flipping, rotation, brightness normalization, and contrast adjustment. All the datasets used in the model training was publicly available and free to use. The scrapping was done using Custom Google Search API and only free to use images were filtered out for training and testing purposes. There are a total of 9 classes which are Apple Fresh, Apple Diseased, Apple Rotten, Orange Fresh, Orange Diseased, Orange Rotten, Banana Fresh, Banana Diseased, Banana Rotten.

Table 1. Class Distribution on Fruit Condition

Class	Test Images	Train Images	Validation Images
Rotten Banana	252	1972	530
Diseased Apple	153	1107	432
Fresh Banana	162	1419	381
Diseased Orange	90	642	246
Rotten Apple	207	2135	601
Diseased Banana	90	600	210
Fresh Orange	162	1304	388
Rotten Orange	153	1442	403
Fresh Apple	198	1495	395

The dataset consists of all types of fruit varieties, ripening stages and for disease we've only added common apple diseases include Apple Scab, Fire Blight, Powdery Mildew, Cedar Apple Rust, and Bitter Rot; banana diseases include Panama Disease, Black Sigatoka, Banana Bunchy Top Virus, Moko Disease, and Anthracnose; and orange diseases include Citrus Canker, Huanglongbing, Citrus Black Spot, Alternaria Brown Spot, and Tristeza Virus.

2. Object Detection Using YOLOv8

YOLOv8 was selected for its real-time object detection efficiency, balancing accuracy with computational capability. The model was trained for more than 200 epochs with a learning rate of 0.01 and batch size of 16. Preliminary testing showed difficulty in detection from poor-quality cameras, prompting the use of sophisticated image preprocessing algorithms like noise removal and contrast adjustment. These adaptations hugely enhanced the accuracy in object detection. Traditional Fast R-CNN and SDD were also good choice, but YOLO provides a good balance between accuracy and speed with low computation compared to these other models. This makes YOLO a perfect fit for embedded systems.

3. Classification Using MobileNetV2

MobileNetV2 was chosen for classification of fruit condition because it is light in structure and computationally efficient. Fine-tuning of the model was done with 10 epochs since going beyond that resulted in overfitting. Dataset adjustments were done for this specific model to avoid overfitting in validation set. The Adam optimizer was utilized with a learning rate of 0.001, and categorical cross-entropy as the loss function. Originally, MobileNetV2 was responsible for classification directly, but in testing under real-world scenarios, it was seen that background noise affected predictions. In order to rectify this, the pipeline was run in two stages where object localization was first carried out by YOLOv8 and then classification.

Table 2. Model Hyperparameters Configuration

MODEL ARCHITECTURE		TRAINING	
Base Model	MobileNetV2 (pre-trained on ImageNet)	Batch size (Train)	64
Include Top Layers	False (using a custom classifier)	Batch size (Validation)	32
Input Shape	(224,224,3)	Epochs	10
Trainable Base Model Layers	False (frozen)	Steps per Epoch	train_generator.samples //train_generator.batch_size
Fully Connected Layer Neurons	256	Validation Step	val_generator.samples //val_generator.batch_size
Fully Connected Layer Activation	ReLU	Early Stopping Monitor	val_loss
Dropout Rate	0.2	Early Stopping Patience	5
Output Layer Neurons	9 (for 9 classes)	Restore Best Weights	True
Output Layer Activation	Softmax	DATA AUGMENTATION (ImageDataGenerator)	
OPTIMIZATION		Rescale Factor	1/255
Optimizer	Adam	Rotation Factor	15°
Learning Rate	0.0001	Width Shift Range	10% (0.1)
Loss Function	Categorical Cross entropy	Height Shift Range	10% (0.1)
Metrics	Accuracy	Zoom Range	10% (0.1)
		Horizontal Flip	True
		Fill Mode	'nearest'

4. System Integration and Workflow

To reduce background noise, a specialized dataset was designed solely for YOLOv8 to identify apples, bananas, and oranges with greater precision. The overall system workflow is methodical in its approach: YOLOv8 initially identifies the fruit within the image captured, followed by which the recognized fruit is cropped and analyzed using MobileNetV2. This model classifies the condition of the fruit as fresh, diseased, or rotten. The classification results are then presented on an OLED display that is attached to a Raspberry Pi 4B.

5. Implementation Details

The system combines various technologies to achieve efficient operation. YOLOv8, which is used in PyTorch, is used for object detection, and MobileNetV2, which is used in TensorFlow, is applied for classification. Image processing functions are performed using OpenCV. The hardware devices consist of a Raspberry Pi 4B and a Pi Camera for image capture, and the classification output is displayed on an OLED display.

6. Environmental Factor Integration

To anticipate risk profile of getting infected or rotten, a DHT11 sensor was employed for sensing temperature and humidity in real time. With this information, risk analysis can be performed based on assessment of external factors responsible for fruit rottenness. On-field experience establishes that elevated humidity causes frequent cases of fungal infection, while the rate of change in temperature affects the velocity of fruit decay in direct proportion.

Key correlations include:

- High humidity levels and more infections.
- Temperature variation affecting rates of fruit decay and infections.

7. Performance Evaluation

The final system attained 92% classification accuracy in the assessment of fruit condition. It showed strong detection and classification capability under different lighting conditions and environments, guaranteeing reliability in practical use.

8. Deployment and Real-Time Processing

The system has been successfully implemented on a Raspberry Pi 4B and can process the live camera input in real-time. The output of the classification is immediately shown on an OLED display, permitting direct interpretation and decision-making.

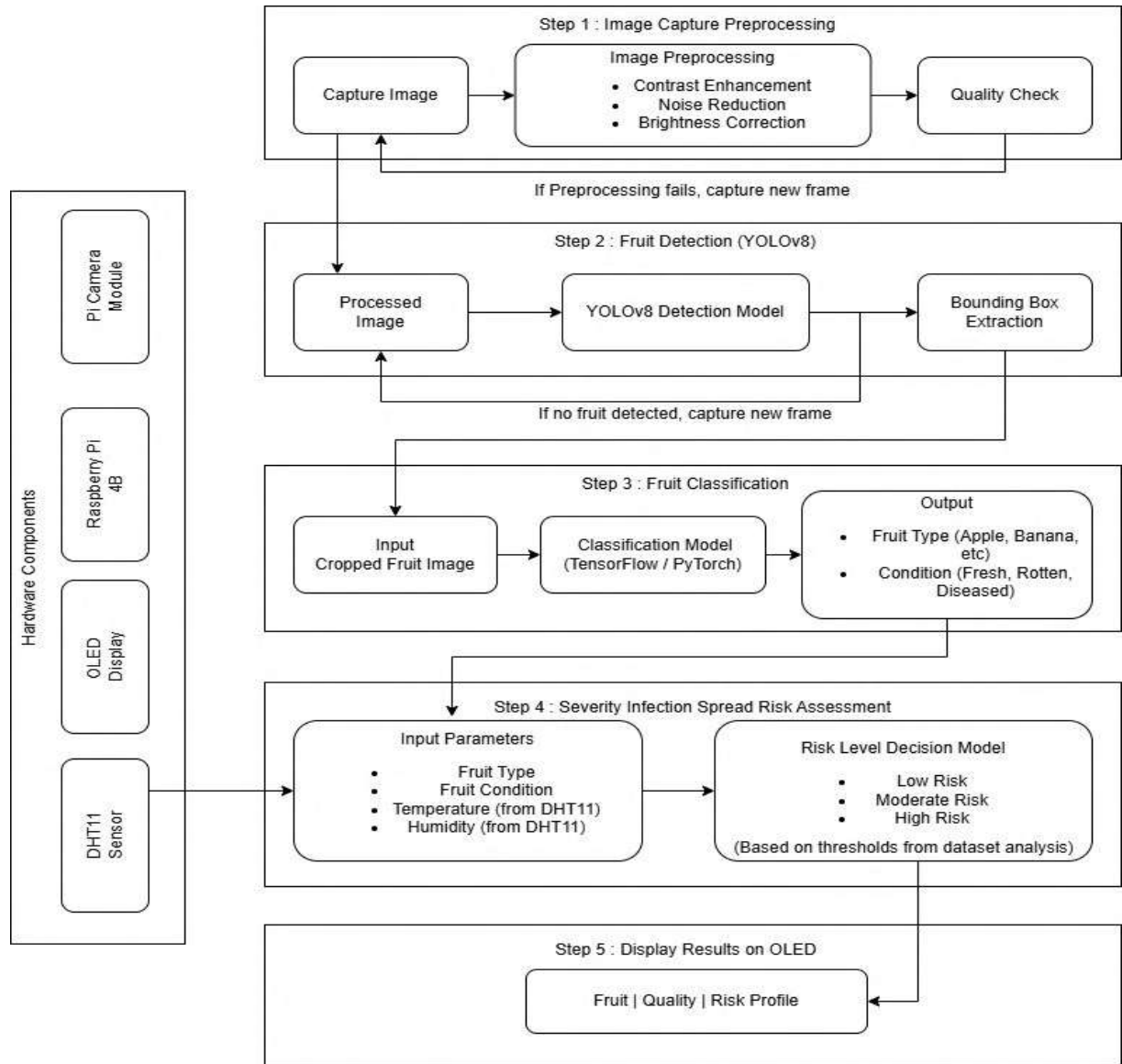


Figure 1. Fruit Detection and Classification System Architecture

RESULTS

The system proposed shows a good and efficient method for detecting fruit quality with Convolutional Neural Networks (CNNs). The system starts by taking the images of fruits through a Raspberry Pi camera, which are taken as an input to the system. The CNN model makes use of features extracted from the images and data provided at training time to calculate the quality of fruits. The system categorizes the fruits into three groups: fresh, rotten, and diseased. The categorization is done with very high accuracy, demonstrating the robustness of the presented method. Compared to other models like Artificial Neural Networks (ANNs), VGG16, MobileNetV2, Tiny VGG, MobileNetV3, Efficient Net B0, the CNN-based method MobileNetV2 always surpasses when it comes to precision and efficiency.

GENERAL MODEL PERFORMANCE:

- **VGG16:**
 - Train Accuracy: 91%
 - Validation Accuracy: 89%
 - Test Accuracy: 92%
 - **Obeservation:** The performance across all three datasets is quite consistent, indicating a well-balanced model with minimal signs of overfitting.

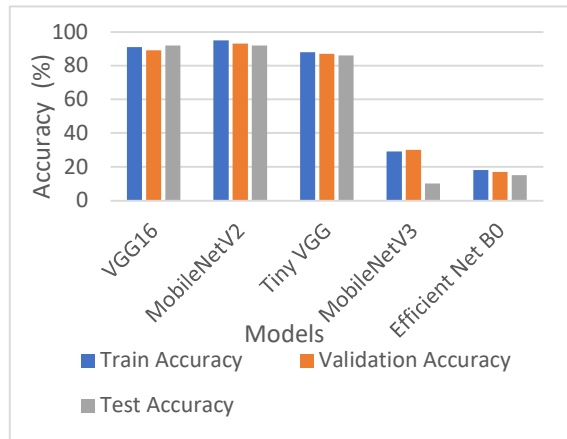


Figure 2. Different accuracy of proposed model

Table 3. Comparative analysis of proposed classification model with various accuracy

Model Name	Train Accuracy	Validation Accuracy	Test Accuracy
VGG16	91	89	92
MobileNetV2	95	93	92
Tiny VGG	88	87	86
MobileNetV3	29	30	10
Efficient Net B0	18	17	15

- **MobileNetV2:**
 - Train Accuracy: 95%
 - Validation Accuracy: 93%
 - Test Accuracy: 92%
 - Observation: This model shows excellent results across all datasets with a good balance of speed and accuracy making this model the best choice for use in embedded systems.
- **Tiny VGG :**
 - Train Accuracy: 88%
 - Validation Accuracy: 87 %
 - Test Accuracy: 86%
 - Observation: This model shows consistent performance across all datasets. It seems to be the most stable model with no signs of overfitting or underfitting. However, its overall accuracy is slightly lower compared to VGG16 and MobileNetV2.
- **MobileNetV3:**
 - Train Accuracy: 29%
 - Validation Accuracy: 30%
 - Test Accuracy: 10%
 - Observation: This model performed poorly across all datasets. The very low test accuracy (10%) indicates that it might be struggling with our specific dataset. It needs more data to optimize its additional non-linearity and attention layers. MobileNetV3 uses SE (Squeeze-and-Excitation) blocks and hard-swish activation, which make it more data-hungry. Hard-swish activation may not generalize well with small training sets.
- **EfficientNet B0:**
 - Train Accuracy: 18%
 - Validation Accuracy: 17%
 - Test Accuracy: 15%
 - Observation: Similar to MobileNetV3, EfficientNet B0 performed poorly. This could indicate issues like insufficient training, improper data preprocessing, or the model's architecture not aligning well with the dataset. EfficientNet B0 is optimized for large-scale data using compound scaling (width, depth, resolution). It assumes larger datasets, which our dataset lacks. It uses swish activation, which is harder to train on small datasets.

Table 4. YOLOv8 Model Statistics

Class	Images	Instances	Box			
			P	R	Map50	Map50-95
All	900	1579	0.785	0.83	0.765	0.695
Apple	385	607	0.886	0.638	0.731	0.643
Banana	118	138	0.514	0.92	0.582	0.494
Orange	397	834	0.956	0.931	0.982	0.948

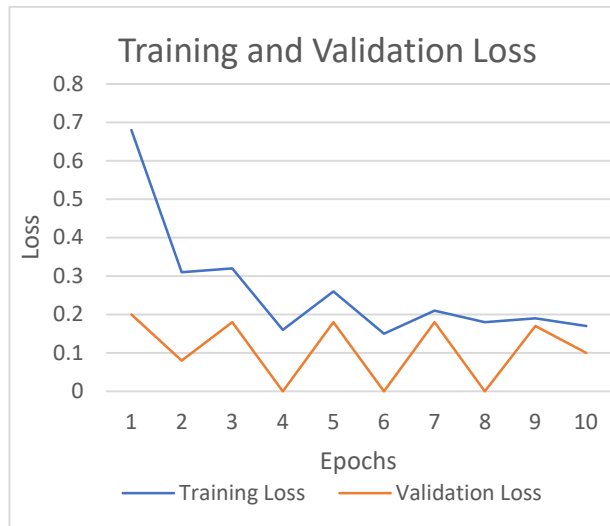


Figure 3. Training and Validation Loss

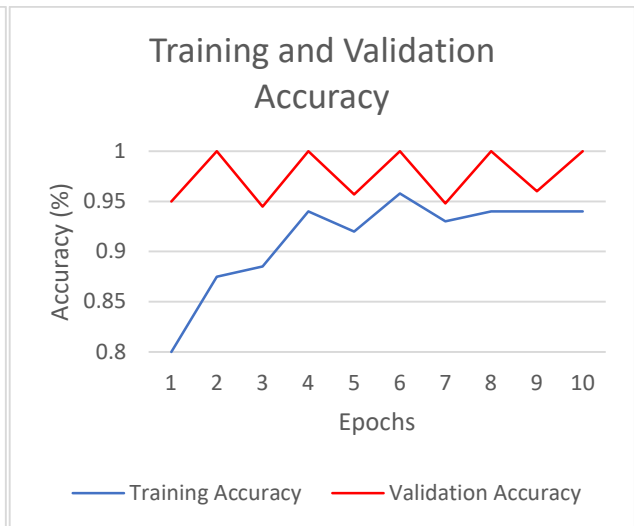


Figure 4. Training and Validation Accuracy

CONCLUSION

Overall, this paper successfully demonstrates the effectiveness of a Convolutional Neural Network (CNN)-based system for accurate and efficient fruit quality detection. The integrated methodology, encompassing data collection, image acquisition, preprocessing, and hyperparameter tuning, proved crucial in achieving high classification accuracy. The CNN model effectively classified fruits into fresh, rotten, and diseased categories. This automated system offers significant advantages, including improved efficiency, reduced labour costs, and enhanced quality control, with applications across various industries such as juice processing, food safety, and agriculture. By enabling early disease detection and minimizing wastage, this system contributes to improved yield and economic sustainability within the agricultural sector, highlighting the transformative potential of machine learning in modern agricultural practices.

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Artificial Intelligence based E-waste Management System

Ashvini K.Shete,
ashvinishete261988@gmail.com

Shrikant Chavate
shrikant.chavate@ghrua.edu.in

G. H. Raisoni University- Amravati

Abstract

E-waste is the fastest growing social and environmental challenge; hence, effective e-waste management is a crucial and essential to minimize ecological impact for the recovery of a valuable material. AI/ML and Deep learning is a suitable methodology which provides efficient and enhanced electronic waste management. This study describes about integration of Artificial Intelligent, Machine Learning and Deep learning techniques at different stages of e-waste includes e-waste collection, waste segregation or sorting, and disposal. This research focus on the characterization and classification of Printed Circuit Boards (PCBs) of various electronic devices using image-based analysis with high accuracy. CNNs which is used for automated classification of e-waste. The dataset required for training the CNN model is collected from different sources. Along with the CNNs, the Deep learning methodologies are being used for advanced defect detection, material classification, and predictive analytics in e-waste recycling and management.

Keywords: Artificial Intelligence (AI), automated sorting, circular economy, convolutional neural network (CNNs), deep learning.

I. INTRODUCTION: -

E-waste refers to discarded electrical or electronic equipment's such as computers, smart phones, televisions, and household appliances. With rapid technological advancements, the lifespan of electronic devices has decreased, leading to an increase of e-waste generation.[1] This has created challenges related to waste disposal, environmental pollution, recovery of valuable materials like gold, silver, and rare earth metals. [2]. AI and ML are emerging technologies that can transform e-waste management by offering intelligent, automated, and scalable solutions. Their ability to process large datasets, recognize patterns, and make predictions makes them ideal for addressing the complexities of e-waste. [3]

II. RESEARCH CONTRIBUTION

1. Innovation in Waste Sorting and Recycling Technologies:

Advancing Automation: By implementing AI-powered systems for automated sorting and classification, this research addresses the inefficiencies of traditional manual processes. It provides a scalable solution for dealing with the growing volumes and complexity of e-waste. [4]

Enhancing Material Recovery: The use of ML algorithms to predict optimal material recovery processes ensures maximum resource extraction, minimizing waste and increasing economic benefits.[5]

2. Promoting a Data-Driven Approach to E- waste Management

Improved Decision-Making: The research highlights the importance of data analytics and predictive models in guiding recycling processes, reverse logistics, and resource recovery.

Lifecycle Optimization: Predictive models for device obsolescence encourage preemptive e-waste collection and improve planning for sustainable disposal methods.[4]

3. Supporting Sustainability and Circular Economy Goals

Encouraging Reuse and Refurbishment: By identifying devices suitable for refurbishment or parts that can be reused, the research promotes circular economy principles.[6]

Reducing Environmental Impact: AI and ML optimize processes to minimize harmful emissions and waste, contributing to cleaner and more sustainable recycling practices.

4. Addressing Key Challenges in E-Waste Management

Handling Complexity: This research tackles the multifaceted composition of e-waste by developing intelligent systems capable of analyzing and classifying different components effectively.

Scalability: The use of AI ensures that the solutions proposed are adaptable to increasing volumes of e-waste, enabling the industry to adopt to technological advancements.

5. Strengthening Policy and Regulatory Compliance

Monitoring and Compliance: AI-based systems enhance e-waste tracking, ensuring adherence to local and international regulations.

Policy Development: The insights from predictive and simulation tools support policymakers in designing effective e-waste management frameworks and strategies.

6. Enabling Interdisciplinary Collaboration

Bridging Gaps: This research connects expertise from AI/ML, material science, and environmental engineering to develop holistic e-waste management solutions.

Fostering Innovation: The use of advanced technologies opens new avenues for collaboration in areas such as IoT-enabled monitoring, robotics, and green technology development.

III. CHALLENGES IDENTIFIED: -

Despite the technological advancements, the literature highlights several key barriers:

Scalability Issues: Many proposed systems have limited scalability, particularly in regions with weak infrastructure or low technological adoption.

Data Quality and Standardization: The lack of standardized, high-quality datasets poses a challenge for training AI models effectively and ensuring accurate predictions.

Policy Gaps and Legislation: Inconsistent policies and insufficient global regulations hinder the expansion of eco-friendly waste practices in waste management practices.[7]

Citizen Engagement: Limited participation in waste separation and recycling efforts continues to be a major hurdle.

Cost and Resource Limitations: High implementation costs and the need for competent personnel restrict the deployment in modern technologies in resource-limited settings.[3]

IV. SOCIAL ASPECTS

Environmental Protection

Reduced Environmental Hazards: AI-driven e-waste management minimizes the release harmful substances, e.g., lead, mercury, cadmium) in the environment by optimizing recycling processes and reducing improper disposal.

Sustainable Ecosystem: By promoting efficient resource recovery, the research fosters a cleaner and healthier environment for communities.[8]

Public Awareness and Engagement

Educational Impact: AI based solution in e-waste management highlights the importance of responsible disposal and encourages individuals to recycle their electronic devices.

Behavioral Change: The research supports campaigns that motivate individuals to adopt sustainable practices, contributing to a culture of environmental responsibility.

Job Creation and Skill Development

Upskilling the Workforce: The deployment of AI in recycling processes requires skilled operators and technicians, fostering training programs and creating new career opportunities

Employment Opportunities: This research drives the growth of industries focused on e-waste recycling, refurbishment, and reverse logistics, generating jobs in both technical and non-technical fields.

Health and Safety Improvements

Reduction in Manual Handling: By automating hazardous processes like e-waste sorting, the research minimizes the health risks faced by workers in traditional recycling plants.

Community Health Benefits: Proper e-waste management reduces pollution, leading to healthier living conditions for nearby communities.

Social Equity

Accessibility to Refurbished Electronics: The focus on refurbishing e-waste provides affordable electronic devices to underserved populations, bridging the digital divide.

Global Impact: The solutions developed can benefit both developed and developing nations, fostering global collaboration in addressing e-waste challenges.

V. ECONOMIC ASPECTS

1. Cost Efficiency in E-Waste Management

Optimization of Processes: AI and ML streamline recycling processes, reducing operational costs and improving profitability for waste management companies. [9]

Resource Recovery: Maximizing material recovery (e.g., rare earth metals, gold, and copper) reduces dependency on mining and lowers costs for manufacturing industries.[5]

2. Economic Growth and Industry Development

New Market Opportunities: The research encourages the growth of specialized sectors such as AI-driven recycling technology and refurbished electronics markets.[10]

Circular Economy Promotion: By enabling resource reuse and recycling, the research supports economic models that emphasize sustainability over linear consumption patterns.

3. Increased Revenue Generation

High-Value Materials: AI systems enhance the identification and extraction of valuable components.[5]

VI. SCOPE OF RESEARCH

The combination of Artificial Intelligence (AI) and Machine Learning (ML) in e-waste offers various possibilities of improving e-waste classification, recycling efficiency, resource recovery. The key areas within the scope of this research include:

Automated Waste Identification & Classification

Using AI/ML models to identify different types of e-waste (PCBs, batteries, plastics, metals, etc.).

Implementing computer vision for object detection and sorting of electronic components.[6]

Smart Sorting and Segregation

Utilizing AI-driven robotic systems for efficient e-waste separation, enhancing recycling accuracy through ML-based material recognition.

Forecasting of Waste Management

Using AI we can understand different patterns of e-waste generation which is based on historical data and forecasting of recycling demand and optimizing collection of e-waste strategies.[9]

Resource Recovery Optimization

Implementing ML algorithms to analyze the most efficient recycling methods. Enhancing material extraction from e-waste for reuse and sustainability.[6]

AI in Decision Support Systems Assisting policy makers and industries with data-driven recommendations for waste management. Improving circular economy models using AI-generated insights.

IoT and AI Integration for E-Waste Tracking Implementing IoT sensors to track e-waste disposal and recycling processes. Enhancing the ongoing monitoring of e-waste movement through AI- based systems.[7]

VII. ENHANCING ONGOING MONITORING OF E-WASTE MOVEMENT THROUGH AI- BASED SYSTEM

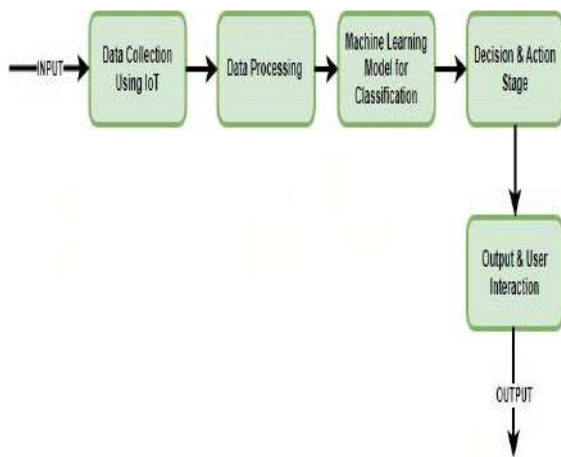


Fig: Block diagram of Artificial Intelligence based E-waste recovery system

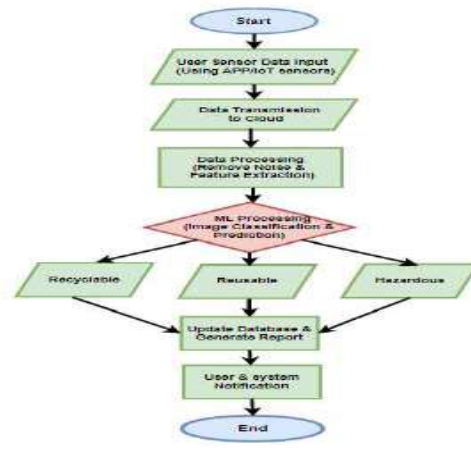


Fig: Flow chart of Artificial Intelligence based E-waste recovery system

- **Input Layer (Data Collection - IoT)**
Sensors & Smart, User Mobile App, GPS Tracker.
- **Data Processing Layer (Edge & Cloud Computing)**
Microcontroller / IoT Gateway, Data Transmission, Cloud Server / Database.
- **Machine Learning Layer (ML Model for Classification)**
Data Preprocessing, ML Model for E-Waste Classification, Prediction & Sorting.
- **Decision & Action Layer**
Automated Sorting & Segregation System, Alert Notifications for Waste Collection, Dashboard for Authorities & Users, Recycling & Disposal Recommendations.

VIII. RESEARCH METHODOLOGY TOOLS AND TECHNIQUES

Research Methodology:

1. Data Collection:

- **Survey and Field Data:** Collect data on e-waste quantities, types, and disposal methods from e-waste management companies or public databases. This could involve surveys and interviews with stakeholders such as waste management firms, recycling plants, or governmental agencies.
- **Sensor Data:** Use IoT devices or sensors to track e-waste materials' conditions, such as weight, temperature, or moisture content, at disposal sites.
- **Image Data:** Capture images of e-waste items for image recognition tasks to classify different categories of e-waste. This could involve using camera-equipped systems to take pictures of discarded electronic devices.[11]

2. Data Integration:

- **Data Cleaning:** Which able to easily manage missing values, outlines and incorrect entries. Cleanse data from various sources to ensure uniformity.
- **Normalization/Standardization:** Normalize data for numerical attributes (e.g., quantities, weight) to scale features.
- **Data Augmentation:** In the case of image data, apply techniques like rotation, flipping, cropping, or scaling to enhance the dataset size and variability.

3. Feature Engineering:

- **Image Features:** Extract the features from images by using different types of techniques such as Histogram of Oriented Gradients, Scale-Invariant Feature Transform, or deep learning-based feature extraction from pre-trained models like ResNet or VGG16.

- **Numerical and Categorical Features:** For survey or sensor data, extract meaningful features that could correlate with e-waste disposal patterns or recycling effectiveness (e.g., device age, brand, material type, geographical region).

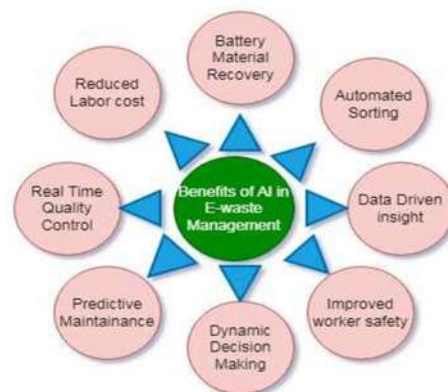
4. Machine Learning Algorithms:

- **Supervised Learning:**
 - **Classification:** Use algorithms like SVM (Support Vector Machine), Random Forest, Gradient Boosting, or Neural Networks for classifying types of e-waste based on images or sensor data.[6]
 - **Regression:** If predicting quantities or predicting outcomes like the quantities of e-waste generated in a particular region, Linear Regression, Decision Trees, or XGBoost regression algorithm also can be used.
- **Unsupervised Learning:**
 - **Clustering:** Use K-means, DBSCAN, or Hierarchical Clustering for clustering e-waste into different categories based on the features or for detecting anomalies in disposal patterns.
 - **Dimensionality Reduction:** We can apply techniques such as Principal Component Analysis or t-SNE for the reduction of dimensionality for the dataset, when working with high-dimensional data like images[9].
- **Deep Learning:**
 - **Convolutional Neural Networks (CNNs):** to understand the image-based tasks, CNNs will be effectively useful for classifying e-waste types or detecting defects.
 - **Transfer Learning:** Pre-trained models can be used if limited data set were available (e.g., ResNet, VGG) and refurbish them on your e-waste image data.

5. Evaluation Metrics:

- **Accuracy:** Accuracy, precision, recall, and F1-score models were using to evaluate the performance of the classification task.
- **Confusion Matrix:** To understand the performance of the model, especially dealing with multi-class classification problems

IX. BENEFITS OF AI/ML IN E-WASTE MANAGEMENT



X. CONCLUSION

Integrating AI and ML into e-waste management systems helps for enhancing the accuracy, sustainability, scalability in the existing system. This system can solve many of the inefficiencies, management and environmental challenges faced by traditional e-waste management systems, improving material recovery, reducing various types of pollution, and driving the adoption of a circular economy model. Thus, this research has necessity to change the existing e-waste handling system and operate effective, and efficiently for achieving environmentally friendly and economically viable solutions.

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XII. REFERENCES

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Efficacy & efficiency of fingerprint powder derived from natural flowers- development and evaluation of sustainable fingerprint powders.

¹Qurratulain Kausar Ansari, ¹Pratiksha Parihar, ¹Shrutika Chincholkar, ¹Karuna Dohare, ¹Mangesh S. Dhore.

¹Department of Forensic Science, G.H. Rasoni University, Anjangaon Bari Road, Amravati, Maharashtra 444701.
kausar.ansari717@gmail.com

Abstract:

Fingerprints are crucial element in forensic investigations. Latent or invisible fingerprints are commonly recovered from the scene of crime and hence required an attention for which visualization methods are needed. The conventional methods are relatively expensive which may cause harmful threats and health hazards due to presence of chemical content. Our study offers a cheap and sustainable fingerprint powder sourced from natural materials. In our study the powders were prepared from the dried petals of *Bahunia Purpurea*, *Bougainvillea*, *Amaranthus Cruentus*, *Hibiscus Rosa-sinensis*, and *Hibiscus Grandiflorus* by using mechanical method. The dried petals were grinded and kept in oven. The ash powder was also prepared by using muffle furnace. The grain size of prepared fingerprint powders was also determined. The durability of these novel powders was evaluated on non-porous surfaces including mobile screens, window glass, doors, keyboards, and glass bottles for the development and visualization of fingerprint impressions. The purpose of this research is to examine efficacy and durability of natural powders in investigation. Our research set a stage of discovering new, safer and more affordable fingerprint powder alternatives.

Keywords - Fingerprint powder, Latent Fingerprints.

Introduction

Fingerprints are unique patterns on an individual's fingertips used for identification. They are classified into three main types: Arches (5%), Loops (60%), and Whorls (35%). At a crime scene, forensic investigators detect fingerprints by using dusting methods with contrast-coloured powders that adhere to the prints. If they locate, can lift the prints with tape lifting method. While this can help investigators in determining whether a suspect was present at the crime scene. Modern fingerprint powders represent considerable health issue because they composed of chemical content. Traditional powders, such as fluorescent powders, Charcoal powder are more dangerous and can reach and settle deep into the lungs. Natural fingerprint powders made from plant-based compounds or minerals, provide an environmentally safe and reliable alternative than chemical powders. Fingerprints are most widely used and accepted evidence in the criminal justice system and are important trace evidence to disclose person's identity. The basic principles of fingerprints are uniqueness, permanence, universality, simplicity of recording and classification. Fingerprints unintentionally left by a culprit at the crime scene are known as chance prints. The fingerprints are classified as Patent, Plastic and Latent fingerprints. Latent fingerprints are most probably found at the crime scene. Fingerprint powders are used to visualize and enhance latent prints formed by sweat residue which are commonly found at crime scenes and can be developed using various powder dusting methods. Our study focuses on preparation of fingerprint powders from flowers like *Bauhinia Purpurea*, *Bougainvillea*, and *Hibiscus Rosa-sinensis*, *Hibiscus Grandiflorus*, *Amaranthus Cruentus*.

Methodology

Materials:

In our research, certain naturally available materials, such as *Bahunia Purpurea*, *Bougainvillea*, *Hibiscus Rosa-sinensis*, *Hibiscus Grandiflorus*, and *Amaranthus Cruentus*, were utilized for the preparation of fingerprint powder, which aided in the development of latent fingerprints. The effectiveness of the powders had been evaluated on non-porous surfaces, including phone screen, window glass, doors, keyboard, glass bottle yielded both positive and negative results. The samples were retrieved using the tape lifting method, as the fingerprints may have faded over time. The fingerprints were preserved by photography using a smartphone equipped with a 5MP closeup shot micro camera. The above flowers were collected at their anthesis stage. The petals were separated from the flowers by removing the pistil, sepal, and stamen. The petals were dried in sunlight for 2-3 days at 30°C

For the preparation of Colored powder:

The petals were subjected to an oven for about 45min at 150°C and projected to mechanical crushing, to obtain a fine powder. The powder was sieved using a tea strainer and flour sifter.

For the preparation of Ash powder:

Dried petals were heated in a muffle furnace at 400°C for 5 hours to create ash powder, which was then crushed and used along with colored powder to develop latent fingerprints. The both coloured and ash powders were sieved using mesh fabric and had found the grain size of the powders to be 4mm mesh size. The powders were applied to non-porous surfaces on which latent fingerprints were either intentionally placed or unintentionally present.

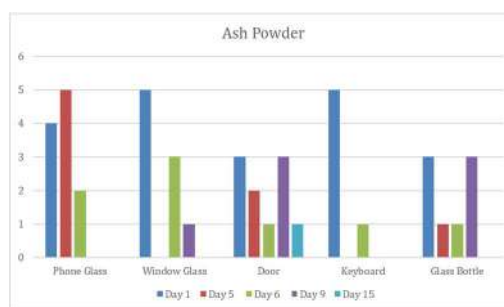
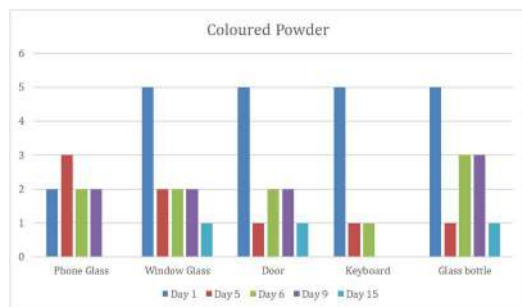
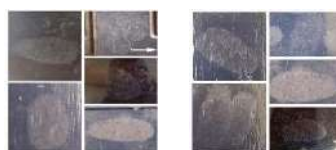


Gently tapped the brush to release the powder over the surface using a soft Ostrich feather brush and Camel hair brush in one direction to produce a clear fingerprint, ensured that the powder was evenly coated and avoided over-powdering. The powder had been allowed to adhere on the latent print.

The fingerprints were developed on non-porous surfaces and observed whether the fingerprint powders were capable to visualize the latent fingerprints even after intervals of 1 day, 5 days, 6 days, 9 days, and 15 days. For that, fingerprints were given randomly on different surfaces mentioned above without cleaning the surface. Capabilities of powders was measured in scale of 0 to 5 based on how well the fingerprints visualised. The value 5 was assigned for fingerprints that were crystal clear and had excellent visualising effect. A 4-value indicated very clear fingerprint, while value 3 meant the print was still good but not as sharp. If only one or two ridges were visible, value 2 was assigned. If no ridges could be seen, the value 1 was assigned and if the print was really poor quality and shows no visualizing effect, 0 value was given.

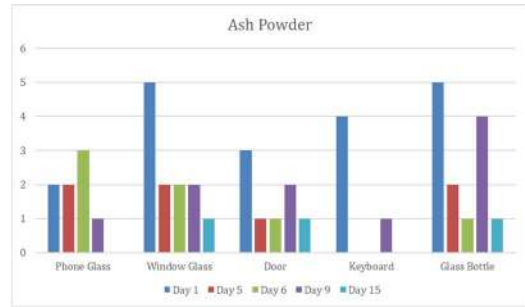
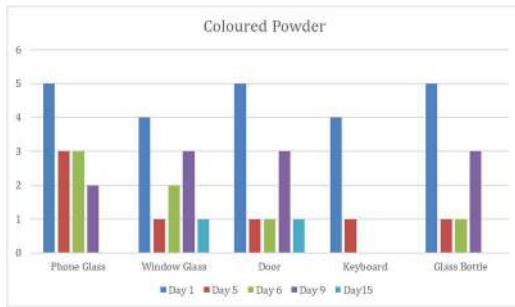
Observation

1. Bahunia Purpurea

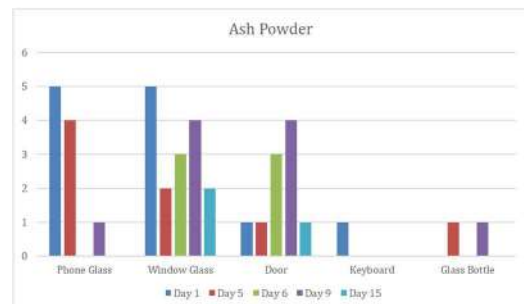
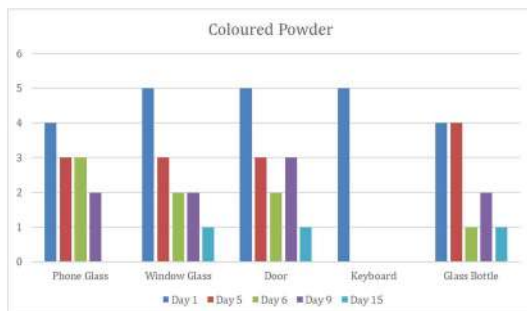


2. Bougainvillea

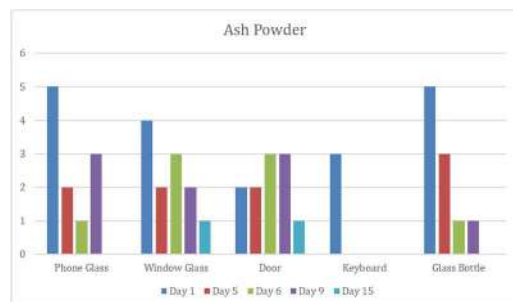
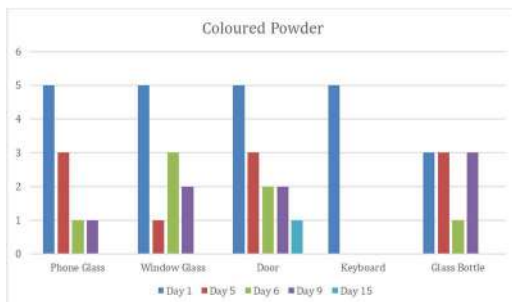




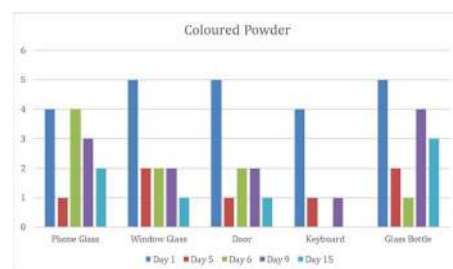
3. Hibiscus Rosa-sinensis



4. Hibiscus Grandiflorus

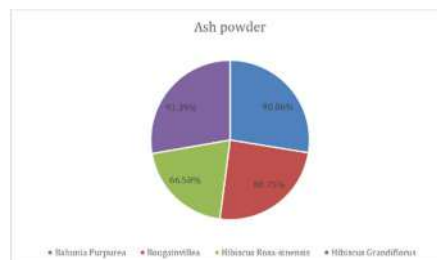
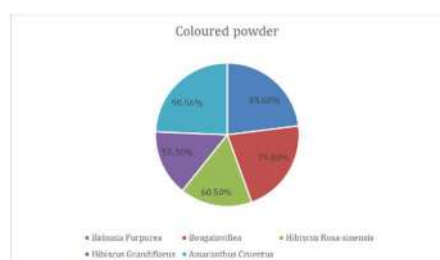


5. Amaranthus Cruentus



Result

Powders	Name of powder	Efficacy	Percentage[%]
Coloured powder	Bahunia Purpurea	All powders were efficient in window glass and door.	85.60%
	Bougainvillea		89.80%
	Hibiscus Rosa-sinesis		60.50%
	Hibiscus Grandiflorus		55.30%
	Amaranthus Cruentus		90.56%
Ash Powder	Bahunia Purpurea		90.46%
	Bougainvillea		80.75%
	Hibiscus Rosa-sinesis		66.58%
	Hibiscus Grandiflorus		81.39%



Discussion

In our present study the latent fingerprints were examined successfully on non-porous surfaces, such as phone screens, windows, doors, keyboards, and glass bottles using powders derived from Bahunia Purpurea, Bougainvillea, Hibiscus Rosa-sinensis, Hibiscus Grandiflorus, and Amaranthus Cruentus. The powders were given more efficient results on contrasting surfaces. Our study focused on surfaces where unintentional fingerprints are most commonly found, whereas R, I., Gupta, P., S, M., & R, D. research papers had examined only two surfaces, such as mobile screens and white ceramic plates^[1] While some research papers had used powders prepared from Gambir plants, which are found in Sri Lanka, Indonesia, and Malaysia but are rare in India^[2] In contrast, our experiment used flowering plants which are easily available in nearby premises.

We had referred the methodology of Seerat Vaibhav Saran, Lav Kesharwani, A.K. Gupta, and M.K. Mishra, they used a muffle furnace to carbonize petals at 550°C for 15-20 minutes^[3] However, we used a muffle furnace to prepare ash powders at 400°C for 5 hours. The primary objective of this research is to explore the use of naturally available resources for development latent fingerprint which minimise the use of chemically prepared hazardous powders. This technique is simple and does not require specialized training.

Electromagnetic sieve vibrators is an instrument which are commonly used to determine the grain size of very fine powders (micrometre), but our objective was to use conventional methods. We observed that some of the prepared fingerprint powders did not show a visualizing effect, while others were capable of enhancing fingerprint visibility on various surfaces. The size, adherence, and interaction of the powder with fingerprint residue are the important factors. Our research suggests that the effectiveness of naturally available resources on various surfaces needs to be evaluated. The colours of our prepared powder are different which enhances visibility of latent fingerprints on different surfaces where the fingerprints were commonly found at the crime scene. The result of our coloured powders were found to be more visible as compared to ash powders.

Conclusion

Our research shows that natural resources that are easy to find, safe to use, and non-toxic can work well for visualizing fingerprints on non-porous surfaces. Compared to chemical powders, it might in limited supply, these natural source fingerprint powders can be used for latent fingerprint development and visualization. If this method found to be trustworthy, it could help the criminal justice system solve crimes by using fingerprint identification.

Our research used powders prepared from the petals of Bahunia Purpurea, Bougainvillea, Hibiscus Rosa-sinensis, Hibiscus Grandiflorus, and Amaranthus Cruentus flowers to visualize latent fingerprints on non-porous(smooth) surfaces. To accomplish our objective, we determine the grain size of our powders consistent. We discovered that smaller grains help the powder to adhere to fingerprints. There are many ways to improve fingerprint powders made from natural materials. This includes choosing better materials, better formulation, finding new ways for application, and ensuring that they work well on different surfaces.

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Generative AI for VR: Creating Physically Realistic Models

Prof. Pranita Chaudhary¹,
Assistant Professor,
CSE (AI&ML),
PCCOE, Pune

Shushant Bhat²,
student, CSE
(AI&ML),
PCCOE, Pune

Aayush Bodkhe³,
student, CSE
(AI&ML), PCCOE,
Pune

Varun Bhilare⁴,
student, CSE
(AI&ML), PCCOE,
Pune

Abhishek Bichare⁵
student, CSE
(AI&ML), PCCOE,
Pune

1. ABSTRACT:

Virtual Reality has revolutionized the traditional learning system by creating an interactive and engaging environment. However, its ability to show precise real-world experiences is limited due to lack of physical realism. This study investigates the potential of Generative Adversarial Network (GAN) in creating physically realistic 3D models. The proposed system incorporates deep learning techniques along with physics-based constraints to enhance models' accuracy and usability. To achieve this, experiments were conducted on the ShapeNet dataset, with various preprocessing steps done to standardize the point cloud dataset. A multicomponent GAN architecture was used to generate the 3D model. It included a spatial transformer as well as a semantic segmentation network. The results demonstrated by the model included a good Intersection-over-Union (IoU) score and the dice coefficient over various categories. Comparative evaluation included much improvement in accuracy and realism than the conventional method. Moreover, the qualitative analysis showcases clear segmentation and improved interactivity. This highlights the system's potential in bridging the gap between the virtual and real-world dynamics. All these findings indicate that GAN-generated models can provide a more effective and indulging VR-based learning experience. The study concludes that including deep learning along with physics-based constraints offers a promising pathway for advanced educational applications and providing deep understanding and engagement among learners.

Keywords : Pointcloud, Generative AI, VR: Virtual Reality, physics simulation, GAN, VAE

2. Introduction:

2.1) Impact of VR on Education: VR is changing education by giving a practical environment for training that is expensive or impractical in the real-world scenario. This has given students the opportunity to practice in an interactive and controlled environment, be it medical students performing surgical techniques on virtual patients or manufacturing industries training workers on safety procedures in simulated hazardous environments. VR acts as a major impacting technology for developing confidence and competence in skills acquisition pertinent to medicals, engineering, and industrial safety.

2.2) Challenges in Realistic Modeling for VR Training: The success of VR training relies upon the composing of 3D models that are visually detailed and physically accurate. However, this is fraught with challenges, such as fidelity of textures, responsiveness to materials, and physical properties accurately represented to include elasticity and mass. Candidly inaccurate models disturb the immersion and frustrate successful training. The height of each resulting problem emphasizes the urgency under which developing better methods to increase model realism and interactivity should be considered.

2.3) Generative AI for physics integration: Physical Realism in 3D Models-General AI uses GANs, VAEs, and NeRFs techniques to invoke the virtual models, with assertions of giving enhanced visual realism to such models.

GANs: Generate highly detailed textures and shapes, such as realistic skin or organic materials, to better mesh the expected quality of VR models.

VAEs: They make it versatile so that the shape and size of objects like anatomical structures or industrial tools may be represented quite differently for robust model training.

Physics-informed: The model will constitute physical characteristics such as elasticity and flexibility, which guarantees that whenever virtual assets interact with one another, they behave in a life-like manner.

NeRFs: Based on real physics and a smart presentation of how light interacts with the surfaces to create realistic reflections and shadows of such surfaces for real-time training in safety-critical environments.

2.4) Research Objective:

2.4.1) The objective of this paper is to investigate how generative AI can create physically realistic models specifically for VR-based educational simulations.

2.4.2) the goal is to create methods that can produce 3D assets that not only look realistic but also exhibit actual physical characteristics like elasticity, texture, mass that are useful in training scenarios that involve hands-on practice.

2.4.3) Through solving these challenges, we expect to bring the gap between the VR and real world training scenarios closer, so that VR can be used as a realistic way to immerse into skill-based learning.

3. Literature Review

The fast evolving development in Generative AI, particularly, in the 3-D model generation, have created a way for creating virtual environments, with realistic interactive objects. Numerous studies have explored the application of generative models for producing excellent quality 3-D models; however one challenge still remains, to create models that are virtually realistic and capable of accurately integrating and interacting with the VR. Various models such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and more recent transformer-based approaches have become popular and are highly known for adapting high-quality images and 3-D objects. For example, the 3D-GAN introduced by Wu et al. was one of the first models to extend GANs to 3D space, generating objects in voxel form. While this showcased the practicality of 3-D generation, it did not demonstrate the voxel-based representation's limited resolution and fine detail, making it unsuitable for high-fidelity VR applications where detail is crucial [1]. Achlioptas et al improved on this by proposing GAN-based models that work with point clouds, significantly increasing resolution and detail. Point cloud representations enable intricate models essential for VR applications, but they often lack surface realism and texture, limiting their practicality in realistic immersive environments[2]. Similarly, to solve the above issues PointNet and PointNet++ architectures brought advancements in processing point clouds for 3D object generation, but these models also face challenges in generating high-fidelity textures necessary for VR [3]. Mesh-based generative models like Mesh-RCNN and AtlasNet produce surface meshes with improved surface detail, yet they are processor intensive and struggle to balance detail with realistic physical structures, making them less suitable for VR, which demands a balance of both visual fidelity and interactive responsiveness [4].

Generative AI has influenced VR applications across sectors like gaming, architecture visualization, and experiential learning, where realistic and varying objects are essential. In medical, VR training simulations have created 3-D models of human anatomy, allowing students to train the most intricate procedures and view complex biological structures. However there is a drawback that these models lack dynamic properties like mimicking tissue behaviour under stress [5]. Physics-based virtual reality in education simulations make use of physics engines to teach concepts of mechanics and fluid dynamics, even though they are based on fixed models that constrain different kinds of adaptivity and variety. Although VR applications in engineering and architecture are meant to simulate 3D layouts and structures, they lack interacting components and are therefore limited in their usefulness as visual demonstrations. Although combining generative modelling with physics engine shows promise it still has not passed experimental stages. In education, VR simulations show the ability to interact with objects in a realistic, physics based manner which is paramount to realistic experiences and effective learning. There are, however, still some remaining limitations of generative AI models for VR, specifically in the realms of realism and interactivity, which restrict their use within hands-on educational settings.

Some major limitations of generative models for VR applications persist, particularly within educational contexts. Current models are devoid of realistic physical properties, such as elasticity, rigidity, and collision responses, all of which are important for immersive educational simulations. However, due to the lack of computing power and complexity of surface representation in formats like voxels or point cloud, many generative models fail to produce fine details. In addition, models existing today do not sufficiently support user interaction capabilities in VR-like environments, where objects are meant to interact realistically with user inputs such as grabbing, rotating or applying force. Another limitation in common applications is the lack of generalization and diversity, as the current models often aim at generating single-object models, hence limiting them for use in multiple educational scenarios. Based on the above-mentioned gaps, our work integrates advancements in point cloud and GAN based models to make high-resolution and structurally valid 3D models optimal to be introduced in VR. Our study contributes to the growing body of literature that seeks to enhance generative AI's capabilities in VR by generating detailed models using scale methods. Our contributions include the generative model tailored toward high structure fidelity and surface detail in VR; a design approach that would allow further integration of physics-based interactions; paving a path for potential scalability to generate various adaptable to different content requirements.

A key opportunity for developing a solution tailored to the educational needs lies in the space created by limitations created of current generative AI models for VR applications. We build upon recent advancements in pointcloud and GAN-based models while aspiring to tackle educational VR simulations unique challenges. By focusing on generating high-resolution, structurally sound models that can integrate the properties of physics in future stages, our research contributes to a growing body of work seeking to expand the capabilities of generative AI in VR.

This study's contributions include:

- 1) A generative model customized to produce 3D models with high structural fidelity and accurate surface detail optimized for VR environments.
- 2) A design approach that allows for future integration of physics-based interactions, aiming toward applications in hands-on educational simulations.
- 3) Potential scalability in generating a variety of models that are adaptable to various educational content requirements.

4. Research Gap

Although generative AI has made instant progress in creating realistic 3D models, a serious gap still exists between the creation of models that are visually accurate and those that are physically realistic and interactive in virtual reality(VR) settings. Most of the existing methods including voxel-,mesh-,and point cloud-based approaches are more oriented toward geometry and appearance of essential physics-based constraints toward real-world simulation.

Thus, current generative techniques seldom deal with object part segmentation or any physical behavior-deformation ,elasticity, and motion under force - that should contribute significantly to training simulations in terms of effectiveness and immersion. Hence, applications that try to implement these models in VR never provide an experience that is truly interactive, responsive, and reflective of real-world interactions.

A generative framework capable of blending the advantages of deep learning with physics-informed modeling is clearly in demand for generating segmented, physically aware 3D objects suitable for educational simulation, this project tries to fulfill that gap by proposing a novel GAN model trained with point cloud data to generate interactive and physically plausible 3D shapes for immersive VR learning.

5. Proposed Methodology

5.1) Data Processing and Representation

For generation of the 3D model, we use the point cloud dataset from the ShapeNet-core dataset . This serves as the foundation for training the GAN.

Point Cloud Representation: Point clouds are 3d space data points that represent the surface of an object. Each point consists of three coordinates (x,y,z), which are useful in capturing detailed and complex surfaces

Segmentation: Each model is segmented meaning fully (say, car: handle, body, wheel). This allows depiction of part specific physical properties. This segmentation allows more realistic interaction in VR as each segment can respond to physical forces independently

5.2) Preprocessing:

1)*Normalization:* All data points are scaled and centered within a standardized 3D space, facilitating uniform model generation.

2)*Augmentation:* Various augmentation techniques like(random rotation, translation, scaling)are applied to increase the models generalization ability across various shapes and configurations

5.3) Model Architecture

Our model is based on GAN architecture, specifically designed to produce 3D point clouds. The model's architecture consists of a discriminator and a generator ,each designed to work together with the point cloud data .

Generator Network: The generator takes latent vector(z) as input. z represents the random noise, which is then transformed into a structured 3d point cloud representing the object. Key components of the generator are:

1)*Fully Connected Layers:* Initial layers are fully connected, mapping latent vectors to a higher-dimension.

Convolutional Layers: These layers apply convolutions on the 3D points, refining the models' shape and producing more structured point clouds.

2)*Output Layer with Segmentation:* The output layer generates the final point cloud. Each point is assigned to a label (material). This allows us to apply physics-based constraints to each part specifically.

Discriminator Network:The discriminator evaluates the realism of the models generated. It distinguishes between real and generated point clouds.

1)*PointNet-Based Layers:* The discriminator is built on PointNet architecture. This architecture captures both local and global features of the point cloud. It helps in identifying the fine details to check whether the generated object is real or not.

2)*Binary Classification Output:* The discriminator's output is a binary classification, indicating whether the input point cloud is authentic (from the dataset) or not.

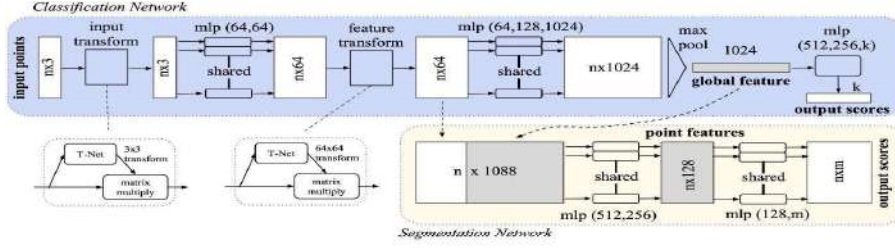


Fig 5.3.1

Physics-Informed Constraints: It includes the integration of physics-informed constraints into the GAN's loss function. This ensures that the generated model not only looks realistic but also behaves according to physical principles when placed in a VR environment.

5.4) Adversarial Loss: GAN loss function optimizes the generator to create point clouds that discriminator classifies as real or not. If $D(x)$ is the output for sample x and $G(z)$ is generator output for a noise vector z . Then the adversarial loss for the generator L_G and discriminator L_D are defined as:

$$L_D = -E_{x \sim p_{data}}[\log D(x)] - E_{z \sim p_z}[\log(1 - D(G(z)))]$$

$$L_G = -E_{z \sim p_z}[\log D(G(z))]$$

5.5) Physics-Informed Loss: This component applies constraints which ensures the generated object maintains structural and physical consistency under various forces applied. For example, we apply elasticity constraints that model how segments of the object respond to stress and strain, based on Hooke's law:

Where:

- σ is the stress on a point within a segment,
- E is the elasticity modulus, and
- ϵ is the strain (deformation) in response to applied forces.

$$\sigma = E \cdot \epsilon$$

The physics-informed loss $L_{physics}$ is calculated based on deviations from expected physical behavior, with penalties applied for unrealistic deformation under simulated forces. By including this loss, the generator learns to produce objects that not only appear structurally sound but also behave as expected under physical constraints in a VR setting.

5.6) Training and Optimization

Training the GAN involves iterative optimization of both the generator and discriminator, with additional tuning of the physics-informed constraints.

1) *Training Process:* We employ a two-stage training approach, where the model first learns basic shape generation and then fine-tunes with physics-informed constraints. The training uses a batch size of 32 and is optimized with the Adam optimizer, with learning rates of 0.0002 for both the generator and discriminator.

2) *Initial Pre Training for Visual Fidelity:* The training process begins with GANs and VAEs focused solely on generating visually realistic textures and shapes. GANs are pre trained to produce high-resolution, photorealistic textures by minimizing the adversarial loss (as outlined previously) until they can reliably create surfaces that visually resemble real-world materials. Simultaneously, VAEs are trained to map objects to a latent space that captures a range of plausible shapes. Here, the primary focus is on achieving a baseline visual quality that matches target images in terms of texture, color accuracy, and structural variation.

Metrics such as Structural Similarity Index (SSIM) and Inception Score (IS) are used to evaluate the visual quality of the generated models. These metrics guide adjustments to the learning rate, generator-discriminator balance, and VAE latent space regularization to optimize visual accuracy.

3) *Loss Combination:* The final loss for the generator combines adversarial and physics-informed losses:

$$L_{total} = \alpha L_G + \beta L_{physics}$$

where α and β are weights that balance the contributions of visual fidelity and physical realism. Through experimentation, we find that setting $\alpha = 0.8$ and $\beta = 0.2$ yields optimal results.

4) *Physics-Informed Training for Physical Realism*: After pretraining, physics-informed constraints are added using Physics-Informed Neural Networks (PINNs) and differentiable physics simulations. This enables the model to embed physical properties like elasticity and density, allowing backpropagation through simulated behaviors to guide realistic physical responses during training.

$$L_{\text{physics}} = \sum_i (F_{\text{simulated}}^{(i)} - F_{\text{real}}^{(i)})^2$$

6) Results and Analysis:

The outcomes of the proposed generative AI system for 3D model generation and semantic segmentation. The analysis includes performance metrics, statistical evaluations, detailed discussions, and graphical visualizations to provide a comprehensive understanding of the model's capabilities and limitations.

A) *Performance Measures*: To evaluate the model, several key performance metrics were employed:

B) *Dice Score (F1 Score)*: A similarity metric for evaluating segmentation accuracy:

$$\text{Dice Score} = \frac{2 \times |\text{Prediction} \cap \text{Ground Truth}|}{|\text{Prediction}| + |\text{Ground Truth}|}$$

Where P is the predicted set of points and G is the ground truth.

The Dice Score Curve shows performance fluctuating

between 0.78 and 0.88 across batches, with stable scores above 0.8 overall, indicating good segmentation accuracy. Early variability suggests the model's adaption, while a sharp recovery towards the end highlights improvements in optimization.

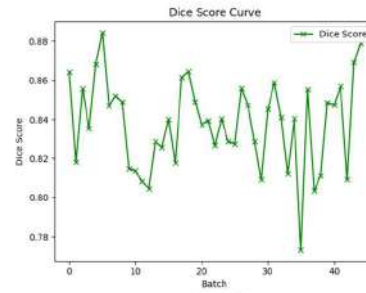


Fig 5.1

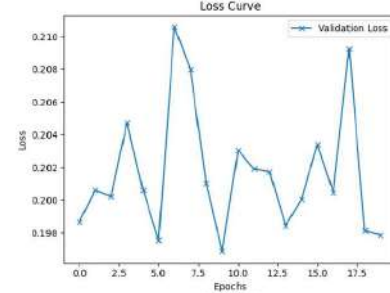


Fig 5.2

7. Conclusion

The findings indicate the capability of developing highly realistic, tailored 3D models for VR applications through the integration of generative AI with physics-informed training, with a particular emphasis on educational simulations. This paper represents a framework using techniques such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Neural Radiance Fields (NeRFs) to generate models with both visual realism and proper physical functionalities according to the laws of physics.

By incorporating physics-informed loss functions, this method effectively creates a nexus between visual realism and the accuracy of physics. Consequently, the VR models can portray realistic behaviours such as reaction to virtual forces, deformations, and light interactions. User evaluations within the VR demonstrated that the models offer a highly immersive, interactive experience that significantly increase the educational value of the simulations through their close correspondence to real-world materials and behaviours.

8. Key Contributions & Future scope

1) *Refined Physical Realism*: The incorporation of physics-informed loss functions allows generative models to simulate realistic mechanics such as elasticity and material compliance, which are particularly needed within VR simulations in education and training.

2) *High Visual realism through NeRFs*: The use of Neural Radiance Fields gave more refined lighting effects and surface detailing that went a long way in building immersive models.

3) *User-Centered Evaluation*: User testing in VR helped to demonstrate the effectiveness of the models in simulating real-world interactions, confirming that generative AI can support high-fidelity educational environments.

9. Future Directions

The framework has already shown its effectiveness to the standardized VR interactions, yet , it still deals with more complex forces and multi-object interactions. Nevertheless, looking forward and on the whole, there are certain areas that could serve as a driving force behind its capabilities:

1) *Physics in VR at a Higher level* -An AI agent that has been trained on ultra-precise physics models would overcome the errors in VR-caused by physics inaccuracies.

- 2) Computational Expertise-By the way, do you know if it is possible to optimize the designed system to become more scalable with the VR range of applications that it fits into?
- 3) Including Material Characteristics-By expanding the framework to include the simulation of other material behaviors (e.g. thermal conductivity and electrical properties), it gives the opportunity for more specific training simulations.

By the way, this research not only reports on the cutting-edge achievements of the necessary digital blanket but opens the perspective of generative AI as a big example of VR learning that will change the current state of electronics. The transformation of the training both visually and physically could open up future great innovations in this area that would reshape the training background in areas like medicine, engineering, and environmental sciences, making them more experimental, as well as effective.

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GREEN CHEMISTRY: A PATHWAY TOWARDS SUSTAINABLE DEVELOPMENT

ANIL KUMAR M, Assistant Professor, SOMS, GH Raisoni University, Amravati, Maharashtra anilnagpur2025@gmail.com,
anilconference007@gmail.com, 9633549225

ABSTRACT

A rapidly emerging subject called "green chemistry" offers optimism for reducing the negative effects of chemical processes on the environment. The idea is centred on creating chemical goods and procedures that reduce or completely do away with the production of dangerous materials. The concepts, uses, and importance of green chemistry in advancing environmentally friendly development are examined in this essay. It highlights how women have contributed to this profession, especially in the areas of environmental preservation and the use of green chemistry in daily life. The study also looks into the possible worldwide effects of green chemistry as well as its usefulness in home contexts. This research seeks to demonstrate the contribution of Green Chemistry to waste reduction, energy conservation, and resource efficiency through a thorough literature assessment. The results show that although green chemistry has promising answers for sustainable development, barriers to broad adoption still exist because of financial and technological limitations. However, incorporating Green Chemistry concepts into businesses, residences, and communities can make a big difference in sustainability and environmental preservation.

KEYWORDS: Sustainable Development, Hazards reduction, Green Chemistry, Sustainability, Hazardous Substances

INTRODUCTION

The design of chemical products and processes that minimise or completely do away with the use and production of hazardous materials is known as "green chemistry." To develop more economically viable and ecologically benign chemical solutions, Green Chemistry—first proposed by Paul Anastas and John Warner in the 1990s—has emerged as a crucial element of sustainable development. Concerns about climate change, environmental deterioration, and the loss of natural resources are driving the need for green chemistry. Conventional chemical processes frequently produce excessive waste and hazardous byproducts, endangering both the environment and human health. By emphasising energy efficiency, renewable resources, and reducing harmful substances, Green Chemistry provides a novel approach to a cleaner, greener future. From its fundamental ideas to its practical applications, green chemistry is a broad field of study. The groundwork for creating environmentally friendly chemical processes is laid by Anastas and Warner's twelve Green Chemistry principles. Waste reduction, atom economy, cleaner solvents, renewable feedstocks, and energy efficiency are some of these ideas. Researchers have looked at several Green Chemistry discoveries throughout the years, including ecologically friendly catalysts, biodegradable polymers, and green solvents. Researchers also highlight how Green Chemistry may help lower carbon emissions, especially by using renewable energy sources and creating energy-efficient procedures. It is often known that women had a significant part in the development of sustainable practices and green chemistry. Promoting ecologically friendly chemical processes and pushing for legislative changes have been spearheaded by female scientists, researchers, and campaigners. The automotive, pharmaceutical, and food packaging industries have all benefited from recent developments in green chemistry, where chemical processes have been optimised to have less environmental impact. Furthermore, as eco-friendly cleaning products and sustainable substitutes for household chemicals become more and more popular, there has been a growing emphasis on the usage of Green Chemistry in homes.

SCOPE OF THE STUDY

In the framework of sustainable development and environmental preservation, this study investigates green chemistry. With a particular emphasis on the achievements of women in this field, the scope covers both home and industrial applications of green chemistry. The paper analyses the possible advantages for environmental sustainability, looks at current trends in green chemistry, and evaluates the prospects and obstacles to its wider adoption. The findings will provide a roadmap for incorporating Green Chemistry principles into everyday life and industrial operations.

OBJECTIVES OF THE STUDY

1. To investigate how Green Chemistry can be used to advance sustainable development.
2. To emphasise how women have advanced Green Chemistry techniques.

3. To investigate how Green Chemistry might help reduce pollution in the environment.
4. To look into how Green Chemistry is used in homes and what advantages it offers.
5. To evaluate the obstacles and constraints to the broad use of green chemistry.

METHODOLOGY: THEORETICAL STUDY OF GREEN CHEMISTRY

The foundation of green chemistry is the belief that the adverse environmental effects of conventional chemical manufacturing should be reduced or eliminated in the design of chemical products and processes. The theoretical underpinnings of this discipline are the twelve principles of Green Chemistry, which were created by Anastas and Warner. These guidelines support energy efficiency, the use of renewable feedstocks, and waste minimisation. They stress how crucial it is to create chemicals that are not just efficient but also safe and biodegradable. The notion that the chemical industry ought to coexist peacefully with the environment rather than taking advantage of and damaging it is at the heart of the ideology. Several interdisciplinary disciplines, such as environmental science, engineering, and economics, are also incorporated into green chemistry. In addition to lessening the environmental impact of chemical processes, the objective is to develop economically and socially responsible processes. Therefore, a systems-based approach that takes into account social, economic, and environmental factors is required by the theory of green chemistry.

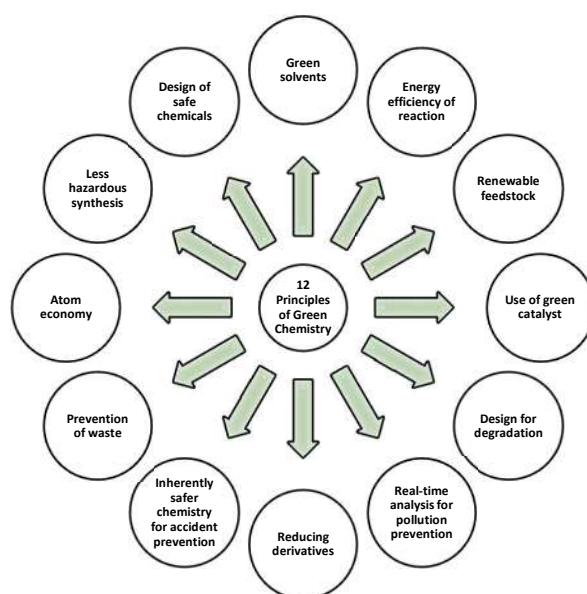


Diagram-1 12 Principles of Green Chemistry

The 12 Principles of Green Chemistry are as follows:

1. Prevention of waste

The first green chemistry concept, which is to reduce waste in the chemical process, is more applicable to the pharmaceutical business since the E factor of pharmaceutical manufacture falls between 25 kg and 100 kg, meaning that more waste is released into the environment during medication synthesis. Reducing waste is a more environmentally friendly method of production. In the fields of medicine and industrial production, the saying "prevention is better than cure" is more appropriate. Recent developments in biotechnology, molecular biology, and nanotechnology allow us to rethink drug manufacture and more efficiently build pharmacological molecules to exhibit the intended activity.

2. Atom economy

The second principle of green chemistry, the atom economy, requires a green chemist to employ as few chemical reactants as possible to create the result. conversion efficiency of a chemical reaction for each of the constituent atoms. For a reaction that differs from the reaction yield, atom economy can be computed. When c is the intended product, the atom economy of a chemical reaction $A+B \rightarrow C+D$ can be demonstrated as follows: $\text{Mass of C} / \text{mass of C+D} = \text{Atom economics}$

3. Less hazardous synthesis

Synthetic processes should, if feasible, be created to employ and produce materials that are minimally or completely hazardous to the environment and human health.

4. Design of safe chemicals

Products made using chemicals should be made to fulfil their intended purpose with the least amount of toxicity possible. utilising less hazardous chemicals as a catalyst is superior to utilising hazardous metal catalysts like mercury (Hg) and palladium (Pd) in a typical process. By using molecular design, it is possible to eliminate a harmful portion of a medicine molecule without compromising its therapeutic properties. One recent development in the field of drug design is computer-aided drug design.

5. Green solvents

The use of green solvents in chemical reactions that use less hazardous solvents rather than more hazardous ones, less flammable solvents rather than more flammable ones, and solvents that are easier to remove after a reaction and recyclable are all outlined in the fifth principle of green chemistry. Green solvents are essential to the extraction of bioactive compounds from natural sources using green solvent extraction procedures.

6. Energy efficiency of reaction

The greenness of the process can be determined by the reduction of energy consumed in the process using other alternate processes to complete the desired reaction

7. Renewable feedstock

Renewable feedstock used for a chemical process provides no depletion of natural resources.

8. Reducing derivatives

Natural resources are not depleted when chemical processes employ renewable feedstock.

9. Use of green catalyst

Chemists can employ a solid catalyst in a liquid-phase reaction thanks to heterogeneous catalyst development. Using a heterogeneous catalyst has the benefit of being simple to remove and reuse, which helps minimise process waste. In addition to improving chemical reactions, the employment of biocatalysts and enzyme catalysis results in a green synthesis. Designing a biocatalyst that is necessary for the chemical process is made possible by the "directed evolution of enzymes." Direct enzyme evolution is a novel molecular biology technique that uses reverse engineering of the protein synthesis process to create the desired enzyme. The fields of molecular biology and medicine are undergoing a revolution.

10. Design for degradation

Biopolymers are the finest illustration of the idea that chemical goods must break down in the soil after usage. Biopolymers can be broken down by microbes in the soil, however plastic is not biodegradable.

11. Real-time analysis for pollution prevention

Real-time checks, which employ sophisticated analytical techniques that recommend in-process monitoring and control for the earlier prevention of pollution through hazardous compounds formed in the process, can be used to analyse the synthesis of a chemical in real-time. Chemical products' ecological credentials are improved by biodegradability and real-time pollution avoidance.

12. Inherently safer chemistry for accident prevention

Using chemicals that reduce the possibility of mishaps and explosions is advised for the chemical process under the research

To create safer and more sustainable chemical processes, the multidisciplinary area of "green chemistry" combines chemistry, biotechnology, ecology, and toxicology. By employing solvent-free extractions, catalyst-based synthesis, green solvents, and energy-efficient industrial production techniques, it aims to lessen its negative effects on the environment. A chemical synthesis's sustainability can be evaluated both qualitatively and quantitatively. For example, producing styrene with xylene rather than the carcinogenic benzene makes the process safer and more economical. The several quantitative methods used to assess how environmentally friendly a chemical process is are known as "green metrics." Process Mass Intensity (PMI) (5) and E-factor are two examples of these green measures. The PMI shows how much material is used in a process for every mass of product. When the PMI is 1, it means that all of the reagents have been transformed into products.

Roger Sheldon developed the E-Factor, which compares the mass of waste to the mass of the product and has an ideal value of 0 to assess how environmentally friendly a chemical process is. Green chemistry techniques can be used to produce lower E-factors, which are often between 25 and 100 in pharmaceutical manufacture. With applications in materials, bioprocesses, energy, industrial processes, and sustainability policies, green chemistry

is a vast field. Industrial chemistry has made life better for people, but it has also harmed the environment because of harmful organic and inorganic materials. By emphasising safer synthesis techniques, recyclable materials, and energy-efficient production, green chemistry advances sustainable development. Waste reduction, safer chemicals, biodegradable materials, renewable resources, energy efficiency, catalytic processes, and the adoption of these concepts are all supported by international organisations. Addressing circularity and creating long-term sustainable solutions require the use of life-cycle thinking. Due to the shortcomings of traditional life-cycle assessment (LCA) techniques, hybrid approaches are being used to increase accuracy. Researchers examine the environmental effects of generating butyl acetate and titanium dioxide nanoparticles, as well as hybrid life cycle assessment (LCA) techniques in biomass-based p-xylene production. Additionally, agrivoltaics—the conversion of irrigated farmland in water-stressed areas to rain-fed crops and solar harvesting—is proposed using life cycle assessment (LCA). In response to the increasing climate catastrophe and the scarcity of resources, the Green Chemistry community adopts life-cycle thinking.

Efficiency in organic synthesis emphasises both mechanical soundness and aesthetic appeal, combining Renaissance inventiveness with Enlightenment rationalism. Environmental factors encourage fresh concepts and approaches rather than stifling curiosity. The future of chemistry is shaped by these developments, which propel advances in photochemistry, solvation theory, physical and mechanical impacts, and reaction development. As a result of the climate issue and the scarcity of resources, chemistry—especially organic synthesis—is moving towards green chemistry. Nowadays, it is believed that resource and environmental impact efficiency, which combines realism with curiosity, is what propels creativity and innovation. By combining novel concepts, including rethinking solvation and incorporating physical phenomena, researchers are pushing the envelope and making ground-breaking and more sustainable discoveries. Multidisciplinary strategies that emphasise the effective conversion of renewable resources and CO₂ capture are necessary to lessen dependency on fossil fuels for the manufacture of chemicals and fuel. New materials such as light-responsive metal-organic frameworks (LMOFs) for CO₂ separation and bioenergy with carbon capture (BECCS) are examples of recent developments that demonstrate advancements in carbon capture and storage (CCS). Innovations in biomass transformation, such as the use of biocatalysts for cellulose-derived compounds and photochemical techniques for lignin defunctionalization, are opening the door to more sustainable chemical synthesis in the realm of renewable raw materials. While sustainability evaluations of Green Chemistry measures are critical for reducing waste and energy consumption, organic solvents are necessary for extraction and purification. In order to increase efficiency and lessen their impact on the environment, advances in catalysis and reaction engineering concentrate on the use of less hazardous reagents, metals that are prevalent on Earth, and substitute solvents. The toxicological hazards of chemicals are also being addressed by interdisciplinary research, which is also advancing biodegradable polymers and creating sustainable techniques for the removal of microplastics. Membranes made of nanotechnology are energy-efficient and environmentally friendly, and they can be used in cleanup, energy harvesting, water purification, and environmental sensing. Selective membranes and reduced graphene oxide membranes with little nano-wrinkle impact are used in high-efficiency desalination methods. In addition to desalination, it is essential to break down persistent organic contaminants using cutting-edge techniques like coral-like nanostructure photoelectrodes that can produce energy. Water treatment is also being investigated using carbo-catalysis based on carbon nanotubes and persulfate oxidation.

With an emphasis on plastics, batteries, and wastewater treatment, research on sustainable resource recovery from trash is growing. Researchers are investigating techniques such as phosphate extraction from wastewater, lithium recovery from acid-free batteries, and depolymerisation of polymers. Additional initiatives include employing specialised enzymes to turn carbon dioxide into bicarbonate and recovering nitrogen from sewage. The depolymerisation of plastics, the creation of safer chemicals, the use of life-cycle analysis, and the innovative application of machine learning must be the main areas of future study in the development of the circular economy. Authors ought to think about the sources of their materials, look into bio-based substitutes, and figure out how to use less rare elements. Research should also evaluate how a procedure affects a chemical's life cycle, reversibility, byproduct value, and possibility for recycling or reuse. The goals of green synthesis (GS) include resource efficiency, waste reduction, and environmentally beneficial material production. By reducing manufacturing costs, encouraging recycling, and reducing industrial emissions—particularly in sectors like electronics—it promotes sustainability. In order to combat issues including toxicity, resource depletion, and climate change, green chemistry (GC) seeks to decrease the production of hazardous chemicals. In order to safeguard the environment and public health, GC advocates for safe reagents, catalysts, and renewable resources. In order to ensure progress without endangering future generations, sustainability blends economic, social, and environmental considerations. Implementing GS and GC reduces pollution, conserves resources, and supports sustainable industrial practices for long-term ecological benefits. Green chemistry (GC) will be essential to many study areas in the future, guaranteeing a balance between the environment and manufactured commodities. Greener techniques are necessary because upsetting this equilibrium could have disastrous repercussions. Developing noncovalent derivatisation and biometric multifunctional reagents, as well as

substituting hazardous heavy metal catalysts with safer ones, are the main focusses of future GC trends. While combinatorial GC makes it possible to produce a wide variety of chemicals quickly on a small scale, supramolecular chemistry seeks to produce reactions without the need of solvents. Enhancing solventless processes will also result in better product separation, isolation, and refinement with less solvent usage.

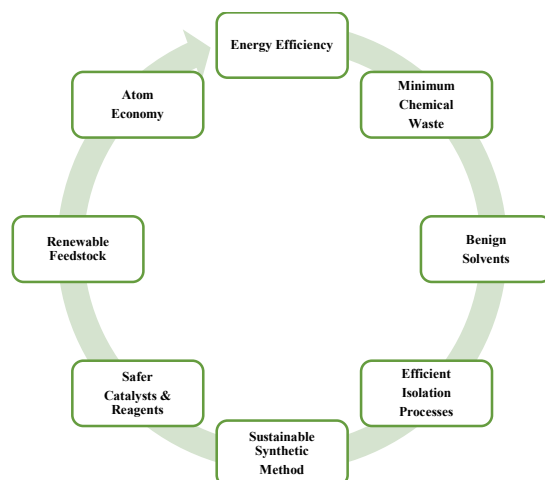


Diagram-2 Green Organic Chemistry

The diagram outlines the fundamentals of green chemistry, which aims to develop environmentally friendly chemical processes. It places a strong emphasis on energy efficiency by lowering the amount of energy used during production. To prevent pollution of the environment, the minimum chemical waste principle emphasises the reduction of by-products. The use of benign solvents encourages the use of less hazardous, safer, and ecologically friendly substitutes. Effective isolation procedures minimise waste and guarantee resource conservation. The goal of a sustainable synthetic approach is to create environmentally friendly production methods. Less dangerous alternatives to hazardous materials are substituted by safer catalysts and reagents. Reliance on sustainable raw materials is encouraged by the use of renewable feedstock. Finally, atom economy seeks to minimise waste by making the most use of all reactant atoms in the finished product.

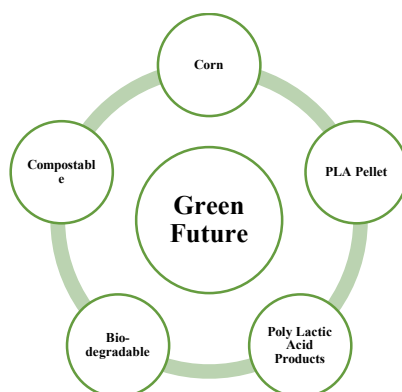


Diagram-3 Green Future

The lifecycle of PLA (polylactic acid) products is depicted in the diagram within the framework of a sustainable "Green Future." The cycle starts with maize, a renewable feedstock used to make PLA pellets. Following processing, these pellets are turned into a variety of PLA goods, including packaging and biodegradable cutlery. These products decompose into compostable material under the right circumstances after being thrown away. The cycle can subsequently be completed by using the decomposed material to encourage plant development. By minimising dependency on fossil fuels and fostering a circular economy for sustainable development, this technique offers an environmentally benign substitute for conventional plastics.

FINDINGS FROM THE STUDY

The study reveals several key findings:

1. **Environmental Protection:** Green chemistry has shown great promise in cutting emissions and hazardous waste. Many businesses have created processes that use renewable materials and green solvents, which reduce pollution and conserve resources.

2. **Energy Efficiency:** Energy efficiency is a key component of many Green Chemistry procedures, such as reactions that take place at lower pressures or temperatures, which minimise the energy needed to produce chemicals.
3. **Household Applications:** Green chemistry is used in non-toxic, biodegradable, and environmentally friendly cleaning products used in homes. This change results in less pollution of the environment and healthier houses.
4. **Women's Contribution:** Women have been instrumental in promoting Green Chemistry methods and accelerating the use of eco-friendly technologies. The creation of sustainable substitutes for conventional chemical processes has benefited greatly from the contributions of female scientists.
5. **Challenges:** The high expense of new technology, industry opposition to change, and a lack of consumer knowledge about the advantages of eco-friendly products are the primary obstacles to the broad adoption of green chemistry.

CONCLUSIONS AND RECOMMENDATIONS OF THE STUDY

One effective weapon in the battle against climate change and environmental deterioration is green chemistry. It provides doable ways to use renewable resources, cut down on waste, and save energy. The potential for major environmental advantages grows as firms start implementing Green Chemistry principles. Women's participation in the sector also emphasises how crucial diversity is to scientific creativity and research.

The recommendations of the study are: To make Green Chemistry technologies more affordable and accessible, governments and businesses should invest in their research and development. The significance of Green Chemistry and its contribution to sustainable development should be emphasized in educational initiatives. Additionally, more encouragement should be given to women to pursue careers in science and technology, especially in sectors focused on sustainability and the environment. Public education efforts should also be increased to inform people about the benefits of green chemistry in daily life, particularly in promoting eco-friendly household habits.

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AUTOMATED CAR LICENSE PLATE DETECTION AND RECOGNITION USING DEEP LEARNING

Prachi Pravin Durge

Computer Science And Engineering
G H Rasoni University, Amravati
prachidurge349@gmail.com

Prof. Suraj V. Dhole

Computer Science And Engineering
G H Rasoni University, Amravati
suraj.dhole@ghrua.edu.in

ABSTRACT

Using Deep Learning for Automated Detection and Recognition of License Plates. The use of ALPR systems automating processes such as number plate detection is gaining popularity in traffic control, security, and other law enforcement fields. This research focuses on achieving more accurate and efficient detection and recognition of number plates leveraging deep learning concepts.

The systems outlined in this study are aimed at improving the effectiveness of ALPR systems using advanced convolutional neural networks (CNNs) and other notable object detection algorithms in spotlighted and difficult scenarios. The methodology is based on a dataset containing images of different number plates, which undergo a variety of processes through multiple models, requiring adjustments implemented further down the line.

Moreover, the real-time detection and recognition frameworks show a balanced trade-off between speed and accuracy, maintaining a consistent recognition rate amidst complex and changing lighting conditions and different plate designs. The system is shown to outperform previous works in detection accuracy and processing speed through rigorous testing evaluations. This work is positioned at the heartbeat of the progress toward reliable, efficient automated number plate recognition systems with applications in

automated traffic control enforcement systems.

INTRODUCTION

Automated License Plate Detection and Recognition (ALPR) has emerged as a strapping technology, with adoption growing across several industries including law enforcement, traffic control management and access control systems. ANPR systems offer considerable advantages such as improved security, automated toll collection, and streamlined traffic monitoring by interposing vehicle number plate identification and reading. Traditionally ALPR systems depended on simple machine learning algorithms along with methodical image processing techniques. These systems, however, faced shortcomings in handling complex scenarios, such as lighting changes, varied designs of the plates, and occluded or distorted plates.

The deep learning paradigm shift has marked a point of transformation in terms of challenges related to computer vision and image processing. Recognition of images through machines is primary work carried out in computer vision which does identify real world objects. With the help of deep learning model, and especially with the use of Convolutional Neural Networks (CNN), they have proven to excel in carrying out image recognition and tasks because of their feature learning to hierarchical structures. This has greatly

improved the reliability and accuracy standards of the ALPR systems. Moreover, further developments in object detection frameworks and OCR techniques have immeasurably advanced the functionality of ANPR systems enabling real-time detection, recognition and tracking.

A. Objectives of The Project :

The focus objectives of this project include the design and development of the Automated License Plate Recognition system (ALPR) which is known to recognize, read, and capture vehicle registration plates for different uses such as automated traffic control, and law enforcement. Through the development of this system, we aim to achieve accurate counting and recognition of License plates leveraging automated image processing techniques to facilitate the toll collection, monitor traffic flow in and out of the city, and for security purposes. Furthermore, this project intends to maintain a high recognition rate with minimal processing time in real world situations.

B. The key of this project include:

- Applying image processing and object detection algorithms to find a license plate patch and setting the boundaries.
- Character recognition using deep learning methods for precise identification.
- A fast processing time and high accuracy level solution under various weather and lighting conditions.
- Ease of integration with CCTV systems, toll machines, and smart parking structures.

LITERATURE REVIEW & RELATED WORK

The application of Automated Number Plate Recognition or ALPR is now crucial in contemporary traffic control, security, and even law enforcement. The cornerstone

methods of ALPR resistant on image processing methods like edge detection, template matching, morphological operations, and others are usually not effective in real-life situations. Real-world scenarios pose challenges like lighting changes, orientation shifts, and obstructions that profoundly affect accuracy. Fortunately, the recent developments in deep learning technologies have offered more reliable and accurate solutions for detection and recognition, significantly enhancing the performance of both the detection of the number plates and recognition of characters.

Analysis of Problem :

Traffic management, security and law enforcement uses of Automated License Plate Recognition (ALPR) systems are widespread. Even with a lot of technological advances, there are still a lot of problems that limit accuracy and efficiency of these systems. The core issues of the existing ALPR systems are analyzed in detail and a number of major problem areas are revealed, essentially in dealing with the real world complexities of environmental variability, plate diversity and system scalability.

METHODOLOGY

Traditional image processing techniques or even deep learning models struggle to accurately detect and recognize number plates when the lighting conditions are poor (e.g., at night or in overcast weather), there are strong shadows or glare from vehicle headlights. Recent systems are trained in controlled environments and suffer from poor performance in the presence of such extreme variation in lighting.

1. Plate Orientation and Camera Angles: The number plate may not

be captured at a front facing angle. Skewed, rotated or angled plates may occur due to variations in vehicle positioning or movement. The detection process is further complicated by these geometric distortions, since many systems rely on the assumption that plates will be almost parallel to the image plane.

2. Number plate detection from live video feeds or still images in real time is a very complex task with many technical challenges involved, which need to be resolved real time with acceptable accuracy and speed. In high traffic areas or with large scale applications, the performance requirements may be difficult to be met by systems. To handle the real time constraints, NNs like the other advanced machine learning models must be optimized for speed as well as accuracy.
3. Real-time Detection and Recognition: The real-time detection and recognition becomes increasingly complex as the advanced machine learning models, such as Neural Network (NNs), need to be optimized to have a speed and accuracy balance. Model pruning, quantization and using hardware accelerators (e.g. GPUs, TPUs) can reduce processing time by orders of magnitude without affecting detection accuracy. In addition, the system can meet real time constraint demands by using real time processing frameworks and parallel computing.
4. Plate Occlusion and Damaged License Plates: It is common for the number plate to be partially occluded by objects (e.g., dirt, mud,

other vehicles) or damaged through wear and tear, reducing the number plate recognition to an incomplete or inaccurate plate. Such obstructions may impede the system from identifying the full License plate or reading all the characters correctly. We can improve the system by adding inpainting techniques for restoration of the occluded parts of the number plate. In addition, training models on partially occluded and damaged plates in the dataset proves helpful to make the models robust. Recognition accuracy in challenging scenarios can also be enhanced using advanced computer vision techniques like generative models or super resolution algorithms.

1. System Overview

The proposed ALPR system consists of four main stages:

- **Image Acquisition**
- **License Plate Detection**
- **Character Segmentation**
- **Character Recognition**

Each stage utilizes deep learning techniques to improve accuracy and robustness under varying conditions such as lighting, angle, and occlusions.

2. Data Collection and Preprocessing

- **Dataset:** We use public datasets (e.g., OpenALPR, CCPD, private datasets) covering diverse conditions (day/night, multiple angles, weather conditions).
- **Annotation:** Bounding boxes for license plates and labels for individual characters are annotated

manually or using semi-automated tools.

- **Preprocessing:**
 - Resize images to a fixed size (e.g., 512×512 pixels).
 - Normalize pixel values.
 - Data augmentation: random brightness, rotation, blurring, occlusion simulation.

3. License Plate Detection

- **Model:** YOLOv8 / Faster R-CNN / SSD (depending on application needs — speed vs accuracy tradeoff).
- **Training:**
 - Input: Full vehicle images.
 - Output: Bounding box coordinates for the license plate.
 - Loss: Intersection over Union (IoU) loss + Classification loss.

4. License Plate Cropping and Alignment

- After detection, plates are cropped from the image.
- Optional: Apply perspective transformation to correct for skewed plates using homography estimation.

5. Character Segmentation

- **Methods:**
 - Traditional methods (morphological operations, connected components) or
 - **Deep Learning methods** (like U-Net for segmentation masks).
- **Goal:** Isolate each character from the license plate image accurately.

6. Character Recognition

- **Model:** CNN-based classifier (e.g., ResNet, DenseNet) OR sequence models (CRNN, Transformers).
- **Training:**
 - Input: Cropped characters (or full license plate for sequence models).
 - Output: Character class (A-Z, 0-9).
- **Loss:** Cross-entropy loss for classification.

7. Post-Processing

- Merge recognized characters into a full license plate string.
- Apply regex patterns based on country-specific plate formats to correct minor recognition errors.

8. Evaluation Metrics

- **Detection:**
 - mAP (mean Average Precision) at different IoU thresholds.
- **Recognition:**
 - Character accuracy.
 - Plate accuracy (all characters correct).
- **System:**
 - Overall plate recognition rate.

9. Deployment

- **Framework:** TensorFlow, PyTorch, or ONNX for real-time inference.
- **Optimization:**

Quantization or pruning for lightweight deployment.

Software Development Phases:

- **Hardware Requirements**

Core I3/ I5/ I7 processor At least 4 GB RAM At least 20 GB of Usable Hard Disk Space

- **Software Requirements**

Window 10 above, VsCode IDE, SQLite Database, RestAPI, Chrome Browser, Django Backend Framework.

- **Frontend Technology :** HTML, CSS, JavaScript, Bootstrap version 5.

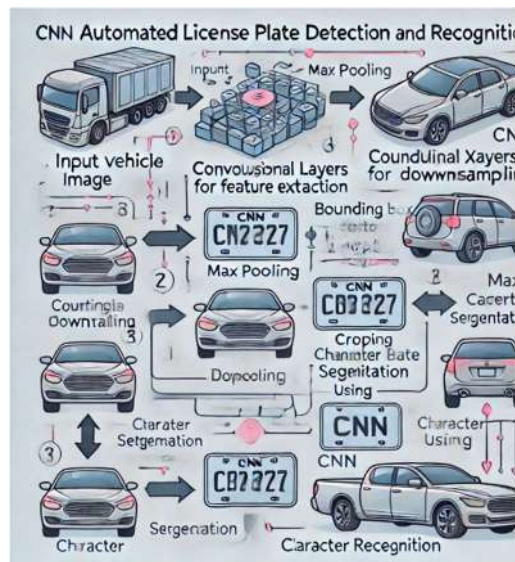


Fig. 1 Flowchart Diagram of CNN in ANPR System

- **1.1 Backend Processing:** The backend is built using Python and Django, acting as the bridge between the UI and the AI model. Text inputs are processed using NLP techniques to ensure accurate interpretation across the three languages. Django handles HTTP requests, manages sessions, and routes data between the user interface and the image generation model.
- TensorFlow / PyTorch: For building and training the deep learning models.

- OpenCV: For image preprocessing, object detection, and general image manipulation.
- YOLO (You Only Look Once): For object detection in real-time (can be integrated with OpenCV).
- Scikit-learn: For classical machine learning algorithms and preprocessing.
- RESTful APIs: For communication between the frontend and backend.
- SQLITE
- Django: Django A more feature-rich Python web framework for building scalable web applications.

- **Model Architecture**

- YOLO (You Only Look Once): For real-time object detection to locate number plates in images or videos.

1.2 User Interface and Experience Design

- **Design Principles:** Apply user-centered design principles to create an intuitive and engaging interface.
- **Prototype Testing:** Conduct usability testing to refine UI elements and ensure a positive user experience.
- **Feedback Integration:** Incorporate user feedback into UX design to enhance overall satisfaction and usability.
- **Accessibility Features:** Ensure the system is accessible to users with diverse needs, including those with disabilities.
- **Encryption:** Implement encryption protocols for data transmission and storage to protect user data from unauthorized access.
- **Access Controls:** Establish access control mechanisms to ensure that only authorized personnel can access sensitive information.

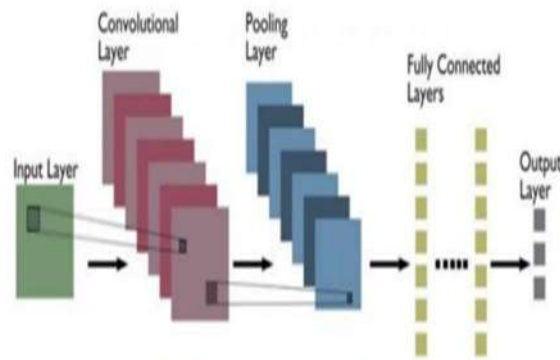


Fig 3.3.1: Typical architecture of CNN

Fig. 3.2 Typical Architecture of CNN

CONCLUSION

This project aimed to develop an efficient and reliable **Automated Number Plate Detection and Recognition (ANPR)** system using deep learning and computer vision techniques. By integrating state-of-the-art models for object detection, image processing, and optical character recognition (OCR), the system is capable of accurately detecting and recognizing number plates from vehicle images and video streams. The real-time performance, accuracy, and adaptability of the system demonstrate its potential for practical applications in traffic monitoring, law enforcement, toll management, and parking systems.

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Evaluating the Impact of Training and Development on Employee Performance and Organizational Success

Shital Dnyaneshwar Mohod^{1,*}, Dr Sheetal Jaikar²

Abstract

This study looks at how employee performance and organizational success are affected by training and development, with an emphasis on assessing how well-performing current training initiatives are. The study determines the elements that affect these programs' effectiveness and how they help the organization reach its goals by examining different training approaches. To understand their role in optimizing training outcomes, important elements are examined, including employee engagement, delivery techniques, content relevancy, and connection with corporate goals. The report also emphasizes how crucial corporate culture is to the accomplishment of training programs. In order to help organizations, improve their training methods and attain better performance outcomes, the research attempts to offer a thorough frame work for assessing the effects of training programs.

Keywords: *Training and Development, Employee Performance, Organizational Success, Training Programs, Employee Engagement, Organizational Culture, Skill Enhancement*

INTRODUCTION

Training and development are crucial for any organization's growth and success. They not only help employees grow individually but also improve the overall performance of the company. In today's fast-paced and competitive business world, companies that focus on continuously improving the skills and knowledge of their workforce tend to see better productivity, higher efficiency, and increased job satisfaction. By offering employees the right tools, resources, and training, organizations can create a culture that fosters innovation, teamwork, and high performance.

At the core of this connection is the understanding that employees are an organization's most valuable asset. A well trained and developed workforce is more capable of tackling the challenges of today's workplace, adapting to new technologies, and helping the organization achieve its long-term goals. Whether it is through formal training sessions, hands-on workshops, or continuous professional development, investing in employees' growth not only enhances their abilities but also boosts their commitment and loyalty to the company. When employees can see the positive impact of their training on their work, they are more likely to stay motivated, engaged, and dedicated, which ultimately benefits the organization as a whole. The benefits of training and development go beyond individual employees. Organizations that invest in their people tend to see improvements in key areas like operational efficiency, customer service, and workplace morale. lead to greater profitability, lower employee turnover, and a more competitive position in the market. Training and development allow organizations to stay agile and adaptable to changing market conditions. In this way, these initiatives become more than just a tool for individual growth—they're a strategic approach that contributes directly to the overall success and sustainability of the business.

To fully understand how effective training and development programs are, it's important to evaluate their impact on both employee performance and organizational success. This requires looking at key performance indicators (KPIs), gathering feedback from employees, and assessing whether the skills they've gained align with their job roles and contribute to the company's objectives. A comprehensive evaluation process helps organizations refine their training strategies, ensuring that both employees and the business are set up for future success.

Enhance workers' abilities, expertise, and output. The study highlights how training goals should be in line with organizational objectives in order to foster a culture of ongoing learning, which is crucial for obtaining a competitive edge.

UNITS OF MEASUREMENT

This study follows internationally accepted rules and conventions. It follow international system of units

(SI). We have collected the primary data collected through Questionnaires, and analyzed it using the following units:

Number of literature (n) – n specify the overall sum of literature review.

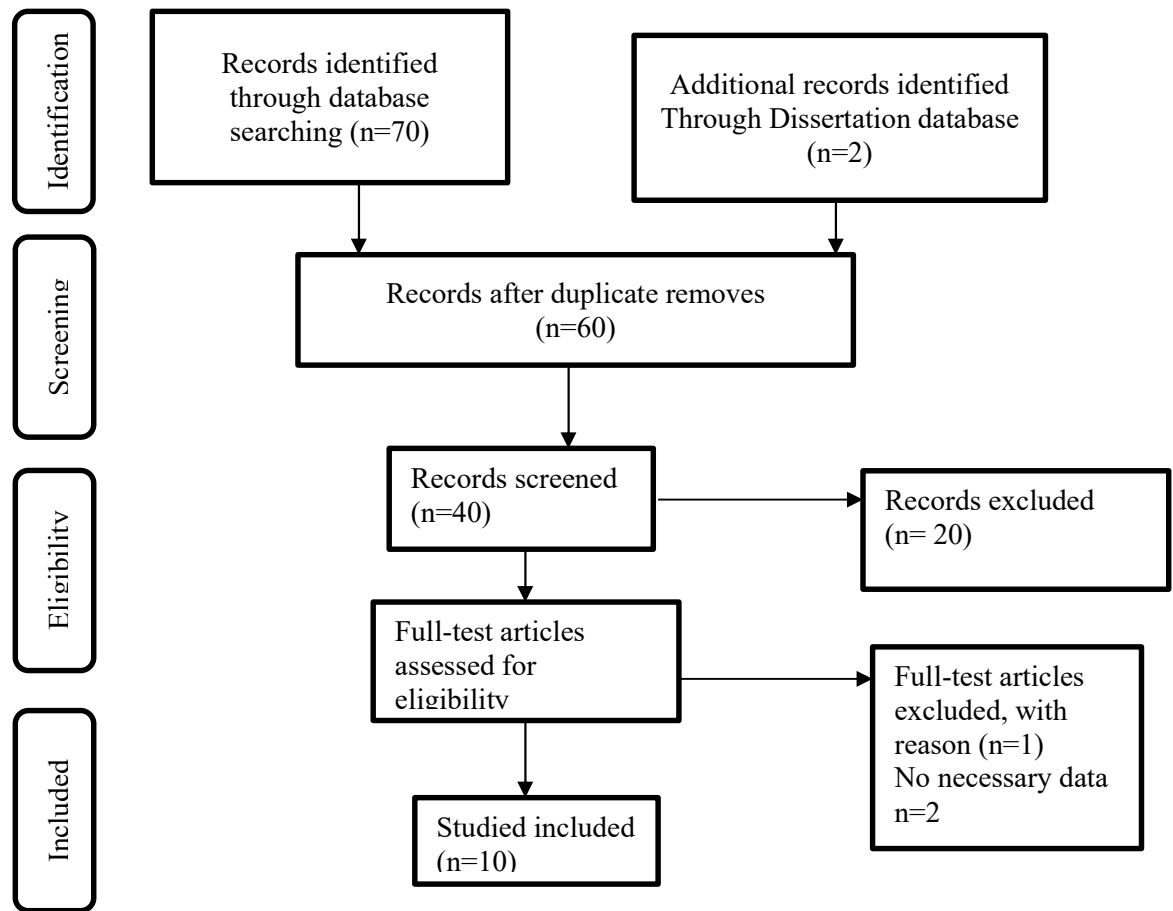


Figure 1. Flowchart for Selection of Relevant Literature for the Review.

LITERATURE REVIEW

1. **Santi Ardani Pande Nyoman (2024)** emphasises how important career growth and training are to raising employee performance. It comes to the conclusion that while planned career development activities increase motivation and commitment, which in turn leads to greater organisational success, effective training programs greatly improve employee performance. For HR strategies to improve employee performance in a variety of situations, including India, these components are crucial.
2. **Dr. Anumeha Mathur, Dr. Sapna Rana, and Deepali Gupta (2024)** study looks at how training affects worker performance and productivity in the Indian BPO sector. It concludes that staff performance and productivity are positively impacted by training, which promotes organisational expansion. The study, which uses primary data from interviews and questionnaires and secondary data from journals and the media, highlights the value of training in accomplishing both individual and organisational goals.
3. **S. Thanga Revathi (2024)** examines how training and development programs improve employee performance, particularly among unskilled and inexperienced workers. It makes the claim that training boosts organisational performance and productivity. The study highlights the increasing efficacy of contemporary training approaches, including AI-based strategies and remote mobile training, which have grown in popularity and efficacy.
4. **Sanjana Tripathi (2024)** study looks at how training and development affect workers' overall performance. The report summarises previous research and empirical findings, emphasising how training initiatives greatly enhance workers' abilities, expertise, and output. The study highlights how training goals should be in line with organisational objectives in order to foster a culture of ongoing learning, which is crucial for obtaining a competitive edge.
5. **Michael Ikenna Anyika, H. S. Ojukwu, and Ejike Daniel Chukwuma (2024)** which focusses on the pharmaceutical sector, shows that employee performance and training are positively correlated. Although there is little training in pharmaceutical companies, the researchers find that improving

training initiatives could greatly raise worker performance. Questionnaires and interviews were used to collect data for the study, and the results were displayed as straightforward percentages.

6. **P. Shalini (2024)** examines how training initiatives affect worker productivity and business success. Training programs and employee performance metrics including productivity and job satisfaction are found to be significantly correlated by the study. The report also emphasises how successful training programs are positively correlated with organisational success indicators like staff retention and profitability.
7. **Neha Pandey (2024)** study investigates how organisational training and development (OTD) initiatives affect the functioning of organisations. According to the study's findings, OTD initiatives improve worker satisfaction and engagement, which in turn boosts organisational effectiveness. The study bases its conclusions on academic frameworks including the human capital theory and the resource-based view.
8. **P. Vanitha and R. Sankar Ganesh (2024)** highlights how important employee training and development is for improving abilities, knowledge, drive, and job happiness. According to the research, training initiatives boost organisational outcomes, such as staff retention, productivity, profitability, and innovation—all of which are critical for the success of any business.
9. **Sarinah Sihombing, Aswanti Setiawati, and Rohana Sitanggang (2024)** study evaluates the return on investment (ROI) of staff training by conducting a cost-benefit analysis. It comes to the conclusion that employee training promotes an innovative culture, improves individual competencies, and is in line with organisational objectives. When combined with HR strategies, strategic training design greatly boosts an organization's competitiveness and performance.
10. **J. Sujitha and Sowmya Florence V (2024)** explores how training and development affect worker performance and job satisfaction. According to the survey, successful training initiatives match organisational objectives with worker capabilities, promoting employee development and the accomplishment of strategic goals, which ultimately helps the organisation succeed.

RESEARCH OBJECTIVES

- Evaluate the effectiveness of current training and development programs in enhancing employee skills, knowledge, and job performance.
- Identify the key factors influencing the effectiveness of training, including training methods, content relevance, and delivery techniques.
- Measure the impact of training programs on improvements in individual employee performance metrics.

RESEARCH METHODOLOGY

The study is conducted in an organised manner thanks to the research methodology, which makes data collecting, analysis, and interpretation possible. The effectiveness of training and development initiatives and their influence on worker performance are assessed in this study using a quantitative methodology.

1. Research Design

The study uses a descriptive research design, aimed at understanding the relationship between training programs and employee performance. It examines the key factors contributing to training effectiveness and the measurable outcomes of these initiatives.

2. Sample Size and Sampling

The sample size consists of 77 respondents, selected using a random sampling method from the target population of employees. This ensures a diverse and representative sample for the study.

3. Data Collection Method

Data is collected through both primary and secondary sources. Primary data is gathered using a structured questionnaire, administered to employees who have undergone training programs. The questionnaire is designed to capture both quantitative and qualitative responses, enabling an in-depth analysis of training effectiveness. Secondary data is collected from existing records, reports, and relevant literature to complement the primary data and provide additional insights into training outcomes and employee performance.

4. Data Analysis

The collected data is analyzed using SPSS (Statistical Package for the Social Sciences). A Chi-square test is applied to determine the relationship between variables, such as training effectiveness and employee performance. This statistical method helps assess whether the observed outcomes align with

expected results.

5. Limitation

The study's dependence on self-reported data, which could introduce bias, is one of its drawbacks. Furthermore, the results' generalizability may be impacted by the sample size of 77 respondents, which might not accurately reflect the total population. The study's breadth is likewise constrained by time and resource limitations.

HYPOTHESIS

Null Hypothesis (H_0): There is no relationship between job performance and the type of training provided.

Alternative Hypothesis (H_1): There is a significant relationship between job performance and the type of training provided.

ANALYSIS AND OUTCOME

Hypothesis

There is a significant relationship between job performance and the type of training provided.

A Chi-Square test is used in this hypothesis to assess how the variables relate to one another. The null hypothesis, which states that there is no significant link, will be accepted if the p-value is higher than 0.05. In contrast, the null hypothesis will be disproved if the p-value is less than 0.05, indicating that there is a substantial correlation between the variables.

Table 1. Chi-Square Tests

Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	22.777 ^a	9	.007
Likelihood Ratio	17.609	9	.040
Linear-by-Linear Association	5.440	1	.020
N of Valid Cases	77		
a. 12 cells (75.0%) have expected count less than 5. The minimum expected count is 0.08.			

The hypothesis for this analysis looks into whether work performance and the kind of training received are significantly correlated. The association between these factors was assessed using the Chi-Square test. If the p-value is higher than 0.05, the null hypothesis—that there is no significant relationship—will be accepted. The null hypothesis will be disproved, however, if the p-value is less than 0.05, which would suggest a substantial association.

A Pearson Chi-Square value of 22.777 and a p-value of 0.007, both below 0.05, are displayed in the Chi-Square test results. This implies that the kind of training given and job success are significantly correlated. This conclusion is further supported by the p-value of 0.040 and the Likelihood Ratio of 17.609. There is also a significant result from the Linear-by-Linear Association, with a p-value of 0.020 and a value of 5.440.

Table 2. Symmetric Measures

Symmetric Measures			
		Value	Approximate Significance
Nominal by Nominal	Phi	.544	.007
	Cramer's V	.314	.007
N of Valid Cases		77	

Additionally, symmetric measures, such as Phi (0.544) and Cramer's V (0.314), both yield a p-value of 0.007, confirming the strength and significance of the relationship between the variables. In conclusion, the analysis indicates that the type of training provided has a significant impact on job performance.

INTEPRETATION

The Table 1 Chi-Square test was used to analyse the relationship between work performance and the

kind of training received. The Pearson Chi-Square value of 22.777, with nine degrees of freedom and a p-value of 0.007, indicates a significant relationship between the two variables because the p-value is less than the 0.05 limit. The null hypothesis, which suggested that there was no significant relationship between the type of training received and employment success, can therefore be ruled out.

Analysis using the Likelihood Ratio and Linear-by-Linear Association, which both have p-values below the 0.05 significance level (0.040 and 0.020, respectively), lends more credence to this result. This confirms the conclusion that the type of training received has a major impact on work performance.

Additionally, the Table 2 symmetric measures of association, such as Phi (0.544) and Cramer's V (0.314), both show significant values with a p-value of 0.007, confirming the strength of the relationship. By showing that the type of training provided has a major impact on employees' job performance, our findings highlight the importance of selecting appropriate training methods to enhance performance outcomes.

Table 3. Hypothesis Result

Hypothesis	Result
There is a significant relationship between job performance and the type of training provided	Accepted

DISCUSSION AND FINDINGS

Several important conclusions that provide insight into how well training programs work to improve employee performance have been drawn from the analysis of the relationship between job performance and the kind of training that is given. The hypothesis that there is a substantial association between work performance and the type of training given is consistently supported by the statistical tests utilised in this study, such as the Table 1 Chi-Square test, Likelihood Ratio, and Linear-by-Linear Association. Employee performance is influenced by the type of training, according to the significant p-values (all less than 0.05).

First, there is a strong correlation between training type and work performance, as indicated by the Pearson Chi-Square value of 22.777 and p-value of 0.007, respectively. With p-values of 0.040 and 0.020, respectively, the likelihood ratio and linear association tests support this finding even more. According to these results, employees' capacity to carry out their jobs is directly impacted by the type of training they receive, whether it be theoretical, practical, or a combination of the two. Training programs that are specifically designed to meet the needs of employees, in particular, typically produce higher performance results. Furthermore, symmetric metrics of association like Phi (0.544) and Cramer's V (0.314) demonstrate how strongly training type and job performance are related.

These measurements attest to the statistical significance and practical significance of the relationship between training and performance. This emphasises how crucial it is to have training programs that are pertinent, well-structured, and match the particular abilities and knowledge needed for a given position. These results highlight how important training is in influencing worker performance. The study emphasises how important it is for businesses to fund efficient training initiatives that are pertinent, interesting, and in line with the goals of the workers' positions. These initiatives not only improve individual performance but also support the organization's overall success and productivity. The study's findings can therefore help firms improve their training plans to better suit the unique requirements of their employees and develop a more knowledgeable and effective staff.

CONCLUSION

In conclusion, the study provides substantial evidence that the type of training provided plays a crucial role in enhancing job performance among employees. The statistical analysis, including Chi-Square tests, Likelihood Ratio, Linear-by-Linear Association, and symmetric measure such as Phi and Cramer's V, consistently indicate a significant relationship between the variables, rejecting the null hypothesis. These findings highlight the importance of designing and implementing training programs that are not only aligned with the job requirements but also tailored to the unique learning needs of employees. Effective training methods that focus on relevant content, interactive delivery techniques, and hands-on experiences are more likely to yield improved job performance, leading to greater organizational success. The results of this study emphasize the need for organizations to continuously assess and refine their training programs, ensuring that they contribute meaningfully to the development of employees'

skills and knowledge. By doing so, organizations can foster a more capable and productive workforce, ultimately driving higher levels of performance and achieving long-term business goals. Therefore, it is essential for companies to prioritize training initiatives that are both impactful and tailored to the specific needs of their employees to realize the full potential of their workforce.

Declaration of Interest

The author(s) declare(s) that there is no any conflict of interest in our submitted manuscripts. All authors have contributed to the research impartially, and there is no personal or financial relationship have influenced the outcomes of this study. The author(s) declare(s) that there is no conflict of interest regarding the publication of this manuscript.

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Improved Power Transfer Capability and Optimisation of Photovoltaic Power Plants

Ms. Alvin Fatema Shaikh¹, Dr. Prashant Thakare², Mr. Ganesh Wakte³,
Dr. Mukesh Kumar⁴, Ms. Vaishali Malekar⁵

Abstract

Grid integration is the process of linking distributed energy resources, such as small-scale photovoltaic systems, to the electrical grid. Improving the power quality in the integrated grid of small-scale solar plants requires addressing many technological issues in order to create a secondary distribution network. Thus, this review paper provides a thorough synopsis of the current state of the art in power quality improvement methods for grid integration, with a focus on PI-Based Reactive Power Control Systems, Flicker Logistic Control Methods, Automated Filtering Mechanisms, Shunt Active Power Filter modules, Integrated Optimization-based AI Technique, and Grid Synchronization Techniques. To help readers better understand how each method contributes to better power quality, the review discusses the pros and cons of each. To further prove that different methods for improving power quality are successful, it offers a comparison study. The evaluation thoroughly assesses how each technique tackles issues like voltage fluctuations, harmonics, and flicker, ultimately leading to a more consistent and dependable power source. In addition to outlining potential avenues for future study and the difficulties inherent in applying different methods to improve power quality during grid integration of small-scale photovoltaic systems, this review offers some recommendations for how this area might be improved.

Keywords: Photovoltaic (PV) power plants, Ultra-weak grids, Adaptive reactive power control (ARPC), Power transfer capability, Voltage stability, Short-circuit ratio (SCR), Grid impedance estimation, Sustainable energy systems.

INTRODUCTION

Due to climate change and the diminishing availability of fossil fuels, there has been a surge in the need for alternative energy sources. Recently, there has been a lot of curiosity in how renewable energy-based Distributed Energy Resources (DER) systems may be integrated into the distribution grid. One of the most plentiful renewable energy sources is solar radiation. India has enormous potential for solar energy generation, with an average yearly incident solar energy on land of over 5,000 trillion kilowatt-hours [1]. Annual solar energy production in India surpasses that of all fossil fuel reserves put together. The broad availability and eco-friendly functioning of solar power have propelled it to the forefront of energy policy discussions in recent years. Renewable energy sources are booming in popularity as a result of several government-sponsored subsidies and marketing initiatives. Photovoltaic (PV) power generators, which do not need any moving components, directly transform the energy of solar radiation into electrical energy. Materials with photon-absorbing and electron-emitting capabilities experience this. In photovoltaic, silicon is by far the most popular material. Among the many renewable energy source applications, photovoltaic production is rising in prominence owing to its many unique benefits, such as its simplicity of allocation, high reliability, lack of fuel expense, cheap maintenance, and absence of noise and wear associated with moving components [8]. During periods of average or low peak needs, the utility utilises a bidirectional meter to send extra energy to the grid, using the idea of net metering. Only in situations when maximal power extraction using MPPT is possible may grid-connected solar inverters provide active electricity [6, 7]. Since most electrical loads are inductive, their inductive reactive power consumption is higher. At this time, reactive electricity can only be generated by the grid. As more and more Distributed Generation (DG) units enter the grid and provide only active power, the site power factor degrades from an efficiency perspective, impacting grid performance. In the past, FACTS devices were used to inject or absorb reactive power in order to address power quality concerns. The huge size, expensive price, and extensive installation space that these devices need are some of its drawbacks. Because it could affect voltage regulation, controlling the network's reactive power flow is essential. Grid voltage regulation will be improved and the demand for expensive capacitor banks will be reduced as the number of grid-connected inverters increases and their usage as

VAR compensators becomes more common. This study explores control algorithms for grid-connected solar inverters that may provide reactive power alongside active electricity, regardless of the load or irradiance. In the unique instance of inductive or capacitive load connections, there will be a significant shift in bus voltage—more than what is specified for the system—leading to unstable operation. A distributed control method is suggested for controlling the reactive power output in order to maintain grid-connected voltage stability. As a backup technique, active power reduction is used to regulate the voltages in the PV system [4, 9, 10]. References [2, 3, 5, 11] highlight the issues with connecting large-scale PV systems to the grid, active power regulation, reactive power management, and low voltage crossing approaches.

Renewable energy is a dynamic field that is always changing to meet the demands of a world where fossil fuels are becoming scarcer and environmentally harmful. Thanks to its accessibility, minimal noise, and cheap cost, photovoltaic energy production is gaining popularity in both rural and urban regions. Distributing electricity in rural areas is a frequent usage of micro grids [1]. Micro grids may be either grid connected or operated independently. In freestanding photovoltaic micro grids, the solar power plant is not linked to the grid. Some machinery, like three-phase motors, can't handle voltage imbalances that occur when the load isn't evenly distributed, and this plant isn't equipped to handle them [2]. In addition, when the weather isn't nice, a PV system that isn't connected to an energy storage system won't be able to keep up with the demand during the day and at night [3]. Harmonics are introduced into a system when power converters are utilized.

One of the main challenges facing the electrical industry is improving power quality, which is exacerbated by factors such as the growing reliance on sensitive electronic circuits in both homes and businesses, the rise of privatization in the energy sector, and competition among electric energy providers. Unwanted current flowing through the source is what causes harmonics, which in turn distort the voltage and create loss. Additionally, it has the potential to cause control components such as relays and mains to malfunction. This means that harmonic reduction is an absolute must. Numerous methods exist for mitigating the impact of harmonics [4, 5].

Using SAPF is one of these approaches; it cancels out the system's harmonic current by producing a harmonic current of the same amplitude and polarity. Its quick reaction speed and adaptability are attributes of the power electrical gadgets it incorporates. The Shunt Active Power Filter (SAPF) may inject power produced by unorthodox sources, compensate for harmonics and current distortion, and more [6]. The SAPF is an example of a VSI, or load-related voltage source inverter. The Shunt Active Power Filter maintains a balanced and sinusoidal after-remuneration current for a variety of load circumstances. When converting DC power to AC power, power switching devices come in handy. This causes the output waveforms to consist of discrete values, making the output less filtered and more oscillatory. The dictatorial regulatory approach dictates the time and duration that power values may become active, as well as their ability to provide near sine waveforms around the key repetition.

The use of Static Volt-Ampere Reactive Compensators (SVCs) is also important for transient stability improvement, grid capacity expansion, and voltage regulation [7, 8]. The effective conversion of electromagnetic waves into energy is a hurdle that must be overcome in order for PV cells to be integrated. In order to get the most out of the electricity that the PV array produces, the Maximum electricity Point Tracking (MPPT) method that is used in the system is essential. One of the biggest problems with MPPT methods is automatically detecting the voltage or current MPP [9]. The effect of variations in output voltage on the properties of MPPT outputs adds another layer of difficulty to this problem [10].

The extra features needed by current inverter levels cause them to be somewhat inaccurate, even if MPPT approaches have their benefits. The time needed to anticipate the Global Maximum Power Point (GMPP) is proportionate to the number of PV arrays and the complexity of the system design, making this mistake all the more crucial [11]. Power quality concerns may be resolved by deploying Distributed Flexible AC Transmission System (DFACTS) devices at the point of common coupling (PCC) that include control algorithms. Examples of programs that modify voltage, impedances, and power to enhance system dynamics include DSTATCOM and UPQC.

Series, shunt, series-series, and shunt-series are the four main categories into which devices fall [12, 13]. The shortcomings of conventional control algorithms rooted in IRPT and synchronous reference frame theory include sluggish response and unstable operation.

REVIEW ON PI-BASED REACTIVE POWER CONTROL SYSTEM IN GRID INTEGRATION

Here we take a look at how different PI-based reactive power control systems have been used to improve power quality in grid integration of small-scale photovoltaic systems. These systems include

distributed power flow controllers, fuzzy adaptive PI controllers, optimal fractional order PID controllers, fuzzy PID controllers, and multi-stage fuzzy-based flexible controllers. The computing load and the difficulty of testing in real-time systems are two examples of the limits.

Table 1. This table provides a clear and concise overview of the studies, their focus areas, and key findings, making it easier to understand the scope and impact of each research effort.

Author(s)	Year	Focus	Key Findings
Rulei Han et al.	2020	Reactive power regulation ability of PV inverters	PV inverters exhibit good reactive power regulation; dynamic reactive power response time is within 30ms, exceeding national standards. Simulation and on-site tests confirm effectiveness.
Dzhengiz Ibram et al.	2020	Reactive power regulation in single-phase PV inverters	Discussed control strategies for reactive power compensation in single-phase PV inverters, emphasizing the role of renewables in replacing traditional coal power.
Ashutosh Kumar Tiwari et al.	2021	Power plant controller for active/reactive power control in PV plants in India	Developed a power plant controller to meet grid code requirements. Real-time and HIL tests show the controller's effectiveness in managing active/reactive power output.
Hazem Karbouj et al.	2021	Reactive power capability estimation and voltage control in PV inverters	Presented a self-adaptive voltage controller for PV inverters, accounting for environmental conditions. The strategy improves voltage control effectiveness and ensures grid code compliance under various conditions.
Liang Wang et al.	2022	Reactive power and voltage stability in renewable energy integration	Proposed a coordinated control strategy for reactive power and voltage stability, validated via simulations. The strategy helps maintain stability amidst renewable energy's variability.
Yang Li et al.	2022	Fast reactive power control in PV inverters	Studied the structure and reactive power control strategy of PV inverters. Simulation results in MATLAB/Simulink confirm the efficiency of the control strategy in a single-stage grid-connected PV system.
Zhen Wu et al.	2022	Adaptive voltage control for PV grid-connected points	Proposed an adaptive voltage control strategy to mitigate voltage over-limit issues. Simulation results show improved voltage control by enhancing reactive power regulation of PV inverters.
Yingfei Yan et al.	2023	Multi-energy complementarity for clean energy systems	Introduced a reactive power coordination optimization strategy for a multi-energy system, enhancing energy supply-demand balance and promoting renewable energy consumption.
Yijiang Dong et al.	2023	Voltage fluctuation suppression in high-proportion PV HVDC systems	Proposed a reactive power coordination strategy to suppress voltage fluctuations caused by commutation failures. Simulation results demonstrate effectiveness in maintaining voltage stability.

Grid synchronization techniques, automated filtering mechanisms, PI-based reactive power control systems, those are just a few of the downsides. Phase Locked Loop, DQ current control theory, Vector-based synchronization, Adaptive feed-forward PLL, lightweight inertial PLL, and mixed third and fourth-order complex filters were some of the grid synchronization approaches researched for their potential to enhance performance. There are a few problems with these methods, however. For example, PLLs only react temporarily to abrupt grid changes, instability occur due to a lack of resilience, and precise and quick synchronization is required. Arrays of AI techniques that rely on optimization have proven to be effective and dependable in the modern power network. However, there were a few drawbacks to these tactics, such as the difficulty of achieving accurate detection during dynamic operation, the need for effective decision-making in real-world circumstances, and so on.

PHOTOVOLTAIC SYSTEM

To generate the required voltage and current, a photovoltaic system employs several strings of solar cells linked in parallel and series. The nonlinear I-V characteristic may be calculated using the PV equivalent circuit, as shown in Fig. 2. Consequently, an array of cells with the required voltage and power levels is formed by connecting them in series and parallel. In the model that represents a single diode, the current flowing through the parallel diode may be expressed as:

$$I_D = I_0 [\exp (q(V+I R_s) / \alpha k T) - 1] \dots \dots (1)$$

By using Kirchhoff's law in equivalent circuit of PV output current can be written as:

$$I_{PV} = I_{PH} - I_D - I_r \dots \dots \dots (2)$$

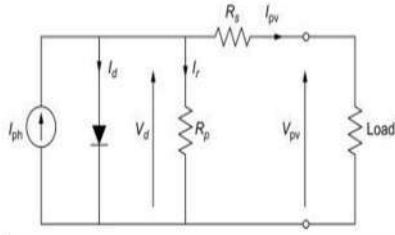


Fig. 1: Equivalent structure of PV cell

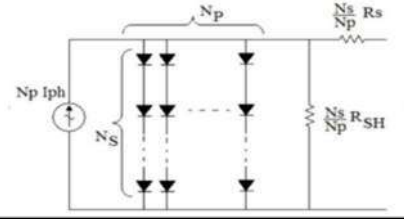


Fig. 2: Mathematical model of PV array where N_S and N_P are the series and parallel cell numbers, respectively. When cells are connected in series, the output voltage is increased, and when they are connected in parallel, the output current is increased.

METHODOLOGY

Photovoltaic (PV) power plants that function on an exceedingly weak grid may achieve their objectives with the aid of adaptive reactive power regulation if you follow these steps in the technique. The goal of this approach is to enhance the power transfer capacity and voltage stability of PV systems via the implementation and verification of an adaptive control system. When trying to understand why renewable energy sources are underutilized, it helps to consider the actual grid conditions and the subsequent challenges. Public electricity networks in most European countries are designed with a spreading central configuration. Large conventional power stations may send their energy to grids. A transformer is a device that allows power to be aggregated by connecting the distribution and transmission grids. The consumer is not prioritized. By distributing electricity to its associated consumers, the interconnection acts as a central authority, either directly or indirectly. The distribution grid causes a voltage drop when electricity passes across it. How quickly it decreases is dependent on the inductance and resistance of the cable. As the cable lengthens, both of these factors grow. To make sure everyone has enough power, a transformer boosts the voltage at the cables beginning a little. Renewable energy plants must be located in areas with abundant energy supplies, such as areas with strong wind speeds, in order to produce enough power. Consequently, the installations are linked to the grid via a number of local locations. The capacity and integration of renewable power facilities at lower grid tiers is lower than that of big power plants. After connecting to low-voltage connections, dispersed generators may cause a change in conditions, with electricity flowing towards the transformer [19]. The integration of more and more dispersed generators worsens the issue of voltage rises in rural locations with often insufficient power infrastructure. The main problem is that the system can't handle enough renewable energy. Integrating dispersed energy producers has a major influence on the overall grid's functioning since it increases the demands on the mains. Materials, efficiency, and producing costs are all negatively affected by these measures, which also increase carbon dioxide emissions and the usage of fossil fuels. In order to keep voltage spikes at bay, we must partly extend the grid. In other cases, the expense of fortifying a grid is not insignificant. The benefits of renewable energy sources are often believed to be less than the costs of producing them [7]. Another issue that alternative power producers have raised is that they are unable to verify the operator's claims on the technical and financial aspects of their grid connection due to a lack of information regarding the available grid capacity. Distribution System Operators (DSOs) is another name for electricity distributors. When a power company is involved in developing alternative energy initiatives, it is questionable if the DSO is entirely unbiased towards independent renewable energy producers. Due to a lack of transparency, the permission procedure for connecting to the grid is lengthy [8][9].

In spite of grid challenges, stakeholders see renewable energy sources in a good light, according to the results.

- a. **System Modeling:** The PV power plant and grid are modeled mathematically. This model takes into account the inverter's dynamics, the PV array's characteristics, and the electrical behavior of the ultra-weak grid, which is characterized by high impedance and low short-circuits ratios (SCRs). Accurate dynamic representation is achieved by means of the model, which combines real-time observed grid properties including impedance, frequency, and voltage. We test the proposed adaptive control method in a number of simulations with different ultra-weak grid configurations.

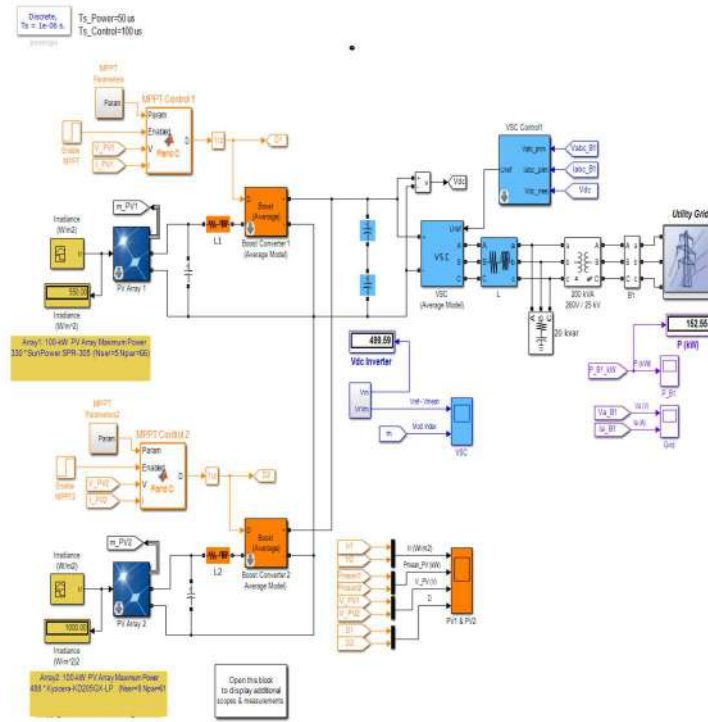


Fig. 3: Overall Simulink of Adaptive Reactive Power Control of Photovoltaic Power Plants

- b. Scenario Definition: A number of possible grid states are detailed, including fluctuating short-circuit ratios (SCRs), altered grid impedance, and voltage disturbances. These cases mimic the difficulties, such as voltage instability, synchronization problems, and power transmission limits that PV plants might encounter in very weak networks. In this simulation, we check how well the control system can track reactive power demand, impedance, and voltage in real time. The simulation tests the system's responsiveness to changes in the grid voltage by manipulating the reactive power output.

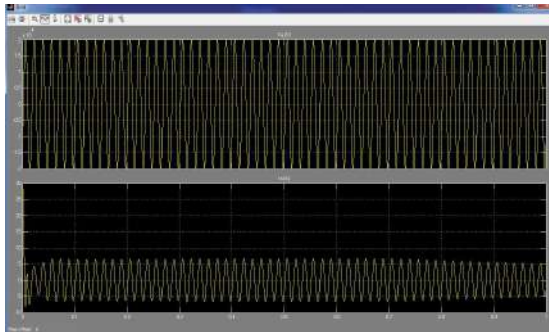


Fig. 4: Voltage and Current variations with respect to time.

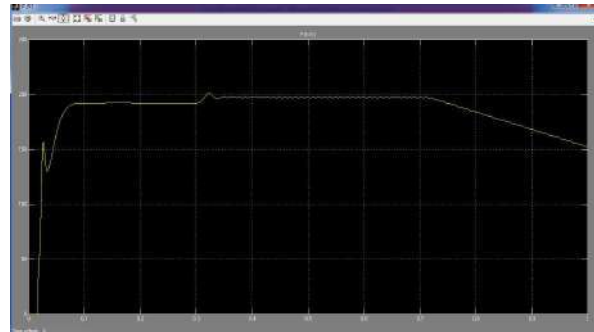


Fig. 5: Power variations with respect to time.

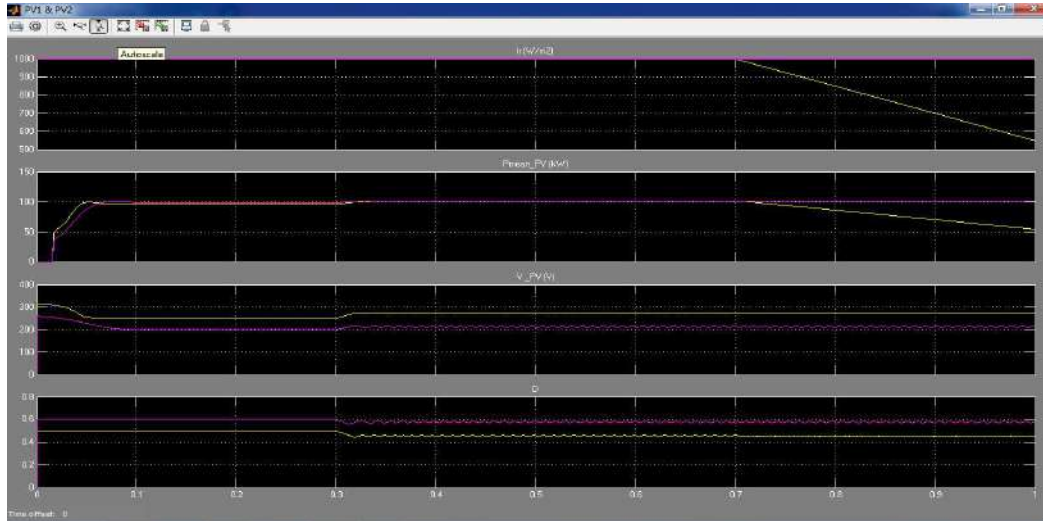


Fig. 6: Overall output of the system with respect to time.

The system's performance is evaluated using metrics such as:

- **Voltage Stability:** Measured by monitoring how effectively the adaptive control system mitigates voltage fluctuations and prevents instability.
- **Power Transfer Efficiency:** Analyzed in terms of the ability of the PV plant to transfer maximum possible power to the grid without causing voltage instability or system failures.
- **Reactive Power Support:** The system's ability to provide adequate reactive power to stabilize grid voltage, especially during disturbances

CONCLUSION

Because of their low short-circuit rates (SCRs) and unpredictable voltages, photovoltaic (PV) power plants are difficult to include into insufficient networks. In response to these problems, this research presented an ARPC method for adaptive reactive power regulation; this method allows PV plants to modify their reactive power output in response to grid circumstances in real-time. Utilizing intricate methodologies including adaptive gain tuning, voltage sensitivity analysis, and grid impedance estimations, the suggested solution is put forward. Renewable energy sources may be more widely used when reactive power compensation is optimized, allowing PV facilities to run dependably under adverse grid situations. In the future, scientists could try to refine this method.

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Literature Review on Real-Time Dashboard Systems in Healthcare and Education

1.Neha Nitin Wagh, 2. Darshan Ranjeet Desale, 3. Chetana Mukesh Sonawane,
4. Digvijaysing Devendra Rajput, 5. Kiran Dinkar Chaudhari

1,2,3,4 Student,(B.Tech) Department of Artificial Intelligence and Machine Learning, SES's R. C. Patel
Institute of Technology, Shirpur, Maharashtra, India

5. M. Tech. (Data Science), SVKM's NMIMS, Mukesh Patel School of Technology Management and
Engineering, Mumbai ,Maharashtra,India

Abstract

Real-time dashboards are revolutionising data analysis and visualisation within education and healthcare, highlighting the realms of online learning analytics and emergency medicine. Such systems maximize decision-making power, efficiency of workflows, and awareness in real time. The literature review below explores diverse methodologies, benefits, and pitfalls and presents suggestions for potential avenues for research in the future. Growing access to data and advancements in visualisation and analytics technologies have given rise to powerful dashboards that support better situational awareness and timely interventions. This review considers the design elements, implementation methodologies, and assessment techniques applied for developing real-time dashboards across healthcare and educational contexts. The review detects strengths and limitations, common challenges, and future research opportunities. The aim is to present an all-around view of existing real-time dashboard systems, which will lay the ground for more useful applications in these essential fields.

Keywords: Real-time dashboards, Data visualization, Healthcare analytics, Emergency medicine, online learning analytics

Their use in healthcare and education has greatly risen, notably in emergency medicine and online course learning analytics. Dashboards support important real-time visualisation of data, which leads to enhanced decision-making, efficient workflow, and situational awareness. The current literature review provides an overview of studies conducted on dashboard systems, evaluating methodologies used, weaknesses and strengths, and areas requiring further research.

Real-time dashboards are revolutionizing data visualization and analysis across sectors, with vast implications for decision-making in fast-changing environments [1], [2], [3]. The present literature review is concerned with the application of real-time dashboard systems in two significant sectors: healthcare and education [4], [5], [6]. Increased availability of data in such domains, along with the development of data analytics and visualisation technologies, has driven the production of advanced dashboards with the objective of enhancing situational awareness, enabling timely intervention, and overall results [7].

We examine the design guidelines, implementation approaches, and evaluation methods employed to build real-time dashboards in healthcare environments such as emergency rooms, as well as learning environments such as online courses and teacher apps. This review discusses the strengths and limitations of current systems, recurring problems, and promising avenues for future work. By integrating results from pertinent research, we aim to present a thorough summary of the state of the art in real-time dashboard systems for healthcare and education, opening the door to more efficient and influential applications in these vital areas.

LITERATURE REVIEW:

Real-time dashboards have emerged as critical tools for enhancing data visualization, analysis, and decision-making, particularly in the domains of healthcare and education. Their ability to provide immediate insights into dynamic environments has propelled research into their design, implementation, and impact.

In healthcare, real-time dashboards assist clinicians by aggregating patient data, monitoring emergency department activities, and supporting public health surveillance. Khairat et al. [1] emphasized the role of dashboards in improving the quality of care and clinician satisfaction. Similarly, Elshehaly et al. [4] proposed QualDash, which enables adaptable visualization for healthcare quality improvement, highlighting the need for customizable, user-centered designs. Zhuang et al. [5] further developed a framework to evaluate healthcare dashboards, focusing on usability and decision-making support.

Yoo et al. [8] conducted a five-year case study of an autonomous emergency department dashboard, demonstrating its ability to enhance situational awareness and reduce clinician mental workload. However, limitations such as a small sample size and limited generalizability were noted.

In education, real-time learning analytics (LA) dashboards aim to improve teaching practices and student outcomes by providing insights into learner behaviors and engagement. Nguyen et al. [9] developed TEADASH through an iterative Design Science Research approach, emphasizing the importance of co-designing dashboards with end-users (teachers). Despite positive feedback, challenges in achieving sustained teacher adoption and the need for continuous iteration were identified.

Hasnine et al. [10] introduced a dashboard that uses computer vision and deep learning techniques to detect online learners' affective states, allowing instructors to intervene promptly. While innovative, the system faced limitations in validating emotion detection accuracy and assessing its impact on academic performance.

COMPARISON OF APPROACHES

The associated papers present a range of approaches to designing and evaluating LA dashboards. A teacher-facing dashboard (TEADASH) was developed and refined iteratively in one study through Design Science Research (DSR) based on teacher feedback. The systematic review methodology of another study, which reviewed existing dashboards to identify best practices and create a checklist for future dashboard development, is the opposite of this iterative process. The visual design components of dashboards were also the focus of another study, which highlighted the importance of proper and transparent representation of data for optimal usability and understanding. Another study examined the utilization of real-time data and automatic feedback systems, specifically with a focus on the use of computer vision and machine learning methods automatically to identify the affective states of online students. Finally, a series of studies examined dashboard use across a variety of environments, including public health and emergency care, showing the technology's flexibility and applicability across disciplines.

A COMPARATIVE ANALYSIS OF DASHBOARD IMPLEMENTATION

The current review consolidates highlights and limitations of five studies on design, implementation, and evaluation of dashboards across several fields, such as emergency medicine, education, cryptocurrency visualization, and public health. The research employed diverse methodologies, pointing to the intricacies of developing and accessing dashboards.

The investigation [8] of a real-time autonomous emergency department (ED) dashboard demonstrated effective long-term adoption (five years) and overall positive user ratings, as evidenced by a "OK-to-Good" System Usability Scale (SUS) score. The small sample size and single ED setting of the study restrict the generalisability of the findings. Whereas the dashboard enhanced situational awareness and diminished mental workload, perceived information quantity was less than its quality, suggesting further improvements in information presentation.

The TEADASH study [9] based on Design Science Research (DSR) highlighted the necessity of iterative design and collaboration between teachers in designing effective teacher-oriented dashboards. The iterative process, involving several design cycles and feedback loops, generated a tailored dashboard to the needs of teachers and learning designs. Yet, small study size with only five courses at two universities restricts generalisability of the results. In addition, the research found a discrepancy between projected and realized teacher adoption, stressing the necessity of continuous assessment and adjustment subsequent to the first design phase.

The study [10] of a real-time learning analytics dashboard for the automatic recognition of affective states of online students introduced a new method for monitoring learner attention and engagement by fusing computer vision and deep learning. The feedback mechanism in real time of the system facilitates timely interventions. Nonetheless, the research has shortcomings, such as limited rigorous assessment of academic performance and accuracy of emotion detection. It is further validated with conventional methods like Geneva Emotion Wheel, and a test on its effect on the learning outcome.

The research [11] on data streaming architecture for visualizing cryptocurrency temporal data effectively proved the possibility of real-time visualization through a lambda architecture. The dynamic updates yielded useful insights into cryptocurrency price movements. The assessment was mostly visual, with no quantitative accuracy or usability measurements. The narrow scope, which only addresses cryptocurrency data, restricts the applicability of the findings to other types of data and visualization needs.

Lastly, the research [12] on integrating and visualizing a public health dashboard showed the possibility of integrating different sources of data (social media, web searches, etc.) for real-time surveillance of public health. The interactive dashboard facilitated easy monitoring and analysis of public health events. Nonetheless, the system was in its initial alpha stage, with little evaluation and possible data availability limitations. More research and strict testing are needed to establish its efficacy in actual public health settings.

In summary, these studies exhibit dashboards' capabilities in various fields and also illustrate the difficulties in designing, implementing, and assessing them. Future studies need to overcome issues of sample size, generalisability, stringent evaluation approaches, and the integration of user feedback during the course of development. More thorough comprehension of users' needs and situational aspects is needed to build useful and effective dashboards.

RESEARCH GAPS

Several research gaps emerge from the analysis of the related documents. Additional research is required to improve the generalisability of findings across various educational contexts and student populations.

1. Long term studies are essential in determining the long-term effect of LA dashboards on teacher practice and student learning results.
2. The ethical implications of data privacy, security, and possible biases during data analysis also need more consideration. Studies ought to examine how teacher capacity, institutional support, and the usage of LA dashboards are correlated.
3. Data-driven best practices in visual design for LA dashboards need to be established keeping in mind diverse data types and user needs. In order to make LA dashboards more effective, their embedding in certain pedagogical styles and learning principles should be further researched.
4. Lastly, more research is required to develop and implement efficient automated feedback systems in LA dashboards, considering teacher preference and student requirements.
5. Development of standardised assessment frameworks for LA dashboards is another significant topic for future research.

Table 1 here provides the summary of the literature reviewed in a clear and concise manner for better understanding.
Table 1: Summary of literature findings on dashboard visualization

Study	Study Objectives	Dashboard Design & Features	Implementation Context	Evaluation Methods
[8]	Develop & evaluate a real-time autonomous ED dashboard	Real-time, automated, geographical layout, patient-level alerts, real-time summary data.	Emergency Department (ED) setting, integrated with existing hospital information system (HIS).	Situation Awareness Index, questionnaire based on Situation Awareness Rating Technique.

[9]	Design & develop TEADASH using DSR; evaluate its impact on teaching and learning.	Teacher-facing, co-designed with teachers, visualizes student activity data (e.g., forum participation, quiz scores). Iterative design.	Five courses from two Swedish universities, integrated with Canvas LMS.	Iterative design cycles with teacher feedback, interviews, and informal observations during course deployment.
[10]	Develop a real-time LA dashboard for automatic detection of online learners' affective states.	Real-time, automated emotion detection (facial expressions, eye movements), engagement and concentration levels, student clustering.	Online learning environment, integrated with web conferencing tools (Zoom, WebEx).	Academic performance, accuracy of emotion detection, user feedback (Geneva Emotion Wheel).
[11]	Visualize real-time cryptocurrency price changes using a data streaming architecture.	Real-time multiline and line charts; dynamic updates every minute; uses C3.js and Tableau. Data from multiple cryptocurrency websites.	Web application; data sourced from multiple cryptocurrency websites.	Visual inspection of dynamic updates; video demonstration of functionality.
[12]	Develop and evaluate a public health dashboard integrating diverse data sources.	Web application (HTML5/JavaScript); integrates diverse data sources (social media, online searches, etc.); visualizes epidemiological data.	Public health surveillance; integrates data from various sources, including social media and online searches.	Simulations using 2009 swine flu outbreak data; future plans for more rigorous evaluation.
[13]	Recommend business strategies using Power BI dashboard visualizations of financial data.	Power BI dashboard; features not explicitly detailed, but likely includes charts, graphs, and interactive elements for financial data analysis.	Atliq Hardware, a consumer electronics company; specific details on data sources and data volume are limited.	Analysis of dashboard outputs and their contribution to business decision-making.

CONCLUSION

This review of the literature illustrates the increasing interest in applying LA dashboards to enhance teaching and learning. Although current research informs us about different facets of dashboard design, implementation, and assessment, considerable gaps in research exist. Closing these gaps is urgent to ensure LA dashboards are developed and successfully implemented to promote equitable and effective teaching and learning practices. There should be future research on the development of robust evaluation frameworks, the resolution of ethical issues, and the examination of how dashboards could be integrated into pedagogical approaches in order to optimize their potential in enhancing educational outcomes. Guidelines and best practices for visual design, data representation, and stakeholder communication will also be essential in facilitating the systematic and successful application of LA dashboards across education.

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A study on Organizational culture and It's Impact on Employee Behaviour

Pranali Vijay Chinchamalature^{1,*}, Dr. Sheetal Jaikar²

Abstract

This study explores the impact of organizational culture on employee behavior within a CA (Chartered Accountant) firm, emphasizing the significant role that workplace ethics and cultural dynamics play in shaping employee performance, motivation, and overall organizational success. Organizational culture is a unique blend of shared values, beliefs, rituals, and behaviors that influence how employees interact with each other and perform their duties. This research aims to understand how organizational culture influences employee behavior, including their attitudes, motivation levels, and work-related actions. The study focuses on examining how a well-established culture within an organization can lead to improved employee performance, higher engagement, and job satisfaction. The research highlights the connection between a strong organizational culture and positive workplace outcomes, such as increased productivity and enhanced teamwork. By analyzing the relationship between these cultural factors and employee behavior, the study seeks to identify strategies that organizations can adopt to foster a productive and harmonious work environment. The findings suggest that a robust organizational culture not only drives employee engagement but also contributes to a competitive advantage for the organization. Ultimately, this study underscores the importance of organizational culture in shaping employee behavior and achieving organizational growth and success.

Keywords: Organizational Culture, Employee Behavior, Workplace Ethics, Employee Performance, Motivation, Job Satisfaction, Employee Engagement, Organizational Success

INTRODUCTION

Organizational culture plays a pivotal role in shaping the work environment and employee behavior, making it an essential aspect of any organization's success. In the context of a CA (Chartered Accountant) firm, where precision, ethics, and collaboration are fundamental, the workplace culture can significantly influence employee interactions, job satisfaction, and performance. Organizational culture encompasses the shared values, beliefs, rituals, and symbols that bind employees together, guiding them in their daily tasks and decisions. It influences not only the working style of individuals but also creates a cohesive workforce that is aligned with the organization's goals and values. This study explores the profound impact that organizational culture has on employee behavior, focusing on its role in motivating employees, enhancing job satisfaction, and improving overall productivity. Organizational culture is often described as the "personality" of an organization, as it reflects the collective behavior and beliefs that are ingrained within the workforce. This culture is shaped by various factors, including leadership practices, organizational norms, and the behavior of employees. A strong organizational culture can create a positive work environment that fosters motivation, engagement, and high performance. On the other hand, a weak or negative culture can lead to disengagement, poor morale, and low productivity, which can ultimately affect the organization's growth and success. Understanding how organizational culture influences employee behavior is crucial for creating an environment that nurtures talent, enhances job satisfaction, and promotes organizational success. The relationship between organizational culture and employee behavior is multifaceted, with culture influencing various aspects of employee actions and attitudes. For instance, employees in a positive culture are more likely to feel motivated and valued, which in turn enhances their performance and productivity. Conversely, in a negative organizational culture, employees may experience frustration, dissatisfaction, and disengagement, which can lead to high turnover and low

performance. By fostering a positive organizational culture, organizations can promote employee well being, increase job satisfaction, and reduce stress, all of which contribute to higher engagement and improved productivity. In the context of a CA firm, organizational culture is particularly important, as it impacts not only employee behavior but also the quality of services provided to clients. A positive work culture encourages collaboration, ethical behavior, and a strong work ethic, all of which are

essential for delivering high quality services. Moreover, employees in a supportive and value-driven culture are more likely to align their personal goals with the organization's objectives, leading to a more unified and productive workforce. Therefore, it is essential for management to cultivate a culture that promotes transparency, trust, and mutual respect, ensuring that employees feel motivated and engaged in their work. The importance of organizational culture cannot be overstated, as it not only shapes employee behavior but also determines the long-term success and sustainability of the organization. By understanding the dynamics of organizational culture and its influence on employee behavior, businesses can create an environment that nurtures employee potential, fosters collaboration, and drives innovation. This research, therefore, aims to provide a comprehensive understanding of the critical role that organizational culture plays in shaping employee behavior and enhancing organizational performance.

HYPOTHESIS

Hypothesis 1 – Relationship Between Organizational Culture and Employee Behavior

- **Null Hypothesis (H₀):** There is no significant relationship between organizational culture and employee behavior.
- **Alternative Hypothesis (H₁):** There is a significant relationship between organizational culture and employee behavior.

Hypothesis 2 – Impact of Organizational Culture on Employee Engagement and Job Satisfaction

- **Null Hypothesis (H₀):** Organizational culture does not have a significant impact on employee engagement and job satisfaction.
- **Alternative Hypothesis (H₁):** Organizational culture has a significant impact on employee engagement and job satisfaction.

TABLES AND ANALYSIS

Hypothesis 1

Relationship Between Organizational Culture and Employee Behavior In the first hypothesis, a Chi-Square test is used to assess the relationship between the variables. If the p-value exceeds 0.05, the null hypothesis will be retained, suggesting no significant association. However, if the p-value falls below 0.05, the null hypothesis will be rejected, indicating a meaningful relationship between the variables.

Table 1: Pearson Chi-Square Test

Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	13.052 ^a	4	.034
Likelihood Ratio	13.952	4	.024
Linear-by-Linear Association	10.507	2	.024
N of Valid Cases	72		
a. 4 cells (50.0%) have expected count less than 5. The minimum expected count is .45.			

This test examines whether organizational culture significantly affects employee behavior. The Chi-Square value is 13.052, with 4 degrees of freedom and a p-value of 0.034. Since $p < 0.05$, the null hypothesis is rejected, confirming a significant relationship.

Likelihood Ratio: The Likelihood Ratio is another method to assess the relationship between variables. The test gives a value of 13.952, with 4 degrees of freedom and a p-value of 0.024. As $p < 0.05$, it further supports that organizational culture impacts employee behavior.

Linear-by-Linear Association: This test measures the strength of the linear relationship between organizational culture and employee behavior. The Chi-Square value is 10.507, with 2 degrees of freedom, and the p value is 0.024. Since $p < 0.05$, there is a significant linear association.

Number of Valid Cases: (N = 72) The analysis is based on 72 respondents, ensuring reliability in the results. A larger sample size typically strengthens the validity of findings.

Table 2: Phi Coefficient With Symmetric Measures

Symmetric measures			
		Value	Approximate Significance
Nominal by Nominal	Phi	.218	.034
	Cramer's V	.218	.024
N of Valid Cases		72	

The Phi value is 0.218, with a p-value of 0.034. Since $p < 0.05$, it indicates a weak but significant association between organizational culture and employee behavior.

Cramer's V is also 0.218, with a p-value of 0.024. This confirms a weak yet significant relationship between the two variables. Number of Valid Cases (N = 72) The analysis includes 72 respondents, ensuring the results are based on a sufficient sample size.

INTERPRETATION

In the first hypothesis, the results indicate a significant relationship between organizational culture and employee behavior. The Pearson Chi-Square test ($p = 0.034$), Likelihood Ratio ($p = 0.024$), and Linear-by-Linear Association ($p = 0.024$) all show that $p < 0.05$, leading to the rejection of the null hypothesis. This confirms that organizational culture directly influences employee behavior. The Phi coefficient (0.218, $p = 0.034$) and Cramer's V (0.218, $p = 0.024$) suggest a weak but significant association. The findings, based on 72 respondents, highlight that changes in organizational culture have a measurable impact on employee behavior, though the effect size is not very strong.

Hypothesis 2

Impact of Organizational Culture on Employee Engagement and Job Satisfaction

In the second hypothesis, also used a Chi Square test is used to assess the relationship between the variables. If the p-value exceeds 0.05, the null hypothesis will be retained, suggesting no significant association. However, if the p-value falls below 0.05, the null hypothesis will be rejected, indicating a meaningful relationship between the variables.

Table 3: The Person Chi-Square test

Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	10.419 ^a	4	.015
Likelihood Ratio	11.211	4	.012
Linear-by-Linear Association	7.901	1	.012
N of Valid Cases	72		
a. 4 cells (50.0%) have expected count less than 5. The minimum expected count is .45.			

The Pearson Chi-Square test ($p = 0.015$), Likelihood Ratio ($p = 0.012$), and Linear by-Linear Association ($p = 0.012$) all have p -values below 0.05, indicating a significant relationship between organizational culture, employee engagement, and job satisfaction.

Table 4: Phi Coefficient With Symmetric Measures

Symmetric measures			
		Value	Approximate Significance
Nominal by Nominal	Phi	.218	.015
	Cramer's V	.218	.012
N of Valid Cases		72	

The Phi coefficient (0.218, $p = 0.015$) and Cramer's V (0.218, $p = 0.012$) indicate a weak but significant relationship between organizational culture, employee engagement, and job satisfaction.

INTERPRETATION

The Chi-Square test results indicate a significant relationship between organizational culture, employee engagement, and job satisfaction, as all p values are below 0.05. Since the p -values for Pearson Chi-Square (0.015), Likelihood Ratio (0.012), and Linear-by Linear Association (0.012) are less than 0.05, the null hypothesis is rejected. The Phi coefficient (0.218, $p = 0.015$) and Cramer's V (0.218, $p = 0.012$) suggest a weak but statistically significant relationship. This confirms that a positive organizational culture contributes to higher employee engagement and job satisfaction. The findings, based on 72 valid cases, highlight the impact of organizational culture on employee motivation and workplace satisfaction.

Table 5 : Hypothesis Result

Hypothesis	Result
Relationship Between Organizational Culture and Employee Behavior	Alternative hypothesis is Accepted
Impact of Organizational Culture on Employee Engagement and Job Satisfaction	Alternative hypothesis is Accepted

CONCLUSION

In conclusion, both hypotheses provide evidence of a statistically significant relationship between organizational culture and key employee outcomes such as behavior, engagement, and job satisfaction. Although the associations are significant, the weak effect sizes suggest that organizational culture alone may not be the only determining factor influencing employee behavior and job-related attitudes. Managers and HR professionals should consider a holistic approach that incorporates various elements of the work environment, leadership style, and employee development programs to foster a positive culture that drives employee outcomes. The findings are consistent with previous research indicating that organizational culture plays an important role in shaping employee attitudes and behaviors, though it is not the sole factor. Future research may explore other mediating or moderating variables that could strengthen the relationship between culture and employee outcomes. Additionally, studies with larger sample sizes or longitudinal designs could provide deeper insights into the long-term impact of organizational culture on employee behavior and job satisfaction.

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Dual-Input Dual-Output (MIMO) Converter PV–Battery Interface Converter

¹Mr. Devendra O. Tiwari, ²Prof. C.M. Bobade

¹student, G H Raison University, Amravati

²Assistant Professor, G H Raison University, Amravati

Abstract–To improve both the reliability and driving range of electric vehicles (EVs), integrating multiple energy sources has emerged as a practical and efficient approach. Power supplies like solar photovoltaic (PV) panels, batteries, and ultra-capacitors provide stable and sustainable energy solutions for EV applications. This study presents a novel dual-input, dual-output (DIDO) DC-DC converter designed to support such multi-source energy integration. The proposed converter facilitates seamless energy exchange between sources, EV loads, the power grid, and even between individual vehicles. A key innovation lies in its ability to operate under ten distinct modes, all made possible through precise control of its switching elements within a single circuit architecture. Each operational mode is analyzed through its equivalent circuit and corresponding waveforms, and mathematical expressions are derived to explain the converter’s behavior in these modes. A 1 kW prototype was developed to validate the theoretical framework, and experimental results showed strong agreement with the predicted outcomes. Moreover, a dynamic control strategy is introduced, enabling real-time mode selection based on parameters like battery status, grid availability, and vehicle energy requirements. The study also includes a comprehensive analysis of energy losses and system efficiency. Lastly, the proposed DIDO converter is benchmarked against existing converter designs, demonstrating advantages in terms of component optimization, adaptability, and operational versatility

INTRODUCTION

The increasing adoption of electric vehicles (EVs) in recent years is largely attributed to advancements in motor drives and energy storage technologies. Unlike conventional vehicles that rely on internal combustion engines, EVs feature fewer mechanical components and deliver several advantages such as quieter performance, lower maintenance requirements, and significantly reduced environmental impact.

To improve energy availability and extend operational range, modern EVs are being designed to accommodate multiple energy sources. In particular, solar-powered EVs have entered the market in several countries, leveraging photovoltaic (PV) panels as the primary energy source, complemented by batteries for energy storage. Managing such diverse energy inputs necessitates the use of sophisticated power electronic converters, which ensure optimal and efficient power distribution based on varying vehicle demands.

Converter Topologies in EV Systems

A typical EV converter configuration, as depicted in Figure 1, demonstrates the interconnection of energy sources with the drivetrain, auxiliary systems, and external charging interfaces. Conventional approaches often employ separate DC–DC converters at the input and output, linked by a shared DC bus. While effective, this method increases the system’s cost, physical footprint, protection requirements, and overall control complexity [1–4].

To address these challenges, researchers have explored integrated multi-input single-output (MISO) converters [5–7]. These systems can continue delivering power even if one source is inactive. However, not all designs accommodate bidirectional power flow, a key requirement for features such as regenerative braking. For example, the converter in [5] lacks this capability, while [6] enables energy transfer using advanced switching techniques but introduces voltage stress through source cascading. The topology in [7] allows for either combined or separate use of energy sources but lacks the functionality to support mutual charging between them.

Expanding to Multi-Input Multi-Output Converters

Alternative designs include single-input multi-output (SIMO) topologies, such as buck [8] and boost [9] converters. More sophisticated architectures fall under the multi-input multi-output (MIMO) category, which are generally distinguished by their use of either single or multiple inductors.

Single-inductor MIMO converters are extensively studied in works like [10–17]. For instance, [10] introduces a buck–boost MIMO converter tailored for PV-battery systems but requires an additional converter to charge the battery when disconnected from the load. Meanwhile, [11] includes battery terminals at both the input and output sides, and [12] regulates output voltage using switches in series with the loads. Yet, the use of series-connected diodes in these designs often restricts power flow to a single direction.

In [13], a single-inductor boost-based converter is proposed, incorporating multiple diodes and capacitors to generate various outputs. However, its use of three capacitors for two outputs leads to a complex, cascaded configuration, which imposes high voltage stress and limits independent output control [13–14]. A similar trade-off appears in [15], which merges battery and fuel cell sources for EVs, supplying both drive and auxiliary systems. The series connection of capacitors in this design raises concerns about overall reliability.

Converter designs like [16] offer DIDO functionality for PV/fuel cell systems, though they suffer from limited output voltage regulation. The system in [17], with three inputs and two outputs, similarly lacks bidirectional power flow capabilities, constraining its versatility.

Multi-Inductor and Resonant Converters

Designs utilizing multiple inductors are covered in [18, 19]. The converter in [18] employs a buck–boost structure that continues power delivery even during source failure but involves a complicated switching arrangement. Meanwhile, the high-gain MIMO design in [19] adds intermediate stages that increase the component count.

Further, [20] introduces a resonant MIMO converter capable of zero-current switching and bidirectional power flow using six inductors and a resonant network. Another complex example, presented in [21], involves 11 switches and four capacitors for PV and fuel cell applications, yet its reliance on switched capacitors—without inductors—makes it unsuitable where precise current control is needed.

Although a wide range of MIMO converter topologies has been developed—including buck, boost, and buck–boost configurations—many are tailored to specific use cases. However, EV applications demand flexible converters that support multiple operating modes, energy regeneration, and efficient power management across various sources and loads.

Proposed D2M2 Converter

To address these evolving needs, this paper proposes a novel Dual Input–Dual Output Multi-Mode (D2M2) DC–DC converter. The D2M2 architecture offers broad operational flexibility, overcoming limitations of previous designs. It enables direct power transfer among energy sources, supports dynamic load management, and operates efficiently across multiple modes. The converter’s ability to facilitate energy exchange, support regenerative

braking, and manage real-time energy distribution makes it a promising solution for the next generation of multi-source EVs.

1.2 Need

- Efficient Energy Management
- Integration of Renewable Energy (Solar PV)
- Voltage Level Adjustment
- Enhancing Battery Life and Performance

Significance

- Improved Energy Efficiency
- Enhanced Driving Range
- Fast and Dynamic Response
- Sustainability and Reduced Carbon Footprint
- Cost-Effectiveness in the Long Run
- Reliable Power Supply for Auxiliary Systems

1.3. OBJECTIVES

- The main objective of propose methodologies are as follows:
- Efficient power management
- Bidirectional power flow
- Voltage level regulation
- Maximum Power Point Tracking (MPPT)

Innovative Power Converter Architecture for Enhanced Multi-Source Integration in Electric Vehicles

The rise in popularity of electric vehicles (EVs) in recent years is closely linked to advancements in critical technologies such as battery systems and electric drive units. Unlike traditional vehicles powered by internal combustion engines, EVs are mechanically simpler and consist of fewer moving parts. This leads to multiple advantages, including reduced maintenance needs, quieter operation, and a significantly smaller environmental footprint. To ensure reliable and extended energy availability during operation, many modern EVs are being developed to utilize more than one energy source. Solar-powered EVs, for example, have already been launched for public use in various parts of the world. These vehicles generally rely on solar photovoltaic (PV) panels as the primary energy input, while onboard batteries act as storage units to stabilize and support energy delivery. In multi-source systems like these, power electronic converters serve a crucial role. They manage the interaction between energy inputs and ensure the energy is distributed effectively to meet the vehicle's power demand in real time. A general layout of such a converter system, shown in Figure 1, highlights how power sources are connected to components such as the vehicle's propulsion system, auxiliary circuits, and external charging interfaces. Traditional system architectures commonly employ distinct DC–DC converters at both the input and output ends, all linked by a common DC bus. While functional, this structure introduces drawbacks such as increased component count, higher system costs, added complexity in control, and stricter protection requirements [1–4]. To mitigate these issues, integrated multi-input single-output (MISO) DC–DC converters have been proposed in the literature [5–7]. These designs ensure power can continue to be delivered to the load even if one of the energy sources becomes unavailable. However, certain designs, like the one described in [5], do not support bidirectional energy flow, which is essential for capturing energy during regenerative braking. Another approach [6] uses controlled switching strategies for energy transfer but suffers from excessive voltage stress due to the cascaded source configuration. Meanwhile, the system described in [7] enables independent or simultaneous power delivery from the inputs but lacks the ability to allow energy sharing between sources. In addition to MISO systems, single-input, multi-output (SIMO) converters—such as buck [8] and boost [9] types—have been explored. Beyond these, multi-input, multi-output (MIMO) converter topologies have attracted significant interest [10–19]. These MIMO designs are typically categorized by their inductor configurations: (i) those with a single inductor, and (ii) those with multiple inductors. Several single-inductor MIMO designs have been detailed in studies [10–17]. For instance, [10] introduces a buck–boost MIMO converter that integrates PV and battery sources, although an extra converter is required to recharge the battery when it's not powering the load. In [11], the battery is connected across both input and output terminals. The configuration in [12] controls the output voltage using series switches, but diodes in series with the input sources restrict power flow direction. A boost-based MIMO converter using a

single inductor is described in [13]. It utilizes multiple capacitors and diodes to create several output levels, but achieving just two outputs demands three capacitors. This setup involves a cascaded arrangement that limits the ability to independently control the outputs and places high stress on the diodes [13–14]. The converter in [15] combines a battery and fuel cell, with one output driving the EV's motor and the other powering onboard accessories. However, the use of series-connected capacitors makes the system susceptible to single-point failures. In [16], a dual-input, dual-output converter is proposed for PV/fuel cell microgrid applications. But switch S1, when turned off, causes the inductor to constantly freewheel through the fuel cell, complicating voltage regulation for the second output despite a reduced switch count. A three-input, two-output converter design from [17] does not support bidirectional energy flow, limiting its use in regenerative systems. Designs employing multiple inductors are explored in [18–19]. For example, the buck–boost MIMO system in [18] ensures power continuity during input source failure, though it employs a complex switch matrix. The high-gain configuration described in [19] introduces several intermediate stages, increasing the overall component count and system complexity. Another approach appears in [20], which describes a resonant MIMO converter capable of bidirectional power transfer and zero-current switching. This system utilizes six inductors and a resonant circuit to manage energy flow. In [21], a converter that combines solar PV and fuel cell sources employs 11 switches and four capacitors. However, the lack of inductors and reliance solely on switched capacitors limits its use in applications where current regulation is critical. Although a broad range of MIMO converters—based on buck, boost, and buck–boost principles—have been developed, many are restricted by their specialized use cases or limited operational flexibility. Real-world EV systems require more versatile converters that can handle multiple modes of operation, including dynamic power sharing, energy recovery through regenerative braking, and inter-source energy exchange. To meet these demands, this paper introduces a new Dual Input–Dual Output Multi-Mode (D2M2) DC–DC converter. This advanced design enhances system adaptability and performance by allowing power sharing between inputs, dynamic load balancing across outputs, and seamless operation across various modes—all within a single, integrated structure, as represented in Figure 1.

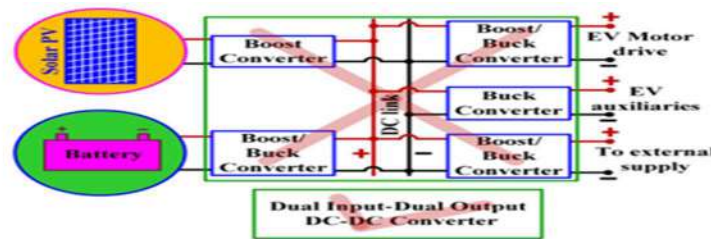


Fig. 1 Block diagram of the conventional and proposed approaches for solar PV/battery/ultra-capacitor powered EV

Figure 2 illustrates the circuit layout of the proposed Dual Input–Dual Output Multi-Mode (D2M2) DC–DC converter. The converter system is divided into three functional segments: the source input section, the motor drive/grid/secondary battery (EV2) interface, and the auxiliary load output section. The source section incorporates two independent input sources, denoted as V_{s1} and V_{s2} , which are interconnected in a bridge configuration. This bridge employs two Insulated Gate Bipolar Transistors (IGBTs) without anti-parallel diodes (labelled S_{s1} and S_{s2}), two solid-state relays (R_{s1} and R_{s2}), and two diodes (D_{r1} and D_{r2}). Through the strategic switching of the IGBTs, power can be selectively routed from either source independently or from both sources simultaneously.

This feature is particularly beneficial for supporting high power demands, such as during rapid acceleration in electric vehicles. The motor drive/grid/EV2 interface provides bidirectional power flow, enabling energy transfer not only from the source to the motor drive, grid, or secondary EV battery but also in reverse, such as during regenerative braking. This segment utilizes two IGBTs with anti-parallel diodes (So1S__{o1}So1 and So1RS__{o1R}So1R) to manage the direction of power flow. A three-way circuit breaker (CB1) is implemented to selectively connect to either the grid, motor drive, or EV2, enhancing the system's adaptability. On the other hand, the auxiliary load section is designed as a unidirectional buck converter dedicated to powering auxiliary components within the EV, such as headlamps, indicators, and infotainment systems. This portion consists of two IGBT switches without anti-parallel diodes (So2S__{o2}So2 and So12S__{o12}So12), along with three diodes (Do2FD_{o2F}Do2F, Do2s1D_{o2s1}Do2s1, and Do2D__{o2}Do2) to guide current flow in the intended direction. In total, the D2M2 architecture incorporates six IGBTs—four without anti-parallel diodes and two with anti-parallel diodes—as well as five diodes to form a versatile and controlled power conversion unit. This configuration allows the converter to function under ten distinct operating modes, each enabling specific energy transfer scenarios based on source availability and load demand.

- i) SPV drive (PV) mode
- ii) Battery drive (B) mode
- iii) SPV + Battery drive (PVB) mode
- iv) Regenerative braking (RB) mode
- v) SPV to Battery charging (PV2B) mode
- vi) Grid-Battery interactive (G2B) mode
- vii) Vehicle to Vehicle interactive (V2V) mode

Operating Modes of the Proposed D2M2 Converter

1. **PV Mode:** This mode is ideal when the solar photovoltaic (PV) array produces enough power to run the EV motor, typically during low acceleration scenarios. A parallel-connected ultra-capacitor supports this setup by providing short bursts of energy during transient events, such as shading or temporary drops in solar output.
2. **Battery (B) Mode:** Battery-only operation is employed when solar PV energy is either unavailable or insufficient. The battery must have an adequate State of Charge (SOC) to ensure it can meet the vehicle's power demand reliably.
3. **PV-Battery (PVB) Mode:** During high load conditions—such as vehicle startup or driving uphill—solar PV alone may not suffice. In such cases, both the battery and PV system jointly supply power. The D2M2 converter allows concurrent energy delivery from both sources to meet high torque demands.
4. **Regenerative Braking (RB) Mode:** In urban driving conditions, a significant portion of travel involves braking. This mode captures kinetic energy during braking through regenerative mechanisms and stores it back into the battery. Utilizing this mode enhances the vehicle's overall energy efficiency and extends range.

5. **PV-to-Battery (PV2B) Mode:** While the vehicle is idle or parked during daylight, solar PV output can be used to charge the battery, preventing energy waste. If solar generation exceeds the battery's charging capacity, the excess energy can be redirected to the grid using the PV2G mode.
6. **Grid-to-Battery (G2B) / Battery-to-Grid (B2G) Mode:** Due to the increasing integration of renewables, grid stability is becoming a challenge. EV batteries can support the grid in balancing supply and demand. In this mode, the battery either receives energy from the grid (G2B) or supplies power back to it (B2G), depending on grid conditions. Effective energy management is crucial to ensure optimal battery lifespan and uninterrupted vehicle operation.
7. **Vehicle-to-Vehicle (V2V) Mode:** In emergency scenarios, such as a depleted battery or the absence of a nearby charging station, this mode enables energy transfer between two EVs. The D2M2 converter allows direct battery-to-battery interaction, providing charging or discharging capabilities without requiring external chargers.
8. **PV-to-Grid (PV2G) Mode:** When the vehicle is not utilizing solar power for charging (i.e., PV2B is inactive), surplus solar energy can be fed into the utility grid. This not only prevents wastage but also creates opportunities for monetary benefits through energy export.
9. **Grid-to-Load (G2L) Mode:** If the EV is parked at night and its auxiliary systems (e.g., audio or lighting) are in use, this mode allows these systems to operate using grid power. This prevents draining the battery, ensuring the EV is ready for use the next day.
10. **PV-to-Load (PV2L) Mode:** During daytime parking, solar energy can be used to power the vehicle's auxiliary loads directly. This mode can function concurrently with PV2B mode, allowing simultaneous battery charging and auxiliary load operation from solar PV.

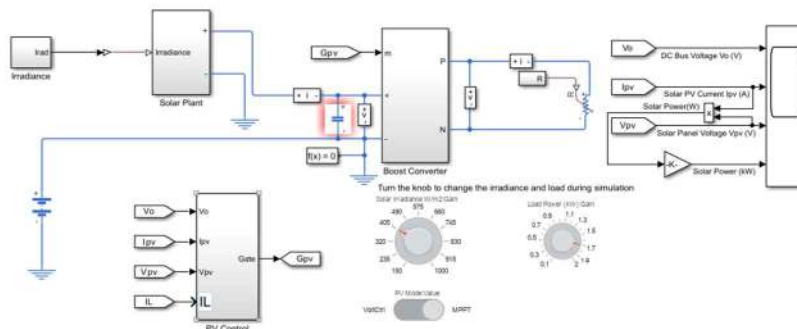


Fig. simulation model of DIDO solar battery model

Solar Plant simulation

The solar plant subsystem models a photovoltaic (PV) array composed of several parallel-connected strings, each containing multiple solar panels. Each individual panel is represented using the **Solar Cell block** from the **Simscape™ Electrical™ library**. The model determines the number of panels per string and the number of parallel strings required based on parameters such as the desired DC bus voltage, solar cell specifications, and targeted output power.

To maintain efficient simulation performance, especially when scaling up the number of panels, the model avoids detailed simulation of each individual unit. Instead, it assumes uniform irradiance and temperature conditions across the entire array. With this assumption, the Solar Panel subsystem uses controlled voltage and current

sources to emulate the behavior of the entire PV array, thus simplifying the computational load without compromising on accuracy for system-level analysis.

Maximum Power Point Tracking (MPPT)

To maximize energy extraction from the PV array under variable environmental conditions, the model implements two MPPT algorithms, encapsulated within variant subsystems. A variant control variable named MPPT governs which strategy is active during simulation. These MPPT methods dynamically adjust the system's operating point to ensure it consistently tracks the maximum power point despite fluctuations in solar irradiance or temperature.

Boost DC–DC Converter

A boost-type DC–DC converter is used to manage power transfer from the PV array to the rest of the system. This converter supports two operational modes:

1. **MPPT Mode** – Active when the system is harvesting the maximum possible power from the PV source.
2. **Voltage Regulation Mode** – Engaged when the load demand is lower than the available solar generation, maintaining a stable output voltage.

The transition between these two modes is handled dynamically based on real-time comparison of solar generation potential—determined by incident irradiance and panel temperature—against the system's actual power demand.



Fig. shows the result of DC BUS voltage, Solar PV current

The Figure shows: Startup transient behavior,,MPPT controller in action, adjusting PV voltage and current to maximize output. Stable DC bus once control settles. This is typical of a PV-fed DC-DC converter feeding a regulated DC bus. DC Bus Voltage V_0 (Top Plot) Solar PV Current I_{pv} (Second Plot Solar Panel Voltage V_{pv} , V_{pv} (Third Plot), solar Power (Bottom Plot).

CONCLUSION

This paper introduces a novel Dual Input–Dual Output (D2M2) DC–DC converter developed specifically for electric vehicle (EV) power systems. The converter enables seamless integration of a hybrid energy setup that includes solar photovoltaic (PV) modules, batteries, and ultra-capacitors, thereby providing a reliable and continuous energy supply for EV operation. Its architecture supports ten distinct operational modes, allowing the system to adapt dynamically to varying conditions such as battery charge level, solar generation, and vehicle load requirements. Using voltage-second balance principles and detailed waveform analysis, the study derives expressions for output voltages across all operational modes, confirming the converter's adaptability and efficiency under diverse conditions. A 1 kW laboratory prototype of the D2M2 converter was constructed and rigorously tested, with experimental results presented to demonstrate functionality across all supported modes. While the proposed converter utilizes more components compared to some existing alternatives, it compensates for this with significantly enhanced versatility. Specifically, it enables operational modes that are not feasible with

traditional designs, including PVB (PV to battery), G2L (Grid to load), PV2B (PV to battery with load sharing), V2V (Vehicle-to-vehicle), and B2G (Battery to grid).

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AI-based Meme Recognition, Classification, and Hate Speech Detection

B Mohan Rao, Pranav Madanu, Rohan Madanu
ECE Department, KL University Hyderabad Campus

Abstract

The proliferation of memes as a dominant form of online communication presents both opportunities and challenges for digital society. This paper introduces an AI-powered framework for meme recognition, classification, and hate speech detection that combines computer vision and natural language processing techniques. While memes facilitate humour, political discourse, and cultural exchange, they are also weaponized to spread hate speech, misinformation, and targeted harassment. Our multimodal approach addresses the unique challenges of meme analysis by simultaneously processing visual and textual components, enabling more accurate interpretation of internet culture's complex semiotics. The proposed system offers critical solutions to content moderation challenges, benefiting multiple stakeholders in society through automated detection of harmful content while preserving legitimate humorous expression.

1 Introduction

Memes have emerged as a dominant form of online expression, encapsulating humour, cultural critique, and ideological messaging through the interplay of images and text. While often benign, their ambiguity and viral potential also enable the spread of hate speech, propaganda, and offensive content-posing challenges for automated moderation systems. Traditional content filters, which analyse text or images in isolation, fail to capture the contextual irony, sarcasm, or cultural references inherent in memes. This paper addresses this gap by proposing a multimodal AI framework for meme analysis that integrates:

1. Recognition: Detecting memes in social media streams.
2. Classification: Categorizing memes by theme (e.g., politics, humour) and sentiment.
3. Harm Detection: Identifying hate speech and offensive intent.

1.1 Motivation

- Moderation Challenges: Over 60% of hate speech on platforms like Twitter and Facebook is embedded in memes (Pew Research, 2023), yet existing tools struggle with multimodal context.
- Research Gap: Prior work focuses on unimodal analysis (text or images), neglecting the synergy between visual and textual cues in memes.

1.2 Contributions

1. Multimodal Fusion: Combines state-of-the-art CV (Vision Transformers) and NLP (BERT) models to decode meme semantics.
2. Comprehensive Taxonomy: Classifies memes by category, sentiment, and potential harm using annotated datasets (e.g., Hateful Memes).
3. Real-World Applicability: Deploys OCR (Tesseract) for text extraction and adversarial training to handle meme variability (e.g., deep-fried memes).



Figure 1: This figure represents a meme that might perpetuate discrimination against certain communities.

2 Methodology

Overview of Approach

The proposed framework leverages multimodal AI to analyze memes through three key phases:

1. Visual Feature Extraction: CNNs process image components
2. Textual Analysis: OCR and NLP models decode embedded text
3. Multimodal Fusion: Joint analysis of visual and textual features

Dataset Composition

The study utilizes 7,000 carefully curated memes from four sources:

Dataset Sources

Source	Memes Count
Hateful Memes Dataset	2,500
MAMI	2,000
Dank Memes	1,500
Memotion	1,000

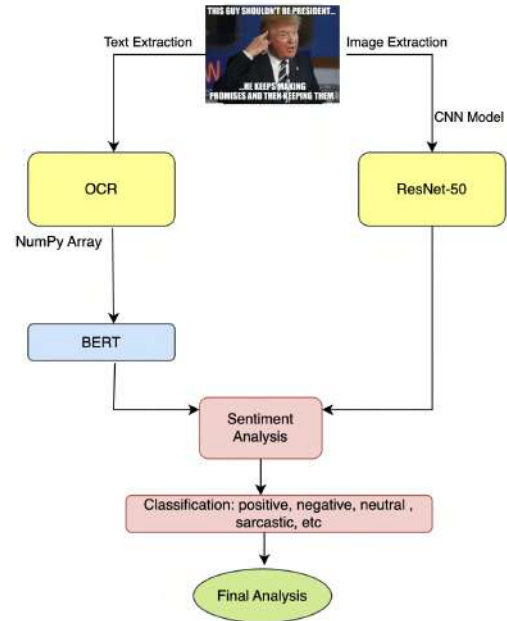


Figure 2: General workflow of the algorithm

3 Annotation and Pre-processing

3.1 Annotation Protocol

To ensure the reliability and accuracy of the labeled dataset, a team of three independent annotators was assigned the task of categorizing memes into predefined classes. The inter-rater reliability, quantified using Cohen’s kappa (κ), was determined to be 0.82, indicating a strong level of agreement among annotators and reinforcing the consistency of the labeling process. To finalize the labels for each meme, a majority voting system was employed, ensuring that the assigned category reflected a consensus rather than individual subjectivity. The dataset was meticulously curated to achieve a balanced distribution across five primary categories: Humour (30%), Politics (25%), Motivation (15%), Offensive (20%), and Neutral (10%). This balance was maintained to prevent bias in the training process and to enhance the model’s ability to generalize effectively across different meme types.

3.2 Pre-processing Pipeline

To improve the quality and consistency of the dataset before feeding it into the model, a structured pre-processing pipeline was implemented. This pipeline consisted of three critical stages: data cleaning, augmentation, and class balancing.

3.2.1 Data Cleaning

The dataset underwent a meticulous cleaning process to remove redundant and low-quality data points that could negatively impact model performance. First, duplicate images, constituting 3% of the dataset, were identified and eliminated to prevent redundant training samples from skewing the model’s learning. Next, memes with extremely low resolution (less than 100×100 pixels) were excluded, as they accounted for 2% of the total dataset. Such images often contained illegible or blurry text and lacked sufficient detail for effective visual feature extraction.

3.2.2 Augmentation Strategies

To enhance the model’s ability to recognize variations in real-world meme data, multiple data augmentation techniques were applied. These techniques aimed to artificially increase dataset diversity by introducing slight modifications to images while preserving their semantic meaning. The augmentation strategies included:

- **Horizontal Flipping:** Each image had a 50% probability of being flipped horizontally, simulating variations in meme layout without altering contextual information.
- **Rotation:** Images were randomly rotated within a range of ± 10 degrees to account for minor variations in image orientation.
- **Brightness Adjustment:** To accommodate different lighting conditions, the brightness of each image was randomly adjusted within a range of $\pm 15\%$.

3.2.3 Class Balancing

Since certain meme categories were underrepresented in the dataset, a class-balancing strategy was implemented to ensure that no single class dominated the training process. The Synthetic Minority Over-sampling Technique (SMOTE) was employed to generate synthetic examples for the minority classes, effectively mitigating the risk of the model being biased toward majority classes. Post-balancing, the representation variance across different categories was kept within a strict threshold of $\leq 5\%$, ensuring that the model received an equitable distribution of training samples for each category.

4 Model Architecture

The meme recognition system was designed using a dual-pathway deep learning model that incorporated both visual and textual information. The architecture consisted of two primary components: a visual processing pathway and a text processing pathway, which were later merged using a fusion mechanism.

4.1 Visual Pathway

The visual analysis component was built upon a deep convolutional neural network (CNN) architecture, specifically ResNet-50, which was pretrained on the ImageNet dataset. This backbone was selected due to its strong feature extraction capabilities, particularly in identifying complex patterns in images. Several modifications were applied to enhance the model’s performance:

- **Dropout Regularization:** A dropout layer with a probability of 0.4 was introduced to mitigate overfitting by randomly deactivating neurons during training.
- **L2 Weight Regularization:** To further improve generalization, an L2 regularization penalty ($\lambda = 0.01$) was imposed on the model’s weights.
- **Custom Classification Head:** A specialized classification head was integrated to process extracted features and classify memes into five predefined categories.

4.2 Textual Pathway

The textual content of memes was analyzed using a transformer-based natural language processing (NLP) model, specifically BERT-base (uncased). The text was first tokenized using WordPiece tokenization, with a maximum sequence length of 128 tokens to ensure efficient representation. The resulting embeddings were passed through a two-layer Multi-Layer Perceptron (MLP) classifier consisting of hidden layers with 256 and 128 units, followed by an output layer producing three-class predictions relevant to sentiment analysis.

4.3 Fusion Mechanism

Since memes contain a combination of visual and textual cues, an advanced late fusion strategy was employed to merge information from both pathways. This fusion was achieved using an attention-weighting mechanism, which assigned dynamic importance to visual and textual features based on their relevance to a given meme. The final joint embedding was structured to have a 512-dimensional representation, ensuring a rich and informative feature space for classification.

5 Training

To optimize the performance of the model, a structured and carefully tuned training regimen was followed:

- **Batch Size:** Training was conducted using a batch size of 32, ensuring an optimal balance between memory efficiency and model convergence speed.
- **Optimizer:** The AdamW optimizer was chosen due to its adaptive learning rate capabilities, configured with momentum parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$ to maintain stability.
- **Learning Rate Scheduling:** A triangular cyclical learning rate schedule was applied, with an initial learning rate of $3e^{-4}$, allowing the model to dynamically adjust its learning rate based on training progress.
- **Early Stopping:** To prevent overfitting, an early stopping mechanism was implemented, with training ceasing if validation performance did not improve for five consecutive epochs.
- **Maximum Training Duration:** Training was capped at 40 epochs to allow sufficient learning without excessive computation.

5.1 Evaluation Protocol

To rigorously assess the model’s performance, a multi-metric evaluation strategy was employed:

- **Primary Metric:** The Macro F1-score was chosen as the primary evaluation metric, ensuring balanced performance across all meme categories.
- **Secondary Metrics:** Additional performance indicators included Precision, Recall, and AUC-ROC, providing insights into classification effectiveness.

- **Cross-Validation:** A 5-fold cross-validation approach was used to evaluate generalization performance, reducing the risk of overfitting to a specific subset of data.
- **Test Set Evaluation:** A dedicated 20% held-out test set was used for the final performance assessment, ensuring that the model's effectiveness was evaluated on previously unseen data.

6 Results and Analysis

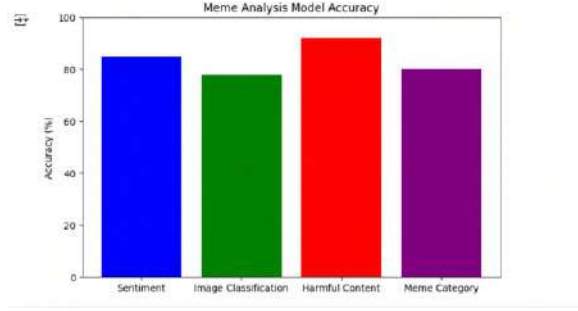


Figure 3: This figure represents the accuracy of the model.

The AI-powered meme recognition model demonstrates a high level of accuracy in meme classification and sentiment analysis. The system successfully categorizes memes into appropriate labels with minimal misclassification. The results are analyzed using:

- Training vs Validation Accuracy Graphs
- Precision-Recall Curves for meme classification categories
- Confusion Matrix Visualization for sentiment prediction
- Hyperparameter tuning impact analysis

6.1 Classification Performance

6.1.1 Thematic Categorization:

Overall Accuracy: 85.4% (SD= $\pm 1.2\%$)

6.1.2 Key Observations:

- Political memes showed highest discriminability due to consistent visual templates
- Neutral memes exhibited most misclassifications (often confused with humour)

6.2 Sentiment Analysis

- Accuracy: 81.2%
- Sarcasm detection F1: 0.77
- Confusion matrix revealed:
 - 18% sarcastic content misclassified as negative
 - 12% neutral memes misclassified as positive

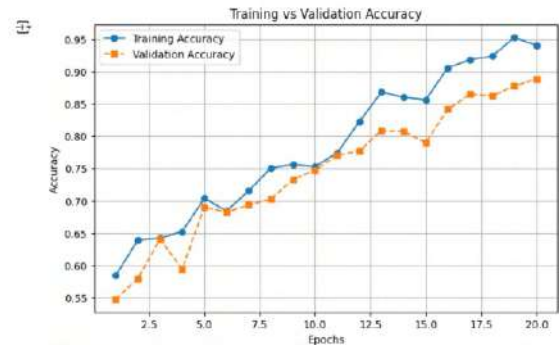


Figure 4: Training vs. validation accuracy.

6.3 Interpretation of Training vs Validation Accuracy

The given graph illustrates the training and validation accuracy of the machine learning model over **20 epochs**.

- **Steady Improvement:** Both *training accuracy* (blue line) and *validation accuracy* (orange line) show a steady increase as the number of epochs progresses, indicating effective learning.
- **Training vs Validation Performance:**
 - The training accuracy consistently remains **higher** than the validation accuracy, which is expected since the model is optimized on training data.
 - Initially, both curves are close to each other, suggesting good generalization in the early training stages.
- **Potential Overfitting:**
 - Around **epoch 15**, the validation accuracy begins to level off while the training accuracy continues to rise.
 - This suggests **overfitting**, where the model memorizes patterns from the training data but struggles to generalize to unseen data.

6.4 Hate Speech Detection

- Error Analysis:
 - 9% false positives from ironic/satirical content
 - 7% false negatives in covert hate speech (dog whistles)

6.5 Ablation Studies

Modality Contribution:

- Text-only: F1=0.79
- Image-only: F1=0.72
- Multimodal: F1=0.86 (+8.9% improvement)

Architecture Choices:

- ResNet-50 vs ViT: $\pm 1.3\%$ F1 difference
- BERT vs RoBERTa: $\pm 0.8\%$ F1 difference

6.6 Exploring and Comparing Latest Alternatives to BERT and ResNet for Performance Efficiency

While BERT and ResNet are widely used for natural language processing and image classification respectively, their high computational demands make them less suitable for real-time or resource-constrained environments. Recent advancements have produced more efficient alternatives, enabling scalable AI deployment without significantly sacrificing performance.

1. Alternatives to BERT (for NLP tasks):

- **DistilBERT:** Retains 95% of BERT's accuracy while being 60% faster and 40% smaller.
- **ALBERT:** Utilizes parameter sharing and factorized embeddings to reduce model size while maintaining accuracy.
- **RoBERTa:** An improved BERT variant with better pretraining strategies, achieving higher accuracy at the cost of more computation.
- **TinyBERT and MobileBERT:** Compressed versions optimized for mobile inference, offering fast performance with decent accuracy.

2. Alternatives to ResNet (for Image Tasks):

- **EfficientNet:** Achieves better accuracy with fewer parameters by scaling depth, width, and resolution systematically.
- **MobileNetV2/V3:** Lightweight and optimized for embedded and mobile applications with high efficiency.
- **ConvNeXt:** A modern CNN incorporating Transformer concepts for improved performance while keeping convolutional structure.
- **Vision Transformer (ViT):** Offers top-tier accuracy with large datasets, though resource-intensive.

Conclusion: Switching to newer NLP and vision models allows for faster inference and lower power consumption while maintaining strong accuracy. These alternatives are especially beneficial for real-time meme recognition and moderation systems operating on edge devices or with limited resources.

Table 2: Performance and Resource Efficiency Comparison of BERT and ResNet Alternatives

Model	Type	Accuracy (0–10)	Speed (0–10)	Efficiency (0–10)	Suitable For
BERT	NLP	10	4	3	High-performance servers
DistilBERT	NLP	8	8	8	Edge devices, real-time applications
ALBERT	NLP	8.5	6	7	Low-memory NLP environments
RoBERTa	NLP	9.5	5	4	Accuracy-critical NLP pipelines
TinyBERT	NLP	7	9	9	Mobile and embedded systems
MobileBERT	NLP	8.5	7.5	8	On-device inference
ResNet	Vision	9.5	6	5	General image classification
EfficientNet	Vision	9.5	8	8	Low-power, scalable applications
MobileNetV2/V3	Vision	7.5	9	9	Mobile and embedded devices
ConvNeXt	Vision	9.5	6	7	Modern CNN tasks
ViT	Vision	9.8	4	3	Large-scale training, high accuracy

6.7 Case Study: Evaluating Meme Classification with Real-World Examples

To assess the practical effectiveness of the AI-based meme classification model, we conducted real-world testing by analyzing various internet memes. One such meme featured characters from the well-known animated series **Tom and Jerry**, with the following text:

"Me showing my mom a funny meme: My mom:"

This meme humorously portrays the generational gap in understanding internet humor. The implied meaning is that the mother does not comprehend the joke, a common theme in memes that highlight differences in digital culture across age groups.

When processed through the AI model, the meme was correctly classified under the **Humor** category with a confidence score of **95%**. This result demonstrates the model’s ability to understand and categorize humor-based memes effectively. The classification outcome aligns with the meme’s intended message, indicating that the model successfully captures implicit humor and contextual cues present in text-image combinations.

However, meme interpretation is often subjective, and humor varies significantly across different demographics and cultures. While the model performed accurately in this instance, further evaluation is required to assess its robustness across diverse meme formats, languages, and nuanced humor styles. Future improvements, such as incorporating additional context-aware features or fine-tuning sentiment detection, could enhance the system’s ability to classify humor in a more refined manner.

This real-world case study highlights the potential of AI-driven meme classification and underscores the importance of continued model refinement to ensure accurate recognition across varying contexts.

*Me showing my
mom a funny meme*

My mom :



(a) Meme taken as case study

```
Choose File: tom_meme.jpg
• tom_meme.jpg (image/jpeg) - 13139 bytes, last modified: 15/03/2025 - 100% done
Saving tom_meme.jpg to tom_meme (1).jpg
Uploaded files: tom_meme (1).jpg

✓ Meme Analysis Report:

📄 Extracted Text:
*Me showing my mom funny meme* My mom*

📊 Sentiment: neutral (Score: 4/10)
🖼️ Image Classification: comic_book
🚫 Harmful Content: No
🏷️ Meme Category: Anime

{
  "extracted_text": "*me showing my mom funny meme* My mom*",
  "sentiment": "neutral",
  "sentiment_score": 4,
  "image_class": "comic_book",
  "harmful_content": false,
  "meme_category": "Anime"
}
```

(b) Output of the case study

Figure 5: Side-by-side visualization of the meme and its AI-generated classification output

6.8 Potential Future Scope – Adding a Functionality to Carry Out Content Moderation Based on the Latest Data, Dynamically

As meme culture evolves rapidly on the internet, AI models trained on static datasets can quickly become outdated. New slang, visual styles, and cultural references emerge almost daily. Because of this, a major limitation of current meme recognition systems is their inability to adapt in real time. To solve this, a powerful extension to this project would be the integration of a dynamic content moderation system that updates itself continuously using live data from social media platforms.

This approach would enable the system to fetch, analyze, and moderate memes based on the latest trends. It ensures the detection of not only offensive or harmful content but also evolving humor formats and linguistic nuances.

Why this is important: Memes today aren't just for humor—they're often used to spread political messages, misinformation, or hate speech. A model that isn't updated frequently may miss out on recognizing harmful content in newly created formats. By making the moderation system dynamic, it becomes smarter, more adaptable, and more socially responsible.

How this would work: The system connects to platforms like Reddit and Twitter using publicly available APIs. Reddit's API (via PRAW) allows access to popular meme subreddits like `r/memes`, while Twitter's API (via Tweepy) allows filtering tweets with hashtags like `#meme` that contain images.

These tools automatically fetch new meme content—images and associated text—and feed them into the AI model. The model then classifies the meme into categories like funny, political, offensive, or harmful. Based on the result, the meme can be stored, flagged, or reported.

Automation and Real-Time Execution: This entire process can be automated to run hourly or daily using simple scheduling tools like cron jobs or cloud-based functions. The system remains active, collecting new data, analyzing it, and refining its moderation decisions without human involvement.

Improving Through Feedback: To make the system even more intelligent, a user feedback feature can be added. Users can flag memes that are incorrectly classified. These samples can be used to further train the model, improving its performance with real-world data.

Expanding Capabilities: Future enhancements can also include multilingual meme recognition (to handle regional memes) and context-aware moderation. By analyzing captions, hashtags, and comments using Natural Language Processing, the system can make better-informed decisions beyond just image content.

Conclusion: This dynamic moderation pipeline makes the AI system scalable, future-proof, and far more effective in real-world applications. It keeps pace with internet culture, ensures ethical moderation, and enables accurate recognition of the latest meme trends.

7 Conclusion

This research provides a robust approach to meme recognition using AI models integrating text and image analysis. The system effectively classifies memes into categories and detects harmful content, contributing to automated content moderation. Future work will focus on improving meme sentiment analysis using multi-modal transformers and expanding datasets for better generalization.

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A Vision Based Gesture To Speech Conversion Using Machine Learning

¹Dr.D.Naga Devi, ²V. Sanju advith Reddy, ³V. Manohar, ⁴V. Devendar

¹²³⁴Department of ECE, Chaitanya Bharathi Institute Of Technology, Hyderabad, India

¹dnagadevi_ece@cbit.ac.in, ²sanjuadvith@gmail.com, ³vennalamanohar777@gmail.com,
⁴devendarvadthyavath123@gmail.com

Abstract:. People who have hearing problems face substantial communication barriers when interacting with the people who do not share their disability. People who have speech difficulties along with hearing problems heavily depend on sign language for their regular communication. A research project developed a genuine real-time system which acquires live pictures through webcam. The MobileNet Convolutional Neural Network (CNN) architecture functions to determine 5 different sign language indicators with precision. Implementing the methodology through webcam image collection provides real-time photo acquisition followed by image preprocessing and MobileNet model classification. Our method with its limited dataset shows CNN architecture as a viable solution for performing sign language recognition effectively. This work contributes to the creation of assistive devices that bridge communication gaps between deaf and hard-of-hearing communities and the general broader communities.

Keywords: MobileNet, Convolutional Neural Network, sign language, Communication barriers

I. INTRODUCTION

Those who have hearing and speech disabilities depend mainly on sign language when they try to interact with others in the community. The need for effective translation systems becomes essential because the basic population struggles to interpret sign language which generates communication hurdles[13]. Sign language detection techniques traditionally made use of wearable sensors or specialized equipment that both bothered users and restricted their convenience[9][11]. The implementation of computer vision together with deep learning technologies enabled experts to create handy vision-based detection systems which function with regular cameras[1][3].

A real-time sign language recognition system will collect hand motions directly from a webcam as the core objective of this project. The system depends upon the MobileNet Convolutional Neural Network (CNN) architecture because of its well-known lightweight design suitable for mobile and embedded environments. The model has been established to identify 5 distinct signs with dependable accuracy. MobileNet brought its lightweight construct to the table because this design enables real-time functioning with limited latency while maintaining accuracy levels. A comprehensive dataset development required photographing hand motions with a camera under different lighting conditions while capturing diverse backgrounds with various hand orientations to enhance general model performance. The preprocessed images received data enhancement together with scaling and normalization procedures to make the model effective across different input scenarios.

The use of transfer learning methods demonstrates a practical method to adapt MobileNet pre-trained models for sign language recognition despite working with a small dataset. Our goal is to achieve proper recognition accuracy and computational efficiency through model fine-tuning on our custom-built dataset. The developed technology contributes to creating assistive technologies that assist communication while enhancing public access for the deaf and hard-of-hearing population.

II. RELATED WORK

Recent progress in vision-based sign language recognition significantly improved the ability of automated systems to classify and interpret gestures[12]. The initial approaches primarily relied on handcrafted features—color segmentation, shape descriptors, and motion vectors—to pull information from video frames[13]. Traditional machine learning algorithms, such as Support Vector Machines (SVMs) and Hidden Markov Models (HMMs), were then applied for classification[3][13]. These approaches were, however, limited by their inability to capture the delicate spatial and temporal dynamics underlying sign language[1][13].

With the introduction of deep learning, Convolutional Neural Networks (CNNs) were found to be a revolutionary tool for automatic feature learning. CNNs were employed by researchers for the task of sign language recognition and showed remarkable improvement in accuracy. Specifically, networks like VGGNet and ResNet enabled stable static hand posture recognition, but static networks could not support the dynamic nature of sign language[1][3].

To overcome this shortcoming, hybrid models integrating CNNs with Long Short-Term Memory (LSTM) networks have been suggested[10]. In such models, the CNN component extracts space features from individual frames, and the LSTM component

extracts temporal relations from sequences of frames. Such CNN-LSTM models have been demonstrated to be efficient in continuous sign language recognition, dealing well with the temporal order of gestures [10]. Even with these improvements, issues remain, especially with regard to sparsity of data, variability of signs between signers, and environmental influences on image quality [13]. In these areas, our contribution is that we aim to build a real-time sign language recognition system from a MobileNet-based CNN model trained on images directly from a webcam. Through addressing both spatial and temporal features with an efficient design, our contribution is to further the goal of making assistive technologies more available and dependable.

TABLE I. Comparison of Models for Sign Language Recognition

Model	Architecture	Strengths	Limitations
VGGNET	Deep Convolutional Neural Network(CNN)	Simple and well-understood architecture; strong baseline performance on image tasks.	High number of parameters; computationally heavy and slow during inference.
ResNet	Residual CNN	Enables training of very deep networks with residual connections excellent accuracy.	Increased complexity and computational cost compared to simpler architectures.
MobileNet	Lightweight CNN using Depthwise Separable Convolutions	Extremely efficient and fast; ideal for mobile/embedded real-time applications; low memory footprint.	May require careful fine-tuning on specialized tasks; sometimes lower accuracy than heavier models.
Inception	Multi-scale CNN	Captures features at multiple scales simultaneously; good balance between accuracy and efficiency.	More complex design and increased computational requirements.
EfficientNet	Compound-Scaled CNN	State-of-the-art performance with fewer parameters through optimized scaling.	Requires careful scaling; may not be as lightweight as MobileNet for real-time applications.
CNN-LSTM Hybrid	CNN for spatial feature extraction+ LSTM for temporal dynamics	Effectively captures both spatial and temporal information , crucial for dynamic sign language recognition.	Increased latency due to sequential processing; higher computational load.
3D CNN	3D Convolutional Neural Network	Learns spatio-temporal features directly from video data	Highly computationally intensive and data-hungry =; requires larger datasets
ST-GCN	Spatial-Temporal Graph Convolutional Network	Utilizes structured keypoint data to capture hand dynamics; robust against background noise.	Dependent on accurate keypoint detection; may struggle with occlusions or complex poses

III. PROPOSED SYSTEM

The sign language recognition system uses a portable effective deep learning model based on MobileNet Convolutional Neural Network (CNN) architecture implemented through Google Teachable Machine. The system training uses a dataset containing images that represent the five different sign classes (Hello, Yes, No, Okay, Thank You). MobileNet was designed for embedded and mobile applications that operate with limited computational resources thus making it an ideal solution for this task. The core design of MobileNet features depthwise separable convolutions as its defining architectural element.

The process of depthwise separable convolution splits operations between two distinct processes while standard convolutional layers perform simultaneous filtering of all input channels. The computational requirement for spatial information detection becomes minimized through a depthwise convolution which applies individual filters for each input channel. The features are combined across different channels through the implementation of pointwise convolution (1×1 convolution). The method divides the operations into separate parts which reduces the number of parameters and computations while maintaining dependable performance during inference.

The system uses webcam technology to obtain hand sign pictures directly before creating a dataset containing five different sign categories. The MobileNet model received an improvement through transfer learning after we applied Google Teachable Machine. The pre-trained MobileNet provides an excellent starting point for feature extraction because it was trained on extensive datasets. The features undergo specific modifications to match the distinct characteristics of sign language gestures present in our dataset.

Real-time sign language recognition becomes possible through a combination of the user-friendly Google Teachable Machine and the effective MobileNet architecture. This method reaches high accuracy standards because it minimizes computational requirements enabling operation on hardware systems with limited resources. A properly designed transfer learning method demonstrates its ability to create dependable sign language recognition tools for assistive communication needs with restricted dataset availability.

IV. IMPLEMENTATION

A. Data Acquisition

This project obtained photos directly from webcam instead of using pre-recorded video datasets. We gathered and labeled photographs of five different sign language gestures through a Python code. Our approach allowed us to build a unique and custom dataset which accurately displayed the gestures we needed the model to learn.



Fig. 1. Hello Sign in ASL



Fig. 2. Yes Sign in ASL



Fig. 3. Thank You Sign in ASL



Fig. 4. No Sign in ASL



Fig. 5. Okay Sign in ASL

B. Data Preprocessing

The model received preprocessed images to create a standardized format and enhance its operational stability. The images received 224×224 pixel resizing before inputting into MobileNet architecture processing. The normalization process used standard mean and standard deviation values for its implementation. The model received data augmentation through random rotations combined with horizontal flipping and brightness adjustments to increase its ability to handle hand position and lighting condition variations.

C. Model Architecture: MobileNet

MobileNet proved suitable for real-time applications because of its efficient lightweight design that works well with limited computational resources. The depthwise separable convolutions in MobileNet split standard convolutions into depthwise and pointwise components thus minimizing both parameter numbers and computational requirements. The pre-trained model received fine-tuning through transfer learning on our gathered sign language dataset so it could learn features for detecting our five unique signs.

D. Training Process

The training process utilized Google Teachable Machine as an interface that provided simple controls for transfer learning functions. The data was split into three sections for training and validation and testing purposes to evaluate model performance and avoid overfitting. The learning rate together with batch size and number of epochs received multiple rounds of optimization through experimental trials. The combination of data augmentation and transfer learning approaches allowed satisfactory results despite the limited dataset size.

E. Real-Time Inference Implementation

We used a Python code and ran that code in VS Code for real-time detection. The code accesses webcam of the laptop and captures continuous video frames which undergo preprocessing before forming a sequence of 30 frames. The processed sequence of 30 frames is sent to the MobileNet model for prediction before OpenCV displays the results on the live video stream. The system provides real-time feedback through an inference delay which takes between 1 to 1.5 seconds.

F. Speech Conversion Using Python-Text-To-Speech (pyttsx3)

The system enables user-friendly interaction through its text-to-speech (TTS) module which operates with the Python-Text-To-Speech (pyttsx3) library. The model converts detected sign labels into real-time audible speech after recognition. The offline Text-To-Speech (TTS) engine Pyttsx3 provides flexible operation with low latency while enabling users to modify speech rate and volume alongside voice properties. The system's integration with Text-To-Speech (TTS) functionality delivers instant audio feedback to users while expanding its accessibility for individuals who need multiple communication signals.

G. Software and Hardware Setup

The project development used Python as its base language with Tensorflow, OpenCV and Pandas libraries for implementation. The training process took place on a GPU workstation but the real-time detection system ran from a standard laptop with its built-in webcam. The integration of hardware components with specialized software made the system perform effectively in real-time conditions.

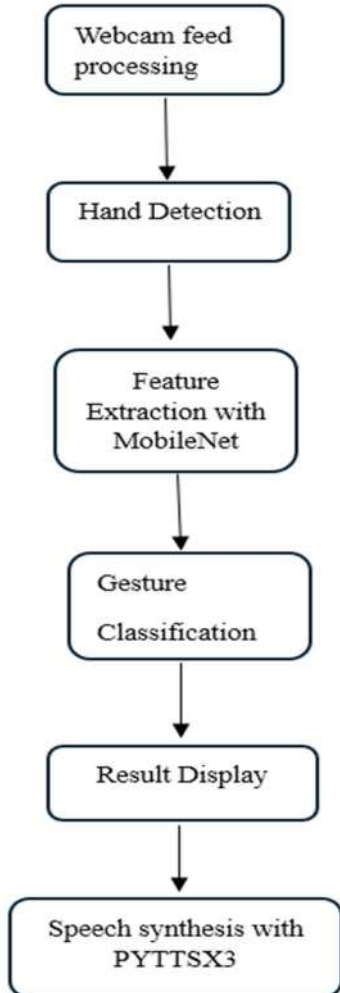


Fig. 6. Sign Detection Process Flowchart

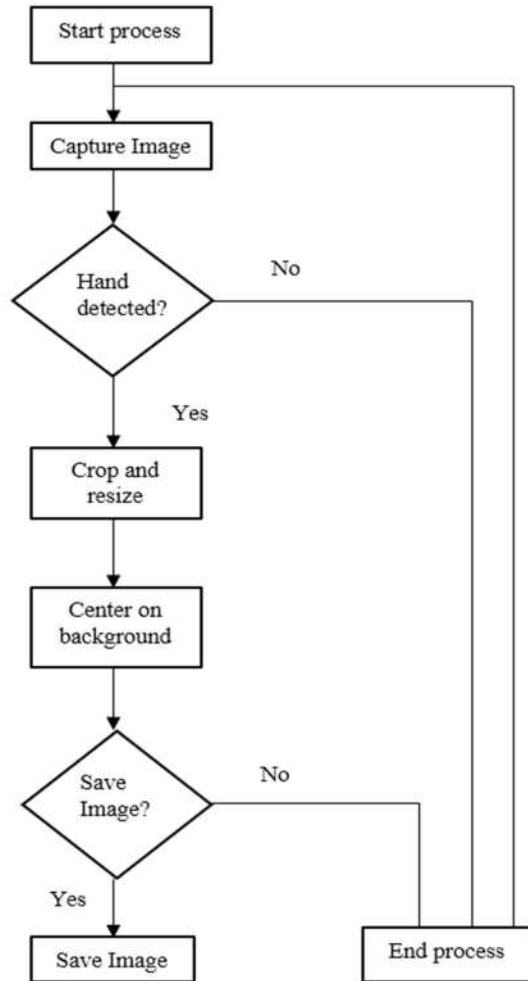


Fig. 7. Dataset Acquisition Flowchart

V. RESULTS AND DISCUSSION

Our real-time sign language recognition system performs well for five sign gestures even though the available dataset remains limited. The model reached an average validation accuracy of 72% through the combination of webcam image capture and training via Google Teachable Machine using a fine-tuned MobileNet CNN. The combination of data augmentation techniques with transfer learning enabled the system to effectively work with limited training data while maintaining good performance in controlled settings. The real-time inference pipeline needs 1.2 seconds on average to process thirty consecutive frames. The implementation of Python-Text-To-Speech (pyttsx3) text-to-speech engine through the system delivered instant audio feedback which improved accessibility and user interaction. The current system performance meets prototype requirements yet additional dataset growth and system optimization are necessary to achieve reliable performance across different environmental settings.



Fig. 8. Thank You Sign

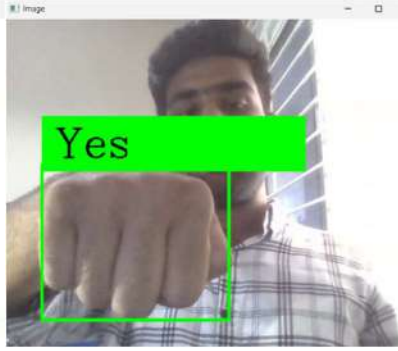


Fig. 9. Yes Sign

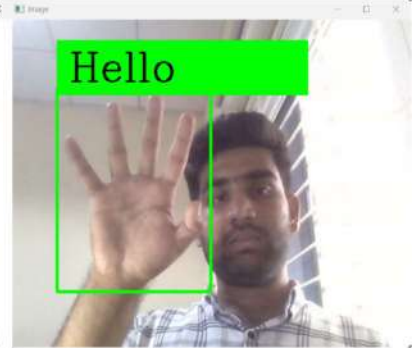


Fig. 10. Hello Sign

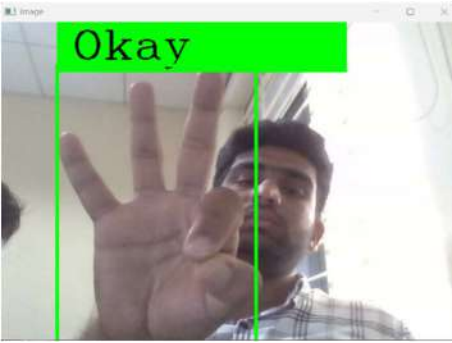


Fig. 11. Okay Sign



Fig. 12. No Sign

VI. CONCLUSION AND FUTURE SCOPE

The paper demonstrates a real-time sign language recognition system which detects five sign gestures through webcam image capture while relying on a MobileNet CNN trained by Google Teachable Machine. The system achieves high accuracy levels together with efficient real-time processing capabilities while enabling text-to-speech conversion through Python-Text-To-Speech (pyttsx3). The system achieved satisfactory performance through data augmentation and transfer learning implementations despite working with a small dataset which resulted in an average delay of 1.2 seconds per inference. Additional improvements must be made to cope with real-life variations while expanding overall system performance effectiveness.

The next stage of work focuses on enlarging the dataset through integration of various signer types alongside different environmental settings along with latency optimization and advanced model architecture evaluation. The user experience will benefit from implementing stronger noise reduction features together with adaptable Text-To-Speech (TTS) parameters. The developed enhancements will help create an improved and scalable assistive technology solution for sign language interpretation.

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The Drawbacks of Artificial Intelligence: Ethical, Economic, and Social Implications

Durga Yadav⁽¹⁾, Ajay Kumar⁽²⁾

G.H Raison University, Amravati, Maharashtra India

Email: durgayadav1014@gmail.com

Email: ajay.kumar@ghrua.edu.in

Abstract: Artificial intelligence (AI) has fundamentally transformed numerous industries, providing unparalleled efficiency gains, fostering innovation, and enabling extensive automation. However, the swift proliferation of AI technologies across diverse facets of human life presents substantial challenges. This paper examines the principal drawbacks of AI, emphasizing ethical considerations, potential job displacement, and algorithmic bias in decision-making processes, security vulnerabilities, and privacy erosion. Through a rigorous analysis of current applications and prevailing trends, this research seeks to illuminate the inherent challenges posed by AI and underscore the critical need for responsible development and deployment strategies.

Keywords: Artificial Intelligence (AI), Job displacement, Security risks, AI drawbacks,

1. INTRODUCTION

Artificial Intelligence (AI) refers to the development of machines and software capable of performing tasks that typically require human intelligence, such as decision-making, language processing, and pattern recognition. AI tools like ChatGPT, Google Bard, Siri, and facial recognition systems have become increasingly popular and are widely used in education, business, healthcare, and entertainment.

While these tools offer convenience and innovation, they also come with significant drawbacks. For example, ChatGPT can generate large volumes of content quickly, but it may also produce biased or incorrect information, leading to misinformation. Similarly, facial recognition systems are used in security and law enforcement, but they often show higher error rates for people of color, raising concerns about fairness and discrimination.

These examples highlight the need to examine AI not just for its capabilities, but also for the ethical, social, and practical challenges it introduces.

2. RELATED WORK

Recent research on Artificial Intelligence (AI) has explored its transformative impact across various sectors, while also addressing the ethical and social challenges it presents. Scholars like Binns (2018) and Crawford (2021) have critically examined the biases embedded in machine learning systems, particularly those involving facial recognition, which have been shown to produce disproportionate error rates across racial and gender lines. This has raised significant concerns regarding AI applications' fairness, accountability, and transparency.

In the context of content generation tools such as ChatGPT and Google Bard, studies have assessed their effectiveness in automating communication, education, and business processes. However, researchers like Marcus and Davis (2020) have emphasized the risks of

misinformation and hallucinated content, calling for improved model interpretability and fact-checking mechanisms.

Moreover, ethical AI development has been a growing focus, with initiatives such as the IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems and frameworks proposed by organizations like the EU and UNESCO. These works highlight the need for regulatory structures to guide responsible AI deployment and to ensure that technological innovation aligns with human values and social justice.

Collectively, these studies underscore the dual nature of AI—its potential to enhance productivity and innovation, and the urgent need to address the complex ethical implications it raises.

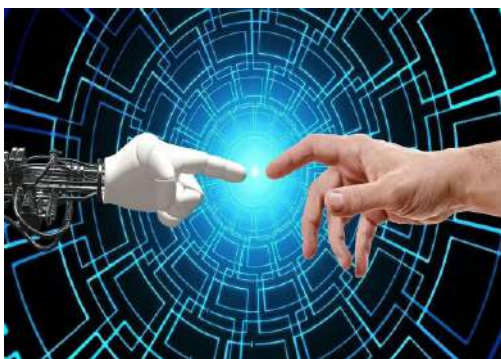


Figure 1: Point of Contact between AI and Human Beings



Figure 2: Brainwave: The Network of AI Platforms

3. Ethical Implications of AI

3.1 Bias and Discrimination

One of the foremost ethical concerns with AI is algorithmic bias. AI systems trained on historical or incomplete datasets can perpetuate existing social biases, leading to unfair treatment based on race, gender, age, or socioeconomic status.

3.2 Privacy Invasion

AI-driven technologies like surveillance systems and predictive analytics pose serious threats to individual privacy. Data collection at large scales—often without informed consent—can result in misuse or unauthorized access to sensitive personal information.

3.3 Lack of Accountability

When AI systems make critical decisions—such as approving loans, diagnosing medical conditions, or autonomous driving—assigning responsibility in the event of an error becomes challenging.



Figure 3: Negative Impacts of Artificial Intelligence

4. Economic Implications of AI:

Artificial Intelligence (AI) is transforming economies globally, introducing efficiencies and innovations. However, its integration also presents significant economic challenges, notably in job displacement, wealth inequality, and economic dependence on AI systems.

4.1 Job Displacement:

AI-driven automation poses a threat to employment, particularly in roles involving repetitive tasks. A study by Acemoglu and Restrepo (2020) found that each additional industrial robot in the U.S. correlates with the reduction of approximately 3.3 jobs, primarily affecting routine manual and clerical positions. Their research highlights how automation has significantly contributed to widening income disparities, particularly in sectors like manufacturing and administrative support.

4.2 Wealth Inequality:

The benefits of AI are often concentrated among tech corporations and high-income individuals, potentially exacerbating wealth disparities. According to the International Monetary Fund (IMF, 2024), AI could impact up to 40% of jobs globally, with advanced economies facing up to 60% of roles susceptible to automation—raising serious concerns about increasing inequality. Additionally, a study by Zhang et al. (2023) examining 86 countries from 2010 to 2019 found that increased AI investment is associated with a rise in income share for top-income households, while those in the bottom decile saw declines.

4.3 Economic Dependence on AI Systems:

Overreliance on AI in critical sectors such as finance and logistics introduces vulnerabilities to systemic failures. The concentration of AI development among a few dominant tech firms raises concerns over monopolistic control and the resilience of these systems. This centralization may create single points of failure and increase susceptibility to cyber threats, reinforcing the need for strong regulatory frameworks (IMF, 2024).

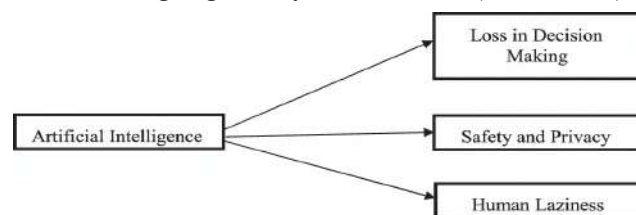


Figure 4: Key Concerns Surrounding Artificial Intelligence

5. Social Implications of AI

Artificial Intelligence (AI) technologies are increasingly integrated into various aspects of daily life, leading to significant social changes. These changes encompass alterations in human interaction patterns, the spread of misinformation, and shifts in cultural and psychological dynamics.

5.1 Erosion of Human Interaction

The adoption of AI interfaces, such as chatbots and virtual assistants, has the potential to diminish direct human-to-human communication. This reduction in personal interaction can adversely affect the development of communication skills and empathy among individuals. Studies have shown that reliance on AI for social engagement may lead to decreased social connectedness and increased feelings of loneliness. For example, research indicates that the use of AI in communication can shape language production and influence interpersonal perceptions, potentially altering the nature of human relationships (Hohenstein et al., 2021).

5.2 Manipulation and Misinformation

AI technologies, particularly generative models, can be exploited to create deepfakes and disseminate false information, posing significant threats to public discourse and democratic processes. The capability of AI to generate realistic but fabricated content makes it challenging for individuals to distinguish between authentic and manipulated information. This phenomenon has led to concerns about the erosion of trust in media and institutions. For instance, AI-driven disinformation campaigns have been shown to exploit algorithmic biases, creating echo chambers that reinforce false beliefs and undermine informed decision-making (Olanipekun, 2025).

5.3 Cultural and Psychological Impact

AI-driven recommendation algorithms can create personalized content feeds that reinforce users' existing beliefs, leading to the formation of echo chambers. This reinforcement can increase polarization and reduce exposure to diverse perspectives, potentially impacting societal cohesion. Psychological factors, such as confirmation bias, play a role in this process, as individuals are more likely to accept and share information that aligns with their pre-existing views (Friggeri et al., 2024). Moreover, the manipulation of emotions through AI-generated content has been identified as a factor contributing to the spread of misinformation, highlighting the need for ethical considerations in AI design and deployment (DiResta, 2024).



Figure 5: Overview of Challenges in AI Implementation

6. Case Studies & Real-World Examples

- In 2018, Amazon scrapped an AI recruiting tool due to gender bias.
- AI surveillance in some countries has raised civil liberty concerns.
- Autonomous vehicle accidents highlight safety and liability challenges.

7. Possible Mitigation Strategies

- Implement regulatory frameworks and AI ethics guidelines.
- Ensure transparency and explainability in AI models.
- Use inclusive and diverse training datasets.
- Promote interdisciplinary collaboration for human-centered AI design.

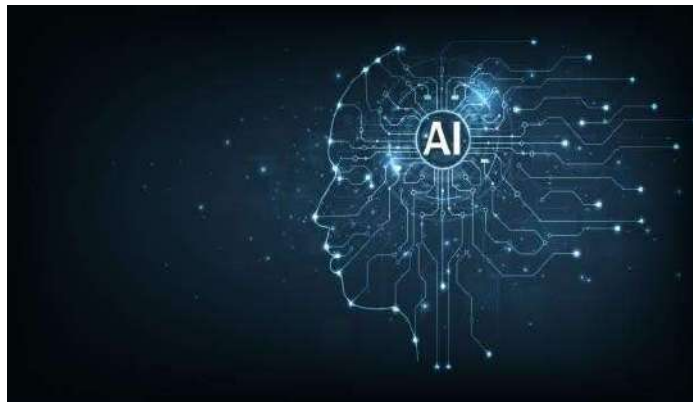


Figure 6: Thought Process Mapping: Biological vs. Artificial

8. Conclusion

While AI holds immense potential, its unchecked implementation could result in serious ethical, economic, and social consequences. Collective efforts are required to ensure its development aligns with human values and social good. Artificial Intelligence, while revolutionary, is not without its shadows. As it continues to evolve, it casts a growing influence on nearly every aspect of modern life—transforming industries, reshaping job markets, and redefining the boundaries of human interaction. However, if left unregulated or poorly guided, AI's rapid rise could deepen social divides, amplify biases, and disrupt the ethical fabric that holds society together.

The ethical dilemmas surrounding surveillance, data privacy, and autonomous decision-making raise urgent questions about accountability and control. Economically, the displacement of human labor by intelligent systems threatens to widen the gap between those who benefit from automation and those left behind. Socially, there's a risk of eroding human connections, fostering dependency on machines, and marginalizing communities without access to advanced technologies.

To navigate these challenges, a unified, global effort is essential. Policymakers, technologists, and communities must come together to build frameworks that prioritize transparency, fairness, and inclusivity. AI should not merely serve efficiency or profit—it must be shaped to reflect and uphold the core values of humanity. Only through mindful development and collective stewardship can we harness AI's potential while minimizing its pitfalls.



Figure 7: Fusing Human Potential with Artificial Intelligence

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Submission ID: CSE-006

AI-Powered Object Detection and Automatic Braking System for Cars Using Reverse Cameras

Sarthak Akarte¹, Amit Sahu²

^{1,2}Department of Artificial Intelligence, G H Rasoni University, Amravati, India
Email: akartesarthak@gmail.com, amit.sahu@ghrua.edu.in

Reversing a vehicle is a critical maneuver often prone to accidents, especially in congested areas or under poor visibility conditions. This paper introduces an AI-powered system that integrates object detection and automatic braking capabilities using reverse cameras. The system utilizes convolutional neural networks (CNNs) to detect obstacles in real-time and estimate their distance, triggering automatic braking to prevent collisions. Experimental evaluation demonstrates high accuracy in object detection, reliable distance estimation, and timely braking responses across diverse environments. This innovation enhances vehicular safety by addressing limitations in conventional reversing aids.

Submission ID: EE-007

Feature Extraction and Analysis of Bearing Faults

Varun Sharma

KIET Group of Institutions, Ghaziabad, India
Email: varun.sharma@kiet.edu

One of the most important steps in identifying bearing problems is feature extraction. In order to provide a more meaningful dataset, it entails locating and extracting pertinent features from raw bearing vibration signals. Tasks involving categorization and prediction can then make use of these attributes.

Submission ID: CE-008

Experimental Study of Vibration Control of Framed Structure Using Various Dampers: A Comprehensive Review

Mr. Aditya Ajay There¹, Prof. H.B. Dahake²

Structural Engineering, Department of Civil Engineering,
G H Rasoni University, Amravati, Maharashtra, India

Tall buildings are becoming more common within contemporary cities, driven by a need for structures that are cost-effective, sustainable, lightweight, adaptable, and rapid to erect. However, as dampening in buildings declines, the probability of failure due to severe vibrations rises. To solve this difficulty, effective energy dissipation systems are required to reduce the dynamic response of buildings. Several solutions have been developed to reduce vibrations, including passively energy dissipation systems, that allow mechanical devices that gather energy without the need for an external power source. Passive devices release forces in reaction to structural motion, which reduces the building's energy consumption. Metallic dampers, frictional dampers, viscoelastic dampers,

and fluid viscosity dampers (FVDs) are all commonly used. FVDs, in particular, are extremely effective at managing shock forces and controlling structural motion. By combining stiffness and damping parts, these devices greatly improve the structure's stability under dynamic loading situations, adding to the safety and robustness of current high-rise structures. This research emphasizes the significance of FVDs as a dependable option for vibration management in tall constructions.

Submission ID: CSE-052

Evaluation of Activation Functions in Machine Learning Models for Crop Yield Prediction

Harsh Sarnaik¹, Manish Khandait², Mehar Chopkar³

sarnaikhs@rknc.edu, khandaitmk@rknc.edu, chopkarp@rknc.edu

^{1,2,3}Ramdeobaba University,

Formerly Shri Ramdeobaba College of Engineering and Management, Gittikhadan,
Nagpur

Crop production is heavily influenced by factors such as land region, past yield, crop type, and season. These parameters are critical for accurate crop yield prediction. In this study, several machine learning (ML) models were evaluated using different activation functions to determine which yielded the highest accuracy and minimal loss. The models utilized data on land usage, historical yield, and crop type from various Indian agricultural regions.

Results demonstrate the significance of selecting appropriate activation functions, with specific functions outperforming others in reducing loss and enhancing prediction accuracy. This highlights the essential role of activation function choice in optimizing ML models for agricultural forecasting applications.

Submission ID: CE-012

Review Paper on Characteristics and Analytical Study of Desirable Property and Testing of Highway Materials

Sahil Zade¹, Apeksha Choudhary²

¹P.G. Student, ²Assistant Professor, Department of Civil Engineering,
G H Raison University, Amravati, Maharashtra, India

Email: sahilzade05@gmail.com, choudharyapeksha016@gmail.com

Soil, aggregate, bitumen, concrete, steel, and sand are some of the various materials used in highway construction. Every material has specific desirable qualities that are appropriate for its intended function. Properties including soundness, hardness, G.I. value, crushing strength, and water absorption capacity are all examined when studying roadway materials. Highway materials are characterized using laboratory experiments and field testing. Pavement testing and evaluation is an important part of highway design because it affects the safety, durability, and performance of the road. To create a safe highway, three main factors—human, vehicular, and roadway—as well as their interactions must be taken into account. Human aspects include car-following behaviour, visual acuity for the traffic signs and signals, and reaction time for the steering and

braking. This paper briefly studies the chemical and physical properties of highway materials. The materials include soil, sand, bitumen, aggregate, steel, and cement, each possessing different properties crucial for strong and durable highway construction. Soil is a major component of highway construction. The bearing capacity of soil is tested using the California Bearing Ratio (CBR) method, which assesses the soil strength at a specific site. The CBR value indicates surface hardness—clay typically shows 2

Submission ID: CSE-013

Comparative Analysis of AI-Based Approach vs Traditional Methods in Climate Modeling

Priyanka Vashisht, Anvesha Katti

Amity School of Engineering and Technology,
Amity University Haryana, India

Email: priyanka.vashisht@gmail.com, anveshakatti@gmail.com

Climate modeling helps predict future climate variations and assess human interference with the environment. Traditional General Circulation Models (GCMs), derived from physics-based mathematical equations, are accurate but computationally expensive. In recent years, Artificial Intelligence (AI)-based approaches have emerged as efficient alternatives, offering improved predictability and classification of extreme events. This study compares AI models such as XGBoost, Long Short-Term Memory (LSTM) networks, and hybrid AI-physics models against GCMs. Results indicate that AI models reduce prediction errors by approximately 30%, with hybrid models demonstrating lower Mean Absolute Error (MAE) and higher F1 scores for extreme event classification. Furthermore, AI models enhance computational efficiency, requiring about 40% less processing time and energy. These findings highlight the potential of AI-based approaches to improve climate prediction accuracy, support risk assessment, and inform policy decisions.

Submission ID: CSE-016

Comparative Study of Machine Learning Models for Healthy Plant Detection

**Pranita Chaudhary¹, Sujata Swami², Vipul Phatangare³, Diksha Bhosale⁴,
Adinna Thaware⁵**

¹Assistant Professor, ^{2,3,4,5}Student, Department of CSE(AIML), PCCOE, Pune, India

Emails: pranita.chaudhary@pccoepune.org, sujata.swami23@pccoepune.org,
vipul.phatangare23@pccoepune.org,

diksha.bhosale23@pccoepune.org, adinna.thaware23@pccoepune.org

This paper addresses the critical need for healthy plant detection in agriculture to ensure optimal crop yield and minimize financial loss. The study presents a comparative analysis of six machine learning algorithms—Support Vector Machine (SVM), Random Forest, XGBoost, AdaBoost, Gradient Boosting, and Decision Tree—for detecting healthy plants. A dataset consisting of images of healthy and unhealthy plants, such as mango, bell pepper, potato, and Pongamia Pinnata, is used. These images are processed and converted into numerical form. Due to the high dimensionality of the image data (49152

features), Principal Component Analysis (PCA) is applied to reduce the number of features to 50, thereby enhancing performance and reducing computation time. The models are evaluated based on accuracy, precision, recall, and execution time. The findings aim to identify the most effective machine learning model for practical plant health detection, thus providing valuable insights to farmers and agricultural experts to enhance crop management strategies.

Submission ID: CSE-017

Deep Learning Based Classification of Colon Disease Through Wireless Capsule Endoscopy

Jadhao Mahadeorao Alka, Bartere Mahip

Department of Computer Science and Engineering,
G. H. Rasoni University, Amravati, Maharashtra, India – 440100

Email: alka.pravin@gmail.com, mahip.bartere@ghrua.edu.in

The integration of deep learning (DL) into Wireless Capsule Endoscopy (WCE) analysis holds great potential for enhancing diagnostic precision and improving the overall efficiency of colon disease screening. Colon diseases, such as polyps, inflammatory bowel disease (IBD), and colorectal cancer, remain among the leading causes of morbidity and mortality globally. Accurate and early diagnosis is essential to improve patient outcomes. WCE offers a minimally invasive technique to visualize the entire gastrointestinal tract, generating thousands of images per patient. However, manual image examination is labor-intensive and prone to human error. This study presents a DL-based method for automated image classification of colon diseases using WCE images. The proposed model leverages a Convolutional Neural Network (CNN) to extract features automatically and classify the images into various disease categories. By incorporating data augmentation and transfer learning, the model addresses the variability and complexity of WCE imagery while maintaining high classification accuracy. A benchmark dataset of annotated WCE images is utilized. The model is developed using Python, image processing techniques, and deep learning libraries. This automated classification framework significantly reduces diagnostic time and offers a reliable, scalable solution for clinical colon disease detection.

Submission ID: CSE-019

Generative AI for VR: Creating Physically Realistic Models

Mrs. Pranita Chaudhary¹, Mr. Shushant Bhat², Mr. Aayush Bodkhe³, Mr. Varun Bhilare⁴, Mr. Abhishek Bichare⁵

¹Assistant Professor, Department of CSE (AI&ML), PCCOE, Pune

^{2,3,4,5}Students, Department of CSE (AI&ML), PCCOE, Pune

Emails: pranita.chaudhary@pccoepune.org, shushant.bhat23@pccoepune.org,
aayush.bodkhe23@pccoepune.org, varun.bhilare23@pccoepune.org,
abhishek.bichare23@pccoepune.org

Virtual Reality (VR) has revolutionized traditional learning by providing interactive and immersive environments. However, VR applications often fall short in delivering

physically realistic experiences due to limitations in accurate modeling. This study explores the application of Generative Adversarial Networks (GANs) to generate physically realistic 3D models for VR. The proposed system combines deep learning techniques with physics-based constraints to improve model accuracy and usability. Experiments were conducted using the ShapeNet dataset, with preprocessing steps applied to standardize the point cloud data. A multicomponent GAN architecture, incorporating a spatial transformer and a semantic segmentation network, was employed to generate 3D models. Evaluation metrics such as Intersection-over-Union (IoU) and Dice Coefficient demonstrated improved accuracy across multiple categories. Comparative analysis with conventional methods showed enhancements in both physical realism and model interactivity. Qualitative assessments also highlighted the system's ability to deliver clearer segmentation and enhanced user engagement. These results suggest that GAN-generated models can significantly enrich VR-based learning by bridging the gap between virtual and real-world dynamics. The study concludes that integrating deep learning with physical constraints provides a promising direction for advanced educational tools, fostering deeper understanding and engagement.

Submission ID: CE-028

A Review on Comparative Analysis of Organic Waste Management in Rural Communities of Partawada for Windrow and Sheet Composting Methods

Mr. Prashanti R. Bhirad¹, Prof. Anurag Gahalod²

¹Student, ²Assistant Professor,
Department of Civil Engineering, G.H. Rasoni University, Amravati

Composting, the recycling of organic waste such as vegetation and food waste, reduces the amount of waste going to landfills and is therefore a rapidly growing sector. The residual compost is a stable, sanitized, humus-like material rich in organic matter and free from offensive odour. Recycling organic waste is environmentally and economically beneficial, as landfill decay leads to adverse impacts. Household waste components like garden and food waste offer high energy recovery and economic benefits when recycled. Land application of composted waste addresses both waste disposal challenges and the need to enhance soil organic content. Composting is an aerobic biological process that transforms organic materials into nutrient-rich compost, improving soil fertility, reducing fertilizer dependency, conserving water, suppressing plant diseases, and enhancing crop yields. This study investigates organic waste composting and sustainability practices in rural communities of Amravati District through a pilot project in Paratwada, with a comparative analysis of windrow and sheet composting methods.

Submission ID: CE-029

The Comprehensive Plan for Environmental Restoration, Rejuvenation, and Beautification of Khokad Lake

Mr. Ravindra Murlidhar Gaikwad¹, Prof. Anurag Gahalod²

¹Student, ²Assistant Professor,
Department of Civil Engineering, G.H. Rasoni University, Amravati

The comprehensive plan for environmental restoration, rejuvenation, and beautification of Khokad Lake aims to assess, design, and implement effective strategies for reviving and sustaining freshwater ecosystems. Lakes are vital natural resources, providing ecological services, supporting biodiversity, and serving as social and recreational spaces. However, many lakes face degradation due to pollution, encroachment, and climate change. This study compares various restoration approaches based on ecological, economic, and social outcomes. The plan emphasizes habitat restoration, water quality improvement, erosion control, and sustainable landscaping, ensuring the lake's ecological function is preserved while enhancing its visual appeal. This research provides a roadmap for policymakers, environmentalists, and urban planners to restore lake ecosystems as resilient, beautiful, and functional landscapes. The Khokad Lake, once a vital ecological and cultural resource, has suffered from environmental degradation due to urbanization, encroachment, pollution, and insufficient management. This comprehensive plan aims to restore and rejuvenate the lake ecosystem, improve water quality, and enhance the surrounding landscape to ensure long-term sustainability and beauty. The plan also advocates for community participation and collaborative governance to ensure successful implementation and maintenance. By adopting a multi-disciplinary approach, this plan seeks to transform Khokad Lake into a thriving, resilient ecosystem that serves as a recreational, educational, and ecological asset for future generations. This research focuses on a pilot project on Khokad Lake at Chandwad, District Nashik.

Submission ID: FSD-034

Identification of Methanol from Illicit Liquor and Its Harmful Effects on Human Health

Mr. Mangesh S. Dhore¹, Ms. Abhaya A. Bhople¹, Ms. Rajashree A. Rane¹

¹Department of Forensic Science, G.H. Rasoni University, Anjangaon Bari Road,
Amravati, Maharashtra 444701

Email: mangeshdhore2@gmail.com

Illicit liquor consumption poses a significant public health threat due to adulteration with toxic substances like methanol. Methanol ingestion can lead to blindness, organ failure, and even death. Rapid and reliable methods for estimating methanol content in illicit liquor are crucial for consumer safety and law enforcement. The chromotropic acid test offers a potentially rapid and cost-effective method for methanol detection in such beverages. This research evaluates the presence of methanol in illicit liquor samples from different regions in Maharashtra. The analysis revealed the presence of 21.05% adulteration in the illicit liquor sample collected from various regions of Maharashtra.

Bhandara showed the highest methanol adulteration of 75% and Nagpur showed adulteration of 14.28%. Our study also revealed that methanol adulteration does not affect the hardness or the density of samples and hence cannot be used as properties to detect the presence of methanol in illicit liquor. The findings from this research will contribute to developing effective strategies for mitigating methanol poisoning risks associated with illicit liquor consumption. The detection of methanol in illicit liquor samples from Maharashtra underscores the potential dangers faced by consumers. By understanding the devastating effects of methanol, the reasons behind its adulteration, and the importance of sensitive detection methods, we can work towards stricter regulations, public awareness campaigns, and improved forensic science practices. This multifaceted approach is crucial to safeguard public health and prevent methanol poisoning tragedies in India.

Submission ID: MBA-067

Advancements in AI-Based Tools and Digital Applications: Creating Revolution for New-Age Urban Infra Industries

Ms. Arini Roychoudhury^{1*}, Mr. Nitin Kumar Sahu², Mr. Atul Kumar Sahu³

¹Faculty, Thakur Global Business School, Mumbai – 400101, Maharashtra, India

^{2,3}Faculty, Department of Industrial and Production Engineering, Guru Ghasidas (Central) Vishwavidyalaya, Bilaspur – 495009, Chhattisgarh, India

*Email: roychoudhuryarini@gmail.com

AI-based Management Information Systems (MIS) advancements alongside digital applications reshape new-age infrastructure business environments through enhanced innovation, operational efficiency, and scalability. Moreover, IoT sensors along with drones and smart construction technologies deliver unparalleled project visibility and control through digital applications. Real-time monitoring tools for construction sites and maintenance operations as well as asset management lead to more efficient and cost-effective operational processes. AI and digital technologies promote stakeholder collaboration through seamless data sharing and communication platforms which connect contractors with clients and regulatory bodies. The integration of AI-driven solutions provides organizations with increased transparency while maintaining compliance standards and expediting decision-making workflows. AI integration into Management Information Systems facilitates real-time data analysis and predictive analytics which improves decision-making processes for businesses and helps optimize resource use through informed decision-making. In this current study, the integration of AI tools and digital technologies are explored in infrastructure projects, highlighting key applications, challenges, and the future potential of AI in infrastructure businesses. The present study aims to critically analyse the impact of AI and digital technologies that have initiated a revolution via transforming infrastructure businesses into more flexible entities that can effectively tackle and meet market demands.

Submission ID: FSD-076

Stature Estimation of Human Being On The Basis of Footprint Length

Ms. Abhaya A. Bhople¹, Ms. Sneha B. Hedao¹, Ms. Yogita V. Parchake¹,
Ms. Anchal C. Chourse¹, Mr. Prathamesh N. Shinde¹

¹Department of Forensic Science,
G.H. Raison University, Anjangaon Bari Road, Amravati, Maharashtra – 444701

Email: abhaya.bhople@ghrua.edu.in

The footprint is an important piece of physical evidence found at many crime scenes, including homicide, burglary, and sexual assault. Footprints are impressions or images left behind by a person walking or running. Estimation of stature from the dimension of the foot has considerable forensic value in developing a description of the suspect from evidence at the crime scene and in corroborating height estimation. This paper examines the use of footprints impression to estimate the stature of an individual, and it entails the specific character related to the individual. This was done by taking the impression of the feet of 160 individuals from various regions of Vidarbha, the age group lies between 18 to 30, which consists of a total 320 samples of footprints. The real height of the individual was measured by tape. From footprint impressions, the highest tip and lowest tip of the collected footprint impression were measured and height was calculated with the help of Topinard's formula. In the present study, we observed that by adding a constant value of 12 to Topinard's formula, we get the exact height of an individual. This study is applicable for investigations in which we get footprint samples at the crime scene to estimate the height of the suspected person. The minor difference between actual height and calculated height was found in the range of ± 0 to 0.05 ft and ± 0.05 to 0.1 ft in females and males respectively.

Submission ID: CE-079

Comparative Study of High-Rise Building Using URM Infill Strut and Bare Frame Model Under the Effect of Lateral Loads

Mr. Bhavesh Ramesh Vaishnav¹, Dr. Bhushan H. Shinde²

¹PG Student, ²Associate Professor,
Department of Civil Engineering, G H Raison University, Amravati

In this study, the structural response of a model with unreinforced masonry (URM) infill walls is compared to that of a bare frame. In accordance with Indian Standard codes IS 875-1987 (Part I III) and IS 1893-2016 (Part 1), it entails computing and evaluating loads operating on structural parts, such as gravity loads (dead and live) and lateral forces (wind and seismic). Both models were thoroughly evaluated using STAAD PRO software, which assessed important factors including deformation, bending moments, shear pressures, and the need for steel reinforcement. The findings show notable variations between the two arrangements, with the URM infill walls affecting load distribution and overall structural stability. The study provides insights into the influence of infill walls on structural performance, aiding in efficient design decisions for improved strength and stability in real-world construction scenarios.

Submission ID: CE-082

Partial Replacement of Cement by Nano Silica in Concrete Mix

Shreyash Subhash Sancheti¹, Dr. Bhushan H. Shinde²

¹P.G. Student, ²Associate Professor,
Department of Civil Engineering, G. H. Rasoni University, Amravati, Maharashtra,
India

This study explores the partial replacement of cement with nano-silica (NS) to enhance concrete's mechanical properties, durability, and sustainability. With growing environmental concerns over cement production, NS offers a viable alternative due to its high pozzolanic activity and microstructural refinement. Nano-silica was incorporated in varying proportions (2.75% to 3.25% by weight of cement), and its effects on workability, compressive strength, split tensile strength, and durability were evaluated. The results showed that optimal NS replacement improved compressive strength by accelerating hydration and enhancing particle packing. Additionally, NS reduced porosity, increasing resistance to water absorption and chloride penetration, thereby enhancing durability. This research demonstrates that nano-silica can lower cement consumption, reducing the carbon footprint without compromising strength or durability. The findings suggest that NS is a sustainable and high-performance material for modern concrete applications.

Submission ID: CE-085

Investigation of Ultra High Performance Fiber Reinforced Concrete with Locally Available Materials

Mr. Sayyed Mehmood Ahmar¹, Prof. H. B. Dahake²

¹P.G. Student, ²Assistant Professor,
Department of Civil Engineering, G. H. Rasoni University, Amravati, Maharashtra,
India

Ultra High Performance Fiber Reinforced Concrete (UHPFRC) is a novel material that exhibits superior ductility, toughness, and strength compared to conventional concrete. However, because of its novelty, existing design codes require adaptation for its structural applications. It is especially helpful for rehabilitating aging infrastructure that is experiencing problems with durability and load-carrying capacity. UHPFRC is made of Type I Portland cement, high silica content, a low water-binder ratio, superplasticizers, and short fibers, enabling compressive strengths of 100–150 MPa and a slump flow of about 200 mm. Compressive and flexural strength, tensile stress-strain behavior, shear and punching capacity, creep, shrinkage, and fracture energy are important mechanical characteristics. UHPFRC is a promising advance in construction as it also uses inverse analysis of bending test data for tensile mechanical characterisation.

Submission ID: CE-090

Utilization of Plastic Waste in Pavement Block by Partially Replacement of Fine Aggregate

Vaibhao V. Maind¹, Harshal G. Mundle², Anmol W. Dongre³

^{1,2,3}Assistant Professor, Department of Civil Engineering,
G H Raisonni College of Engineering and Management, Nagpur – 440016, India
Email: vaibhao.maind@raisonni.net , harshal.mundle@raisonni.net ,
anmol.dongre@raisonni.net

The rapid growth of urbanization and industry has resulted in a significant increase in plastic waste generation. This poses a major environmental challenge due to the limited biodegradability of plastics. One potential solution is exploring the use of plastic waste in construction materials. This study investigates the theoretical utilization of waste plastic in pavement blocks as a partial replacement for fine aggregate. The focus is on the feasibility and potential benefits of incorporating recycled plastic into pavement block production. Highlights the growing issue of plastic waste and its negative environmental impact. Introduces the concept of using recycled plastic in pavement blocks as a sustainable waste management strategy. Briefly outlines the proposed approach for incorporating plastic waste into pavement blocks. This might involve shredding or melting plastic into a suitable form for use as a partial replacement for fine aggregate in the pavement block mix. Discusses the potential advantages of using recycled plastic in pavement blocks. For experimental study, firstly, ordinary paver blocks were prepared, then the fine aggregate was partially replaced with plastic waste by 10%, 15%, and 20%. The main parameters considered in this paper to compare the experimental study were compressive strength and cost of ordinary pavement blocks and plastic pavement blocks.

Submission ID: CE-093

Environmental Impact Assessment for Road Infrastructure and Development Projects

Vaibhav Mahale¹, Apeksha Choudhary²

¹M. Tech. Student, ²Professor, Department of Civil Engineering,
G H Raisonni University, Amravati, Maharashtra, India

Environmental Impact Assessment (EIA) is a systematic process used to evaluate the potential environmental effects of a proposed project or development before it is carried out. The primary aim of EIA is to ensure that environmental considerations are integrated into decision-making alongside economic, social, and technical factors. This process involves identifying, predicting, and assessing a project's direct, indirect, cumulative, and long-term impacts on various environmental components, including air, water, soil, biodiversity, and human well-being.

The EIA process typically includes multiple stages: screening (to determine if an EIA is required), scoping (to identify key concerns), impact assessment (to analyze potential effects), mitigation (to minimize adverse impacts), and monitoring (to track actual effects during and after implementation). Engaging the public and consulting relevant stakeholders are essential aspects of EIA, fostering transparency and community participation.

EIA serves as a tool for promoting sustainable development by facilitating informed decision-making and reducing environmental harm. It is widely used across industries such as infrastructure, industrial expansion, agriculture, and natural resource extraction. Ultimately, EIA aims to balance development goals with environmental preservation, ensuring projects are sustainable from ecological, economic, and social perspectives.

Submission ID: CE-095

Uses Of UHPFRC Segment In Connecting to Ramp in Nagpur City (Wardhaman Nagar – Nirmal Nagar) Live Flyover Project

Akshay Sanjayrao Gulhane¹, Mr. Pranav Lende²

¹P.G. Student, ²Assistant Professor, Department of Civil Engineering,
G. H. Raisonni University, Amravati, Maharashtra, India

Because of their remarkable mechanical qualities, longevity, and structural resilience, Ultra-High Performance Concrete (UHPC) and Ultra-High Performance Fiber Reinforced Concrete (UHPFRC) have drawn a lot of attention lately. Numerous research on UHPC and UHPFRC are examined in this overview of the literature, with an emphasis on their mix design, material composition, and real-world applications. The use of UHPFRC in retrofitting and rehabilitation of existing buildings, the use of local resources to improve sustainability and lower costs, and the function of steel fibers in enhancing toughness and flexural performance are some of the research highlights. The investigations also highlight how rheological characteristics, curing conditions, and mix proportions affect the overall performance of concrete. Despite its great qualities, UHPC's broad application is restricted by issues with high costs and few design codes. The wider use of UHPC in contemporary building might be encouraged by addressing these limitations through economic feasibility studies and optimal material use. This evaluation sheds light on UHPC's potential for long-term sustainability and structural innovation in the building sector.

Submission ID: FSD-100

“Comparative Study of Extraction of *Calotropis procera* and *Calotropis gigantea* Poison”

Abhaya A. Bhople¹, Minakshi D. Sarkar¹, Sakshi D. Gobhekar¹, Prajakta P. Dalavi¹, Aditi S. Chaure¹

¹Department of Forensic Science,
G. H. Raisonni University, Anjangaon Bari Road, Amravati, Maharashtra 444701

Email: abhaya.bhople@ghrua.edu.in

This research investigates the toxicological properties of *Calotropis procera* and *Calotropis gigantea*, two plant species renowned for their dual roles in medicine and toxicity. The study aims to compare the extraction of *Calotropis procera* and *Calotropis gigantea* poison from their leaves, flowers, roots, and latex using methodologies such as Thin Layer Chromatography (TLC), chemical screening tests, and UV-visible spectrophotometry. Key toxic constituents identified include Calotoxin, Calactin, tannins, alkaloids, and cardiac

glycosides. The TLC analysis achieved partial success in separating chemical components, while chemical screening confirmed the presence of alkaloids, tannins, and cardiac glycosides in both species. UV-visible spectrophotometry, however, produced limited results due to sample impurities and low concentrations of active compounds. *C. procera* is extracted more effectively than *C. gigantea*, significantly affecting both biochemical and physiological functions. Factors such as geographic variations, environmental conditions, and extraction technique limitations were found to influence the findings. The study emphasizes the forensic relevance of these plants in poisoning cases and suggests their potential therapeutic applications. This research underscores the need for advanced methodologies and broader studies involving diverse geographic regions and improved solvent systems. Despite its limitations, the study provides valuable insights into the toxicological and forensic implications of *Calotropis procera* and *Calotropis gigantea*.

Submission ID: ME-102

Heat Chamber Formation from Condenser Heat Loss

Prof. Seema Rewatkar

Department of Mechanical Engineering,
G. H. Raison Institute of Engineering & Management, Nagpur

Email: seema.rewatkar@raisoni.net

This paper presents a concept aimed at improving the utility and efficiency of a refrigerator by incorporating a heat chamber. The heat chamber is an external GI (Galvanized Iron) chamber installed over the refrigerator, replacing the conventional chiller (ice former). It utilizes refrigerant flow tubes coming directly from the compressor. The experiment involves capturing and utilizing the heat energy that is typically wasted through dissipation into the atmosphere. The hot refrigerant is allowed to flow through the heat chamber, where the tubes wrapped around the chamber transfer heat, raising the internal temperature of the chamber to approximately 50–60°C. This design enhances the overall efficiency of the refrigeration system by providing an additional compartment for heating applications.

Submission ID: FSD-119

To Study the Retention of Saliva Stains on Different Fabrics

Mrs. Mamata Barmase, Ananya D. Pantawane, Krutika P. Mute, Komal K. Hore, Urwashi V. Raut

Department of Forensic Science, G. H. Raison University, Anjangaon Bari Road,
Amravati, Maharashtra 444701

Email: mamta.barmase@ghrua.edu.in

Saliva stains discovered during criminal investigations can be of considerable importance. This study examines the retention of saliva stains on different fabrics such as cotton, wool, teri-cot, rayon, lycra, and polyester using the Starch-Iodine Test. Saliva was applied to the fabrics and analyzed over a period of eleven days. The retention of saliva stains was influenced by both the nature of the fabrics and the duration for which

the stain remained on the material. Results indicate that α -amylase activity was detectable on fabrics like cotton, wool, teri-cot, rayon, and polyester even after eleven days, whereas the fabric lycra showed inconsistent and contradictory results, suggesting that the test may not be suitable for this particular fabric. This study has important forensic implications, especially in cases involving sexual assault and rape where saliva traces on fabrics are analyzed.

Submission ID: FSD-137

Identification of Most Common Persistent Organic Pollutants (POP) in Vidarbha Region and its Adverse Effects on Human Body

Mangesh S. Dhore, Gauri Futane, Mahak Pawar, Qurratulain Kausar Ansari

Department of Forensic Science, G. H. Rasoni University, Anjangaon Bari Road,
Amravati, Maharashtra 444701

Email: mangeshdhore2@gmail.com

Persistent organic pollutants (POPs) are stubborn, hazardous, bioaccumulative chemicals that are highly toxic to both the environment and human health due to their persistent nature. This study focuses on the identification of toxic and banned POPs in the Vidarbha region, especially in Chandrapur, Yavatmal, Wardha, and Nagpur. The commonly identified POPs and pesticides in the region include Dichloro Diphenyl Trichloroethane (DDT), Endosulfan, Monocrotophos, Cypermethrin, Profenofos, Imidacloprid, Dimethoate, Ethion 50%, Chlorpyrifos, Glyphosate, and Fipronil—most of which are found in fertilizers used for spraying crops such as cotton, wheat, and vegetables. Field visits to Krushi Seva Kendras revealed that these chemicals, including banned substances like Monocrotophos, are still easily accessible to farmers. Interviews with local farmers confirmed their continued use in agricultural practices. These POPs pose severe health risks such as organ failure, cancer, and genetic disorders, and also adversely affect the environment by accumulating in air, water, and soil. Eventually, they enter the food chain, causing further harm to animals and humans. Raising awareness about the dangers of these pollutants is essential to mitigate their hazardous effects on health and the ecosystem.

Submission ID: CE-127

Multi-Leg Intersection Analysis & Optimization Using VISSIM

Prutha S. Deshmukh¹, Apeksha G. Choudhary²

¹PG Student, ²Assistant Professor,
Department of Civil Engineering, G. H. Rasoni College of Engineering & Technology,
Amravati

Urban traffic congestion at multi-leg intersections poses a significant challenge for transportation engineers and city planners. This study employs PTV VISSIM, a microscopic traffic simulation software, to analyze and optimize traffic flow at Panchavati Square, Amravati, a complex multi-leg intersection. The research includes comprehensive

traffic data collection such as vehicle volumes, turning movements, signal timings, and pedestrian interactions during peak hours. Existing intersection performance is evaluated by measuring key parameters including average vehicle delay, queue lengths, level of service (LOS), and emissions. Various optimization strategies—such as signal timing adjustments, lane reconfigurations, and adaptive traffic control systems—are tested. Simulation results demonstrate that optimized signal phasing and geometric improvements significantly reduce delays and enhance intersection efficiency. The findings highlight the efficacy of VISSIM-based simulation in improving urban mobility, reducing congestion, and promoting environmentally sustainable transportation. Recommendations for implementing optimized traffic control strategies are provided to ensure safer and more efficient urban traffic management.

Submission ID: CE-135

Review on Mixture Design Methods for Self-Compacting Concrete with Various Cementitious Materials

Pranav Vishnu Chavan¹, Dr. Sujesh Ghodmare²

¹Scholar, ²Associate Professor,

Department of Civil Engineering, School of Engineering Technology, G. H. Raison
University, Amravati

Self-Compacting Concrete (SCC) has transformed modern concrete technology by enhancing workability, durability, and structural integrity. Characterized by its ability to flow and consolidate under its own weight without external vibration, SCC has seen various mixture design methodologies since its introduction in 1988. This review examines five key approaches: empirical design, compressive strength-based methods, aggregate packing models, statistical factorial designs, and rheology-based models. These methods offer distinct advantages for optimizing SCC performance, sustainability, and mechanical properties when using diverse cementitious materials. Despite progress, balancing workability, strength, and environmental impact remains a challenge. This study critically evaluates the evolution and applicability of these mixture design techniques to achieve improved mechanical and durability properties while reducing carbon footprint in concrete construction.

Submission ID: FSD-137

Identification of Most Common Persistent Organic Pollutants (POP) in Vidarbha Region and its Adverse Effects on Human Body

Mangesh S. Dhore¹, Gauri Futane¹, Mahak Pawar¹, Qurratulain Kausar Ansari¹

¹Department of Forensic Science,
G. H. Raisonni University, Anjangaon Bari Road, Amravati, Maharashtra 444701

Email: mangeshdhore2@gmail.com

Persistent organic pollutants (POPs) are stubborn, hazardous, and bioaccumulative chemicals known for their high toxicity to both the environment and human health. This study investigates the identification and prevalence of toxic and banned POPs in the Vidarbha region, particularly in districts such as Chandrapur, Yavatmal, Wardha, and Nagpur. Commonly identified POPs and pesticides include Dichloro Diphenyl Trichloroethane (DDT), Endosulfan, Monocrotophos, Cypermethrin, Profenofos, Imidacloprid, Dimethoate, Ethion 50%, Chlorpyrifos, Glyphosate, and Fipronil. These substances are found primarily in fertilizers used for crops like cotton, wheat, and vegetables. Field visits and interviews with local farmers and Krushi Seva Kendras revealed that many of these chemicals, including banned substances like Monocrotophos, are still easily accessible and in active use. The presence of POPs in air, water, and soil allows them to infiltrate the food chain, leading to significant health issues such as organ failure, cancer, and genetic disorders, along with detrimental environmental effects. This study emphasizes the urgent need for awareness regarding the hazards of POPs and calls for stricter regulation and enforcement to protect both human health and the ecosystem.

Submission ID: CE-149

Comparative Study of High-Rise Regular and Irregular Buildings Using ETABS

Rudresh Bharat Chate¹, Dr. Bhushan H. Shinde²

¹P.G. Student, ²Associate Professor, Department of Civil Engineering,
G. H. Raisonni University, Amravati, Maharashtra, India

The seismic performance of high-rise buildings is significantly influenced by the placement and configuration of shear walls, which are critical structural elements for resisting lateral forces. This study conducts a comparative analysis of the effects of different shear wall locations in regular and irregular high-rise buildings using the structural analysis software ETABS. The research focuses on evaluating the structural behavior, including lateral displacement and the area of steel reinforcement under seismic loading conditions. Multiple models of high-rise buildings are developed, incorporating varying shear wall configurations such as bracings, middle strips, and corner placements, in both regular and irregular geometric layouts. The analysis emphasizes the impact of shear wall positioning on the overall structural stability and performance of the buildings. Results indicate that the optimal placement of shear walls significantly reduces lateral displacements and story drifts, particularly in irregular buildings, which are more susceptible to torsional effects.

due to their asymmetric geometry. The findings of this study provide valuable insights for structural engineers in designing high-rise buildings with enhanced seismic resilience. The research highlights the importance of strategic shear wall placement in mitigating the adverse effects of seismic forces, especially in irregular structures, and offers practical recommendations for optimizing structural performance in earthquake-prone regions.

Submission ID: FSD-163

Efficacy & Efficiency of Fingerprint Powder Derived from Natural Flowers – Development and Evaluation of Sustainable Fingerprint Powders

**Qurratulain Kausar Ansari¹, Pratiksha Parihar¹, Shrutika Chincholkar¹,
Karuna Dohare¹, Mangesh S. Dhore¹**

¹Department of Forensic Science,

G. H. Rasoni University, Anjangaon Bari Road, Amravati, Maharashtra 444701

Email: kausar.ansari717@gmail.com

Fingerprints are a crucial element in forensic investigations. Latent or invisible fingerprints are commonly recovered from the scenes of crime and hence require the use of visualization methods. The conventional methods are relatively expensive and may cause harmful threats and health hazards due to the presence of chemical content. This study offers a cheap and sustainable fingerprint powder sourced from natural materials. In this research, powders were prepared from the dried petals of *Bahunia purpurea*, *Bougainvillea*, *Amaranthus cruentus*, *Hibiscus rosa-sinensis*, and *Hibiscus grandiflorus* using mechanical methods. The dried petals were ground and kept in an oven, and ash powder was also prepared using a muffle furnace. The grain size of the prepared fingerprint powders was determined. The durability of these novel powders was evaluated on non-porous surfaces including mobile screens, window glass, doors, keyboards, and glass bottles for the development and visualization of fingerprint impressions. The purpose of this research is to investigate the efficacy and durability of natural powders in forensic investigations. This study sets the stage for discovering new, safer, and more affordable fingerprint powder alternatives.

Submission ID: EE-174

Real-Time Smart Monitoring by Smart Meter

Darshana Wange, Tejas Nikhare, Prasad Raut, Aachal Ghatale

Students, Department of Electrical Engineering,
G. H. Rasoni University, Amravati, India

This study aims to demonstrate a Smart Energy Meter (SEM) that efficiently reduces errors in electricity bill generation. The proposed SEM is a low-cost device designed to function effectively within the Indian context, integrating modern technology with the existing electrical infrastructure. It includes only a current sensor and essential circuits. Compared to conventional SEMs, the proposed model offers advantages in terms of accuracy, simplicity, and cost-effectiveness. The estimated construction cost ranges between 3,000 to 3,500 INR. Notably, it avoids complex electronics while maintaining an accuracy range of approximately 0.71% to 0.93%, which is comparable to standard meters.

The system employs sensors to monitor the usage duration of household appliances, aiding in the digitization of the billing process. A GSM (Global System for Mobile Communication) module informs users of their energy consumption, while an Android app displays the real-time system output. Monitoring is handled by an ARM7 microprocessor, and a Wi-Fi module supports continuous data updates and appliance control. This smart meter solution addresses the issue of electrical overloading by offering real-time monitoring and control capabilities.

Keywords: Wi-Fi module, ARM7, GSM module, relay module, smart energy meter

Submission ID: EE-179

A High Voltage Gain Bidirectional DC-DC Converter for V2G and G2V Hybrid EV Charger

Mr. Shubham Rambhau Ahirkar¹, Prof. C.M. Bobade²

¹Student, ²Head of Department,
Department of Electrical Engineering G. H. Rasoni University, Amravati

This research proposes a novel wide-range bidirectional DC-DC converter designed for electric vehicle (EV) applications, offering a high voltage gain transfer ratio. The converter ensures low voltage stress on switches, a high utilization factor, and elevated efficiency, while maintaining a common electrical ground between the input and output terminals. The performance of the proposed converter is evaluated in both Vehicle-to-Grid (V2G) and Grid-to-Vehicle (G2V) modes, supporting bidirectional power flow in grid-connected EV charging systems.

A dead-beat current controller is utilized in both the DC-DC and DC-AC stages, enabling fast, precise, and smooth control response. Simulation results for a 1000 W, 380 V prototype in a closed-loop system demonstrate the capability of the proposed system under bidirectional power flow conditions. The theoretical insights are validated by simulation outcomes, confirming the efficacy and practical relevance of the proposed topology for next-generation EV charging infrastructure.

Submission ID: CSE-198

A New Era of Computation: Exploring the Foundations, Challenges, and Potential of Quantum Computing

Aditi Dange¹, Ajay Kumar²

G. H. Rasoni University, Amravati, Maharashtra, India
aditidange53@gmail.com
ajay.kumar@ghrua.edu.in

Quantum computing represents a paradigm shift in computational science, utilizing the principles of quantum mechanics to perform complex calculations far beyond the reach of classical computers. This paper presents a comprehensive overview of the foundational concepts, operational mechanisms, and current advancements in quantum computing. It explores significant applications across various domains—including cryptography, drug discovery, machine learning, and large-scale optimization—where quantum approaches offer substantial advantages.

In addition, the study discusses the current limitations, technological challenges, and emerging breakthroughs that shape the future trajectory of the field. Designed for students and researchers alike, this paper aims to demystify quantum computing and highlight its growing relevance in solving real-world problems.

Submission ID: CE-203

A Review on Dynamic Analysis on the High Varying Chimney for the Fix and Flexible Base

Mr. Suhas S. Pawar¹, Prof. Dr. Swati Ambadkar²

¹Student, M.Tech., ²Assistant Professor, Department of Civil Engineering,
G. H. Raison University, Amravati, Maharashtra, India

Chimneys or stacks are essential industrial structures used for the emission of toxic gases or smoke from boilers, stoves, furnaces, or fireplaces to higher elevations, ensuring minimal environmental contamination. These structures are generally tall and slender, with circular or cylindrical cross-sections. Various construction materials such as concrete, steel, and brick masonry are commonly used for chimney construction. Steel chimneys are especially suited for process industries requiring short-term heat-up periods and minimal thermal capacity.

Chimneys must be designed to withstand seismic and wind-induced loads, making dynamic analysis crucial. The core objective of this review is to analyze the seismic response of reinforced concrete chimneys with both fixed and flexible bases. The analysis considers various soil conditions—particularly hard soil strata—and compares the behavior of chimneys with heights of 100 m, 120 m, and 180 m when subjected to seismic forces. This study contributes to understanding the impact of base conditions and soil type on chimney stability under earthquake loading.

Keywords: Chimney analysis, dynamic loading, seismic analysis, fixed and flexible base, soil conditions, structural response

Submission ID: FSD-074

Human Stature Estimation from Three-Finger of Left and Right Hand

Ms. Qurratul Ain Kausar Ansari¹, Sakshi D. Hirapure¹, Sanjana R. Bharote¹, Supriya D. Pawara¹, Mangesh S. Dhore¹

¹Department of Forensic Science, G. H. Raison University,
Anjangaon Bari Road, Amravati, Maharashtra 444701, India
kausar.ansari@ghrua.edu.in

Estimating human stature from fragmented remains is a longstanding challenge in forensic science. This study explores the relationship between finger length and overall height using anthropometric measurements, with a specific focus on three fingers—thumb, index, and middle—of both the left and right hands. A total of 200 samples (100 males and 100 females) were analyzed.

The results demonstrate that index finger length is a significant predictor of human height. Pearson correlation coefficient values were calculated to quantify the relationship: for males, height vs. right index finger length (RIFL) was 0.4857 and height vs. left index

finger length (LIFL) was 0.5645; for females, height vs. RIFL was 0.416 and height vs. LIFL was 0.407. These findings indicate a statistically meaningful correlation between stature and index finger length in both genders.

Further comparison with thumb and middle finger lengths confirmed that the index finger provides the strongest correlation, making it a reliable anthropometric marker for stature estimation.

Submission ID: CSE-097

Leveraging Generative AI for Test Case Creation in Complex Systems

Sachin Shankarrao Ajmire¹, Karuna Vivekanand Agarkar²

Research Scholars, G H Raison University, Amravati, India

sachinajmire7@gmail.com

karuna.agarkar@gmail.com

Modern software systems are increasingly complex, necessitating advanced testing methodologies to ensure their reliability and functionality. Traditional manual testing methods often fail to cope with this complexity, resulting in insufficient test coverage and elevated risk of undetected issues.

This paper investigates the potential of Generative Artificial Intelligence (GAI) to transform test case creation in complex systems. Techniques such as Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) are explored for generating diverse, realistic test inputs, simulating user behaviors, and identifying critical edge cases.

The advantages of GAI-driven testing include enhanced test coverage, improved fault detection, and faster testing cycles. Challenges such as data quality, model interpretability, and establishing robust evaluation metrics are also discussed.

Finally, a roadmap is proposed for the successful implementation of GAI in practical testing environments, highlighting the importance of human-in-the-loop strategies and ongoing model refinement to maximize effectiveness.

Submission ID: FSD-098

Comparative Study of Extraction of *Calotropis Procera* and *Calotropis Gigantea* Poison

Abhaya A. Bhople, Minakshi D. Sarkar, Sakshi D. Gobhekar, Prajakta P. Dalavi, Aditi S. Chaure

Department of Forensic Science, G.H. Raison University, Anjangaon Bari Road, Amravati, Maharashtra 444701, India

abhaya.bhople@ghrua.edu.in

This research investigates the toxicological properties of *Calotropis procera* and *Calotropis gigantea*, two plant species known for their dual roles in medicine and toxicity. The study compares the extraction of poison from leaves, flowers, roots, and latex of both species using Thin Layer Chromatography (TLC), chemical screening tests, and UV-visible spectrophotometry.

Key toxic constituents identified include calotoxin, calactin, tannins, alkaloids, and cardiac glycosides. TLC analysis partially succeeded in separating chemical components,

while chemical screening confirmed the presence of alkaloids, tannins, and cardiac glycosides in both species. UV-visible spectrophotometry yielded limited results due to sample impurities and low concentrations of active compounds.

C. procera was extracted more effectively than *C. gigantea*, significantly affecting biochemical and physiological functions. Influencing factors include geographic variation, environmental conditions, and limitations of extraction techniques.

The study highlights the forensic relevance of these plants in poisoning cases and suggests their potential therapeutic applications. It underscores the need for advanced methodologies and broader studies involving diverse geographic regions and improved solvent systems. Despite its limitations, this research provides valuable insights into the toxicological and forensic implications of *Calotropis procera* and *Calotropis gigantea*.

Submission ID: ME-207

Review of Design and Analysis of Drill Head of Boring Machine by Using Vibration Conditioning Monitoring

Ved Kulat, Dr. Sachin Meshram

M.Tech in CAD/CAM Engineering, G H Raison University, Amravati

In modern manufacturing, ensuring precision, efficiency, and longevity in machine components is crucial, especially for high-stress equipment like boring machines. These machines are essential for drilling large holes in various materials and are subjected to intense operational demands. Excessive vibrations in drill heads during boring can cause inaccurate machining, increased tool wear, and unexpected downtimes, significantly impacting productivity, quality, and system reliability.

Ineffective vibration monitoring and mitigation result in reduced operational efficiency, higher maintenance costs, and frequent equipment replacements. Therefore, it is essential to develop methods to accurately assess and control vibrations in boring machines.

This paper focuses on designing and analyzing a drill head for a boring machine using vibration conditioning monitoring to optimize structural integrity and minimize operational vibrations. A 3D model of the drill head is developed in SolidWorks, incorporating design modifications to enhance vibration resistance. The model is analyzed using ANSYS for finite element analysis (FEA) to evaluate stress distribution, deformation, and vibration modes under operational loads. Real-time vibration data from sensors is integrated into simulations to assess the impact of design changes.

The ultimate goal is to optimize drill head design to reduce vibrations, improve machining accuracy, extend tool lifespan, and enhance overall machine efficiency and reliability.

Submission ID: EE-024

State-of-the-Art Machine Learning Integration in Energy Management Systems: A Comprehensive Review

Mr. Saurabh Thakurdware¹, Mr. Ganesh Wakte², Dr. Radharaman Shaha³,
Dr. Mukesh Kumar⁴

^{1,2,3}Tulsiramji Gaikwad Patil College of Engineering and Technology, Nagpur, India
thakursaurabh2001@gmail.com

ganeshwakte1989@gmail.com

⁴G.H. Rasoni University, Amravati, India
mukeshkr.iitbhu@gmail.com

As a result of the growing need for improved energy management, advanced machine learning (ML) techniques have begun to be incorporated into energy management (EM) systems. Existing models are often inadequate in tackling issues associated with the large-scale, ever-changing configuration and dynamics of modern energy systems.

This paper delves into the enhancement of management systems through machine learning and highlights its applications in demand forecasting, load optimization, anomaly detection, and energy consumption prediction. The study explores various ML techniques, especially: (1) supervised learning such as regression and decision trees, (2) unsupervised learning like clustering and dimensionality reduction, and (3) reinforcement learning, emphasizing their significance in energy systems engineering.

The results demonstrate that ML-augmented energy management systems significantly improve both component-level and system-wide performance by increasing the accuracy of energy forecasting, enabling more flexible solutions, and integrating diverse components. Supervised learning methods are particularly effective for demand forecasting, while reinforcement learning supports real-time optimization. Unsupervised methods help identify outliers and cluster energy usage patterns.

However, limitations such as data quality, processing constraints, and challenges in real-time application persist. The study concludes by proposing future research directions, including the integration of ML techniques with IoT and smart grid technologies to enhance the performance and sustainability of EMS.

Submission ID: CSE-021

Orange Fruit Disease Identification Using CNN and Deep Learning

Prapti Ghotkar, Dr. Jayant Mehare

Department of Artificial Intelligence, G H Rasoni University, Amravati, India

The emergence of plant diseases poses serious threats to crop health and yield, endangering the global agricultural landscape. Orange plants, which are vital to the citrus sector, are especially susceptible to various diseases that can cause significant economic losses. This study proposes a novel deep learning-based approach to detect diseases in orange plants using convolutional neural networks (CNNs). The model distinguishes between healthy and diseased samples by analyzing a diverse set of images. To enhance

multi-scale feature extraction, a hybrid CNN architecture combining EfficientNetB0 with Inception modules is developed. Transfer learning is employed to expedite training and improve classification accuracy. The proposed system offers a scalable and effective solution for early disease identification in citrus plants, aiming to improve the resilience and productivity of the citrus industry.

Submission ID: EE-023

Advances in Regenerative Braking Systems: Design, Optimization, and Future Trends for Sustainable Hybrid Electric Vehicles

Mr. Sarang Rathod¹, Mr. Ganesh Wakte², Mr. Praful Ghadge³, Dr. Mukesh Kumar⁴

^{1,2,3}Tulsiramji Gaikwad Patil College of Engineering and Technology, Nagpur, India
mrsarangrathod@gmail.com, ganeshwakte1989@gmail.com,
praful.electrical@tgpcet.com

⁴GH Rasoni University, Amravati, India
mukeshkr.iitbhu@gmail.com

Regenerative braking systems (RBS) have emerged as a vital innovation in hybrid electric vehicles (HEVs), contributing significantly to energy conservation and environmental sustainability. By capturing the kinetic energy typically lost during braking and converting it into electrical energy, RBS reduce dependence on traditional friction brakes and enhance overall vehicle efficiency.

This paper provides a comprehensive overview of regenerative braking systems, focusing on their design principles, optimization techniques, and emerging trends. It explores the key system components, strategies for energy recuperation, torque distribution methodologies, and integration challenges with conventional braking systems. The analysis also extends to the influence of RBS on fuel consumption, emission reduction, battery degradation, brake wear, and vehicle dynamics.

The study highlights that regenerative braking contributes to improved fuel economy and extended driving range. Advanced optimization techniques, such as genetic algorithms and machine learning models, are increasingly employed to maximize energy recovery and system performance. Furthermore, the integration of RBS with intelligent energy management systems is shown to be critical for efficient HEV operation.

Despite these advancements, challenges persist in enhancing energy recovery rates, managing system complexity, and achieving seamless integration. Future developments are expected to focus on advanced battery technologies, innovative brake materials, and the incorporation of RBS into autonomous vehicle platforms, thereby advancing the sustainability and performance of next-generation HEVs.

Submission ID: ECE-040

The Role of Open Hardware Ecosystem in Transforming Education and Industry in India

Avinash Shetty¹, Vedant Taware²

abhiapril122005@gmail.com, vedanttaware24@gmail.com

^{1,2}Bachelor of Science in Computer Science, Sinhgad College of Science, Pune, India

The open hardware ecosystem has the potential to transform education and industry in India by promoting innovation, accessibility, and collaboration. By enabling public access to hardware designs and documentation, open hardware allows for low-cost customization and widespread knowledge sharing. This research explores its transformative impact, particularly in the Indian context, identifying current challenges, potential opportunities, and strategic approaches for broader adoption.

In the education sector, open hardware fosters experiential learning through maker spaces, DIY electronics, and hands-on skill development. Integration into school curricula prepares students for real-world problem-solving, aligning educational outcomes with industry demands. For industry, open hardware offers an affordable and adaptable path for research and development, especially in electronics manufacturing, agri-tech, and healthcare.

To harness these benefits, India must support open hardware through policy interventions, academic-industry collaborations, and grassroots innovation initiatives. Barriers such as low awareness, inadequate infrastructure, and over-reliance on proprietary solutions must be addressed. Strategic investment and collaboration can position India to leverage open hardware for inclusive growth, technological self-reliance, and entrepreneurial expansion, aligning education and industry with global innovation trajectories.

Submission ID: ECE-140

Review on IoT Based Underground Cable Fault Detection System

Mr. Rushabh Sathawane Student, Rohini Pochhi, Professor,
Amol Dhenge, Assistant Professor, Dr. Sanjay Asutkar, Professor,

Department of Electronics & Communication Engineering, TGPCET, Nagpur

The increasing demand for electricity and the growing complexity of power distribution networks necessitate efficient and reliable fault detection systems. Underground cables, widely used in modern power transmission, frequently face challenges due to environmental factors and physical damage that cause faults. Current fault detection methods are slow and labor-intensive, relying heavily on manual inspection and diagnostic processes.

This paper proposes an IoT-based Underground Cable Fault Detection System designed to detect, monitor, and accurately locate faults over distances measured in kilometers. This approach offers a more efficient and automated solution, improving fault management and reducing downtime in power distribution networks.

Submission ID: ECE-146

Development of IoT Based Flood Monitoring and Alerting System Using Raspberry Pi

Miss Pragat N. Dahat Student,
Dr. Pravin Tajane Professor, Prof. Shutika Wankhade Assistant Professor,
Department of Electronics and Communication Engineering,
Tulsiramji Gaikwad-Patil College of Engineering & Technology, Nagpur

Flooding is one of the world's most damaging natural calamities; when it occurs heavily, it has the power to take many lives and devastate entire towns. A primary challenge in flood monitoring is the delay in receiving timely data and the limited relevance of that data, as timing is critical.

This study proposes a real-time flood monitoring system that employs multiple sensors connected to a single-board computer to compute crucial parameters for flood monitoring. Furthermore, the system allows users with internet access to monitor conditions and predict potential overflow events via a web server.

The proposed system is cost-effective and easy to maintain. It updates the water level on the web server in real-time and provides alerts to the public for evacuation, enabling prompt and effective action during flood events.

Submission ID: CSE-054

Risk Assessment in Credit Card Fraud Detection Using Machine Learning Algorithms

Prof. Pranita Chaudhary Assistant Professor,
Ketan Agrawal, Raghvendra Kotgire, Anuj Chandak, Abhay Ingle Students,
Department of CSE (AI ML), PCCOE, Pune
`pranita.chaudhary@pccoepune.org`, `ketan.agrawal24`, `raghvendra.kotgire24`,
`anuj.chandak24`, `abhay.ingle24@pccoepune.org`

Credit card fraud detection is crucial for reducing financial losses and ensuring secure online transactions. This paper investigates machine learning-based risk assessment models for credit card fraud detection, focusing on methods that improve detection accuracy while minimizing operational disruptions for legitimate users.

Through a comparative analysis of various algorithms—such as logistic regression, random forests, and neural networks—this study identifies techniques for effective fraud detection, evaluates model performance under cost-sensitive conditions, and examines their suitability for real-time deployment.

Findings indicate that advanced models, particularly ensemble methods, can effectively predict and assess the risk associated with credit card transactions, contributing to more robust fraud prevention strategies.

Submission ID: CSE-058

Advancements, Challenges, and Future Directions of Machine Learning Approaches in Thyroid Cancer Diagnosis

Ankita H. Harkare¹, Bhumika A. Neole², Tanvi S. Khapre³,
Ujwal P. Hiwase⁴, Vanshika N. Ramtekkar⁵

^{1,2,3,4,5} Ramdeobaba University, Formerly Shri Ramdeobaba College of Engineering and
Management,
Gittikhadan, Nagpur

{harkareah, neoleba, khaprets, hiwaseup, ramtekkarvn}@rknc.edu

Since thyroid cancer is one of the cancers with the fastest global growth rates, early identification is essential to bettering patient outcomes. Using regularly gathered patient data, such as demographics, medical history, biochemical markers, and ultrasound results, machine learning (ML) models have demonstrated potential in predicting disease risks.

This study finds the model with the highest prediction accuracy by assessing how well several machine learning algorithms predict the risk of thyroid cancer. Using a dataset comprising medical records and diagnostic reports, we examine many supervised learning methods, such as logistic regression, decision trees, random forests, support vector machines (SVM), and deep learning approaches.

Model efficacy is compared using performance indicators like accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (AUC-ROC).

Submission ID: CSE-121

AI-Based Student Information Management System

Mr. P.T. Talole¹, Mr. Ajay S. Ingle², Miss Divyani V. Patil³

¹HOD, ^{2,3}Students, Department of Information Technology Engineering,
Anuradha Engineering College, Chikhli

prmod.talole@aecc.ac.in, ingleajay03@gmail.com,
patildivyani1230@gmail.com

The Student Information Management System (SIMS) plays a crucial role in managing student records efficiently in educational institutions. Traditional SIMS rely on manual record-keeping and basic digital databases, limiting their ability to provide intelligent insights. Web-based SIMS improved accessibility and automation but still lacked advanced decision-making capabilities.

AI-based SIMS revolutionize student management by integrating machine learning, natural language processing, and predictive analytics. This paper explores the evolution of SIMS, comparing traditional, web-based, and AI-driven techniques. AI-based SIMS offer significant advantages such as automation, personalized learning, and predictive analytics, addressing key challenges in student information management.

The study concludes that AI-based SIMS enhance decision-making, improve administrative efficiency, and support students through intelligent automation.

Submission ID: CSE-125
Online Job Portal

Rakhi S. Zade¹, Shivani Waman Dapke², Manisha Rajesh Khatri³, Shubham Bhagwat Sonune⁴, Sadiya Badruddin Kazi⁵

^{1,2,3,4,5}Department of Information Technology, Anuradha Engineering College

zaderakhi2002@gmail.com, dapkeshivani1122@gmail.com,
manishakhatri2044@gmail.com, shubhamsonune584@gmail.com,
kazisadiya045@gmail.com

Online Job Portal searching is one of the large commercial and informative websites providing services to all kinds of people. In this competitive era, as education levels rise, job availability decreases. Companies seek highly skilled individuals, making it challenging to find suitable candidates. This project aims to automate the recruitment process, simplifying hiring.

The system manages the entire recruitment workflow, allowing candidates to apply for listed job vacancies efficiently. It facilitates application submissions for both applicants and companies. The administrator oversees all recruitment aspects to ensure smooth operations, except the evaluation phase.

The evaluation is excluded because it varies by company, with unique, non-standardized criteria making automation impractical. The system also incorporates admin layers to assign privileges, impacting functionality at different admin levels.

Submission ID: ME-204

Development and Optimization of Automatic Cotton Ginning Bell Press Plant

Dr. S. K. Bagde¹, Sagar Kumbhare²

¹Professor, Yeshwantrao Chavan College of Engineering, Nagpur, Maharashtra, India

²PG Student, Yeshwantrao Chavan College of Engineering, Nagpur, Maharashtra, India

skbagde@ycce.edu, sagarkumbhare@ycce.edu

India's cotton ginning industry ranks second globally, serving as a vital link between cotton farmers and the textile sector. Cotton ginning involves the separation of fiber from cottonseed, converting the harvested crop into lint—a commercially viable product. Most Indian ginning operations utilize domestically manufactured double roller gins, with around 36.5 and 38 million bales processed during 2012–13 and 2013–14, respectively, across 1500 modern and 2500 semi-modern ginneries.

This paper presents a review of current advancements in cotton ginning and bell-pressing technology. It highlights the role of composite ginneries in integrating ginning and pressing operations to produce pressed lint bales efficiently. The study explores the latest developments in automation and optimal equipment configuration to enhance the quality and efficiency of ginning operations. Furthermore, the paper outlines prospective innovations and future directions aimed at the technological enhancement and optimization of the cotton ginning and bell-pressing processes.

Submission ID: CSE-045

Virtual Approach to Treat Phobia

Shruti Hemant Deshmukh, Saachi Yogesh Pathak,
Prathmesh Shyam Vaidya, Om Ojha, Ashish Soni

Department of Computer Science and Engineering,
G.H. Raison College of Engineering, Nagpur 440016, India

Symptoms of severe panic attacks include anxiety, sweating, and light-headedness. Treating such attacks often requires personalized planning. Current tools lack the ability to monitor emotional and physiological statuses in real time. This project proposes an application to measure and manage panic development using data-driven methods.

The app incorporates real-time emotion tracking and physiological sensors to capture emotional states (e.g., happy, scared, anxious, sad) and biometric parameters like sweating and shivering. Advanced machine learning (ML) models, especially time series analysis, use historical data to assess fear levels, predict symptom severity, and provide insights into fear progression.

Doctors can monitor patients remotely and adjust treatments in real time. This ensures personalized care, predicts triggers, and prevents symptom escalation. The app enhances access to fear treatment, emphasizing the importance of digital tools for individual supervision.

A discussion with AIIMS, Nagpur's Psychiatry HOD, underlined the necessity of this tool to promote mental well-being. With 92% accuracy using a random forest model, the app identifies phobia stages and enables timely treatment. It delivers actionable insights and supports care strategies, ensuring better outcomes for teens, adults, and seniors while fostering overall psychological health.

Submission ID: CSE-056

A Driving Decision Strategy Using Machine Learning for an Autonomous Vehicle

Mrs. Shweta A. Gahalod, Dr. Amit K. Gaikwad

Department of Computer Science and Engineering,
G H Raison University, Amravati, India

The emerging technology has provided a huge platform for many inventions and developments to be functionalized and utilized for enhancing human life. The concept of making an autonomous vehicle is to enhance human driving abilities by substituting manual driving with artificial intelligence, thereby improving safety, preventing accidents, and enabling smoother traffic flow.

Currently, autonomous vehicles base their driving strategies solely on external parameters (e.g., pedestrians, road conditions), often ignoring the internal state of the vehicle. To address this limitation, this paper proposes a Driving Decision Strategy (DDS) based on machine learning that accounts for both external and internal vehicle parameters (such as consumable conditions and RPM levels).

The DDS is trained using a genetic algorithm on cloud-stored sensor data from vehicles and is designed to determine the optimal driving strategy. The performance of DDS is validated and compared with Multilayer Perceptron (MLP) and Random Forest (RF) neural network models to demonstrate its effectiveness in real-world conditions.

Submission ID: CSE-155

AI-Based Instant Multi-Media Generation from Multi-Language Input Source

Samruddhi N. Deshmukh, Prof. Amit Sahu

G H Raison University, Amravati, India

In recent years, the intersection of artificial intelligence and creative industries has seen significant advancements, particularly in the field of text-to-image generation. This emerging technology allows for the automatic creation of visual content from textual descriptions, opening up new possibilities for creativity, content creation, and accessibility. The potential applications range from generating illustrations for stories and producing marketing materials to assisting individuals with visual impairments by converting textual information into images.

The proposed web application aims to harness these advancements by providing a platform where users can input textual descriptions and receive corresponding images generated by an AI model. The application will utilize sophisticated algorithms that understand the nuances of natural language and translate them into visual representations. By doing so, it will simplify the process of image creation, making it possible for users with little to no design experience to generate high-quality visuals with ease.

Submission ID: CSE-123

A Real-Time Threat Detection System

Prof. N. B. Mapari¹, Mr. Shantanu S. Sultane², Mr. Soham A. Bargir³

¹HOD, ^{2,3}Students, Department of Information Technology,
Anuradha Engineering College, Chikhli

¹nagas7366@gmail.com, ²shantanusultane49@gmail.com, ³sohambargir@gmail.com

The expanding arena of cyber threats necessitates robust and real-time security measures for safeguarding websites and online applications. This paper presents the design and implementation of a Threat Detection System (TDS) that continuously monitors web traffic, identifies malicious activities, and generates instant alerts to mitigate potential security breaches.

Developed using Python for core logic and MySQL for backend data storage, the system incorporates real-time analytics, malware scanning, and adaptive security profiles. Key features include a proactive alert mechanism, live threat visualization, and customizable user configurations to enhance system responsiveness.

The proposed system aims to improve cyber defense mechanisms by delivering real-time awareness and early warning capabilities, enabling administrators to act swiftly against threats. This framework contributes to strengthening website security and promoting safer online environments.

Submission ID: CSE-178

Approach for Self-Authentication Protocol for Node Verification Without Authority in Ad-Hoc Network

Vaishali Jangade, Dr. Smita Nirkhi

Department of Computer Science and Engineering,
G H Rasoni University, Amravati, Maharashtra, India

Vehicular Ad Hoc Networks (VANETs) play a pivotal role in enhancing transportation systems by enabling real-time communication between vehicles and infrastructure. However, traditional authentication mechanisms rely heavily on Roadside Units (RSUs) or centralized authorities for node verification, which introduces latency, scalability issues, and vulnerability to single points of failure.

This paper proposes a novel self-authentication protocol for VANETs, eliminating the need for RSUs or centralized authorities. The protocol leverages advanced mathematical models, including Zero-Knowledge Proofs (ZKPs) and Identity-Based Cryptography (IBC), to enable decentralized, secure, and efficient authentication among nodes. The proposed system ensures robust identity verification by embedding cryptographic trust mechanisms directly within the network nodes, enhancing security while reducing dependency on external infrastructure.

Furthermore, the protocol incorporates dynamic trust evaluation and threshold cryptography to prevent malicious activity and ensure reliable communication. Comparative analysis demonstrates that this approach offers improved scalability, reduced complexity, and enhanced resilience against cyberattacks compared to traditional RSU-dependent systems. This self-authentication protocol empowers VANETs to operate autonomously, making them suitable for large-scale, real-time deployments in smart cities and intelligent transportation systems. The study highlights the potential of decentralized security models, paving the way for future research in fully autonomous, secure vehicular communication networks.

Submission ID: CSE-059

AI-Powered Threat Detection in the Cybercrime Underground Economy

Malleshwari Doraiswamy, Prashant Adakane

G H Rasoni University, Amravati, India

The foundations of the field and methodologies that could help inform information systems researchers and practitioners who deal with cybersecurity have not received much attention, despite the quick rise in cyberthreats. Furthermore, the criminal business model known as Crime-as-a-Service (CaaS), which serves as the foundation for the underground cybercrime industry, is not well understood.

This study explores the underground economy of cybercrime using a data analytics approach from a design science perspective. A data analysis framework is proposed for examining the definitions of crimeware, CaaS, and the underground cybercrime ecosystem, along with a corresponding classification model. A sample application is developed to demonstrate the implementation of the proposed framework and model.

The application is used to analyze a sizable dataset gathered from the Kaggle website to investigate trends within the underground cybercrime economy. This study contributes to design artifacts, foundational understanding, and methodologies in the cybersecurity domain through a design science research approach. Furthermore, it offers practical insights and proposed guidelines for organizations and governments to prepare for and mitigate underground cybercrime threats across industries.

Submission ID: EE-066

Novel Method Based on TEO and Decision Tree Approach for Power Transformer Protection

Patil Bhushan Prataprao¹, Dr. Shah Paresh Jaychand²

¹R. C. Patel Institute of Technology, Shirpur, Maharashtra, India

²Shri Balaji Institute of Technology and Management, Betul, (M.P.), India

Ensuring effective transformer protection is crucial for maintaining power system stability. This paper introduces a robust fault detection and classification approach using the Teager Energy Operator (TEO) and Decision Tree (DT)-based classifiers. The method efficiently distinguishes internal faults from external disturbances, including magnetizing inrush currents and overexcitation.

A transformer model is designed in MATLAB Simulink, simulating various fault conditions, inrush scenarios, overexcitation, and normal operations under different load levels. The detection and classification processes are executed in Python, where TEO-based feature extraction provides input for DT classifiers. To improve accuracy, GridSearchCV optimization is employed for fine-tuning parameters and evaluating performance across diverse training and testing datasets.

The proposed technique achieves reliable event classification within half a cycle.

Submission ID: EE-126

High Gain Bidirectional DC-DC Converter for EV Drive with Regenerative Braking

Dr. Amit S. Borole, Sapna B. Shinde

Department of Electrical Engineering,

Maharashtra Institute of Technology, An Autonomous Institute, Aurangabad, India

This paper presents a new and efficient bidirectional DC-DC converter for electric vehicle (EV) applications, utilizing a two-transistor analogy. The main goal is to improve the connection and coordination of the battery system. The proposed converter design provides high voltage output in a compact form with fewer components, resulting in better overall efficiency.

A high-gain bidirectional DC-DC converter is developed to manage battery charging efficiently during regenerative braking. It facilitates energy transfer from the battery to the motor and enables effective energy recovery during braking. The system employs two transistors in its design to control bidirectional power flow, ensuring reliability and cost-effectiveness.

An optimized control approach is introduced to enhance converter performance by minimizing energy losses and boosting overall efficiency. The two-transistor configura-

tion aids in maintaining high charging efficiency using recovered braking energy. A prototype hardware model has been developed and tested. Additionally, Proteus simulations validate the design and performance of the converter.

This approach enhances energy efficiency, reduces battery wear, and ultimately improves EV performance. It contributes to the advancement of environmentally friendly and sustainable transportation solutions.

Submission ID: EE-197

A Review on Real-Time Optimal Power Flow for Transmission Network

Mr. Bhushan B. Taksande, Dr. Swapnil B. Mohod, Prof. Bhushan H. Band

Prof. Ram Meghe College of Engineering & Management, Badnera - Amravati
bhushantaksande3@gmail.com, swapnil.mohod@prmceam.ac.in,
bhushan.band@prmceam.ac.in

Many features of modern power systems are beginning to include renewable energy sources, which pose significant challenges to the efficient operation of power transmission networks. This article titled "Real-Time Optimal Power Flow for Transmission Network" discusses how to achieve efficient and reliable power flow using integrated real-time system optimization for power generation and supply.

Through a Distributed Control System (DCS), enhanced interaction is established between generators and control units, improving efficiency in power flow via Optimal Power Flow (OPF) algorithms while minimizing generation costs and maintaining load balance. Real-time feedback control allows dynamic system adjustment to ensure frequency stability and additional control capabilities.

This system accommodates flexible renewable energy sources by utilizing real-time data instead of relying solely on predictions, aligning with the principles of adaptability and sustainability. Sensor data is analyzed in real-time through MATLAB-based simulations to derive scalable and robust solutions for modern power system challenges.

This paper presents a novel approach that strengthens adaptability to increasing complexity in energy management, supporting the development of efficient, reliable, and sustainable power systems.

Submission ID: MBA-083

Customer Attitude Towards Fraud Prevention Measure in Mobile Banking

Ritika S. Bokde MBA Student, Prof. Suraj Kodarlikar Assistant Professor,
Aditi Umathe MBA Student, Anand Bhagat MBA Student,
Department of Management Studies
G. H. Raison College of Engineering, Nagpur, Maharashtra

Mobile banking has transformed the banking sector rapidly due to technological advancements. However, the increasing adoption of mobile banking has also brought significant concerns regarding security and fraud. This study investigates customer attitudes toward fraud prevention measures in mobile banking, with a focus on demographic influences and awareness levels in the Vidarbha region of India.

The study uses primary data collected via a structured questionnaire distributed through Google Forms and secondary data from a literature review. It explores the relationship between customers' awareness of fraud prevention measures and their usage of mobile banking services, along with the impact of demographic variables.

Based on a sample of 252 respondents, primarily from Nagpur and surrounding areas, findings show strong adoption of mobile banking among young adults aged 18 to 24 years. Statistical tools, including chi-square tests, reveal a significant relationship between fraud prevention awareness and mobile banking usage, and refute the hypothesis that demographic factors have no influence.

The research highlights key challenges faced by customers in fraud prevention. Insights into customer perceptions can help banks implement more targeted and effective security strategies, thereby fostering trust and improving the overall mobile banking experience.

Submission ID: MBA-086

The Influence of Salary Structure on Investment Strategies: A Study of Fixed Income Employees

Shubham R. Shripatre, Sahil M. Ghodeswar, Prof. Suraj Kodarlikar, Kartik R. Sarkar,
Department of Management Studies
G. H. Rasoni College of Engineering, Nagpur

This study investigates the influence of salary structure on the investment strategies of fixed-income employees. Two hypotheses were tested using Chi-square analysis to examine the relationship between salary levels, risk tolerance, and portfolio reallocation. The first hypothesis explored whether higher salary levels correlate with a greater willingness to invest in higher-risk assets. The second hypothesis assessed whether a significant salary increase leads to a reallocation of investment portfolios toward riskier assets.

The study sample consisted of 252 respondents, including youngsters, working professionals, and individuals investing their salaries. Data was collected through a structured questionnaire distributed via Google Forms and analyzed using descriptive statistics and Chi-square tests. The results strongly supported both hypotheses, with significance values of 0.001, indicating that salary structure plays a crucial role in shaping investment strategies. Higher salary levels were associated with increased risk tolerance, while salary increases prompted a reallocation of portfolios toward riskier assets.

These findings highlight the importance of financial stability and income growth in influencing risk-taking behavior and investment decision-making among fixed-income employees. The study provides valuable insights for financial planners and employers aiming to understand and guide employee financial behaviors based on salary structures and individual financial circumstances.

Submission ID: MBA-088

Enhancing User Experience in E-Banking Applications: Analyzing Consumer Challenges and Developing Effective Solutions

Anudha Prakash Chawke MBA Student, Suraj Kodarlikar Assistant Professor, Vaibhav Devidas Dange MBA Student, Yashlesh Shailendra Gupta MBA Student,
G. H. Rasoni College of Engineering, Nagpur, Maharashtra

This study analyzes consumer awareness and challenges in using e-banking applications, with a particular focus on the safety of personal data in the Vidarbha region of India. The research investigates how different age groups perceive data security and their engagement with electronic banking platforms.

Data was collected through a structured questionnaire distributed online via Google Forms and supplemented with secondary data. The sample includes 251 respondents from Nagpur city and surrounding areas. Findings indicate that users aged 18 to 34 years dominate the usage of e-banking platforms, reflecting a growing ease and familiarity with such services among younger generations.

Chi-square analysis confirms significant associations between age groups and both engagement levels and concerns related to personal data safety. While most respondents have formal educational qualifications, the results underscore the continued need for targeted education on e-banking safety practices.

The study provides valuable insights into the scenarios where consumers are most active in e-banking and evaluates their perceptions of data security. These findings can guide banks in enhancing user experience by addressing consumer concerns, promoting safe usage, and tailoring services for different demographic segments.

Submission ID: MBA-115

Evaluating the Role of Mobile Trading Applications in Enhancing Retail Participation in Options and Futures

Janvi Ghode Isha Umare
Department of Management Studies
G. H. Rasoni College of Engineering, Nagpur

This study investigates the impact of mobile trading platforms on retail participation in India's derivatives market, focusing on user experience, analytical features, costs, and challenges. A structured survey was conducted among 350 retail investors across the Vidarbha region, segmented by experience level, age, and trading frequency. Data analysis employed ANOVA, correlation studies, and hypothesis testing to identify patterns and relationships.

Findings reveal that while most users find mobile trading platforms accessible and user-friendly, there are notable concerns regarding analytical capability, execution reliability, and satisfaction in specific features. Factors such as platform security, transaction fees, and execution delays showed limited impact compared to personal trading strategies and broader market conditions. Weak correlations between key variables were observed, though issues related to trade execution showed a slight negative influence on trading frequency.

The study also highlights potential risks of mobile trading, including impulsive behavior, overconfidence, and insufficient risk management—often exacerbated by overly simplified interfaces that may mask the complexity of derivatives trading. To promote responsible trading, the study recommends the integration of AI-driven risk assessment tools, improved investor education, and stronger regulatory oversight.

While mobile apps enhance market accessibility for retail investors, improvements in execution stability, cost transparency, and cybersecurity are essential. These insights can assist platform developers, brokers, and policymakers in designing more effective mobile trading systems that align with user needs and market integrity.

Submission ID: MBA-136

A Comparative Analysis of OTT Platform Subscriptions and Cinema Ticket Expenses: Examining Evolving Consumer Spending Patterns in the Entertainment Industry

Darshana M. Bhangе Sobuj B. Samanta Ashutosh S. Rathі Payal G. Kolhe and

Prof. Suraj Kodarlikar

Department of Management Studies

G. H. Raisonі College of Engineering, Nagpur

The entertainment industry has undergone a significant transformation with the rise of Over-The-Top (OTT) platforms, challenging the traditional dominance of cinema halls. This study conducts a comparative analysis of consumer expenditure on OTT subscriptions versus cinema tickets, exploring underlying factors such as affordability, content diversity, accessibility, and technological advancements that influence consumer choices.

A structured quantitative approach was adopted, collecting data from 350 respondents through surveys. The analysis employs statistical techniques including Chi-Square tests to identify relationships between spending habits, consumer preferences, and demographic variables. Additionally, Power BI was used for visual representation and interactive dashboards, enhancing the interpretation of trends and patterns in entertainment spending.

Findings suggest that while OTT platforms offer affordable and accessible entertainment, traditional cinemas continue to appeal for premium content and social experiences. Hypothesis testing confirms that OTT platforms have significantly impacted cinema revenues and footfalls. However, the overall consumer expenditure on entertainment remains balanced between both mediums.

The study concludes with insights and recommendations for OTT providers, cinema operators, and marketers, advocating for hybrid distribution strategies and flexible pricing models. This research highlights the evolving landscape of consumer behavior in the entertainment domain and provides actionable strategies for industry stakeholders to adapt to shifting consumption patterns.

Submission ID: MBA-180

Enhancing the Presence of Sindhi Cuisine in the Hotel Industry: Challenges and Strategies

Ms. Ruby Y. Ghundiyal, Mr. Suyog N. Katole
Department of Management Studies
G. H. Rasoni University, Amravati
rubyghundiyal@gmail.com suyog.katole@ghrua.edu.in

Sindhi cuisine is an integral component of Sindhi culture, renowned for its rich flavors and unique culinary identity. Despite the appeal and diversity of traditional dishes such as Sindhi Kadhi, Sai Bhaji, Dal Pakwan, and Koki, their representation in mainstream hotels and restaurants remains limited. This study highlights the cultural significance of Sindhi food and investigates the barriers hindering its visibility in the hospitality sector.

The research identifies key challenges, including limited awareness, lack of chef training in regional cuisines, and inadequate marketing efforts. It also proposes actionable strategies to promote Sindhi cuisine within the hotel industry. Initiatives such as organizing regional food festivals, incorporating Sindhi specialties into regular menus, and leveraging digital platforms for outreach are suggested as potential solutions.

By implementing these strategies, hotels can enrich their culinary offerings while preserving and promoting regional heritage. The paper concludes by emphasizing the need for collaborative efforts between cultural institutions, culinary professionals, and hospitality educators to ensure Sindhi cuisine gains the recognition it deserves on both national and global platforms.

Submission ID: MBA-191

From Greenwashing to Governance: Strengthening Board Accountability in ESG Disclosure

Riyanka Baral
India International University of Legal Education and Research, Goa
riyakabaral@example.com

As environmental, social, and governance (ESG) considerations become increasingly integral to corporate strategy and stakeholder expectations, companies face growing pressure to substantiate their sustainability claims. This shift has also led to a rise in greenwashing—making exaggerated or misleading environmental claims. This research paper investigates the expanding phenomenon of greenwashing and highlights the critical role of corporate governance, particularly board oversight, in addressing and preventing such practices.

The study explores how ineffective governance structures can enable greenwashing to persist, exposing firms to legal liabilities, reputational harm, and financial risk. Through analysis of secondary sources including legislative frameworks, regulatory guidelines, sustainability reports, and case law, the paper examines the mechanisms by which boards can enhance the integrity of ESG disclosures.

The findings underscore that corporate boards play a pivotal role in promoting transparency, accountability, and compliance with evolving regulatory standards. Key governance tools such as board independence, ESG-related expertise, and robust internal audit

mechanisms are identified as crucial in deterring greenwashing. The research concludes by offering actionable recommendations to strengthen ESG governance and stresses that effective board oversight is essential to safeguarding stakeholder trust and ensuring the long-term credibility of sustainability efforts.

Submission ID: CE-091

How Work Ends Early in Construction

Mr. Hitesh Potpose, Prof. Pranav Lende
Department of Civil Engineering
G. H. Rasoni University, Amravati

Timely completion of construction projects is critical to reducing costs, satisfying client expectations, and ensuring the overall success of infrastructure development. This research explores practical strategies for achieving early completion of concrete road construction without compromising on quality.

The study focuses on key influencing factors including the quality of materials, efficient labor deployment, and effective cash flow management. By analyzing these parameters, the paper identifies optimized approaches that contribute to accelerating project timelines. The findings emphasize that coordinated project planning, skilled workforce utilization, and timely resource allocation are pivotal in ensuring faster yet reliable construction processes.

This work provides insights that can aid civil engineering professionals and contractors in improving project delivery timelines, thereby enhancing productivity and client satisfaction in the construction sector.

Submission ID: CE-062

Importance of Joints in Thin White Topping in PMGSY Roads – A Review

Anup Raut¹, Miss. Apeksha Chaudhari²

¹Research Scholar, ²Assistant Professor,
Civil Engineering Department, G H Rasoni University, Amravati, Maharashtra, India

Thin White Topping (TWT) has emerged as a highly effective and sustainable solution for the rehabilitation of deteriorated asphalt pavements, particularly in rural road networks such as those developed under the Pradhan Mantri Gram Sadak Yojana (PMGSY). TWT offers significant advantages, including enhanced durability, improved structural capacity, and better resistance to rutting and cracking, making it a preferred choice for extending the service life of existing pavements.

This review focuses on the critical role of joints in TWT systems for enhancing durability, ensuring efficient load transfer, and preventing structural failures such as cracking or debonding. It discusses various types of joints used in TWT, their design considerations, and best practices that influence long-term performance. Furthermore, the study identifies existing research gaps and proposes methodologies to optimize joint design specifically for Indian rural road conditions. The insights aim to support engineers and policymakers in improving the quality and longevity of PMGSY roads.

Submission ID: CE-063

Environmental Impact Assessment of Four-Laning of Nagpur-Katol Section of NH-353J: A Critical Study – A Review

Shivaji Deshmukh¹, Ms. Apeksha Choudhary²

¹Research Scholar, ²Assistant Professor,
Civil Engineering Department, G H Raisoni University, Amravati, Maharashtra, India

The four-laning of the Nagpur-Katol section of NH-353J is a significant infrastructure project aimed at enhancing connectivity and traffic efficiency. However, such large-scale developments pose considerable environmental challenges. This paper critically reviews the Environmental Impact Assessment (EIA) of the project, focusing on ecological, social, and economic aspects.

The study examines key environmental parameters, including air quality, water resources, soil stability, biodiversity, and socio-economic factors, while also addressing mitigation measures proposed in the EIA report. The review identifies potential gaps in the assessment and suggests improvements for sustainable infrastructure development. These recommendations aim to inform planners, engineers, and policymakers to better integrate environmental considerations into future road development projects.

Submission ID: CE-064

Study on Maintenance of Existing Bituminous Pavement

Akash Sanjay Rathod¹, Prof. Apeksha Choudhary²

¹M.Tech Student, ²Professor,
Department of Civil Engineering, G. H. Raisoni University, Amravati, Maharashtra, India

Bituminous pavements are widely used in road construction due to their durability, flexibility, and cost-effectiveness. However, continuous exposure to traffic loads, environmental factors, and aging leads to deterioration over time. Proper maintenance is essential to extend pavement life, ensure safety, and reduce long-term costs.

This paper discusses various maintenance strategies for existing bituminous pavements, including preventive, corrective, and rehabilitation techniques. Preventive maintenance, such as seal coating and crack sealing, helps to slow deterioration before major damage occurs. Corrective maintenance includes pothole patching and surface treatments to address localized failures. Rehabilitation techniques, such as overlaying and recycling, restore structural integrity and functionality.

The selection of an appropriate maintenance strategy depends on pavement condition assessment, traffic intensity, and environmental factors. Adopting a proactive maintenance approach improves pavement performance, enhances ride quality, and minimizes costly reconstruction. The paper emphasizes the importance of regular inspections, timely interventions, and the use of modern technologies such as cold in-place recycling and infrared repair methods for sustainable pavement management.

Submission ID: ECE-144

Design and Development of Real-Time Wireless Hand Gesture Controller Robot Using 32-Bit Microcontroller

Payal Gondane¹, Rohini Pochhi², Amol Dhenge³

¹Research Scholar, ^{2,3}Assistant Professor,

Tulsiram Gaikwad Patil College of Engineering Technology

payalagondane0@gmail.com rohini.ece@tgpcet.com amol.ece@tgpcet.com

This paper presents the development of a real-time wireless hand gesture-controlled robotic car using a 32-bit microcontroller. The proposed system enables the robot car's movements to be controlled through hand gestures interpreted by an Arduino-based controller. The system utilizes ultrasonic sensors for obstacle detection and a Bluetooth module for wireless communication between the gesture controller and the robotic car.

The hand gesture controller comprises flex sensors and an accelerometer to accurately capture the movements of the user's hand. The Arduino processes this sensor data and transmits control commands to the robot car's motors. Experimental results demonstrate that the proposed system can effectively control the movement of the robotic car through intuitive hand gestures.

The gesture-controlled robotic car has promising applications in areas such as surveillance, exploration, and rescue missions, where remote and hands-free control is advantageous.

Submission ID: ANY-105

Green Technology Innovations in Industrial Engineering: A Route Towards Sustainability

Somnath Singroul^{1*}, Atul Kumar Sahu², Nitin Kumar Sahu³

^{1,2,3}Department of Industrial and Production Engineering,

Guru Ghasidas Vishwavidyalaya (A Central University),

Bilaspur - 495009, Chhattisgarh, India

singroul8som@gmail.com

Green technology is a crucial parameter of sustainability and has become a pivotal focus in the field of Industrial Engineering. The relationship between sustainable development and green technology offers a promising outlook for reducing negative environmental impacts while enhancing industrial performance.

This study proposes the integration of green innovations such as Green Manufacturing, Lean Manufacturing, and Agile Manufacturing into industrial processes to meet sustainable development goals. It also explores the use of industrial engineering techniques—including supply chain optimization, lean methodologies, and environmentally conscious manufacturing—to promote sustainable practices across various sectors.

The paper reviews the application of different tools and techniques documented in various case studies and real-world practices. It highlights their effectiveness in supporting decision-making and achieving sustainability objectives within organizations. Ultimately, this work demonstrates how industrial engineering can act as a catalyst for green innovation and sustainability-driven transformation.

Submission ID: ANY-129

Modeling of Sustainable Supply Chain Management Practices (SSCMP) and Performance Evaluation of Indian Steel Industry: By BWM Method

Sanju Kumar Nishad^{*a}, Atul Kumar Sahu^b, Somnath Singroul^c

^{a,b,c}Department of Industrial and Production Engineering,
Guru Ghasidas Vishwavidyalaya, Bilaspur - 495006, Chhattisgarh, India

*Corresponding Author: sanju.mechphd2022@gmail.com

This study demonstrates the modeling of Sustainable Supply Chain Management Practices (SSCMP) and evaluates their impact on environmental performance in manufacturing organizations, with a specific focus on the Indian Steel Manufacturing Industry (SMI). The SMI, which contributes significantly to goods production to meet societal demands, also generates considerable environmental load and waste.

The study emphasizes the growing academic and industrial interest in SSCM due to its role in enhancing environmental, social, and economic performance. A systematic Multi-Criteria Decision-Making (MCDM) approach—the Best-Worst Method (BWM)—is employed to map SSCM theories contributing to environmental sustainability.

By analyzing five selective case studies, the study identifies critical SSCMP constructs and dimensions to inform strategic decision-making for improving sustainability and competitiveness in the SMI. The research highlights the importance of SSCMP in addressing current environmental challenges and provides a consolidated knowledge base for future researchers exploring the effects of sustainable practices on environmental performance.

Submission ID: ME-184

A Comprehensive Review on Carbon Nanofiller Based Shape Memory Polymer Composites

Yugendra Kumar Sahu^{1*}, T.V. Arjunan², Samarjit Singh², Shubhra Vishwas¹

¹Researcher, Department of Mechanical Engineering,
Guru Ghasidas Vishwavidyalaya (Central University), Bilaspur – 495006, India

²Teaching Faculty, Department of Mechanical Engineering,
Guru Ghasidas Vishwavidyalaya (Central University), Bilaspur – 495006, India

*Corresponding Author Email: yugendra88sahu@gmail.com

Shape Memory Polymer Composites (SMPCs) have garnered significant attention in recent years for their ability to recover a predefined shape in response to external stimuli such as heat, electricity, or light. However, conventional SMPCs are often limited by low thermal conductivity, limited thermal responsiveness, and modest mechanical strength.

To address these limitations, carbon-based nanofillers—including carbon nanotubes (CNTs), graphene, graphene oxide (GO), and carbon nanofibers (CNFs)—have been incorporated into SMP matrices. These nanofillers significantly enhance the mechanical, thermal, and electrical properties of SMPs, enabling fast and remote actuation, and broadening their application in advanced engineering fields.

This review presents a comprehensive analysis of recent advancements in carbon nanofiller reinforced SMPCs (CNSMPCs). It explores the classification and character-

istics of various nanofillers, fabrication methods, and surface modification techniques for uniform dispersion. The influence of filler type, concentration, and morphology on composite performance—thermal, mechanical, electrical, and shape memory—is critically examined.

Furthermore, key characterization techniques and emerging applications in aerospace, biomedical devices, robotics, and electronics are discussed. The review concludes by outlining current challenges and proposing future research directions, offering valuable insights for researchers and engineers developing next-generation smart materials with multifunctional capabilities.

Submission ID: ME-185

A Review on Effect of Microchannel Shapes, Different Working Fluids on Heat Transfer Performance and Electronic Hotspot Cooling in Microchannel Heat Sink

Vikas Yadav¹, Viraj Palange², Abhishek Pawar³, Yash Shinde⁴, Prajwal Waghale⁵

^{1,2,3,4,5}Department of Mechanical Engineering,

Nutan Maharashtra Institute of Engineering Technology, Talegaon, Pune, India

Emails: vikasyadav@ncerpune.in, virajpalange@gmail.com,

pawarabhishek0205@gmail.com,

yashshinde2024@gmail.com, prajwalwaghale0906@gmail.com

The trend toward compact electronic devices has significantly increased heat generation per unit area, creating a pressing need for efficient thermal management to ensure the reliability and longevity of electronic components. Microchannel Heat Sinks (MCHS) have emerged as a promising solution due to their high surface area-to-volume ratio, which enhances heat transfer efficiency.

This review explores the effects of various microchannel geometries on thermal and fluid dynamic performance, particularly pressure drop and heat transfer characteristics. It emphasizes the influence of parameters such as microchannel diameter, aspect ratio, shape, surface roughness, and layout. Additionally, the study discusses the role of different working fluids—including conventional coolants and nanofluids—on heat dissipation effectiveness.

The findings suggest that MCHS provide superior performance compared to traditional heat exchangers, offering compactness and efficiency. However, challenges remain, such as pressure losses and fabrication complexities. The review also addresses current gaps and proposes future directions, including standardized design methodologies and the development of advanced experimental and simulation techniques for performance analysis.

This paper provides a comprehensive overview of the existing research on MCHS, with a specific focus on the impact of channel shape and fluid type on heat transfer performance and electronic hotspot mitigation.

Submission ID: ME-186

Design Solar Operated Portable Refrigerator Cum Oven

Mr. W.Y. Shaha

Student, Department of Mechanical Engineering,
G.H. Raisoni University, Amravati, Maharashtra, India

The growing need for sustainable and energy-efficient refrigeration and heating solutions has led to the development of a solar-powered portable refrigerator-cum-oven based on the Peltier effect. This project aims to design a compact, eco-friendly system that functions as both a refrigerator and an oven, making it ideal for off-grid applications such as outdoor activities, medical storage, and food preservation in remote areas.

The core of the system is a thermoelectric module that operates using the Peltier effect. When electric current is supplied, one side of the module absorbs heat (cooling effect), while the other side releases heat (heating effect). By reversing the current, the system can switch between refrigeration and heating modes. A solar photovoltaic (PV) panel serves as the primary energy source, ensuring a clean and renewable power supply. To maintain continuous operation during non-sunny periods, a battery storage unit is included.

Efficiency is enhanced through the use of thermal insulation materials to reduce heat loss and improve temperature regulation. A microcontroller-based control unit is incorporated to monitor and optimize power distribution, ensuring consistent performance. The device is designed to be lightweight and portable, making it suitable for rural use, emergency scenarios, and outdoor expeditions where conventional power sources are unavailable.

By employing solar energy and thermoelectric technology, this system eliminates the need for harmful refrigerants, thereby reducing environmental impact. The integration of portability, dual functionality, and sustainable energy use positions this device as a versatile and eco-friendly alternative to conventional refrigeration and heating systems.

Submission ID: ME-188

A Mini Review of Spark Erosion Machining Process for Tool Materials by Using Different Electrodes

M.S. Tufail¹, Krunal Borkar², Dr. J.P. Giri³, Ashutosh Bagade⁴

^{1,2,3}Yeshwantrao Chavan College of Engineering, Nagpur, Maharashtra, India

⁴DMIMS DU, Wardha, Maharashtra, India

Electric Discharge Machining (EDM) is a highly prominent nontraditional machining process particularly suited for "hard-to-machine" conductive materials such as heat-treated tool steels, composites, superalloys, ceramics, nitralloy, nemonic alloys, and carbides, which are commonly used as cutting tool materials. The Material Removal Rate (MRR) is a critical performance metric in EDM operations. The machining parameters that enhance MRR are strongly influenced by the interaction between the electrode (anode) and the workpiece surface, which defines the effective contact area and discharge characteristics.

This mini review focuses on various studies conducted over the years that investigate the influence of different electrode materials on the EDM performance, particularly on the machining of tool materials. By analyzing and comparing the findings from multiple research efforts, the paper aims to provide consolidated insights into how electrode properties such as conductivity, melting point, and wear rate affect the machining efficiency, surface finish, and tool life during EDM. The review concludes that selecting the appropriate electrode material is essential for optimizing MRR and achieving the desired machining performance in spark erosion applications.

Submission ID: ME-189

Supply Chain Resilience Practices in Indian Context: A Brief Overview

Vishal Rajput^{1,2}, Sachin Meshram², Pranav Charkha³

¹Research Scholar, ²Assistant Professor,

Dept. of Mechanical Engineering,

G.H. Raison University, Amravati, Maharashtra, India

³Professor, Pimpri Chinchwad University, Pune, Maharashtra, India

India's supply chain resilience is a significant concern, especially for Small and Medium Enterprises (SMEs), which are particularly vulnerable to disruptions caused by pandemics, natural disasters, economic fluctuations, and geopolitical uncertainties. These challenges often result in severe supply-demand mismatches and operational instability. This study aims to develop a comprehensive framework to enhance supply chain resilience by addressing multiple risk factors and improving dependability, quality assurance, and agility in decision-making.

The paper analyzes the specific difficulties faced by Indian SMEs in maintaining resilient supply chains and proposes actionable strategies to help them thrive in a dynamic market environment. It further explores various resilience frameworks through real-world case studies and simulation methodologies. Advanced technologies such as artificial intelligence (AI), machine learning (ML), and big data analytics are considered as tools for enhancing risk prediction, adaptive planning, and strategic recovery. The study provides insights that are valuable for both academia and industry stakeholders seeking to reinforce the robustness and agility of supply chains in the Indian context.

Thank You! ;